

General Relativity and Cosmology

Leonardo Cerasi



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Notation

Conventions

The totally-antisymmetric n -dimensional Levi-Civita symbol $\epsilon^{i_1 \dots i_n}$ is defined with the convention $\epsilon^{1 \dots n} = +1$.

Given $\alpha \in \mathbb{C}^{n \times n}$, with $n \in \mathbb{N}_0$, its complex conjugate is denoted as α^* , its transpose as $\alpha^\top$ and its Hermitian conjugate as $\alpha^\dagger := (\alpha^*)^\top$.

The Landau symbol for a function $f : D \subseteq \mathbb{R} \rightarrow \mathbb{C}$ is defined by the condition $\exists M \in \mathbb{R} : |o(f(x))| \leq M |f(x)| \quad \forall x \in D$.

The imaginary unit is denoted by i , so that $\mathbb{C}(\mathbb{R}) = \text{span}_{\mathbb{R}}(1, i)$.

Topology

The empty set is denoted by $\emptyset$ and the power set of a set A by $\mathcal{P}(A) := \{B : B \subseteq A\}$. The counting numbers are $\mathbb{N} \equiv \{1, 2, 3, \dots\}$, and the natural numbers are defined by $\mathbb{N}_0 \equiv \{0\} \cup \mathbb{N}$.

The n -dimensional sphere is denoted by $\mathbb{S}^n$, while the n -dimensional disk by $\mathbb{D}^n$, so that $\partial \mathbb{D}^n = \mathbb{S}^{n-1}$, where in general $\partial \Sigma$ denotes the boundary of Σ . The measure on $\mathbb{S}^n$ is denoted by $d\Omega_n$, while its line element by $d\Omega_n^2$.

The permutation group of n objects, i.e. the n^{th} symmetric group, is denoted by S^n .

Multilinear Algebra

Given two $\mathbb{K}$ -vector spaces V and W , with $\mathbb{K}$ a generic field, the space of all $\mathbb{K}$ -linear applications $f : V \rightarrow W$ is denoted by $\text{Hom}_{\mathbb{K}}(V, W)$: in particular, $\text{Hom}_{\mathbb{K}}(V, V) \equiv \text{End}(V)$. The subset of $\text{End}(V)$ of all automorphisms of V is the automorphism group $\text{Aut}(V)$, which is a group under composition of morphisms.

Given a $\mathbb{K}$ -vector space V , the space of all rank- (k, l) tensors on V is denoted by $T_l^k(V) = \bigotimes^k(V) \otimes \bigotimes^l(V^*)$, the space of exterior p -forms by $\bigwedge^p(V) \subseteq T_k^0(V)$ and the Grassmann algebra of V by $\bigwedge(V) = \bigoplus_{k=0}^n \bigwedge^k(V)$.

Differential Geometry

Given a fiber bundle $\pi : E \rightarrow M$, the space of smooth sections of E on M is denoted by $\Gamma(E)$.

Given a smooth manifold $\mathcal{M}$, the space of all smooth scalar functions on $\mathcal{M}$ is denoted by $\mathcal{C}^\infty(\mathcal{M})$, its tangent bundle by $T\mathcal{M}$, its tangent rank- (k, l) tensor bundle by $T_l^k \mathcal{M} = \bigsqcup_{p \in \mathcal{M}} T_l^k(T_p \mathcal{M})$ and its alternating covariant k -tensor bundle by $\Lambda^k(\mathcal{M}) = \bigsqcup_{p \in \mathcal{M}} \Lambda^k(T_p \mathcal{M})$. Then, the space of all smooth vector fields on $\mathcal{M}$ is $\mathfrak{X}(\mathcal{M}) = \Gamma(T\mathcal{M})$, the space of all smooth (k, l) -tensors on $\mathcal{M}$ by $\mathfrak{T}_l^k(\mathcal{M}) = \Gamma(T_l^k \mathcal{M})$, the space of all smooth p -forms on $\mathcal{M}$ by $\Omega^p(\mathcal{M}) = \Gamma(\Lambda^p(\mathcal{M}))$ and the space of all smooth forms on $\mathcal{M}$ by $\Omega(\mathcal{M}) := \bigoplus_{k=0}^n \Omega^k(\mathcal{M})$.

The exterior derivative is denoted by d , the covariant exterior derivative by D , the partial derivative by ∂ , the nabla operator by ∇ and the Laplace–Beltrami operator by ∇^2 . The codifferential is denoted by $d^\dagger$, and the Laplace–de Rham operator by $\Delta_d = dd^\dagger + d^\dagger d$.

A list of “important” Lie groups:

$\text{GL}(n, \mathbb{K}) := \{A \in \mathbb{K}^{n \times n} : \det A \neq 0\}$ general linear group (Lie group for $\mathbb{K} = \mathbb{R}, \mathbb{C}$)

$\text{SL}(n, \mathbb{K}) := \{A \in \text{GL}(n, \mathbb{K}) : \det A = 1\}$ special linear group (Lie group for $\mathbb{K} = \mathbb{R}, \mathbb{C}$)

$\text{O}(n) := \{A \in \mathbb{R}^{n \times n} : AA^\top = A^\top A = I_n\}$ orthogonal group

$\text{SO}(n) := \{A \in \text{O}(n) : \det A = 1\}$ special orthogonal group

$\text{U}(n) := \{A \in \mathbb{C}^{n \times n} : AA^\dagger = A^\dagger A = I_n\}$ unitary group

$\text{SU}(n) := \{A \in \text{U}(n) : \det A = 1\}$ special unitary group

Given a Lie group G , its associated Lie algebra is usually denoted by $\mathfrak{g}$.

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Part I

Differential Geometry

Smooth Manifolds

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§1.1 Smooth manifolds

§1.1.1 Topological manifolds

Definition 1.1.1 (Topological space)

The **topology** $\mathcal{T}$ of a set X is a family of subsets of X , i.e. $\mathcal{T} \subseteq \mathcal{O}(X)$, defined as **open sets**, with the following properties:

- (1) $\emptyset, X \in \mathcal{T}$;
- (2) $O_\alpha, O_\beta \in \mathcal{T} \implies O_\alpha \cap O_\beta \in \mathcal{T}$;
- (3) $\{O_\alpha\}_{\alpha \in \mathcal{I}} \subset \mathcal{T} \implies \bigcup_{\alpha \in \mathcal{I}} O_\alpha \in \mathcal{T}$.

A **topological space** M is a set of points, endowed with a topology $\mathcal{T}$.

Given a topological space $(M, \mathcal{T})$, $O \in \mathcal{T}$ is a **neighbourhood** of a point $p \in M$ if $p \in O$: then, $(M, \mathcal{T})$ is **Hausdorff** if $\forall p, q \in M \exists O_1, O_2 \in \mathcal{T} : p \in O_1, q \in O_2 \wedge O_1 \cap O_2 = \emptyset$.

Topological spaces allow for the introduction of the concept of continuity: given two topological spaces $(M_1, \mathcal{T}_1)$ and $(M_2, \mathcal{T}_2)$, a map $f : M_1 \rightarrow M_2$ is **continuous** if $O \in \mathcal{T}_2 \implies f^{-1}(O) \in \mathcal{T}_1$.

Definition 1.1.2 (Homeomorphism)

Given two topological spaces $(M_1, \mathcal{T}_1)$ and $(M_2, \mathcal{T}_2)$, a map $f : M_1 \rightarrow M_2$ is a **homeomorphism** if it is bijective and bicontinuous, i.e. both f and f^{-1} are continuous.

The composition of two homeomorphisms $f : M_1 \rightarrow M_2$ and $g : M_2 \rightarrow M_3$ is still a homeomorphism $g \circ f : M_1 \rightarrow M_3$, since both bijectivity and continuity are preserved by composition.

Definition 1.1.3 (Topological manifold)

An n -dimensional **topological manifold** $\mathcal{M}$ is a Hausdorff topological space such that:

- (1) $\mathcal{M}$ is *second-countable*, i.e. there exists a countable basis for $\mathcal{T}(\mathcal{M})$;
- (2) $\mathcal{M}$ is locally homeomorphic to $\mathbb{R}^n$, i.e. $\forall p \in \mathcal{M} \exists U \in \mathcal{T}(\mathcal{M}) : p \in U \wedge \exists \varphi : U \rightarrow O \in \mathcal{T}(\mathbb{R}^n)$ homeomorphism;

Note that, by definition, every topological manifold has a well-defined dimension. Indeed, using the theory of de Rham cohomology, it is possible to prove the following theorem (Th. 3.1.7).

Theorem 1.1.1 (Topological invariance of dimension)

A non-empty n -dimensional topological manifold cannot be homeomorphic to an m -dimensional topological manifold, unless $m = n$.

The empty set is excluded since it is an n -dimensional topological manifold for all $n \in \mathbb{N}$.

§1.1.1.1 Coordinate charts**Definition 1.1.4** (Coordinate chart)

Let M be a topological n -manifold. A **coordinate chart** (or just a *chart*) on M is a pair (U, φ) , where $U \in \mathcal{T}(M)$ and $\varphi : U \rightarrow \varphi(U) \in \mathcal{T}(\mathbb{R}^n)$ is a homeomorphism.

Given a chart (U, φ) , the set U is called a *coordinate domain*, while φ is a (*local*) *coordinate map*, whose component functions $\varphi = (x^1, \dots, x^n)$ are called *local coordinates* on U .

By Def. 1.1.3, each point $p \in M$ is contained in the coordinate domain of a chart: if $\varphi(p) = 0$ in a chart (U, φ) , then this chart is said to be “centered at p ”.

Example 1.1.1 (Spheres)

Consider the unit n -sphere $\mathbb{S}^n$, with $n \in \mathbb{N}_0$. It is a second-countable Hausdorff topological space, since it is a topological subspace of $\mathbb{R}^{n+1}$. To show local Euclidean, for each index $i = 1, \dots, n+1$ let U_i^+ denote the subset of $\mathbb{R}^{n+1}$ where the i^{th} coordinate is positive:

$$U_i^+ \equiv \{(x^1, \dots, x^{n+1}) \in \mathbb{R}^{n+1} : x^i > 0\}$$

and similarly U_i^- the set with $x^i < 0$. Then, let $f : \mathbb{D}^n \rightarrow \mathbb{R}$ be the continuous function:

$$f(u) = \sqrt{1 - \|u\|^2}$$

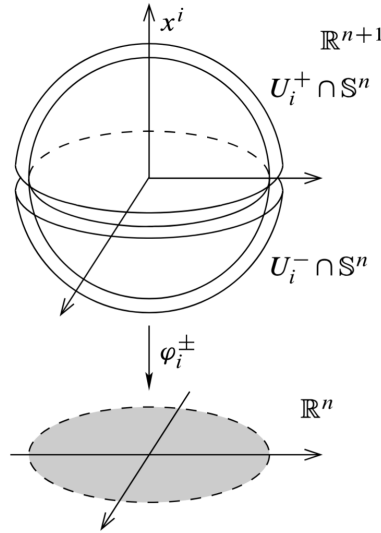
For each $i = 1, \dots, n+1$, it is easy to check that $U_i^\pm \cap \mathbb{S}^n$ is the graph of the function:

$$x^i = \pm f(x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^{n+1})$$

Thus, each subset $U_i^\pm \cap \mathbb{S}^n \subset \mathbb{S}^n$ is locally Euclidean of dimension n , and the local coordinate maps are given by (see Fig. 1.1):

$$\varphi_i^\pm : U_i^\pm \cap \mathbb{S}^n \rightarrow \mathbb{D}^n \quad \varphi_i^\pm(x^1, \dots, x^{n+1}) = (x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^{n+1})$$

Since each point of $\mathbb{S}^n$ is in the domain of at least one of these $2n+2$ charts, $\mathbb{S}^n$ is a topological n -manifold.

Figure 1.1: Charts on $\mathbb{S}^n$.**Example 1.1.2** (Tori and product manifolds)

For $n \in \mathbb{N}$, the n -torus is the product space $\mathbb{T}^n := \mathbb{S}^1 \times \cdots \times \mathbb{S}^1$ of n copies of $\mathbb{S}^1$. By Ex. 1.1.1, this is a topological n -space.

Indeed, suppose $M_1, \dots, M_k$ are topological manifolds of dimensions $n_1, \dots, n_k$ respectively. The product space $M_1 \times \cdots \times M_k$ is itself a second-countable Hausdorff space, since it is a finite product of second-countable Hausdorff spaces, hence only local Euclidean-ity needs to be checked. Given any point $(p_1, \dots, p_k) \in M_1 \times \cdots \times M_k$, choose a chart (U_i, φ_i) on each M_i such that $p_i \in U_i$. Then, the product map:

$$\varphi_1 \times \cdots \times \varphi_k : U_1 \times \cdots \times U_k \rightarrow \varphi_1(U_1) \times \cdots \times \varphi_k(U_k) \in \mathcal{T}(\mathbb{R}^{n_1 + \cdots + n_k})$$

is a homeomorphism, thus $M_1 \times \cdots \times M_k$ is a topological manifold of dimension $n_1 + \cdots + n_k$.

A set of charts $\mathcal{A} = \{U_\alpha, \varphi_\alpha\}$ which cover $\mathcal{M}$, i.e. such that $\bigcup_\alpha U_\alpha = \mathcal{M}$ is called an **atlas**. In general, an atlas may be contained in a larger atlas: a *maximal atlas* is an atlas which is not properly contained in any larger atlas.

§1.1.2 Smooth manifolds

Given $U \in \mathcal{T}(\mathbb{R}^n)$ and $V \in \mathcal{T}(\mathbb{R}^m)$, a function $f : U \rightarrow V$ is **smooth** if each of its components is $\mathcal{C}^\infty(U)$. If f is also bijective and has a smooth inverse function, then it is a *diffeomorphism*.

Definition 1.1.5 (Smooth compatibility)

Let $\mathcal{M}$ be a topological n -manifold. Two charts (U, φ) and (V, ψ) of $\mathcal{M}$ are **smoothly compatible** if either $U \cap V = \emptyset$ or $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is a diffeomorphism.

If any two charts in an atlas are smoothly compatible, then the atlas is a **smooth atlas**.

The same definition applies to weaker conditions on the differentiability of a function, i.e. $\mathcal{C}^k$ functions which are continuously differentiable up to the k^{th} derivative, and to real-analytic functions, i.e. $\mathcal{C}^\omega$ functions which can be expressed as a convergent power series in a neighbourhood of each point.

Definition 1.1.6 (Smooth manifold)

An n -dimensional **smooth manifold** is an n -dimensional topological manifold equipped with a maximal smooth atlas.

This definition can be extended to $\mathcal{C}^k$ manifolds and *real-analytic manifolds* by considering $\mathcal{C}^k$ or $\mathcal{C}^\omega$ atlases, respectively.

§1.1.2.1 Smooth charts

Each smooth coordinate map φ_α provides a coordinate system on the smooth coordinate domain U_α : $\varphi_\alpha(p) = (x^1(p), \dots, x^n(p))$. The *transition functions* $\varphi_\beta \circ \varphi_\alpha^{-1}$ are therefore coordinate transformations on overlapping regions of the manifold, represented as diffeomorphism between open subsets of $\mathbb{R}^n$.

Example 1.1.3 (Finite-dimensional vector spaces)

Let V be an n -dimensional $\mathbb{R}$ -vector space. Any norm on V determines a topology, which can be shown to be independent on the choice of norm. With this topology, V is a topological n -manifold, and a natural smooth structure on it can be defined as follows.

Each ordered basis $(E_1, \dots, E_n)$ of V defines a basis isomorphism $\beta : \mathbb{R}^n \rightarrow V$ as:

$$\beta(x) = \sum_{i=1}^n x^i E_i$$

which is clearly continuous, i.e. a homeomorphism, so (V, β^{-1}) is a chart on V . If $(\tilde{E}_1, \dots, \tilde{E}_n)$ is any other ordered basis of V , denoting with $\tilde{\beta}$ the corresponding homeomorphism, then $\exists A \in \text{GL}(n, \mathbb{R})$ such that (denoting the elements of A as $A_{ij} \equiv A_i^j$):

$$E_i = \sum_{j=1}^n A_i^j \tilde{E}_j$$

The transition map between the two charts (V, β^{-1}) and $(V, \tilde{\beta}^{-1})$ is then given by $\tilde{\beta}^{-1} \circ \beta(x) = \tilde{x}$, where:

$$\sum_{j=1}^n \tilde{x}^j \tilde{E}_j = \sum_{i=1}^n x^i E_i = \sum_{i,j=1}^n x^i A_i^j \tilde{E}_j \quad \Rightarrow \quad \tilde{x}^j = \sum_{i=1}^n x^i A_i^j$$

Since A is invertible, this is clearly a diffeomorphism, so any two charts on V are smoothly compatible, and the collection of all such charts thus defines a maximal smooth atlas on V , called the *standard smooth structure* on V .

The construction of a smooth manifold is structured in two stages: starting from a topological space, first check that it is a topological manifold, and then specify a smooth atlas on it. However, it is convenient to combine these steps into a single one, especially when starting with a set which is not already equipped with a topology.

Lemma 1.1.1 (Smooth chart lemma)

Let M be a set, and consider a collection $\{U_\alpha\}$ of subsets of M together with maps $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$ such that:

- (1) φ_α is a bijection between U_α and $\varphi_\alpha(U_\alpha) \in \mathcal{T}(\mathbb{R}^n)$ for all α ;
- (2) $\varphi_\alpha(U_\alpha \cap U_\beta), \varphi_\beta(U_\alpha \cap U_\beta) \in \mathcal{T}(\mathbb{R}^n)$ for all α, β ;
- (3) if $U_\alpha \cap U_\beta \neq \emptyset$, then $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$ is smooth;
- (4) $M = \bigcup_{\alpha \in \mathcal{I}} U_\alpha$, with $\mathcal{I}$ countable;
- (5) given $p \neq q \in M$, either $\exists \alpha : p, q \in U_\alpha$, or $\exists \alpha, \beta : p \in U_\alpha, q \in U_\beta \wedge U_\alpha \cap U_\beta = \emptyset$.

Then, M has a unique smooth manifold structure with $\mathcal{T} = \{\varphi_\alpha^{-1}(V) : V \in \mathcal{T}(\mathbb{R}^n)\}$ and $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$.

Proof. To define a topology on M , take all sets of the form $\varphi_\alpha^{-1}(V)$, with $V \in \mathcal{T}(\mathbb{R}^n)$, as a basis. To prove that this is indeed the basis for a topology, consider two basis sets $\varphi_\alpha^{-1}(V)$ and $\varphi_\beta^{-1}(W)$ with non-empty intersection: WTS $\forall p \in \varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W) \exists B$ basis set such that $p \in B \wedge B \subseteq \varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W)$.

It suffices to show that $\varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W)$ is itself a basis set (see Fig. 1.2). To see this, note that (3) implies that $(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)$ is an open subset of $\varphi_\alpha(U_\alpha \cap U_\beta)$ (as $W \in \mathcal{T}(\mathbb{R}^n)$), while (2) implies that $\varphi_\alpha(U_\alpha \cap U_\beta) \in \mathcal{T}(\mathbb{R}^n)$. It follows that (see Fig. 1.2):

$$\varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W) = \varphi_\alpha^{-1}(V \cap (\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W))$$

is also a basis set, as $V \cap (\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W) \in \mathcal{T}(\mathbb{R}^n)$. Therefore, M is a topological space with topology $\mathcal{T} = \{\varphi_\alpha^{-1}(V) : V \in \mathcal{T}(\mathbb{R}^n)\}$.

By this definition, each map $\varphi_\alpha : U_\alpha \rightarrow \varphi_\alpha(U_\alpha) \in \mathcal{T}(\mathbb{R}^n)$ is a homeomorphism, so $(M, \mathcal{T})$ is a topological n -manifold, since the Hausdorff property follows directly from (5), while second-countability from (4). Finally, (3) guarantees that the collection $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ is a smooth atlas over M , making $(M, \mathcal{T}, \mathcal{A})$ a smooth n -manifold. $\square$

This lemma allows to define both a topology and a smooth structure on a set with suitable “charts” that overlap smoothly.

§1.1.2.2 Smooth maps

With the introduction of a smooth structure on a manifold, it is possible to import the concepts of Analysis on $\mathbb{R}^n$ to functions between manifolds. In the reminder of these notes, the term “function” typically denotes maps whose codomain is $\mathbb{R}^k$, with $k \in \mathbb{N}$.

Definition 1.1.7 (Smooth function)

Let $(\mathcal{M}, \mathcal{A})$ be a smooth n -manifold. A function $f : \mathcal{M} \rightarrow \mathbb{R}^k$, with $k \in \mathbb{N}$, is a **smooth function** if $\forall p \in \mathcal{M} \exists (U, \varphi) \in \mathcal{A} : p \in U \wedge f \circ \varphi^{-1} \in \mathcal{C}^\infty(\varphi(U))$.

The function $\hat{f}_\varphi \equiv f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}^k$ is the *coordinate representation* of f : by definition, a function is smooth if and only if its coordinate representation is smooth in some smooth chart around each point. By smooth compatibility, a smooth function $f : \mathcal{M} \rightarrow \mathbb{R}^k$ is smooth on every smooth chart of $\mathcal{M}$, hence smooth functions have smooth coordinate representations in every smooth chart.

The most important case is that of smooth real-valued functions $f : \mathcal{M} \rightarrow \mathbb{R}$: the set of all such functions is denoted by $\mathcal{C}^\infty(\mathcal{M})$, and it is an $\mathbb{R}$ -vector space under pointwise addition and constant multiplication.

Definition 1.1.8 (Smooth map)

Let $(\mathcal{M}_1, \mathcal{A}_1), (\mathcal{M}_2, \mathcal{A}_2)$ be smooth manifolds. A map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a **smooth map** if $\forall p \in \mathcal{M}_1 \exists (U, \varphi) \in \mathcal{A}_1, (V, \psi) \in \mathcal{A}_2 : p \in U, f(p) \in V \wedge f(U) \subseteq V \wedge \psi \circ f \circ \varphi^{-1} \in \mathcal{C}^\infty(\varphi(U))$.

From this definition of continuity it follows that smoothness implies continuity.

Proposition 1.1.1 (Smoothness and continuity)

Every smooth map is continuous.

Proof. Suppose $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is smooth. Given $p \in \mathcal{M}_1$, by the smoothness of f there are smooth charts $(U, \varphi) : p \in U$ and $(V, \psi) : f(p) \in V$ such that $f(U) \subseteq V$ and $\psi \circ f \circ \varphi^{-1} : \varphi(U) \rightarrow \psi(V)$ is smooth, hence continuous. Since $\varphi : U \rightarrow \varphi(U)$ and $\psi : V \rightarrow \psi(V)$ are homeomorphisms:

$$f|_U = \psi^{-1} \circ (\psi \circ f \circ \varphi^{-1}) \circ \varphi : U \rightarrow V$$

which is a composition of continuous maps. Since f is continuous in a neighbourhood of each point, it is continuous on all $\mathcal{M}_1$. $\square$

The condition that, for a map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ to be smooth, for all $p \in \mathcal{M}_1$ there need to be coordinate domains $U \ni p$ and $V \ni f(p)$ such that $f(U) \subseteq V$ is included in the definition of smoothness precisely so that smoothness automatically implies continuity.

There are simple classes of maps which are automatically smooth, since they reduce to composition of smooth maps between subsets of Euclidean spaces. In particular:

- every constant map is smooth;
- $\mathbb{1}_{\mathcal{M}}$ is smooth;

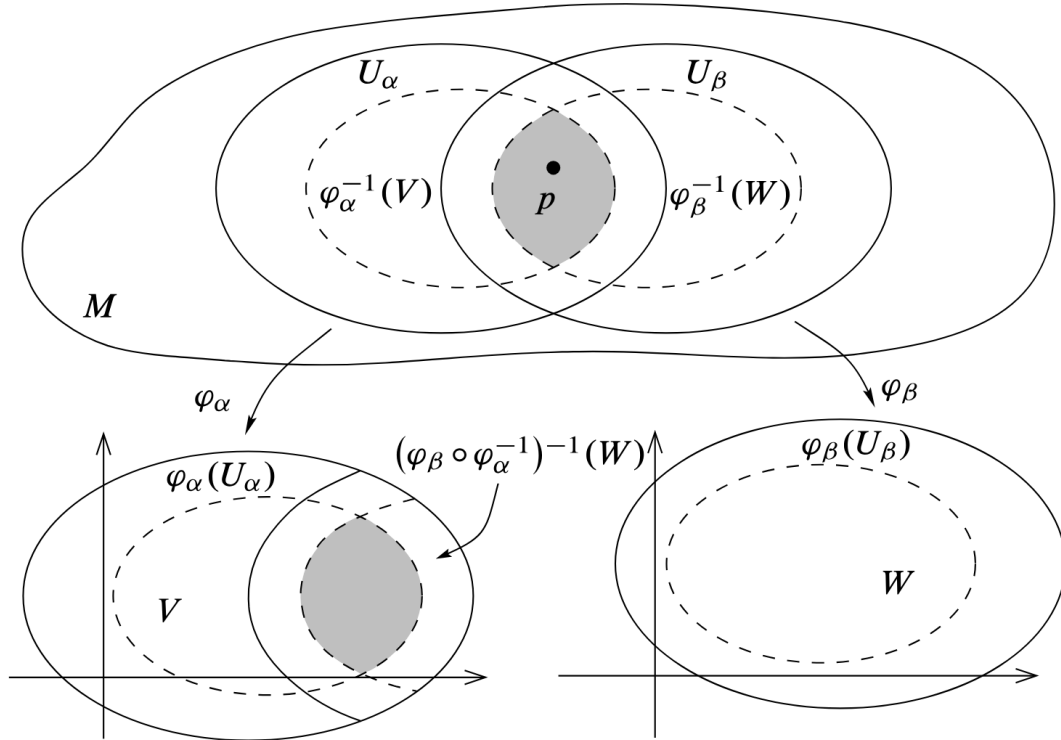


Figure 1.2: Smooth chart lemma.

- given an open submanifold $\mathcal{U} \subseteq \mathcal{M}$, the inclusion map $\mathcal{U} \hookrightarrow \mathcal{M}$ is smooth;
- the composition of smooth maps is a smooth map.

Proposition 1.1.2 (Smooth maps on product manifolds)

Let $\mathcal{M}_1, \dots, \mathcal{M}_k$ and $\mathcal{M}$ be smooth manifolds, and for each $i = 1, \dots, k$ let $\pi_i : \mathcal{M}_1 \times \dots \times \mathcal{M}_k \rightarrow \mathcal{M}_i$ be the projection onto $\mathcal{M}_i$. A map $f : \mathcal{M} \rightarrow \mathcal{M}_1 \times \dots \times \mathcal{M}_k$ is smooth if and only if each of the component maps $f_i = \pi_i \circ f : \mathcal{M} \rightarrow \mathcal{M}_i$ is smooth.

In general, to check that a map is smooth, it suffices to show that each of its component map is either smooth or a composition of smooth maps.

Example 1.1.4 (Circles and tori)

If the circle $\mathbb{S}^1$ is given its standard smooth structure, the map $\varepsilon : \mathbb{R} \rightarrow \mathbb{S}^1$ defined by $\varepsilon(t) = e^{2\pi it}$ is smooth, because with respect to any angle coordinate θ on $\mathbb{S}^1$ it has a coordinate representation $\hat{\varepsilon}_\theta = 2\pi t + c$, for some constant $c \in \mathbb{R}$.

By Prop. 1.1.2, the map $\varepsilon_n : \mathbb{R}^n \rightarrow \mathbb{T}^n$ defined as $\varepsilon_n(t_1, \dots, t_n) = (e^{2\pi it_1}, \dots, e^{2\pi it_n})$ is smooth.

Example 1.1.5 (Inclusions on spheres)

Consider the n -sphere $\mathbb{S}^n$ with its standard smooth structure. The inclusion map $\iota : \mathbb{S}^n \hookrightarrow \mathbb{R}^{n+1}$ is clearly continuous, since it is the inclusion map of a topological space. Its coordinate representation with respect to any of the charts introduced in Ex. 1.1.1 is:

$$\hat{\iota}_\pm(u^1, \dots, u^n) = \iota \circ (\varphi_i^\pm)^{-1}(u^1, \dots, u^n) = (u^1, \dots, u^{i-1}, \pm\sqrt{1 - \|u\|^2}, u^{i+1}, \dots, u^n)$$

which is smooth on its domain $\mathbb{D}^n$, since $\|u\| < 1$.

The most interesting class of smooth maps are smooth bijections with a smooth inverse, known as **diffeomorphisms**. Since smoothness implies continuity, a diffeomorphism is also a homeomorphism; moreover, every composition of diffeomorphisms is a diffeomorphism, since composition preserves both continuity and smoothness, and by Prop. 1.1.2 every finite product of diffeomorphisms is a diffeomorphism.

Two smooth manifolds are said to be *diffeomorphic* if there exists a diffeomorphism between them: this is clearly an equivalence relation on the class of all smooth manifolds. A weaker version of Th. 1.1.1 holds in the case of diffeomorphic manifolds.

Theorem 1.1.2 (Diffeomorphism invariance of dimension)

A non-empty n -dimensional smooth manifold cannot be diffeomorphic to an m -dimensional smooth manifold, unless $m = n$.

Like two topological spaces are considered to be “the same” if they are homeomorphic, two smooth manifolds are essentially indistinguishable if they are diffeomorphic. A natural question to ask, then, is whether a given topological manifold admits smooth structures which are not diffeomorphic to each other.

Proposition 1.1.3 (Uniqueness of smooth structures in low dimension)

Every n -dimensional topological manifold, with $n \in \mathbb{N}_{\leq 3}$, has a smooth structure that is unique up to diffeomorphism.

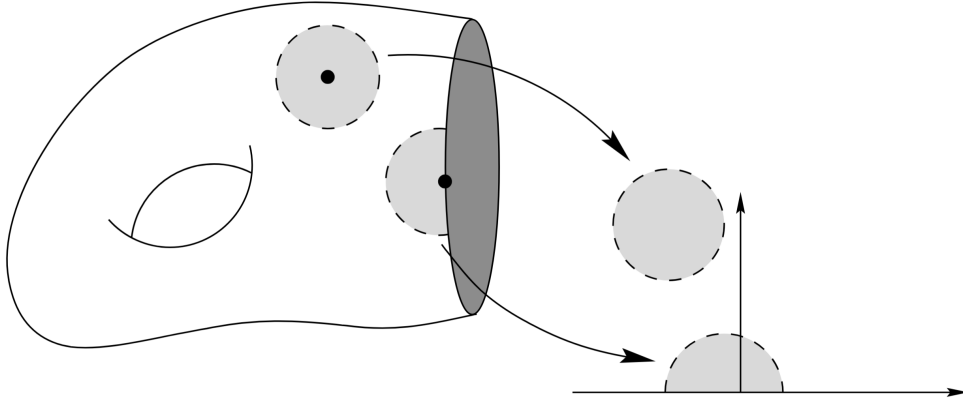


Figure 1.3: A manifold with boundary.

On the contrary, in higher dimensions it turns out that there are cases of topological manifolds which admit several non-diffeomorphic smooth structures. For example, the Euclidean space $\mathbb{R}^n$ admits a unique smooth structure (up to diffeomorphism), except for $n = 4$: in this case, $\mathbb{R}^4$ admits uncountably-many non-diffeomorphic distinct smooth structures (seldom called *fake* $\mathbb{R}^4$'s). Another example is $\mathbb{S}^7$, which admits exactly 15 diffeomorphism classes of smooth structures. Finally, in any dimension $n \in \mathbb{N}_{\geq 4}$ there exist compact topological manifolds that have no smooth structures at all.

Example 1.1.6 (0-dimensional manifolds)

A topological 0-manifold $\mathcal{M}$ is just a countable discrete space. For each point $p \in \mathcal{M}$, the only neighbourhood of p that is homeomorphic to an open subset of $\mathbb{R}^0$ is $\{p\}$ itself, and there is exactly one coordinate map $\varphi : \{p\} \rightarrow \mathbb{R}^0$. Thus, the set of all charts on $\mathcal{M}$ trivially satisfies the smooth compatibility condition, since they have no overlapping coordinate domains, hence each 0-dimensional manifold has a unique smooth structure.

§1.1.3 Manifolds with boundary

In the analysis of integration on smooth manifolds (§2.3), one often encounters spaces which would be smooth except for a “boundary” of some sort, e.g. closed balls in $\mathbb{R}^n$ or closed hemispheres in $\mathbb{S}^n$. To describe these objects, the definition of manifold needs to be extended. To do so, define the *closed n -dimensional upper half-plane* $\mathbb{U}^n \subseteq \mathbb{R}^n$ as:

$$\mathbb{U}^n := \{(x^1, \dots, x^n) \in \mathbb{R}^n : x^n \geq 0\} \quad (1.1.1)$$

Clearly, for $n > 0$ this means that $\text{Int } \mathbb{U}^n = \{x^n > 0\}$ and $\partial \mathbb{U}^n = \{x^n = 0\}$, while for $n = 0$ trivially $\mathbb{U}^0 = \mathbb{R}^0 = \{0\}$, hence $\text{Int } \mathbb{U}^0 = \mathbb{R}^0$ and $\partial \mathbb{U}^0 = \emptyset$.

Definition 1.1.9 (Topological manifold with boundary)

An n -dimensional **topological manifold with boundary** $\mathcal{M}$ is a second-countable Hausdorff space in which every point has a neighbourhood homeomorphic to either an open subset of $\mathbb{R}^n$ or a (relatively) open subset of $\mathbb{U}^n$.

The intuition behind this definition is depicted in Fig. 1.3. A chart (U, φ) of $\mathcal{M}$ is then classified as a *interior chart* if $\varphi(U) \in \mathcal{T}(\mathbb{R}^n) : \varphi(U) \cap \partial \mathbb{U}^n = \emptyset$, or as a *boundary chart* if $\varphi(U) \in \mathcal{T}(\mathbb{U}^n) : \varphi(U) \cap \partial \mathbb{U}^n \neq \emptyset$. Boundary chart whose image is of the form $B_r(x) \cap \mathbb{U}^n$ for some $x \in \partial \mathbb{U}^n$ and $r \in \mathbb{R}^+$ is called a coordinate half-ball.

A point $p \in \mathcal{M}$ is an interior point of $\mathcal{M}$ if it is in the domain of some interior chart, while it is a boundary point of $\mathcal{M}$ if it is in the domain of some boundary chart $(U, \varphi) : \varphi(p) \in \partial\mathbb{U}^n$: the set of all boundary points of $\mathcal{M}$ is the **boundary** $\partial\mathcal{M}$ of $\mathcal{M}$.

Theorem 1.1.3 (Topological invariance of boundary)

Let $\mathcal{M}$ be a topological manifold with boundary. Then, each point of $\mathcal{M}$ is either a boundary point of an interior point, but not both.

Therefore, $\partial\mathcal{M} \cap \text{Int } \mathcal{M} = \emptyset$ and $\mathcal{M} = \text{Int } \mathcal{M} \sqcup \partial\mathcal{M}$.

Observation 1.1.1 (Topological boundary and manifold boundary)

The notions of topological boundary and manifold boundary are well-distinct: a manifold with boundary may have a non-empty manifold boundary, irrespective of whether it has a topological boundary as a subset of some other topological space.

A clear example is the closed unit ball $\mathbb{D}^n$. Indeed, it is a topological manifold with boundary whose manifold boundary is $\partial\mathbb{D}^n = \mathbb{S}^{n-1}$. When viewed as a subset of $\mathbb{R}^n$, the topological boundary coincides with the manifold boundary; however, if $\mathbb{D}^n$ is thought of as a subset of $\mathbb{R}^{n+1}$, its topological boundary is $\mathbb{D}^n$ itself.

Note that a topological manifold with a boundary does not in general satisfy the definition of topological manifold. On the contrary, all topological manifolds are topological manifolds with a boundary, whose boundary is empty. In the following, the term manifold is used to denote a manifold that may or may not have a boundary, with further specification given only if required.

Proposition 1.1.4 (Characterization of manifolds with boundary)

Let $\mathcal{M}$ be a topological n -manifold with boundary. Then:

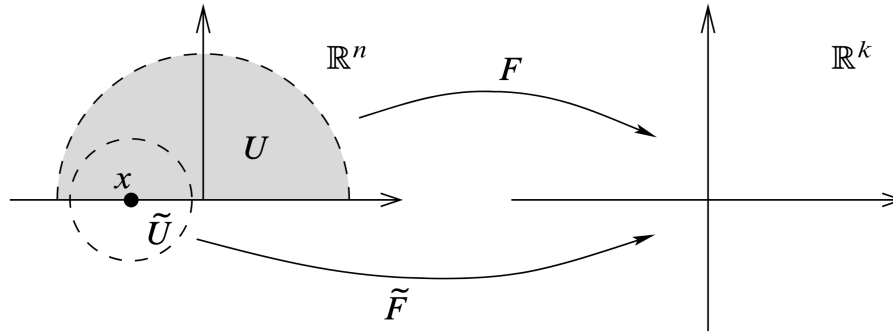
- (1) $\text{Int } \mathcal{M} \subseteq \mathcal{M}$ is open and a topological n -manifold without boundary;
- (2) $\partial\mathcal{M} \subseteq \mathcal{M}$ is closed and a topological $(n - 1)$ -manifold without boundary;
- (3) $\mathcal{M}$ is a topological manifold if and only if $\partial\mathcal{M} = \emptyset$;
- (4) if $n = 0$, then $\partial\mathcal{M} = \emptyset$ and $\mathcal{M}$ is a topological 0-manifold.

§1.1.3.1 Smooth manifolds with boundary

Recall that a map from an arbitrary $A \subseteq \mathbb{R}^n$ to $\mathbb{R}^k$ is said to be *smooth* if in a neighbourhood of each point of A it admits an extension to a smooth map defined on an open subset of $\mathbb{R}^n$. Thus, given $U \in \mathcal{T}(\mathbb{U}^n)$, a map $f : U \rightarrow \mathbb{R}^k$ is smooth if for each $x \in U$ there exist $\tilde{U} \in \mathcal{T}(\mathbb{U}^n) : x \in \tilde{U}$ and a smooth map $\tilde{f} : \tilde{U} \rightarrow \mathbb{R}^k$ that agrees with f on $\tilde{U} \cap \mathbb{U}^n$ (see Fig. 1.4). Clearly, the restriction of f to $U \cap \text{Int } \mathbb{U}^n$ is smooth in the usual sense. Moreover, all partial derivatives of f at points of $U \cap \partial\mathbb{U}^n$ are determined by continuity by their values in $\text{Int } \mathbb{U}^n$, hence they are independent on the choice of extension.

Example 1.1.7 (Functions on the plane)

Let $\text{Int } \mathbb{D}^2 \subseteq \mathbb{R}^2$ be the open unit disk, let $U = \mathbb{U}^2 \cap \text{Int } \mathbb{D}^2$ and $f : U \rightarrow \mathbb{R}$ defined as $f(x, y) = \sqrt{1 - x^2 - y^2}$. This expression naturally extends to all $\text{Int } \mathbb{D}^2$ in a smooth way, hence f is smooth

Figure 1.4: Smoothness of maps on open subsets $\mathbb{U}^n$.

on U . On the other hand, $g(x, y) = \sqrt{y}$ has no smooth extension to any neighbourhood of $(0, 0) \in \mathbb{R}^2$, since $\partial_y g \rightarrow \infty$ as $y \rightarrow 0$. Therefore, g is not smooth on U .

With this notion of smoothness, Def. 1.1.6 directly generalizes to smooth manifolds with boundary. As noted before, every smooth manifold without boundary is automatically a smooth manifold with boundary, whose boundary is empty. Moreover, if $\mathcal{M}$ is a smooth manifold with boundary, then $\text{Int } \mathcal{M}$ is an open smooth submanifold of $\mathcal{M}$ without boundary.

The construction of smooth structures on finite products illustrated in Ex. 1.1.2 does not extend directly to smooth manifolds with boundary. This is due to the fact that $\mathbb{U}^n \times \mathbb{U}^m$ is not itself a half-space, hence a finite product of smooth manifolds with boundary cannot generally be considered as a smooth manifold with boundary (it is instead a smooth manifold with corners, see §2.3.2.2). However, it can be shown that if $\mathcal{M}_1, \dots, \mathcal{M}_k$ are smooth manifolds without boundary and $\mathcal{M}$ is a smooth manifold with boundary, then $\mathcal{M}_1 \times \dots \times \mathcal{M}_k \times \mathcal{M}$ is a smooth manifold with boundary, and $\partial(\mathcal{M}_1 \times \dots \times \mathcal{M}_k \times \mathcal{M}) = \mathcal{M}_1 \times \dots \times \mathcal{M}_k \times \partial\mathcal{M}$.

Theorem 1.1.4 (Smooth invariance of boundary)

Let $\mathcal{M}$ be a smooth manifold with boundary, and let $p \in \mathcal{M}$. If there exists a chart (U, φ) of $\mathcal{M}$ such that $\varphi(U) \subseteq \mathbb{U}^n$ and $\varphi(p) \in \partial\mathbb{U}^n$, then the same is true for every smooth chart of $\mathcal{M}$ whose domain contains p .

§1.2 Tangent vectors

The central idea of calculus is *linear approximation*: for example, the approximations of a single-variable function with its tangent line, of a parametrized curve with its velocity vector, or of a surface with its tangent plane. In order to generalize calculus on smooth manifolds, then, one needs to introduce the notion of a “tangent space” to a manifold at each of its points.

§1.2.1 Derivations

Before considering generic smooth manifolds, it is convenient to analyze the Euclidean case first.

Definition 1.2.1 (Geometric tangent space)

Given a point $a \in \mathbb{R}^n$, the **geometric tangent space** to $\mathbb{R}^n$ at a is the set $\mathbb{R}_a^n := \{a\} \times \mathbb{R}^n = \{(a, \mathbf{v}) : \mathbf{v} \in \mathbb{R}^n\}$. A **geometric tangent vector** in $\mathbb{R}^n$ is an element $\mathbf{v}_a \equiv (a, \mathbf{v}) \in \mathbb{R}_a^n$, for some $a \in \mathbb{R}^n$.

Geometric tangent vectors can be thought as “arrows” with initial point at a . The set $\mathbb{R}_a^n$ is an $\mathbb{R}$ -vector space under the natural operations of pointwise addition and constant multiplication.

Given a basis $\{e_i\}_{i=1,\dots,n}$ of $\mathbb{R}^n$, then $\{\mathbf{e}_i|_a\}_{i=1,\dots,n}$ is a basis of $\mathbb{R}_a^n$: indeed, $\mathbb{R}_a^n$ and $\mathbb{R}^n$ are essentially the same space, with the index in $\mathbb{R}_a^n$ only needed to distinguish between geometric tangent spaces at different points, since they are disjoint sets.

The same geometric construction does not generalize to smooth manifolds, which are in general not embedded in any Euclidean space. However, a suitable construction can be defined starting from geometric vectors. Indeed, any geometric tangent vector $\mathbf{v}_a \in \mathbb{R}_a^n$ defines a map $D_{\mathbf{v}}|_a : \mathcal{C}^\infty \mathbb{R}^n \rightarrow \mathbb{R}$ which takes the directional derivative along $\mathbf{v}$ at a :

$$D_{\mathbf{v}}|_a f \equiv D_{\mathbf{v}} f(a) := \left. \frac{d}{dt} \right|_{t=0} f(a + t\mathbf{v}) \quad (1.2.1)$$

In terms of the standard basis, by the chain rule the directional derivative can be written more concretely as:

$$D_{\mathbf{v}}|_a f = v^i \frac{\partial f}{\partial x^i}(a) \quad (1.2.2)$$

where the Einstein’s summation convention is assumed. With this construction in mind, it is possible to give a general definition.

Definition 1.2.2 (Derivation)

Let $\mathcal{M}$ be a smooth manifold and $p \in \mathcal{M}$. Then, a linear map $X_p : \mathcal{C}^\infty(\mathcal{M}) \rightarrow \mathbb{R}$ is a **derivation at p** if it satisfies the *Leibniz rule*:

$$X_p(fg) = f(p)X_p(g) + g(p)X_p(f) \quad \forall f, g \in \mathcal{C}^\infty(\mathcal{M}) \quad (1.2.3)$$

The set of all derivations of $\mathcal{C}^\infty(\mathcal{M})$ at a point $p \in \mathcal{M}$, denoted by $T_p \mathcal{M}$, is a $\mathbb{R}$ -vector space called the **tangent space** to $\mathcal{M}$ at p , and its elements are called *tangent vectors* to $\mathcal{M}$ at p .

Lemma 1.2.1 (Properties of derivations)

Let $\mathcal{M}$ be a smooth manifold and $p \in \mathcal{M}$. Then, given $X_p \in T_p \mathcal{M}$ and $f, g \in \mathcal{C}^\infty(\mathcal{M})$:

- (1) if f is a constant function, then $X_p f = 0$;
- (2) if $f(p) = g(p) = 0$, then $X_p(fg) = 0$;

Proof. (2) follows trivially from the Leibniz rule. To prove (1), it suffices to check it for the constant function $f_1(x) \equiv 1$, since others follow from linearity. For f_1 , the product rule gives:

$$X_p(f_1) = X_p(f_1 f_1) = 2f_1(p)X_p(f_1) = 2X_p(f_1)$$

which implies $X_p(f_1) = 0$. □

§1.2.2 Differentials

To relate the abstract definition of tangent spaces through derivations to the geometric tangent spaces on $\mathbb{R}^n$, one needs to study the relation between smooth maps and derivations.

In the case of a smooth map between Euclidean spaces, the total derivative of the map at a point (represented by its Jacobian matrix) is a linear map that represents the linear approximation of the

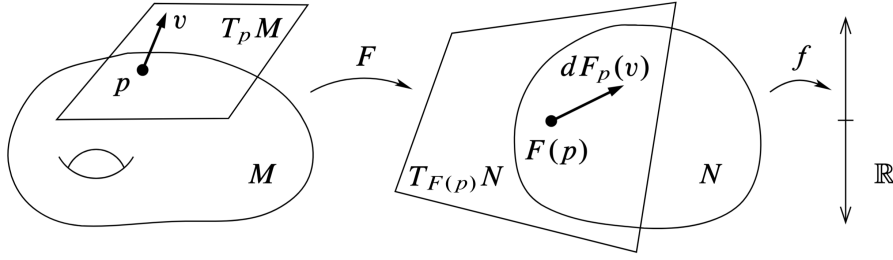


Figure 1.5: Visualization of the differential of a smooth map.

map near the given point. In the manifold case there is a similar map, but it makes no sense to talk about a linear map between manifolds: it will instead be a linear map between tangent spaces.

Definition 1.2.3 (Differential)

Let $\mathcal{M}_1, \mathcal{M}_2$ be smooth manifolds and $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ a smooth map. The **differential** of f at $p \in \mathcal{M}_1$ is the map $df_p : T_p \mathcal{M}_1 \rightarrow T_{f(p)} \mathcal{M}_2$ defined as follows: given $X_p \in T_p \mathcal{M}_1$, let $df_p(X_p)$ be the derivation at $f(p)$ that acts on $g \in \mathcal{C}^\infty(\mathcal{M}_2)$ as:

$$df_p(X_p)(g) := X_p(g \circ f) \quad (1.2.4)$$

The nature of $df_p(X_p)$ as a derivation is naturally derived from the nature of X_p as a derivation:

$$df_p(X_p)(gh) = X_p((gh) \circ f) = X_p((g \circ f)(h \circ f)) = g(f(p))df_p(X_p)(h) + h(f(p))df_p(X_p)(g)$$

Lemma 1.2.2 (Properties of differentials)

Let $\mathcal{M}, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$ be smooth manifolds and $p \in \mathcal{M}_1$. Then, given smooth maps $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ and $g : \mathcal{M}_2 \rightarrow \mathcal{M}_3$:

- (1) $df_p : T_p \mathcal{M}_1 \rightarrow T_{f(p)} \mathcal{M}_2$ is linear;
- (2) $d(g \circ f)_p = dg_{f(p)} \circ df_p : T_p \mathcal{M}_1 \rightarrow T_{g \circ f(p)} \mathcal{M}_3$;
- (3) $d(\mathbb{1}_{\mathcal{M}}) = \mathbb{1}_{T_p \mathcal{M}}$;
- (4) if f is a diffeomorphism, then df_p is an isomorphism, and $(df_p)^{-1} = d(f^{-1})_{f(p)}$.

The differential of a smooth map can be visualized as in Fig. 1.5. Its main application is to relate the tangent space to a smooth manifold with a Euclidean tangent space.

There is, however, an important issue to be addressed first: while tangent spaces are defined in terms of smooth functions on the whole manifold, coordinate charts are in general defined only on open subsets. The key point is that tangent vectors (i.e. derivations) act locally.

Proposition 1.2.1 (Tangent space to an open submanifold)

Let $\mathcal{M}$ be a smooth manifold, $U \in \mathcal{T}(\mathcal{M})$ and $\iota : U \hookrightarrow \mathcal{M}$ the inclusion map. For every $p \in U$, the differential $d\iota_p : T_p U \rightarrow T_p \mathcal{M}$ is an isomorphism.

Proof. To prove injectivity, suppose $X_p \in T_p \mathcal{M}$ and $d\iota_p(X_p) = 0 \in T_p \mathcal{M}$. Let $B \in \mathcal{T}(\mathcal{M}) : p \in \mathcal{M} \cap \overline{B} \subseteq U$. If $f \in \mathcal{C}^\infty$ is arbitrary, it can be extended as $\tilde{f} \in \mathcal{C}^\infty(\mathcal{M}) : \tilde{f}|_{\overline{B}} \equiv f$.

Lemma 1.2.3 (Locality of derivations)

Let $\mathcal{M}$ be a smooth manifold and $p \in \mathcal{M}$. Given $X_p \in T_p\mathcal{M}$, if $f, g \in \mathcal{C}^\infty(\mathcal{M})$ agree on some neighbourhood of p , then $X_p(f) = X_p(g)$.

By Lemma 1.2.3, since f and $\tilde{f}|_U$ are smooth functions on U which agree on a neighbourhood of p :

$$X_p(f) = X_p(\tilde{f}|_U) = X_p(\tilde{f} \circ \iota) = d\iota_p(X_p)(\tilde{f}) = 0$$

Since this holds for every $f \in \mathcal{C}^\infty(U)$, then $X_p \equiv 0$, so $d\iota_p$ is injective as $\ker d\iota_p = \{0\}$.

To prove surjectivity, suppose $Y_p \in T_p\mathcal{M}$ is arbitrary. Define an operator $X_p : \mathcal{C}^\infty(U) \rightarrow \mathbb{R}$ by setting $X_p(f) = Y_p(\tilde{f})$, where $\tilde{f}$ is any smooth function on all $\mathcal{M}$ which agrees with f on $\bar{B}$. By Lemma 1.2.3, $X_p(f)$ is independent of the choice of $\tilde{f}$, hence X_p is well-defined, and it is easily checked to be a derivation of $\mathcal{C}^\infty(U)$ at p . For any $g \in \mathcal{C}^\infty(\mathcal{M})$, then:

$$d\iota_p(X_p)(g) = X_p(g \circ \iota) = Y_p(\tilde{g} \circ \iota) = Y_p(g)$$

where the last two equalities follow from the fact that $g \circ \iota$, $\tilde{g} \circ \iota$ and g all agree on B . Hence, $d\iota_p$ is surjective, and therefore an isomorphism. $\square$

Given an open set $U \subseteq \mathcal{M}$, the isomorphism $d\iota_p : T_pU \rightarrow T_p\mathcal{M}$ is canonically defined, independently on any choices: it is then possible to identify $T_pU \cong T_p\mathcal{M}$ for any point $p \in U$. This identification comes from the fact that $d\iota_p(X_p)$ is the same derivation as X_p , just acting on functions on the whole manifold instead of on U only: since by Lemma 1.2.3 the action of a derivation on a function depends only on the values of the function on arbitrarily small neighbourhoods, this identification is well-posed. In particular, this means that any tangent vector $X_p \in T_p\mathcal{M}$ can be unambiguously applied to functions defined only in a neighbourhood of p , not necessarily on the whole $\mathcal{M}$.

Theorem 1.2.1 (Dimension of tangent spaces)

Let $\mathcal{M}$ be a smooth n -manifold and $p \in \mathcal{M}$. Then $T_p\mathcal{M}$ is an n -dimensional $\mathbb{R}$ -vector space.

Proof. Given $p \in \mathcal{M}$, let (U, φ) be a smooth coordinate chart containing p . Since $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$ is a diffeomorphism, by (4) of Lemma 1.2.2 it follows that $d\varphi_p : T_pU \rightarrow T_{\varphi(p)}\varphi(U)$ is an isomorphism. Prop. 1.2.1 guarantees that $T_pU \cong T_p\mathcal{M}$ and $T_{\varphi(p)}\varphi(U) \cong T_{\varphi(p)}\mathbb{R}^n$, hence $\dim_{\mathbb{R}} T_p\mathcal{M} = \dim_{\mathbb{R}} T_{\varphi(p)}\mathbb{R}^n = n$. $\square$

In the case of finite-dimensional vector spaces, a stronger result holds. Given an n -dimensional $\mathbb{R}$ -vector field V (with its standard smooth structure) and a point $a \in V$, for each $\mathbf{v} \in V$ consider the map $D_{\mathbf{v}}|_a : \mathcal{C}^\infty(V) \rightarrow \mathbb{R}$ defined as in Eq. 1.2.1. Then, it can be shown that the map $\mathbf{v} \mapsto D_{\mathbf{v}}|_a$ is a canonical isomorphism $V \cong T_aV$, so that given $L \in \text{Hom}_{\mathbb{R}}(V, W)$ the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\cong} & T_aV \\ L \downarrow & & \downarrow dL_a \\ W & \xrightarrow{\cong} & T_{L(a)}W \end{array} \quad (1.2.5)$$

The fact that $V \cong T_aV$ is a canonical isomorphism (i.e. independent of the choice of basis) allows to identify the tangent vectors to a finite-dimensional vector space with the elements of the space itself. In particular, since isomorphisms map bases to bases, given the canonical basis $\{\mathbf{e}_i\}_{i=1, \dots, n}$ of

V , the canonical basis of $T_a V$ is formed by:

$$D_{\mathbf{e}_i}|_a = \left. \frac{\partial}{\partial x^i} \right|_a \quad (1.2.6)$$

More generally, if $\mathcal{M}$ is an open submanifold of a vector space V , then the isomorphisms $T_p \mathcal{M} \cong T_p V \cong V$ allow to canonically identify each tangent space to $\mathcal{M}$ with V .

Proposition 1.2.2 (Tangent space to a product manifold)

Let $\mathcal{M}_1, \dots, \mathcal{M}_k$ be smooth manifolds, and for each $j = 1, \dots, k$ let $\pi_j : \mathcal{M}_1 \times \dots \times \mathcal{M}_k \rightarrow \mathcal{M}_j$ be the projection onto $\mathcal{M}_j$. For any point $p = (p_1, \dots, p_k) \in \mathcal{M}_1 \times \dots \times \mathcal{M}_k$, the map $\alpha : T_p(\mathcal{M}_1 \times \dots \times \mathcal{M}_k) \rightarrow T_{p_1} \mathcal{M}_1 \oplus \dots \oplus T_{p_k} \mathcal{M}_k$ defined as:

$$\alpha(\mathbf{v}) = (\mathrm{d}(\pi_1)_p(\mathbf{v}), \dots, \mathrm{d}(\pi_k)_p(\mathbf{v}))$$

is a canonical isomorphism.

The canonical identification $T_{p,q}(\mathcal{M}_1 \times \mathcal{M}_2) \cong T_p \mathcal{M}_1 \oplus T_q \mathcal{M}_2$ is then justified.

§1.2.3 Coordinate bases

To make tangent spaces less abstract, it is possible to introduce coordinates on them.

§1.2.3.1 Derivations

Consider a smooth n -manifold $\mathcal{M}$ and let (U, φ) be a smooth chart on $\mathcal{M}$. Since $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$ is a diffeomorphism, combining (4) of Lemma 1.2.2 and Prop. 1.2.1, it is clear that $\mathrm{d}\varphi_p : T_p \mathcal{M} \rightarrow T_{\varphi(p)} \mathbb{R}^n$ is an isomorphism. By Eq. 1.2.6, the canonical basis of $T_{\varphi(p)} \mathbb{R}^n$ is $\partial_1|_{\varphi(p)}, \dots, \partial_n|_{\varphi(p)}$, therefore the preimages of these vectors under the isomorphism $\mathrm{d}\varphi_p$ form a basis of $T_p \mathcal{M}$:

$$\left. \frac{\partial}{\partial x^i} \right|_p \equiv (\mathrm{d}\varphi_p)^{-1} \left(\left. \frac{\partial}{\partial x^i} \right|_{\varphi(p)} \right) = \mathrm{d}(\varphi^{-1})_{\varphi(p)} \left(\left. \frac{\partial}{\partial x^i} \right|_{\varphi(p)} \right)$$

Then, by Def. 1.2.3, these basis vector act on $\mathcal{C}^\infty(U)$ as:

$$\left. \frac{\partial}{\partial x^i} \right|_p f = \left. \frac{\partial}{\partial x^i} \right|_{\varphi(p)} (f \circ \varphi^{-1}) = \frac{\partial \hat{f}}{\partial x^i}(\hat{p}) \quad (1.2.7)$$

where the hat denotes the coordinate representation, as usual. Therefore, $\partial_i|_p$ is the derivation that takes the i^{th} partial derivative of (the coordinate representation of) $f \in \mathcal{C}^\infty(U)$ at (the coordinate representation of) $p \in U$: these are called the **coordinate vectors** at p , forming a *coordinate basis* of $T_p \mathcal{M}$ associated to the given coordinate system. In the special case $\mathcal{M} = \mathbb{R}^n$, the coordinate vectors are precisely the partial derivative operators. Tangent vectors $X_p \in T_p \mathcal{M}$ can thus be written uniquely as:

$$X_p = X_p^i \left. \frac{\partial}{\partial x^i} \right|_p \quad (1.2.8)$$

Since $p_i|_p x^j = \delta_i^j$, the components of X_p can be easily computed by its action the coordinate functions: $X_p^i = X_p(x^i)$.

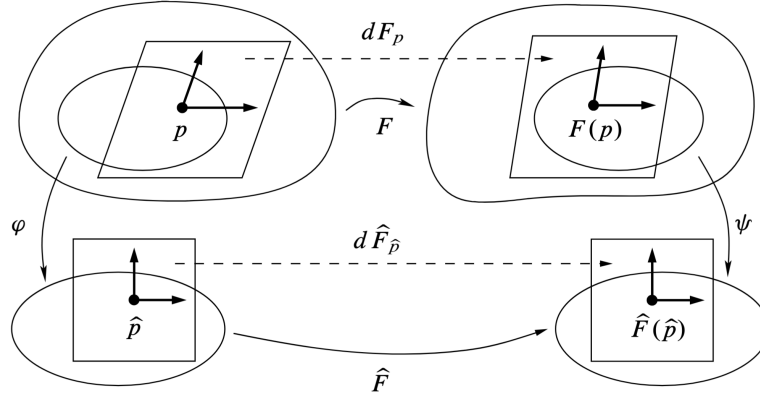


Figure 1.6: The differential of a smooth map in coordinates.

§1.2.3.2 Differentials

To see how differentials are expressed in coordinates, consider first the special case of a smooth map $f : U \rightarrow V$, with $U \in \mathcal{T}(\mathbb{R}^n)$ and $V \in \mathcal{T}(\mathbb{R}^m)$. Denoting the coordinates in the domain U as $(x^1, \dots, x^n)$ and the coordinates in the codomain V as $(y^1, \dots, y^m)$, by the chain rule:

$$df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) (g) = \frac{\partial}{\partial x^i} \Big|_p (g \circ f) = \frac{\partial g}{\partial y^j} (f(p)) \frac{\partial f^j}{\partial x^i} (p) = \left(\frac{\partial f^j}{\partial x^i} (p) \frac{\partial}{\partial y^j} \Big|_{f(p)} \right) g$$

Therefore:

$$df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) = \frac{\partial f^j}{\partial x^i} (p) \frac{\partial}{\partial y^j} \Big|_{f(p)} \quad (1.2.9)$$

In other words, the matrix representation of the action of df_p on $T_p U$ is given by:

$$[df_p] = \begin{bmatrix} \frac{\partial f^1}{\partial x^1} (p) & \cdots & \frac{\partial f^1}{\partial x^n} (p) \\ \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x^1} (p) & \cdots & \frac{\partial f^m}{\partial x^n} (p) \end{bmatrix}$$

This is precisely the Jacobian of f at p , which is the matrix representation of the total derivative: hence, the differential $df_p : T_p \mathbb{R}^n \rightarrow T_{f(p)} \mathbb{R}^m$ corresponds to the total derivative $[df_p] : \mathbb{R}^n \rightarrow \mathbb{R}^m$, under the usual identification $T_p \mathbb{R}^n \cong \mathbb{R}^n$.

Now, consider the general case of a smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ between smooth manifolds. Choosing smooth charts $(U, \varphi) : U \ni p$ for $\mathcal{M}_1$ and $(V, \psi) : V \ni f(p)$ for $\mathcal{M}_2$, the coordinate representation of f reads $\hat{f} = \psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \rightarrow \psi(V)$ (see Fig. 1.6). Denoting $\hat{p} = \varphi(p)$, by Eq. 1.2.9 $d\hat{f}_{\hat{p}}$ is represented with respect to the standard coordinate basis by the Jacobian matrix of $\hat{f}$ at $\hat{p}$. Using $f \circ \varphi^{-1} = \psi^{-1} \circ \hat{f}$, then:

$$df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) = df_p \left(d(\varphi^{-1})_{\hat{p}} \left(\frac{\partial}{\partial x^i} \Big|_{\hat{p}} \right) \right) = d(\psi^{-1})_{\hat{f}(\hat{p})} \left(d\hat{f}_{\hat{p}} \left(\frac{\partial}{\partial x^i} \Big|_{\hat{p}} \right) \right)$$

Using Eq. 1.2.9:

$$df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) = d(\psi^{-1})_{\hat{f}(\hat{p})} \left(\frac{\partial \hat{f}^j}{\partial x^i} (\hat{p}) \frac{\partial}{\partial y^j} \Big|_{\hat{f}(\hat{p})} \right) = \frac{\partial \hat{f}^j}{\partial x^i} (\hat{p}) \frac{\partial}{\partial y^j} \Big|_{f(p)} \quad (1.2.10)$$

Therefore, the representation of the differential df_p in the coordinate basis is the Jacobian matrix of (the coordinate representation of) f at (the coordinate representation of) p . Indeed, the differential is defined precisely to give a coordinate-independent meaning to the Jacobian. Moreover, since the differential “pushes” tangent vectors forward from the domain manifold to the codomain manifold (as per Def. 1.2.3), it is also called the *(pointwise) push-forward*.

§1.2.3.3 Change of coordinates

Suppose (U, φ) and (V, ψ) are two smooth charts on $\mathcal{M}$, and let $p \in U \cap V$. Denoting the coordinate functions of φ by x^i and those of ψ by $\tilde{x}^i$, any tangent vector at p can be expressed either on the basis $\partial_i|_p$ or $\tilde{\partial}_i|_p$. To see how these two representations are related, consider the transition map $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$, which is expressed as (with an abuse of notation):

$$\psi \circ \varphi^{-1}(x) = (\tilde{x}^1(x), \dots, \tilde{x}^n(x))$$

Its differential $d(\psi \circ \varphi^{-1})_{\varphi(p)} : T_{\varphi(p)}\mathbb{R}^n \rightarrow T_{\psi(p)}\mathbb{R}^n$ can then be written using Eq. 1.2.9:

$$d(\psi \circ \varphi^{-1})_{\varphi(p)} \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) = \frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p)) \frac{\partial}{\partial \tilde{x}^j} \Big|_{\psi(p)}$$

A visualization of this map is pictured in Fig. 1.7. Using the definition of coordinate vectors:

$$\begin{aligned} \frac{\partial}{\partial x^i} \Big|_p &= d(\varphi^{-1})_{\varphi(p)} \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) \\ &= d(\psi^{-1})_{\psi(p)} \circ d(\psi \circ \varphi^{-1})_{\varphi(p)} \left(\frac{\partial}{\partial x^i} \Big|_{\varphi(p)} \right) = d(\psi^{-1})_{\psi(p)} \left(\frac{\partial \tilde{x}^j}{\partial x^i}(\varphi(p)) \frac{\partial}{\partial \tilde{x}^j} \Big|_{\psi(p)} \right) \end{aligned}$$

Using the chain rule and introducing the coordinate representation $\hat{p} = \varphi(p)$:

$$\frac{\partial}{\partial x^i} \Big|_p = \frac{\partial \tilde{x}^j}{\partial x^i}(\hat{p}) \frac{\partial}{\partial \tilde{x}^j} \Big|_p \quad (1.2.11)$$

Applying this equation to Eq. 1.2.8, the transformation rule for the components of a tangent vector is found:

$$\tilde{X}_p^j = \frac{\partial \tilde{x}^j}{\partial x^i}(\hat{p}) X_p^i \quad (1.2.12)$$

It is then clear that the components of tangent vectors transform inversely with respect to coordinate basis vectors: for this reason, they are known as *contravariant vectors*. Eq. 1.2.11 and Eq. 1.2.12 respectively give the *alibi* (passive) and *alias* (active) interpretation for the change of basis: either the components are left unchanged while the basis vector are transformed, or vice versa.

§1.2.4 Tangent bundle

Definition 1.2.4 (Tangent bundle)

Given a smooth manifold $\mathcal{M}$, its **tangent bundle** is defined by the following disjoint union:

$$T\mathcal{M} := \bigsqcup_{p \in \mathcal{M}} T_p\mathcal{M} \equiv \bigcup_{p \in \mathcal{M}} \{p\} \times T_p\mathcal{M} = \{(p, X_p) : p \in \mathcal{M}, X_p \in T_p\mathcal{M}\} \quad (1.2.13)$$

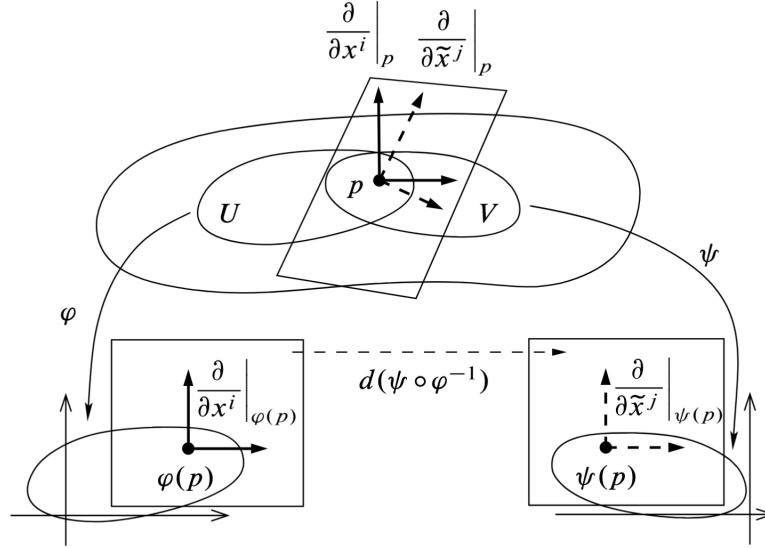


Figure 1.7: Change of coordinates.

This set comes equipped with a natural projection map $\pi : T\mathcal{M} \rightarrow \mathcal{M}$ which sends each vector in $T_p\mathcal{M}$ to the point $p \in \mathcal{M}$ at which it is tangent: $\pi(p, X_p) = p$. In the following, with an abuse of notation, the tangent space $T_p\mathcal{M}$ is often identified with its image under the canonical injection $T_p\mathcal{M} \hookrightarrow T\mathcal{M} : X_p \mapsto (p, X_p)$, so as to write simply X_p in place of (p, X_p) .

Note that, in general, the tangent bundle can be identified as a Cartesian product only locally, while globally its structure may be more complex (see §1.5.1).

Proposition 1.2.3 (Tangent bundle as smooth manifold)

Given a smooth n -manifold $\mathcal{M}$, its tangent bundle $T\mathcal{M}$ has a natural structure of $2n$ -dimensional smooth manifold, with respect to which the projection $\pi : T\mathcal{M} \rightarrow \mathcal{M}$ is smooth.

Proof. First, show that $T\mathcal{M}$ is a smooth $2n$ -manifold. Since $T\mathcal{M}$ does not come endowed with a topology, it is convenient to use Lemma 1.1.1.

Given any smooth chart (U, φ) over $\mathcal{M}$, note that $\pi^{-1}(U) \subseteq T\mathcal{M}$ is the set of all tangent vectors to $\mathcal{M}$ at all points of U . Denote the coordinate functions of φ as $(x^1, \dots, x^n)$, and define the map $\tilde{\varphi} : \pi^{-1}(U) \rightarrow \mathbb{R}^{2n}$ as:

$$\tilde{\varphi} \left(X^i \frac{\partial}{\partial x^i} \Big|_p \right) \equiv (x^1(p), \dots, x^n(p), X^1, \dots, X^n)$$

Clearly, $\text{Im } \tilde{\varphi} = \varphi(U) \times \mathbb{R}^n \in \mathcal{T}(\mathbb{R}^{2n})$, hence it is continuous. Moreover, it is a bijection onto its image, since its inverse can be written explicitly as:

$$\tilde{\varphi}^{-1}(x^1, \dots, x^n, X^1, \dots, X^n) = X^i \frac{\partial}{\partial x^i} \Big|_{\varphi^{-1}(p)}$$

Therefore, $\tilde{\varphi}$ is a homeomorphism, and $(\pi^{-1}(U), \tilde{\varphi})$ is a chart over $T\mathcal{M}$. Then, consider two smooth charts $(U, \varphi), (V, \psi)$ over $\mathcal{M}$, and the associated charts $(\pi^{-1}(U), \tilde{\varphi}), (\pi^{-1}(V), \tilde{\psi})$ over $T\mathcal{M}$. The following sets:

$$\tilde{\varphi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \varphi(U \cap V) \times \mathbb{R}^n$$

$$\tilde{\psi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \psi(U \cap V) \times \mathbb{R}^n$$

are in $\mathcal{T}(\mathbb{R}^{2n})$, and the transition map $\tilde{\psi} \circ \tilde{\varphi}^{-1} : \varphi(U \cap V) \times \mathbb{R}^n \rightarrow \psi(U \cap V) \times \mathbb{R}^n$ between them can be written explicitly as:

$$\tilde{\psi} \circ \tilde{\varphi}^{-1}(x^1, \dots, x^n, X^1, \dots, X^n) = \left(y^1(x), \dots, y^n(x), \frac{\partial y^1}{\partial x^i} X^i, \dots, \frac{\partial y^n}{\partial x^i} X^i \right)$$

where $(y^1, \dots, y^n)$ are the coordinate functions of ψ . This is clearly smooth, hence the charts $(\pi^{-1}(U), \tilde{\varphi})$ and $(\pi^{-1}(V), \tilde{\psi})$ are smoothly compatible.

Choosing a countable cover $\{U_\alpha\}$ of $\mathcal{M}$ from the smooth coordinate domains of $\mathcal{M}$, a countable cover of $T\mathcal{M}$ is obtained as $\{\pi^{-1}(U_\alpha)\}$: this countable cover satisfies the conditions (1)-(4) of Lemma 1.1.1. To check the Hausdorff condition (5), just note that any two points in same fiber of π lie in the same chart, while if (p, X) and (q, Y) lie in different fibers, there exist disjoint smooth coordinate domains $U, V \subseteq \mathcal{M}$ such that $p \in U$ and $q \in V$, and $\pi^{-1}(U) \ni (p, X)$ and $\pi^{-1}(V) \ni (q, Y)$ are disjoint coordinate neighbourhoods. Therefore, by Lemma 1.1.1, $T\mathcal{M}$ is a smooth $2n$ -manifold.

Now, to see that $\pi : T\mathcal{M} \rightarrow \mathcal{M}$ is smooth, note that, with respect to charts (U, φ) of $\mathcal{M}$ and $(\pi^{-1}(U), \tilde{\varphi})$ of $T\mathcal{M}$, its coordinate representation is $\pi(p, X) = (x^1, \dots, x^n)$. $\square$

The coordinates (x^i, X^i) are called the *natural coordinates* on $T\mathcal{M}$.

A particular case in which the tangent bundle can in fact be thought as a Cartesian product is that of a smooth n -manifold $\mathcal{M}$ which can be covered by a single smooth chart $(\mathcal{M}, \varphi)$. In this case, φ is a diffeomorphism from $\mathcal{M}$ to $\varphi(\mathcal{M}) \subseteq \mathbb{R}^n$: defining $\tilde{\varphi} : \pi^{-1}(\mathcal{M}) = T\mathcal{M} \rightarrow \mathcal{M} \times \mathbb{R}^n$ as in the previous proof, it retains the nature of diffeomorphism of φ , hence $T\mathcal{M}$ is diffeomorphic to $\mathcal{M} \times \mathbb{R}^n$.

Given a smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$, the introduction of the tangent bundle allows to define a **global differential** $df : T\mathcal{M}_1 \rightarrow T\mathcal{M}_2$, which is just the map whose restriction to each $T_p\mathcal{M}_1 \subseteq T\mathcal{M}_1$ is df_p . The global differential $df : T\mathcal{M}_1 \rightarrow T\mathcal{M}_2$ is a smooth map, since in terms of natural coordinates on $T\mathcal{M}_1$ it is expressed as:

$$df(x^1, \dots, x^n, X^1, \dots, X^n) = \left(f^1(x), \dots, f^m(x), \frac{\partial f^1}{\partial x^i}(x) X^i, \dots, \frac{\partial f^m}{\partial x^i}(x) X^i \right)$$

Lemma 1.2.4 (Properties of global differentials)

Let $\mathcal{M}, \mathcal{M}_1, \mathcal{M}_2$ be smooth manifolds. Then, given smooth maps $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ and $g : \mathcal{M}_2 \rightarrow \mathcal{M}_3$:

- (1) $d(g \circ f) = dg \circ df$;
- (2) $d(\mathbb{1}_{\mathcal{M}}) = \mathbb{1}_{T\mathcal{M}}$;
- (3) if f is a diffeomorphism, the df is also a diffeomorphism, and $(df)^{-1} = d(f^{-1})$.

(3) allows to set unambiguous $df^{-1} \equiv (df)^{-1} = d(f^{-1})$ when f is a diffeomorphism.

§1.2.5 Vector fields

The notion of a vector field over $U \in \mathcal{T}(\mathbb{R}^n)$ as a continuous map $U \mapsto \mathbb{R}^n$ which assigns vector to each point $p \in U$ can be extended directly to smooth manifolds.

Definition 1.2.5 (Vector field)

Let $\mathcal{M}$ be a smooth manifold. A **vector field** over $\mathcal{M}$ is a continuous map $X : \mathcal{M} \rightarrow T\mathcal{M}$ such that $\pi \circ X = \mathbb{1}_{\mathcal{M}}$, where $\pi : T\mathcal{M} \rightarrow \mathcal{M}$ is the natural projection.

The map which defines a vector field is usually written as $p \mapsto X_p$, hence the condition $\pi \circ X = \mathbb{1}_{\mathcal{M}}$ is equivalent to $X_p \in T_p\mathcal{M}$. If this map is smooth, it is called a **smooth vector field**: the set of all the smooth vector fields on a smooth manifold $\mathcal{M}$ is denoted by $\mathfrak{X}(\mathcal{M})$. This is trivially a vector space under pointwise addition and constant multiplication.

Smooth vector fields can also be multiplied by smooth functions. Given $f \in \mathcal{C}^\infty(\mathcal{M})$ and $X \in \mathfrak{X}(\mathcal{M})$, then $fX : \mathcal{M} \rightarrow T\mathcal{M} : (fX)_p = f(p)X_p$ is still a smooth vector field¹. Then, as per Eq. 1.2.8, a vector field can be expressed as:

$$X = X^i \frac{\partial}{\partial x^i}$$

where $X^i : p \mapsto X_p^i$ are its *component functions* in the given chart: a vector field is smooth if and only if its component functions are smooth.

Example 1.2.1 (Coordinate vector fields)

If (U, φ) is a smooth chart on the smooth manifold $\mathcal{M}$ and $(x^1, \dots, x^n)$ are the coordinate functions of φ , then the map:

$$p \mapsto \left. \frac{\partial}{\partial x^i} \right|_p$$

determines a vector field on U , called the i^{th} **coordinate vector field**, which is trivially smooth since its component functions are constants.

Example 1.2.2 (Angle coordinates on tori)

Let θ be an angle coordinate on $U \in \mathcal{T}(\mathbb{S}^1)$, and let $\frac{d}{d\theta}$ denote the corresponding coordinate vector field. Since any other angle coordinate may differ from θ only by an additive constant, by Eq. 1.2.11 $\frac{d}{d\theta} = \frac{d}{d\theta}$ on their common domain. Thus, there is a globally-defined smooth vector field on $\mathbb{S}^1$ whose coordinate representation is $\frac{d}{d\theta}$ with respect to any angle coordinate.

Generalizing to $\mathbb{T}^n = \mathbb{S}^1 \times \dots \times \mathbb{S}^1$, the globally-defined smooth vector fields are $\frac{d}{d\theta^i}$, for each $i = 1, \dots, n$.

§1.2.5.1 Frames

Coordinate vector fields in a smooth chart are not the only way to represent vector fields.

Suppose $\mathcal{M}$ is a smooth n -manifold. An ordered k -tuple $(X_1, \dots, X_k)$ of vector fields defined on some subset $U \subseteq \mathcal{M}$ is said to be **linearly independent** if $(X_1|_p, \dots, X_k|_p)$ is a linearly independent k -tuple in $T_p\mathcal{M}$ for each $p \in U$, and is said to *span the tangent bundle* if $(X_1|_p, \dots, X_k|_p)$ spans $T_p\mathcal{M}$ at each $p \in U$.

Definition 1.2.6 (Frame)

Let $\mathcal{M}$ be a smooth n -manifold. A **local frame** on $\mathcal{M}$ is an ordered n -tuple $(E_1, \dots, E_n)$ of vector fields on $U \in \mathcal{T}(\mathcal{M})$ that is linearly independent and spans the tangent bundle $T\mathcal{M}$. If $U = \mathcal{M}$, it is a **global frame**, and a **smooth frame** if the vector fields E_i are smooth.

¹This means that $\mathfrak{X}(\mathcal{M})$ is a module over the ring $\mathcal{C}^\infty(\mathcal{M})$.

If $(E_1, \dots, E_n)$ is a local frame, then the tangent vectors $(E_1|_p, \dots, E_n|_p)$ form a basis of $T_p\mathcal{M}$ for each $p \in U$. The coordinate vector fields of some chart (U, φ) form a *coordinate frame*.

Proposition 1.2.4 (Completion of local frames)

Let $\mathcal{M}$ be a smooth n -manifold. Then:

- (1) if $(X_1, \dots, X_k)$ is a linearly independent k -tuple of smooth vector fields on $U \in \mathcal{T}(\mathcal{M})$, with $1 \leq k < n$, then for each $p \in U$ there exist $X_{k+1}, \dots, X_n$ smooth vector fields on a neighbourhood $V \ni p$ such that $(X_1, \dots, X_n)$ is a smooth local frame on $U \cap V$;
- (2) if $(v_1, \dots, v_k)$ is a linearly independent k -tuple of vectors in $T_p\mathcal{M}$ for some $p \in \mathcal{M}$, with $1 \leq k \leq n$, then there exists a smooth local frame $(X_1, \dots, X_n)$ on a neighbourhood of p such that $X_i|_p = v_i \ \forall i = 1, \dots, k$;

For subsets of $\mathbb{R}^n$, there is a special type of frames. A k -tuple of vector fields $(E_1, \dots, E_k)$ on some $A \subseteq \mathbb{R}^n$ is *orthonormal* if for each $p \in A$ the vectors $(E_1|_p, \dots, E_k|_p)$ are orthonormal with respect to the Euclidean scalar product (with the identification $T_p\mathbb{R}^n \cong \mathbb{R}^n$). A (local or global) frame consisting of orthonormal vector fields is called an **orthonormal frame**.

Starting from a smooth local frame $(X_1, \dots, X_n)$ on $T\mathbb{R}^n$ over $U \in \mathcal{T}(\mathbb{R}^n)$, it is possible to find a smooth orthonormal frame $(E_1, \dots, E_n)$ on $T\mathbb{R}^n$ over U such that:

$$\text{span}(E_1|_p, \dots, E_j|_p) = \text{span}(X_1|_p, \dots, X_j|_p)$$

for each $j = 1, \dots, n$ and each $p \in U$, using the *Gram–Schmidt algorithm*:

$$E_j = \frac{X_j - \sum_{i=1}^{j-1} (X_j \cdot E_i) E_i}{\left\| X_j - \sum_{i=1}^{j-1} (X_j \cdot E_i) E_i \right\|} \quad (1.2.14)$$

This is a recursive definition, and it is well-posed since $X_j|_p \notin \text{span}(E_1|_p, \dots, E_{j-1}|_p)$ for each $j = 1, \dots, n$ and each $p \in U$, so the denominator is a nowhere-vanishing smooth function on U .

While smooth local frames are easily found, using Prop. 1.2.4, global ones are not. Indeed, smooth manifolds which admit a smooth local frames are said to be **parallelizable**: for example, $\mathbb{R}^n$ and $\mathbb{T}^n$ are all parallelizable, while the only parallelizable spheres are $\mathbb{S}^1$, $\mathbb{S}^3$ and $\mathbb{S}^7$. In general, the parallelizability of a smooth n -manifold $\mathcal{M}$ is linked to whether its tangent bundle is diffeomorphic to $\mathcal{M} \times \mathbb{R}^n$.

§1.2.5.2 Vector fields as derivations

A fundamental property of vector fields is that they define linear operators on the space smooth real-valued functions. If $X \in \mathfrak{X}(\mathcal{M})$ and $f \in \mathcal{C}^\infty(U)$, with $U \in \mathcal{T}(\mathcal{M})$, then $Xf : U \rightarrow \mathbb{R}$ defined by:

$$(Xf)(p) = X_p(f) \quad (1.2.15)$$

which can be shown to be smooth over U . Since the action of a tangent vector on a function is determined by the values of the function on an arbitrarily small neighbourhood, it follows that Xf is locally determined: on any open subset $V \subseteq U$, it is $(Xf)|_V = X(f|_V)$.

Since $X \in \mathfrak{X}(\mathcal{M})$ defines an endomorphism in $\mathcal{C}^\infty(\mathcal{M})$ as $f \mapsto Xf$ which obeys the Leibniz rule, it generalizes Def. 1.2.2.

Definition 1.2.7 (Derivation)

Let $\mathcal{M}$ be a smooth manifold. Then, $X \in \text{End } \mathcal{C}^\infty(\mathcal{M})$ is a **derivation** of $\mathcal{C}^\infty(\mathcal{M})$ if it satisfies the Leibniz rule:

$$X(fg) = fXg + gXf \quad \forall f, g \in \mathcal{C}^\infty(\mathcal{M}) \quad (1.2.16)$$

In general, derivations of $\mathcal{C}^\infty(\mathcal{M})$ can be identified with smooth vectors, since it can be shown that a map $D : \mathcal{C}^\infty(\mathcal{M}) \rightarrow \mathcal{C}^\infty(\mathcal{M})$ is a derivation if and only if it is of the form $Df = Xf$, for some $X \in \mathfrak{X}(\mathcal{M})$. For this reason, the same symbol X denotes both maps $X : \mathcal{M} \rightarrow T\mathcal{M}$ and $X : \mathcal{C}^\infty(\mathcal{M}) \rightarrow \mathcal{C}^\infty(\mathcal{M})$.

§1.2.5.3 Push-forward of vector fields

If $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a smooth map and X is a vector field on $\mathcal{M}_1$, then for each point $p \in \mathcal{M}_1$ a tangent vector to $\mathcal{M}_2$ is obtained as $df_p(X_p) \in T_{f(p)}\mathcal{M}_2$. However, this does not in general define a vector field on $\mathcal{M}_2$: for example, if f is not surjective, there is no way to decide how to assign tangent vectors to points $q \in \mathcal{M}_2 - f(\mathcal{M}_1)$, while if f is not injective there are some points $q \in \mathcal{M}_2$ where many tangent vectors can be assigned, one for each point in $f^{-1}(q)$.

Definition 1.2.8 (f -relation)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map, $X \in \mathfrak{X}(\mathcal{M}_1)$ and $Y \in \mathfrak{X}(\mathcal{M}_2)$. Then, X and Y are **f -related** if $df_p(X_p) = Y_{f(p)} \quad \forall p \in \mathcal{M}_1$.

f -related vector fields can be characterized by their action on smooth functions.

Proposition 1.2.5

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map, $X \in \mathfrak{X}(\mathcal{M}_1)$ and $Y \in \mathfrak{X}(\mathcal{M}_2)$. Then, X and Y are f -related if and only if for every $g \in \mathcal{C}^\infty(U)$, with $U \in \mathcal{T}(\mathcal{M}_2)$:

$$X(g \circ f) = Y(g) \circ f \quad (1.2.17)$$

Proof. For any $p \in \mathcal{M}_1$ and any $g \in \mathcal{C}^\infty(U)$, with $U \in \mathcal{T}(\mathcal{M}_2) : U \ni f(p)$:

$$X(g \circ f)(p) = X_p(g \circ f) = df_p(X_p)(g)$$

while:

$$Y(g) \circ f(p) = (Y(g))(f(p)) = Y_{f(p)}(g)$$

Therefore $X(g \circ f) = Y(g) \circ f \iff df_p(X_p) = Y_{f(p)} \quad \forall p \in \mathcal{M}_1$. □

Note that, for a given smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$, there may be no f -related vector fields. In a special case, however, the existence is guaranteed.

Theorem 1.2.2 (Push-forward)

Let $\mathcal{M}_1, \mathcal{M}_2$ be smooth manifolds and $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ a diffeomorphism. For every $X \in \mathfrak{X}(\mathcal{M}_1)$ there is a unique $Y \in \mathfrak{X}(\mathcal{M}_2)$ that is f -related to X , called the **push-forward** of X .

Proof. Given $X \in \mathfrak{X}(\mathcal{M}_1)$, define the vector field $Y_p \equiv df_{f^{-1}(q)}(X_{f^{-1}(q)}) \quad \forall q \in \mathcal{M}_2$. Clearly, this is the unique vector that is f -related to X . To show that it is smooth, note that $Y : \mathcal{M}_2 \rightarrow T\mathcal{M}_2$

is the following composition:

$$\mathcal{M}_2 \xrightarrow{f^{-1}} \mathcal{M}_1 \xrightarrow{X} T\mathcal{M}_1 \xrightarrow{df} T\mathcal{M}_2$$

which is a composition of smooth maps, hence it is smooth. $\square$

The push-forward of X by a diffeomorphism f is usually denoted by f_*X , thus by the preceding proof:

$$(f_*X)_{f(p)} := df_p(X_p) \quad (1.2.18)$$

An alternative characterization, in terms of vector fields directly, comes from Prop. 1.2.5. For any $g \in \mathcal{C}^\infty(\mathcal{M}_2)$:

$$(f_*X)(g) \circ f = X(g \circ f) \quad (1.2.19)$$

§1.2.5.4 Lie brackets

Consider a smooth manifold $\mathcal{M}$ and $X, Y \in \mathfrak{X}(\mathcal{M})$. Since smooth vector fields are derivations of $\mathcal{C}^\infty(\mathcal{M})$, given $f \in \mathcal{C}^\infty(\mathcal{M})$ it is possible to apply Y to $X(f)$, obtaining another smooth function $Y(X(f)) \equiv YX(f) \in \mathcal{C}^\infty(\mathcal{M})$. The operator $f \mapsto YX(f)$, however, does not satisfy Leibniz rule, hence it does not define a vector field.

To obtain another vector field from two smooth vector fields, define the **Lie bracket** of X and Y as $[X, Y] : \mathcal{C}^\infty(\mathcal{M}) \rightarrow \mathcal{C}^\infty(\mathcal{M})$ such that:

$$[X, Y](f) := XY(f) - YX(f) \quad (1.2.20)$$

Lemma 1.2.5 (Lie bracket)

Let $X, Y \in \mathfrak{X}(\mathcal{M})$. Then $[X, Y] \in \mathfrak{X}(\mathcal{M})$.

Proof. Since smooth vector fields are identified with derivations of $\mathcal{C}^\infty(\mathcal{M})$, it suffices to show that $[X, Y]$ is a derivation of $\mathcal{C}^\infty(\mathcal{M})$. For arbitrary $f, g \in \mathcal{C}^\infty(\mathcal{M})$:

$$\begin{aligned} [X, Y](fg) &= X(Y(fg)) - Y(X(fg)) = X(fY(g) - gY(f)) - Y(fX(g) - gX(f)) \\ &= X(f)Y(g) + fXY(g) - X(g)Y(f) - gXY(f) + \\ &\quad - Y(f)X(g) - fYX(g) + Y(g)X(f) + gYX(f) \\ &= fXY(g) + gXY(f) - fYX(g) - gYX(f) = f[X, Y](g) + g[X, Y](f) \end{aligned}$$

which concludes the proof. $\square$

In general, the expression for $[X, Y]_p(f)$ contains second derivatives of f at $p \in \mathcal{M}$ that cancel out, leaving a simple expression. Since $[X, Y]$ is a smooth vector field, its action on function is determined locally $[X, Y](f)|_U = [X, Y](f|_U)$, thus its expression needs only to be computed in a single smooth chart $(U, (x^i))$:

$$[X, Y](f) = X^i \frac{\partial}{\partial x^i} \left(Y^j \frac{\partial f}{\partial x^j} \right) - Y^j \frac{\partial}{\partial x^j} \left(X^i \frac{\partial f}{\partial x^i} \right) = X^i \frac{\partial Y^j}{\partial x^i} \frac{\partial f}{\partial x^j} - Y^j \frac{\partial X^i}{\partial x^j} \frac{\partial f}{\partial x^i}$$

where the Schwarz lemma $\partial_i \partial_j f = \partial_j \partial_i f$ has been used, as $f \in \mathcal{C}^\infty(\mathcal{M})$. Renaming dummy indices, the coordinate expression of the Lie bracket is found:

$$[X, Y] = \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right) \frac{\partial}{\partial x^j} \quad (1.2.21)$$

It is then clear, using again the Schwarz lemma, that for coordinate vector fields:

$$\left[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right] \equiv 0$$

Lemma 1.2.6 (Properties of Lie brackets)

Let $\mathcal{M}$ be a smooth manifold and $X, Y, Z \in \mathfrak{X}(\mathcal{M})$. Then:

(1) (bilinearity) for $a, b \in \mathbb{R}$:

$$[aX + bY, Z] = a[X, Z] + b[Y, Z] \quad [Z, aX + bY] = a[Z, X] + b[Z, Y]$$

(2) (antisymmetry):

$$[X, Y] = -[Y, X]$$

(3) (Jacobi identity):

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0 \quad (1.2.22)$$

(4) for $f, g \in \mathcal{C}^\infty(\mathcal{M})$:

$$[fX, gY] = fg[X, Y] + (fX(g))Y - (gY(f))X \quad (1.2.23)$$

Proof. Bilinearity and antisymmetry are trivial by definition. The Jacobi identity and (4) are direct computations. $\square$

The importance of Eq. 1.2.23 will be clear once the Lie derivative is introduced (see §1.4.3).

Proposition 1.2.6 (Naturality of Lie brackets)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map, $X_1, X_2 \in \mathfrak{X}(\mathcal{M}_1)$ and $Y_1, Y_2 \in \mathfrak{X}(\mathcal{M}_2)$ such that X_i is f -related to Y_i , for $i = 1, 2$. Then, $[X_1, X_2]$ is f -related to $[Y_1, Y_2]$.

Proof. Using Prop. 1.2.5 and the fact that X_i and Y_i are f -related, given $g \in \mathcal{C}^\infty(\mathcal{M}_1)$:

$$X_1 X_2 (g \circ f) = X_1 (Y_2 (g) \circ f) = (Y_1 Y_2 (g)) \circ f$$

and in a similar way $X_2 X_1 (g \circ f) = (Y_2 Y_1 (g)) \circ f$, therefore:

$$[X_1, X_2] (g \circ f) = (Y_1 Y_2 (g)) \circ f - (Y_2 Y_1 (g)) \circ f = [Y_1, Y_2] (g) \circ f$$

which shows that $[X_1, X_2]$ and $[Y_1, Y_2]$ are f -related. $\square$

The special case of $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ diffeomorphism yields the push-forward of a Lie bracket:

$$f_*[X_1, X_2] = [f_*X_1, f_*X_2] \quad (1.2.24)$$

§1.3 Submanifolds

In the study of integration (§2.3), a fundamental class of objects are submanifolds, i.e. manifolds which are defined inside other manifolds through special kinds of functions: immersions and embeddings.

§1.3.1 Submersions, immersions and embeddings

Definition 1.3.1 (Rank of smooth map)

Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be smooth manifolds. Given a smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$, its **rank** at $p \in \mathcal{M}_1$ is the rank² of the linear map $df_p : T_p\mathcal{M}_1 \rightarrow T_{f(p)}\mathcal{M}_2$.

Since the rank of f is the dimension of $\text{Im } df_p \subseteq T_{f(p)}\mathcal{M}_2$, and by Eq. 1.3.1, the rank of f is always bounded above by $\min_{\mathbb{R}}\{\dim_{\mathbb{R}} \mathcal{M}_1, \dim_{\mathbb{R}} \mathcal{M}_2\}$: in particular, if the rank of df_p is equal to this upper bound at some point $p \in \mathcal{M}_1$, f is said to have **full rank** at p . The interesting class of smooth maps are those with **constant rank**, which have the same rank r at each point of their domain: in this case, it is written $\text{rk } f = r$.

Definition 1.3.2 (Submersions and immersions)

A smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a **smooth submersion** if df_p is surjective at each point, i.e. $\text{rk } f = \dim_{\mathbb{R}} \mathcal{M}_2$, while it is a **smooth immersion** if df_p is injective at each point, i.e. $\text{rk } f = \dim_{\mathbb{R}} \mathcal{M}_1$.

By continuity, if df_p is surjective/injective, then there exists a neighbourhood $U \in \mathcal{T}(\mathcal{M}_1) : p \in U$ such that $f|_U$ is a submersion/immersion.

Example 1.3.1 (Submersions and immersions)

Suppose $\mathcal{M}_1, \dots, \mathcal{M}_k$ are smooth manifolds. Then, each projection map $\pi_i : \mathcal{M}_1 \times \dots \times \mathcal{M}_k \rightarrow \mathcal{M}_i$ is a smooth submersion. In particular, the projection $\pi : \mathbb{R}^{n+k} \rightarrow \mathbb{R}^n$ on the first n coordinates is a smooth submersion.

The following smooth map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ (torus of revolution) is a smooth immersion of $\mathbb{R}^2$ into $\mathbb{R}^3$:

$$f(u, v) = ((2 + \cos 2\pi u) \cos 2\pi v, (2 + \cos 2\pi u) \sin 2\pi v, \sin 2\pi u)$$

The inverse function theorem has a particular corollary for constant-rank smooth maps.

Theorem 1.3.1 (Rank theorem)

Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be smooth manifolds without boundary of dimensions m and n , respectively, and let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map of constant rank r . For each $p \in \mathcal{M}_1$, there exist smooth charts (U, φ) of $\mathcal{M}_1$ and $(V, \psi) : f(U) \subseteq V$ of $\mathcal{M}_2$ centered at p and $f(p)$, respectively, such that f has a coordinate representation of the form:

$$\hat{f}(x^1, \dots, x^r, x^{r+1}, \dots, x^m) = (x^1, \dots, x^r, 0, \dots, 0) \quad (1.3.2)$$

Although the rank theorem has a local nature, it has an important global corollary.

Corollary 1.3.1.1 (Global rank theorem)

Given a smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ between smooth manifolds without boundary, then:

²Given a linear map $T : V \rightarrow W$ between finite-dimensional $\mathbb{K}$ -vector spaces, the *rank* of T is defined as $\text{rk } T := \dim_{\mathbb{K}} \text{Im } T$, while its *nullity* as $\text{null } T := \dim_{\mathbb{K}} \ker T$. They are linked by the rank-nullity theorem:

$$\dim_{\mathbb{K}} V = \text{rk } T + \text{null } T \quad (1.3.1)$$

- (1) if f is surjective, then it is a smooth submersion;
- (2) if f is injective, then it is a smooth immersion;
- (3) if f is bijective, then it is a diffeomorphism.

In the case of smooth manifolds with boundary, the rank theorem applies unchanged to interior points. For boundary points, the result holds for smooth immersions.

Theorem 1.3.2 (Local immersion theorem for boundaries)

Let $\mathcal{M}_1$ be a smooth m -manifold with boundary and $\mathcal{M}_2$ a smooth n -manifold without boundary, and let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth immersion. For any $p \in \partial\mathcal{M}_1$, there exists a smooth boundary chart (U, φ) of $\mathcal{M}_1$ and a smooth chart $(V, \psi) : f(U) \subseteq V$ centered at p and $f(p)$, respectively, such that f has a coordinate representation of the form:

$$\hat{f}(x^1, \dots, x^m) = (x^1, \dots, x^m, 0, \dots, 0) \quad (1.3.3)$$

No such theorem can be formulated for smooth maps whose codomain has a boundary, since the image of the map could intersect the boundary in unpredictable ways, hence the map cannot be put in a simple canonical forms without imposing strong constraints on it.

§1.3.1.1 Embeddings

A continuous map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a **topological embedding** if it is a homeomorphism $\mathcal{M}_1 \rightarrow f(\mathcal{M}_1) \subseteq \mathcal{M}_2$ in the subspace topology of $f(\mathcal{M}_1)$ determined by that of $\mathcal{M}_2$.

Definition 1.3.3 (Smooth embedding)

A smooth immersion $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a **smooth embedding** if it is also a topological embedding.

Clearly, every composition of smooth embeddings is still a smooth embedding. Note that a smooth embedding is *both* a topological embedding and a smooth immersion, and not just a topological embedding that happens to be smooth.

Proposition 1.3.1 (Criteria for smooth embeddings)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be an injective smooth immersion. If any of the following conditions hold, then f is a smooth embedding:

- (1) f is an open or closed map^a;
- (2) f is a proper map^b;
- (3) $\mathcal{M}_1$ is compact;
- (4) $\partial\mathcal{M}_1 = \emptyset$ and $\dim_{\mathbb{R}} \mathcal{M}_1 = \dim_{\mathbb{R}} \mathcal{M}_2$.

^aAn arbitrary map $f : X \rightarrow Y$ between topological spaces is said to be an *open/closed map* if it maps open/closed subsets to open/closed subsets. Note that continuous maps can be open, closed, both or neither.

^bAn arbitrary map $f : X \rightarrow Y$ between topological spaces is said to be a *proper map* if, for every compact subset $K \subseteq Y$, the preimage $f^{-1}(K) \subseteq X$ is compact.

This topological conditions allow to formulate a local condition for smooth immersions.

Theorem 1.3.3 (Local embedding theorem)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map. Then, f is a smooth immersion if and only if every point in $\mathcal{M}_1$ has a neighbourhood $U \in \mathcal{T}(\mathcal{M}_1)$ such that $f|_U : U \rightarrow \mathcal{M}_2$ is a smooth embedding.

Proof. ($\Leftarrow$) If every point in $\mathcal{M}_1$ has a neighbourhood on which f is a smooth embedding, then f has full rank everywhere on $\mathcal{M}_1$, so it is a smooth immersion.

($\Rightarrow$) Suppose f is a smooth immersion, and let $p \in \mathcal{M}_1$. First, show that p has a neighbourhood on which f is injective. If $f(p) \notin \partial\mathcal{M}_2$, then either by Th. 1.3.1 (if $p \notin \partial\mathcal{M}_1$) or by Th. 1.3.2 (if $p \in \partial\mathcal{M}_1$) there is a neighbourhood U_1 of p on which f has coordinate representation of the form Eq. 1.3.3, hence $f|_{U_1}$ is injective. If, on the other hand, $f(p) \in \partial\mathcal{M}_2$, let (W, ψ) be any smooth boundary chart of $\mathcal{M}_2$ centered at $f(p)$. Let $U_0 = f^{-1}(W) \ni p$, and let $\iota : \mathbb{U}^n \hookrightarrow \mathbb{R}^n$ be the inclusion map, so that the preceding argument can be applied to $\iota \circ \psi \circ f|_{U_0} : U_0 \rightarrow \mathbb{R}^n$ to show that p has a neighbourhood $U_1 \subseteq U_0$ such that $\iota \circ \psi \circ f|_{U_1}$ is injective, from which it follows that $f|_{U_1}$ is injective.

Now, let $p \in \mathcal{M}_1$ be arbitrary, and let U be a neighbourhood of p on which f is injective. It can be shown that $f|_U$ is a topological embedding, hence, being f a smooth immersion, $f|_U$ is a smooth embedding. $\square$

This local embedding theorem hints to a topological definition of immersions: given two topological spaces X, Y , a continuous map $f : X \rightarrow Y$ is a **topological immersion** if every point of X has a neighbourhood $U \in \mathcal{T}(X)$ such that $f|_U : U \rightarrow Y$ is a topological embedding. Therefore, every smooth immersion is also a topological immersion, but, just like embeddings, a smooth immersion is not a topological immersion that happens to be smooth.

§1.3.1.2 Submersions

If $\pi : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is any continuous map, a *section* of π is a continuous right inverse for π , i.e. a continuous map $\sigma : \mathcal{M}_2 \rightarrow \mathcal{M}_1$ such that $\pi \circ \sigma = \mathbb{1}_{\mathcal{M}_2}$. Similarly, a local section is a section only defined on some $U \in \mathcal{T}(\mathcal{M}_2)$. A key property of submersions is that they admit a plethora of smooth local sections.

Theorem 1.3.4 (Local section theorem)

Let $\pi : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map. Then, π is a smooth submersion if and only if every point of $\mathcal{M}_1$ is in the image of a smooth local section of π .

Proof. ($\Rightarrow$) Suppose π is a smooth submersion. Given $p \in \mathcal{M}_1$, let $q = \pi(p) \in \mathcal{M}_2$. By Th. 1.3.1, it is possible to choose coordinates $(x^1, \dots, x^m)$ centered at p and $(y^1, \dots, y^n)$ centered at q in which π has coordinate representation:

$$\hat{\pi}(x^1, \dots, x^n, \dots, x^{n+1}, \dots, x^m) = (x^1, \dots, x^n)$$

Given a sufficiently small $\varepsilon \in \mathbb{R}^+$, the coordinate cube defined as $C_\varepsilon \equiv \{x \in \mathbb{R}^m : |x^i| \leq \varepsilon \forall i = 1, \dots, m\}$ is a neighbourhood of p , whose image under π is the cube $C'_\varepsilon \equiv \{y \in \mathbb{R}^n : |y^i| \leq \varepsilon \forall i = 1, \dots, n\}$. Then, the map $\sigma : C'_\varepsilon \rightarrow C_\varepsilon$ whose coordinate representation is:

$$\sigma(x^1, \dots, x^n) = (x^1, \dots, x^n, 0, \dots, 0)$$

is a smooth local section of π such that $\sigma(q) = p$ (see Fig. 1.8).

($\Leftarrow$) Assume that each point of $\mathcal{M}_1$ is in the image of a smooth local section of π . Then, given $p \in \mathcal{M}_1$, let $\sigma : U \rightarrow \mathcal{M}_1$ be a smooth local section of π such that $\sigma(q) = p$, where $q = \pi(p) \in$

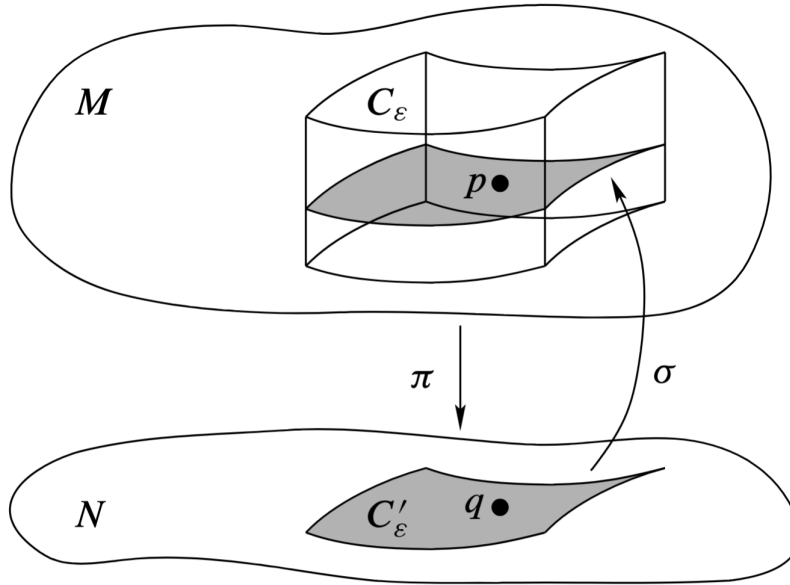


Figure 1.8: Local section of a smooth submersion.

$U \subseteq \mathcal{M}_2$. The equation $\pi \circ \sigma = \text{id}_U$ implies $d\pi_p \circ d\sigma_q = \text{id}_{T_q \mathcal{M}_2}$, which in turn implies that $d\pi_p$ is surjective, i.e. that π is a smooth submersion. $\square$

This theorem hints to a topological definition of submersions: given two topological spaces X, Y , a continuous map $\pi : X \rightarrow Y$ is a **topological submersion** if every point of X is in the image of a continuous local section of π . Therefore, every smooth submersion is also a topological submersion.

§1.3.2 Embedded submanifolds

Definition 1.3.4 (Embedded submanifold)

Let $\mathcal{M}$ be a smooth manifold. An **embedded submanifold** of $\mathcal{M}$ is a subset $S \subseteq \mathcal{M}$ that is a manifold in the subspace topology, endowed with a smooth structure with respect to which the inclusion map $S \hookrightarrow \mathcal{M}$ is a smooth embedding.

The **codimension** of an embedded submanifold $S \subseteq \mathcal{M}$ is defined as $\text{codim}_{\mathbb{R}} S := \dim_{\mathbb{R}} \mathcal{M} - \dim_{\mathbb{R}} S$, and $\mathcal{M}$ is called the *ambient manifold* of S . A codimension-1 embedded submanifold is an *embedded hypersurface*. If the inclusion map $S \hookrightarrow \mathcal{M}$ is only a smooth immersion, the S is an *immersed submanifold*.

Proposition 1.3.2 (Embeddings and submanifolds)

Let $\mathcal{M}_1$ be a smooth submanifold with/without boundary and $\mathcal{M}_2$ a smooth submanifold, and let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth embedding. With the subspace topology, $f(\mathcal{M}_1)$ is a topological manifold with/without boundary, and it has a unique smooth structure which makes it an embedded submanifold with/without boundary of $\mathcal{M}_2$ such that $f : \mathcal{M}_1 \rightarrow f(\mathcal{M}_1)$ is a diffeomorphism.

Proof. Set $S = f(\mathcal{M}_1)$ and give it the subspace topology inherited from $\mathcal{M}_2$. Then, f being an embedding means that $f : \mathcal{M}_1 \rightarrow S$ is a homeomorphism, and thus S is a topological manifold. Give S a smooth structure by taking smooth charts of the form $(f(U), \varphi \circ f^{-1})$, where (U, φ) is any smooth chart of $\mathcal{M}_1$: smooth compatibility of these charts follows from smooth compatibility of

charts of $\mathcal{M}_1$. With this smooth structure on S , the map $f : \mathcal{M}_1 \rightarrow f(\mathcal{M}_1)$ is a diffeomorphism essentially by definition, and this is clearly the only smooth structure on S with this property. Then, the inclusion map $\iota : S \hookrightarrow \mathcal{M}_2$ is just the composition of $S \xrightarrow{f^{-1}} \mathcal{M}_1 \xrightarrow{f} \mathcal{M}_2$, i.e. the composition of a diffeomorphism with a smooth embedding, and it therefore is a smooth embedding. $\square$

By Prop. 1.3.2, every embedded submanifold is the image of a smooth embedding: its own inclusion map. An embedded submanifold $S \subseteq \mathcal{M}$ is said to be *properly embedded* if the inclusion $S \hookrightarrow \mathcal{M}$ is a proper map. By topological considerations, it can be shown that $S \subseteq \mathcal{M}$ is properly embedded if and only if it is a closed subset of $\mathcal{M}$: in particular, every compact embedded submanifold is properly embedded, as compact subsets of Hausdorff spaces are closed.

Lemma 1.3.1 (Codimension-0 embedded submanifolds)

Every open subset of a smooth manifold is an embedded codimension-0 submanifold.

Given a smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$, it is crucial to know whether f is still smooth when its domain or codomain is restricted to a submanifold. The case of a restricted domain is not difficult, while the case of a restricted codomain requires more care.

Theorem 1.3.5 (Restricting maps to submanifolds)

Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be smooth manifolds, and let $S \subseteq \mathcal{M}_1$ be an embedded submanifold. Then:

- (1) if $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a smooth map, then $f|_S : S \rightarrow \mathcal{M}_2$ is smooth;
- (2) if $f : \mathcal{M}_2 \rightarrow \mathcal{M}_1$ is a smooth map such that $f(\mathcal{M}_2) \subseteq S$, then $f : \mathcal{M}_2 \rightarrow S$ is smooth.

§1.3.2.1 Slice charts

Given $U \in \mathcal{T}(\mathbb{R}^n)$ and $k \in \{0, \dots, n\}$, a *k-dimensional slice of U* (or simply a *k-slice*) is any subset of U of the form:

$$S = \{(x^1, \dots, x^k, x^{k+1}, \dots, x^n) \in U : x^{k+1} = c^{k+1}, \dots, x^n = c^n\}$$

for some constants $c^{k+1}, \dots, c^n \in \mathbb{R}$. Trivially, the only n -slice is $S = U$. Moreover, it is also possible to defined slices by setting different coordinates other than the last ones to constants: in any case, every k -slice is homeomorphic to an open subset of $\mathbb{R}^k$.

Let $\mathcal{M}$ be a smooth n -manifold, and let (U, φ) be a smooth chart on $\mathcal{M}$. If $S \subseteq U$ such that $\varphi(S)$ is a k -slice of $\varphi(U)$, then S is a *k-slice of U* . A typical k -slice is defined by setting the $n - k$ constants to zero: this can always be achieved by subtracting said constants from the respective coordinate functions. Given $S \subseteq \mathcal{M}$ and $k \in \{0, \dots, n\}$, then S satisfies the *local k-slice condition* if each point of S is contained in the domain of a smooth chart (U, φ) of $\mathcal{M}$ such that $S \cap U$ is a single k -slice of U (see Fig. 1.9): any such chart is a **slice chart** of S in $\mathcal{M}$, and the corresponding coordinates are called *slice coordinates*.

Theorem 1.3.6 (Local slice criterion for embedded submanifolds)

Let $\mathcal{M}$ be a smooth n -submanifold without boundary. If $S \subseteq \mathcal{M}$ is an embedded k -dimensional submanifold without boundary, then S satisfies the local k -slice condition. Conversely, if $S \subseteq \mathcal{M}$ satisfies the local k -slice condition, then it is a topological k -manifold without boundary with the subspace topology, and it has a unique smooth structure making it a k -dimensional embedded submanifold without boundary of $\mathcal{M}$.

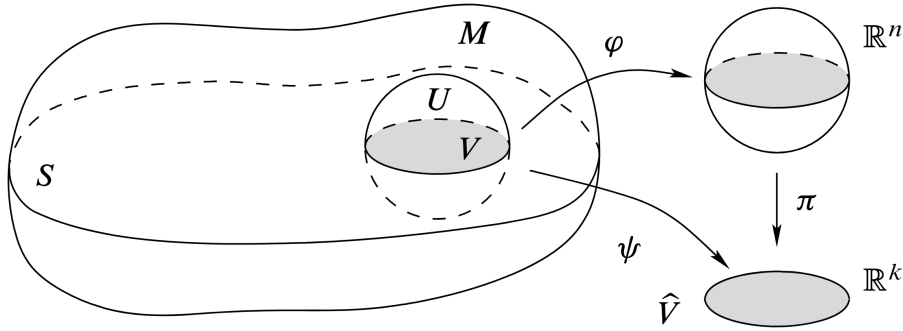


Figure 1.9: A slice chart of a subset satisfying the k -chart condition.

Note that the local slice condition is a condition on the subset $S \subseteq \mathcal{M}$ only: it does not presuppose any particular topology or smooth structure on S .

Example 1.3.2 (Spheres as submanifolds)

To show that $\mathbb{S}^n$ is an embedded submanifold of $\mathbb{R}^{n+1}$, recall Ex. 1.1.1: the intersection of $\mathbb{S}^n$ with the open subset $\{x \in \mathbb{R}^{n+1} : x^i > 0\}$ is the graph of a smooth function:

$$x^i = f(x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^{n+1})$$

Similarly, the intersection of $\mathbb{S}^n$ with the open subset $\{x \in \mathbb{R}^{n+1} : x^i < 0\}$ is the graph of $-f$. This means that every point in $\mathbb{S}^n$ is in one of these sets, hence $\mathbb{S}^n$ satisfies the local n -slice condition: therefore, $\mathbb{S}^n$ is an embedded submanifold of $\mathbb{R}^{n+1}$, with slice charts exactly given by Ex. 1.1.1.

If $\mathcal{M}$ is a smooth manifold with non-empty boundary, then S might intersect $\partial\mathcal{M}$ in complicated ways, hence it is very difficult to prove general results about the existence of slice charts of S in $\mathcal{M}$ in this case. However, there is a special case: if $\mathcal{M}$ is a smooth n -manifold with boundary, then $\partial\mathcal{M}$ is a topological $(n-1)$ -manifold without boundary with the subspace topology, and it has a unique smooth structure such that it is a properly embedded submanifold of $\mathcal{M}$.

§1.3.2.2 Level sets

Embedded manifolds are often presented as solution sets of system of equations.

Definition 1.3.5 (Level set)

Let $\Phi : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be any map between smooth manifolds. A **level set** of Φ is a set $\Phi^{-1}(c)$, with $c \in \mathcal{M}_2$.

Obviously, there are plenty of level sets of smooth functions which are not smooth submanifolds. However, a particular case is that of constant-rank smooth maps between manifolds without boundary, in which case the rank theorem applies.

Proposition 1.3.3 (Constant-rank level sets)

Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be smooth manifolds without boundary, and let $\Phi : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map with constant rank r . Then, each level set of Φ is a properly embedded submanifold of codimension r in $\mathcal{M}_1$.

Proof. Setting $m = \dim_{\mathbb{R}} \mathcal{M}_1$, $n = \dim_{\mathbb{R}} \mathcal{M}_2$ and $k = m - r$, let $c \in \mathcal{M}_2$ be arbitrary, and let $S = \Phi^{-1}(c) \subseteq \mathcal{M}_1$. From Th. 1.3.1, for each $p \in S$ there are smooth charts (U, φ) centered at p and (V, ψ) centered at $c = \Phi(p)$ in which Φ has the coordinate expression given in Eq. 1.3.2, so that $S \cap U$ is the following k -slice:

$$S \cap U = \{(x^1, \dots, x^r, x^{r+1}, \dots, x^m) \in U : x^1 = \dots = x^r = 0\}$$

Therefore, S satisfies the local k -slice condition, so it is an embedded submanifold of dimension k . It is closed in $\mathcal{M}_1$ by continuity, hence it is properly embedded. $\square$

In the case of a smooth submersion, which by definition has constant rank equal to the dimension of the codomain, the level sets are properly embedded submanifolds of dimension $\dim_{\mathbb{R}} \mathcal{M}_1 - \dim_{\mathbb{R}} \mathcal{M}_2$ in $\mathcal{M}_1$. This is analogous to the fact that the kernel of a linear map $L : \mathbb{R}^m \rightarrow \mathbb{R}^r$ is a linear subspace of codimension r , by the rank-nullity theorem (Eq. 1.3.1).

Given a smooth map $\Phi : \mathcal{M}_1 \rightarrow \mathcal{M}_2$, a point $p \in \mathcal{M}_1$ can be classified as follows: if $d\Phi_p : T_p \mathcal{M}_1 \rightarrow T_{\Phi(p)} \mathcal{M}_2$ is surjective, then p is a **regular point** of Φ , otherwise it is a **critical point**. In particular, every point of $\mathcal{M}_1$ is critical if $\dim_{\mathbb{R}} \mathcal{M}_1 < \dim_{\mathbb{R}} \mathcal{M}_2$, while every point of $\mathcal{M}_1$ is regular if and only if Φ is a submersion.: in any case, the set of regular points of Φ is always an open subset of $\mathcal{M}_1$. The same denominations can be imported on points of $\mathcal{M}_2$: a point $c \in \mathcal{M}_2$ is a **regular value** of Φ if every point of the level set $\Phi^{-1}(c)$ is a regular point, otherwise it is a **critical value**. By convention, if $\Phi^{-1}(c) = \emptyset$, then c is a regular value of Φ . If c is a regular value of Φ , then $\Phi^{-1}(c)$ is a **regular level set**: in other words, a regular level set consists entirely of regular points of Φ (points p where $d\Phi_p$ is surjective).

Theorem 1.3.7 (Regular level set theorem)

Every regular level set of a smooth map between smooth manifolds without boundary is a properly embedded submanifold of codimension is equal to the dimension of the codomain.

Proof. Let $\Phi : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map and let $c \in \mathcal{M}_2$ be a regular value of Φ : the set U of points $p \in \mathcal{M}_1$ where $\text{rk } d\Phi_p = \dim_{\mathbb{R}} \mathcal{M}_2$ is open in $\mathcal{M}_1$ and contains $\Phi^{-1}(c)$ by hypothesis. Therefore, $\Phi|_U : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a smooth submersion, and by Prop. 1.3.3 $\Phi^{-1}(c)$ is an embedded submanifold of U . Since the composition of smooth embeddings $\Phi^{-1}(c) \hookrightarrow U \hookrightarrow \mathcal{M}_1$ is again a smooth embedding, it follows that $\Phi^{-1}(c)$ is an embedded submanifold of $\mathcal{M}_1$, and it is closed by continuity. $\square$

Note that Th. 1.3.7 also applies to empty level sets, which are both regular level sets and properly embedded submanifolds.

Definition 1.3.6 (Defining map)

Let $S \subseteq \mathcal{M}$ be an embedded submanifolds. A smooth map $\Phi : \mathcal{M} \rightarrow \mathcal{M}'$ is a **defining map** of S if S is a regular level set of Φ .

More generally, if $U \in \mathcal{T}(\mathcal{M})$ and $\Phi : U \rightarrow \mathcal{M}'$ is a smooth map such that $S \cap U$ is a regular level set of Φ , then Φ is a **local defining map** of S . It can be shown that every embedded submanifold admits a local defining function (i.e. $\mathcal{M}' = \mathbb{R}^{\dim_{\mathbb{R}} \mathcal{M} - \text{rk } d\Phi}$) in a neighbourhood of each one of its points.

Example 1.3.3 (Surfaces of revolution)

Let $H = \{(r, z) \in \mathbb{R}^2 : r > 0\}$ and suppose $C \subseteq H$ is an embedded 1-dimensional submanifold

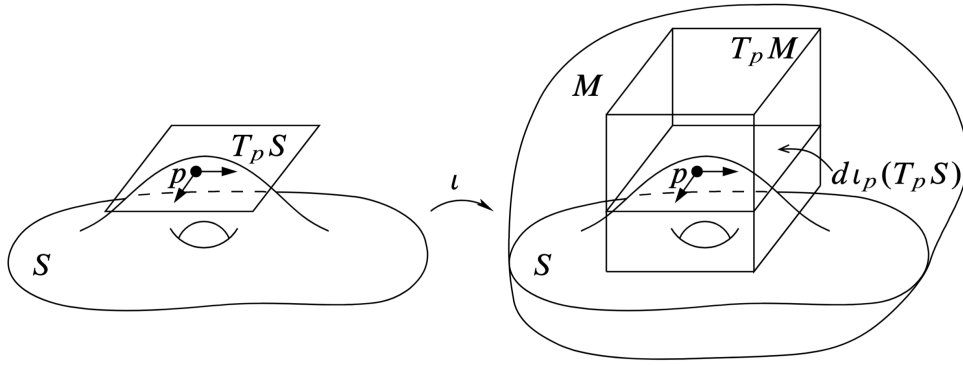


Figure 1.10: Tangent space to an embedded submanifold.

of H . The **surface of revolution** determined by C is the subset $S_C \subseteq \mathbb{R}^3$ defined as:

$$S_C = \{(x, y, z) \in \mathbb{R}^3 : (\sqrt{x^2 + y^2}, z) \in C\}$$

The set C is the *generating curve* of S_C . If $\varphi : U \rightarrow \mathbb{R}$ is any local defining function of C in H , a local defining function Φ for S_C is found as:

$$\Phi(x, y, z) = \varphi(\sqrt{x^2 + y^2}, z)$$

defined on:

$$\tilde{U} = \{(x, y, z) \in \mathbb{R}^3 : (\sqrt{x^2 + y^2}, z) \in U\}$$

A straightforward computation shows that the Jacobian of Φ is:

$$\mathcal{J}_\Phi(x, y, z) = \left(\frac{x}{r} \frac{\partial \varphi}{\partial r}(r, z), \frac{y}{r} \frac{\partial \varphi}{\partial r}(r, z), \frac{\partial \varphi}{\partial z}(r, z) \right)$$

As $\mathcal{J}_\Phi|_{S_C} \neq 0$, S_C is a regular level set of Φ and is an embedded 2-submanifold of $\mathbb{R}^3$.

If $\mathcal{M}$ is a smooth manifold with boundary, a **boundary-defining function** is a smooth function $f : \mathcal{M} \rightarrow [0, \infty)$ such that $f^{-1}(0) = \partial \mathcal{M}$ and $df_p \neq 0$ at all points $p \in \partial \mathcal{M}$. For example, $f(x) = 1 - \|x\|^2$ is a boundary-defining function for the unit disk $\mathbb{D}^n$. It can be shown that every smooth manifold with boundary admits a boundary-defining function.

§1.3.2.3 Tangent space to a submanifold

Let $\mathcal{M}$ be a smooth manifold, and let $S \subseteq \mathcal{M}$ be an embedded submanifold. Since the inclusion map $\iota : S \hookrightarrow \mathcal{M}$ is a smooth immersion, at each point $p \in S$ the linear map $d\iota_p : T_p S \rightarrow T_p \mathcal{M}$ is injective: for any vector $v \in T_p S$, the image vector $\tilde{v} = d\iota_p(v) \in T_p \mathcal{M}$ acts on $f \in C^\infty(\mathcal{M})$ as:

$$\tilde{v}(f) = d\iota_p(v)(f) = v(f \circ \iota) = v(f|_S)$$

Identifying $T_p S$ with its image under $d\iota_p$, from now on $T_p S$ is thought of as a linear subspace of $T_p \mathcal{M}$, as illustrated in Fig. 1.10. There are several ways of characterizing $T_p S \subseteq T_p \mathcal{M}$.

Proposition 1.3.4

Let $\mathcal{M}$ be a smooth manifold and $S \subseteq \mathcal{M}$ an embedded submanifold, and let $p \in S$. Then, $T_p S$

is characterized as a subspace of $T_p\mathcal{M}$ as:

$$T_pS = \{v \in T_p\mathcal{M} : v(f) = 0 \ \forall f \in \mathcal{C}^\infty(\mathcal{M}) : f|_S = 0\} \quad (1.3.4)$$

Proof. First, suppose $v \in T_pS \subseteq T_p\mathcal{M}$: more precisely, $v = d\iota_p(w)$ for some $w \in T_pS$, where $\iota : S \hookrightarrow \mathcal{M}$ is the inclusion map. Given $f \in \mathcal{C}^\infty(\mathcal{M}) : f|_S = 0$, then $f \circ \iota \equiv 0$, so:

$$v(f) = d\iota_p(w)(f) = w(f \circ \iota) = 0$$

Conversely, if $v \in T_p\mathcal{M}$ satisfies $v(f) = 0$ whenever $f \in \mathcal{C}^\infty(\mathcal{M}) : f|_S = 0$, WTS $\exists w \in T_pS : v = d\iota_p(w)$. Let $(x^1, \dots, x^n)$ be slice coordinates of S in some neighbourhood U of p , so that $U \cap S$ is the subset of U where $x^{k+1} = \dots = x^n = 0$, and $(x^1, \dots, x^k)$ are coordinates for $U \cap S$. In these coordinates, the inclusion map $\iota : S \cap U \hookrightarrow \mathcal{M}$ has the following coordinate expression:

$$\iota(x^1, \dots, x^k) = (x^1, \dots, x^k, 0, \dots, 0)$$

Thus, T_pS (to be precise $d\iota_p(T_pS)$) is exactly the subspace of $T_p\mathcal{M}$ spanned by $\partial_1|_p, \dots, \partial_k|_p$. Writing the coordinate expression of $v \in T_p\mathcal{M}$:

$$v = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i} \Big|_p$$

hence $v \in T_pS$ if and only if $v^i = 0$ for all $i > k$. Now, consider $\hat{f} = x^j$, for an index $j > k$: clearly $f|_S = 0$, hence:

$$0 = v(f) = \sum_{i=1}^n v^i \frac{\partial x^j}{\partial x^i} = v^j$$

Therefore, $v^i = 0$ for all $i > k$, thus $v \in T_pS$. □

If an embedded submanifold is defined by a defining map, this map gives a straightforward characterization of the tangent space to each point of the submanifold.

Proposition 1.3.5

Let $\mathcal{M}$ be a smooth manifold and $S \subseteq \mathcal{M}$ an embedded submanifold. If $\Phi : U \rightarrow \mathcal{M}'$ is any local defining map of S , with $U \subseteq \mathcal{M}$, then $T_pS = \ker d\Phi_p$ for each $p \in S \cap U$.

Proof. Identify T_pS with the subspace $d\iota_p(T_pS) \subseteq T_p\mathcal{M}$, where $\iota : S \hookrightarrow \mathcal{M}$ is the inclusion map. Since $\Phi \circ \iota$ is constant on $S \cap U$, it follows that $d\Phi_p \circ d\iota_p$ is the zero map $T_pS \rightarrow T_{\Phi(p)}\mathcal{M}'$, hence $\text{Im } d\iota_p \subseteq \ker d\Phi_p$. On the other hand, $d\Phi_p$ is surjective by definition of defining map, hence Eq. 1.3.1 implies:

$$\dim_{\mathbb{R}} \ker d\Phi_p = \dim_{\mathbb{R}} T_p\mathcal{M} - \dim_{\mathbb{R}} T_{\Phi(p)}\mathcal{M}' = \dim_{\mathbb{R}} T_pS = \dim_{\mathbb{R}} \text{Im } d\iota_p$$

Therefore, $\text{Im } d\iota_p = \ker d\Phi_p$. □

In the special case of $\mathcal{M}' = \mathbb{R}^k$, this means that a vector $v \in T_p\mathcal{M}$ is tangent to S if and only if $v(\Phi^i) = 0$ for each $i = 1, \dots, k$.

§1.3.2.4 Regular domains

Definition 1.3.7 (Regular domain)

Let $\mathcal{M}$ be a smooth manifold. A **regular domain** is a properly embedded codimension-0 submanifold with boundary of $\mathcal{M}$.

Familiar examples of regular domains are $\mathbb{U}^n \subseteq \mathbb{R}^n$ and $\mathbb{D}^n \subseteq \mathbb{R}^n$.

Given a topological space X , an *exhaustion function* of X is a continuous function $f : X \rightarrow \mathbb{R}$ such that the set $f^{-1}((-\infty, c])$, called a *sublevel set* of f , is compact for each $c \in \mathbb{R}$.

Proposition 1.3.6

Suppose $\mathcal{M}$ is a smooth manifold and $f \in \mathcal{C}^\infty(\mathcal{M})$. Then:

- (1) for each regular value b of f , the sublevel set $f^{-1}((-\infty, b])$ is a regular domain in $\mathcal{M}$;
- (2) if a and b are two regular values of f , with $a < b$, then $f^{-1}([a, b])$ is a regular domain in $\mathcal{M}$.

A set of the form $f^{-1}((-\infty, b])$ for b regular value of $f \in \mathcal{C}^\infty(\mathcal{M})$ is called a *regular sublevel set* of f : (1) of Prop. 1.3.6 shows that every regular sublevel set of a smooth $\mathbb{R}$ -valued function is a regular domain. If $D \subseteq \mathcal{M}$ is a regular domain and $f \in \mathcal{C}^\infty(\mathcal{M})$ is a smooth function such that D is a regular sublevel set of f , then f is called a *defining function* of D .

Theorem 1.3.8 (Defining function of regular domain)

Let $\mathcal{M}$ be a smooth manifold and $D \subseteq \mathcal{M}$ a regular domain. Then, there exists a defining function for D , which can be taken to be a smooth exhaustion function of $\mathcal{M}$ if D is compact.

§1.3.3 Sard's and Whitney's theorems

Fundamental theorems of differential topology are found extending the notion of sets of measure zero to manifolds.

§1.3.3.1 Sets of measure zero

A key notion in Analysis is that of measure-zero sets, i.e. sets which are negligible in integrals.

A *closed rectangle* in $\mathbb{R}^n$ is a product set $[a_1, b_1] \times \cdots \times [a_n, b_n]$, with $a_i < b_i$ for each $i = 1, \dots, n$, and an *open rectangle* is defined analogously with open intervals. In either case, the **volume** of a rectangle A is defined as:

$$\text{Vol}_n(A) := \prod_{i=1}^n (b_i - a_i) \quad (1.3.5)$$

Definition 1.3.8 (Set of measure zero)

A subset $X \subseteq \mathbb{R}^n$ is said to have ***n-dimensional measure zero*** if for every $\delta \in \mathbb{R}^+$ there exists a countable cover of X by open rectangles $\{C_i\}_{i \in \mathcal{I}}$ such that:

$$\sum_{i \in \mathcal{I}} \text{Vol}_n(C_i) < \delta \quad (1.3.6)$$

The same definition applies if open rectangles are replaced with open cubes or open balls. Note that the property of being measure-zero depends on the particular Euclidean space considered: for example, an open interval in the x -axis has measure zero in $\mathbb{R}^2$, but not in $\mathbb{R}^1$.

Lemma 1.3.2 (Properties of measure-zero sets)

- (1) Every subset of a measure-zero set in $\mathbb{R}^n$ has measure zero in $\mathbb{R}^n$.
- (2) A countable union of sets of measure zero in $\mathbb{R}^n$ has measure zero in $\mathbb{R}^n$.
- (3) If $k < n$, then every subset of $\mathbb{R}^k$ has measure zero in $\mathbb{R}^n$.

Proposition 1.3.7 (Measure and smooth maps)

Suppose $A \subseteq \mathbb{R}^n$ has measure zero in $\mathbb{R}^n$ and $f : A \rightarrow \mathbb{R}^n$ is a smooth map. Then, $f(A)$ has measure zero in $\mathbb{R}^n$.

The notion of measure-zero set can be extended in a diffeomorphism-invariant way to smooth manifolds: a subset $A \subseteq \mathcal{M}$ of a smooth n -manifold has **measure zero in $\mathcal{M}$** if, for every smooth chart (U, φ) on $\mathcal{M}$, the subset $\varphi(A \cap U) \subseteq \mathbb{R}^n$ has n -dimensional measure zero.

Lemma 1.3.3

Let $\mathcal{M}$ be a smooth n -manifold and $A \subseteq \mathcal{M}$, and let $\{U_\alpha, \varphi_\alpha\}$ be a countable collection of smooth charts on $\mathcal{M}$ such that $A \subseteq \bigcup_\alpha U_\alpha$ and $\varphi_\alpha(A \cap U_\alpha)$ has measure zero in $\mathbb{R}^n$ for each α . Then, A has measure zero in $\mathcal{M}$.

Proof. Let (V, ψ) be an arbitrary smooth chart on $\mathcal{M}$. WTS $\psi(A \cap V)$ has measure zero in $\mathbb{R}^n$. For each U_α :

$$\psi(A \cap V \cap U_\alpha) = (\psi \circ \varphi_\alpha^{-1}) \circ \varphi_\alpha(A \cap V \cap U_\alpha)$$

Now, $\varphi_\alpha(A \cap V \cap U_\alpha) \subseteq \varphi_\alpha(A \cap U_\alpha)$, which has measure zero in $\mathbb{R}^n$ by hypothesis. By Prop. 1.3.7, then, $\psi(A \cap V \cap U_\alpha)$ has measure zero. Since $\psi(V \cap A)$ is the union of countably many such sets, by (2) of Lemma 1.3.2 it too has measure zero. $\square$

(2) of Lemma 1.3.2 holds in the manifold case too: a countable union of sets of measure zero in $\mathcal{M}$ has measure zero in $\mathcal{M}$.

Proposition 1.3.8 (Measure and smooth maps)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map between smooth n -manifolds, and let $A \subseteq \mathcal{M}_1$ be a set of measure zero in $\mathcal{M}_1$. Then, $f(A)$ has measure zero in $\mathcal{M}_2$.

Proof. Let $\{(U_i, \varphi_i)\}$ be a countable cover of $\mathcal{M}_1$ by smooth charts, and let (V, ψ) be an arbitrary smooth chart on $\mathcal{M}_2$. WTS $\psi(f(A) \cap V)$ has measure zero in $\mathbb{R}^n$: this set is the union of countably many subsets of the form $\psi \circ f \circ \varphi_i^{-1}(\varphi_i(A \cap U_i \cap f^{-1}(V)))$, each of which has measure zero in $\mathbb{R}^n$ by Prop. 1.3.7. $\square$

§1.3.3.2 Sard's theorem

Sard's theorem is fundamental in the study of embedded manifolds.

Theorem 1.3.9 (Sard's theorem)

The set of critical values of a smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ has measure zero in $\mathcal{M}_2$.

Proof. Let $m = \dim_{\mathbb{R}} \mathcal{M}_1$ and $n = \dim_{\mathbb{R}} \mathcal{M}_2$: the proof is by induction on m .

($m = 0$) In this case the result is immediate: if $n = 0$, the f has no critical points, while if $n > 0$ the entire image of f has measure zero by (3) of Lemma 1.3.2.

($m \leq 1$) Assume the thesis to be true for maps whose domains have dimension less than m . By covering $\mathcal{M}_1$ and $\mathcal{M}_2$ with countably many smooth charts, the proof reduces to the case of f being a smooth map from $U \in \mathcal{T}(\mathbb{R}^m)$ (or $\mathbb{U}^m$) to $\mathbb{R}^n$. Let $(x^1, \dots, x^m)$ be smooth coordinates on the domain U and $(y^1, \dots, y^n)$ smooth coordinates on the codomain, and let $C \subseteq U$ denote the set of critical points of f . Define a decreasing sequence of subsets $C \supseteq C_1 \supseteq C_2 \supseteq \dots$ as:

$$C_k \equiv \{x \in C : \text{all } i^{\text{th}} \text{ partial derivatives of } f \text{ vanish at } x, \text{ for } 1 \leq i \leq k\}$$

By continuity, C and C_k are closed in U , for all $k \in \mathbb{N}$. The proof of $f(C)$ has measure zero in $\mathcal{M}_2$ is in three steps:

- (1) WTS $f(C - C_1)$ has measure zero. Since C_1 is closed in U , it is possible to replace U with $U - C_1$ and assume $C_1 = \emptyset$. Let $a \in C$: this assumption means that some first partial derivative of f is not zero at a . WLOG let $\partial_1 f^1(a) \neq 0$: this means that it is possible to defined new smooth coordinates $(u, v) = (u, v^2, \dots, v^m)$ in some neighbourhood $V_a \subseteq U$ of a as:

$$u = f^1 \quad v^2 = x^2 \quad \dots \quad v^m = x^m$$

since the Jacobian of the coordinate transformation is non-singular at a . Shrinking V_a if necessary, assume that $\bar{V}_a$ is a compact set of U and that this coordinates extend smoothly to $\bar{V}_a$. The coordinate representation of f and its jacobian in this coordinates is:

$$f(u, v^2, \dots, v^m) = (u, f^2(u, v), \dots, f^n(u, v)) \quad \partial f(u, v) = \begin{bmatrix} 1 & 0 \\ \cdot & \frac{\partial F^i}{\partial v^j} \end{bmatrix}$$

Therefore, $C \cap \bar{V}_a$ consists of exactly those points where the $(n-1) \times (m-1)$ matrix $[\partial F^i / \partial v^j]$ has rank less than $n-1$. WTS $f(C \cap \bar{V}_a)$ has measure zero in $\mathbb{R}^n$.

Lemma 1.3.4 (Compact sets of measure zero)

Let $K \subseteq \mathbb{R}^n$ be a compact subset such that $K \cap (\{c\} \times \mathbb{R}^{n-1})$ has $(n-1)$ -dimensional measure zero for every $c \in \mathbb{R}$. Then K has n -dimensional measure zero.

Since $f(C \cap \bar{V}_a)$ is compact, by Lemma 1.3.4 it suffices to show that its intersection with the hyperplane $y^1 = c$ has $(n-1)$ -dimensional measure zero. For $c \in \mathbb{R}$, let $B_c \equiv \{v \in \mathbb{R}^{m-1} : (c, v) \in \bar{V}_a\}$ and define $f_c : B_c \rightarrow \mathbb{R}^{n-1}$ as:

$$f_c(v) = (f^2(c, v), \dots, f^n(c, v))$$

Since $f(c, v) = (c, f_c(v))$, the critical values of $f|_{\bar{V}_a}$ that lie in the hyperplane $y^1 = c$ are of the form (c, w) , with w critical value of f_c . By the inductive hypothesis, the set of critical values of each f_c has $(n-1)$ -dimensional measure zero: thus, by Lemma 1.3.4, $f(C \cap \bar{V}_a)$ has measure zero. As U is covered by countably many sets of the form $\bar{V}_a$, $f(C \cap U)$ is a countable union of sets of measure zero and thus has measure zero.

- (2) WTS $f(C_k - C_{k+1})$ has measure zero for each $k \in \mathbb{N}$. Again, since C_{k+1} is closed in U , it is possible to set $C_{k+1} = \emptyset$ and assume that at every point of C_k there is some $(k+1)^{\text{th}}$ partial derivative of f that does not vanish. Let $a \in C_k$, and let $y : U \rightarrow \mathbb{R}$ denote the k^{th} partial derivative of f that has at least one non-vanishing first partial derivative at a . Then, a is a regular point of the smooth map y , so there is a neighbourhood V_a of a consisting of regular points of y . Let $Y = y^{-1}(0) \cap V_a$ be the zero set of y in V_a , which is a smooth hypersurface by Th. 1.3.7: by definition of C_k , all k^{th} derivative of f (including y) vanish on C_k , so $C_k \cap V_a \subseteq Y$. At any $p \in C_k \cap V_a$, df_p is not surjective, so $d(f|_Y)_p = df_p|_{T_p Y}$ is not surjective either. Thus, $f(C_k \cap V_a)$ is contained in the set of critical values of $f|_Y : Y \rightarrow \mathbb{R}^n$, which has measure zero by the induction hypothesis. As U can be covered by countably many sets V_a , $f(C_k - C_{k+1})$ is contained in a countable union of sets $f(C_k \cap V_a)$, thus it has measure zero.
- (3) WTS $f(C_k)$ has measure zero for $k > m/n - 1$. For each $a \in U$, there is a closed cube $E \subseteq U : a \in E$: since U can be covered by countably many such cubes, it suffices to show that $f(C_k \cap E)$ has measure zero. Let A be a constant that bounds the absolute values of all the $(k+1)^{\text{th}}$ derivatives of f in E , let R denote the side length of E and let $K \in \mathbb{N}$ large (see later). Subdividing E into K^m cubes of side R/K , denoted by $(E_1, \dots, E_{K^m})$, if there is a point $a_i \in C_k \cap E_i$, then it can be shown (using Taylor's remainder theorem) that:

$$\|f(x) - f(a_i)\| \leq A' \|x - a_i\|^{k+1}$$

for some constant A' which depends on A , k and m . Thus, $f(E_i)$ is contained in a ball of radius $A'(R/K)^{k+1}$, implying that $f(C_k \cap E)$ is contained in a union of K^m balls whose total n -dimensional volume is no more than:

$$K^m (A')^n (E/K)^{n(k+1)} \equiv A'' K^{m-nk-n}$$

Since $k > m/n - 1$, i.e. $m - nk - n < 0$, by choosing K sufficiently large this volume can be made as small as desired, so $f(C_k \cap E)$ has measure zero by Def. 1.3.8.

These steps cover all possible subsets of $C \cap U$, hence $f(C \cap U)$ has measure zero. $\square$

Corollary 1.3.9.1

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map. If $\dim_{\mathbb{R}} \mathcal{M}_1 < \dim_{\mathbb{R}} \mathcal{M}_2$, then $f(\mathcal{M}_1)$ has measure zero in $\mathcal{M}_2$.

Proof. In this case each point of $\mathcal{M}_1$ is a critical point of f . $\square$

Corollary 1.3.9.2

Let $\mathcal{M}$ be a smooth manifold and $S \subseteq \mathcal{M}$ an embedded submanifold. If $\text{codim}_{\mathbb{R}} S > 0$, then S has measure zero in $\mathcal{M}$.

Proof. Apply Cor. 1.3.9.1 to the inclusion map $S \hookrightarrow \mathcal{M}$. $\square$

§1.3.3.3 Whitney's embedding theorem

An important application of Sard's theorem is to show that every smooth manifold can be embedded into a Euclidean space.

Theorem 1.3.10 (Whitney's embedding theorem)

Every smooth n -manifold admits a proper smooth embedding into $\mathbb{R}^{2n+1}$.

Proof. The proof can be considerably simplified with the following lemma.

Lemma 1.3.5

Let $\mathcal{M}$ be a smooth n -manifold. If $\mathcal{M}$ admits a smooth embedding into $\mathbb{R}^N$ for some N , then it admits a proper smooth embedding into $\mathbb{R}^{2n+1}$.

By Lemma 1.3.5, it suffices to show that $\mathcal{M}$ admits a smooth embedding into some Euclidean space.

First, suppose $\mathcal{M}$ is compact, i.e. it admits a finite cover $\{B_1, \dots, B_m\}$ in which each B_i is a regular coordinate ball (or half-ball). This means that for each $i = 1, \dots, m$ there exist a coordinate domain $B'_i \supseteq \overline{B}_i$ and a smooth coordinate map $\varphi_i : B'_i \rightarrow \mathbb{R}^n$ that restricts to a diffeomorphism from $\overline{B}_i$ to a compact subset of $\mathbb{R}^n$. For each i , let $\rho_i : \mathcal{M} \rightarrow \mathbb{R}$ be a smooth bump function³ for $\overline{B}_i$ supported in B'_i , and define a smooth map $f : \mathcal{M} \rightarrow \mathbb{R}^{nm+m}$ as:

$$f(p) = (\rho_1(p)\varphi_1(p), \dots, \rho_m(p)\varphi_m(p), \rho_1(p), \dots, \rho_m(p))$$

where $\rho_i\varphi_i$ is extended to be zero outside of the support of ρ_i . WTS f is an injective smooth immersion, since, by compactness of $\mathcal{M}$, this implies that f is a smooth embedding.

To see that f is injective, suppose $f(p) = f(q)$. Since $\{B_1, \dots, B_m\}$ covers $\mathcal{M}$, there is some i such that $p \in B_i$. Then $\rho_i(p) = 1$, and $\rho_i(q) = \rho_i(p) = 1$ implies that $q \in \text{supp } \rho_i \subseteq B'_i$, so:

$$\varphi_i(q) = \rho_i(q)\varphi_i(q) = \rho_i(p)\varphi_i(p) = \varphi_i(p)$$

Since φ_i is injective on B'_i , this means that $p = q$, hence f is injective.

Next, to see that f is a smooth immersion, let $p \in \mathcal{M}$ be arbitrary and choose i such that $p \in B_i$. As $\rho_i \equiv 1$ on a neighbourhood of p , $d(\rho_i\varphi_i)_p = d(\varphi_i)_p$, which is injective. It follows that df_p is injective, thus f is an injective smooth immersion, and hence an embedding.

Now, suppose $\mathcal{M}$ is non-compact. Let $f : \mathcal{M} \rightarrow \mathbb{R}$ be a smooth exhaustion function of $\mathcal{M}$: Th. 1.3.9 shows that for each $i \in \mathbb{N}_0$ there are regular values of f such that $i < a_i < b_i < i + 1$. Define the subsets $D_i, E_i \subseteq \mathcal{M}$ as:

$$D_i = \begin{cases} f^{-1}((-\infty, 1]) & i = 0 \\ f^{-1}([i, i + 1]) & i \leq 1 \end{cases} \quad E_i = \begin{cases} f^{-1}((-\infty, a_i]) & i = 0 \\ f^{-1}([b_{i-1}, a_{i+1}]) & i \leq 1 \end{cases}$$

By Prop. 1.3.6, each E_i is a compact regular domain. Clearly $D_i \subseteq \text{Int } E_i$, $\mathcal{M} = \bigcup_{i \in \mathbb{N}_0} B_i$ and $E_i \cap E_j = \emptyset$ unless $j = i - 1, i, i + 1$. By the above discussion, since for each $i \in \mathbb{N}_0$ there is a smooth embedding $\varphi_i : E_i \rightarrow \mathbb{R}^{2n+1}$. For each i , let $\rho_i : \mathcal{M} \rightarrow \mathbb{R}$ be a smooth bump function on D_i and supported in $\text{Int } E_i$, and define $F : \mathcal{M} \rightarrow \mathbb{R}^{2n+1} \times \mathbb{R}^{2n+1} \times \mathbb{R}$ as:

$$F(p) = \left(\sum_{k \in \mathbb{N}_0} \rho_{2k}(p)\varphi_{2k}(p), \sum_{k \in \mathbb{N}_0} \rho_{2k+1}(p)\varphi_{2k+1}(p), f(p) \right)$$

F is smooth because only one term in each sum is non-zero in a neighbourhood of each point, and F is proper because f is.

WTS F is an injective smooth immersion, and hence a smooth embedding. Suppose $F(p) = F(q)$. Then, $p \in D_j$ for some j , and $f(p) = f(q)$ implies $q \in D_j$ as well: as in the compact case, then, $p = q$, i.e. F is injective. Now, let $p \in \mathcal{M}$ arbitrary, and choose j such that $p \in D_j$: then $\rho_j \equiv 1$ on a neighbourhood of p . Assuming j odd, for all q in this neighbourhood $F(q) = (\varphi_j(q), \dots, \dots)$, which implies that df_p is injective, and similarly for j even: this shows that f is a smooth immersion, concluding the proof. $\square$

This means that every smooth n -manifold is diffeomorphic to a properly embedded submanifold of $\mathbb{R}^{2n+1}$. Moreover, if only immersions rather than embeddings are required, it is possible to lower the dimensionality of the ambient Euclidean space by one.

Theorem 1.3.11 (Whitney's immersion theorem)

Every smooth n -manifold admits a smooth immersion into $\mathbb{R}^{2n}$.

Using much more sophisticated techniques of algebraic geometry, it is possible to prove that, in the case $n > 0$, in both theorems the dimensionality of the ambient Euclidean space can be further decreased by one (Whitney's *strong* theorems).

§1.3.3.4 Whitney's approximation theorems

It is possible to show that continuous maps can be approximated by smooth ones. To formalize the concept of approximation, given a smooth manifold $\mathcal{M}$, if $\delta : \mathcal{M} \rightarrow \mathbb{R}^+$ is a positive continuous function, two functions $f, g : \mathcal{M} \rightarrow \mathbb{R}^k$ are δ -**close** if $\|f(x) - g(x)\| < \delta(x) \forall x \in \mathcal{M}$.

Theorem 1.3.12 (Whitney's approximation theorem for functions)

Let $\mathcal{M}$ be a smooth manifold and $f : \mathcal{M} \rightarrow \mathbb{R}^k$ a continuous function. Given any positive continuous function $\delta : \mathcal{M} \rightarrow \mathbb{R}$, there exists a smooth function $g : \mathcal{M} \rightarrow \mathbb{R}^k$ that is δ -close to f . If f is smooth on a closed subset $A \subseteq \mathcal{M}$, then g can be chosen to be equal to f on A .

Proof. If f is smooth on a closed subset $A \subseteq \mathcal{M}$, it can be extended to a smooth function $f_0 : \mathcal{M} \rightarrow \mathbb{R}^k : f_0|_A = f|_A$. Then, let:

$$U_0 \equiv \{y \in \mathcal{M} : \|f_0(y) - f(y)\| < \delta(y)\}$$

Clearly $U_0 \in \mathcal{T}(\mathcal{M}) : U_0 \supseteq A$ (if $A = \emptyset$, just take $U_0 = \emptyset$ and $f_0 \equiv 0$).

WTS there exist $\{x_i\}_{i \in \mathbb{N}} \subseteq \mathcal{M} - A$ and neighbourhoods $U_i \subseteq \mathcal{M} - A : x_i \in U_i$ such that $\{U_i\}_{i \in \mathbb{N}}$ is an open countable cover of $\mathcal{M} - A$, and $\|f(y) - f(x_i)\| < \delta(y) \forall y \in U_i$. To see this, for any $x \in \mathcal{M} - A$ let U_x be a neighbourhood of x contained in $\mathcal{M} - A$ and small enough that, for all $y \in U_x$:

$$\delta(y) > \frac{1}{2}\delta(x) \quad \|f(y) - f(x)\| < \frac{1}{2}\delta(x)$$

Such a neighbourhood exists by continuity of δ and f . Then, if $y \in U_x$:

$$\|f(y) - f(x)\| < \frac{1}{2}\delta(x) < \delta(y)$$

The collection $\{U_x : x \in \mathcal{M} - A\}$ is an open cover of $\mathcal{M} - A$: choosing a countable subcover $\{U_{x_i}\}_{i \in \mathbb{N}}$ and setting $U_i = U_{x_i}$, the desired countable open cover is found.

³If X is a topological space, $A \subseteq X$ is a closed set and $U \in \mathcal{T}(X) : U \supseteq A$, a continuous function $\psi : X \rightarrow \mathbb{R}$ is a *bump function for A supported in U* if $0 \leq \psi|_X \leq 1$, $\psi|_A \equiv 1$ and $\text{supp } \psi \subseteq U$.

Now, let $\{\varphi_0, \varphi_i\}$ be a partition of unity subordinate to the cover $\{U_0, U_i\}$ of $\mathcal{M}$, and define $g : \mathcal{M} \rightarrow \mathbb{R}^k$ as:

$$g(y) = \varphi_0(y)f_0(y) + \sum_{i \in \mathbb{N}} \varphi_i(y)f(x_i)$$

Clearly g is smooth, and $g|_A = f|_A$. For any $y \in \mathcal{M}$, the condition $\sum_{i \in \mathbb{N}_0} \varphi_i \equiv 1$ implies:

$$\begin{aligned} \|g(y) - f(y)\| &= \left\| \varphi_0(y)f_0(y) + \sum_{i \in \mathbb{N}} \varphi_i(y)f(x_i) - \left(\varphi_0(y) + \sum_{i \in \mathbb{N}} \varphi_i(y) \right) f(y) \right\| \\ &\leq \varphi_0(y) \|f_0(y) - f(y)\| + \sum_{i \in \mathbb{N}} \varphi_i(y) \|f(x_i) - f(y)\| \\ &< \varphi_0(y)\delta(y) + \sum_{i \in \mathbb{N}} \varphi_i(y)\delta(y) = \delta(y) \end{aligned}$$

which concludes the proof. $\square$

Corollary 1.3.12.1

Let $\mathcal{M}$ be a smooth manifold and $\delta : \mathcal{M} \rightarrow \mathbb{R}^+$ a positive continuous function. Then, there is a smooth function $e : \mathcal{M} \rightarrow \mathbb{R}^+$ such that $0 < e(x) < \delta(x) \forall x \in \mathcal{M}$.

Proof. Use Th. 1.3.12 to construct a smooth function $e : \mathcal{M} \rightarrow \mathbb{R}^+$ such that:

$$\left\| e(x) - \frac{1}{2}\delta(x) \right\| < \frac{1}{2}\delta(x)$$

for all $x \in \mathcal{M}$. $\square$

To find an analogous theorem for continuous maps between smooth manifolds, it is necessary to introduce additional concepts.

For each $x \in \mathbb{R}^n$, recall the canonical identification $T_x \mathbb{R}^n \cong \mathbb{R}^n$ and the canonical diffeomorphism $T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$: by these identifications, each $T_x \mathbb{R}^n$ inherits a Euclidean inner product. Let $\mathcal{M} \subseteq \mathbb{R}^n$ be an embedded m -submanifold without boundary: for each $x \in \mathcal{M}$, the *normal space* to $\mathcal{M}$ at x is defined as the $(n - m)$ -dimensional subspace $N_x \mathcal{M} \subseteq T_x \mathbb{R}^n$ consisting of all vectors that are orthogonal to $T_x \mathcal{M}$ with respect to the Euclidean dot product. The **normal bundle** of $\mathcal{M}$, denoted by $N\mathcal{M}$, is the subset of $T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$ consisting of vectors normal to $\mathcal{M}$:

$$N\mathcal{M} = \{(x, v) \in \mathbb{R}^n \times \mathbb{R}^n : x \in \mathcal{M}, v \in N_x \mathcal{M}\} \quad (1.3.7)$$

There is a natural projection $\pi_N : N\mathcal{M} \rightarrow \mathcal{M}$ defined as the restriction to $N\mathcal{M}$ of $\pi : T\mathbb{R}^n \rightarrow \mathbb{R}^n$. Moreover, it is possible to show that $N\mathcal{M}$ is an embedded n -submanifold of $T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$. Then, defining the smooth map $E : N\mathcal{M} \subseteq \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $E(x, v) = x + v$, each normal space $N_x \mathcal{M}$ is affinely mapped onto the affine subspace through x and orthogonal to $T_x \mathcal{M}$: a *tubular neighbourhood* of $\mathcal{M}$ is a neighbourhood U of $\mathcal{M}$ in $\mathbb{R}^n$ that is the diffeomorphic image under E of an open $V \subseteq N\mathcal{M}$ of the form:

$$V = \{(x, v) \in N\mathcal{M} : \|v\| < \delta(x)\} \quad (1.3.8)$$

for some positive continuous function $\delta : \mathcal{M} \rightarrow \mathbb{R}^+$ (see Fig. 1.11). It can be shown that every embedded submanifold of $\mathbb{R}^n$ has a tubular neighbourhood.

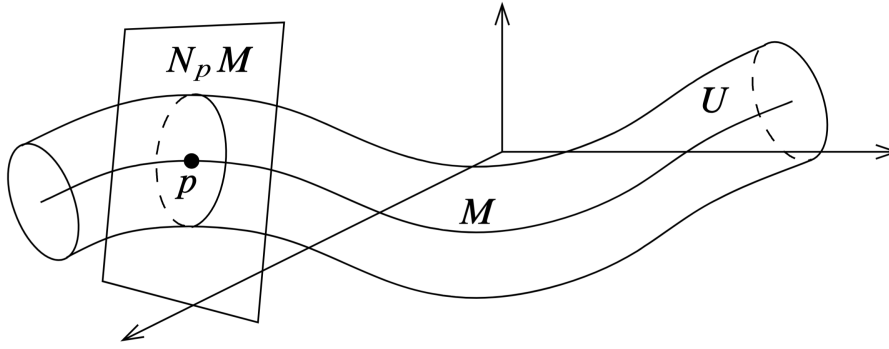


Figure 1.11: Tubular neighbourhood.

Proposition 1.3.9 (Tubular neighbourhoods and submersions)

Let $\mathcal{M} \subseteq \mathbb{R}^n$ be an embedded submanifold without boundary. If U is any tubular neighbourhood of $\mathcal{M}$, there exists a smooth map $r : U \rightarrow \mathcal{M}$ that is both a retraction⁴ and a smooth submersion.

Proof. Let $N\mathcal{M} \subseteq T\mathbb{R}^n$ be the normal bundle of $\mathcal{M}$, and let $\mathcal{M}_0 \equiv \{(x, 0) \in N\mathcal{M} : x \in \mathcal{M}\}$. By definition of tubular neighbourhood, there is an open $V \subseteq N\mathcal{M} : V \supseteq \mathcal{M}_0$ such that $E : V \rightarrow U$ is a diffeomorphism. Define $r \equiv \pi_N \circ E^{-1} : U \rightarrow \mathcal{M}$, which is smooth by composition: for $x \in \mathcal{M}$, note that $E(x, 0) = x$, so $r(x) = \pi \circ E^{-1}(x) = \pi(x, 0) = x$, which shows that r is a retraction. Since π is a smooth submersion and E^{-1} is a diffeomorphism, it follows that r is a smooth submersion. $\square$

It is then possible to generalize Whitney's approximation theorem to maps between manifolds. Reference §A.1 for properties of homotopies.

Theorem 1.3.13 (Whitney's approximation theorem)

Let $\mathcal{M}_1$ be a smooth manifold, $\mathcal{M}_2$ a smooth manifold without boundary and $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ a continuous map. Then, f is homotopic to a smooth map. If f is smooth on a closed subset $A \subseteq \mathcal{M}_1$, then the homotopy can be taken to be relative to A .

Proof. By Th. 1.3.10, assume $\mathcal{M}_2$ is a properly embedded submanifold of $\mathbb{R}^n$. Let U be a tubular neighbourhood of $\mathcal{M}_2$ in $\mathbb{R}^n$, and let $r : U \rightarrow \mathcal{M}_2$ be the smooth retraction given by Prop. 1.3.9. For any $x \in \mathcal{M}_2$, let:

$$\delta(x) \equiv \sup_{\mathbb{R}} \{\varepsilon \leq 1 : B_\varepsilon(x) \subseteq U\}$$

Using the triangle inequality, it is possible to show that $\delta : \mathcal{M}_2 \rightarrow \mathbb{R}^+$ is continuous. Let $\tilde{\delta} \equiv \delta \circ f : \mathcal{M}_1 \rightarrow \mathbb{R}^+$: by Th. 1.3.12, there exists a smooth function $g : \mathcal{M}_1 \rightarrow \mathbb{R}^n$ that is $\tilde{\delta}$ -close to f , and is equal to f on A (which might be empty). Let $H : \mathcal{M}_1 \times [0, 1] \rightarrow \mathcal{M}_2$ be the composition of r with the straight-line homotopy between f and g :

$$H(p, t) = r((1 - t)f(p) + tg(p))$$

This is well-defined, as $\|g(p) - f(p)\| < \tilde{\delta}(p) = \delta(f(p)) \forall p \in \mathcal{M}_1$ by $\tilde{\delta}$ -closeness, which means that $g(p)$ is contained in a ball of radius $\delta(f(p))$ around $f(p)$: since this ball is contained in U , so is the entire line segment from $f(p)$ to $g(p)$.

⁴Given a topological space X , a *retraction* of X onto a subspace $Y \subseteq X$ is a continuous map $r : X \rightarrow Y$ such that $r|_Y = \mathbb{1}_Y$.

Therefore, H is a homotopy between $H(p, 0) = f(p)$ and $H(p, 1) = r(g(p))$, which is a smooth map by composition. Moreover, it satisfies $H(p, t) = f(p)$ for $p \in A$, since $f|_A = g|_A$. $\square$

Corollary 1.3.13.1 (Extension lemma for smooth maps)

Let $\mathcal{M}_1$ be a smooth manifold, $\mathcal{M}_2$ a smooth manifold without boundary and $f : A \rightarrow \mathcal{M}_2$ a smooth map, with $A \subseteq \mathcal{M}_1$ closed. Then, f has a smooth extension on $\mathcal{M}_1$ if and only if it has a continuous extension on $\mathcal{M}_1$.

Proof. If $F : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a continuous extension of f on all $\mathcal{M}_1$, Th. 1.3.13 guarantees the existence of a smooth map $\tilde{F}$ that agrees with f on A , i.e. a smooth extension of f . The converse is trivial since smoothness implies continuity. $\square$

If $\mathcal{M}_1, \mathcal{M}_2$ are smooth manifolds, a homotopy $H : \mathcal{M}_1 \times [0, 1] \rightarrow \mathcal{M}_2$ is a *smooth homotopy* if it is also a smooth map, i.e. if it extends to a smooth map on some neighbourhood of $\mathcal{M}_1 \times [0, 1]$ in $\mathcal{M}_1 \times \mathbb{R}$. Two maps are *smoothly homotopic* if there is a smooth homotopy between them: smooth homotopy is an equivalence relation on the set of all smooth maps $\mathcal{M}_1 \rightarrow \mathcal{M}_2$.

Proposition 1.3.10 (Smooth homotopy of smooth maps)

Let $\mathcal{M}_1$ be a smooth manifold, $\mathcal{M}_2$ a smooth manifold without boundary and $f, g : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ two smooth maps. If f and g are homotopic, then they are smoothly homotopic. If f and g are homotopic relative to some closed subset $A \subseteq \mathcal{M}_1$, then they are smoothly homotopic relative to A .

Proof. Let $H : \mathcal{M}_1 \times [0, 1] \rightarrow \mathcal{M}_2$ be a homotopy from f to g (relative to A , which may be empty): WTS H can be replaced with a smooth homotopy. To do so, define $\bar{H} : \mathcal{M}_1 \times \mathbb{R} \rightarrow \mathcal{M}_2$ as:

$$\bar{H}(x, t) = \begin{cases} H(x, t) & t \in [0, 1] \\ H(x, 0) & t \leq 0 \\ H(x, 1) & t \geq 1 \end{cases}$$

which is continuous. The restriction of $\bar{H}$ to $(\mathcal{M}_1 \times \{0\}) \cup (\mathcal{M}_1 \times \{1\})$ is smooth, since it is equal to $f \circ \pi_1$ on $\mathcal{M}_1 \times \{0\}$ and to $g \circ \pi_1$ on $\mathcal{M}_1 \times \{1\}$, where $\pi_1 : \mathcal{M}_1 \times \mathbb{R} \rightarrow \mathcal{M}_1$ is the projection on the first component. If H is a homotopy relative to A , then $\bar{H}$ is also smooth on $A \times [0, 1]$. Since $\mathcal{M}_1 \times \mathbb{R}$ is a smooth manifold, Th. 1.3.13 guarantees that there is a smooth map $\tilde{H} : \mathcal{M}_1 \times \mathbb{R} \rightarrow \mathcal{M}_2$ whose restriction to $(\mathcal{M}_1 \times \{0\}) \cup (\mathcal{M}_1 \times \{1\}) \cup (A \times [0, 1])$ agrees with $\bar{H}$, and thus with H : then, $\tilde{H}|_{\mathcal{M}_1 \times [0, 1]}$ is a smooth homotopy (relative to A) between f and g . $\square$

When the codomain manifold has non-empty boundary, Th. 1.3.13 and Prop. 1.3.10 do not hold, since it might not be possible to find a smooth map that agrees with f on A . However, discarding homotopy relative to a subset, the other results can be extended to maps between manifolds with boundary.

§1.4 Curves

The familiar notion of curve in $\mathbb{R}^n$ can be extended to generic manifolds.

Definition 1.4.1 (Curve)

Let $\mathcal{M}$ be a topological manifold. A **curve** on $\mathcal{M}$ is a continuous map $\gamma : J \rightarrow \mathcal{M}$, where $J \subseteq \mathbb{R}$ is an interval.

The definition of tangent spaces leads to a natural interpretation of velocity vectors of curves. Given a smooth curve $\gamma : J \rightarrow \mathcal{M}$ and $t_0 \in J$, the **velocity** of γ at t_0 is the vector:

$$\gamma'(t_0) := d\gamma \left(\frac{d}{dt} \Big|_{t_0} \right) \in T_{\gamma(t_0)}\mathcal{M} \quad (1.4.1)$$

where $\frac{d}{dt}|_{t_0}$ is the standard coordinate basis vector in $T_{t_0}\mathbb{R}$. This tangent vector acts on functions as:

$$\gamma'(t_0)(f) = d\gamma \left(\frac{d}{dt} \Big|_{t_0} \right) f = \frac{d}{dt} \Big|_{t_0} (f \circ \gamma) = (f \circ \gamma)'(t_0)$$

Therefore, $\gamma'(t_0)$ is the derivation at $\gamma(t_0)$ obtained by taking the derivative of a function along the curve⁵. If (U, φ) is a smooth chart with coordinate functions $(x^1, \dots, x^n)$ and $\gamma(t_0) \in U$, writing the coordinate representation of γ as $\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$, the coordinate expression for the velocity vector is:

$$\gamma'(t_0) = \frac{d\gamma^i}{dt}(t_0) \frac{\partial}{\partial x^i} \Big|_{\gamma(t_0)} \quad (1.4.2)$$

Proposition 1.4.1 (Tangent vectors and curves)

Let $\mathcal{M}$ be a smooth manifold and $p \in \mathcal{M}$. Every $X_p \in T_p\mathcal{M}$ is the velocity of some smooth curve on $\mathcal{M}$.

Proof. Let (U, φ) be a smooth chart centered at $p \in \mathcal{M}$, and write $X_p = X_p^i \partial_i|_p$. For a sufficiently small $\epsilon > 0$, let $\gamma : (-\epsilon, \epsilon) \rightarrow U$ be the curve whose coordinate representation is:

$$\gamma(t) = (tX_p^1, \dots, tX_p^n)$$

Clearly, this is a smooth curve with $\gamma(0) = p$ and $\gamma'(0) = X_p$. □

Given a smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ and a smooth curve $\gamma : J \rightarrow \mathcal{M}_1$, the velocity of the composite curve $f \circ \gamma : J \rightarrow \mathcal{M}_2$ if given, for any $t_0 \in J$, by:

$$(f \circ \gamma)'(t_0) = d(f \circ \gamma) \left(\frac{d}{dt} \Big|_{t_0} \right) = df \circ d\gamma \left(\frac{d}{dt} \Big|_{t_0} \right) = df(\gamma'(t_0)) \quad (1.4.3)$$

Since curves behave well under composition of smooth maps, it is possible to compute the action of the smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ on $X_p \in T_p\mathcal{M}_1$ as:

$$df_p(X_p) = (f \circ \gamma)'(0) \quad (1.4.4)$$

for any smooth curve $\gamma : J \rightarrow \mathcal{M}_1$ such that $\gamma(0) = p$ and $\gamma'(0) = X_p$.

§1.4.1 Integral curves

A smooth curve $\gamma : J \rightarrow \mathcal{M}$ naturally determines a tangent vector $\gamma'(t) \in T_{\gamma(t)}\mathcal{M}$ for each $t \in J$. It is also possible to work backward, associating a curve to a vector field.

⁵If J has one or two endpoints, the t -derivative is interpreted as one-sided.

Definition 1.4.2 (Integral curve)

Let $\mathcal{M}$ be a smooth manifold and X be a vector field on $\mathcal{M}$. An **integral curve** of X is a differentiable curve $\gamma : J \rightarrow \mathcal{M}$ such that $\gamma'(t) = X_{\gamma(t)} \forall t \in J$.

Finding an integral curve generally reduces to solving a system of ordinary differential equations (ODEs) in a smooth chart. Indeed, on a smooth coordinate domain $U \subseteq \mathcal{M}$, writing $\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$, the condition for γ to be an integral curve of $X \in \mathfrak{X}(\mathcal{M})$ becomes:

$$\dot{\gamma}^i(t) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)} = X_{\gamma(t)}^i \frac{\partial}{\partial x^i} \Big|_{\gamma(t)}$$

which results in the following autonomous⁶ system of ODEs (setting $X_p^i \equiv X^i(p)$):

$$\begin{cases} \dot{\gamma}^1(t) = X^1(\gamma^1(t), \dots, \gamma^n(t)) \\ \vdots \\ \dot{\gamma}^n(t) = X^n(\gamma^1(t), \dots, \gamma^n(t)) \end{cases} \quad (1.4.5)$$

An important result of Real Analysis guarantees the existence of a solution to this system.

Theorem 1.4.1 (Fundamental theorem of autonomous ODEs)

Let $U \in \mathcal{T}(\mathbb{R}^n)$ and $\mathbf{f} : U \rightarrow \mathbb{R}^n$ is a smooth vector-valued function. Consider the following Cauchy problem:

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t)) \\ \mathbf{x}(t_0) = \mathbf{c} \end{cases} \quad (1.4.6)$$

for arbitrary $t_0 \in \mathbb{R}$ and $\mathbf{c} \in U$. Then:

- (1) (existence) for any $t_0 \in \mathbb{R}$ and $\mathbf{x}_0 \in U$, there exist an open interval $J_0 \ni t_0$ and an open subset $U_0 \subseteq U : U_0 \ni \mathbf{x}_0$ such that for each $\mathbf{c} \in U$ there is a $\mathcal{C}^1$ map $\mathbf{x} : J_0 \rightarrow U$ which solves Eq. 1.4.6;
- (2) (uniqueness) any two differentiable solutions agree on their common domain;
- (3) (smoothness) given J_0 and U_0 as in (1), and given $\theta_0 : J_0 \times U_0 \rightarrow U$ defined as $\theta(t, \mathbf{c}) = \mathbf{x}(t)$, where $\mathbf{x} : J_0 \rightarrow U$ is the unique solution to Eq. 1.4.6 with $\mathbf{x}(t_0) = \mathbf{c}$, then θ is smooth.

Therefore, given $X \in \mathfrak{X}(\mathcal{M})$, for each $p \in \mathcal{M}$ there exist $\epsilon > 0$ and a smooth curve $\gamma : (-\epsilon, \epsilon) \rightarrow \mathcal{M}$ which is an integral curve of X .

Proposition 1.4.2 (Affine reparametrization invariance)

Let $X \in \mathfrak{X}(\mathcal{M})$ and $\gamma : J \rightarrow \mathcal{M}$ an integral curve of X . For any $a, b \in \mathbb{R} : a \neq 0$, the curve $\tilde{\gamma} : \tilde{J} \rightarrow \mathcal{M} : \tilde{\gamma}(t) = a\gamma(t) + b$, with $\tilde{J} = \{t \in \mathbb{R} : at + b \in J\}$, is still an integral curve of X .

Moreover, it can be shown that, given a smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$, then $X \in \mathfrak{X}(\mathcal{M}_1)$ and $Y \in \mathfrak{X}(\mathcal{M}_2)$ are f -related if and only if f takes integral curves of X to integral curves of Y , i.e. for each integral curve γ of X , $f \circ \gamma$ is an integral curve of Y .

⁶A system $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t))$ is *autonomous* if $\mathbf{f}$ does not depend explicitly on t .

§1.4.2 Flows

Another way to visualize the family of integral curves associated with a vector field is the following. Let $\mathcal{M}$ be a smooth manifold and $X \in \mathfrak{X}(\mathcal{M})$, and suppose that, for each point $p \in \mathcal{M}$, there is a unique integral curve $\theta^{(p)} : \mathbb{R} \rightarrow \mathcal{M}$ of X starting at p (i.e. with $\gamma(0) = p$) and defined for all $t \in \mathbb{R}$. For each $t \in \mathbb{R}$, define $\theta_t : \mathcal{M} \rightarrow \mathcal{M}$ which sends each $p \in \mathcal{M}$ to the point obtained by following for a time t the integral curve starting at p , i.e.:

$$\theta_t(p) = \theta^{(p)}(t)$$

Therefore, each map θ_t “slides” the manifold along the integral curves of X for a time t . By affine reparametrization invariance, for each $s \in \mathbb{R}$ the map $t \mapsto \theta^{(p)}(t + s)$ is still an integral curve of X starting at $q = \theta^{(p)}(s)$: by the assumed uniqueness of integral curves, then, $\theta^{(q)}(t) = \theta^{(p)}(s + t)$. Translating to the maps θ_t :

$$\theta_t \circ \theta_s = \theta_{t+s}$$

Together with $\theta_0 = 1_{\mathcal{M}}$, which holds by definition, this implies that the map $\theta : \mathbb{R} \times \mathcal{M} \rightarrow \mathcal{M}$ is an action of the additive group $\mathbb{R}_+$ on $\mathcal{M}$.

Definition 1.4.3 (Global flow)

Let $\mathcal{M}$ be a topological manifold. A **global flow** on $\mathcal{M}$ is a continuous left $\mathbb{R}_+$ -action on $\mathcal{M}$, i.e. a continuous map $\theta : \mathbb{R} \times \mathcal{M} \rightarrow \mathcal{M}$ such that, for any $s, t \in \mathbb{R}$ and $p \in \mathcal{M}$:

$$\theta(t, \theta(s, p)) = \theta(t + s, p) \quad \theta(0, p) = p \quad (1.4.7)$$

$\theta : \mathbb{R} \times \mathcal{M} \rightarrow \mathcal{M}$ is also called a *one-parameter group action*. Given a global flow θ on $\mathcal{M}$, two collections of maps are defined:

- for each $t \in \mathbb{R}$, the map $\theta_t : \mathcal{M} \rightarrow \mathcal{M}$ is defined as $\theta_t(p) \equiv \theta(t, p)$;
- for each $p \in \mathcal{M}$, the map $\theta^{(p)} : \mathbb{R} \rightarrow \mathcal{M}$ is defined as $\theta^{(p)}(t) = \theta(t, p)$.

As already noted, θ_t follows the group laws, while the image of $\theta^{(p)}$ is the orbit of p under the group action. Moreover, on a smooth manifold it is possible to associate a vector field X to the global flow as $p \mapsto X_p = \theta^{(p)'}(0) \in T_p\mathcal{M}$: this is called the **infinitesimal generator** of θ .

Proposition 1.4.3 (Smooth infinitesimal generator)

Let $\theta : \mathbb{R} \times \mathcal{M} \rightarrow \mathcal{M}$ be a smooth global flow on a smooth manifold $\mathcal{M}$. The infinitesimal generator X of θ is a smooth vector field on $\mathcal{M}$, and each curve $\theta^{(p)}$ is an integral curve of X .

Proof. To show that $X \in \mathfrak{X}(\mathcal{M})$, it suffices to show that $X(f) \in \mathcal{C}^\infty(U) \forall f \in \mathcal{C}^\infty(U)$, with $U \in \mathcal{T}(\mathcal{M})$. For any such f and any $p \in U$:

$$X(f)(p) = X_p(f) = \theta^{(p)'}(0)(f) = \left. \frac{d}{dt} \right|_{t=0} f(\theta^{(p)}(t)) = \left. \frac{\partial}{\partial t} \right|_{(0,p)} f(\theta(t, p))$$

Since $f(\theta(t, p))$ is a smooth function of (t, p) by composition, $X(f)(p)$ depends smoothly on p , hence $X \in \mathfrak{X}(\mathcal{M})$.

Next, show that $\theta^{(p)}$ is an integral curve of X for all $p \in \mathcal{M}$. This means that $\theta^{(p)'}(t) = X_{\theta^{(p)}(t)} \forall p \in \mathcal{M}, t \in \mathbb{R}$. Let $t_0 \in \mathbb{R}$ be arbitrary, and set $q = \theta^{(p)}(t_0) = \theta_{t_0}(p)$, so WTS

$\theta^{(p)'}(t_0) = X_q$. By group law, for all $t \in \mathbb{R}$:

$$\theta^{(q)}(t) = \theta_t(q) = \theta_t(\theta_{t_0}(p)) = \theta_{t+t_0}(p) = \theta^{(p)}(t + t_0)$$

therefore, for any smooth real-valued function f defined on a neighbourhood of q :

$$X_q(f) = \theta^{(q)'}(0)(f) = \left. \frac{d}{dt} \right|_{t=0} f(\theta^{(q)}(t)) = \left. \frac{d}{dt} \right|_{t=0} f(\theta^{(p)}(t + t_0)) = \theta^{(p)'}(t_0)(f)$$

which completes the proof. $\square$

While every smooth global flow gives rise to a smooth vector field whose integral curves are precisely the curves defined by the flow, the converse is not true: indeed, there are smooth vector fields whose integral curves are not defined for all $t \in \mathbb{R}$.

Definition 1.4.4 (Local flow)

Let $\mathcal{M}$ be a smooth manifold. A *flow domain* is $\mathfrak{D} \in \mathcal{I}(\mathbb{R} \times \mathcal{M})$ such that, for each $p \in \mathcal{M}$, the set $\mathfrak{D}^{(p)} \equiv \{t \in \mathbb{R} : (t, p) \in \mathfrak{D}\}$ is an open interval with $0 \in \mathfrak{D}^{(p)}$. A **local flow** on $\mathcal{M}$ is a continuous map $\theta : \mathfrak{D} \rightarrow \mathcal{M}$ such that, for all $p \in \mathcal{M}$:

- (1) $\theta(0, p) = p$;
- (2) $\theta(t, \theta(s, p)) = \theta(t + s, p) \quad \forall s \in \mathfrak{D}^{(p)}, t \in \mathfrak{D}^{(\theta(s, p))} : s + t \in \mathfrak{D}^{(p)}$.

Just as for a global flow, whenever $(t, p) \in \mathfrak{D}$, the maps $\theta_t(p) = \theta^{(p)}(t) = \theta(t, p)$ are defined. Moreover, for each $t \in \mathbb{R}$, the set $\mathcal{M}_t \equiv \{p \in \mathcal{M} : (t, p) \in \mathfrak{D}\}$ is defined, so that:

$$p \in \mathcal{M}_t \iff t \in \mathfrak{D}^{(p)} \iff (t, p) \in \mathfrak{D}$$

Given a smooth local flow θ , its infinitesimal generator X is defined by $X_p = \theta^{(p)'}(0)$. Identically to Prop. 1.4.3, if $\theta : \mathfrak{D} \rightarrow \mathcal{M}$ is a smooth local flow, then the infinitesimal generator X of θ is a smooth vector field, and each curve $\theta^{(p)}$ is an integral curve of X . From now on, local flows are simply called “flows”.

A **maximal integral curve** is an integral curve which cannot be extended to an integral curve on any larger open interval, while a **maximal flow** is a flow which admits no extension to a flow on a larger flow domain.

Theorem 1.4.2 (Fundamental theorem of flows)

Let $X \in \mathfrak{X}(\mathcal{M})$. There is a unique smooth maximal flow $\theta : \mathfrak{D} \rightarrow \mathcal{M}$, called the **flow of X** , whose infinitesimal generator is X and such that, for each $p \in \mathcal{M}$:

- (1) $\theta^{(p)} : \mathfrak{D}^{(p)} \rightarrow \mathcal{M}$ is the unique maximal integral curve of X starting at p ;
- (2) if $s \in \mathfrak{D}^{(p)}$, then $\mathfrak{D}^{(\theta(s, p))} = \mathfrak{D}^{(p)} - s \equiv \{t - s : t \in \mathfrak{D}^{(p)}\}$.

It can be shown that, given a smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ and two vector fields $X \in \mathfrak{X}(\mathcal{M}_1), Y \in \mathfrak{X}(\mathcal{M}_2)$, denoting the flow of X as θ and the flow of Y as η , if X and Y are f -related, then for each

$t \in \mathbb{R}$ the following diagram commutes:

$$\begin{array}{ccc}
 & \mathcal{M}_{1,t} & \xrightarrow{f} & \mathcal{M}_{2,t} \\
 f(\mathcal{M}_{1,t}) \subseteq \mathcal{M}_{2,t} & \theta_t \downarrow & & \downarrow \eta_t \\
 & \mathcal{M}_{1,-t} & \xrightarrow{f} & \mathcal{M}_{2,-t}
 \end{array} \tag{1.4.8}$$

If f is a diffeomorphism, then clearly the flow of f_*X has the domain $\mathcal{M}_{2,t} = f(\mathcal{M}_{1,t})$, for each $t \in \mathbb{R}$, and is given by:

$$\eta_t = f \circ \theta_t \circ f^{-1} \tag{1.4.9}$$

A smooth vector field whose flow is a global flow is said to be **complete**. Equivalently, the maximal integral curves of a complete vector field are defined for all $t \in \mathbb{R}$. While it is difficult, in general, to determine a priori whether a vector field is complete or no, it turns out that a sufficient condition can be imposed on its *support*, i.e. on the closure of the set $\{p \in \mathcal{M} : X_p \neq 0\}$.

Proposition 1.4.4 (Compactly support and completeness)

Every compactly-supported smooth vector field on a smooth manifold is complete.

Trivially, then, on a compact smooth manifold every smooth vector field is complete.

§1.4.3 Lie derivative

In Euclidean space $\mathbb{R}^n$, the directional derivative is trivially generalized to smooth vector fields. Indeed, given $X \in \mathfrak{X}(\mathbb{R}^n)$ and $\mathbf{v} \in T_p\mathbb{R}^n$, the directional derivative of X along $\mathbf{v}$ is:

$$D_{\mathbf{v}}X(p) = \left. \frac{d}{dt} \right|_{t=0} X_{p+t\mathbf{v}} = \lim_{t \rightarrow 0} \frac{X_{p+t\mathbf{v}} - X_p}{t} = D_{\mathbf{v}}X_p^i \frac{\partial}{\partial x^i} \Big|_p \tag{1.4.10}$$

This definition, however, relies on the fact that $T_{p+t\mathbf{v}}\mathbb{R}^n \cong \mathbb{R}^n \cong T_p\mathbb{R}^n$, being it a vector space, while for a general manifold $T_p\mathcal{M}$ and $T_q\mathcal{M}$ are entirely different vector spaces for $p \neq q$. The solution is to consider a vector field $V \in \mathfrak{X}(\mathcal{M})$, in place of $\mathbf{v} \in T_p\mathcal{M}$, so as to use the flow of V to push values of X to the same point.

Definition 1.4.5 (Lie derivative of vector field)

Let $\mathcal{M}$ be a smooth manifold and $X, V \in \mathfrak{X}(\mathcal{M})$. Denoting as θ the flow of V on $\mathcal{M}$, the **Lie derivative** of X with respect to V is the vector field (provided it exists):

$$(\mathcal{L}_V X)_p := \left. \frac{d}{dt} \right|_{t=0} d(\theta_{-t})_{\theta_t(p)}(X_{\theta_t(p)}) \tag{1.4.11}$$

Since it is defined from smooth maps, it can be shown that $\mathcal{L}_V X \in \mathfrak{X}(\mathcal{M})$. A visualization of the Lie derivative is pictured in Fig. 1.12.

Def. 1.4.5 is not particularly useful for computations, since it requires first to find the flow of $V \in \mathfrak{X}(\mathcal{M})$. Fortunately, a simple formula can be proved to hold for $X, Y \in \mathfrak{X}(\mathcal{M})$:

$$\mathcal{L}_V X = [V, X] \tag{1.4.12}$$

which relates directly the Lie derivative to the two vector fields considered through their Lie bracket. This formula allows for an interpretation of the Lie bracket: it is the directional derivative of the second vector field along the flow of the first one.

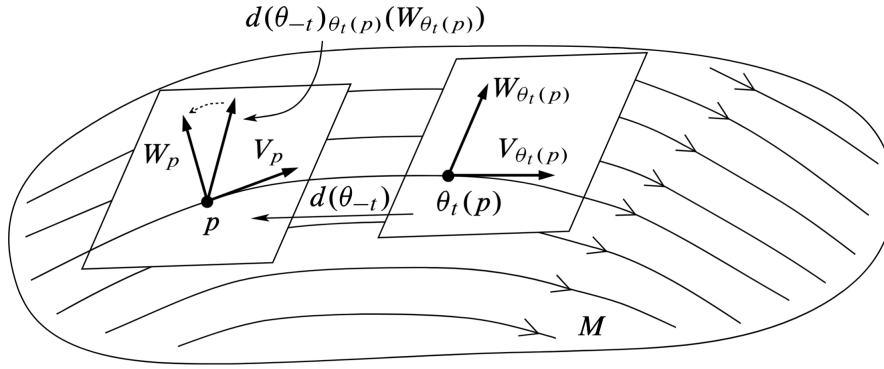


Figure 1.12: Action of the Lie derivative on a vector field.

Moreover, a number of non-obvious properties of Lie derivatives follow from the properties of Lie brackets.

Lemma 1.4.1 (Properties of Lie derivative)

Let $\mathcal{M}$ be a smooth manifold and $V, X, Y \in \mathfrak{X}(\mathcal{M})$. Then:

- (1) $\mathcal{L}_V X = -\mathcal{L}_X V$;
- (2) $\mathcal{L}_V [X, Y] = [\mathcal{L}_V X, Y] + [X, \mathcal{L}_V Y]$;
- (3) $\mathcal{L}_{[V, X]} Y = \mathcal{L}_V \mathcal{L}_X Y - \mathcal{L}_X \mathcal{L}_V Y$;
- (4) if $f \in C^\infty(\mathcal{M})$, then $\mathcal{L}_V (fX) = V(f)X + f\mathcal{L}_V X$;
- (5) if $f : \mathcal{M} \rightarrow \mathcal{M}'$ is a diffeomorphism, then $f_*(\mathcal{L}_V X) = \mathcal{L}_{f_*V} f_*X$.

If $V, X \in \mathfrak{X}(\mathcal{M})$ and θ is the flow of V , the Lie derivative $(\mathcal{L}_V X)_p$ expressed, by definition, the t -derivative of $d(\theta_{-t})_{\theta_t(p)}(X_{\theta_t(p)})$ at $t = 0$. To compute this t -derivative at later times, the following formula is found:

$$\left. \frac{d}{dt} \right|_{t=t_0} d(\theta_{-t})_{\theta_t(p)}(X_{\theta_t(p)}) = d(\theta_{-t})((\mathcal{L}_V X)_{\theta_{t_0}(p)}) \quad (1.4.13)$$

§1.4.3.1 Commuting vector fields

Let $\mathcal{M}$ be a smooth manifold. $X, Y \in \mathfrak{X}(\mathcal{M})$ are said to **commute** if $[X, Y] \equiv 0$. Given a smooth flow θ over $\mathcal{M}$, a vector field is **invariant under θ** if X is θ_t -related to itself for each t : this means that $X|_{\mathcal{M}_t}$ is θ_t -related to $X|_{\mathcal{M}_{-t}}$ for each t , or equivalently $d(\theta_t)_p(X_p) = X_{\theta_t(p)}$ for all (t, p) in the domain of θ . These two concepts are closely related by the Lie derivative.

Proposition 1.4.5 (Commutativity and flow invariance)

Let $X, Y \in \mathfrak{X}(\mathcal{M})$. Then the following conditions are equivalent:

- (1) X and Y commute;
- (2) X is invariant under the flow of Y ;
- (3) Y is invariant under the flow of X .

Proof. Denote the flow of Y with θ . If (2) holds, then $X_{\theta_t(p)} = d(\theta_t)_p(X_p)$ for all (t, p) in the domain of θ . Applying $d(\theta_{-t})_{\theta_t(p)}$ to both sides, $X_p = d(\theta_{-t})_{\theta_t(p)}(X_{\theta_t(p)})$ which directly implies $[Y, X] = \mathcal{L}_Y X = 0$ from Def. 1.4.5. The same argument shows that (3) $\implies$ (1).

To prove that (1) $\implies$ (2), assume $[Y, X] = \mathcal{L}_Y X = 0$. Let $p \in \mathcal{M}$ be arbitrary, and let $W(t) = d(\theta_{-t})_{\theta_t(p)}(X_{\theta_t(p)})$ for $t \in \mathfrak{D}^{(p)}$. By Eq. 1.4.13 $W'(t) \equiv 0$, but since $W(0) = X_p$, this implies $W(t) = X_p \forall t \in \mathfrak{D}^{(p)}$. Applying $d(\theta_t)_p$ to both sides shows that X is invariant under θ . The same argument shows that (1) $\implies$ (3). $\square$

Trivially, then, every smooth vector field is invariant under its own flow, as $[X, X] \equiv 0$.

A deeper characterization of commuting vector fields is in terms of the relationship between their respective flows. Suppose $X, Y \in \mathfrak{X}(\mathcal{M})$, and let θ and ψ denote their respective flows. If X and Y are complete, these flows are said to *commute* if $\theta_t \circ \psi_s = \psi_s \circ \theta_t \forall s, t \in \mathbb{R}$. In general, however, the naive imposition of this relation on the intersection of the flow domains does not give the right result (i.e. the flows might not commute in this naive sense, even though the vector field do). A generalization of the commutation condition is possible.

Definition 1.4.6 (Commuting flows)

Let $\mathcal{M}$ be a smooth manifold and let θ, ψ be smooth flows on $\mathcal{M}$. Then, θ and ψ are said to **commute** if, for every $p \in \mathcal{M}$, whenever $J, K \ni 0$ are open intervals such that one of the expressions $\theta_t \circ \psi_s$ or $\psi_s \circ \theta_t$ is defined for all $(s, t) \in J \times K$, then both are defined and they are equal.

Theorem 1.4.3 (Commutativity)

Smooth vector fields commute if and only if their flows commute.

An important application of commuting vector fields is in the study of frames on smooth manifolds. Given a smooth n -manifold $\mathcal{M}$, a smooth local frame $(E_1, \dots, E_n)$ is a **commuting frame** (or *holonomic frame*) if $[E_i, E_j] = 0 \forall i, j = 1, \dots, n$.

Since every coordinate frame is a commuting frame, given that Lie brackets are defined independently of local charts, it follows that a necessary condition for a smooth frame to be expressible as a coordinate frame in some smooth chart is that it must be a commuting frame. It turns out that this is, in fact, a sufficient condition too.

Proposition 1.4.6 (Commuting and coordinate frames)

Let $\mathcal{M}$ be a smooth n -manifold, and let $(X_1, \dots, X_k)$ be a linearly independent k -tuple of smooth commuting vector fields on $V \in \mathcal{T}(\mathcal{M})$. Then, for each $p \in V$, there exists a smooth coordinate chart $(U, (x^i))$ centered at p such that $V_i = \partial_i$ for $i = 1, \dots, k$.

§1.4.3.2 Time-dependent vector fields

The analysis of integral curves has been so far limited to time-independent vector fields, for which the system of autonomous ODEs Eq. 1.4.5 is granted to have a solution by Th. 1.4.1. Now, the case of non-autonomous ODEs is explored.

Definition 1.4.7 (Time-dependent vector field)

Let $\mathcal{M}$ be a smooth manifold. A **time-dependent vector field** is a continuous map $X : J \times \mathcal{M} \rightarrow$

$T\mathcal{M}$, where $J \subseteq \mathbb{R}$ is an interval, such that $X(t, p) \in T_p\mathcal{M}$ for each $(t, p) \in J \times \mathcal{M}$.

By this definition, the map $X_t : \mathcal{M} \rightarrow T\mathcal{M} : X_t(p) = X(t, p)$ is clearly a vector field on $\mathcal{M}$ for each $t \in J$. Then, given an interval $J_0 \subseteq J$, a curve $\gamma : J_0 \rightarrow \mathcal{M}$ is an integral curve of X if:

$$\gamma'(t) = X(t, \gamma(t)) \quad (1.4.14)$$

for all $t \in J_0$. This is a natural extension of Def. 1.4.2: indeed, note that any ordinary vector field X on $\mathcal{M}$ determines a time-dependent vector field on $\mathbb{R} \times \mathcal{M}$ just by setting $X(t, p) \equiv X_p$.

A time-dependent vector field might not generate a flow, since two integral curves starting at the same point but at different times might follow a different path, while all integral curves of a flow through a given point must have the same image. In this context, the fundamental theorem of flows Th. 1.4.2 is generalized as follows.

Theorem 1.4.4 (Fundamental theorem of time-dependent flows)

Let $\mathcal{M}$ be a smooth manifold, let $J \subseteq \mathbb{R}$ be an interval, and let $X : J \times \mathcal{M} \rightarrow T\mathcal{M}$ be a smooth time-dependent vector field on $\mathcal{M}$. There exist $\mathcal{E} \in \mathcal{T}(J \times J \times \mathcal{M})$ and a smooth map $\psi : \mathcal{E} \rightarrow \mathcal{M}$, called the **time-dependent flow of X** , such that:

- (1) for each $t_0 \in J$ and $p \in \mathcal{M}$, the set $\mathcal{E}^{(t_0, p)} = \{t \in J : (t, t_0, p) \in \mathcal{E}\} \ni t_0$ is an open interval, and the smooth curve $\psi^{(t_0, p)} : \mathcal{E}^{(t_0, p)} \rightarrow \mathcal{M}$ defined by $\psi^{(t_0, p)}(t) = \psi(t, t_0, p)$ is the unique maximal integral curve of X with initial condition $\psi^{(t_0, p)}(t_0) = p$;
- (2) if $t_1 \in \mathcal{E}^{(t_0, p)}$ and $q = \psi^{(t_0, p)}(t_1)$, then $\mathcal{E}^{(t_1, p)} = \mathcal{E}^{(t_0, p)}$ and $\psi^{(t_1, p)} = \psi^{(t_0, p)}$;
- (3) for each $(t_1, t_0) \in J \times J$, the set $\mathcal{M}_{t_1, t_0} = \{p \in \mathcal{M} : (t_1, t_0, p) \in \mathcal{E}\} \in \mathcal{T}(\mathcal{M})$, and the map $\psi_{t_1, t_0} : \mathcal{M}_{t_1, t_0} \rightarrow \mathcal{M}$ defined by $\psi_{t_1, t_0}(p) = \psi(t_1, t_0, p)$ is a diffeomorphism from $\mathcal{M}_{t_1, t_0}$ to $\mathcal{M}_{t_0, t_1}$ with inverse ψ_{t_0, t_1} ;
- (4) if $p \in \mathcal{M}_{t_1, t_0}$ and $\psi_{t_1, t_0}(p) \in \mathcal{M}_{t_2, t_1}$, then $p \in \mathcal{M}_{t_2, t_0}$ and $\psi_{t_2, t_1} \circ \psi_{t_1, t_0}(p) = \psi_{t_2, t_0}(p)$.

Example 1.4.1 (Euclidean example)

Define a time-dependent vector field X on $\mathbb{R}^n$ as:

$$X(t, x) = \frac{1}{t} x^i \frac{\partial}{\partial x^i} \Big|_x \quad (t, x) \in \mathbb{R}^+ \times \mathbb{R}^n$$

Consider arbitrary $(t_0, x_0) \in \mathbb{R}_+ \times \mathbb{R}^n$, and let $\gamma(t) = (x^1(t), \dots, x^n(t))$ denote the integral curve of X with initial condition $\gamma(t_0) = x_0$, described by the following system of non-autonomous ODEs: $\dot{x}^i(t) = x^i(t)/t$ and $x^i(t_0) = x_0^i$, for $i = 1, \dots, n$.

The maximal solution is easily found as $x^i(t) = tx_0^i/t_0$, for $t \in \mathbb{R}^+$, hence the time-dependent flow of X is given by $\psi(t, t_0, x) = tx/t_0$, for $(t, t_0, x) \in \mathbb{R}^+ \times \mathbb{R}^+ \times \mathbb{R}^n$.

§1.5 Vector bundles

It is possible to define additional structure on a topological manifold, in addition to a smooth structure.

Definition 1.5.1 ($\mathbb{R}$ -vector bundle)

Given a topological space M , an $\mathbb{R}$ -**vector bundle of rank k** over M is a topological space E together with a surjective continuous map $\pi : E \rightarrow M$ such that:

- (1) $\forall p \in M$ the *fiber* $E_p = \pi^{-1}(p)$ over p is endowed with the structure of a k -dimensional $\mathbb{R}$ -vector space;
- (2) $\forall p \in M \exists U \subseteq M : p \in U \wedge \chi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$ homeomorphism, called *local trivialization of E over U* , such that:
 - $\pi_1 \circ \chi = \pi$, with the projector $\pi_1 : U \times \mathbb{R}^k \rightarrow U$;
 - $\forall q \in U$ the restriction of χ to E_q is a vector-space isomorphism between E_q and $\{q\} \times \mathbb{R}^k \cong \mathbb{R}^k$.

This definition generalizes trivially to complex vector bundle substituting $\mathbb{R}$ with $\mathbb{C}$. In the following, only real vector bundles are considered.

If M and E are smooth manifolds, π is a smooth map and the local trivializations can be chosen to be diffeomorphisms, then E is called a **smooth vector bundle**. If there exists a local trivialization of E over the whole M , i.e. $\chi : E \rightarrow M \times \mathbb{R}^k$, then E is said to be a *trivial bundle*: clearly, any non-trivial bundle requires more than one trivialization.

Lemma 1.5.1

Let $\pi : E \rightarrow \mathcal{M}$ be a smooth vector bundle of rank k over $\mathcal{M}$, and suppose $\chi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$ and $\psi : \pi^{-1}(V) \rightarrow V \times \mathbb{R}^k$ are smooth local trivializations of E such that $U \cap V \neq \emptyset$. Then, there exists a smooth map $\tau : U \cap V \rightarrow \text{GL}(k, \mathbb{R})$ such that $\chi \circ \psi^{-1} : (U \cap V) \times \mathbb{R}^k \rightarrow (U \cap V) \times \mathbb{R}^k$ has the form:

$$\chi \circ \psi^{-1}(p, v) = (p, \tau(p)v) \quad (1.5.1)$$

Proof. By hypothesis, the following diagram commutes:

$$\begin{array}{ccccc} (U \cap V) \times \mathbb{R}^k & \xleftarrow{\psi} & \pi^{-1}(U \cap V) & \xrightarrow{\chi} & (U \cap V) \times \mathbb{R}^k \\ & \searrow \pi_1 & \downarrow \pi & \swarrow \pi_1 & \\ & & U \cap V & & \end{array}$$

where the local trivialization are interpreted as the respective restrictions on $\pi^{-1}(U \cap V)$. It follows that $\pi_1 \circ (\chi \circ \psi^{-1}) = \pi_1$, hence:

$$\chi \circ \psi^{-1}(p, v) = (p, \sigma(p, v))$$

for some smooth map $\sigma : (U \cap V) \times \mathbb{R}^k \rightarrow \mathbb{R}^k$. Moreover, for each $p \in U \cap V$, the map $\mathbb{R}^k \rightarrow \mathbb{R}^k$ defined as $v \mapsto \sigma(p, v)$ is an invertible linear map, hence the thesis. $\square$

The smooth map $\tau : U \cap V \rightarrow \text{GL}(n, \mathbb{R})$ is called the **transition function** between χ and ψ .

A natural construction for a smooth vector bundle on a smooth manifold is to define a collection of vector spaces, one for each point of the manifold: this is the same idea as the tangent bundle, which is in fact a vector bundle (see §1.5.1). In order to make such a collection a smooth vector bundle, one first needs to define a manifold topology and a smooth structure on the total space of the bundle, i.e. on the disjoint union of said collection, and then construct the smooth local trivializations. However,

these steps can be condensated, provided that local trivializations overlap with smooth transition functions.

Theorem 1.5.1 (Vector bundle construction theorem)

Let $\mathcal{M}$ be a smooth manifold, and let $\{U_\alpha\}_{\alpha \in \mathcal{I}}$ be an open cover of $\mathcal{M}$. Suppose that, for each $\alpha, \beta, \gamma \in \mathcal{I}$, there exists a smooth map $\tau_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow \text{GL}(k, \mathbb{R})$ such that:

$$\tau_{\alpha\beta}(p)\tau_{\beta\gamma}(p) = \tau_{\alpha\gamma}(p) \quad \forall p \in U_\alpha \cap U_\beta \cap U_\gamma \quad (1.5.2)$$

Then, there exists a smooth rank- k vector bundle $\pi : E \rightarrow \mathcal{M}$ with smooth local trivializations $\chi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^k$, whose transition functions are the given maps $\tau_{\alpha\beta}$.

§1.5.1 Tangent bundle

Proposition 1.5.1 (Tangent bundle as vector bundle)

Given a smooth n -manifold $\mathcal{M}$, then its tangent bundle $T\mathcal{M}$ has the structure of smooth vector bundle of rank n over $\mathcal{M}$.

Proof. Since the fibers $\pi^{-1}(p) = T_p\mathcal{M}$ are n -dimensional $\mathbb{R}$ -vector spaces by Th. 1.2.1, only the smooth local trivializations need to be constructed.

Given any smooth chart (U, φ) of $\mathcal{M}$ with coordinates $(x^1, \dots, x^n)$, define the map $\chi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^n$ as:

$$\chi : X^i \frac{\partial}{\partial x^i} \Big|_p \mapsto (p, X^1, \dots, X^n)$$

This is linear on fibers and satisfies $\pi_1 \circ \chi = \pi$, i.e. the following diagram is commutative:

$$\begin{array}{ccc} \pi^{-1}(U) & \xrightarrow{\chi} & U \times \mathbb{R}^n \\ & \searrow \pi & \downarrow \pi_1 \\ & & U \end{array}$$

Now, note that the composite map $(\varphi \times \mathbb{1}_{\mathbb{R}^n}) \circ \chi : \pi^{-1}(U) \rightarrow \varphi(U) \times \mathbb{R}^n$, i.e.:

$$\pi^{-1}(U) \xrightarrow{\chi} U \times \mathbb{R}^n \xrightarrow{\varphi \times \mathbb{1}_{\mathbb{R}^n}} \varphi(U) \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$$

is equal to the coordinate map $\tilde{\varphi} : \pi^{-1}(U) \rightarrow \mathbb{R}^{2n}$ defined above. Both $\varphi \times \mathbb{1}_{\mathbb{R}^n}$ and $\tilde{\varphi}$ are diffeomorphisms, hence χ is a smooth local trivialization of $T\mathcal{M}$ over $U \subseteq \mathcal{M}$. $\square$

It can be shown that the smooth structure defined in Prop. 1.2.3 is the only smooth structure on $T\mathcal{M}$ such that $\pi : T\mathcal{M} \rightarrow \mathcal{M}$ is smooth vector bundle. The coordinates (x^i, X^i) are called *natural coordinates* on $T\mathcal{M}$.

§1.5.1.1 Sections and frames

The natural structure of $T\mathcal{M}$ as a vector bundle over $\mathcal{M}$ allows to import the notions of vector bundles on the tangent bundle.

Definition 1.5.2 (Section of vector bundle)

Let $\pi : E \rightarrow M$ be a vector bundle. A **section** of E is a continuous map $\sigma : M \rightarrow E$ such that:

$$\pi \circ \sigma = \mathbb{1}_M \quad (1.5.3)$$

This means that σ associate to each point $p \in M$ an element of its fiber, i.e. $\sigma : p \mapsto \sigma(p) \in E_p$. In general, it is possible to define *local* sections of E as continuous maps $\sigma : U \rightarrow E$ defined on $U \in \mathcal{T}(M)$ and satisfying $\pi \circ \sigma = \mathbb{1}_U$: sections of E defined over all M are then called *global* sections (clearly a local section over U is a global section of the restricted bundle $E|_U$). If the vector bundle considered is a smooth vector bundle and σ is a smooth map from its domain to E , then it is a *smooth* (local or global) section of E .

The set of all smooth global sections of a smooth vector bundle $\pi : E \rightarrow M$, denoted by $\Gamma(E)$, has the natural structure of $\mathbb{R}$ -vector space under pointwise addition and scalar multiplication:

$$(c_1\sigma_1 + c_2\sigma_2)(p) = c_1\sigma_1(p) + c_2\sigma_2(p) \quad \forall c_1, c_2 \in \mathbb{R}, \sigma_1, \sigma_2 \in \Gamma(E), p \in M$$

The multiplication of $\sigma \in \Gamma(E)$ by $f \in \mathcal{C}^\infty(M)$ yields a new section $(f\sigma)(p) = f(p)\sigma(p)$.

The identification of sections of $T\mathcal{M}$ with vector fields over $\mathcal{M}$ is then clear, and in particular:

$$\Gamma(T\mathcal{M}) = \mathfrak{X}(\mathcal{M}) \quad (1.5.4)$$

To introduce frames on vector bundles, it is first necessary to import the notion of linear independence on sections. Given $U \in \mathcal{T}(M)$, an m -tuple of local sections $(\sigma_1, \dots, \sigma_m)$ of E over U is linearly independent if $(\sigma_1(p), \dots, \sigma_m(p))$ is a linearly independent m -tuple in E_p for all $p \in U$.

Definition 1.5.3 (Frame for vector bundle)

Let $\pi : E \rightarrow M$ be a vector bundle of rank k . A **local frame** for E over $U \in \mathcal{T}(M)$ is an ordered k -tuple $(\sigma_1, \dots, \sigma_k)$ of linearly independent local sections over U such that:

$$E_p = \text{span}\{\sigma_1(p), \dots, \sigma_k(p)\} \quad \forall p \in U$$

If $U = M$, then this is a **global frame**.

If $E \rightarrow M$ is a smooth vector bundle and each σ_i is a smooth (local or global) section, then $(\sigma_1, \dots, \sigma_k)$ is a *smooth* (local or global) frame.

Local frames are linked to local trivializations. To see this, consider a smooth vector bundle $\pi : E \rightarrow M$ of rank k and a smooth local trivialization $\chi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$, with $U \in \mathcal{T}(M)$. Take the standard basis $\{e_1, \dots, e_k\}$ for $\mathbb{R}^k$ and define $\tilde{e}_i : U \rightarrow U \times \mathbb{R}^k$ as $\tilde{e}_i(p) = (p, e_i)$. Moreover, define the maps $\sigma_1, \dots, \sigma_k : U \rightarrow E$ by:

$$\sigma_i(p) = \chi^{-1}(p, e_i) = \chi^{-1} \circ \tilde{e}_i(p)$$

Then, σ_i is smooth since χ is a diffeomorphism, and it is a section of E :

$$\pi_1 \circ \chi = \pi \quad \implies \quad \pi \circ \sigma_i(p) = \pi \circ \chi^{-1}(p, e_i) = \pi_1(p, e_i) = p$$

To see that $(\sigma_1(p), \dots, \sigma_k(p))$ is a basis of E_p , note that χ is an isomorphism $E_p \cong \{p\} \times \mathbb{R}^k$ such that $\chi(\sigma_i(p)) = (p, e_i)$, hence χ maps $\{\sigma_1(p), \dots, \sigma_k(p)\}$ to the standard basis of $\{p\} \times \mathbb{R}^k \cong \mathbb{R}^k$.

Therefore, $(\sigma_1, \dots, \sigma_k)$ is a *local frame associated to* χ . It can be shown that every smooth local frame for a smooth vector bundle is associated to a smooth local trivialization in this way.

On $T\mathcal{M}$, local frames determine local bases for tangent spaces. An important case is that of *coordinate frames*: given a smooth chart (U, φ) with coordinate functions $(x^1, \dots, x^n)$, then the coordinate vector fields $(\partial_1, \dots, \partial_n)$ form a smooth local frame on U .

§1.5.1.2 Bundle homomorphisms

Definition 1.5.4 (Bundle homomorphism)

Let $\pi : E \rightarrow \mathcal{M}$ and $\pi' : E' \rightarrow \mathcal{M}'$ be vector bundles. A continuous map $f : E \rightarrow E'$ is a **bundle homomorphism** if there exists a map $F : \mathcal{M} \rightarrow \mathcal{M}'$ satisfying:

$$\pi' \circ f = F \circ \pi \quad \begin{array}{ccc} E & \xrightarrow{f} & E' \\ \pi \downarrow & & \downarrow \pi' \\ \mathcal{M} & \xrightarrow{F} & \mathcal{M}' \end{array} \quad (1.5.5)$$

and such that, for each $p \in \mathcal{M}$, the restriction $f|_{E_p} : E_p \rightarrow E'_{F(p)}$ is linear.

The relation between f and F is expressed by saying that f *covers* F . The covered map F is continuous and uniquely by f as:

$$F = \pi' \circ f \circ \zeta \quad (1.5.6)$$

where $\zeta : \mathcal{M} \rightarrow E$ is the **zero section**, defined as $\zeta(p) = 0 \in E_p \forall p \in \mathcal{M}$. Clearly, if f is smooth between smooth bundles, then F is smooth too. A bijective bundle homomorphism whose inverse is also a bundle homomorphism is called a *bundle isomorphism*, and the two bundles are said to be isomorphic.

Example 1.5.1 (Tangent bundle homomorphism)

Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be smooth manifolds, and let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map. Then, the global differential $df : T\mathcal{M}_1 \rightarrow T\mathcal{M}_2$ is a smooth bundle homomorphism covering f .

A case of particular interest is that of vector bundles with the same basis. In this case, if $\pi : E \rightarrow \mathcal{M}$ and $\pi' : E' \rightarrow \mathcal{M}$ are homomorphic, the bundle homomorphism $f : E \rightarrow E'$ is said to be *over* $\mathcal{M}$ if it covers $\mathbb{1}_{\mathcal{M}}$, i.e. if the following diagram commutes:

$$\begin{array}{ccc} E & \xrightarrow{f} & E' \\ \pi \searrow & & \swarrow \pi' \\ & \mathcal{M} & \end{array}$$

and if the restriction of f to each fiber is linear. Given smooth bundles $\pi : E \rightarrow \mathcal{M}$ and $\pi' : E' \rightarrow \mathcal{M}$, if $f : E \rightarrow E'$ is a smooth bundle homomorphism over $\mathcal{M}$, then composition with f induces a map $\tilde{f} : \Gamma(E) \rightarrow \Gamma(E')$ defined as $\tilde{f}(\sigma) = f \circ \sigma$. Since a bundle homomorphism is linear on fibers, the resulting map $\tilde{f}$ is linear over $\mathbb{R}$. In fact, it is linear over $\mathcal{C}^\infty(\mathcal{M})$:

$$\tilde{f}(g_1\sigma_1 + g_2\sigma_2) = g_1\tilde{f}(\sigma_1) + g_2\tilde{f}(\sigma_2)$$

for any $g_1, g_2 \in \mathcal{C}^\infty(\mathcal{M})$ and $\sigma_1, \sigma_2 \in \Gamma(E)$. The converse is true as well.

Lemma 1.5.2 (Characterization of bundle homomorphisms)

Let $\pi : E \rightarrow \mathcal{M}$ and $\pi' : E' \rightarrow \mathcal{M}$ be smooth vector bundles. A map $\tilde{f} : \Gamma(E) \rightarrow \Gamma(E')$ is linear over $\mathcal{C}^\infty(\mathcal{M})$ if and only if there is a smooth bundle homomorphism $f : E \rightarrow E'$ over $\mathcal{M}$ such that $\tilde{f} = f \circ \sigma \forall \sigma \in \Gamma(E)$.

Given this lemma, it is customary to identify $\tilde{f}$ with f , hence writing directly $f : \Gamma(E) \rightarrow \Gamma(E')$, since the action on sections is obtained simply applying the bundle homomorphism pointwise: for example, if $a \in \mathbb{R}$, then the same notation $X \mapsto aX$ denotes both the operation of multiplying vectors in each tangent space $T_p\mathcal{M}$ by a , and the operation of multiplying vector fields by a (note that multiplying by a is indeed a bundle homomorphism $T\mathcal{M} \rightarrow T\mathcal{M}$).

§1.5.2 Cotangent bundle

While tangent vectors naturally give a coordinate-free interpretation of derivatives of curves, the same can be achieved for real-valued functions on manifolds introducing functionals on the tangent space at each point.

§1.5.2.1 Covectors**Definition 1.5.5** (Covector)

Let V be a finite-dimensional $\mathbb{R}$ -vector space. A **covector** on V is a linear map $\omega : V \rightarrow \mathbb{R}$.

The space of all covectors is itself a real vector space under pointwise addition and scalar multiplication, and it is usually denoted by $V^* := \text{Hom}_{\mathbb{R}}(V, \mathbb{R})$.

Proposition 1.5.2 (Dual basis)

Let V be a finite-dimensional $\mathbb{R}$ -vector space. Given any basis $(E_1, \dots, E_n)$ of V , its **dual basis** $(\varepsilon^1, \dots, \varepsilon^n) \subseteq V^*$ is defined by:

$$\varepsilon^i(E_j) = \delta_j^i \quad (1.5.7)$$

and it is a basis of V^* . Therefore, $\dim_{\mathbb{R}} V^* = \dim_{\mathbb{R}} V$.

If $(e_1, \dots, e_n)$ is the standard basis of $\mathbb{R}^n$, the dual basis $(e^1, \dots, e^n)$ is called the *standard dual basis* of V^* . In general, if (E_i) is a basis of V and (ε^i) is its dual basis, then given $v = v^j E_j \in V$:

$$\varepsilon^i(v) = v^j \varepsilon^i(E_j) = v^j \delta_j^i = v^i$$

By Prop. 1.5.2, any covector can be expressed as $\omega = \omega_i \varepsilon^i \in V^*$, hence its action on $v = v^j E_j \in V$ is clearly determined:

$$\omega(v) = \omega_i v^i \quad (1.5.8)$$

Given two finite-dimensional $\mathbb{R}$ -vector spaces V, W and $A \in \text{Hom}_{\mathbb{R}}(V, W)$, it is possible to define a **dual map** (or transposed map) $A^* \in \text{Hom}_{\mathbb{R}}(W^*, V^*)$ as:

$$A^* \circ \omega = \omega \circ A \quad (1.5.9)$$

The dual map clearly satisfies $(A \circ B)^* = B^* \circ A^*$ and $\mathbb{1}_V^* = \mathbb{1}_{V^*}$.

Proposition 1.5.3 (Second dual space)

For any finite-dimensional $\mathbb{R}$ -vector space V , there exists a canonical isomorphism $\xi : V \rightarrow V^{**} \equiv (V^*)^*$ such that, for each $v \in V$, $\xi(v) : V^* \rightarrow \mathbb{R}$ acts on $\omega \in V^*$ as:

$$\xi(v)(\omega) = \omega(v) \quad (1.5.10)$$

Proof. By Prop. 1.5.2, $\dim_{\mathbb{R}} V^{**} = \dim_{\mathbb{R}} V$, hence it suffices to verify that ξ is injective. Suppose $v \in V - \{0\}$, extend it to a basis $(v \equiv E_1, E_2, \dots, E_n)$ of V and let $(\varepsilon^1, \dots, \varepsilon^n)$ be its dual basis. Then, $\xi(v) \neq 0$ since:

$$\xi(v)(\varepsilon^1) = \varepsilon^1(v) = \varepsilon^1(E_1) = 1$$

Therefore, $\ker \xi = \{0\}$, i.e. it is injective. $\square$

The existence of this canonical isomorphism requires $\dim_{\mathbb{R}} V$ to be finite: in the infinite-dimensional case, Prop. 1.5.2 does not hold in the first place, i.e. there exists in general no isomorphism $V \cong V^*$, hence no canonical isomorphism $V \cong V^{**}$ either. Moreover, note that the isomorphism $V \cong V^*$ can never be canonical, since it depends explicitly on the choice of basis.

Given Prop. 1.5.3, $\omega(v) \in \mathbb{R}$ is usually denoted with a more symmetric notation $\langle \omega, v \rangle = \langle v, \omega \rangle$, which can be thought of either as the action of $\omega \in V^*$ on $v \in V$, or as the action of $\xi(v) \in V^{**}$ on $\omega \in V^*$. In the following, the map ξ is usually omitted, and $v \in V$ is thought of either as a vector or as a linear functional on V^* , depending on the context.

§1.5.2.2 Tangent covectors and covector fields**Definition 1.5.6** (Cotangent space)

Let $\mathcal{M}$ be a smooth manifold. For each $p \in \mathcal{M}$, the **cotangent space** at p is $T_p^* \mathcal{M} := (T_p \mathcal{M})^*$.

The elements of $T_p^* \mathcal{M}$ are called the *tangent covectors* at p . Given a smooth chart $(U, (x^i))$ of $\mathcal{M}$, for each $p \in U$ the coordinate basis $(\partial_i|_p)$ of $T_p \mathcal{M}$ determines a *coordinate dual basis* $(\lambda^i|_p)$ of $T_p^* \mathcal{M}$. Hence, any covector $\omega \in T_p^* \mathcal{M}$ can be written as:

$$\omega = \omega_i \lambda^i|_p \quad \omega_i = \omega \left(\frac{\partial}{\partial x^i} \Big|_p \right)$$

Now, let $(V, (\tilde{x}^i))$ be another smooth chart that contains p , and let $(\tilde{\lambda}^i|_p)$ be the dual basis of $(\partial_i|_p)$. Applying Eq. 1.2.11, by the linearity of $\omega \in V^*$, the transformation rule for covectors is found:

$$\omega_i = \frac{\partial \tilde{x}^j}{\partial x^i}(\hat{p}) \tilde{\omega}_j \quad (1.5.11)$$

Since they transform as basis vectors, i.e. according to the Jacobian of the coordinate transformation, covectors are also called *covariant vectors*.

As for tangent vectors, it is possible to define the **cotangent bundle** of a smooth manifold $\mathcal{M}$ as the disjoint union:

$$T^* \mathcal{M} := \bigsqcup_{p \in \mathcal{M}} T_p^* \mathcal{M} \quad (1.5.12)$$

which comes with a natural projection $\pi : T^* \mathcal{M} \rightarrow \mathcal{M}$ which sends $(p, \omega) \mapsto p$.

Proposition 1.5.4 (Cotangent bundle as vector bundle)

Given a smooth n -manifold $\mathcal{M}$, then its cotangent bundle $T^*\mathcal{M}$ has the structure of smooth vector bundle of rank n over $\mathcal{M}$.

Proof. The proof is analogous to that of Prop. 1.5.1. □

As in the case of the tangent bundle, the smooth structure on $T^*\mathcal{M}$ determined by that of $T\mathcal{M}$ is the unique one which gives to $\pi : T^*\mathcal{M} \rightarrow \mathcal{M}$ the structure of smooth rank- n vector bundle on $\mathcal{M}$. Moreover, smooth local coordinates on $\mathcal{M}$ yield smooth local coordinates on $T^*\mathcal{M}$: indeed, if (x^i) are smooth coordinates on $U \in \mathcal{T}(\mathcal{M})$, then the map $\pi^{-1}(U) \rightarrow \mathbb{R}^{2n}$ defined as:

$$\xi_i \lambda^i|_p \mapsto (x^1(p), \dots, x^n(p), \xi_1, \dots, \xi_n) \quad (1.5.13)$$

is a smooth local coordinate map on $\pi^{-1}(U) \in \mathcal{T}(T^*\mathcal{M})$. These are called the *natural coordinates* on $T^*\mathcal{M}$ associated to (x^i) .

As for vector fields, a (local or global) section of $T^*\mathcal{M}$ is called a **covector field** (or *differential 1-form*). In any smooth local coordinates on $U \in \mathcal{T}(\mathcal{M})$, a covector field ω can be written in terms of coordinate covector fields λ^i as $\omega = \omega_i \lambda^i$, with the n functions $\omega_i : U \rightarrow \mathbb{R}$ being the *component functions* of ω :

$$\omega_i(p) = \omega_p \left(\frac{\partial}{\partial x^i} \Big|_p \right)$$

Given a vector field X and a covector field ω over $\mathcal{M}$, it is possible to form a function $\omega(X) : \mathcal{M} \rightarrow \mathbb{R}$ such that, for any $p \in \mathcal{M}$:

$$\omega(X)(p) = \omega_p(X_p) \quad (1.5.14)$$

In terms of the respective component functions, this function has the coordinate representation $\omega(X) = \omega_i X^i$.

Lemma 1.5.3 (Smoothness criteria for covector fields)

Let $\mathcal{M}$ be a smooth manifold, and let $\omega : \mathcal{M} \rightarrow T^*\mathcal{M}$ be a covector field. Then the following conditions are equivalent:

- (1) ω is smooth;
- (2) in every coordinate chart, the component functions of ω are smooth;
- (3) each point of $\mathcal{M}$ is contained in some coordinate chart in which ω has smooth component functions;
- (4) for every $X \in \mathfrak{X}(\mathcal{M})$, the function $\omega(X)$ is smooth on $\mathcal{M}$;
- (5) for every $U \in \mathcal{T}(\mathcal{M})$ and $X \in \mathfrak{X}(U)$, the function $\omega(X) : U \rightarrow \mathbb{R}$ is smooth on U .

Since any open subset of a smooth manifold is again a smooth manifold, this lemma applies equally well on covector fields defined only on some open subset of $\mathcal{M}$.

Definition 1.5.7 (Coframe)

Let $\mathcal{M}$ be a smooth manifold, and let $U \in \mathcal{T}(\mathcal{M})$. A **local coframe** over U is an ordered n -tuple of covector fields $(\varepsilon^1, \dots, \varepsilon^n)$ defined on U such that $(\varepsilon^i|_p)$ forms a basis of $T_p^*\mathcal{M}$ at each $p \in U$. If $U = \mathcal{M}$, this is a **global coframe**.

Clearly, a local coframe of $\mathcal{M}$ is just a local frame of the vector bundle $T^*\mathcal{M}$, as per Def. 1.5.3. Given a local frame $(E_1, \dots, E_n)$ for $T\mathcal{M}$ over $U \in \mathcal{T}(\mathcal{M})$, there is a uniquely determined local coframe $(\varepsilon^1, \dots, \varepsilon^n)$ for $T^*\mathcal{M}$ over U defined by $\varepsilon^i(E_j) = \delta_j^i$: this coframe is the *dual coframe*.

Lemma 1.5.4

Let $\mathcal{M}$ be a smooth manifold. If (E_i) is a local frame over $U \in \mathcal{T}(\mathcal{M})$ and (ε^i) is its dual coframe, then (E^i) is smooth if and only if (ε^i) is smooth.

Given a local coframe (ε^i) over $U \in \mathcal{T}(\mathcal{M})$, every covector field ω over U can be expressed as $\omega = \omega_i \varepsilon^i$, where the n -functions $\omega_i : U \rightarrow \mathbb{R}$ are the component functions of ω with respect to the given coframe and are determined by $\omega_i = \omega(E_i)$, where (E_i) is the dual frame of (ε^i) .

The $\mathbb{R}$ -vector space of all smooth covector fields on $\mathcal{M}$ is denoted by $\mathfrak{X}^*(\mathcal{M}) \equiv \Gamma(T^*\mathcal{M})$. As smooth sections of a vector bundle, elements of $\mathfrak{X}^*(\mathcal{M})$ can be multiplied by $\mathbb{R}$ -valued functions, yielding new elements⁷ of $\mathfrak{X}^*(\mathcal{M})$: given $f \in \mathcal{C}^\infty(\mathcal{M})$ and $\omega \in \mathfrak{X}^*(\mathcal{M})$, then $f\omega \in \mathfrak{X}^*(\mathcal{M})$ is defined as $(f\omega)_p = f(p)\omega_p$.

§1.5.2.3 Differential of functions

The partial derivatives of a smooth function cannot be interpreted in a coordinate-free way as the components of a vector fields, but they can be as the components of a covector field.

Definition 1.5.8 (Differential)

Let $f \in \mathcal{C}^\infty(\mathcal{M})$. The **differential of f** is the covector field $df : \mathcal{M} \rightarrow T^*\mathcal{M}$ defined by:

$$df_p(X_p) = X_p(f) \quad \forall X_p \in T_p\mathcal{M} \quad (1.5.15)$$

By (4) of Lemma 1.5.3, the differential of a smooth function is a smooth covector field. To compute the coordinate expression of df , consider a smooth map $(U, (x^i))$ on $\mathcal{M}$: on the coordinate basis, the components of df_p are $df_p(\partial_i|_p) = \partial_i|_p f$, hence:

$$df_p = \frac{\partial f}{\partial x^i}(p) \lambda^i|_p \quad (1.5.16)$$

The differential can then be thought of as the analogue of the gradient, reinterpreted in a coordinate-free way. In the special case $f = x^j : U \rightarrow \mathbb{R}$, the differential is:

$$dx^j|_p = \frac{\partial x^j}{\partial x^i}(p) \lambda^i|_p = \delta_i^j \lambda^i|_p = \lambda^j|_p$$

This shows that the coordinate dual basis $\lambda^i|_p$ is composed just by the differentials of the coordinate functions. Hence, in terms of covector fields:

$$df = \frac{\partial f}{\partial x^i} dx^i \quad (1.5.17)$$

which is the usual expression, reinterpreted in terms of covector fields. The so-defined differential has the usual properties expected from Analysis.

⁷This means that $\mathfrak{X}^*(\mathcal{M})$ is a module over the ring $\mathcal{C}^\infty(\mathcal{M})$.

Lemma 1.5.5 (Properties of the differential)

Let $\mathcal{M}$ be a smooth manifold, and let $f, g \in \mathcal{C}^\infty(\mathcal{M})$. Then:

- (1) $d(af + bg) = adf + bdf \forall a, b \in \mathbb{R}$;
- (2) $d(fg) = fdf + gdf$;
- (3) if $h : J \rightarrow \mathbb{R}$ is smooth over the interval $J \subseteq \mathbb{R} : J \supseteq \text{Im } f$, then $d(h \circ f) = (h' \circ f)df$.

The differential can also be used to express derivatives along curves. Consider a smooth manifold $\mathcal{M}$, a curve $\gamma : J \rightarrow \mathcal{M}$ and $f \in \mathcal{C}^\infty(\mathcal{M})$. Then, the derivative of the function $f \circ \gamma : J \rightarrow \mathbb{R}$ is (from definitions):

$$(f \circ \gamma)'(t) = df_{\gamma(t)}(\gamma'(t)) \quad (1.5.18)$$

Moreover, the notions of differential of a function $f \in \mathcal{C}^\infty(\mathcal{M})$ given by Def. 1.2.3 and by Def. 1.5.8 are truly the same object, once the canonical identification $\mathbb{R} \cong T_{f(p)}\mathbb{R}$ is taken into account, and they are both represented by a row vector whose components are the partial derivatives of f . Similarly, given a smooth curve $\gamma : J \rightarrow \mathcal{M}$, the expression $(f \circ \gamma)'(t)$ has two different meanings. On the one hand, $f \circ \gamma$ can be interpreted as a smooth curve over $\mathbb{R}$, and thus $(f \circ \gamma)'(t)$ is the velocity at the point $f \circ \gamma(t)$, which is an element of $T_{f \circ \gamma(t)}\mathbb{R}$ and equal to $df_{\gamma(t)}(\gamma'(t))$ as per Eq. 1.4.3. On the other hand, $f \circ \gamma$ can also be interpreted simply as a single-variable $\mathbb{R}$ -valued function, and $(f \circ \gamma)'(t)$ is thus its ordinary derivative and equal to $df_{\gamma(t)}(\gamma'(t))$ as per Eq. 1.5.18, now seen as an ordinary real number. The two interpretations are again equivalent as $\mathbb{R} \cong T_{f \circ \gamma(t)}\mathbb{R}$.

§1.5.2.4 Pull-back of covector fields

Since the differential is a linear map on tangent vectors, its dual is a linear map on covectors in the opposite direction.

Definition 1.5.9 (Pull-back)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be smooth map, and let $p \in \mathcal{M}_1$. The differential $df_p : T_p\mathcal{M}_1 \rightarrow T_{f(p)}\mathcal{M}_2$ yields a dual linear map $df_p^* : T_{f(p)}^*\mathcal{M}_2 \rightarrow T_p^*\mathcal{M}_1$, called the **(pointwise) pull-back** by f at p , defined as:

$$df_p^*(\omega_{f(p)})(v) = \omega_{f(p)}(df_p(v)) \quad \forall \omega_{f(p)} \in T_{f(p)}^*\mathcal{M}_2, v \in T_p\mathcal{M}_1 \quad (1.5.19)$$

Contrary to vector fields, whose push-forwards are only defined in presence of diffeomorphisms (Th. 1.2.2), covector fields are always pulled-back to covector fields. Given a smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ and a covector field ω on $\mathcal{M}_2$, the pull-back of ω by f is defined as:

$$(f^*\omega)_p := df_p^*(\omega_{f(p)}) \quad (1.5.20)$$

which is a covector in $T_p^*\mathcal{M}_1$ and, as per Def. 1.5.9, acts on $v \in T_p\mathcal{M}_1$ as:

$$(f^*\omega)_p(v) = \omega_{f(p)}(df_p(v))$$

Proposition 1.5.5 (Smoothness of pull-back)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map, and let ω be a covector field on $\mathcal{M}_2$. Then, $f^*\omega$ is a (continuous) covector field on $\mathcal{M}_1$. Moreover, if ω is smooth, then $f^*\omega$ is smooth.

Proof. Let $p \in \mathcal{M}_1$ arbitrary, and choose a smooth chart $(V, (y^i))$ on $\mathcal{M}_2$ which contains $f(p)$, so that $U = f^{-1}(V)$ is a neighbourhood of p .

Lemma 1.5.6

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map, and suppose that $u : \mathcal{M}_2 \rightarrow \mathbb{R}$ is a continuous function and that ω is a covector field on $\mathcal{M}_2$. Then:

$$f^*(u\omega) = (u \circ f)f^*\omega \quad (1.5.21)$$

In addition, if u is smooth:

$$f^*du = d(u \circ f) \quad (1.5.22)$$

Proof. By direct computation, using the definitions:

$$\begin{aligned} (f^*(u\omega))_p &= df_p^*((u\omega)_{f(p)}) = df_p^*(u(f(p))\omega_{f(p)}) \\ &= u(f(p))df_p^*(\omega_{f(p)}) = u(f(p))(f^*\omega)_p = ((u \circ f)f^*\omega)_p \end{aligned}$$

Letting $v \in T_p\mathcal{M}_1$, then:

$$(f^*du)_p(v) = (df_p^*(du_{f(p)}))(v) = du_{f(p)}(df_p(v)) = df_p(v)u = v(u \circ f) = d(u \circ f)_p(v)$$

This completes the proof. $\square$

Writing ω in coordinate as $\omega = \omega_j dy^j$, with continuous component functions ω_j on V , and using Lemma 1.5.6 twice (applied to $f|_U$):

$$f^*\omega = f^*(\omega_j dy^j) = (\omega_j \circ f)f^*dy^j = (\omega_j \circ f)d(y^j \circ f)$$

which is clearly continuous, and smooth if ω is smooth. $\square$

The formula derived in this proof provides a straightforward expression to compute the pull-back of a covector field, using the fact that $y^j \circ f = f^j$ is the j^{th} component function of f in the (y^j) coordinates:

$$f^*\omega = (\omega_j \circ f)df^j \quad (1.5.23)$$

Example 1.5.2 (Euclidean spaces)

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the map:

$$(u, v) = f(x, y, z) = (x^2y, y \sin z)$$

and let $\omega \in \mathfrak{X}^*(\mathbb{R}^2)$ be the covector field:

$$\omega = v du + u dv$$

Using Eq. 1.5.23, the pull-back of ω by f reads:

$$\begin{aligned} f^*\omega &= (v \circ f)d(u \circ f) + (u \circ f)d(v \circ f) \\ &= (x^2y)d(y \sin z) + (y \sin z)d(x^2y) = 2xy^2 \sin z dx + 2x^2y \sin z dy + x^2y^2 \cos z dz \end{aligned}$$

§1.6 Tensors

So far, the analysis of smooth manifold has been carried by importing concepts of Linear Algebra to them. Now, this approach is extended by considering multilinear maps.

§1.6.1 Multilinear algebra

Let $V_1, \dots, V_k, W$ be $\mathbb{R}$ -vector spaces. A map $f : V_1 \times \dots \times V_k \rightarrow W$ is said to be **multilinear** if it is linear in each variable, i.e. for each $i = 1, \dots, k$ and arbitrary $a, b \in \mathbb{R}$:

$$f(v_1, \dots, av_i + bv'_i, \dots, v_k) = af(v_1, \dots, v_i, \dots, v_k) + bf(v_1, \dots, v'_i, \dots, v_k)$$

The set of multilinear maps from $V_1 \times \dots \times V_k$ to W is denoted by $\text{Mult}(V_1, \dots, V_k; W)$: it clearly is a $\mathbb{R}$ -vector space under pointwise addition and scalar multiplication.

Example 1.6.1 (Familiar multilinear maps)

Both the dot product and the determinant are multilinear functions on $\mathbb{R}^n$: the first one is a bilinear function, while the second one is an n -linear function. In the special case of $\mathbb{R}^3$, another bilinear map is encountered: the cross product, which has $\mathbb{R}^3$ itself as codomain.

An extremely important example of multilinear map is the following one.

Definition 1.6.1 (Tensor product)

Let $V_1, \dots, V_k, W_1, \dots, W_\ell$ be $\mathbb{R}$ -vector spaces, and suppose $f \in \text{Mult}(V_1, \dots, V_k; \mathbb{R})$ and $g \in \text{Mult}(W_1, \dots, W_\ell; \mathbb{R})$. The **tensor product** of f and g is a function $f \otimes g : V_1 \times \dots \times V_k \times W_1 \times \dots \times W_\ell \rightarrow \mathbb{R}$ defined as:

$$f \otimes g(v_1, \dots, v_k, w_1, \dots, w_\ell) := f(v_1, \dots, v_k)g(w_1, \dots, w_\ell) \quad (1.6.1)$$

which is an element of $\text{Mult}(V_1, \dots, V_k, W_1, \dots, W_\ell; \mathbb{R})$.

It is easy to check that the tensor product is bilinear and associative, hence tensor products of three or more multilinear maps can be written unambiguously without parentheses.

Since linear maps on vector spaces are covectors, a natural description of multilinear maps is through tensor products of covectors.

Proposition 1.6.1 (Basis for multilinear functions)

Let $V_1, \dots, V_k$ be $\mathbb{R}$ -vector spaces of dimensions $n_1, \dots, n_k$, respectively, and, for each $j = 1, \dots, k$, let $(E_1^{(j)}, \dots, E_{n_j}^{(j)})$ be a basis for V_j , with $(\varepsilon_1^{(j)}, \dots, \varepsilon_{n_j}^{(j)})$ the corresponding dual basis of V_j^* . Then, the set:

$$\mathcal{B} := \{\varepsilon_{(1)}^{i_1} \otimes \dots \otimes \varepsilon_{(k)}^{i_k} : 1 \leq i_1 \leq n_1, \dots, 1 \leq i_k \leq n_k\}$$

is a basis of $\text{Mult}(V_1, \dots, V_k; \mathbb{R})$, which therefore has dimension $n_1 \dots n_k$.

Proof. Show that $\mathcal{B}$ is linearly independent and spans $\text{Mult}(V_1, \dots, V_k; \mathbb{R})$. Suppose $f \in \text{Mult}(V_1, \dots, V_k; \mathbb{R})$ is arbitrary. For each ordered k -tuple $(i_1, \dots, i_k)$ of integers such that $1 \leq i_j \leq n_j$, define the real number $f_{i_1 \dots i_k} = f(E_{i_1}^{(1)}, \dots, E_{i_k}^{(k)})$. WTS $f = f_{i_1 \dots i_k} \varepsilon_{(1)}^{i_1} \otimes \dots \otimes \varepsilon_{(k)}^{i_k}$. For

any k -tuple $(v_1, \dots, v_k) \in V_1 \times \dots \times V_k$, write $v_j = v_j^{i_j} E_{i_j}^{(j)}$:

$$f_{i_1 \dots i_k} \varepsilon_{(1)}^{i_1} \otimes \dots \otimes \varepsilon_{(k)}^{i_k} (v_1, \dots, v_k) = f_{i_1 \dots i_k} \varepsilon_{(1)}^{i_1} (v_1) \dots \varepsilon_{(k)}^{i_k} (v_k) = f_{i_1 \dots i_k} v_1^{i_1} \dots v_k^{i_k}$$

which is just $f(v_1, \dots, v_k)$ by multilinearity. This proves that $\mathcal{B}$ spans $\text{Mult}(V_1, \dots, V_k; \mathbb{R})$. To show that $\mathcal{B}$ is linearly independent, suppose some linear combination vanishes:

$$f_{i_1 \dots i_k} \varepsilon_{(1)}^{i_1} \otimes \dots \otimes \varepsilon_{(k)}^{i_k} = 0$$

Apply this to any ordered k -tuple of basis vectors $(E_{j_1}^{(1)}, \dots, E_{j_k}^{(k)})$. By the same computation above, this implies that all the coefficients vanish: $f_{i_1 \dots i_k} \equiv 0$. Thus, $\mathcal{B}$ is linearly independent. $\square$

This proof shows that the components of $f \in \text{Mult}(V_1, \dots, V_k; \mathbb{R})$ in the basis $\mathcal{B}$ are:

$$f_{i_1 \dots i_k} = f(E_{i_1}^{(1)}, \dots, E_{i_k}^{(k)}) \quad (1.6.2)$$

Thus, f is completely determined by its action on all possible sequences of basis vectors.

§1.6.1.1 Abstract tensor products

The idea that $\text{Mult}(V_1, \dots, V_k; \mathbb{R})$ can be viewed as the set of all linear combinations of objects of the form $\omega^1 \otimes \dots \otimes \omega^k$, where $\omega^1, \dots, \omega^k$ are covectors, can be extended with the introduction of “formal linear combinations”.

Let S be a set. A *formal linear combination* of elements of S is an expression of the form $\sum_{i=1}^m a_i x_i$, where $a_1, \dots, a_m \in \mathbb{R}$ and $x_1, \dots, x_m \in S$. Since no algebraic structure is being imposed on S , its elements cannot literally be added or multiplied by numbers, however such an expression is completely determined by which elements of S appear in the sum and with which coefficients, hence this definition can in fact be formalized.

Definition 1.6.2 (Free real vector space)

Let S be a set. A **formal linear combination** of elements of S is a function $f : S \rightarrow \mathbb{R}$ such that $f(s) = 0$ for all but finitely many $s \in S$. The **free $\mathbb{R}$ -vector space** on S , denoted by $\mathfrak{F}(S)$, is the set of all formal linear combinations of elements of S .

$\mathfrak{F}(S)$ is indeed a $\mathbb{R}$ -vector space under pointwise addition and scalar multiplication. For each $x \in S$, there is a function $\delta_x \in \mathfrak{F}(S)$ that takes the value 1 on x and vanishes on $S - \{x\}$: typically, the element x is identified with δ_x , hence S can be thought as a subset of $\mathfrak{F}(S)$, and every $f \in \mathfrak{F}(S)$ can be written uniquely in the form $f = \sum_{i=1}^m a_i x_i$, where $x_1, \dots, x_m$ are the elements of S such that $a_i = f(x_i) \neq 0$. Thus, S is a basis of $\mathfrak{F}(S)$, which is therefore finite-dimensional if and only if S is a finite set.

Proposition 1.6.2 (Characteristic property of free vector spaces)

For any set S and any $\mathbb{R}$ -vector space W , every map $A : S \rightarrow W$ has a unique extension to a linear map $\bar{A} : \mathfrak{F}(S) \rightarrow W$.

It is now possible to generalize the tensor product. Let $V_1, \dots, V_k$ be $\mathbb{R}$ -vector spaces, and let $\mathfrak{F}(V_1 \times \dots \times V_k)$ be the set of all finite formal combinations of k -tuples $(v_1, \dots, v_k)$, with $v_i \in V_i$ for $i = 1, \dots, k$. Then, define $\mathfrak{F}_0$ as the subspace of $\mathfrak{F}(V_1 \times \dots \times V_k)$ spanned by all elements of the forms:

$$(v_1, \dots, av_i, \dots, v_k) - a(v_1, \dots, v_i, \dots, v_k)$$

$$(v_1, \dots, v_i + v'_i, \dots, v_k) - (v_1, \dots, v_i, \dots, v_k) - (v_1, \dots, v'_i, \dots, v_k)$$

with $v_j, v'_j \in V_j$ for $j = 1, \dots, k$ and $a \in \mathbb{R}$. The *tensor product* of the spaces $V_1, \dots, V_k$, denoted as $V_1 \otimes \dots \otimes V_k$, is defined as the following quotient vector space:

$$V_1 \otimes \dots \otimes V_k := \mathfrak{F}(V_1 \times \dots \times V_k) / \mathfrak{F}_0 \quad (1.6.3)$$

Moreover, define the natural projection $\Pi : \mathfrak{F}(V_1 \times \dots \times V_k) \rightarrow V_1 \otimes \dots \otimes V_k$, so that the equivalence class of an element $(v_1, \dots, v_k) \in V_1 \times \dots \times V_k$ is $V_1 \otimes \dots \otimes V_k$ is denoted by:

$$v_1 \otimes \dots \otimes v_k \equiv \Pi(v_1, \dots, v_k) \quad (1.6.4)$$

This is called the **(abstract) tensor product** of $v_1, \dots, v_k$. It follows from definition that abstract tensor products satisfy:

$$v_1 \otimes \dots \otimes (av_i + bv'_i) \otimes \dots \otimes v_k = a(v_1 \otimes \dots \otimes v_i \otimes \dots \otimes v_k) + b(v_1 \otimes \dots \otimes v'_i \otimes \dots \otimes v_k)$$

Note that, although by definition every element of $V_1 \otimes \dots \otimes V_k$ can be expressed as a linear combination of elements of the form $v_1 \otimes \dots \otimes v_k$, it is not true in general that every element of the tensor product space is of the form $v_1 \otimes \dots \otimes v_k$.

Theorem 1.6.1 (Characteristic property of tensor product spaces)

Let $V_1, \dots, V_k$ be finite-dimensional $\mathbb{R}$ -vector spaces. If $A : V_1 \times \dots \times V_k \rightarrow X$ is any multilinear map into a vector space X , then there is a unique linear map $\tilde{A} : V_1 \otimes \dots \otimes V_k \rightarrow X$ such that the following diagram commutes:

$$\begin{array}{ccc} V_1 \times \dots \times V_k & \xrightarrow{A} & X \\ \pi \downarrow & \nearrow \tilde{A} & \\ V_1 \otimes \dots \otimes V_k & & \end{array} \quad (1.6.5)$$

where $\pi : V_1 \times \dots \times V_k \rightarrow V_1 \otimes \dots \otimes V_k$ is the map $\pi(v_1, \dots, v_k) = v_1 \otimes \dots \otimes v_k$.

Proof. First, note that any map $A : V_1 \times \dots \times V_k \rightarrow X$ extends uniquely to a linear map $\bar{A} : \mathfrak{F}(V_1 \times \dots \times V_k) \rightarrow X$ by Prop. 1.6.2, which is characterized by $\bar{A}(v_1, \dots, v_k) = A(v_1, \dots, v_k)$ whenever $(v_1, \dots, v_k) \in V_1 \times \dots \times V_k \subseteq \mathfrak{F}(V_1 \times \dots \times V_k)$. Since A is multilinear, the subspace $\mathfrak{F}_0$ is contained in $\ker \bar{A}$, therefore $\bar{A}$ determines a linear map $\tilde{A} : V_1 \otimes \dots \otimes V_k = \mathfrak{F}(V_1 \times \dots \times V_k) / \mathfrak{F}_0 \rightarrow X$ satisfying $\tilde{A} \circ \Pi = \bar{A}$. Since π is equal to the inclusion $V_1 \times \dots \times V_k \hookrightarrow \mathfrak{F}(V_1 \times \dots \times V_k)$ followed by Π , this implies $\tilde{A} \circ \pi = A$, which is the thesis.

Uniqueness follows from the fact that every element of $V_1 \otimes \dots \otimes V_k$ can be written as a linear combination of elements of the form $v_1 \otimes \dots \otimes v_k$, and $\tilde{A}$ is uniquely determined on such elements by $\tilde{A}(v_1 \otimes \dots \otimes v_k) = \bar{A}(v_1, \dots, v_k) = A(v_1, \dots, v_k)$. $\square$

This is called the “characteristic property” of the tensor product since it uniquely characterizes the tensor product up to isomorphism. Using this characteristic property, it is possible to show that the tensor product is associative. i.e. there are unique isomorphisms:

$$V_1 \otimes (V_2 \otimes V_3) \cong V_1 \otimes V_2 \otimes V_3 \cong (V_1 \otimes V_2) \otimes V_3$$

under which elements of the forms $v_1 \otimes (v_2 \otimes v_3)$, $v_1 \otimes v_2 \otimes v_3$ and $(v_1 \otimes v_2) \otimes v_3$ all correspond.

Proposition 1.6.3 (Basis of tensor product space)

Let $V_1, \dots, V_k$ be $\mathbb{R}$ -vector spaces of dimensions $n_1, \dots, n_k$, respectively, and, for each $j = 1, \dots, k$, let $(E_1^{(j)}, \dots, E_{n_j}^{(j)})$ be a basis for V_j . Then, the set:

$$\mathcal{B} := \{E_{i_1}^{(1)} \otimes \dots \otimes E_{i_k}^{(k)} : 1 \leq i_1 \leq n_1, \dots, 1 \leq i_k \leq n_k\}$$

is a basis for $V_1 \otimes \dots \otimes V_k$, which therefore has dimension equal to $n_1 \dots n_k$.

Proof. Elements of the form $v_1 \otimes \dots \otimes v_k$ span the tensor space by definition: expanding each v_i in such an expression in terms of its basis representation shows that $\mathcal{B}$ spans $V_1 \otimes \dots \otimes V_k$.

To prove that $\mathcal{B}$ is linearly independent, assume that some linear combination vanishes:

$$a^{i_1 \dots i_k} E_{i_1}^{(1)} \otimes \dots \otimes E_{i_k}^{(k)} = 0$$

For each ordered k -tuple of indices $(m_1, \dots, m_k)$, define a multilinear map $\tau^{m_1 \dots m_k} : V_1 \times \dots \times V_k \rightarrow \mathbb{R}$ by:

$$\tau^{m_1 \dots m_k}(v_1, \dots, v_k) = \varepsilon_{(1)}^{m_1}(v_1) \dots \varepsilon_{(k)}^{m_k}(v_k)$$

where $(\varepsilon_{(j)}^j)$ is the basis of V_j^* dual to $(E_j^{(j)})$. Since $\tau^{m_1 \dots m_k}$ is multilinear, it determines a linear map $\tilde{\tau}^{m_1 \dots m_k} : V_1 \otimes \dots \otimes V_k \rightarrow \mathbb{R}$ by Th. 1.6.1. It follows that:

$$0 = \tilde{\tau}^{m_1 \dots m_k}(a^{i_1 \dots i_k} E_{i_1}^{(1)} \otimes \dots \otimes E_{i_k}^{(k)}) = a^{i_1 \dots i_k} \tau^{m_1 \dots m_k}(E_{i_1}^{(1)}, \dots, E_{i_k}^{(k)}) = a^{m_1 \dots m_k}$$

which shows that $\mathcal{B}$ is linearly independent. $\square$

It is now possible to show the connection between “abstract” tensor products and “concrete” tensor products.

Lemma 1.6.1

If $V_1, \dots, V_k$ are finite-dimensional vector spaces, there is a canonical isomorphism:

$$V_1^* \otimes \dots \otimes V_k^* \cong \text{Mult}(V_1, \dots, V_k; \mathbb{R}) \quad (1.6.6)$$

under which the $\omega^1 \otimes \dots \otimes \omega^k \in \text{Mult}(V_1, \dots, V_k; \mathbb{R})$, defined by $\omega^1 \otimes \dots \otimes \omega^k(v_1, \dots, v_k) = \omega^1(v_1) \dots \omega^k(v_k)$, and $\omega^1 \otimes \dots \otimes \omega^k \in V_1^* \otimes \dots \otimes V_k^*$, defined by $\omega^1 \otimes \dots \otimes \omega^k = \Pi(\omega^1, \dots, \omega^k)$, correspond.

Proof. Define the map $\Phi : V_1^* \times \dots \times V_k^* \rightarrow \text{Mult}(V_1, \dots, V_k; \mathbb{R})$ as:

$$\Phi(\omega^1, \dots, \omega^k)(v_1, \dots, v_k) = \omega^1(v_1) \dots \omega^k(v_k)$$

which is multilinear in $(v_1, \dots, v_k)$. Moreover, $\Phi(\omega^1, \dots, \omega^k)$ is multilinear in $(\omega^1, \dots, \omega^k)$, hence by the characteristic property of the tensor product it determined a unique linear map $\tilde{\Phi} : V_1^* \otimes \dots \otimes V_k^* \rightarrow \text{Mult}(V_1, \dots, V_k; \mathbb{R})$ which satisfies:

$$\tilde{\Phi}(\omega^1 \otimes \dots \otimes \omega^k)(v_1, \dots, v_k) = \omega^1(v_1) \dots \omega^k(v_k)$$

By this definition, $\tilde{\Phi}$ clearly maps abstract tensor products to tensor products of covectors, and it also maps the basis of $V_1^* \otimes \dots \otimes V_k^*$ given by Prop. 1.6.3 to the basis of $\text{Mult}(V_1, \dots, V_k; \mathbb{R})$ given by Prop. 1.6.1, therefore it is an isomorphism. $\square$

In presence of this canonical isomorphism, $\text{Mult}(V_1, \dots, V_k; \mathbb{R})$ is henceforth identified with $V_1^* \otimes \dots \otimes V_k^*$. Moreover, given that the vector spaces considered are finite-dimensional, the canonical isomorphism $V \cong V^{**}$ allows for another canonical identification:

$$V_1 \otimes \dots \otimes V_k \cong \text{Mult}(V_1^*, \dots, V_k^*; \mathbb{R}) \quad (1.6.7)$$

which emphasizes again the symmetry between V and V^* for finite-dimensional vector spaces.

§1.6.1.2 Mixed tensors

Having defined tensor products of vector spaces, it is now possible to characterize multilinear maps on a $\mathbb{R}$ -vector space V . Elements of the k -fold tensor product $V^* \otimes \dots \otimes V^*$ are called *covariant k -tensors* on V : these are just multilinear maps $\alpha : V \times \dots \times V \rightarrow \mathbb{R}$. Similarly, elements of the k -fold product $V \otimes \dots \otimes V$ are called *contravariant k -tensors* on V , which are multilinear maps $\alpha : V^* \times \dots \times V^* \rightarrow \mathbb{R}$.

Definition 1.6.3 (Tensor)

Let V be a $\mathbb{R}$ -vector space. Given $k, l \in \mathbb{N}_0$, the space of **tensors of rank (k, l)** is defined as:

$$T_l^k(V) := \underbrace{V \otimes \dots \otimes V}_{k \text{ times}} \otimes \underbrace{V^* \otimes \dots \otimes V^*}_{l \text{ times}} \quad (1.6.8)$$

Clearly, $T_0^0(V) = \mathbb{R}$, $T_0^1(V) = V$ and $T_1^0(V) = V^*$, hence scalars, vectors and covectors can be thought as tensors. Moreover, by Prop. 1.6.3, when V is finite-dimensional, any choice of basis (E_i) of V yields automatically a basis of $T_l^k(V)$:

$$\mathcal{B} = \{E_{i_1} \otimes \dots \otimes E_{i_k} \otimes \varepsilon^{j_1} \otimes \dots \otimes \varepsilon^{j_l} \in T_l^k(V) : 1 \leq i_1, \dots, i_k, j_1, \dots, j_l \leq n\} \quad (1.6.9)$$

where (ε^i) is the dual basis of V^* and $n \equiv \dim_{\mathbb{R}} V$. Therefore, $\dim_{\mathbb{R}} T_l^k(V) = n^{k+l}$, and given $\alpha \in T_l^k(V)$, its n^{k+l} components are found as:

$$\alpha^{i_1 \dots i_k}_{j_1 \dots j_l} = \alpha(\varepsilon^{i_1}, \dots, \varepsilon^{i_k}, E_{j_1}, \dots, E_{j_l}) \quad (1.6.10)$$

§1.6.2 Tensor fields

Definition 1.6.4 (Tensor bundles)

Given a smooth manifold $\mathcal{M}$, its **tensor bundle of rank (k, l)** is defined as:

$$T_l^k \mathcal{M} := \bigsqcup_{p \in \mathcal{M}} T_l^k(T_p \mathcal{M}) \quad (1.6.11)$$

There are some natural identifications: $T_0^0 \mathcal{M} = \mathcal{M} \times \mathbb{R}$, $T_0^1 \mathcal{M} = T\mathcal{M}$ and $T_1^0 \mathcal{M} = T^* \mathcal{M}$. As for vector fields, sections of $T_l^k \mathcal{M}$ are **tensor fields of rank (k, l)** over $\mathcal{M}$, and in any smooth chart (x^i) on $\mathcal{M}$ they can be written as:

$$A = A^{i_1 \dots i_k}_{j_1 \dots j_l} \frac{\partial}{\partial x^{i_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{i_k}} \otimes dx^{j_1} \otimes \dots \otimes dx^{j_l} \quad (1.6.12)$$

This expression clearly determined the alias transformation rule for tensor fields. Consider two overlapping charts $(U, (x^i))$, $(V, (\tilde{x}^i))$, and suppose that $A^{i_1 \dots i_k}_{j_1 \dots j_l}$ are the components of a rank- (k, l)

tensor field A over $\mathcal{M}$ in the first chart. Then, the components of A in the second chart are:

$$\tilde{A}^{i_1 \dots i_k}_{j_1 \dots j_l} = \frac{\partial \tilde{x}^{i_1}}{\partial x^{\alpha_1}} \cdots \frac{\partial \tilde{x}^{i_k}}{\partial x^{\alpha_k}} \frac{\partial x^{\beta_1}}{\partial \tilde{x}^{j_1}} \cdots \frac{\partial x^{\beta_l}}{\partial \tilde{x}^{j_l}} A^{\alpha_1 \dots \alpha_k}_{\beta_1 \dots \beta_l} \quad (1.6.13)$$

The space of smooth sections of $T_l^k \mathcal{M}$ is an infinite-dimensional $\mathbb{R}$ -vector space and a module over $\mathcal{C}^\infty(\mathcal{M})$, denoted by:

$$\mathfrak{T}_l^k(\mathcal{M}) \equiv \Gamma(T_l^k \mathcal{M}) \quad (1.6.14)$$

Note that, given $A \in \mathfrak{T}_l^k(\mathcal{M})$ and $B \in \mathfrak{T}_q^p(\mathcal{M})$, then trivially $A \otimes B \in \mathfrak{T}_{l+q}^{k+p}(\mathcal{M})$. Moreover, if $A \in \mathfrak{T}_l^k(\mathcal{M})$, $\omega^1, \dots, \omega^k \in \mathfrak{X}^*(\mathcal{M})$ and $X_1, \dots, X_l \in \mathfrak{X}(\mathcal{M})$, then $A(\omega^1, \dots, \omega^k, X_1, \dots, X_l) \in \mathcal{C}^\infty(\mathcal{M})$, hence a rank- (k, l) tensor field $A \in \mathfrak{T}_l^k(\mathcal{M})$ induces a map:

$$A : \underbrace{\mathfrak{X}^*(\mathcal{M}) \times \cdots \times \mathfrak{X}^*(\mathcal{M})}_{k \text{ times}} \times \underbrace{\mathfrak{X}(\mathcal{M}) \times \cdots \times \mathfrak{X}(\mathcal{M})}_{l \text{ times}} \rightarrow \mathbb{R} \quad (1.6.15)$$

which is multilinear⁸ over $\mathcal{C}^\infty(\mathcal{M})$.

§1.6.2.1 Pull-back of tensor fields

Recall that a diffeomorphism $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ induces two maps: the push-forward of vector fields through the differential $df_p : T_p \mathcal{M}_1 \rightarrow T_{f(p)} \mathcal{M}_2$ and the pull-back of covector fields through the dual of the differential $df_p^* : T_{f(p)}^* \mathcal{M}_2 \rightarrow T_p^* \mathcal{M}_1$. These are respectively defined by Eq. 1.2.4, 1.5.19, and are denoted as $(f_* X)_{f(p)} := df_p(X_p)$ and $(f^* \omega)_p := df_p^*(\omega_{f(p)})$, respectively.

Definition 1.6.5 (Pull-back of tensor field)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a diffeomorphism. Then, the **pull-back** by f at p is defined as the map $f_p^* : T_l^k(T_{f(p)} \mathcal{M}_2) \rightarrow T_l^k(T_p \mathcal{M}_1)$ which acts on $A \in T_l^k(T_{f(p)} \mathcal{M}_2)$ as:

$$(f^* A)_p(\omega^1, \dots, \omega^k, X_1, \dots, X_l) := A((f^* \omega^1)_p, \dots, (f^* \omega^k)_p, (f_*^{-1} X_1)_p, \dots, (f_*^{-1} X_l)_p) \quad (1.6.16)$$

with $\omega^1, \dots, \omega^k \in T^* \mathcal{M}_2$, $X_1, \dots, X_l \in T \mathcal{M}_2$ and $f_*^{-1} \equiv (f^{-1})_*$.

Note that the vector fields are “pushed-back” by the inverse diffeomorphism $f^{-1} : \mathcal{M}_2 \rightarrow \mathcal{M}_1$, so that all fields are mapped from $\mathcal{M}_2$ to $\mathcal{M}_1$.

The definition of a pull-back on tensor fields allows for the extension of the Lie derivative on them (which was defined in Def. 1.4.5 on vector fields through their “push-back”).

Definition 1.6.6 (Lie derivative of tensor field)

Let $\mathcal{M}$ be a smooth manifold, and let $V \in \mathfrak{X}(\mathcal{M})$, $A \in \mathfrak{T}_l^k(\mathcal{M})$. Denoting as θ the flow of V on $\mathcal{M}$, the **Lie derivative** of A with respect to V is the tensor field (provided it exists):

$$(\mathcal{L}_V A)_p := \left. \frac{d}{dt} \right|_{t=0} (\theta_t^* A)_p \quad (1.6.17)$$

Note that the expression being differentiated lies in $T_l^k(T_p \mathcal{M})$ for all t , hence correctly $(\mathcal{L}_V A)_p \in T_l^k(T_p \mathcal{M})$. Moreover, since it is defined from smooth maps, it can be shown that indeed $\mathcal{L}_V A \in \mathfrak{T}_l^k(\mathcal{M})$.

⁸A map $A : \mathfrak{X}(\mathcal{M}) \rightarrow \mathbb{R}$ is linear over $\mathcal{C}^\infty(\mathcal{M})$ if $A(fX + gY) = fA(X) + gA(Y) \forall f, g \in \mathcal{C}^\infty(\mathcal{M}), X, Y \in \mathfrak{X}(\mathcal{M})$, with obvious generalization to $\mathfrak{X}^*(\mathcal{M})$ and to the multilinear case.

Lemma 1.6.2 (Properties of Lie derivative)

Let $V \in \mathfrak{X}(\mathcal{M})$, $f \in \mathcal{C}^\infty(\mathcal{M})$, $A \in \mathfrak{T}_l^k(\mathcal{M})$ and $B \in \mathfrak{T}_q^p(\mathcal{M})$. Then:

- (1) $\mathcal{L}_V f = V(f)$;
- (2) $\mathcal{L}_V(fA) = \mathcal{L}_V(f)A + f\mathcal{L}_V A$;
- (3) $\mathcal{L}_V(A \otimes B) = (\mathcal{L}_V A) \otimes B + A \otimes (\mathcal{L}_V B)$;

A fundamental property of the Lie derivative is that it obeys a convenient product rule.

Proposition 1.6.4 (Product rule)

Let $V \in \mathfrak{X}(\mathcal{M})$, $A \in \mathfrak{T}_l^k(\mathcal{M})$ and $\omega^1, \dots, \omega^k \in \mathfrak{X}^*(\mathcal{M})$, $X_1, \dots, X_l \in \mathfrak{X}(\mathcal{M})$. Then:

$$\begin{aligned} \mathcal{L}_V(A(\omega^1, \dots, \omega^k, X_1, \dots, X_l)) &= (\mathcal{L}_V A)(\omega^1, \dots, \omega^k, X_1, \dots, X_l) + \\ &\quad + \sum_{i=1}^k A(\omega^1, \dots, \mathcal{L}_V \omega^i, \dots, \omega^k, X_1, \dots, X_l) + \\ &\quad + \sum_{j=1}^l A(\omega^1, \dots, \omega^k, X_1, \dots, \mathcal{L}_V X_j, \dots, X_l) \end{aligned} \quad (1.6.18)$$

Corollary 1.6.4.1 (Components of Lie derivative)

Let $V \in \mathfrak{X}(\mathcal{M})$ and $A \in \mathfrak{T}_l^k(\mathcal{M})$. In a smooth chart $(U, (x^i))$ on $\mathcal{M}$, the components of $\mathcal{L}_V A \in \mathfrak{T}_l^k(\mathcal{M})$ are:

$$(\mathcal{L}_V A)^{i_1 \dots i_k}_{j_1 \dots j_l} = V^\ell \partial_\ell A^{i_1 \dots i_k}_{j_1 \dots j_l} - \sum_{i=1}^k A^{i_1 \dots \ell \dots i_k}_{j_1 \dots j_l} \partial_\ell V^{i_i} + \sum_{j=1}^l A^{i_1 \dots i_k}_{j_1 \dots \ell \dots j_l} \partial_{j_j} V^\ell \quad (1.6.19)$$

Proof. Recall Eq. 1.6.10. By Eq. 1.6.18, it is necessary to compute $\mathcal{L}_V \partial_i$ and $\mathcal{L}_V dx^i$. The first one is straightforward, using $\mathcal{L}_V \partial_i = [V, \partial_i]$ and Eq. 1.2.21:

$$\mathcal{L}_V \partial_i = -\partial_i V^\ell \partial_\ell \quad (1.6.20)$$

The second one needs more care.

Lemma 1.6.3 (Lie derivative and differential)

If $V \in \mathfrak{X}(\mathcal{M})$ and $f \in \mathcal{C}^\infty(\mathcal{M})$, then:

$$\mathcal{L}_V(df) = d(\mathcal{L}_V f) \quad (1.6.21)$$

Proof. By Eq. 1.6.18, for any $X \in \mathfrak{X}(\mathcal{M})$:

$$(\mathcal{L}_V(df))(X) = V(df(X)) - df([V, X]) = VX(f) - [V, X](f) = XV(f)$$

Since $XV(f) = d(V(f))(X) = d(\mathcal{L}_V f)(X)$, the proof is complete. $\square$

By Lemma 1.6.3, then, $\mathcal{L}_V dx^i = d(\mathcal{L}_V x^i) = d(V^j \partial_j x^i) = dV^i$, so using Eq. 1.5.17:

$$\mathcal{L}_V dx^i = \partial_\ell V^i dx^\ell \quad (1.6.22)$$

Inserting Eq. 1.6.21-1.6.22 into Eq. 1.6.18 yields the thesis. $\square$

In particular, the Lie derivative of a covector field $\omega \in \mathfrak{X}^*(\mathcal{M})$ reads (in components):

$$\mathcal{L}_V \omega = (V^\ell \partial_\ell \omega_i + \omega_\ell \partial_i V^\ell) dx^i \quad (1.6.23)$$

Finally, it is possible to characterize tensor fields which remain unchanged under certain flows.

Proposition 1.6.5

Let $\mathcal{M}$ be a smooth manifold and $V \in \mathfrak{X}(\mathcal{M})$ with flow θ on $\mathcal{M}$. If $\partial\mathcal{M} \neq \emptyset$, assume V is tangent to $\partial\mathcal{M}$. Then, for any tensor field A over $\mathcal{M}$ and any (t_0, p) in the domain of θ :

$$\left. \frac{d}{dt} \right|_{t=t_0} (\theta_t^* A)_p = (\theta_{t_0}^* (\mathcal{L}_V A))_p \quad (1.6.24)$$

Given a flow θ on $\mathcal{M}$, a tensor field A over $\mathcal{M}$ is said to be **invariant under** θ if, for any (t, p) in the domain of θ :

$$(\theta_t^* A)_p = A_p$$

Theorem 1.6.2 (Invariance under flow)

Let $\mathcal{M}$ be a smooth manifold, and let $V \in \mathfrak{X}(\mathcal{M})$ and $A \in \mathfrak{T}_l^k(\mathcal{M})$. Then, A is invariant under the flow of V if and only if $\mathcal{L}_V A = 0$.

This theorem gives a clear interpretation to the Lie derivative.

Riemannian Manifolds

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§2.1 Metric manifolds

To import concepts of Geometry on smooth manifolds, it is necessary to define additional structure on them.

Consider an n -dimensional $\mathbb{R}$ -vector space V and a symmetric tensor $g \in T_2^0(V)$, i.e. such that $g(v, w) = g(w, v) \forall v, w \in V$. Then, g is **non-degenerate** if the linear map $\hat{g} : V \rightarrow V^* : \hat{g}(v)(w) = g(v, w)$ is an isomorphism, or equivalently if $\forall v \in V : v \neq 0 \exists w \in V : g(v, w) \neq 0$. With an adaptation of the Gram–Schmidt algorithm, every non-degenerate symmetric rank- $(0, 2)$ tensor can be transformed, by a change of basis, to one whose (local) matrix representation, i.e. $[g_{ij}]_{i,j=1,\dots,n}$, is diagonal with all entries equal to ± 1 . By a result known as *Sylvester’s law of inertia*, the numbers r and s of negative and positive entries, respectively, are basis-independent, thus the ordered pair (r, s) is an invariant of g , called its **signature**.

Definition 2.1.1 (Metric)

Let $\mathcal{M}$ be a smooth manifold. A symmetric tensor field $g \in \mathfrak{T}_2^0(\mathcal{M})$ is a **metric** on $\mathcal{M}$ if it is non-degenerate at each point $p \in \mathcal{M}$ and if it has the same signature on all $\mathcal{M}$. A smooth manifold equipped with a metric is a **metric manifold**.

The two most important cases for the signature of the metric tensor are $(0, n)$, in which case it is a *Riemannian metric*, and $(1, n - 1)$, which is a *Lorentzian metric*. In general, any metric with $r > 0$ is called a pseudo-Riemannian metric.

Observation 2.1.1 (Symmetric tensors)

The expression of the metric in a smooth chart $(U, (x^i))$ on $\mathcal{M}$ does not make its symmetry clear:

$$g = g_{ij} dx^i \otimes dx^j$$

To make the symmetry explicit, it is first necessary to further characterize symmetric tensors. Let V be an n -dimensional $\mathbb{R}$ -vector space. A covariant k -tensor is a rank- $(0, k)$ tensor on V , and it is said to be symmetric if its value is unchanged under the interchange of any pair of arguments:

the set of symmetric covariant k -tensors on V is denoted as $\Sigma^k(V) \subseteq T_k^0(V)$. It is possible to define a natural projection $\text{Sym} : T_k^0(V) \rightarrow \Sigma^k(V)$ as:

$$(\text{Sym } \alpha)(v_1, \dots, v_k) := \frac{1}{k!} \sum_{\sigma \in S^k} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \quad (2.1.1)$$

which is clearly a symmetric k -tensor. Note that $\text{Sym}|_{\Sigma^k(V)} = 1_{\Sigma^k(V)}$. In general, the tensor product of two symmetric tensors is not symmetric, but the symmetrization operator allows to define a **symmetric product** of $\alpha \in \Sigma^k(V)$ and $\beta \in \Sigma^l(V)$:

$$\alpha\beta := \text{Sym}(\alpha \otimes \beta) \quad (2.1.2)$$

Clearly $\alpha\beta \in \Sigma^{k+l}(V)$, since its explicit action is:

$$\alpha\beta(v_1, \dots, v_{k+l}) = \sum_{\sigma \in S^{k+l}} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \beta(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)}) \quad (2.1.3)$$

It can be shown that the symmetric product is symmetric and bilinear, and it has a particularly simple action on covectors $\alpha, \beta \in V^*$:

$$\alpha\beta = \frac{1}{2}(\alpha \otimes \beta + \beta \otimes \alpha) \quad (2.1.4)$$

This last expression allows to make the symmetry of the metric transparent:

$$g = g_{ij} dx^i \otimes dx^j = \frac{1}{2}(g_{ij} dx^i \otimes dx^j + g_{ji} dx^j \otimes dx^i) = \frac{1}{2}g_{ij}(dx^i \otimes dx^j + dx^j \otimes dx^i)$$

where $g_{ij} = g_{ji}$ has been used. This is just the symmetric product of dx^i and dx^j , hence:

$$g = g_{ij} dx^i dx^j \quad (2.1.5)$$

which is now manifestly symmetric.

§2.1.1 Riemannian manifolds

Since the diagonal representation of a Riemannian metric is the identity matrix, g_p clearly defines an inner product on $T_p\mathcal{M}$, since it is positive-definite: therefore, given $v, w \in T_p\mathcal{M}$, the notation $g_p(v, w) \equiv \langle v, w \rangle_g$ is set.

Example 2.1.1 (Euclidean metric)

The simplest example of a Riemannian metric is the Euclidean metric $\bar{g}$ on $\mathbb{R}^n$, which is expressed in standard coordinates as:

$$\bar{g} = \delta_{ij} dx^i dx^j$$

It is common to abbreviate the symmetric product as $\alpha\alpha \equiv \alpha^2$, hence the Euclidean metric can be written as:

$$\bar{g} = (dx^1)^2 + \dots + (dx^n)^2$$

Applied to $v, w \in T_p \mathbb{R}^n$, it yields the Euclidean inner product:

$$\bar{g}(v, w) = \delta_{ij} v^i w^j = \sum_{i=1}^n v^i w^i \equiv v \cdot w$$

A peculiar feature of Riemannian metrics, which is not present in pseudo-Riemannian metrics, is their abundance.

Theorem 2.1.1 (Existence of Riemannian metric)

Every smooth manifold admits a Riemannian metric.

Proof. Let $\mathcal{M}$ be a smooth manifold with a smooth atlas $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$. In each coordinate domain there is a Riemannian metric given by the pull-back of the Euclidean metric $g_\alpha = \varphi_\alpha^* \bar{g}$, whose coordinate expression is $\delta_{ij} dx^i dx^j$. Then, consider a smooth partition of unity¹ $\{\psi_\alpha\}$ subordinate to the cover $\{U_\alpha\}$, and define:

$$g = \sum_{\alpha} \psi_{\alpha} g_{\alpha}$$

By local finiteness, this sum has only finitely many non-zero terms in the neighbourhood of each point $p \in \mathcal{M}$, hence this expression defines a smooth tensor field on $\mathcal{M}$. Since it is obviously symmetric, only positive-definiteness needs to be checked. Given any $v \in T_p \mathcal{M} : v \neq 0$:

$$g_p(v, v) = \sum_{\alpha} \psi_{\alpha}(p) g_{\alpha}|_p(v, v)$$

This sum is non-negative, as each term is non-negative. Moreover, at least one function ψ_{α} is strictly positive at p (as they must sum to 1), and $g_{\alpha}|_p(v, v) > 0$ since $v \neq 0$, therefore $g_p(v, v) > 0$. $\square$

This proof shows that there is a great amount of arbitrariness in the construction of a metric on a smooth manifold: indeed, the same manifold can be equipped with extremely different metrics, yielding extremely different geometric properties. The main geometric constructions induced by a Riemannian metric are:

- the **norm** of a tangent vector $v \in T_p \mathcal{M}$:

$$\|v\|_g := \sqrt{\langle v, v \rangle_g} \quad (2.1.6)$$

- the **angle** between two non-zero tangent vectors $v, w \in T_p \mathcal{M}$, which is the unique $\theta \in [0, \pi]$ such that:

$$\cos \theta := \frac{\langle v, w \rangle_g}{\|v\|_g \|w\|_g} \quad (2.1.7)$$

¹Let $\mathcal{M}$ be a topological space, and let $\mathcal{X} = \{X_{\alpha}\}_{\alpha \in \mathcal{I}}$ be an arbitrary open cover of $\mathcal{M}$. A *partition of unity* subordinate to $\mathcal{X}$ is a family $\{\psi_{\alpha}\}_{\alpha \in \mathcal{I}}$ of continuous functions $\psi_{\alpha} : \mathcal{M} \rightarrow [0, 1] \subseteq \mathbb{R}$ such that:

- (1) $\text{supp } \psi_{\alpha} \subseteq X_{\alpha} \ \forall \alpha \in \mathcal{I}$;
- (2) the family $\{\text{supp } \psi_{\alpha}\}_{\alpha \in \mathcal{I}}$ is locally finite, i.e. every point has a neighbourhood that intersects $\text{supp } \psi_{\alpha}$ only for finitely many $\alpha \in \mathcal{I}$;
- (3) $\sum_{\alpha \in \mathcal{I}} \psi_{\alpha}(p) = 1 \ \forall p \in \mathcal{M}$.

The sum in (3) has no convergence problems, since it has only a finite number of non-zero terms due to (2). If $\mathcal{M}$ is a smooth manifold and each function ψ_{α} is smooth, then this is a smooth partition of unity.

- two tangent vectors $v, w \in T_p\mathcal{M}$ are **orthogonal** if $\langle v, w \rangle_g = 0$.

Note that two vectors are orthogonal either if one or both are zero or if the angle between them is $\frac{\pi}{2}$.

Definition 2.1.2 (Orthonormal frame)

Let $(\mathcal{M}, g)$ be a Riemannian manifold. A local frame $(E_1, \dots, E_n)$ on $U \in \mathcal{T}(\mathcal{M})$ is an **orthonormal frame** if $(E_1|_p, \dots, E_n|_p)$ is an orthonormal basis of $T_p\mathcal{M}$ at each point $p \in U$, or equivalently if $\langle E_i, E_j \rangle_g = \delta_{ij}$.

In general, a smooth orthonormal frame can be found starting from a smooth local frame and applying the Gram–Schmidt algorithm: therefore, for each point $p \in \mathcal{M}$, there is a smooth orthonormal frame on a neighbourhood of p , found applying the Gram–Schmidt algorithm to the coordinate frame on said neighbourhood.

§2.1.1.1 Pull-back metrics

Suppose $\mathcal{M}_1, \mathcal{M}_2$ are smooth manifolds, g is a Riemannian metric on $\mathcal{M}_2$ and $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a smooth map. Then, the pull-back f^*g is a smooth covariant 2-tensor field on $\mathcal{M}_1$: if it is positive-definite, f^*g is a Riemannian metric on $\mathcal{M}_1$, called the **pull-back** metric on $\mathcal{M}_1$ determined by f .

Example 2.1.2 (Helicoid)

Consider the smooth map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by:

$$f(u, v) = (u \cos v, u \sin v, v)$$

whose image is a surface known as a **helicoid**. The pull-back metric $f^*\bar{g}$ on the helicoid can be computed substituting the component functions of f in the coordinate expression of $\bar{g}$:

$$f^*\bar{g} = d(u \cos v)^2 + d(u \sin v)^2 + dv^2 = du^2 + (1 + u^2)dv^2$$

The same procedure can be applied to transform a Riemannian metric under a change of coordinates, which is thought as the identity map expressed in terms of different coordinates for the domain and the codomain.

Another important application of pull-back metrics is the study of geometrical invariants under isometries.

Definition 2.1.3 (Riemannian isometry)

Let $(\mathcal{M}, g)$ and $(\tilde{\mathcal{M}}, \tilde{g})$ be Riemannian manifolds. A diffeomorphism $f : \mathcal{M} \rightarrow \tilde{\mathcal{M}}$ is a **Riemannian isometry** if $f^*\tilde{g} = g$.

This definition can be generalized to *local Riemannian isometries*, where the condition of isometry is only enforced on some $U \in \mathcal{T}(\tilde{\mathcal{M}})$. If there exists a Riemannian isometry between two Riemannian manifolds, they are said to be *isometric*.

§2.1.1.2 Riemannian distance

The introduction of a metric on a smooth manifold allows to measure the lengths of curves.

Definition 2.1.4 (Length of curve)

Let $(\mathcal{M}, g)$ be a Riemannian manifold. If $\gamma : [a, b] \rightarrow \mathcal{M}$ is a piecewise-smooth curve, the **length of γ** is defined as:

$$L_g(\gamma) := \int_a^b dt \|\gamma'(t)\|_g \quad (2.1.8)$$

This integral is well-defined since $\|\gamma'(t)\|_g$ is continuous on $[a, b]$, except for finitely many values of t . In general, given $c \in (a, b)$:

$$L_g(\gamma) = L_g(\gamma|_{[a, c]}) + L_g(\gamma|_{[c, b]}) \quad (2.1.9)$$

The lengths of curves are local isometry invariants of Riemannian manifolds: given a Riemannian isometry $f : \mathcal{M} \rightarrow \tilde{\mathcal{M}}$, for every piecewise-smooth curve γ on $\mathcal{M}$ it is trivial to show that $L_{\tilde{g}}(f \circ \gamma) = L_g(\gamma)$.

Moreover, the length of a curve is also invariant under reparametrizations of the curve. Given a piecewise-smooth curve $\gamma : [a, b] \rightarrow \mathcal{M}$ on a Riemannian manifold, a **reparametrization** of γ induced by a diffeomorphism $\varphi : [c, d] \rightarrow [a, b]$ is a piecewise-smooth curve $\tilde{\gamma} = \gamma \circ \varphi$.

Proposition 2.1.1 (Reparametrization-invariance of length)

Let $(\mathcal{M}, g)$ be a Riemannian manifold, and let $\gamma : [a, b] \rightarrow \mathcal{M}$ be a piecewise-smooth curve. If $\tilde{\gamma}$ is a reparametrization of γ , then $L_g(\tilde{\gamma}) = L_g(\gamma)$.

Proof. Assume first that γ is smooth, and let $\varphi : [c, d] \rightarrow [a, b]$ be a diffeomorphism such that $\tilde{\gamma} = \gamma \circ \varphi$: since it is a diffeomorphism, either $\varphi' > 0$ or $\varphi' < 0$ everywhere. Inserting the reparametrization in the integral:

$$\begin{aligned} L_g(\tilde{\gamma}) &= \int_c^d dt \|\tilde{\gamma}'(t)\|_g = \int_c^d dt \left\| \frac{d}{dt}(\gamma \circ \varphi)(t) \right\|_g \\ &= \int_c^d dt \|\varphi'(t) \gamma'(\varphi(t))\|_g = \int_c^d dt |\varphi'(t)| \|\gamma'(\varphi(t))\|_g \end{aligned}$$

Now, if $\varphi' > 0$, then $|\varphi'(t)| = \varphi'(t)$ and, under the change of integration variable $s = \varphi(t)$, the integral becomes:

$$L_g(\tilde{\gamma}) = \int_a^b ds \|\gamma'(s)\|_g = L_g(\gamma)$$

On the other hand, if $\varphi' < 0$, then $|\varphi'(t)| = -\varphi'(t)$, and the same transformation yields:

$$L_g(\tilde{\gamma}) = - \int_d^c ds \|\gamma'(s)\|_g = L_g(\gamma)$$

If γ is only piecewise-smooth, apply the same reasoning to each subinterval on which it is smooth. $\square$

Having defined the notion of length for curves, it is then possible to introduce the notion of distance between points of the manifold. Consider a connected patch $U \in \mathcal{T}(\mathcal{M})$ of a Riemannian manifold (U, g) : for any $p, q \in U$, the **(Riemannian) distance** from p to q is defined as:

$$d_g(p, q) := \inf_{\mathbb{R}} \{L_g(\gamma) \in \mathbb{R}_0^+ : \gamma : [a, b] \rightarrow \mathcal{M} : \gamma(a) = p, \gamma(b) = q\} \quad (2.1.10)$$

which is well-defined since $p, q \in U$ connected. If $\mathcal{M}$ is connected, the distance function is defined on the whole $\mathcal{M}$, and in fact it gives $\mathcal{M}$ the structure of metric space.

Theorem 2.1.2 (Riemannian manifolds as metric spaces)

Let $(\mathcal{M}, g)$ be a connected Riemannian manifold. Then, $(\mathcal{M}, d_g)$ is a metric space whose topology is the same as the original manifold topology.

Proof. By definition $d_g(p, q) \leq 0 \ \forall p, q \in \mathcal{M}$. Since every constant curve has vanishing length, $d_g(p, p) = 0 \ \forall p \in \mathcal{M}$, while $d_g(p, q) = d_g(q, p)$ by reparametrization invariance.

Now, suppose γ_1 and γ_2 are piecewise-smooth curves from p to q and from q to r , respectively, and let γ be the piecewise-smooth curve that first follows γ_1 and then follows γ_2 (suitably reparametrized). Then:

$$d_g(p, r) \leq L_g(\gamma) = L_g(\gamma_1) + L_g(\gamma_2)$$

Taking the infimum over all such γ_1 and γ_2 , the triangle inequality $d_g(p, r) \leq d_g(p, q) + d_g(q, r)$ is found.

To prove that $(\mathcal{M}, g)$ is a metric space, there only remains to show that $d_g(p, q) > 0$ for $p \neq q \in \mathcal{M}$. To do so, let U be a smooth coordinate domain containing p and not q , and identify it with $U \in \mathcal{T}(\mathbb{R}^n)$ through the corresponding smooth coordinate map.

Lemma 2.1.1

Let g be a Riemannian metric on $U \in \mathcal{T}(\mathbb{R}^n)$. Given a compact subset $K \subseteq U$, there exist $c, C \in \mathbb{R}^+$ such that:

$$c \|v\|_g \leq \|v\|_g \leq C \|v\|_g \quad \forall v \in T_x \mathbb{R}^n \ \forall x \in K \quad (2.1.11)$$

Proof. For any compact subset $K \subseteq U$, let $L \subseteq T\mathbb{R}^n$ defined as:

$$L = \{(x, y) \in T\mathbb{R}^n : x \in K, \|y\|_g = 1\}$$

Given the canonical identification $T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$, then $L \cong K \times \mathbb{S}^{n-1}$, hence it is compact. Since the norm is continuous and strictly positive over L , there exist $c, C \in \mathbb{R}^+$ such that $c \leq \|v\|_g \leq C$ whenever $(x, v) \in L$. If $x \in K$ and $v \neq 0 \in T_x \mathbb{R}^n$, let $\lambda = \|v\|_g$, so that $(x, \lambda^{-1}v) \in L$ and, by homogeneity of the norm:

$$\|v\|_g = \lambda \|\lambda^{-1}v\|_g \leq \lambda C = C \|v\|_g$$

and similarly for the lower bound. The same inequalities become trivial for $v = 0$. $\square$

Consider a regular coordinate ball V of radius $\varepsilon > 0$ and centered at p such that $\bar{V} \subseteq U$. Applying Lemma 2.1.1, there are $c, C \in \mathbb{R}^+$ such that Eq. 2.1.11 holds for all $x \in \bar{V}$ and $v \in T_x \mathcal{M}$. Then, for any piecewise-smooth curve γ lying entirely in $\bar{V}$:

$$cL_g(\gamma) \leq L_g(\gamma) \leq CL_g(\gamma)$$

Now, suppose $\gamma : [a, b] \rightarrow \mathcal{M}$ is a piecewise-smooth curve from p to q , and let t_0 be the infimum of $t \in [a, b] : \gamma(t) \notin \bar{V}$: by continuity $\gamma(t_0) \in \partial V$ (indeed $\partial V \cap \text{supp } \gamma = \{t_0\}$), and $\gamma(t) \in \bar{V} \ \forall t \in [a, t_0]$. Thus:

$$L_g(\gamma) \geq L_g(\gamma|_{[a, t_0]}) \geq cL_g(\gamma|_{[a, t_0]}) \geq cd_g(p, \gamma(t_0)) = c\varepsilon > 0$$

Taking the infimum over all such curves, $d_g(p, q) \geq c\varepsilon > 0$ is concluded.

Finally, there remains to show that the metric topology generated by d_g is the same as the given manifold topology on $\mathcal{M}$. To do so, WTS $U \in \mathcal{T}_{\text{man}}(\mathcal{M}) \iff U \in \mathcal{T}_{\text{dist}}(\mathcal{M})$.

Suppose first that $U \in \mathcal{T}_{\text{man}}(\mathcal{M})$, let $p \in U$ and let V be a regular coordinate ball of radius $\varepsilon > 0$ centered at p such that $\bar{V} \subseteq U$. The previous argument shows that $d_g(p, q) \geq c\varepsilon$ for $q \notin \bar{V}$, hence $d_g(p, q) < c\varepsilon \implies q \in \bar{V} \subseteq U$, i.e. the metric ball of radius $c\varepsilon$ centered at p is contained in U . Hence, $U \in \mathcal{T}_{\text{dist}}(\mathcal{M})$.

Conversely, suppose $W \in \mathcal{T}_{\text{dist}}(\mathcal{M})$, and let $p \in W$. Furthermore, let V be a regular coordinate ball of radius $r > 0$ centered at p , let $\bar{g}$ be the Euclidean metric on $\bar{V}$ determined by the given coordinates, and let $c, C \in \mathbb{R}^+$ such that Eq. 2.1.11 is satisfied for $v \in T_q\mathcal{M}, q \in \bar{V}$. Consider $\varepsilon \in (0, r)$ small enough that the metric ball centered at p of radius $C\varepsilon$ is contained in W , and let V_ε be the set of points in $\bar{V}$ whose Euclidean distance from p is less than ε . If $q \in V_\varepsilon$, let γ be the straight-line segment in coordinates from p to q . By Lemma 2.1.1 as above:

$$d_g(p, q) \leq L_g(\gamma) \leq CL_{\bar{g}}(\gamma) \leq C\varepsilon$$

This shows that V_ε is contained in the metric ball of radius $C\varepsilon$ centered at p , and therefore in W . Since $V_\varepsilon \in \mathcal{T}_{\text{man}}(\mathcal{M})$ is a neighbourhood of p , this shows that $W \in \mathcal{T}_{\text{man}}(\mathcal{M})$. $\square$

By this theorem, all of the terminology of metric spaces can be carried over to connected Riemannian manifolds. A connected Riemannian manifold $(\mathcal{M}, g)$ is complete if $(\mathcal{M}, d_g)$ is a complete metric space, i.e. if every Cauchy sequence in $\mathcal{M}$ converges to a point in $\mathcal{M}$, and g is then said to be a **complete Riemannian metric**: it can be shown that every connected smooth manifold admits a complete Riemannian metric.

§2.1.2 Lorentzian manifolds

The Lorentzian manifold is more subtle to study, since there is no analogue of Th. 2.1.1 for Lorentzian metrics, a consequence of the fact that it is not generally true that a linear combination of non-degenerate covariant 2-tensors with positive coefficients is necessarily non-degenerate: indeed, not every smooth manifold admits a Lorentzian metric.

Example 2.1.3 (Minkowski space)

The simplest example of Lorentzian manifold is $\mathbb{R}^n$ with the **Lorentz–Minkowski metric**:

$$\eta = -(dx^0)^2 + (dx^1)^2 + \cdots + (dx^{n-1})^2 \quad (2.1.12)$$

Note that it is conventional to define the coordinate corresponding to the negative entry of the metric as the 0th coordinate.

A Lorentzian metric does not induce naturally an inner product on tangent vectors, since the negative entry makes it non-positive-definite. Nonetheless, it is possible to define one on restricted classes of tangent vectors. To do so, classify $v \in T_p\mathcal{M}$ as follows:

- **timelike** if $g_p(v, v) < 0$;
- **lightlike** (or **null**) if $g_p(v, v) = 0$;
- **spacelike** if $g_p(v, v) > 0$.

Similarly, a **timelike curve** is a curve whose tangent vector is everywhere timelike, and analogously for the other cases. Then, keeping the three classes of tangent vectors and curves separate, it is

possible to import the same geometric constructions defined on Riemannian manifolds on spacelike vectors in a direct way, and on timelike vectors changing the sign of the inner product. Hence, for example, the length of a timelike piecewise-smooth curve $\gamma : [a, b] \rightarrow \mathcal{M}$ is defined as:

$$L_g(\gamma) := \int_a^b dt \sqrt{-\langle \gamma'(t), \gamma'(t) \rangle_g}$$

This is known as the *proper time* of the timelike curve γ elapsed between $\gamma(a)$ and $\gamma(b)$. In general, the affine parameter of a timelike curve is its proper time. Indeed, defining the proper time of a timelike curve $\gamma : [t_0, t_f] \rightarrow \mathcal{M}$ as:

$$\tau(t) := \int_{t_0}^t dt \sqrt{-\langle \gamma'(t), \gamma'(t) \rangle_g} \quad (2.1.13)$$

then it is clear that $L_g(\gamma) = \int_0^{\tau_f} d\tau$, with $\tau_f \equiv \tau(t_f)$.

Observation 2.1.2 (Negative Lorentzian metric)

By Def. 2.1.1, if g is a metric on a smooth manifold $\mathcal{M}$, then so is $\tilde{g} = -g$. Obviously, the geometric description determined by $\tilde{g}$ is completely equivalent to that of g , provided that the sign change is accounted in all expressions involving an odd number of metric coefficients. In the Lorentzian case, the negative metric has the opposite signature of the “ordinary” metric: this means that spacelike and timelike vectors are exchanged with each other, while null vectors remain unchanged.

§2.1.3 Tangent-cotangent isomorphism

The definition of a metric on a smooth manifold also provides a natural isomorphism between the tangent and cotangent bundles.

Let $(\mathcal{M}, g)$ be a metric manifold, and define a bundle homomorphism (Def. 1.5.4) $\hat{g} : T\mathcal{M} \rightarrow T^*\mathcal{M}$ as follows: for each $p \in \mathcal{M}$ and $v \in T_p\mathcal{M}$, let $\hat{g}(v) \in T_p^*\mathcal{M}$ defined as:

$$\hat{g}(v)(w) := g_p(v, w) \quad \forall w \in T_p\mathcal{M} \quad (2.1.14)$$

To see that this is a smooth bundle homomorphism, consider its action on smooth vector fields: $\hat{g}(X)(Y) = g(X, Y)$, with $X, Y \in \mathfrak{X}(\mathcal{M})$. Recall Eq. 1.6.15: as $\hat{g}(X)(Y)$ is multilinear over $\mathcal{C}^\infty(\mathcal{M})$, it is linear over $\mathcal{C}^\infty(\mathcal{M})$ as a function of Y , thus $\hat{g}(X)$ is a smooth covector (i.e. rank-(0, 1) tensor) field; then, since $\hat{g}(X)$ is linear over $\mathcal{C}^\infty(\mathcal{M})$ as a function of X , $\hat{g}$ is a smooth bundle homomorphism by Lemma 1.5.2. Note that the same symbol is used for the pointwise bundle homomorphism $\hat{g} : T\mathcal{M} \rightarrow T^*\mathcal{M}$ and for the linear map on sections $\hat{g} : \mathfrak{X}(\mathcal{M}) \rightarrow \mathfrak{X}^*(\mathcal{M})$.

Since the metric is non-degenerate, $\hat{g}(v) = 0 \implies 0 = \hat{g}(v)(v) = \langle v, v \rangle_g \iff v = 0 \in T_p\mathcal{M}$, hence $\ker \hat{g}|_{T_p\mathcal{M}} = 0$, i.e. $\hat{g}$ is injective. For dimensional reasons, therefore, $\hat{g}$ is bijection, so it is a canonical smooth bundle isomorphism.

Given the canonical isomorphism $T\mathcal{M} \cong T^*\mathcal{M}$, it is conventional to set a simpler notation through the so-called *musical isomorphisms*:

$$T\mathcal{M} \ni X \xrightarrow{\hat{g}} X^\flat \equiv \hat{g}(X) \in T^*\mathcal{M} \quad T^*\mathcal{M} \ni \omega \xrightarrow{\hat{g}^{-1}} \omega^\sharp \equiv \hat{g}^{-1}(\omega) \in T\mathcal{M}$$

In any smooth chart on $\mathcal{M}$, the metric tensor is $g = g_{ij}dx^i dx^j$, thus, given $X, Y \in \mathfrak{X}(\mathcal{M})$, the action of the bundle isomorphism is $\hat{g}(X)(Y) = g_{ij}X^i X^j$, which implies that:

$$X^\flat = \hat{g}(X) = g_{ij}X^i dx^j \equiv X_j dx^j \quad (2.1.15)$$

It is then clear that the action of $\hat{g}$ on a vector field lowers its contravariant index to a covariant one. The action of $\hat{g}$ on $\omega \in \mathfrak{X}^*(\mathcal{M})$ is similar, since g_{ij} is invertible at each point (being the matrix representation of an isomorphism, which is invertible) with inverse denoted by g^{ij} (i.e. such that $g_{ij}g^{jk} = g^{jk}g_{ij} = \delta_i^k$):

$$\omega^\sharp = \hat{g}^{-1}(\omega) = g^{ij}\omega_i \frac{\partial}{\partial x^j} \equiv \omega^j \frac{\partial}{\partial x^j} \quad (2.1.16)$$

Hence, the action of $\hat{g}^{-1}$ on a covector field raises its covariant index to a contravariant one.

Example 2.1.4 (Gradient operator)

Given $f \in C^\infty(\mathcal{M})$, its **gradient** is defined as:

$$\nabla f := (df)^\sharp \quad (2.1.17)$$

Note that ∇f is the unique vector field such that, for any vector field X over $\mathcal{M}$, satisfies:

$$\langle \nabla f, X \rangle_g = \hat{g}(\nabla f)(X) = df(X) = X(f)$$

In a smooth coordinate chart $(U, (x^i))$, the gradient has the expression:

$$\nabla f = g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^j}$$

which reduces to the usual expression in the Euclidean case.

§2.2 Differential forms

§2.2.1 Grassmann algebra

Let V be an n -dimensional $\mathbb{R}$ -vector space. A covariant k -tensor is said to be *alternating* if its value changes sign under any interchange of two arguments, or equivalently if any permutation of its arguments results in an overall multiplication by the sign of said permutation. Alternating covariant k -tensors are called **exterior forms**, and they form a vector space $\bigwedge^k(V) \subseteq T_k^0(V)$.

Lemma 2.2.1 (Characterization of alternating tensors)

Let $\alpha \in T_k^0(V)$. Then the following conditions are equivalent:

- (1) α is alternating;
- (2) $\alpha(v_1, \dots, v_k) = 0$ whenever $(v_1, \dots, v_k)$ is linearly dependent;
- (3) α vanishes whenever two of its arguments are equal.

Proof. (1) $\Rightarrow$ (3) and (2) $\Rightarrow$ (3) trivially. WTS (3) $\Rightarrow$ (1) and (3) $\Rightarrow$ (2). Assume that α satisfies (3). For any k -tuple $(v_1, \dots, v_k)$ of vectors, then:

$$\begin{aligned} 0 &= \alpha(v_1, \dots, v_i + v_j, \dots, v_i + v_j, \dots, v_k) \\ &= \alpha(v_1, \dots, v_i, \dots, v_i, \dots, v_k) + \alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) \\ &\quad + \alpha(v_1, \dots, v_j, \dots, v_i, \dots, v_k) + \alpha(v_1, \dots, v_j, \dots, v_j, \dots, v_k) \\ &= \alpha(v_1, \dots, v_i, \dots, v_j, v_k) + \alpha(v_1, \dots, v_j, \dots, v_i, \dots, v_k) \end{aligned}$$

Thus α is alternating. On the other hand, if $(v_1, \dots, v_k)$ is linearly independent, then one vector, WLOG v_k , can be written as a linear combination of the others, hence $\alpha(v_1, \dots, v_k)$ breaks down into a sum of vanishing terms by (3). $\square$

Similarly to Obs. 2.1.1, it is possible to define a natural projection $\text{Alt} : T_k^0(V) \rightarrow \bigwedge^k(V)$ as:

$$(\text{Alt } \alpha)(v_1, \dots, v_k) := \frac{1}{k!} \sum_{\sigma \in S^k} (\text{sgn } \sigma) \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \quad (2.2.1)$$

which is clearly an alternating k -tensor. Note that $\text{Alt}|_{\bigwedge^k(V)} = 1_{\bigwedge^k(V)}$. For computations involving alternating tensors, it is convenient to define the following notation: given $k \in \mathbb{N}$, an ordered k -tuple $I = (i_1, \dots, i_k) \in \mathbb{N}^k$ is a *multi-index of length k* . Moreover, given $\sigma \in S^n$, the permutation of I under σ is denoted by $I_\sigma \equiv (i_{\sigma(1)}, \dots, i_{\sigma(k)})$, with the convention $I_{\sigma\tau} \equiv (I_\sigma)_\tau$ for $\sigma, \tau \in S^k$. Multi-indices allow for the generalization of the determinant in $\mathbb{R}^n$. Let V be an n -dimensional $\mathbb{R}$ -vector space, and suppose $(\varepsilon^1, \dots, \varepsilon^n)$ is a basis of V^* . For each multi-index $I = (i_1, \dots, i_k)$ such that $1 \leq i_1, \dots, i_k \leq n$, define $\varepsilon^I \equiv \varepsilon^{i_1 \dots i_k} \in T_k^0(V)$ by:

$$\varepsilon^I(v_1, \dots, v_k) := \begin{vmatrix} \varepsilon^{i_1}(v_1) & \dots & \varepsilon^{i_1}(v_k) \\ \vdots & \ddots & \vdots \\ \varepsilon^{i_k}(v_1) & \dots & \varepsilon^{i_k}(v_k) \end{vmatrix} = \begin{vmatrix} v_1^{i_1} & \dots & v_k^{i_1} \\ \vdots & \ddots & \vdots \\ v_1^{i_k} & \dots & v_k^{i_k} \end{vmatrix} \quad (2.2.2)$$

Since the determinant changes sign under the interchange of any two columns, $\varepsilon^I \in \bigwedge^k(V)$: this is called an *elementary exterior form*.

Lemma 2.2.2 (Properties of elementary exterior forms)

Let V be an n -dimensional $\mathbb{R}$ -vector space, and let (E_i) be a basis of V . If (ε^i) is the dual basis of V^* and $I = (i_1, \dots, i_k)$ is a multi-index, then:

- (1) if I has a repeated index, then $\varepsilon^I \equiv 0$;
- (2) if $J = I_\sigma$ for some $\sigma \in S^k$, then $\varepsilon^J = (\text{sgn } \sigma) \varepsilon^I$;
- (3) defining the **generalized Kronecker delta** as:

$$\delta_J^I := \begin{vmatrix} \delta_{j_1}^{i_1} & \dots & \delta_{j_k}^{i_1} \\ \vdots & \ddots & \vdots \\ \delta_{j_1}^{i_k} & \dots & \delta_{j_k}^{i_k} \end{vmatrix} \quad (2.2.3)$$

then:

$$\varepsilon^I(E_{j_1}, \dots, E_{j_k}) = \delta_J^I \quad (2.2.4)$$

Proof. If I has a repeated index, then ε^I vanishes since the determinant of a matrix with two equal columns is zero, proving (1). On the other hand, (2) is trivial by the alternating property of the determinant. Finally, (3) follows directly from the definition of ε^I . $\square$

Although $\{\varepsilon^I\}_{I \in [0, n]^k}$ is not linearly independent by (1)-(2) of Lemma 2.2.2, it is possible to define a basis of $\bigwedge^k(V)$ choosing an appropriate subset of multi-indices.

Proposition 2.2.1 (Basis of exterior forms)

Let V be an n -dimensional $\mathbb{R}$ -vector space. If (ε^i) is a basis of V^* , then for each $k \in \mathbb{N}_{\leq n}$ a basis of $\bigwedge^k(V)$ is found as:

$$\mathcal{B}_k = \left\{ \varepsilon^I \in \bigwedge^k(V) : I = (i_1, \dots, i_k) : 1 \leq i_1 < \dots < i_k \leq n \right\} \quad (2.2.5)$$

Therefore $\dim_{\mathbb{R}} \bigwedge^k(V) = \binom{n}{k}$.

Proof. The fact that $\dim_{\mathbb{R}} \bigwedge^k(V) = 0$ for $k > n$ follows immediately from (2) of Lemma 2.2.1. In the $k \leq n$ case, let (E^i) be the dual basis of V . To show that $\bigwedge^k(V) = \text{span}_{\mathbb{R}} \mathcal{B}_k$, let $\alpha \in \bigwedge^k(V)$ and, for each arbitrary multi-index $I = (i_1, \dots, i_k)$, define a real number α_I as:

$$\alpha_I = \alpha(E_{i_1}, \dots, E_{i_k})$$

Since α is alternating, by Lemma 2.2.2 $\alpha_I = 0$ if I contains a repeated index, and $\alpha_J = (\text{sgn } \sigma) \alpha_I$ if $J = I_{\sigma}$ for some $\sigma \in S^k$. Therefore:

$$\sum_{I: 1 \leq i_1 < \dots < i_k \leq n} \alpha_I \varepsilon^I(E_{j_1}, \dots, E_{j_k}) = \sum_{I: 1 \leq i_1 < \dots < i_k \leq n} \alpha_I \delta_J^I = \alpha_J = \alpha(E_{j_1}, \dots, E_{j_k})$$

Denoting the sum over *increasing* multi-indices as $\sum_I'$, this shows that $\sum_I' \alpha_I \varepsilon^I = \alpha$, hence $\bigwedge^k(V) = \text{span}_{\mathbb{R}} \mathcal{B}_k$.

Then, to show that $\mathcal{B}_k$ is linearly independent, suppose that $\sum_I' \alpha_I \varepsilon^I = 0$ for some coefficients α_I , and let J be an increasing multi-index. Applying both sides to $(E_{j_1}, \dots, E_{j_k})$, by Lemma 2.2.2:

$$0 = \sum_I' \alpha_I \varepsilon^I(E_{j_1}, \dots, E_{j_k}) = \alpha_J$$

Thus, each coefficient α_J vanishes, proving linear independence. $\square$

Prop. 2.2.1 shows that there is a particular class of exterior forms: those with $k = n$, known as **top forms**. Indeed, $\bigwedge^n(V)$ is 1-dimensional and spanned by $\varepsilon^{1\dots n}$: on $\mathbb{R}^n$, this is just the determinant function $e^{1\dots n}$. Top forms behave simply under linear transformations. Consider a linear map $T : V \rightarrow V$, and define $\det T$ as the determinant of the representation matrix of T (in any basis, by the Binet theorem): then, $\omega \in \bigwedge^n(V)$ transforms as:

$$\omega(Tv_1, \dots, Tv_n) = (\det T) \omega(v_1, \dots, v_n) \quad (2.2.6)$$

Note that, like $\Sigma^0(V) = \mathbb{R}$ and $\Sigma^1(V) = V^*$, so $\bigwedge^0(V) = \mathbb{R}$ and $\bigwedge^1(V) = V^*$.

§2.2.1.1 Exterior product

Analogously to the symmetric product of symmetric tensors $\alpha\beta = \text{Sym}(\alpha \otimes \beta)$, it is possible to define an alternating product of alternating tensors.

Definition 2.2.1 (Exterior product)

Let V an n -dimensional $\mathbb{R}$ -vector space. Given $\omega \in \bigwedge^k(V)$ and $\eta \in \bigwedge^l(V)$, their **exterior**

product is the alternating tensor $\omega \wedge \eta \in \bigwedge^{k+l}(V)$ defined as:

$$\omega \wedge \eta := \frac{(k+l)!}{k!l!} \text{Alt}(\omega \otimes \eta) \quad (2.2.7)$$

The choice of the coefficient is motivated by the simple form taken by exterior product of elementary exterior forms. Given multi-indices $I = (i_1, \dots, i_k)$ and $J = (j_1, \dots, j_l)$, it can be shown that:

$$\varepsilon^I \wedge \varepsilon^J = \varepsilon^{IJ} \quad (2.2.8)$$

with $IJ \equiv (i_1, \dots, i_k, j_1, \dots, j_l)$, i.e. a concatenation.

Lemma 2.2.3 (Properties of the exterior product)

Suppose ω, η, ξ are exterior forms on a finite-dimensional $\mathbb{R}$ -vector space V . Then:

(1) (bilinearity) for $a, a' \in \mathbb{R}$:

$$(a\omega + a'\eta) \wedge \xi = a(\omega \wedge \xi) + a'(\eta \wedge \xi)$$

$$\xi \wedge (a\omega + a'\eta) = a(\xi \wedge \omega) + a'(\xi \wedge \eta)$$

(2) (associativity):

$$\omega \wedge (\eta \wedge \xi) = (\omega \wedge \eta) \wedge \xi$$

(3) (anticommutativity) for $\omega \in \bigwedge^k(V)$ and $\eta \in \bigwedge^l(V)$:

$$\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega \quad (2.2.9)$$

(4) if (ε^i) is a basis of V^* and $I = (i_1, \dots, i_k)$ is a multi-index, then:

$$\varepsilon^I = \varepsilon^{i_1} \wedge \dots \wedge \varepsilon^{i_k} \quad (2.2.10)$$

(5) for any $\omega_1, \dots, \omega_k \in V^*$ and $v_1, \dots, v_k \in V$:

$$\omega^1 \wedge \dots \wedge \omega^k(v_1, \dots, v_k) = \det[\omega^j(v_i)] \quad (2.2.11)$$

Proof. Bilinearity follows immediately from definition, since the tensor product is bilinear and Alt is linear. To prove associativity, use Eq. 2.2.8:

$$(\varepsilon^I \wedge \varepsilon^J) \wedge \varepsilon^K = \varepsilon^{IJK} = \varepsilon^I \wedge (\varepsilon^J \wedge \varepsilon^K)$$

and the general case following from multilinearity. Similarly, using Eq. 2.2.8 again:

$$\varepsilon^I \wedge \varepsilon^J = \varepsilon^{IJ} = (\text{sgn } \sigma) \varepsilon^{JI} = (\text{sgn } \sigma) \varepsilon^J \wedge \varepsilon^I$$

where $\sigma \in S^{k+l} : IJ \mapsto JI$. Since $\text{sgn } \sigma = (-1)^{kl}$, as it can be decomposed into kl transpositions (each index of I must be moved past each of the indices of J), anticommutativity is found from multilinearity. Finally, (4) follows by induction from Eq. 2.2.8, while to prove (5) note that, when $\omega^j = \varepsilon^{i_j}$, it reduces to (4), and then use multilinearity. $\square$

In general, an exterior k -form is said to be decomposable if it can be written as the exterior product

of k exterior 1-forms. Note that, although not every exterior k -form is decomposable when $k > 1$, every exterior k -form can be written as a linear combination of decomposable exterior k -forms (elementary exterior k -forms).

Since it can be shown that the exterior product is the unique associate bilinear anticommutative map $\bigwedge^k(V) \times \bigwedge^l(V) \rightarrow \bigwedge^{k+l}(V)$, the ensemble of $\bigwedge^k(V)$ spaces form a natural structure on V .

Definition 2.2.2 (Grassmann algebra)

Let V be an n -dimensional $\mathbb{R}$ -vector space. The **Grassmann algebra** (or *exterior algebra*) of V is defined as:

$$\bigwedge(V) := \bigoplus_{k=0}^n \bigwedge^k(V) \quad (2.2.12)$$

By Prop. 2.2.1, it follows that $\dim_{\mathbb{R}} \bigwedge(V) = 2^n$. The Grassmann algebra has the structure of 2^n -dimensional $\mathbb{R}$ -vector space and of anticommutative $\mathbb{N}_0$ -graded² unital associative algebra³.

§2.2.1.2 Interior product

While the exterior products allows to build higher-rank exterior forms from lower-rank one, there is another operation which has the opposite effect, resulting in a lower-rank exterior form from a higher-rank one.

Definition 2.2.3 (Interior product)

Let V be an n -dimensional $\mathbb{R}$ -vector space. For each $v \in V$, the linear map $\iota_v : \bigwedge^k(V) \rightarrow \bigwedge^{k-1}(V)$, called the **interior product** with v , is defined as:

$$\iota_v \omega(w_1, \dots, w_{k-1}) := \omega(v, w_1, \dots, w_{k-1}) \quad (2.2.13)$$

Therefore, $\iota_v \omega$ is obtained inserting v as the first argument of ω . By convention, $\iota_v|_{\bigwedge^0(V)} \equiv 0$.

Lemma 2.2.4 (Properties of the interior product)

Let V be an n -dimensional $\mathbb{R}$ -vector space and $v \in V$. Then:

- (1) $\iota_v \circ \iota_v = 0$;
- (2) if $\omega \in \bigwedge^k(V)$ and $\eta \in \bigwedge^l(V)$, then:

$$\iota_v(\omega \wedge \eta) = (\iota_v \omega) \wedge \eta + (-1)^k \omega \wedge (\iota_v \eta) \quad (2.2.14)$$

²Given an index set $\mathcal{I}$, a $\mathbb{K}$ -vector space V is said to be $\mathcal{I}$ -graded if there exists a family of subspaces $\{V_i\}_{i \in \mathcal{I}}$ such that $V = \bigoplus_{i \in \mathcal{I}} V_i$.

³An n -dimensional *associative* $\mathbb{K}$ -algebra is an n -dimensional $\mathbb{K}$ -vector space $\mathcal{A}$ equipped with a bilinear map $\mathcal{A} \times \mathcal{A} \ni (a, b) \mapsto ab \in \mathcal{A}$ such that:

- (1) $(ab)c = a(cb) \quad \forall a, b, c \in \mathcal{A}$;
- (2) $a(b+c) = ab + ac \wedge (a+b)c = ac + ab \quad \forall a, b, c \in \mathcal{A}$;
- (3) $\lambda(ab) = (\lambda a)b = a(\lambda b) \quad \forall \lambda \in \mathbb{K}, a, b \in \mathcal{A}$.

An algebra is said to be *unital* if $\exists 1 \in \mathcal{A} : 1a = a1 = a \quad \forall a \in \mathcal{A}$. Moreover, given an index set $\mathcal{I}$, an algebra $\mathcal{A}$ is $\mathcal{I}$ -graded if $\mathcal{A}$ is an $\mathcal{I}$ -graded vector space and the product satisfies $(\mathcal{A}_k)(\mathcal{A}_l) \subseteq \mathcal{A}_{k+l} \quad \forall k, l \in \mathcal{I}$, and it is anticommutative if $ab = (-1)^{kl}ba \quad \forall a \in \mathcal{A}_k, b \in \mathcal{A}_l$.

Proof. For exterior k -forms with $k \geq 2$, (1) is immediate from definition by the alternating nature of exterior forms, while for $k = 1, 0$ it follows from the convention $\iota_v|_{\Lambda^0(V)} \equiv 0$.

To prove (2), it suffices to consider the case of ω, η decomposable, since every exterior k -form can be written as a linear combination of decomposable ones. For general $\omega^1, \dots, \omega^k \in \Lambda^1(V)$ WTS:

$$\iota_v(\omega^1 \wedge \dots \wedge \omega^k) = \sum_{i=1}^k (-1)^{i-1} \omega^i(v) \omega^1 \wedge \dots \wedge \omega^{i-1} \wedge \omega^{i+1} \wedge \dots \wedge \omega^k \quad (2.2.15)$$

which clearly implies (2). To prove this equation, write $v_1 \equiv v$ and apply both sides to the $(k-1)$ -tuple of vectors $(v_2, \dots, v_k)$:

$$\begin{aligned} (\omega^1 \wedge \dots \wedge \omega^k)(v_1, \dots, v_k) \\ = \sum_{i=1}^k (-1)^{i-1} \omega^i(v_1) (\omega^1 \wedge \dots \wedge \omega^{i-1} \wedge \omega^{i+1} \wedge \dots \wedge \omega^k)(v_2, \dots, v_k) \end{aligned}$$

The left-hand side is just $\det[\omega^i(v_j)] \equiv \det v$. Now, define v_j^i as the $(k-1) \times (k-1)$ submatrix of v obtained removing the i^{th} row and the j^{th} column. Then, this equation becomes:

$$\det v = \sum_{i=1}^k (-1)^{i-1} \omega^i(v_1) \det v_1^i$$

which is just the Laplace expansion of $\det v$ by minors along the first column, hence it is an identity. $\square$

§2.2.2 Differential forms

Consider an n -dimensional smooth manifold $\mathcal{M}$. It is possible to define the subset of $T_k^0 \mathcal{M}$ of alternating rank- k covariant tensors on $\mathcal{M}$ as:

$$\Lambda^k(\mathcal{M}) := \bigsqcup_{p \in \mathcal{M}} \bigwedge^k (T_p^0 \mathcal{M}) \quad (2.2.16)$$

It can be shown that $\Lambda^k(\mathcal{M})$ is a smooth rank- $\binom{n}{k}$ vector subbundle of $T_k^0 \mathcal{M}$. Sections of $\Lambda^k(\mathcal{M})$ are called **differential k -forms** (or just k -forms): therefore, a k -form is a (continuous) tensor field whose value at each point is an alternating tensor. The $\mathbb{R}$ -vector space of smooth k -forms is denoted as:

$$\Omega^k(\mathcal{M}) := \Gamma(\Lambda^k(\mathcal{M})) \quad (2.2.17)$$

The wedge product is extended to k -forms in a pointwise manner, i.e. $(\omega \wedge \eta)_p = \omega_p \wedge \eta_p$. Then, it is possible to define the following direct product:

$$\Omega(\mathcal{M}) := \bigoplus_{k=0}^n \Omega^k(\mathcal{M}) \quad (2.2.18)$$

which, equipped with the wedge product, is an anticommutative $\mathbb{N}_0$ -graded associative unital algebra. In any smooth chart $(U, (x^i))$, a k -form ω can be written locally as:

$$\omega = \frac{1}{k!} \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} = \sum_I' \omega_I dx^I \quad (2.2.19)$$

where the coefficients ω_I are continuous functions defined on the coordinate domain, and $dx^I \equiv dx^{i_1} \wedge \cdots \wedge dx^{i_k}$. Using Lemma 2.2.2, the component functions are found as:

$$\omega_I = \omega \left(\frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_k}} \right)$$

If $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a smooth map and ω is a k -form on $\mathcal{M}_2$, its pull-back on $\mathcal{M}_1$ is given by Eq. 1.6.16:

$$(f^*\omega)_p(v_1, \dots, v_k) = \omega_{f(p)}(df_p(v_1), \dots, df_p(v_k))$$

Lemma 2.2.5 (Pull-back of differential forms)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map. Then:

- (1) $f^* : \Omega^k(\mathcal{M}_2) \rightarrow \Omega^k(\mathcal{M}_1)$ is linear over $\mathbb{R}$;
- (2) $f^*(\omega \wedge \eta) = (f^*\omega) \wedge (f^*\eta)$;
- (3) in any smooth chart $(U, (y^i))$ of $\mathcal{M}_2$:

$$f^* \left(\frac{1}{k!} \omega_{i_1 \dots i_k} dy^{i_1} \wedge \cdots \wedge dy^{i_k} \right) = \frac{1}{k!} (\omega_{i_1 \dots i_k} \circ f) d(y^{i_1} \circ f) \wedge \cdots \wedge d(y^{i_k} \circ f) \quad (2.2.20)$$

Hence, the pull-back of differential forms is computed similarly to that of arbitrary tensor fields.

Proposition 2.2.2 (Pull-back of top forms)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map between smooth n -manifolds. If $(U, (x^i))$ and $(V, (y^i))$ are smooth charts on $\mathcal{M}_1$ and $\mathcal{M}_2$, respectively, and u is a continuous function $u : \mathcal{M}_2 \rightarrow \mathbb{R}$, then on $U \cap f^{-1}(V)$:

$$f^*(u dy^1 \wedge \cdots \wedge dy^n) = (u \circ f)(\det \mathcal{J}_f) dx^1 \wedge \cdots \wedge dx^n \quad (2.2.21)$$

where $\mathcal{J}_f$ is the Jacobian matrix of f in these coordinates.

Proof. Since the fiber of $\Lambda^n(\mathcal{M})$ is spanned by $dx^1 \wedge \cdots \wedge dx^n$ at each point, it suffices to show that both sides of Eq. 2.2.21 give the same result when applied to $(\partial_1, \dots, \partial_n)$. From Lemma 2.2.5:

$$f^*(u dy^1 \wedge \cdots \wedge dy^n) = (u \circ f) df^1 \wedge \cdots \wedge df^n$$

Then, using (5) of Lemma 2.2.3:

$$df^1 \wedge \cdots \wedge df^n \left(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n} \right) = \det \left[df^j \left(\frac{\partial}{\partial x^i} \right) \right] = \det \mathcal{J}_f$$

The left-hand side then gives $(u \circ f) \det \mathcal{J}_f$, which is equal to the right-hand side since $dx^1 \wedge \cdots \wedge dx^n(\partial_1, \dots, \partial_n) = 1$. $\square$

An obvious application of Prop. 2.2.2 is to coordinate changes. Indeed, given two overlapping smooth charts $(U, (x^i))$ and $(\tilde{U}, \tilde{x}^i)$ on a smooth n -manifold $\mathcal{M}$, then on $U \cap \tilde{U}$:

$$d\tilde{x}^1 \wedge \cdots \wedge d\tilde{x}^n = \det \left[\frac{\partial \tilde{x}^j}{\partial x^i} \right] dx^1 \wedge \cdots \wedge dx^n$$

The interior product too extends naturally to vector fields and differential forms by letting it act pointwise. If $X \in \mathfrak{X}(\mathcal{M})$ and $\omega \in \Omega^k(\mathcal{M})$, then $\iota_X \omega \in \Omega^{k-1}(\mathcal{M})$ is defined by $(\iota_X \omega)_p = \iota_{X_p} \omega_p$: therefore, $\iota_X : \Omega^k(\mathcal{M}) \rightarrow \Omega^{k-1}(\mathcal{M})$ is linear over $\mathcal{C}^\infty(\mathcal{M})$ and defines a smooth bundle homomorphism $\iota_X : \Lambda^k(\mathcal{M}) \rightarrow \Lambda^{k-1}(\mathcal{M})$.

§2.2.2.1 Exterior derivative

It is possible to define a natural differential operator on smooth forms, which generalizes the differential of a function, but first it is necessary to analyze the Euclidean case.

If $\omega = \sum_J' \omega_J dx^J$ is a smooth k -form on $U \in \mathcal{T}(\mathbb{R}^n)$, its **exterior derivative** $d\omega$ is the $(k+1)$ -form defined as:

$$d\omega := \sum_J' d\omega_J \wedge dx^J = \sum_J' \sum_{i=1}^n \frac{\partial \omega_J}{\partial x^i} dx^i \wedge dx^J \quad (2.2.22)$$

where $d\omega_J$ is the differential of the component function ω_J . On $\mathbb{R}$ -valued functions (0-forms), the exterior derivative reduces to the differential.

Lemma 2.2.6 (Properties of the Euclidean exterior derivative)

- (1) d is linear over $\mathbb{R}$;
- (2) if $\omega \in \Omega^k(U)$ and $\eta \in \Omega^l(U)$, with $U \in \mathcal{T}(\mathbb{R}^n)$, then:

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta \quad (2.2.23)$$

- (3) $d \circ d \equiv 0$;

- (4) d commutes with pull-backs, i.e. given a smooth map $f : U \rightarrow V$, with $U \in \mathcal{T}(\mathbb{R}^n)$ and $V \in \mathcal{T}(\mathbb{R}^m)$, and $\omega \in \Omega^k(V)$, then:

$$f^*(d\omega) = d(f^*\omega) \quad (2.2.24)$$

Proof. Linearity over $\mathbb{R}$ is trivial from definition. To prove (2), by linearity it suffices to consider $\omega = u dx^I \in \Omega^k(U)$ and $\eta = v dx^J \in \Omega^l(U)$, for smooth functions u and v :

$$\begin{aligned} d(\omega \wedge \eta) &= d(uv dx^I \wedge dx^J) = (u dv + v du) \wedge dx^I \wedge dx^J \\ &= (du \wedge dx^I) \wedge (v dx^J) + (-1)^k (u dx^I) \wedge (dv \wedge dx^J) \end{aligned}$$

where the $(-1)^k$ comes from $dv \wedge dx^I = (-1)^k dx^I \wedge dv$, since dx^I is a k -form and dv is a 1-form. WTS $d(u dx^I) = du \wedge dx^I$ for any multi-index I , not just increasing ones: if I has repeated indices, then $d(u dx^I) = 0 = du \wedge dx^I$, otherwise let σ be a permutation of I which sends it to an increasing multi-index J , so that:

$$d(u dx^I) = (\text{sgn } \sigma) d(u dx^J) = (\text{sgn } \sigma) du \wedge dx^J = du \wedge dx^I$$

Inserting this into the above expression yields (2). Now, consider (3) in the case of a 0-form:

$$d(du) = d\left(\frac{\partial u}{\partial x^i} dx^i\right) = \frac{\partial^2 u}{\partial x^i \partial x^j} dx^i \wedge dx^j = 0$$

which vanishes since the first factor is symmetric under $i \leftrightarrow j$ by the Schwartz lemma on smooth functions $\partial_i \partial_j u = \partial_j \partial_i u$, while the second factor is antisymmetric under $i \leftrightarrow j$ by definition. In

the general case:

$$d(d\omega) = d \sum_J' d\omega_J \wedge dx^J = \sum_J' d(d\omega_J) \wedge dx^J + \sum_J' \sum_{i=1}^k (-1)^i dx^{j_1} \wedge \cdots \wedge d(dx^{j_i}) \wedge \cdots \wedge dx^{j_k}$$

The first term vanishes since ω_J is a 0-form, while the second vanishes as $\partial_i 1 = 0$. Finally, to prove (4), again it suffices to consider $\omega = u dx^{i_1} \wedge \cdots \wedge dx^{i_k}$:

$$f^*(d(u dx^I)) = f^*(du \wedge dx^I) = d(u \circ f) \wedge d(x^{i_1} \circ f) \wedge \cdots \wedge d(x^{i_k} \circ f)$$

$$d(f^*(u dx^I)) = d((u \circ f) d(x^{i_1} \circ f) \wedge \cdots \wedge d(x^{i_k} \circ f))$$

which are equal. □

Usually $d \circ d \equiv d^2$. It is now possible to extend the definition of exterior derivative to manifolds.

Theorem 2.2.1 (Existence and uniqueness of exterior derivative)

Let $\mathcal{M}$ be a smooth n -manifold. There exists a unique operator $d : \Omega^k(\mathcal{M}) \rightarrow \Omega^{k+1}(\mathcal{M})$ for all $k = 0, \dots, n-1$, called **exterior derivative**, which satisfies:

- (1) d is linear over $\mathbb{R}$;
- (2) if $\omega \in \Omega^k(\mathcal{M})$ and $\eta \in \Omega^l(\mathcal{M})$, then:

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta \quad (2.2.25)$$

- (3) $d^2 \equiv 0$;
- (4) given $f \in \mathcal{C}^\infty(\mathcal{M}) = \Omega^0(\mathcal{M})$, then df is the differential of f , defined by $df(X) = X(f)$;
- (5) in any smooth chart $(U, (x^i))$ on $\mathcal{M}$, given $\omega \in \Omega^k(\mathcal{M})$:

$$d\omega = \sum_J' d\omega_J \wedge dx^J \quad (2.2.26)$$

Proof. To prove existence, suppose $\omega \in \Omega^k(\mathcal{M})$, and define $d\omega$ by Eq. 2.2.26 in each smooth chart on $\mathcal{M}$. More precisely, in each smooth chart (U, φ) on $\mathcal{M}$, set:

$$d\omega = \varphi^* d((\varphi^{-1})^* \omega)$$

To see that this is well-defined, note that, for any other smooth chart (V, ψ) , the map $\varphi \circ \psi^{-1}$ is a diffeomorphism $V \rightarrow U$, hence by (4) of Lemma 2.2.6:

$$(\varphi \circ \psi^{-1})^* d((\varphi^{-1})^* \omega) = d((\varphi \circ \psi^{-1})^* (\varphi^{-1})^* \omega)$$

Together with $(\varphi \circ \psi^{-1})^* = (\psi^{-1})^* \circ \varphi^*$, this implies $\varphi^* d((\varphi^{-1})^* \omega) = \psi^* d((\psi^{-1})^* \omega)$, so $d\omega$ is well-defined, and it satisfies (1)-(4) by Prop. 2.2.6.

To prove uniqueness, suppose d is any operator satisfying (1)-(4). It can be proved that $d\omega$ is determined locally, i.e. if ω_1 and ω_2 are k -forms which agree on $U \in \mathcal{T}(\mathcal{M})$, then $d\omega_1 = d\omega_2$ on U . With this, let $\omega \in \Omega^k(\mathcal{M})$ arbitrary, and let (U, φ) be any smooth coordinate chart on $\mathcal{M}$: then, ω can be written on U as $\omega = \sum_I' \omega_I dx^I$. For any $p \in U$, it is possible to construct global smooth

functions $\tilde{\omega}_I$ and $\tilde{x}^i$ on $\mathcal{M}$ which agree with ω_I and x^i in a neighbourhood of p . Using (1)-(4) and the preceding observation, it follows that d is given by Eq. 2.2.26 at p . Since p is arbitrary, then this d must be equal to the one defined above. $\square$

If $\mathcal{A} = \bigoplus_{k \in \mathcal{I}} \mathcal{A}_k$ is an $\mathcal{I}$ -graded algebra, a linear map $T : \mathcal{A} \rightarrow \mathcal{A}$ is a *map of degree m* if $T(\mathcal{A}_k) \subseteq \mathcal{A}_{m+k} \forall k \in \mathcal{I}$, and it is an *antiderivation* if $T(xy) = T(x)y + (-1)^k xT(y) \forall x \in \mathcal{A}_k, y \in \mathcal{A}_l$. Therefore, in the language of graded algebras, d is an antiderivation on $\Omega(\mathcal{M})$ of degree $+1$ whose square is zero, while ι_X is an antiderivation of $\Omega(\mathcal{M})$ of degree -1 whose square is zero, for $X \in \mathfrak{X}(\mathcal{M})$.

Note that (4) of Lemma 2.2.6 holds for the general exterior derivative too. If $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ is a smooth map, then for each k the pull-back map $f^* : \Omega^k(\mathcal{M}_2) \rightarrow \Omega^k(\mathcal{M}_1)$ commutes with d , i.e. for all $\omega \in \Omega^k(\mathcal{M}_2)$:

$$f^*(d\omega) = d(f^*\omega) \quad (2.2.27)$$

The exterior derivative can be thought, in a certain sense, as the dual of the Lie bracket, as knowing all the Lie brackets of basis vector fields in a smooth local frame allows to compute the exterior derivatives of dual covector fields, and vice versa.

Proposition 2.2.3 (Duality between exterior derivative and Lie brackets)

Let $\mathcal{M}$ be a smooth n -manifold, let (E_i) be a smooth local frame on $\mathcal{M}$ and let (ε^i) be its dual coframe. For each $i = 1, \dots, n$, let b_{jk}^i and c_{jk}^i be defined so that:

$$d\varepsilon^i = \sum_{j < k=1}^n b_{jk}^i \varepsilon^j \wedge \varepsilon^k \quad [E_j, E_k] = c_{jk}^i E_i$$

Then, $b_{jk}^i = -c_{jk}^i$.

With Prop. 2.2.3, it is possible to prove an invariant expression for the exterior derivative of $\omega \in \Omega^k(\mathcal{M})$, in addition to the coordinate formula Eq. 2.2.26: for any $X_1, \dots, X_{k+1} \in \mathfrak{X}(\mathcal{M})$:

$$\begin{aligned} d\omega(X_1, \dots, X_{k+1}) &= \sum_{1 \leq i \leq k+1} (-1)^{i-1} X_i(\omega(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_{k+1})) + \\ &+ \sum_{1 \leq i < j \leq k+1} (-1)^{i+j} \omega([X_i, X_j], X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_{j-1}, X_{j+1}, \dots, X_{k+1}) \end{aligned} \quad (2.2.28)$$

The restriction of this formula to 1-forms is particularly useful:

$$d\omega(X, Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y]) \quad (2.2.29)$$

§2.2.2.2 Lie derivatives of differential forms

Eq. 1.6.19 applies to differential forms too. However, the Lie derivative takes a particularly simple form on $\Omega(\mathcal{M})$ due to the exterior derivative.

Theorem 2.2.2 (Cartan's theorem)

Let $\mathcal{M}$ be a smooth n -manifold. Then, for any $V \in \mathfrak{X}(\mathcal{M})$ and $\omega \in \Omega(\mathcal{M})$:

$$\mathcal{L}_V \omega = (\iota_V \circ d + d \circ \iota_V) \omega \quad (2.2.30)$$

Proof. To show the thesis for $\Omega^k(\mathcal{M})$, with $k = 0, \dots, n$, use induction on k . First, consider a 0-form $f \in \mathcal{C}^\infty(\mathcal{M})$:

$$\iota_V(df) + d(\iota_V f) = \iota_V df = df(V) = V(f) = \mathcal{L}_V f$$

Now, let $k \geq 1$, and suppose that Eq. 2.2.30 holds for all forms of degree less than k . Let $\omega \in \Omega^k(\mathcal{M})$ arbitrary, which in smooth local coordinates is $\omega = \sum_I \omega_I dx^I$, and write e.g. $u = x^{i_1}$ and $\beta = \omega_I dx^{i_2} \wedge \dots \wedge dx^{i_k}$. Then, it is clear that each term of the sum can be written as $du \wedge \beta$, for some $u \in \mathcal{C}^\infty(\mathcal{M})$ and $\beta \in \Omega^{k-1}(\mathcal{M})$.

Lemma 2.2.7

Let $\mathcal{M}$ be a smooth manifold, $V \in \mathfrak{X}(\mathcal{M})$ and $\omega, \eta \in \Omega(\mathcal{M})$. Then:

$$\mathcal{L}_V(\omega \wedge \eta) = (\mathcal{L}_V \omega) \wedge \eta + \omega \wedge (\mathcal{L}_V \eta) \quad (2.2.31)$$

Since on smooth functions $\mathcal{L}_V du = d(\mathcal{L}_V u) = d(V(u))$, by the inductive hypothesis and Lemma 2.2.7:

$$\mathcal{L}_V(du \wedge \beta) = (\mathcal{L}_V du) \wedge \beta + du \wedge \mathcal{L}_V \beta = d(V(u)) \wedge \beta + du \wedge (\iota_V d\beta + d(\iota_V \beta))$$

On the other hand, as both d and ι_V are antiderivations on $\Omega(\mathcal{M})$ and $\iota_V du = du(V) = V(u)$:

$$\begin{aligned} \iota_V \circ d(du \wedge \beta) + d \circ \iota_V(du \wedge \beta) \\ = \iota_V(-du \wedge d\beta) + d(V(u)\beta - du \wedge \iota_V \beta) \\ = -V(u)d\beta + du \wedge \iota_V d\beta + d(V(u)) \wedge \beta + V(u)d\beta + du \wedge d(\iota_V \beta) \end{aligned}$$

After cancellations, this expression is equal to the above one, proving the thesis. $\square$

Since $d^2 = 0$, the Lie derivative commutes with the exterior derivative: $\mathcal{L}_V(d\omega) = d(\mathcal{L}_V \omega)$.

§2.3 Integration

§2.3.1 Orientation

In order to properly define integration on manifolds, it is first necessary to introduce the notion of orientation.

Definition 2.3.1 (Consistent orientation)

Let V be an n -dimensional $\mathbb{R}$ -vector space, with $n \geq 1$. Two ordered bases $(E_1, \dots, E_n)$ and $(\tilde{E}_1, \dots, \tilde{E}_n)$ of V are said to be **consistently oriented** if the transition matrix $[B_i^j] \in \mathbb{R}^{n \times n} : E_i = B_i^j \tilde{E}_j$ has positive determinant.

Being consistently oriented is an equivalence relation on the set of all ordered bases of V : it is clearly reflexive (as $\det I_n = 1 > 0$), symmetric (as $\det B^{-1} = (\det B)^{-1}$) and transitive (by Binet's theorem). Moreover, the equivalence classes are only two, since the sign of the transition matrix between two bases is either positive or negative: these two equivalence classes are the two possible **orientations** of V . Given an ordered basis $(E_1, \dots, E_n)$ of V , the orientation determined by this basis (i.e. its equivalence class) is denoted as $[E_1, \dots, E_n]$, while the opposite orientation as $-[E_1, \dots, E_n]$. In

the case of 0-dimensional vector spaces, the orientation is simply defined to be a choice of one of the numbers ± 1 .

Proposition 2.3.1

Let V be an n -dimensional $\mathbb{R}$ -vector space. Each $\omega \in \bigwedge^n(V) : \omega \neq 0$ determines an orientation $\mathcal{O}_\omega$ of V as follows:

- if $n \geq 1$, then $\mathcal{O}_\omega$ is the set of ordered bases $(E_1, \dots, E_n) : \omega(E_1, \dots, E_n) > 0$;
- if $n = 0$, then $\mathcal{O}_\omega$ is $+1$ if $\omega > 0$ and -1 if $\omega < 0$.

Two non-zero exterior n -forms determine the same orientation if and only if each is a positive multiple of the other.

Proof. The 0-dimensional case is trivial, since $\bigwedge^0(V) = \mathbb{R}$. If $n \geq 1$, let $\omega \in \bigwedge^n(V) - \{0\}$, and let $\mathcal{O}_\omega$ denote the set of ordered bases $(E_1, \dots, E_n) : \omega(E_1, \dots, E_n) > 0$. WTS $\mathcal{O}_\omega$ is exactly one equivalence class.

Suppose (E_i) and $(\tilde{E}_i)$ are any two ordered bases of V , and let $B : V \rightarrow V$ be a linear map sending $E_i \mapsto \tilde{E}_i$, i.e. $\tilde{E}_j = B_j^i E_i$. By Eq. 2.2.6:

$$\omega(\tilde{E}_1, \dots, \tilde{E}_n) = (\det B)\omega(E_1, \dots, E_n)$$

Therefore, if $(\tilde{E}_i)$ is consistently oriented with $(E_1, \dots, E_n)$, i.e. $\det B > 0$, then $\omega(\tilde{E}_1, \dots, \tilde{E}_n)$ and $\omega(E_1, \dots, E_n)$ have the same sign, i.e. $\mathcal{O}_\omega$ is one equivalence class. $\square$

Since $\dim_{\mathbb{R}} \bigwedge^n(V) = 1$ for any n -dimensional $\mathbb{R}$ -vector space V , choosing an orientation of V is equivalent to choosing one of the two disconnected components of $\bigwedge^n(V) - \{0\}$, which are called the *positive* and *negative orientations*.

The concept of orientation is trivially imported on smooth manifolds in a pointwise manner: the *pointwise orientation* of a smooth manifold is a choice of orientation of each tangent space. However, this concept is only useful if nearby tangent spaces have consistent orientations. Thus, if $\mathcal{M}$ is a smooth n -manifold, a local frame (E_i) on $U \in \mathcal{T}(\mathcal{M})$ is **(positively) oriented** if $(E_1|_p, \dots, E_n|_p)$ is a positively oriented basis of $T_p\mathcal{M}$ at each point $p \in U$. Negatively oriented frames are defined analogously. A pointwise orientation is said to be *continuous* if every point of $\mathcal{M}$ is in the domain of an oriented local frame (since vector fields which determine a local frame are continuous by definition).

Definition 2.3.2 (Oriented manifold)

Let $\mathcal{M}$ be a smooth n -manifold. An **orientation** $\mathcal{O}$ of $\mathcal{M}$ is a continuous pointwise orientation, and $(\mathcal{M}, \mathcal{O})$ is said to be an **oriented manifold**.

If such a pointwise orientation does not exist, $\mathcal{M}$ is called a *non-orientable manifold*. Given the chosen orientation $\mathcal{O}$, the orientation of $T_p\mathcal{M}$ is denoted by $\mathcal{O}_p$, for each $p \in \mathcal{M}$.

Proposition 2.3.2 (Orientation and top forms)

Let $\mathcal{M}$ be a smooth n -manifold. Any nowhere-vanishing n -form ω determines a unique orientation on $\mathcal{M}$ for which ω is positively oriented at each point. Conversely, if $\mathcal{M}$ has a given orientation, then there is $\omega \in \Omega^n(\mathcal{M}) : \omega \neq 0$ that is positively oriented at each point.

Proof. Let ω be a nowhere-vanishing n -form on $\mathcal{M}$. By Prop. 2.3.1, ω defines a pointwise orientation on $\mathcal{M}$, hence only continuity needs to be checked. The case $n = 0$ is trivial, so assume $n \geq 1$. Given $p \in \mathcal{M}$, let (E_i) be a local frame on a connected neighbourhood U of p , and let (ε^i) be the dual coframe. Then, $\omega = f \varepsilon^1 \wedge \cdots \wedge \varepsilon^n$ on U , for some continuous $f : \mathcal{M} \rightarrow \mathbb{R}$. Since ω is nowhere-vanishing, f is nowhere-vanishing too, hence:

$$\omega(E_1, \dots, E_n) = f \neq 0$$

at all points of U . By the connectedness of U , this means that f is either always positive or always negative on U , therefore the given frame is either positively or negatively oriented. If negatively, it suffices to replace E_1 with $-E_1$ to get a positively oriented frame. Thus, the pointwise orientation determined by ω is continuous.

Conversely, suppose $\mathcal{M}$ is oriented, and let $\Lambda_+^n(\mathcal{M}) \subseteq \Lambda^n(\mathcal{M})$ be the open subset of positively oriented n -forms at all points of $\mathcal{M}$. Then, at each $p \in \mathcal{M}$, the intersection of $\Lambda_+^n(\mathcal{M})$ with the fiber $\bigwedge^n(T_p\mathcal{M})$ is an open half-line, and therefore convex. Thus, with a suitable partition of unity, there exists a smooth global section of $\Lambda_+^n(\mathcal{M})$, i.e. a positively oriented smooth global n -form. $\square$

Because of Prop. 2.3.2, any nowhere-vanishing n -form on $\mathcal{M}$ is called an *orientation form*, and if it determines the given orientation on $\mathcal{M}$ it is said to be *(positively) oriented*. Clearly, if $\omega, \tilde{\omega} \in \Omega^n(\mathcal{M})$ are both positively oriented, then $\tilde{\omega} = f\omega$ for some strictly positive $f \in C^\infty(\mathcal{M})$.

The same analysis can be made with charts. A smooth coordinate chart (x^i) on an oriented smooth manifold is said to be *(positively) oriented* if the coordinate frame (∂_i) is positively oriented, and idem for a negatively oriented one. A smooth atlas $\{(U_\alpha, \varphi_\alpha)\}$ is said to be *consistently oriented* if $\det \mathcal{J}_{\varphi_\beta \circ \varphi_\alpha^{-1}} > 0$ on $\varphi_\alpha(U_\alpha \cap U_\beta)$, for each α, β .

Proposition 2.3.3 (Orientation and coordiante atlases)

Let $\mathcal{M}$ be a smooth n -manifold, with $n \geq 1$. Given a consistently oriented smooth atlas of $\mathcal{M}$, there is a unique orientation of $\mathcal{M}$ such that each chart in the atlas is positively oriented. Conversely, if $\mathcal{M}$ is oriented and either $\partial\mathcal{M} = \emptyset$ or $n > 1$, then the collection of all oriented smooth charts is a consistently oriented atlas of $\mathcal{M}$.

Proof. First, suppose $\mathcal{M}$ has a consistently oriented smooth atlas. Each chart determines a pointwise orientation at each point of its domain, and wherever two charts overlap, the transition matrix between their respective coordinate frames is the Jacobian matrix of the transition map, which has positive determinant by hypothesis, hence they determine the same orientation at each point. The orientation thus determined is continuous since each point is in the domain of an oriented coordinate frame.

Conversely, assume $\mathcal{M}$ is oriented: each point is in the domain of a smooth chart, and if the chart is negatively oriented the replacement $x^1 \mapsto -x^1$ turns it into a positively oriented chart. Since all these charts are positively oriented, their transition maps are guaranteed to have positive Jacobian determinants, hence they form a consistently oriented atlas. Note that the replacement $x^1 \mapsto -x^1$ cannot happen in a 1-manifold with boundary, since by Eq. 1.1.1 the last coordinate of boundary charts must be non-negative, hence either $\partial\mathcal{M} = \emptyset$ or $\dim_{\mathbb{R}} \mathcal{M} > 1$ is required. $\square$

If $\mathcal{M}_1, \dots, \mathcal{M}_k$ are orientable smooth manifolds with orientation forms ω_i on $\mathcal{M}_i$, for each $i = 1, \dots, k$, then it is possible to show that the product manifold $\mathcal{M}_1 \times \cdots \times \mathcal{M}_k$ has a unique orientation, and the orientation form is given by $\pi_1^* \omega_1 \wedge \cdots \wedge \pi_k^* \omega_k$, where $\pi_i : \mathcal{M}_1 \times \cdots \times \mathcal{M}_k \rightarrow \mathcal{M}_i$ is the projection on the i^{th} component.

Proposition 2.3.4 (Connected manifolds)

Let $\mathcal{M}$ be a connected orientable smooth manifold. Then, $\mathcal{M}$ has exactly two orientations, and if two orientations of $\mathcal{M}$ agree at one point, they are equal.

Proposition 2.3.5 (Codimension-0 submanifolds)

Let $\mathcal{M}$ be an oriented smooth manifold, and let $D \subseteq \mathcal{M}$ be a codimension-0 smooth submanifold. Then, the orientation of $\mathcal{M}$ restricts to an orientation of D , and if ω is an orientation form on $\mathcal{M}$, then $\iota_D^* \omega$ is an orientation form on D .

Let $\mathcal{M}_1, \mathcal{M}_2$ be smooth n -manifolds, with $n \geq 1$, and let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a local diffeomorphism⁴: f is **orientation-preserving** if, for each $p \in \mathcal{M}_1$, the isomorphism $df_p : T_p \mathcal{M}_1 \rightarrow T_{f(p)} \mathcal{M}_2$ takes oriented bases to oriented bases, while it is **orientation-reversing** if it takes oriented bases to negatively oriented bases. If $n = 0$, instead, f is orientation-preserving if p and $f(p)$ have the same orientation, and orientation-reversing if they have opposite orientation. The composition of orientation-preserving maps is orientation preserving.

Proposition 2.3.6 (Pull-back orientation)

Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a local diffeomorphism between smooth manifolds. If $\mathcal{M}_2$ is oriented, then $\mathcal{M}_1$ has a unique orientation, called the **pull-back orientation** by f , such that f is orientation-preserving.

Proof. For each $p \in \mathcal{M}_1$, there is a unique orientation on $T_p \mathcal{M}_1$ that makes the isomorphism $df_p : T_p \mathcal{M}_1 \rightarrow T_{f(p)} \mathcal{M}_2$ orientation-preserving, thus defining a pointwise orientation on $\mathcal{M}_1$. To see that this is continuous, choose a smooth orientation form $\omega \in \Omega^n(\mathcal{M}_2)$ and note that $f^* \omega$ is a smooth orientation form on $\mathcal{M}_1$: since this pointwise orientation on $\mathcal{M}_1$ is continuous, it is the unique orientation on $\mathcal{M}_1$ with respect to which f is orientation-preserving. $\square$

If $\mathcal{O}$ is the given orientation on $\mathcal{M}_2$, the pull-back orientation on $\mathcal{M}_1$ is denoted by $f^* \mathcal{O}$. It can be shown that if $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ and $g : \mathcal{M}_2 \rightarrow \mathcal{M}_3$ are local diffeomorphisms and $\mathcal{O}$ is an orientation on $\mathcal{M}_3$, then the pull-back orientation on $\mathcal{M}_1$ is $(g \circ f)^* \mathcal{O} = f^*(g^* \mathcal{O})$.

Proposition 2.3.7 (Parallelizability and orientation)

Every parallelizable smooth manifold is orientable.

Proof. Suppose $\mathcal{M}$ is parallelizable, i.e. it admits a global smooth frame $(E_1, \dots, E_n)$. Define a pointwise orientation on $\mathcal{M}$ by setting $(E_1|_p, \dots, E_n|_p)$ to be positively oriented at each $p \in \mathcal{M}$: this is clearly continuous, since every $p \in \mathcal{M}$ is in the domain of the global oriented frame (E_i) , hence it is an orientation on $\mathcal{M}$. $\square$

§2.3.1.1 Orientation of hypersurfaces

If $\mathcal{M}$ is an oriented smooth manifold and S is a smooth submanifold of $\mathcal{M}$, then S might not inherit an orientation from $\mathcal{M}$, even if S is embedded: indeed, it is not sufficient to restrict an orientation form from $\mathcal{M}$ to S , since the restriction of a top form to a lower-dimensional manifold must necessarily vanish (and the case of codimension-0 submanifolds is treated in Prop. 2.3.5). However, with

⁴A map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ between smooth manifolds is a *local diffeomorphism* if every $p \in \mathcal{M}_1$ has a neighbourhood $U \in \mathcal{T}(\mathcal{M}_1)$ such that $f(U) \in \mathcal{T}(\mathcal{M}_2)$ and $f|_U : U \rightarrow f(U)$ is a diffeomorphism.

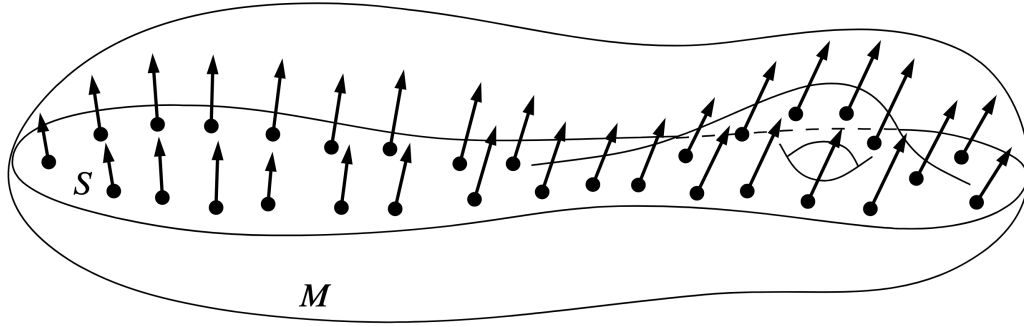


Figure 2.1: A vector field along a submanifold.

an additional restriction, it is possible to induce an orientation on a hypersurface (a codimension-1 submanifold).

Let $\mathcal{M}$ be a smooth manifold and $S \subseteq \mathcal{M}$ an immersed or embedded submanifold. A *vector field along S* is a section of the ambient tangent bundle $T\mathcal{M}|_S$ (see Fig. 2.1), i.e. a continuous map $N : S \rightarrow T\mathcal{M}$ such that $N_p \in T_p\mathcal{M}$ for each $p \in S$ (do not confuse this with vector fields on S , i.e. sections of TS). In general, any vector field on $\mathcal{M}$ restricts to a vector field along S .

Proposition 2.3.8 (Orientation of hypersurface)

Let $\mathcal{M}$ be an oriented smooth n -manifold, $S \subseteq \mathcal{M}$ an immersed hypersurface and N a vector field along S that is nowhere tangent to S . Then, S has a unique orientation such that, for each $p \in S$, $(E_1, \dots, E_{n-1})$ is an oriented basis of $T_p S$ if and only if $(N_p, E_1, \dots, E_{n-1})$ is an oriented basis of $T_p \mathcal{M}$. If ω is an orientation form on $\mathcal{M}$, then $\iota_S^*(\iota_N \omega)$ is an orientation form on S with respect to this orientation.

Proof. Let ω be an orientation form on $\mathcal{M}$. Then, $\sigma = \iota_S^*(\iota_N \omega)$ is an $(n-1)$ -form on S , as the pull-back ι_S^* is just a restriction on vectors tangent to S . If σ is nowhere vanishing, then it is an orientation form on S . To show this, consider a basis $(E_1, \dots, E_{n-1})$ of $T_p S$: since N is nowhere tangent to S , then $(N_p, E_1, \dots, E_{n-1})$ is a basis of $T_p \mathcal{M}$. The fact that ω is non-vanishing implies:

$$\sigma_p(E_1, \dots, E_{n-1}) = \omega_p(N_p, E_1, \dots, E_{n-1}) \neq 0$$

Moreover, since $\sigma_p(E_1, \dots, E_{n-1})$ and $\omega_p(N_p, E_1, \dots, E_{n-1})$ are equal, they have the same sign, hence σ is the desired orientation form. $\square$

Example 2.3.1 (Standard orientation on spheres)

$\mathbb{S}^n$ is a hypersurface in $\mathbb{R}^{n+1}$ to which the vector field $N = x^i \partial_i$ is nowhere tangent, so by Prop. 2.3.8 N induces an orientation on $\mathbb{S}^n$, called the **standard orientation**: this shows that all spheres are orientable.

Not every hypersurface admits a nowhere tangent vector field, however a sufficient condition holds in many cases.

Lemma 2.3.1 (Orientation of regular level set)

Let $\mathcal{M}$ be an oriented smooth manifold, and let $S \subseteq \mathcal{M}$ be a regular level set of a smooth function $f : \mathcal{M} \rightarrow \mathbb{R}$. Then, S is orientable.

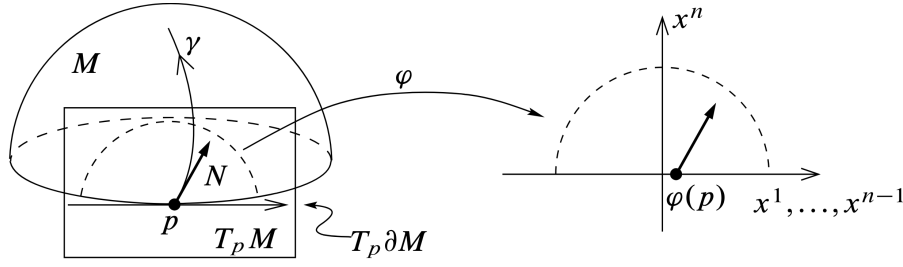


Figure 2.2: Inward-pointing vector.

Proof. Choose any Riemannian metric on $\mathcal{M}$, which always exists by Th. 2.1.1, and let $N = \nabla f|_S$. The hypotheses imply that N is a nowhere tangent vector field along S , so the result follows from Prop. 2.3.8. $\square$

§2.3.1.2 Orientation of boundaries

Recall that, if $\mathcal{M}$ is a smooth manifold with boundary, then $\partial\mathcal{M}$ is an embedded hypersurface in $\mathcal{M}$. Given $p \in \mathcal{M}$, vectors in $T_p\mathcal{M}$ can be separated into three classes: those tangent to the boundary, those pointing inward and those pointing outward. Formally, if $p \in \mathcal{M}$, a vector $v \in T_p\mathcal{M} - T_p\partial\mathcal{M}$ is said to be *inward-pointing* if for some $\varepsilon > 0$ there exists a smooth curve $\gamma : [0, \varepsilon) \rightarrow \mathcal{M}$ such that $\gamma(0) = p$ and $\gamma'(0) = v$, and it is *outward-pointing* if such curve has domain $(-\varepsilon, 0]$ (see Fig. 2.2).

Lemma 2.3.2 (Characterization of tangent vectors to a boundary)

Let $\mathcal{M}$ be a smooth n -manifold with boundary, $p \in \partial\mathcal{M}$ and (x^i) any smooth boundary coordinates in a neighbourhood of p . The inward-pointing vectors in $T_p\mathcal{M}$ are precisely those with positive x^n -component, the outward-pointing ones those with negative x^n -component and the ones tangent to $\partial\mathcal{M}$ those with zero x^n -component.

Therefore, $T_p\mathcal{M}$ with $p \in \mathcal{M}$ is the disjoint union of $T_p\partial\mathcal{M}$, of the set of inward-pointing vectors and the set of outward-pointing vectors. Moreover, $v \in T_p\mathcal{M}$ is inward-pointing if and only if $-v$ is outward-pointing. Since an outward-pointing vector field is nowhere tangent to $\partial\mathcal{M}$, it determines an orientation on $\partial\mathcal{M}$.

Proposition 2.3.9 (Induced orientation on boundary)

Let $\mathcal{M}$ be a smooth oriented n -manifold with boundary, with $n \geq 1$. Then, $\partial\mathcal{M}$ is orientable, and all outward-pointing vector fields along $\partial\mathcal{M}$ determine the same orientation on $\partial\mathcal{M}$, known as the *induced* or **Stokes orientation**.

Proof. Let ω be an orientation form on $\mathcal{M}$, and let N be a smooth outward-pointing vector field along $\partial\mathcal{M}$. The $(n-1)$ -form $\iota_{\partial\mathcal{M}}^*(\iota_N\omega)$ is an orientation on $\partial\mathcal{M}$ by Prop. 2.3.8, so $\partial\mathcal{M}$ is orientable.

WTS this orientation is independent of the choice of N . Let $p \in \mathcal{M}$, and let (x^i) be smooth boundary coordinates on M on a neighbourhood of p . If N and $\tilde{N}$ are two outward-pointing vector fields along $\partial\mathcal{M}$, the last components $N^n(p)$ and $\tilde{N}^n(p)$ are both negative, hence the transition matrix between bases $(N_p, \partial_1|_p, \dots, \partial_{n-1}|_p)$ and $(\tilde{N}_p, \partial_1|_p, \dots, \partial_{n-1}|_p)$ of $T_p\mathcal{M}$ has determinant $N^n(p)/\tilde{N}^n(p) > 0$. Therefore, both bases determine the same orientation on $T_p\mathcal{M}$, so N and $\tilde{N}$ determine the same orientation on $T_p\partial\mathcal{M}$. $\square$

The most useful way of describing submanifolds is by means of local parametrizations.

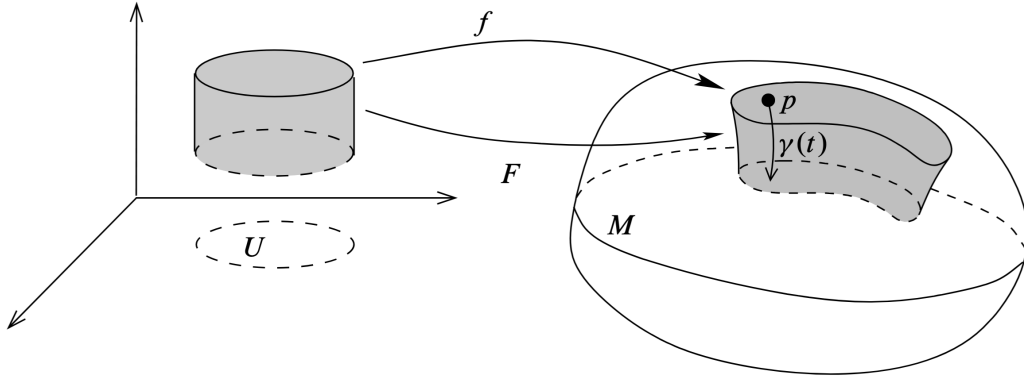


Figure 2.3: Orientation criterion for a boundary parametrization.

Lemma 2.3.3 (Orientation-preserving local parametrization)

Let $\mathcal{M}$ be a smooth oriented n -manifold with boundary. Suppose $U \in \mathcal{T}(\mathbb{R}^{n-1})$ and $F : (a, b] \times U \rightarrow \mathcal{M}$ is a smooth embedding, with $a, b \in \mathbb{R} : a < b$, such that $F : \{b\} \times U \rightarrow \partial\mathcal{M}$ is an embedding. Then, the parametrization $f : U \rightarrow \partial\mathcal{M}$ given by $f(x) = F(b, x)$ is orientation-preserving for $\partial\mathcal{M}$ if and only if f is orientation-preserving for $\mathcal{M}$.

Proof. Let $x \in U$ arbitrary, and let $p = f(x) = F(b, x) \in \partial\mathcal{M}$ (see Fig. 2.3). Since F is an embedding, the linear map $dF_{(b,x)} : (T_b\mathbb{R} \oplus T_x\mathbb{R}^{n-1}) \rightarrow T_p\mathcal{M}$ is bijective. The restriction of $dF_{(b,x)}$ to $T_x\mathbb{R}^{n-1}$ reduces to $df_x : T_x\mathbb{R}^{n-1} \rightarrow T_p\partial\mathcal{M}$, which is already bijective, hence $dF(\frac{\partial}{\partial s}|_{(b,x)}) \notin T_p\partial\mathcal{M}$, where s is the coordinate on $(a, b]$. Then, define the smooth curve $\gamma : [0, \varepsilon) \rightarrow \mathcal{M}$ as:

$$\gamma(t) = F(b - t, x)$$

This curve satisfies $\gamma(0) = p$ and $\gamma'(0) = dF(\frac{\partial}{\partial s}|_{(b,x)}) \notin T_p\partial\mathcal{M}$, therefore $-dF(\frac{\partial}{\partial s}|_{(b,x)})$ is inward-pointing, so $dF(\frac{\partial}{\partial s}|_{(b,x)})$ is outward-pointing. F is orientation-preserving for $\mathcal{M}$ if and only if $(dF(\frac{\partial}{\partial s}), dF(\frac{\partial}{\partial x^1}), \dots, dF(\frac{\partial}{\partial x^{n-1}}))$ is oriented in $T\mathcal{M}$, i.e. (by Prop. 2.3.9) if and only if $(dF(\frac{\partial}{\partial x^1}), \dots, dF(\frac{\partial}{\partial x^{n-1}}))$ is oriented in $T\partial\mathcal{M}$: this is true if and only if $(df(\frac{\partial}{\partial x^1}), \dots, df(\frac{\partial}{\partial x^{n-1}}))$ is oriented in $T\partial\mathcal{M}$, i.e. if and only if f is orientation-preserving for $\partial\mathcal{M}$. $\square$

Example 2.3.2 (Spherical coordinates)

Spherical coordinates yield a smooth local parametrization of $\mathbb{S}^2$ as follows. Let $U = (0, \pi) \times (0, 2\pi) \subseteq \mathbb{R}^2$ and $X : U \rightarrow \mathbb{R}^3$ defined as:

$$X(\varphi, \vartheta) = (\sin \varphi \cos \vartheta, \sin \varphi \sin \vartheta, \cos \varphi)$$

To check whether this preserves or reverses orientation, notice that X is the restriction of the 3-dimensional spherical coordinate parametrization $f : (0, 1] \times U \rightarrow \mathbb{D}^3$ defined as:

$$f(\varrho, \varphi, \vartheta) = (\varrho \sin \varphi \cos \vartheta, \varrho \sin \varphi \sin \vartheta, \varrho \cos \varphi)$$

Clearly $X(\varphi, \vartheta) = f(1, \varphi, \vartheta)$ and, by direct computation, $\mathcal{J}_f(\varrho, \varphi, \vartheta) = \varrho^2 \sin \varphi$ is positive on $(0, 1] \times U$. Therefore, by Lemma 2.3.3, X is orientation-preserving.

§2.3.1.3 Riemannian volume form

Let $(\mathcal{M}, g)$ be a Riemannian n -manifold, with $n \leq 1$. By the Gram–Schmidt algorithm, there is a smooth orthonormal frame $(E_1, \dots, E_n)$ in a neighbourhood of each point of $\mathcal{M}$, therefore it is always possible to find an oriented orthonormal frame in a neighbourhood of each point (replacing $E_1 \mapsto -E_1$ if necessary).

Proposition 2.3.10 (Riemannian volume form)

Let $(\mathcal{M}, g)$ be an oriented Riemannian n -manifold, with $n \geq 1$. There is a unique smooth orientation for $\omega_g \in \Omega^n(\mathcal{M})$, called the **Riemannian volume form**, such that:

$$\omega_g(E_1, \dots, E_n) = 1 \quad (2.3.1)$$

for every local oriented orthonormal frame (E_i) on $\mathcal{M}$.

Proof. First, suppose such a form $\omega_g \in \Omega^n(\mathcal{M})$ exists. If $(E_1, \dots, E_n)$ is any local oriented orthonormal frame on $U \in \mathcal{T}(\mathcal{M})$ and $(\varepsilon^1, \dots, \varepsilon^n)$ is its dual frame, the volume form can be written as $\omega_g = f \varepsilon^1 \wedge \dots \wedge \varepsilon^n$ on U , for some $f \in \mathcal{C}^\infty(U)$: then, Eq. 2.3.1 reduces to $f \equiv 1$, so:

$$\omega_g = \varepsilon^1 \wedge \dots \wedge \varepsilon^n$$

Therefore, such a form is uniquely determined.

To prove existence, define $\omega_g \in \Omega^n(\mathcal{M})$ in a neighbourhood of each point by $\omega_g = \varepsilon^1 \wedge \dots \wedge \varepsilon^n$, so WTS this definition is independent of the choice of oriented orthonormal frame. If $(\tilde{E}_1, \dots, \tilde{E}_n)$ is another oriented orthonormal frame with dual coframe $(\tilde{\varepsilon}^1, \dots, \tilde{\varepsilon}^n)$, let $\tilde{\omega}_g = \tilde{\varepsilon}^1 \wedge \dots \wedge \tilde{\varepsilon}^n$. The transformation between these smooth orthonormal frames can be written as:

$$\tilde{E}_i = A_i^j E_j$$

for some matrix of smooth functions $[A_i^j]$. Since these frames are both orthonormal, $[A_i^j](p) \in \text{O}(n)$ for each $p \in \mathcal{M}$, so $\det[A_i^j] = \pm 1$, and the fact that these frames are consistently oriented restricts to the positive sign. Therefore:

$$\omega_g(\tilde{E}_1, \dots, \tilde{E}_n) = \det[\varepsilon^j(\tilde{E}_i)] = \det[A_i^j] = 1 = \tilde{\omega}_g(\tilde{E}_1, \dots, \tilde{E}_n)$$

thus $\omega_g = \tilde{\omega}_g$. Defining such ω_g in a neighbourhood of each point hence yields a global smooth n -form which satisfies Eq. 2.3.1. $\square$

Moreover, if $(\mathcal{M}, g)$ and $(\tilde{\mathcal{M}}, \tilde{g})$ are two positive-dimensional Riemannian manifolds and $f : \mathcal{M} \rightarrow \tilde{\mathcal{M}}$ is a local isometry, it can be shown that $f^* \omega_{\tilde{g}} = \omega_g$.

Although the expression of the Riemannian form with respect to an oriented orthonormal frame is particularly simple, it is more useful to express it in smooth coordinates.

Proposition 2.3.11 (Coordinate expression of Riemannian form)

Let $(\mathcal{M}, g)$ be an oriented Riemannian n -manifold, with $n \geq 1$, and let (x^i) be any oriented smooth coordinates. Then, the local expression of the Riemannian volume form is:

$$\omega_g = \sqrt{\det[g_{ij}]} dx^1 \wedge \dots \wedge dx^n \quad (2.3.2)$$

Proof. Let $(U, (x^i))$ be an oriented smooth chart on $\mathcal{M}$, and let $p \in U$. In these coordinates $\omega_g = f dx^1 \wedge \cdots \wedge dx^n$ for some positive $f \in \mathcal{C}^\infty(\mathcal{M})$. Let (E_i) be any smooth oriented orthonormal frame on a neighbourhood of p , and let (ε^i) be the dual coframe, so that the coordinate frame can be written as:

$$\frac{\partial}{\partial x^i} = A_i^j E_j$$

Then, by direct computation:

$$f = \omega_g \left(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n} \right) = \varepsilon^1 \wedge \cdots \wedge \varepsilon^n \left(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n} \right) = \det \left[\varepsilon^j \left(\frac{\partial}{\partial x^i} \right) \right] = \det A$$

On the other hand:

$$g_{ij} = \left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle_g = \langle A_i^k E_k, A_j^l E_l \rangle_g = A_i^k A_j^l \langle E_k, E_l \rangle_g = \sum_{k=1}^n A_i^k A_j^k = [A^\top A]_{ij}$$

Therefore:

$$\det[g_{ij}] = \det(A^\top A) = (\det A)^2$$

from which it follows that $f = \det A = \pm \sqrt{\det[g_{ij}]}$. Since both frames (∂_i) and (E_i) are oriented, the sign must be positive. $\square$

An expression which holds for Lorentzian manifolds too is found setting $\sqrt{g} \equiv \sqrt{|\det[g_{ij}]|}$, so that the Riemannian form is expressed as:

$$dV_g \equiv \omega_g = \sqrt{g} dx^1 \wedge \cdots \wedge dx^n \quad (2.3.3)$$

Let $(\mathcal{M}, g)$ be an oriented Riemannian manifold, and let $S \subseteq \mathcal{M}$ be an immersed hypersurface. Any unit normal vector field along S is nowhere tangent to S , so it determines an orientation on S by Prop. 2.3.8.

Proposition 2.3.12 (Induced volume form on immersed hypersurface)

Let $(\mathcal{M}, g)$ be an oriented Riemannian manifold, let $S \subseteq \mathcal{M}$ be an immersed hypersurface, and let $\tilde{g}$ denote the induced metric on S . If N is a smooth unit normal vector field along S , the volume form of $(S, \tilde{g})$ with respect to the orientation of S determined by N is:

$$\omega_{\tilde{g}} = \iota_S^* (\iota_N \omega_g) \quad (2.3.4)$$

Proof. By Prop. 2.3.8, the $(n-1)$ -form $\iota_S^* (\iota_N \omega_g)$ is an orientation form for S : WTS this is the volume form for the induced Riemannian metric on S . To prove this, it suffices to show that it gives 1 whenever it is applied to an oriented orthonormal frame of S , thus let $(E_1, \dots, E_{n-1})$ be such a frame. At each $p \in S$, the basis $(N_p, E_1|_p, \dots, E_{n-1}|_p)$ is orthonormal and oriented for $T_p \mathcal{M}$ (by definition of orientation determined by N), thus:

$$\iota_S^* (\iota_N \omega_g)(E_1, \dots, E_{n-1}) = \omega_g(N, E_1, \dots, E_{n-1}) = 1$$

which concludes the proof. $\square$

A particularly important case is that of Riemannian manifold with boundaries.

Proposition 2.3.13 (Unit normal vector along boundary)

Let $\mathcal{M}$ be a Riemannian manifold with boundary. There is a unique smooth outward-pointing unit normal vector field N along $\partial\mathcal{M}$.

Proof. First, prove uniqueness. At any point $p \in \partial\mathcal{M}$, the subspace $(T_p\partial\mathcal{M})^\perp \subseteq T_p\mathcal{M}$ is 1-dimensional, so there are exactly two unit vectors at p that are normal to $\partial\mathcal{M}$. Since any unit normal vector N is nowhere tangent to $\partial\mathcal{M}$, it must have non-zero x^n -component in any smooth boundary chart, thus exactly one of the two choices of unit normal vector has negative x^n -component, which is equivalent to being outward-pointing.

Now, prove existence. Let $f : \mathcal{M} \rightarrow \mathbb{R}$ a boundary-defining function⁵ for $\mathcal{M}$, and let N be the restriction to $\partial\mathcal{M}$ of the unit normal vector $-\nabla f / \|\nabla f\|_g$: since $df \neq 0$ on $\partial\mathcal{M}$, N is well-defined and smooth on $\partial\mathcal{M}$. Then, it can be shown that N is normal to $\partial\mathcal{M}$ and outward-pointing, since $N(f) = -\langle \nabla f, \nabla f \rangle_g / \|\nabla f\|_g = -\|\nabla f\|_g < 0$. $\square$

By Prop. 2.3.12, if $(\mathcal{M}, g)$ is an oriented Riemannian manifold with boundary and $\tilde{g}$ is the induced Riemannian metric on $\partial\mathcal{M}$, the the volume form of $\tilde{g}$ is $\omega_{\tilde{g}} = \iota_N^* \omega_g$, where N is the outward-pointing unit norm vector field along $\partial\mathcal{M}$.

§2.3.2 Integration on manifolds

To introduce a coordinate-independent notion of integration, first note that there is no way to integrate $\mathbb{R}$ -values functions in a coordinate-independent way on a manifold. Indeed, given a closed ball $B \subseteq C$ and $f : B \rightarrow \mathbb{R}$ the constant function $f \equiv 1$, then:

$$\int_B dV f = \text{Vol}(B)$$

where the shorthand $dV \equiv dx^1 \wedge \cdots \wedge dx^n$ has been introduced. This is clearly not invariant under coordinate transformations. To obtain a coordinate-independent integral on a k -dimensional submanifold, differential k -forms are needed.

§2.3.2.1 Integration of differential forms

Recall that a **domain of integration** in $\mathbb{R}^n$ is a bounded subset whose boundary has measure zero. Let $D \subseteq \mathbb{R}^n$ be a domain of integration, and let ω be a continuous n -form on $\overline{D}$: any such form can be written as $\omega = f dx^1 \wedge \cdots \wedge dx^n$ for some continuous function $f : \overline{D} \rightarrow \mathbb{R}$, hence the *integral* of ω over D is defined to be:

$$\int_D \omega = \int_D dV f$$

In simple terms, the integral is computed by just “erasing the wedges”. In a more general way, if $U \in \mathcal{T}(\mathbb{R}^n)$ (or $\mathbb{U}^n$) and ω is a compactly supported n -form on U , then:

$$\int_U \omega = \int_D \omega$$

where $D \subseteq \mathbb{R}^n$ (or $\mathbb{U}^n$) is any domain of integration such that $\text{supp } \omega \subseteq D$, and ω is extended to be zero on the complement of its support. This definition is independent of the domain D chosen.

⁵If $\mathcal{M}$ is a smooth manifold with boundary, a *boundary-defining function* for $\mathcal{M}$ is a smooth function $f : \mathcal{M} \rightarrow \mathbb{R}_0^+$ such that $f^{-1}(0) = \partial\mathcal{M}$ and $df_p \neq 0$ for all $p \in \partial\mathcal{M}$, e.g. $f(x) = 1 - \|x\|^2$ for $\mathbb{D}^n$. Every smooth manifold with boundary admits a boundary-defining function.

Proposition 2.3.14 (Orientation and integration)

Let $D, E \in \mathcal{T}(\mathbb{R}^n)$ (or $\mathbb{U}^n$) be domain of integration, and let $f : \overline{D} \rightarrow \overline{E}$ be a smooth map that restricts to an orientation-preserving or reversing diffeomorphism from D to E . If ω is an n -form on $\overline{E}$, then:

$$\int_D f^* \omega = \pm \int_E \omega$$

with $+$ if f is orientation-preserving and $-$ if it is orientation-reversing.

Proof. Let (y^i) denote standard coordinates on E and (x^i) those on D . Then, writing $\omega = g \, dy^1 \wedge \cdots \wedge dy^n$ on E and recalling Eq. 2.2.21, changing variables in the integral with f yields:

$$\int_E \omega = \int_E dV_y g = \int_D dV_x |\det \mathcal{J}_f| (g \circ f) = \pm \int_D dV (g \circ f) (\det \mathcal{J}_f) dx^1 \wedge \cdots \wedge dx^n = \int_D f^* \omega$$

with $+$ if f is orientation-preserving and $-$ if it is orientation-reversing. $\square$

This result also extends to compactly supported n -forms defined on open subsets.

Proposition 2.3.15 (Orientation and integration)

Let $U, V \in \mathcal{T}(\mathbb{R}^n)$ (or $\mathbb{U}^n$), and let $f : U \rightarrow V$ an orientation-preserving or reversing diffeomorphism. If ω is a compactly supported n -form on V , then:

$$\int_V \omega = \pm \int_U f^* \omega$$

with $+$ if f is orientation-preserving and $-$ if it is orientation-reversing.

Proof. Let E be an open domain of integration such that $\text{supp } \omega \subseteq E \subseteq \overline{E} \subseteq V$, which always exists by the following lemma.

Lemma 2.3.4 (Compact subsets and open domains)

Let $U \in \mathcal{T}(\mathbb{R}^n)$ (or $\mathbb{U}^n$) and $K \subseteq U$ compact. Then, there is an open domain of integration D such that $K \subseteq D \subseteq \overline{D} \subseteq U$.

Proof. For each $p \in K$, there is an open ball or half-ball containing p whose closure is contained in U : by compactness, finitely many such sets $B_1, \dots, B_m$ cover K . Since the boundary of an open ball is a codimension-1 submanifold and since the boundary of an half-ball is contained in the union of two such submanifolds, the boundary of each has measure zero by Cor. 1.3.9.2, hence $D = B_1 \cup \cdots \cup B_m$ is the required domain of integration. $\square$

Since diffeomorphisms take interiors to interiors, boundaries to boundaries and measure-zero sets to measure-zero sets, $D = f^{-1}(E) \subseteq U$ is an open domain of integration containing $\text{supp } f^* \omega$, so the result follows from Prop. 2.3.14. $\square$

With these results on Euclidean integration, it is possible to generalize the integration of differential forms to manifolds.

Definition 2.3.3 (Integral over manifold)

Let $\mathcal{M}$ be an oriented smooth n -manifold, and let ω be an n -form over $\mathcal{M}$. If ω is compactly supported in the domain of a single smooth chart (U, φ) that is either positively or negatively oriented, then the **integral** of ω over $\mathcal{M}$ is defined as:

$$\int_{\mathcal{M}} \omega := \pm \int_{\varphi(U)} (\varphi^{-1})^* \omega \quad (2.3.5)$$

with $+$ if the chart is positively oriented and $-$ otherwise.

If, on the other hand, ω is compactly supported in $\mathcal{M}$, let $\{U_i\}$ be a finite open cover of $\text{supp } \omega$ by domains of positively or negatively oriented smooth charts, and let $\{\psi_i\}$ be a subordinate smooth partition of unity. Then, the **integral** of ω over $\mathcal{M}$ is defined as:

$$\int_{\mathcal{M}} \omega = \sum_i \int_{\mathcal{M}} \psi_i \omega \quad (2.3.6)$$

Both these integrals are well-defined, since the former is defined as a Euclidean integral and does not depend on the choice of smooth chart $U \supseteq \text{supp } \omega$, while the latter is defined in terms of the former and does not depend on the choice of open cover or partition of unity.

In the 0-dimensional case, a compactly supported 0-form is just an $\mathbb{R}$ -valued function which is non-zero only on finitely many points of a 0-dimensional manifold $\mathcal{M}$, hence:

$$\int_{\mathcal{M}} f = \sum_{p \in \mathcal{M}} \pm f(p)$$

where the sign depends on the local orientation of each point.

If $S \subseteq \mathcal{M}$ is an oriented immersed k -dimensional submanifold and ω is a k -form over $\mathcal{M}$ whose restriction to S is compactly supported, then:

$$\int_S \omega \equiv \int_S \iota_S^* \omega$$

where $\iota_S : S \hookrightarrow \mathcal{M}$ is the inclusion map. In particular, if $\mathcal{M}$ is a compact oriented smooth n -manifold with boundary and ω is a $(n-1)$ -form on $\mathcal{M}$, then:

$$\int_{\partial \mathcal{M}} \omega \equiv \int_{\partial \mathcal{M}} \iota_{\partial \mathcal{M}}^* \omega$$

where the orientation of $\partial \mathcal{M}$ is always understood to be the induced orientation.

Lemma 2.3.5 (Properties of integration of forms)

Let $\mathcal{M}, \mathcal{M}'$ be non-empty oriented smooth n -manifolds, and let ω, η be compactly supported n -forms on $\mathcal{M}$. Then:

(1) (linearity) if $a, b \in \mathbb{R}$, then:

$$\int_{\mathcal{M}} a\omega + b\eta = a \int_{\mathcal{M}} \omega + b \int_{\mathcal{M}} \eta$$

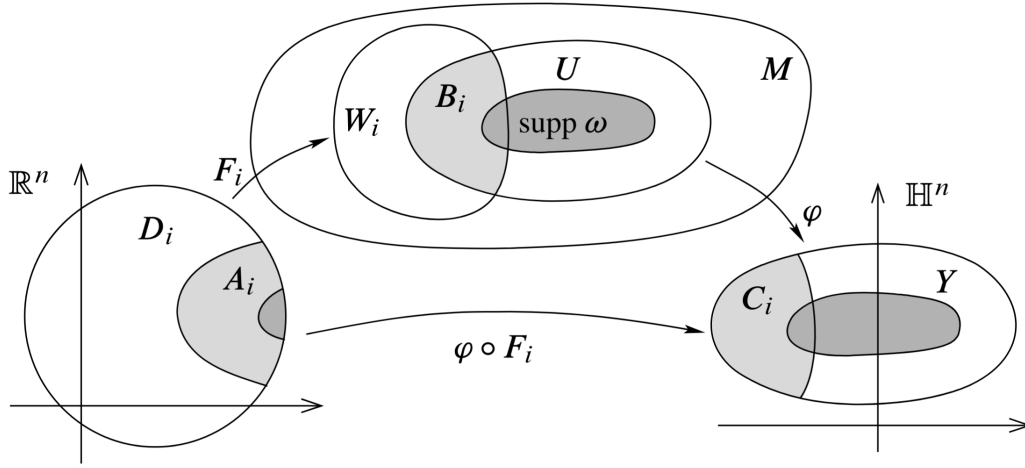


Figure 2.4: Integration over local parametrizations.

(2) (orientation reversal) if $-\mathcal{M}$ denoted $\mathcal{M}$ with opposite orientation, then:

$$\int_{-\mathcal{M}} \omega = - \int_{\mathcal{M}} \omega$$

(3) (positivity) if ω is a positively oriented orientation form, then $\int_{\mathcal{M}} \omega > 0$;

(4) (diffeomorphism invariance) if $f : \mathcal{M}' \rightarrow \mathcal{M}$ is an orientation-preserving or reversing diffeomorphism, then:

$$\int_{\mathcal{M}} \omega = \pm \int_{\mathcal{M}'} f^* \omega$$

with $+$ if f is orientation-preserving and $-$ if it is orientation-reversing.

The definition of integral of forms based on partitions of unity is convenient for theoretical purposes, but quite useless for actual computations: for these, it is much more convenient to separate the manifold into a finite number of pieces whose boundaries are sets of measure zero, and then compute the integral on each piece separately with local parametrizations.

Proposition 2.3.16 (Integration and local parametrizations)

Let $\mathcal{M}$ be an oriented smooth n -manifold, and let ω be a compactly supported n -form on $\mathcal{M}$. Suppose $D_1, \dots, D_k$ are open domains of integrations in $\mathbb{R}^n$, and for each $i = 1, \dots, k$ let $f_i : \overline{D_i} \rightarrow \mathcal{M}$ be smooth maps satisfying:

- (1) f_i restricts to an orientation-preserving diffeomorphism from D_i onto some $W_i \in \mathcal{T}(\mathcal{M})$;
- (2) $W_i \cap W_j = \emptyset$ when $i \neq j$;
- (3) $\text{supp } \omega \subseteq \overline{W_1} \cup \dots \cup \overline{W_k}$.

Then:

$$\int_{\mathcal{M}} \omega = \sum_{i=1}^k \int_{D_i} f_i^* \omega \quad (2.3.7)$$

The hypotheses of Prop. 2.3.16 can be somewhat relaxed. The requirement that each map f_i is

smooth on $\overline{D}_i$ ensures that the boundaries of the image sets W_i have measure zero and that the pull-back forms $f_i^*\omega$ are continuous on $\overline{D}_i$. However, provided that the open subsets W_i together cover all of $\mathcal{M}$ except for a measure-zero set, the maps f_i are allowed to not extend smoothly to the boundary, thus interpreting the resulting integrals of unbounded forms either as improper Riemannian integrals or as Lebesgue integrals.

Example 2.3.3 (Integration on the 2-sphere)

Let ω be the following 2-form on $\mathbb{R}^3$:

$$\omega = x \, dy \wedge dz + y \, dz \wedge dx + z \, dx \wedge dy$$

Let $D = (0, \pi) \times (0, 2\pi)$ (open rectangle) and $f : \overline{D} \rightarrow \mathbb{S}^2$ the spherical coordinate parametrization $f(\varphi, \vartheta) = (\sin \varphi \cos \vartheta, \sin \varphi \sin \vartheta, \cos \varphi)$. By Ex. 2.3.2, $f|_D$ is orientation preserving, so it satisfies the hypotheses of Prop. 2.3.16. Note that:

$$\begin{aligned} f^*dx &= \cos \varphi \cos \vartheta \, d\varphi - \sin \varphi \sin \vartheta \, d\vartheta \\ f^*dy &= \cos \varphi \sin \vartheta \, d\varphi + \sin \varphi \cos \vartheta \, d\vartheta \\ f^*dz &= -\sin \varphi \, d\varphi \end{aligned}$$

Therefore:

$$\begin{aligned} \int_{\mathbb{S}^2} \omega &= \int_D [(-\sin^3 \varphi \cos^2 \vartheta - \cos^2 \varphi \sin \varphi \sin^2 \vartheta) d\vartheta \wedge d\varphi \\ &\quad + (\sin^3 \varphi \sin^2 \vartheta + \cos^2 \varphi \sin \varphi \cos^2 \vartheta) d\varphi \wedge d\vartheta] \\ &= \int_D \sin \varphi \, d\varphi \wedge d\vartheta = \int_0^{2\pi} d\vartheta \int_0^\pi \sin \varphi \, d\varphi = 4\pi \end{aligned}$$

It is now possible to prove the central theorem of integration on manifolds, which generalizes many classical results.

Theorem 2.3.1 (Stokes' theorem)

Let $\mathcal{M}$ be an oriented smooth n -manifold with boundary, and let $\omega \in \Omega^{n-1}(\mathcal{M})$ be compactly supported on $\mathcal{M}$. Then:

$$\int_{\mathcal{M}} d\omega = \int_{\partial \mathcal{M}} \omega \quad (2.3.8)$$

Proof. Consider first a special case: suppose $\mathcal{M} = \mathbb{U}^n$. Then, since ω has compact support, there exists $R \in \mathbb{R}^+$ such that $\text{supp } \omega \subseteq A = [-R, R]^{n-1} \times [0, R]$ (see Fig. 2.5). In the standard basis, ω can be written as:

$$\begin{aligned} \omega &= \sum_{i=1}^n \omega_i dx^1 \wedge \cdots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \cdots \wedge dx^n \\ \Rightarrow d\omega &= \sum_{i,j=1}^n \frac{\partial \omega_i}{\partial x^j} dx^j \wedge dx^1 \wedge \cdots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \cdots \wedge dx^n \\ &= \sum_{i=1}^n (-1)^{i-1} \frac{\partial \omega_i}{\partial x^i} dx^1 \wedge \cdots \wedge dx^n \end{aligned}$$

Therefore, by direct computation:

$$\begin{aligned}
 \int_{\mathbb{U}^n} d\omega &= \sum_{i=1}^n (-1)^{i-1} \int_A \frac{\partial \omega_i}{\partial x^i} dx^1 \wedge \cdots \wedge dx^n = \sum_{i=1}^n (-1)^{i-1} \int_0^R dx^n \int_{-R}^R dx^{n-1} \cdots \int_{-R}^R dx^1 \frac{\partial \omega_i}{\partial x^i} \\
 &= \sum_{i=1}^n (-1)^{i-1} \int_0^R dx^n \int_{-R}^R dx^{n-1} \cdots \int_{-R}^R dx^1 \int_{-R}^R dx^i \frac{\partial \omega_i}{\partial x^i} + (i = n) \\
 &= \sum_{i=1}^n (-1)^{i-1} \int_0^R dx^n \int_{-R}^R dx^{n-1} \cdots \int_{-R}^R dx^1 [\omega_i(x)]_{x^i=-R}^{x^i=R} + (i = n) = 0 + (i = n)
 \end{aligned}$$

since $R \in \mathbb{R}^+$ can be chosen sufficiently large so that $\omega|_{x^i=\pm R} = 0$ (see Fig. 2.5). The only term which might be non-zero is the $i = n$ term:

$$\begin{aligned}
 \int_{\mathbb{U}^n} d\omega &= (-1)^{n-1} \int_{-R}^R dx^{n-1} \cdots \int_{-R}^R dx^1 \int_0^R dx^n \frac{\partial \omega_n}{\partial x^n} \\
 &= (-1)^{n-1} \int_{-R}^R dx^{n-1} \cdots \int_{-R}^R dx^1 [\omega_n(x)]_{x^n=0}^{x^n=R} \\
 &= (-1)^n \int_{-R}^R dx^{n-1} \cdots \int_{-R}^R dx^1 \omega_n(x^1, \dots, x^{n-1}, 0)
 \end{aligned}$$

since, as above, $\omega_n|_{x^n=R} = 0$. Then, compute:

$$\int_{\partial \mathbb{U}^n} \omega = \sum_{i=1}^n \int_{A \cap \partial \mathbb{U}^n} \omega_i(x^1, \dots, x^{n-1}, 0) dx^1 \wedge \cdots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \cdots \wedge dx^n$$

Note that $x^n = 0$ on $\partial \mathbb{U}^n$, hence the pull-back of dx^n to the boundary identically vanishes, and the above sum has only the $i = n$ term which is non-zero:

$$\int_{\partial \mathbb{U}^n} \omega = \int_{A \cap \partial \mathbb{U}^n} \omega_n(x^1, \dots, x^{n-1}, 0) dx^1 \wedge \cdots \wedge dx^{n-1}$$

Observation 2.3.1 (Induced orientation on $\partial \mathbb{U}^n$)

Consider $\mathbb{U}^n$ with the standard orientation inherited by $\mathbb{R}^n$, and identify $\partial \mathbb{U}^n \cong \mathbb{R}^{n-1}$ under the correspondence $(x^1, \dots, x^{n-1}, 0) \leftrightarrow (x^1, \dots, x^{n-1})$. The vector field $-\partial_n$ is outward-pointing along $\partial \mathbb{U}^n$, hence the standard coordinate frame on $\mathbb{R}^{n-1}$ is positively oriented for $\partial \mathbb{U}^n$ if and only if $[-\partial_n, \partial_1, \dots, \partial_{n-1}]$ is the standard orientation of $\mathbb{R}^n$. This orientation satisfies:

$$[-\partial_n, \partial_1, \dots, \partial_{n-1}] = -[\partial_n, \partial_1, \dots, \partial_{n-1}] = (-1)^n [\partial_1, \dots, \partial_n]$$

Therefore, the induced orientation on $\partial \mathbb{U}^n$ is equal to the standard orientation of $\mathbb{R}^{n-1}$ when n is even, while it is opposite to the standard orientation when n is odd: in particular, the standard coordinates on $\partial \mathbb{U}^n \cong \mathbb{R}^{n-1}$ are positively oriented if and only if n is even.

By Obs. 2.3.1, the standard coordinates $(x^1, \dots, x^{n-1})$ are positively oriented on $\partial \mathbb{U}^n$ when n is even and negatively oriented when n is odd, hence correctly:

$$\int_{\mathbb{U}^n} d\omega = \int_{\partial \mathbb{U}^n} \omega$$

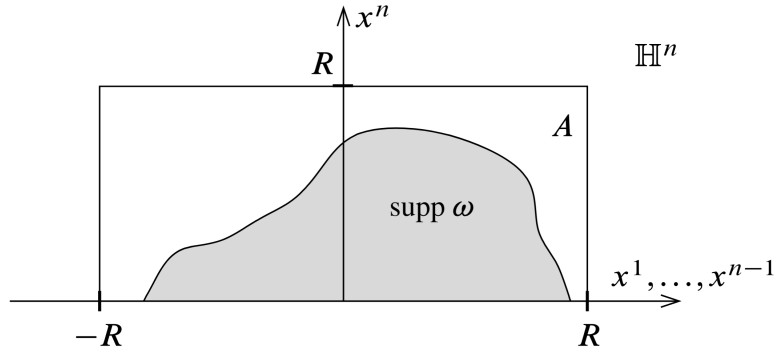


Figure 2.5: Proof of Stokes' theorem.

Next, consider $\mathcal{M} = \mathbb{R}^n$. In this case $\text{supp } \omega \subseteq A = [-R, R]^n$, hence the same computation as above shows that the left-hand side of Stokes' theorem vanishes, while the right-hand side vanishes too since $\partial\mathbb{R}^n = \emptyset$.

Now, consider an arbitrary smooth manifold $\mathcal{M}$ with boundary, and consider $\omega \in \Omega^{n-1}(\mathcal{M})$ which is compactly supported in the domain of a single positively or negatively oriented smooth chart (U, φ) . Assuming φ is a positively oriented boundary chart, by Def. 2.3.3:

$$\int_{\mathcal{M}} d\omega = \int_{\mathbb{U}^n} (\varphi^{-1})^* d\omega = \int_{\mathbb{U}^n} d((\varphi^{-1})^* \omega) = \int_{\partial\mathbb{U}^n} (\varphi^{-1})^* \omega$$

where the last equality follows from the computation above (with $\partial\mathbb{U}^n$ assumed with the induced orientation). Since $d\varphi$ takes outward-pointing vectors on $\partial\mathcal{M}$ to outward-pointing vectors on $\mathbb{U}^n$, it follows that $\varphi|_{U \cap \partial\mathcal{M}}$ is an orientation-preserving diffeomorphism onto $\varphi(U) \cap \partial\mathbb{U}^n$, and thus:

$$\int_{\mathcal{M}} d\omega = \int_{\partial\mathbb{U}^n} (\varphi^{-1})^* \omega = \int_{\partial\mathcal{M}} \omega$$

For a negatively oriented smooth boundary chart, the argument applies verbatim with an additional negative sign on both sides of the equation. For an interior chart, the calculation is performed equivalently with $\mathbb{R}^n$ in place of $\mathbb{U}^n$.

Finally, let $\omega \in \Omega^{n-1}(\mathcal{M})$ be arbitrarily compactly supported. Choosing a cover of $\text{supp } \omega$ by finitely many domains of positively or negatively oriented smooth charts $\{U_i\}$ and choosing a subordinate smooth partition of unity $\{\psi_i\}$, the preceding argument can be applied to $\psi_i \omega$ for each i :

$$\begin{aligned} \int_{\partial\mathcal{M}} \omega &= \sum_i \int_{\partial\mathcal{M}} \psi_i \omega = \sum_i \int_{\mathcal{M}} d(\psi_i \omega) = \sum_i \int_{\mathcal{M}} d\psi_i \wedge \omega + \psi_i d\omega \\ &= \int_{\mathcal{M}} d\left(\sum_i \psi_i\right) \wedge \omega + \int_{\mathcal{M}} \left(\sum_i \psi_i\right) d\omega = 0 + \int_{\mathcal{M}} d\omega \end{aligned}$$

since $\sum_i \psi_i \equiv 1$. □

This result generalizes many classical integral results. For example, let $\mathcal{M}$ be a smooth manifold and $\gamma : [a, b] \rightarrow \mathcal{M}$ a smooth embedding, so that $S = \gamma([a, b])$ is an embedded 1-submanifold with boundary in $\mathcal{M}$. If S is given an orientation such that γ is orientation preserving, then for any

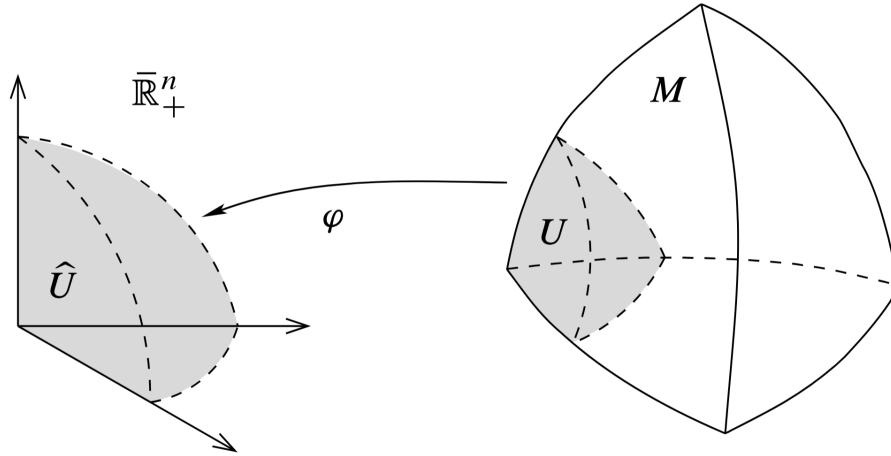


Figure 2.6: A chart with corners.

$f \in \mathcal{C}^\infty(\mathcal{M})$ Stokes' theorem states that:

$$\int_S df = \int_{\partial S} f = f(\gamma(b)) - f(\gamma(a))$$

which is just the fundamental theorem of line integrals Eq. 2.3.17. In particular, when $\gamma : [a, b] \hookrightarrow \mathbb{R}$ is the inclusion map in $\mathcal{M} = \mathbb{R}$, this is the fundamental theorem of calculus.

§2.3.2.2 Manifolds with corners

Many applications of Stokes' theorem involve geometric objects such as triangles, squares or cubes which, although topological manifolds, are not smooth manifolds with boundary, since they have “corners”. It is therefore necessary to generalize Stokes' theorem to these manifolds. Let $\overline{\mathbb{R}}_+^n$ be the model for this kind of corners:

$$\overline{\mathbb{R}}_+^n := \{(x^1, \dots, x^n) \in \mathbb{R}^n : x^i \geq 0 \ \forall i = 1, \dots, n\} \quad (2.3.9)$$

It can be shown that $\overline{\mathbb{R}}_+^n$ is homeomorphic to $\mathbb{U}^n$. The boundary of $\partial\overline{\mathbb{R}}_+^n$ in $\mathbb{R}^n$ is the set of points at which at least one coordinate vanishes, while points at which more than one coordinate vanishes are called its *corner points*.

Given a topological n -manifold $\mathcal{M}$, a **chart with corners** is a pair (U, φ) with $U \in \mathcal{T}(\mathcal{M})$ and a homeomorphism $\varphi : U \rightarrow \varphi(U) \in \mathcal{T}(\overline{\mathbb{R}}_+^n)$, as illustrated in Fig. 2.6. Two charts with corners (U, φ) and (V, ψ) are smoothly compatible if the composite map $\varphi \circ \psi^{-1} : \psi(U \cap V) \rightarrow \varphi(U \cap V)$ is smooth, i.e. if it admits a smooth extension in an open neighbourhood of each point. A *smooth structure with corners* on a topological manifold with boundary is a maximal collection of smoothly compatible interior charts and charts with corners whose domains cover $\mathcal{M}$: a topological manifold with boundary equipped with a smooth structure with corners is a **smooth manifold with corners**, and any chart with corners in said smooth structure with corners is a *smooth chart with corners* on $\mathcal{M}$. For example, any closed rectangle in $\mathbb{R}^n$ is a smooth n -manifold with corners.

With the homeomorphism between $\overline{\mathbb{R}}_+^n$ and $\mathbb{U}^n$, charts with corners are topologically indistinguishable from boundary charts, thus from a topological point of view there is no difference between manifolds with boundary and manifolds with corners. The difference is in the smooth structure, because for $\dim_{\mathbb{R}} \mathcal{M} > 1$ the compatibility condition for charts with corners is different from that for boundary charts: since $\overline{\mathbb{R}}_+^1 = \mathbb{U}^1$, smooth 1-manifolds with corners are no different from smooth manifolds with boundary.

Proposition 2.3.17 (Invariance of corner points)

Let $\mathcal{M}$ be a smooth n -manifold with corners, with $n \geq 2$, and let $p \in \mathcal{M}$. If $\varphi(p)$ is a corner point of $\overline{\mathbb{R}}_+^n$ for some smooth chart with corners (U, φ) , then the same is true for every such chart whose domain contains p .

Proof. Let (U, φ) and (V, ψ) be smooth charts with corners such that $\varphi(p)$ is a corner point by $\psi(p)$ is not: WLOG assume $\varphi(p) = (x^1, \dots, x^k, 0, \dots, 0)$, with $k \leq n-2$. Then, $\psi(V)$ contains an open subset of some $(n-1)$ -dimensional linear subspace $S \subseteq \mathbb{R}^n$, with $\psi(p) \in S$: if $\psi(p) \in \partial \overline{\mathbb{R}}_+^n$, take S to be the unique subspace defined by an equation of the form $x^i = 0$ that contains $\psi(p)$, otherwise if $\psi(p)$ is an interior point, and $(n-1)$ -dimensional subspaces containing $\psi(p)$ works. Let $S' = S \cap \psi(U \cap V)$, and let $\alpha : S' \rightarrow \mathbb{R}^n$ be the restriction of $\varphi \circ \psi^{-1}$ to S' . Since $\varphi \circ \psi^{-1}$ is a diffeomorphism from $\psi(U \cap V)$ to $\varphi(U \cap V)$, it follows that $\psi \circ \varphi^{-1} \circ \alpha = 1_{S'}$, therefore $d\alpha_{\psi(p)}$ is an injective linear map. Let $T = d\alpha_{\psi(p)}(T_{\psi(p)}S) \subseteq \mathbb{R}^n$: since T is $(n-1)$ -dimensional, it must contain a vector v such that one of the last two components, v^{n-1} or v^n , is non-zero, otherwise T would be contained in a codimension-2 subspace. Renumbering the coordinates and replacing v with $-v$ if necessary, $v^n < 0$ can be assumed.

Now, let $\gamma : (-\varepsilon, \varepsilon) \rightarrow S$ be a smooth curve such that $\gamma(0) = p$ and $d\alpha(\gamma'(0)) = v$. Then, $\alpha \circ \gamma(t)$ has negative x^n coordinate for small $t > 0$, which contradicts the fact that α takes its values in $\overline{\mathbb{R}}_+^n$. $\square$

If $\mathcal{M}$ is a smooth manifold with corners, a point $p \in \mathcal{M}$ is called a *corner point* if $\varphi(p)$ is a corner point of $\overline{\mathbb{R}}_+^n$ with respect to some (and hence every) smooth chart with corners (U, φ) ; similarly, p is a *boundary point* if $\varphi(p) \in \partial \overline{\mathbb{R}}_+^n$ with respect to some (and hence every) such chart.

Every smooth manifold is also a smooth manifold with corners (with no corner points), but a smooth manifold with corners is a smooth manifold with boundary if and only if it has no corner points. On the contrary, the boundary of a smooth manifold with corners is not, in general, a smooth manifold with corners (e.g. the boundary of a cube). The boundary of $\overline{\mathbb{R}}_+^n$ can, however, be expressed as the union of finitely many smooth manifolds with corners: $\partial \overline{\mathbb{R}}_+^n = H_1 \cup \dots \cup H_n$, with:

$$H_i := \{(x^1, \dots, x^n) \in \overline{\mathbb{R}}_+^n : x^i = 0\} \quad (2.3.10)$$

which is an $(n-1)$ -dimensional smooth manifold with corners contained in the subspace defined by $x^i = 0$.

To generalize Stokes' theorem, it is necessary to define integration on the boundary of a smooth manifold with corners: this needs to be defined separately since this boundary is not itself a smooth manifold with boundary. Let $\mathcal{M}$ be an oriented smooth n -manifold with corners, and suppose ω is an $(n-1)$ -dimensional form on $\partial \mathcal{M}$ that is compactly supported in the domain of a single oriented smooth chart with corners (U, φ) . The integral of ω over $\partial \mathcal{M}$ is defined as:

$$\int_{\partial \mathcal{M}} \omega := \sum_{i=1}^n \int_{H_i} (\varphi^{-1})^* \omega$$

where each H_i , defined in Eq. 2.3.10, is given the induced orientation as part of the boundary of the subset where $x^i \geq 0$. This means that ω is integrated in coordinates over the codimension-1 of the boundary. If ω is instead compactly supported on $\mathcal{M}$, the integral of ω over $\partial \mathcal{M}$ is defined piecing together with a partition of unity, just like Eq. 2.3.6. In practice, one always uses Prop. 2.3.16, which holds on compact oriented smooth manifolds with corners too.

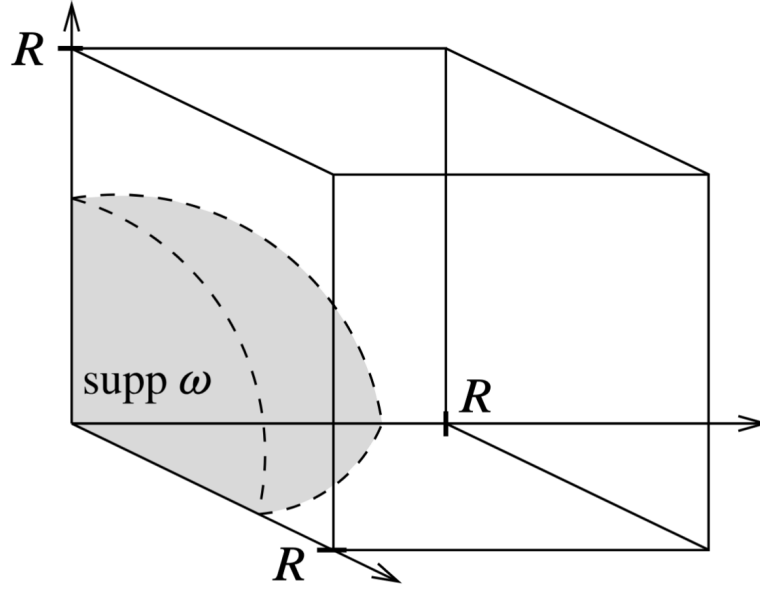


Figure 2.7: Proof of Stokes' theorem on manifolds with corners.

Theorem 2.3.2 (Stokes' theorem on manifolds with corners)

Let $\mathcal{M}$ be an oriented smooth n -manifold with corners, and let $\omega \in \Omega^{n-1}(\mathcal{M})$ be compactly supported on $\mathcal{M}$. Then:

$$\int_{\mathcal{M}} d\omega = \int_{\partial\mathcal{M}} \omega \quad (2.3.11)$$

Proof. The proof is identical to that of Th. 2.3.1, with a few minor changes. By means of smooth charts and a partition of unity, the theorem can be reduced to two cases: $\mathcal{M} = \mathbb{R}^n$ and $\mathcal{M} = \mathbb{R}_+^n$. The $\mathbb{R}^n$ yields zero on both sides just as before. In the case of a chart with corners, ω is supported on the cube $[0, R]^n$, as illustrated in Fig. 2.7, hence as before:

$$\begin{aligned} \int_{\mathbb{R}_+^n} d\omega &= \sum_{i=1}^n (-1)^{i-1} \int_0^R dx^n \dots \int_0^R dx^1 \frac{\partial \omega_i}{\partial x^i} = \sum_{i=1}^n (-1)^{i-1} \int_0^R dx^n \dots \int_0^R dx^1 \int_0^R dx^i \frac{\partial \omega_i}{\partial x^i} \\ &= \sum_{i=1}^n (-1)^{i-1} \int_0^R dx^n \dots \int_0^R dx^1 [\omega_i(x)]_{x^i=0}^{x^i=R} \\ &= \sum_{i=1}^n (-1)^i \int_0^R dx^n \dots \int_0^R dx^1 \omega_i(x^1, \dots, 0, \dots, x^n) = \sum_{i=1}^n \int_{H_i} \omega = \int_{\partial\mathbb{R}_+^n} \omega \end{aligned}$$

where the $(-1)^i$ factor has been absorbed into ω , since the induced orientation on H_i is $(-1)^i$ times that of the standard coordinates $[x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^n]$. $\square$

§2.3.2.3 Line integrals

A particular case of integration on manifolds is that of line integrals, i.e. the integration of a 1-form over the support of a curve.

Let $\mathcal{M}$ be a smooth manifold. A **curve segment** is a continuous curve $\gamma : [a, b] \rightarrow \mathcal{M}$ whose domain is a compact interval in $\mathbb{R}$. A **smooth curve segment** is a curve segment which is smooth when $[a, b]$

is considered as a manifold with boundary, or equivalently if γ has an extension to a smooth curve defined in a neighbourhood of each endpoint: a *piecewise smooth curve segment* is a curve segment on which there exists a partition $a = a_0 < a_1 < \dots < a_k = b$ of $[a, b]$ such that $\gamma|_{[a_{i-1}, a_i]}$ is smooth for each $i = 1, \dots, k$.

It can be shown that if $\mathcal{M}$ is a connected smooth manifold, then any two points of $\mathcal{M}$ can be joined by a piecewise smooth curve segment.

Definition 2.3.4 (Line integral)

Let $\mathcal{M}$ be a smooth manifold, $\gamma : [a, b] \rightarrow \mathcal{M}$ a smooth curve segment and $\omega \in \Omega^1(\mathcal{M})$. Then, the **line integral** of ω over γ is defined as:

$$\int_{\gamma} \omega := \int_{[a, b]} \gamma^* \omega \quad (2.3.12)$$

Line integrals inherit the linearity of Euclidean integrals. Moreover, if γ is a constant map, then $\int_{\gamma} \omega = 0$, while if γ is piecewise smooth with a partition $a = a_0 < a_1 < \dots < a_k = b$:

$$\int_{\gamma} \omega = \sum_{i=1}^k \int_{[a_{i-1}, a_i]} \gamma^* \omega \quad (2.3.13)$$

A result similar to (4) of Lemma 2.3.5 holds for line integrals. Given a smooth map $f : \mathcal{M} \rightarrow \mathcal{M}'$ and $\omega \in \Omega^1(\mathcal{M})(\mathcal{M}')$, then:

$$\int_{\gamma} f^* \omega = \int_{f \circ \gamma} \omega \quad (2.3.14)$$

A key feature of line integrals is that they are independent of the parametrization. If $\gamma : [a, b] \rightarrow \mathcal{M}$ and $\tilde{\gamma} : [c, d] \rightarrow \mathcal{M}$ are piecewise smooth curve segments, then $\tilde{\gamma}$ is a **reparametrization** of γ if $\tilde{\gamma} = \gamma \circ \varphi$ for some diffeomorphism $\varphi : [c, d] \rightarrow [a, b]$: if φ is an increasing function, this is a *forward reparametrization*, while if φ is a decreasing function, it is a *backward reparametrization*. It is then trivial to show that, given $\omega \in \Omega^1(\mathcal{M})$:

$$\int_{\tilde{\gamma}} \omega = \pm \int_{\gamma} \omega \quad (2.3.15)$$

with $+$ for a forward parametrization and $-$ otherwise.

Proposition 2.3.18 (Line integrals and Euclidean integrals)

Let $\mathcal{M}$ be a smooth manifold, $\gamma : [a, b] \rightarrow \mathcal{M}$ a piecewise smooth curve segment and $\omega \in \Omega^1(\mathcal{M})$. Then:

$$\int_{\gamma} \omega = \int_a^b dt \omega_{\gamma(t)}(\gamma'(t)) \quad (2.3.16)$$

Proof. First, suppose γ is smooth and that its image is contained in the domain of a single smooth chart. Writing the coordinate representation of:

$$\omega_{\gamma(t)}(\gamma'(t)) = \omega_i(\gamma(t)) dx^i(\dot{\gamma}^j(t) \partial_j) = \omega_i(\gamma(t)) \dot{\gamma}^i(t)$$

Using Eq. 1.5.23, then:

$$(\gamma^* \omega)_t = (\omega_i \circ \gamma)(t) d(\gamma^i)_t = \omega_i(\gamma(t)) \dot{\gamma}^i(t) dt = \omega_{\gamma(t)}(\gamma'(t)) dt$$

Therefore, by Def. 2.3.4:

$$\int_{\gamma} \omega = \int_{[a,b]} \gamma^* \omega = \int_a^b dt \omega_{\gamma(t)}(\gamma'(t))$$

If γ is an arbitrary smooth curve segment, by compactness there exists a finite partition $a = a_0 < a_1 < \dots < a_k = b$ of $[a, b]$ such that $\gamma([a_{i-1}, a_i])$ is contained in the domain of a single smooth chart for each $i = 1, \dots, k$, thus the preceding argument can be applied on each such subinterval. Finally, if γ is only piecewise smooth, simply apply the same argument on each subinterval on which γ is smooth. $\square$

A special case of line integral is that of a differential. If $\gamma : [a, b] \rightarrow \mathcal{M}$ is a piecewise smooth curve segment and $f \in \mathcal{C}^\infty(\mathcal{M})$, then by Prop. 2.3.18:

$$\int_{\gamma} df = f(\gamma(b)) - f(\gamma(a)) \quad (2.3.17)$$

which is known as the *fundamental theorem of line integrals*.

Theorem 2.3.3 (Path-homotopy invariance of line integrals)

Let $\mathcal{M}$ be a smooth manifold, $\gamma_0, \gamma_1 : [a, b] \rightarrow \mathcal{M}$ path-homotopic piecewise smooth curve segments and $\omega \in \Omega^1(\mathcal{M})$. Then:

$$\int_{\gamma_0} \omega = \int_{\gamma_1} \omega$$

§2.4 Connections

A key geometrical character of metric manifolds is curvature. In order to study the curvature of a metric manifold, one first needs to geodesics, i.e. the generalization of straight lines in Euclidean space. These have two defining properties: they are the unique shortest path between their end-points, and they are the only curves parametrized with zero acceleration. In the context of metric manifolds, the first definition is rather complicated, so the second is adopted: in order to make it well-defined, it is necessary to introduce a coordinate-independent notion of directional derivative. In the Euclidean case, the *directional derivative* of $Y \in \mathfrak{X}(\mathbb{R}^n)$ along $\mathbf{v} \in T_p \mathbb{R}^n$ is defined by Eq. 1.4.10 and Eq. 1.2.2:

$$D_{\mathbf{v}} Y = v^i \frac{\partial Y^j}{\partial x^i}(p) \frac{\partial}{\partial x^j} \Big|_p$$

which is still $D_{\mathbf{v}} Y \in T_p \mathbb{R}^n$. This definition can be generalized to a directional derivative along $X \in \mathfrak{X}(\mathbb{R}^n)$ as:

$$D_X Y = X(Y^i) \frac{\partial}{\partial x^i} \quad (2.4.1)$$

which is instead $D_X Y \in \mathfrak{X}(\mathbb{R}^n)$. This notion can be applied to curves on Euclidean submanifolds. Consider a curve $\gamma : I \rightarrow \mathcal{M}$, where $\mathcal{M} \subseteq \mathbb{R}^n$ is an embedded submanifold, so that its acceleration can be computed as:

$$\gamma''(t) = \ddot{\gamma}^i(t) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)}$$

To think of γ as a geodesic in $\mathcal{M}$, it needs to be “as straight as possible”: defining the *tangential projector* $\pi^\top : T_{\gamma(t)} \mathbb{R}^n \rightarrow T_{\gamma(t)} \mathcal{M}$ as an orthogonal projection, it is reasonable to state that γ is as straight as possible if its tangential acceleration $\gamma''^\top(t) \equiv \pi^\top(\gamma''(t))$ vanishes.

Clearly, this technique cannot be generalized to abstract metric manifolds, in which there is no ambient Euclidean space. Indeed, in a generic metric manifold $\mathcal{M}$, the velocity vector of a curve $\gamma : I \rightarrow \mathcal{M}$ is a well-defined vector $\gamma'(t) \in T_{\gamma(t)}\mathcal{M}$, while its acceleration has no such coordinate-independent interpretation.

Example 2.4.1 (Coordinate-dependence of $\gamma''(t)$ in $\mathbb{R}^2$)

Consider the parametrization of the unit circle in $\mathbb{R}^2$ as $\gamma(t) = (x(t), y(t)) = (\cos t, \sin t)$, so that its acceleration vector is:

$$\gamma''(t) = x''(t) \frac{\partial}{\partial x} \Big|_{\gamma(t)} + y''(t) \frac{\partial}{\partial y} \Big|_{\gamma(t)} = -\cos t \frac{\partial}{\partial x} \Big|_{\gamma(t)} - \sin t \frac{\partial}{\partial y} \Big|_{\gamma(t)}$$

On the contrary, in polar coordinates $\gamma(t) = (r(t), \vartheta(t)) = (1, t)$, so:

$$\gamma''(t) = r''(t) \frac{\partial}{\partial r} \Big|_{\gamma(t)} + \vartheta''(t) \frac{\partial}{\partial \vartheta} \Big|_{\gamma(t)} = 0$$

which clearly shows the coordinate-dependence of $\gamma''(t)$.

As shown in Ex. 2.4.1, the problem with defining $\gamma''(t)$ by differentiating $\gamma'(t)$ with respect to t is that it is necessary to take the folling limit:

$$\gamma''(t) = \lim_{h \rightarrow 0} \frac{\gamma'(t+h) - \gamma'(t)}{h}$$

However, $\gamma'(t+h) \in T_{\gamma(t+h)}\mathcal{M}$ while $\gamma'(t) \in T_{\gamma(t)}\mathcal{M}$, but these are entirely different vector spaces, so it does not make sense to subtract vectors from them. In order to define a coordinate-independent way to differentiate vector fields along curves, then, it is necessary to “connect” nearby tangent spaces.

§2.4.1 Koszul connections

It is easiest to define first a way to differentiate sections of vector bundles.

Definition 2.4.1 (Koszul connection)

Let $\pi : E \rightarrow \mathcal{M}$ be a smooth vector bundle over a smooth manifold $\mathcal{M}$. A **Koszul connection** in E is a map:

$$\nabla : \mathfrak{X}(\mathcal{M}) \times \Gamma(E) \rightarrow \Gamma(E) : (X, Y) \mapsto \nabla_X Y \quad (2.4.2)$$

with the following properties:

- (1) $\nabla_X Y$ is $\mathcal{C}^\infty(\mathcal{M})$ -linear in X , i.e.:

$$\nabla_{f_1 X_1 + f_2 X_2} Y = f_1 \nabla_{X_1} Y + f_2 \nabla_{X_2} Y \quad \forall f_1, f_2 \in \mathcal{C}^\infty(\mathcal{M}), X_1, X_2 \in \mathfrak{X}(\mathcal{M}), Y \in \Gamma(E)$$

- (2) $\nabla_X Y$ is $\mathbb{R}$ -linear in Y , i.e.:

$$\nabla_X (c_1 Y_1 + c_2 Y_2) = c_1 \nabla_X Y_1 + c_2 \nabla_X Y_2 \quad \forall c_1, c_2 \in \mathbb{R}, X \in \mathfrak{X}(\mathcal{M}), Y_1, Y_2 \in \Gamma(E)$$

- (3) for all $f \in \mathcal{C}^\infty(\mathcal{M}), X \in \mathfrak{X}(\mathcal{M}), Y \in \Gamma(E)$:

$$\nabla_X (fY) = f \nabla_X Y + X(f)Y \quad (2.4.3)$$

Since no other kinds of connections are considered in these notes, Koszul connections are also called simply connections. $\nabla_X Y$ is the *covariant derivative of Y along X* .

Although the Koszul connection is defined by its action on global sections of the vector bundle, it is truly a local operator.

Lemma 2.4.1 (Locality of Koszul connections)

Let ∇ be a Koszul connection in a smooth vector bundle $\pi : E \rightarrow \mathcal{M}$. For every $X \in \mathfrak{X}(\mathcal{M})$, $Y \in \Gamma(E)$ and $p \in \mathcal{M}$, the covariant derivative $\nabla_X Y|_p$ depends only on the values of X and Y in an arbitrarily small neighbourhood of p : more precisely, if $X = \tilde{X}$ and $Y = \tilde{Y}$ in a neighbourhood of p , then $\nabla_X Y|_p = \nabla_{\tilde{X}} \tilde{Y}|_p$.

Proof. Replacing Y with $Y - \tilde{Y}$, it suffices to prove that $\nabla_X Y|_p = 0$ if $Y = 0$ on a neighbourhood of p . Suppose $Y \in \Gamma(E) : Y|_U \equiv 0$ on $U \in \mathcal{T}(\mathcal{M}) : U \ni p$, and consider a bump function $\varphi \in \mathcal{C}^\infty(\mathcal{M})$ with support in U such that $\varphi(p) = 1$. By hypothesis, then, $\varphi Y \equiv 0$ on all of $\mathcal{M}$, so for every $X \in \mathfrak{X}(\mathcal{M})$:

$$\nabla_X(\varphi Y) = 0 \quad \implies \quad X(\varphi)Y + \varphi \nabla_X Y = 0$$

Since $Y \equiv 0$ on $\text{supp } \varphi$, the first term is identically zero, so evaluating the remaining expression at p shows that $\nabla_X Y|_p = 0$. The argument for X is similar. $\square$

Proposition 2.4.1 (Restrictions of Koszul connections)

Let ∇ be a Koszul connection in a smooth vector bundle $\pi : E \rightarrow \mathcal{M}$. For every $U \in \mathcal{T}(\mathcal{M})$, there is a unique Koszul connection ∇^U on the restricted bundle $E|_U$ that satisfies the following relation for every $X \in \mathfrak{X}(\mathcal{M})$, $Y \in \Gamma(E)$:

$$\nabla_{X|_U}^U (Y|_U) = (\nabla_X Y)|_U \quad (2.4.4)$$

Proof. First, prove uniqueness. Suppose ∇^U is any such connections, and consider arbitrary $X \in \mathfrak{X}(U)$, $Y \in \Gamma(E|_U)$. Given $p \in U$, it is possible to use a bump function to construct $\tilde{X} \in \mathfrak{X}(\mathcal{M}) : \tilde{X}|_U = X$ and $\tilde{Y} \in \Gamma(E) : \tilde{Y}|_U = Y$, and then by Lemma 2.4.1 and Eq. 2.4.4:

$$\nabla_X^U Y|_p = \nabla_{\tilde{X}|_U}^U (\tilde{Y}|_U)|_p = (\nabla_{\tilde{X}} \tilde{Y})|_p$$

The right-hand side is completely determined by ∇ , so this shows that ∇^U is uniquely defined if it exists.

To prove existence, given $X \in \mathfrak{X}(U)$ and $Y \in \Gamma(E|_U)$, for every $p \in U$ construct $\tilde{X} \in \mathfrak{X}(\mathcal{M})$ and $\tilde{Y} \in \Gamma(E)$ as above, and define $\nabla_X^U Y|_p$ by the above relation. This is independent of the choice of $\tilde{X}$ and $\tilde{Y}$ by Lemma 2.4.1, moreover it is smooth since the same formula holds on some neighbourhood of p and it can be shown to satisfy the conditions of Def. 2.4.1. $\square$

Typically, the restriction of ∇ to U is simply denoted by $\nabla^U \equiv \nabla$ for ease of notation, since by Prop. 2.4.1 there is no ambiguity in doing so. With this result, it is possible to prove a stronger result than Lemma 2.4.1, as it can be shown that only the value X_p is necessary to compute $\nabla_X Y|_p$.

Proposition 2.4.2 (Stronger locality of Koszul connections)

Let ∇ be a Koszul connection in a smooth vector bundle $\pi : E \rightarrow \mathcal{M}$. For every $X \in \mathfrak{X}(\mathcal{M})$,

$Y \in \Gamma(E)$ and $p \in \mathcal{M}$, the covariant derivative $\nabla_X Y|_p$ depends only on X_p and on the values Y in an arbitrarily small neighbourhood of p .

Proof. The claim on Y has been proved in Lemma 2.4.1. To prove the claim on X , it suffices by linearity to assume $X_p = 0$ and $\nabla_X Y|_p = 0$. Choosing coordinates (x^i) on $U \in \mathcal{T}(\mathcal{M}) : U \ni p$ and writing $X = X^i \partial_i$, by Prop. 2.4.1, for every $Y \in \Gamma(E|_U)$:

$$\nabla_X Y|_p = \nabla_{X^i \partial_i} Y|_p = X^i \nabla_{\partial_i} Y|_p = 0$$

which completes the proof. $\square$

By Prop. 2.4.1 and Prop. 2.4.2, it is possible to make sense of $\nabla_v Y$ with $v \in T_p \mathcal{M}$ and Y a smooth local section defined only on some neighbourhood of p : let X be a smooth vector field on a neighbourhood of p such that $X_p = v$, and set $\nabla_v Y \equiv \nabla_X Y|_p$, which does not depend on the extension chosen by Prop. 2.4.2.

§2.4.1.1 Connections on tangent bundles

Let $\mathcal{M}$ be a smooth manifold. A *connection on $\mathcal{M}$* is a Koszul connection in $T\mathcal{M}$, i.e. a map $\nabla : \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \rightarrow \mathfrak{X}(\mathcal{M})$ satisfying Def. 2.4.1.

In order to perform computations, it is necessary to examine how a connection appears in terms of a local frame, so let (E^i) be a smooth local frame for $T\mathcal{M}$ on some $U \in \mathcal{T}(\mathcal{M})$. Then, $\nabla_{E_i} E_j$ can be expanded as follows:

$$\nabla_{E_i} E_j = \Gamma_{ij}^k E_k \quad (2.4.5)$$

where $\Gamma_{ij}^k : U \rightarrow \mathbb{R}$ are n^3 smooth functions on U , called the **connection coefficients** of ∇ in the given frame.

Lemma 2.4.2 (Connection coefficients)

Let $\mathcal{M}$ be a smooth n -manifold, and let ∇ be a connection in $T\mathcal{M}$. Let (E^i) be a smooth local frame over some $U \in \mathcal{T}(\mathcal{M})$, and let $\{\Gamma_{ij}^k\}_{i,j,k=1,\dots,n}$ be the connection coefficients of ∇ in this frame. Writing $X, Y \in \mathfrak{X}(U)$ as $X = X^i E_i$ and $Y = Y^j E_j$:

$$\nabla_X Y = [X(Y^k) + \Gamma_{ij}^k X^i Y^j] E_k \quad (2.4.6)$$

Proof. By Def. 2.4.1:

$$\nabla_X Y = \nabla_X (Y^j E_j) = X(Y^j) E_j + Y^j \nabla_{X^i E_i} E_j = X(Y^j) E_j + X^i Y^j \Gamma_{ij}^k E_k$$

Renaming the first dummy index yields the thesis. $\square$

Under a frame transformations, connections coefficients have a non-homogeneous transformation law, thus showing that they are not the components of a tensor.

Proposition 2.4.3 (Transformation law for connection coefficients)

Let $\mathcal{M}$ be a smooth manifold, and let ∇ be a connection in $T\mathcal{M}$. Let (E_i) and $(\tilde{E}_i)$ be two smooth local frames for $T\mathcal{M}$ on some $U \in \mathcal{T}(\mathcal{M})$ related by $\tilde{E}_i = A^j_i E_j$, for some matrix of smooth functions $[A^j_i]$, and denote the connection coefficients of ∇ in these frames respectively as Γ_{ij}^k

and $\tilde{\Gamma}_{ij}^k$. Then:

$$\tilde{\Gamma}_{ij}^k = (A^{-1})^k_p A^q_i [A^r_j \Gamma_{qr}^p + E_q(A^p_j)] \quad (2.4.7)$$

Proof. By direct computation:

$$\begin{aligned} \tilde{\Gamma}_{ij}^k \tilde{E}_k &= \nabla_{\tilde{E}_i} \tilde{E}_j = \nabla_{A^q_i E_q} (A^r_j E_r) = A^q_i \nabla_{E_q} (A^r_j E_r) \\ &= A^q_i [A^r_j \Gamma_{qr}^p E_p + E_q(A^r_j) E_r] = A^q_i [A^r_j \Gamma_{qr}^p + E_q(A^p_j)] E_p \end{aligned}$$

Writing $E_p = (A^{-1})^k_p \tilde{E}_k$ completes the proof. $\square$

The importance of connections comes from the fact that it is always possible to find one on a smooth manifold.

Theorem 2.4.1 (Existence of connections on tangent bundles)

The tangent bundle of every smooth manifold admits a Koszul connection.

Proof. First, an auxiliary result is needed.

Lemma 2.4.3 (Construction of connections)

Let $\mathcal{M}$ be a smooth n -manifold which admits a global frame (E_i) . Eq. 2.4.6 gives a one-to-one correspondence between connections in $T\mathcal{M}$ and choices of n^3 smooth $\mathbb{R}$ -functions $\{\Gamma_{ij}^k\}_{i,j,k=1,\dots,n}$ on $\mathcal{M}$.

Proof. Every connection determines a set of smooth functions $\{\Gamma_{ij}^k\}_{i,j,k=1,\dots,n}$, defined by Eq. 2.4.5, which satisfy Eq. 2.4.6 by Lemma 2.4.2.

On the other hand, given $\{\Gamma_{ij}^k\}_{i,j,k=1,\dots,n}$, it is possible to define $\nabla_X Y$ by Eq. 2.4.6: the resulting expression is clearly smooth if X and Y are smooth, $\mathbb{R}$ -linear in Y and $\mathcal{C}^\infty(\mathcal{M})$ -linear in X , and the product rule in Def. 2.4.1 follows from a straightforward calculation. $\square$

Let $\mathcal{M}$ be a smooth manifold, and cover $\mathcal{M}$ with coordinate charts $\{U_\alpha\}$: by Lemma 2.4.3, the existence of a connection ∇^α on each U_α is guaranteed. Choosing a partition of unity $\{\varphi_\alpha\}$ subordinate to U_α , define:

$$\nabla_X Y \equiv \sum_\alpha \varphi_\alpha \nabla_X^\alpha Y$$

Since the set of supports of the φ_α functions is locally finite, the right-hand side contains only finitely-many non-zero terms in a neighbourhood of each point, so it defines a smooth vector field on $\mathcal{M}$. Moreover, this definition is manifestly $\mathbb{R}$ -linear in Y and $\mathcal{C}^\infty(\mathcal{M})$ -linear in X . For the product rule:

$$\nabla_X(fY) = \sum_\alpha \varphi_\alpha [X(f)Y + f \nabla_X^\alpha Y] = X(f)Y \sum_\alpha \varphi_\alpha + f \sum_\alpha \varphi_\alpha \nabla_X^\alpha Y = X(f) \nabla_X Y + f \nabla_X Y$$

which completes the proof. $\square$

Since the tangent bundle of any smooth manifold $\mathcal{M}$ always admits a connection ∇^0 , the set $\mathcal{A}(T\mathcal{M})$ of all connections in $T\mathcal{M}$ can be shown to be the following affine space:

$$\mathcal{A}(T\mathcal{M}) = \{\nabla^0 + D : D \in \mathfrak{T}_2^1(\mathcal{M})\} \quad (2.4.8)$$

where $D \in \mathfrak{T}_2^1(\mathcal{M})$ is interpreted as a map $\mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \rightarrow \mathfrak{X}(\mathcal{M})$, and $\nabla^0 + D : \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \rightarrow \mathfrak{X}(\mathcal{M})$ acts on $X, Y \in \mathfrak{X}(\mathcal{M})$ like:

$$(\nabla^0 + D)_X Y = \nabla_X^0 Y + D(X, Y)$$

§2.4.1.2 Covariant derivatives of tensor fields

By Def. 2.4.1, a connection in $T\mathcal{M}$ gives a way of computing covariant derivatives of smooth vector fields: it can be shown that every connection in $T\mathcal{M}$ also automatically induces connections in all tensor bundles over $\mathcal{M}$, thus allowing to compute covariant derivatives of arbitrary-rank smooth tensor fields.

Proposition 2.4.4 (Induced connection on tensor bundles)

Let $\mathcal{M}$ be a smooth manifold, and let ∇ be a connection in $T\mathcal{M}$. Then, ∇ uniquely determines a connection in each tensor bundle $T_l^k \mathcal{M}$, also denoted by ∇ , satisfying the following conditions:

- (1) the two connections agree in $T_0^1 \mathcal{M} = T\mathcal{M}$;
- (2) ∇ in $T_0^0 \mathcal{M} = \mathcal{M} \times \mathbb{R}$ is given by ordinary differentiation of functions:

$$\nabla_X f = X(f)$$

- (3) ∇ obeys the following product rule with respect to tensor products:

$$\nabla_X (F \otimes G) = (\nabla_X F) \otimes G + F \otimes (\nabla_X G) \quad (2.4.9)$$

- (4) ∇ commutes with all contractions: if tr denotes a trace on any pairs of indices (one covariant and one contravariant), then:

$$\nabla_X (\text{tr } F) = \text{tr}(\nabla F)$$

- (5) ∇ obeys the following product rule with respect to the natural pairing between a covector field $\omega \in \mathfrak{X}^*(\mathcal{M})$ and a vector field $Y \in \mathfrak{X}(\mathcal{M})$:

$$\nabla_X \langle \omega, Y \rangle = \langle \nabla_X \omega, Y \rangle + \langle \omega, \nabla_X Y \rangle \quad (2.4.10)$$

- (6) for all $F \in \mathfrak{T}_l^k(\mathcal{M})$, $\omega^1, \dots, \omega^k \in \mathfrak{X}^*(\mathcal{M})$ and $Y_1, \dots, Y_l \in \mathfrak{X}(\mathcal{M})$:

$$\begin{aligned} (\nabla_X F)(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l) &= X(F(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l)) \\ &\quad - \sum_{i=1}^k F(\omega^1, \dots, \nabla_X \omega^i, \dots, \omega^k, Y_1, \dots, Y_l) \\ &\quad - \sum_{j=1}^l F(\omega^1, \dots, \omega^k, Y_1, \dots, \nabla_X Y_j, \dots, Y_l) \end{aligned} \quad (2.4.11)$$

The connection so defined is simply called a “connection in $\mathcal{M}$ ”. In particular, Eq. 2.4.10 allows to compute the covariant derivative of a covector field $\omega \in \mathfrak{X}^*(\mathcal{M})$:

$$(\nabla_X \omega)(Y) = X(\omega(Y)) - \omega(\nabla_X Y) \quad (2.4.12)$$

All of these expressions, however, are not very practical for explicit computations, since computing the value of $\nabla_X F$ at a point requires extending all of its arguments to vector/covector fields on an open set, and then computing a great number of derivatives. A more useful computation strategy makes use of local frames.

Lemma 2.4.4 (Computation of covariant derivatives)

Let $\mathcal{M}$ be a smooth manifold, and let ∇ be a connection in $T\mathcal{M}$. Let (E_i) be a local frame on $\mathcal{M}$, with dual coframe (ε^i) , and let $\{\Gamma_{ij}^k\}$ be the connection coefficients of ∇ with respect to this frame. Let $X \in \mathfrak{X}(\mathcal{M})$ be locally expressed in this frame as $X = X^i E_i$. Then:

- (1) the covariant derivative of $\omega = \omega_i \varepsilon^i \in \mathfrak{X}^*(\mathcal{M})$ is given locally by:

$$\nabla_X \omega = [X(\omega_k) - \Gamma_{jk}^i X^j \omega_i] \varepsilon^k \quad (2.4.13)$$

- (2) given $F \in \mathfrak{T}_l^k(\mathcal{M})$ expressed locally as:

$$F = F_{j_1 \dots j_l}^{i_1 \dots i_k} E_{i_1} \otimes \dots \otimes E_{i_k} \otimes \varepsilon^{j_1} \otimes \dots \otimes \varepsilon^{j_l}$$

then its covariant derivative is given locally in components by:

$$(\nabla_X F)_{j_1 \dots j_l}^{i_1 \dots i_k} = \left[X(F_{j_1 \dots j_l}^{i_1 \dots i_k}) + \sum_{s=1}^k \Gamma_{mp}^{i_s} X^m F_{j_1 \dots j_l}^{i_1 \dots p \dots i_k} - \sum_{s=1}^l \Gamma_{mj_s}^p X^m F_{j_1 \dots p \dots j_l}^{i_1 \dots i_k} \right] \quad (2.4.14)$$

Since the covariant derivative $\nabla_X F$ of a tensor field is $\mathcal{C}^\infty(\mathcal{M})$ -linear in X , the covariant derivatives of F in all directions can be encoded in a single tensor whose rank is one more than the rank of F .

Proposition 2.4.5 (Total covariant derivative)

Let $\mathcal{M}$ be a smooth manifold, and let ∇ be a connection in $T\mathcal{M}$. For every $F \in \mathfrak{T}_l^k(\mathcal{M})$, the following map:

$$\nabla F : \underbrace{\mathfrak{X}^*(\mathcal{M}) \times \dots \times \mathfrak{X}^*(\mathcal{M})}_{k \text{ times}} \times \underbrace{\mathfrak{X}(\mathcal{M}) \times \dots \times \mathfrak{X}(\mathcal{M})}_{l+1 \text{ times}} \rightarrow \mathcal{C}^\infty(\mathcal{M})$$

$$(\nabla F)(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l, X) = (\nabla_X F)(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l)$$

defines a smooth $(k, l+1)$ -tensor on $\mathcal{M}$ called the **total covariant derivative** of F .

To make expression easier to read, usually only the components with respect to the local frame are shown, and the covariant derivative is represented by a semicolon. For example, the (total) covariant derivative of a vector field $Y = Y^i E_i \in \mathfrak{X}(\mathcal{M})$ is a $(1, 1)$ -tensor $\nabla Y = Y_{;j}^i E_i \otimes \varepsilon^j$ with components:

$$Y_{;j}^i = E_j(Y^i) + \Gamma_{jk}^i Y^k \quad (2.4.15)$$

In a similar way, given $\omega = \omega_i \varepsilon^i \in \mathfrak{X}^*(\mathcal{M})$ is $\nabla \omega = \omega_{i;j} \varepsilon^i \otimes \varepsilon^j$, with components:

$$\omega_{i;j} = E_j(\omega_i) - \Gamma_{ji}^k \omega_k \quad (2.4.16)$$

The generalization to an arbitrary $F \in \mathfrak{T}_l^k(\mathcal{M})$ is then trivial:

$$F_{j_1 \dots j_l; m}^{i_1 \dots i_k} = E_m(F_{j_1 \dots j_l}^{i_1 \dots i_k}) + \sum_{s=1}^k \Gamma_{mp}^{i_s} F_{j_1 \dots j_l}^{i_1 \dots p \dots i_k} - \sum_{s=1}^l \Gamma_{mj_s}^p F_{j_1 \dots p \dots j_l}^{i_1 \dots i_k} \quad (2.4.17)$$

§2.5 Levi–Civita connections

§2.5.1 Torsion and curvature

Even though the connection is not a tensor, it is used to construct two important tensors.

Definition 2.5.1 (Torsion tensor)

The **torsion** is a (1,2) tensor defined on $\Omega^1(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M})$ as:

$$\mathsf{T}(\omega, X, Y) := \omega(\nabla_X Y - \nabla_Y X - [X, Y]) \quad (2.5.1)$$

Alternatively, the torsion can be viewed as a map $\mathsf{T} : \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \rightarrow \mathfrak{X}(\mathcal{M})$ such that:

$$\mathsf{T}(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y] \quad (2.5.2)$$

Definition 2.5.2 (Curvature tensor)

The **curvature** is a (1,3) tensor defined on $\Omega^1(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M})$ as:

$$\mathsf{R}(\omega, Z, X, Y) := \omega(\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z) \quad (2.5.3)$$

Alternatively, the curvature can be viewed as a map from $\mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M})$ to the space of differential operators on $\mathfrak{X}(\mathcal{M})$ such that:

$$\mathsf{R}(X, Y) = \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]} \quad (2.5.4)$$

The fact that these are indeed tensors, i.e. they are linear in each argument, can be shown by direct calculation, recalling that $[fX, Y] = f[X, Y] - Y(f)X$.

Proposition 2.5.1

On the coordinate basis $\{\partial_\mu\}$ and $\{dx^\mu\}$ the torsion components are:

$$T^\rho_{\mu\nu} = \Gamma^\rho_{\mu\nu} - \Gamma^\rho_{\nu\mu} \quad (2.5.5)$$

Proof. By direct calculation:

$$\begin{aligned} T^\rho_{\mu\nu} &= \mathsf{T}(dx^\rho, \partial_\mu, \partial_\nu) = dx^\mu(\nabla_\mu \partial_\nu - \nabla_\nu \partial_\mu - [\partial_\mu, \partial_\nu]) \\ &= dx^\mu(\partial_\mu \partial_\nu - \Gamma^\sigma_{\mu\nu} \partial_\sigma - \partial_\nu \partial_\mu + \Gamma^\sigma_{\nu\mu} \partial_\sigma) \\ &= [\Gamma^\sigma_{\mu\nu} - \Gamma^\sigma_{\nu\mu}] \delta^\mu_\sigma = \Gamma^\rho_{\mu\nu} - \Gamma^\rho_{\nu\mu} \end{aligned}$$

which is the thesis. $\square$

Interestingly, even though $\Gamma^\rho_{\mu\nu}$ is not a tensor, its anti-symmetric part $\Gamma^\rho_{[\mu\nu]} = \frac{1}{2}T^\sigma_{\mu\nu}$ is. Clearly, the torsion tensor is anti-symmetric in its lower indices, thus for connections which are symmetric in their lower indices the torsion is null: such connections are said to be **torsion-free**.

Proposition 2.5.2

On the coordinate basis $\{\partial_\mu\}$ and $\{dx^\mu\}$ the curvature components are:

$$R^\sigma_{\rho\mu\nu} = \partial_\mu \Gamma^\sigma_{\nu\rho} - \partial_\nu \Gamma^\sigma_{\mu\rho} + \Gamma^\lambda_{\nu\rho} \Gamma^\sigma_{\mu\lambda} - \Gamma^\lambda_{\mu\rho} \Gamma^\sigma_{\nu\lambda} \quad (2.5.6)$$

Proof. By direct calculation:

$$\begin{aligned} R(dx^\sigma, \partial_\rho, \partial_\mu, \partial_\nu) &= dx^\sigma (\nabla_\mu \nabla_\nu \partial_\rho - \nabla_\nu \nabla_\mu \partial_\rho - \nabla_{[\partial_\mu, \partial_\nu]} \partial_\rho) \\ &= dx^\sigma (\nabla_\mu \nabla_\nu \partial_\rho - \nabla_\nu \nabla_\mu \partial_\rho) \\ &= dx^\sigma (\nabla_\mu (\Gamma^\lambda_{\nu\rho} \partial_\lambda) - \nabla_\nu (\Gamma^\lambda_{\mu\rho} \partial_\lambda)) \\ &= dx^\sigma ((\partial_\mu \Gamma^\lambda_{\nu\rho}) \partial_\lambda + \Gamma^\lambda_{\nu\rho} \Gamma^\tau_{\mu\lambda} \partial_\tau - (\partial_\nu \Gamma^\lambda_{\mu\rho}) \partial_\lambda - \Gamma^\lambda_{\mu\rho} \Gamma^\tau_{\nu\lambda} \partial_\tau) \\ &= \partial_\mu \Gamma^\sigma_{\nu\rho} - \partial_\nu \Gamma^\sigma_{\mu\rho} + \Gamma^\lambda_{\nu\rho} \Gamma^\sigma_{\mu\lambda} - \Gamma^\lambda_{\mu\rho} \Gamma^\sigma_{\nu\lambda} \end{aligned}$$

which completes the proof. $\square$

Clearly, the curvature tensor is anti-symmetric in its last two lower indices, i.e. $R^\sigma_{\rho\mu\nu} = R^\sigma_{\rho[\mu\nu]}$, and it can in fact be written as:

$$R^\sigma_{\rho\mu\nu} = 2\partial_{[\mu} \Gamma^\sigma_{\nu]\rho} + 2\Gamma^\sigma_{[\mu|\lambda|} \Gamma^\lambda_{\nu]\rho} \quad (2.5.7)$$

Theorem 2.5.1 (Ricci identity)

$$2\nabla_{[\mu} \nabla_{\nu]} Z^\sigma = R^\sigma_{\rho\mu\nu} Z^\rho - T^\rho_{\mu\nu} \nabla_\rho Z^\sigma \quad (2.5.8)$$

Proof. By direct calculation:

$$\begin{aligned} \nabla_{[\mu} \nabla_{\nu]} Z^\sigma &= \partial_{[\mu} (\nabla_{\nu]} Z^\sigma) + \Gamma^\sigma_{[\mu|\lambda|} \nabla_{\nu]} Z^\lambda - \Gamma^\rho_{[\mu\nu]} \nabla_\rho Z^\sigma \\ &= \partial_{[\mu} \partial_{\nu]} Z^\sigma + (\partial_{[\mu} \Gamma^\sigma_{\nu]\rho}) Z^\rho + (\partial_{[\mu} Z^\rho) \Gamma^\sigma_{\nu]\rho} + \Gamma^\sigma_{[\mu|\lambda|} \partial_{\nu]} Z^\lambda + \Gamma^\sigma_{[\mu|\lambda|} \Gamma^\lambda_{\nu]\rho} Z^\rho - \frac{1}{2} T^\rho_{\mu\nu} \nabla_\rho Z^\sigma \\ &= (\partial_{[\mu} \Gamma^\sigma_{\nu]\rho} + \Gamma^\sigma_{[\mu|\lambda|} \Gamma^\lambda_{\nu]\rho}) Z^\rho - \frac{1}{2} T^\rho_{\mu\nu} \nabla_\rho Z^\sigma \\ &= \frac{1}{2} R^\sigma_{\rho\mu\nu} Z^\rho - \frac{1}{2} T^\sigma_{\mu\nu} \nabla_\rho Z^\sigma \end{aligned}$$

which is the thesis. $\square$

§2.5.2 Levi-Civita connection

The discussion on the connection has so far been independent of the metric. Starting to consider it, an important result is the fundamental theorem of Riemannian geometry.

Theorem 2.5.2 (Riemann's theorem)

On a metric manifold $(\mathcal{M}, g)$, there exists a unique torsion-free connection that is compatible with the metric, i.e. for all $X \in \mathfrak{X}(\mathcal{M})$:

$$\nabla_X g = 0 \quad (2.5.9)$$

This is called the **Levi-Civita connection**.

Proof. WTS uniqueness: suppose such a connection exists. Then, by Leibniz rule:

$$X(g(Y, Z)) = \nabla_X(g(Y, Z)) = (\nabla_X g)(Y, Z) + g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

Since $\nabla_X g = 0$, by cyclic permutations of X, Y and Z :

$$\begin{aligned} X(g(Y, Z)) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z) \\ Y(g(Z, X)) &= g(\nabla_Y Z, X) + g(Z, \nabla_Y X) \\ Z(g(X, Y)) &= g(\nabla_Z X, Y) + g(X, \nabla_Z Y) \end{aligned}$$

Since the connection is torsion-free, $\nabla_X Y - \nabla_Y X = [X, Y]$, thus these equations become:

$$\begin{aligned} X(g(Y, Z)) &= g(\nabla_Y X, Z) + g(\nabla_X Z, Y) + g([X, Y], Z) \\ Y(g(Z, X)) &= g(\nabla_Z Y, X) + g(\nabla_Y X, Z) + g([Y, Z], X) \\ Z(g(X, Y)) &= g(\nabla_X Z, Y) + g(\nabla_Z Y, X) + g([Z, X], Y) \end{aligned}$$

Adding the first two and subtracting the third:

$$\begin{aligned} g(\nabla_Y X, Z) &= \frac{1}{2} [X(g(Y, Z)) + Y(g(Z, X)) + Z(g(X, Y)) \\ &\quad - g([X, Y], Z) - g([Y, Z], X) + g([Z, X], Y)] \end{aligned}$$

The metric is non-degenerate, thus this uniquely specifies the connection. By direct calculation it can be shown that it indeed satisfies all the properties of a connection. $\square$

It is possible to explicit the components of the Levi-Civita connection in terms of the metric.

Proposition 2.5.3 (Christoffel symbols)

On the coordinate basis $\{\partial_\mu\}$ and $\{dx^\mu\}$ the Levi-Civita connection's components, called **Christoffel symbols**, are:

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\rho} (\partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu}) \quad (2.5.10)$$

Proof. Recalling that $[\partial_\mu, \partial_\nu] = 0$, $\Gamma_{\mu\nu}^\lambda g_{\lambda\rho} = g(\nabla_\mu \partial_\nu, \partial_\rho) = \frac{1}{2} (\partial_\nu g_{\mu\rho} + \partial_\mu g_{\nu\rho} - \partial_\rho g_{\mu\nu})$. $\square$

Example 2.5.1 (Euclidean space)

In flat space $\mathbb{R}^n$, endowed with either Euclidean or Minkowski metric, it is always possible to choose Cartesian coordinates, in which case the Christoffel symbols vanish. Being the Riemann tensor a genuine tensor, it therefore will vanish in all possible coordinate systems on $\mathbb{R}^n$, even in those with $\Gamma_{\mu\nu}^\rho \neq 0$: this expresses the flatness of $\mathbb{R}^n$.

§2.5.3 Einstein tensor

In addition to the anti-symmetry in its last two indices, the Riemann tensor presents additional subtle symmetries when using the Levi-Civita connection.

Proposition 2.5.4 (Symmetries of the Riemann tensor)

On a metric manifold with a Levi–Civita connection:

$$R_{\sigma\rho\mu\nu} = -R_{\sigma\rho\nu\mu} = -R_{\rho\sigma\mu\nu} = R_{\mu\nu\sigma\rho} \quad R_{\sigma[\rho\mu\nu]} = 0 \quad (2.5.11)$$

Proof. Set normal coordinates centered at a point p : then, $\Gamma_{\nu\rho}^\mu = 0$ and $\partial_\mu g^{\lambda\sigma} = 0$ at that point. At p , the Riemann tensor can be written as:

$$\begin{aligned} R_{\sigma\rho\mu\nu} &= g_{\sigma\lambda} R_{\rho\mu\nu}^\lambda = g_{\sigma\lambda} [\partial_\mu \Gamma_{\nu\rho}^\lambda - \partial_\nu \Gamma_{\mu\rho}^\lambda] \\ &= \frac{1}{2} [\partial_\mu (\partial_\nu g_{\sigma\rho} + \partial_\rho g_{\nu\sigma} - \partial_\sigma g_{\nu\rho}) - \partial_\nu (\partial_\mu g_{\sigma\rho} + \partial_\rho g_{\mu\sigma} - \partial_\sigma g_{\mu\rho})] \\ &= \frac{1}{2} [\partial_\mu \partial_\rho g_{\nu\sigma} - \partial_\mu \partial_\sigma g_{\nu\rho} - \partial_\nu \partial_\rho g_{\mu\sigma} + \partial_\nu \partial_\sigma g_{\mu\rho}] \end{aligned}$$

The symmetries are then manifest, and being these tensor equations they are valid in all coordinate systems. $\square$

An important computation tool is the Bianchi identity.

Theorem 2.5.3 (Bianchi identity)

On a metric manifold with a Levi–Civita connection:

$$\nabla_{[\lambda} R_{\sigma\rho]\mu\nu} = 0 \quad \Longleftrightarrow \quad R_{\rho[\mu\nu;\lambda]}^\sigma = 0 \quad (2.5.12)$$

Proof. The two equations are equivalent, so the proof is of the first one. In normal coordinates $\nabla_\mu = \partial_\mu$ at p , so schematically: $R = \partial\Gamma + \Gamma\Gamma$, thus $\nabla R = \partial R = \partial^2\Gamma + \Gamma\partial\Gamma = \partial^2\Gamma$. Explicitly:

$$\partial_\lambda R_{\sigma\rho\mu\nu} = \frac{1}{2} \partial_\lambda [\partial_\mu \partial_\rho g_{\nu\sigma} - \partial_\mu \partial_\sigma g_{\nu\rho} - \partial_\nu \partial_\rho g_{\mu\sigma} + \partial_\nu \partial_\sigma g_{\mu\rho}]$$

Anti-symmetrizing the appropriate indices yields the result. $\square$

Note that Eq. 2.5.11-2.5.12 do not require that the connection is a Levi–Civita connection, but are valid for general torsion-free connections.

Definition 2.5.3 (Ricci tensor and scalar)

On a metric manifold, the **Ricci tensor** is defined as:

$$R_{\mu\nu} := R_{\mu\rho\nu}^\rho \quad (2.5.13)$$

The **Ricci scalar** is defined as:

$$R := g^{\mu\nu} R_{\mu\nu} \quad (2.5.14)$$

From Eq. 2.5.11, the Ricci tensor is clearly symmetric: $R_{\mu\nu} = R_{\nu\mu}$.

Proposition 2.5.5

On a metric manifold:

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu R \quad (2.5.15)$$

Proof. Writing explicitly Bianchi identity:

$$\nabla_\lambda R_{\sigma\rho\mu\nu} + \nabla_\sigma R_{\rho\lambda\mu\nu} + \nabla_\rho R_{\lambda\sigma\mu\nu} = 0$$

Contracting with $g^{\mu\lambda}g^{\rho\nu}$:

$$\nabla^\mu R_{\mu\sigma} - \nabla_\sigma R + \nabla^\nu R_{\nu\sigma} = 0$$

which yields the thesis. $\square$

An important tensor obtained as a combination of the Ricci tensor and scalar is the **Einstein tensor**, which encodes information on the curvature of the metric manifold on which it is defined:

$$G_{\mu\nu} := R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} \quad (2.5.16)$$

Most importantly, the Einstein tensor is covariantly conserved, as of Eq. 2.5.15:

$$\nabla^\mu G_{\mu\nu} = 0 \quad (2.5.17)$$

§2.5.4 Gauss' theorem

The divergence theorem (or Gauss' theorem) states that the integral of a total derivative is a boundary term. It is possible to express this theorem on curved manifolds in a convenient way.

Lemma 2.5.1

$$\Gamma_{\mu\nu}^\mu = \frac{1}{\sqrt{g}}\partial_\nu\sqrt{g} \quad (2.5.18)$$

Proof. A useful identity for invertible matrices: $\text{tr} \log A = \log \det A$. Thus (WLOG $\det g > 0$):

$$\Gamma_{\mu\nu}^\mu = \frac{1}{2}g^{\mu\rho}\partial_\nu g_{\mu\rho} = \frac{1}{2}\text{tr}(g^{-1}\partial_\nu g) = \frac{1}{2}\text{tr}(\partial_\nu \log g) = \frac{1}{2}\partial_\nu \log \det g = \frac{1}{\sqrt{\det g}}\partial_\nu \sqrt{\det g}$$

which is the thesis. $\square$

Theorem 2.5.4 (Gauss' theorem)

Given a Riemannian manifold $(\mathcal{M}, g)$, consider a region $M \subseteq \mathcal{M}$ with boundary ∂M and let n^μ be an outward-pointing unit vector orthogonal to ∂M . Then, for any vector field X^μ on M :

$$\int_M d^n x \sqrt{g} \nabla_\mu X^\mu = \int_{\partial M} d^{n-1} x \sqrt{\gamma} n_\mu X^\mu \quad (2.5.19)$$

where γ_{ij} is the pull-back of the metric to ∂M and $\gamma \equiv \det \gamma_{ij}$.

Proof. From Lemma 2.5.1:

$$\sqrt{g} \nabla_\mu X^\mu = \sqrt{g} (\partial_\mu X^\mu + \Gamma_{\mu\nu}^\mu X^\nu) = \sqrt{g} \left(\partial_\mu X^\mu + X^\nu \frac{1}{\sqrt{g}} \partial_\nu \sqrt{g} \right) = \partial_\mu (\sqrt{g} X^\mu)$$

The integral becomes:

$$\int_M d^n x \sqrt{g} \nabla_\mu X^\mu = \int_M d^n x \partial_\mu (\sqrt{g} X^\mu)$$

This is the integral of an ordinary partial derivative, so the ordinary divergence theorem applies. To evaluate the integral on the boundary, it is convenient to pick coordinates so that ∂M is a

surface at constant x^n . Moreover, to simplify the proof, the possible metrics will be restricted to $g_{\mu\nu} = \text{diag}(\gamma_{ij}, N^2)$. By usual integration rules:

$$\int_M d^n x \partial_\mu (\sqrt{g} X^\mu) = \int_{\partial M} d^{n-1} x \sqrt{\gamma N^2} X^n$$

The unit normal vector is $n^\mu = (0, \dots, 0, \frac{1}{N})$, so that $g_{\mu\nu} n^\mu n^\nu = 1$, therefore $n_\mu = g_{\mu\nu} n^\nu = (0, \dots, 0, N)$. The proof is then concluded because:

$$\int_{\partial M} d^{n-1} x \sqrt{\gamma N^2} X^n = \int_{\partial M} d^{n-1} x \sqrt{\gamma} n_\mu X^\mu$$

which is the thesis. $\square$

Note that this theorem holds on Lorentzian manifolds too, with the condition that ∂M must be purely timelike or purely spacelike, ensuring that $\gamma \neq 0$ at any point.

§2.6 Parallel transport

The connection connects tangent spaces, or more generally any tensor vector space, at different points of the manifold: this map is called parallel transport and it is necessary for the definition of differentiation.

Definition 2.6.1 (Parallel transport)

Consider a vector field X and some associated integral curve γ , with coordinates $x^\mu(\tau)$ such that:

$$X^\mu|_\gamma = \frac{dx^\mu(\tau)}{d\tau}$$

A tensor field T is said to be **parallelly transported** along γ if:

$$\nabla_X T = 0 \tag{2.6.1}$$

Suppose that γ connects two points $p, q \in \mathcal{M}$: Eq. 2.6.1 provides a map from the tensor vector space defined at p to that defined at q . To illustrate this, consider the parallel transport of a vector field Y :

$$X^\nu (\partial_\nu Y^\mu + \Gamma_{\nu\rho}^\mu Y^\rho) = 0$$

Evaluating this equation on γ , considering $Y^\mu = Y^\mu(x(\tau))$:

$$\frac{dY^\mu}{d\tau} + X^\nu \Gamma_{\nu\rho}^\mu Y^\rho = 0 \tag{2.6.2}$$

These are a set of coupled ODEs, thus, given an initial condition (e.g. at $\tau = 0$, i.e. at p), these equations can be solved to find a unique vector at each point along the curve.

Note that the parallel transport depends both on the path (characterized by the vector field X) and on the connection.

Definition 2.6.2 (Geodesic)

Given a vector field X , a **geodesic** is an integral curve to X such that:

$$\nabla_X X = 0 \quad (2.6.3)$$

Proposition 2.6.1 (Geodesic equation)

A geodesic is described by:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0 \quad (2.6.4)$$

Proof. From the above calculations, along γ :

$$0 = \frac{dX^\mu}{d\tau} + \Gamma_{\nu\rho}^\mu X^\nu X^\rho = \frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau}$$

which is the geodesic equation. $\square$

For the Levi-Civita connection $\nabla_X g = 0$. If $Y \in \mathfrak{X}(\mathcal{M})$ is parallelly transported along a geodesic associated to $X \in \mathfrak{X}(\mathcal{M})$, then $\nabla_X Y = \nabla_X X = 0$, therefore $\frac{d}{d\tau}g(X, Y) = 0$: this ensures that the two tangent vectors always determine the same angles along the geodesic.

§2.6.1 Normal coordinates

Theorem 2.6.1 (Equivalence principle)

Given a Riemannian manifold $(\mathcal{M}, g)$ and $p \in \mathcal{M}$, in a neighbourhood of p it is always possible to find coordinates, called **normal coordinates**, such that $g_{\mu\nu}(p) = \delta_{\mu\nu}$ and $g_{\mu\nu,\rho}(p) = 0$.

Proof. By brute force, consider initial coordinates y^μ and find a change of coordinates x^μ which satisfy the requirements:

$$\frac{\partial y^\rho}{\partial x^\mu} \frac{\partial y^\sigma}{\partial x^\nu} \tilde{g}_{\rho\sigma} = g_{\mu\nu}$$

WLOG p is the origin of both sets of coordinates, so:

$$y^\rho = \frac{\partial y^\rho}{\partial x^\mu} \Big|_{x=0} x^\mu + \frac{1}{2} \frac{\partial^2 y^\rho}{\partial x^\mu \partial x^\nu} \Big|_{x=0} x^\mu x^\nu + \dots$$

This, together with the Taylor expansion of $\tilde{g}_{\rho\sigma}$, can be inserted in the transformation equation of the metric, thus finding a set of PDEs for each power of x , which can be solved to characterize the normal coordinates. For example, the first condition is:

$$\frac{\partial y^\rho}{\partial x^\mu} \Big|_{x=0} \frac{\partial y^\sigma}{\partial x^\nu} \Big|_{x=0} \tilde{g}_{\rho\sigma}(p) = \delta_{\mu\nu}$$

Given any $\tilde{g}_{\rho\sigma}(p)$, it is always possible to find $\frac{\partial y}{\partial x}$ that satisfies this condition. In fact, if $\dim_{\mathbb{R}} \mathcal{M} = n$, the Jacobian of the transformation has n^2 independent elements and the equation above puts $\frac{1}{2}n(n+1)$ constraints: the remaining free parameters are $\frac{1}{2}n(n-1)$, which is precisely the dimension of $\text{SO}(n)$, the symmetry group of the flat metric. A similar counting shows that $g_{\mu\nu,\rho}(p) = 0$ puts $\frac{1}{2}n^2(n+1)$ constraints, precisely the number of independent elements of the Hessian of the transformation. $\square$

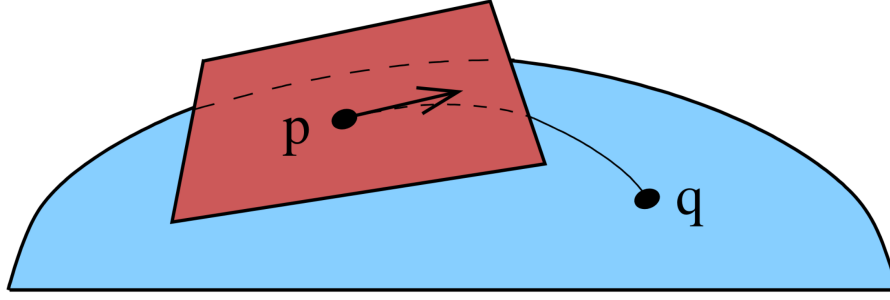


Figure 2.8: Visualization of the exponential map.

This theorem holds for Lorentzian manifolds too, but the flat metric is now $\eta_{\mu\nu}$ and its symmetry group is $\text{SO}(1, n-1)$. The condition $g_{\mu\nu,\rho}(p) = 0$ implies that $\Gamma_{\nu\rho}^{\mu}(p) = 0$, but generally the Christoffel symbols won't vanish away from p . Note, however, that it is not generally possible to ensure the vanishing of second derivatives too: indeed, $g_{\mu\nu,\rho\sigma}(p) = 0$ would put $\frac{1}{4}n^2(n+1)^2$ constraints, but the independent $\frac{\partial^3 y}{\partial^3 x}$ terms are $\frac{1}{6}n^2(n+1)(n+2)$, thus leaving $\frac{1}{12}n^2(n^2-1)$ free terms: this is precisely the number of independent components of the Riemann tensor, therefore in general it is not possible to pick coordinates as to make the Riemann tensor vanish too.

§2.6.1.1 Exponential map

A simple way to construct normal coordinates is the following: given a tangent vector $X_p \in T_p \mathcal{M}$, there is a unique affinely parametrized geodesic through p with tangent vector X_p at p ; then, any point q in the neighbourhood of p is labelled by the coordinates of the geodesic that takes from p to q a fixed amount of time.

Analytically, introducing a coordinate system (not necessarily normal) $\tilde{x}^{\mu}$ in the neighbourhood of p , an affinely parametrized geodesic solves Eq. 2.6.4, with initial conditions $\frac{\partial \tilde{x}^{\mu}}{\partial \tau}|_{\tau=0} = \tilde{X}_p^{\mu}$ and $\tilde{x}^{\mu}(\tau=0) = 0$ that make the solution unique. The uniqueness of the solution allows to define a map $\text{Exp} : T_p \mathcal{M} \rightarrow \mathcal{M}$, called the **exponential map**, which acts as follows: given $X_p \in T_p \mathcal{M}$, construct the appropriate geodesic as above and follow it for a fixed affine distance, conventionally $\tau = 1$, to get a new point $q \in \mathcal{M}$. See Fig. Fig. 2.8 for visual aid. Obviously, there may be points which cannot be reached from p by geodesics, or there may be tangent vectors X_p for which Exp is ill-defined: in General Relativity, this occurs when spacetime has singularities, but these are not relevant issues.

Pick a basis $\{e_{\mu}\}$ of $T_p \mathcal{M}$. Then $\text{Exp} : T_p \mathcal{M} \ni X^{\mu} e_{\mu} \mapsto q \in \mathcal{M}$, thus it is possible to assign coordinates in the neighbourhood of p such that $x^{\mu}(q) = X^{\mu}$: these are the normal coordinates. To show this, note that if $\{e_{\mu}\}$ is orthonormal, then the geodesics will point in orthogonal directions, ensuring that $g_{\mu\nu}(p) = \delta_{\mu\nu}$. Now, fix a point q associated to a given tangent vector $X_p \in T_p \mathcal{M}$: this means that q is at distance $\tau = 1$ from p along the given geodesic. Note that the geodesic equation is homogeneous in τ , thus in general $\text{Exp} : \tau X_p \mapsto x^{\mu}(\tau) = \tau X^{\mu}$, which means that geodesics take a simple form in these coordinates:

$$x^{\mu}(\tau) = \tau X^{\mu}$$

Being these geodesics, they must solve Eq. 2.6.4, that is:

$$\Gamma_{\nu\rho}^{\mu}(x(\tau))X^{\nu}X^{\rho} = 0$$

which holds at any point along the geodesic, i.e. at any $\tau \in \mathbb{R}^+$. At most points $x(\tau)$, this equation only holds for those choices of X^{μ} tangent to the geodesics. However, at $x(0) = 0$, i.e. at p , it must hold for any tangent vector: this means that $\Gamma_{(\nu\rho)}^{\mu}(p) = 0$ which, for a torsion-free connection,

ensures that $\Gamma_{\nu\rho}^{\mu}(p) = 0$. But vanishing Christoffel symbols imply a vanishing first derivative of the metric: for the Levi-Civita connection $2g_{\mu\sigma}\Gamma_{\nu\rho}^{\sigma} = g_{\mu\nu,\rho} + g_{\mu\rho,\nu} - g_{\nu\rho,\mu}$, thus symmetrizing $(\mu\nu)$ cancels the last two terms, leaving an identity that, evaluated at p , gives $g_{\mu\nu,\rho}(p) = 0$. Hence, these are indeed normal coordinates.

§2.6.1.2 Equivalence principle

Normal coordinates are conceptually important in General Relativity: an observer at point p who parametrizes their immediate surroundings using coordinates constructed by geodesics will experience a locally flat metric. This is precisely Einstein's equivalence principle: any free-falling observer, performing local experiments, will not experience a gravitational field. The formal definition of free-falling observer is an observer which follows geodesics, while the local lack of gravitational field means $g_{\mu\mu}(p) = \eta_{\mu\nu}$. In this context, normal coordinates are called **local inertial frame**.

To understand what "local" means, note that there is a way to distinguish whether a gravitational field is present at p : a non-vanishing Riemann tensor. This depends on the second derivatives of the metric, which in general will be non-vanishing. However, to measure the effects of the Riemann tensor, one needs to compare the results of experiments at p and at a nearby point q : this is a non-local observation.

§2.6.2 Curvature and torsion

With reference to Fig. 2.9, consider a tangent vector $Z_p \in T_p\mathcal{M}$ and two vector fields $X, Y \in \mathfrak{X}(\mathcal{M}) : [X, Y] = 0$, i.e. they are linearly independent. Construct two curves γ, γ' as in figure, both leading to a point $r \in \mathcal{M}$ which, for simplicity, is close to p . It is possible to impose normal coordinates centered at p such that $x^\mu = (\tau, \sigma, \dots)$, so that $X = \frac{\partial}{\partial \tau}$ and $Y = \frac{\partial}{\partial \sigma}$: then $x^\mu(p) = (0, 0, 0, \dots)$, $x^\mu(q) = (\delta\tau, 0, 0, \dots)$, $x^\mu(s) = (0, \delta\sigma, 0, \dots)$ and $x^\mu(r) = (\delta\tau, \delta\sigma, 0, \dots)$, with $\delta\tau, \delta\sigma$

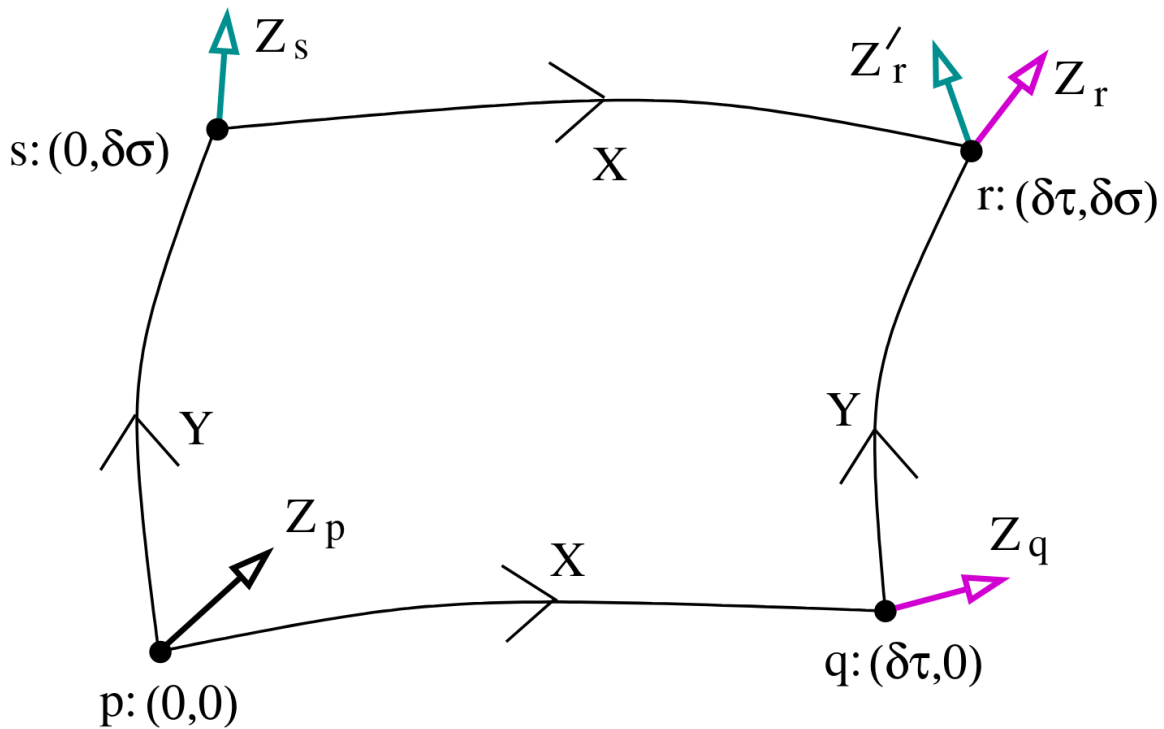


Figure 2.9: Parallel transport along different paths.

small.

First, parallel transport Z_p along X to Z_q , so that Z^μ solves:

$$\frac{dZ^\mu}{d\tau} + X^\nu \Gamma_{\nu\rho}^\mu Z^\rho = 0$$

In normal coordinates $\Gamma_{\nu\rho}^\mu(p) = 0$, thus $\frac{dZ^\mu}{d\tau}\big|_{\tau=0} = 0$ and the Taylor expansion is:

$$\begin{aligned} Z_q^\mu &= Z_p^\mu + \frac{\delta\tau^2}{2} \frac{d^2 Z^\mu}{d\tau^2} \bigg|_{\tau=0} + o(\delta\tau^3) \\ &= Z_p^\mu - \frac{\delta\tau^2}{s} \left[X^\nu Z^\rho \frac{d\Gamma_{\nu\rho}^\mu}{d\tau} + \frac{dX^\nu}{d\tau} Z^\rho \Gamma_{\nu\rho}^\mu + X^\nu \frac{dZ^\rho}{d\tau} \Gamma_{\nu\rho}^\mu \right]_{\tau=0} + o(\delta\tau^3) \\ &= Z_p^\mu - \frac{\delta\tau^2}{2} X^\nu Z^\rho \frac{d\Gamma_{\nu\rho}^\mu}{d\tau} \bigg|_{\tau=0} + o(\delta\tau^3) = Z_p^\mu - \frac{\delta\tau^2}{2} [X^\nu X^\sigma Z^\rho \Gamma_{\nu\rho,\sigma}^\mu]_p + o(\delta\tau^3) \end{aligned}$$

where $\frac{d}{d\tau} = X^\sigma \partial_\sigma$. Now, Z_q needs to be parallelly transported along Y to Z_r , but this time $\frac{dZ^\mu}{d\sigma}\big|_{\sigma=0}$ does not vanish, in general. From the parallel transport equation:

$$\frac{dZ^\mu}{d\sigma} \bigg|_{\sigma=0} = - [Y^\nu Z^\rho \Gamma_{\nu\rho}^\mu]_q = - [Y^\nu Z^\rho X^\sigma \Gamma_{\nu\rho,\sigma}^\mu]_p \delta\tau + o(\delta\tau^2)$$

The expansions of Y^ν and Z^ρ at leading order multiply $\Gamma_{\nu\rho}^\mu(p) = 0$, thus only contribute to higher order terms. Next order in $\delta\sigma$:

$$\frac{d^2 Z^\mu}{d\sigma^2} \bigg|_{\sigma=0} = - \left[\left(\frac{dY^\nu}{d\sigma} Z^\rho + Y^\nu \frac{dZ^\rho}{d\sigma} \right) \Gamma_{\nu\rho}^\mu + Y^\nu Z^\rho \frac{d\Gamma_{\nu\rho}^\mu}{d\sigma} \right]_q = - [Y^\nu Y^\sigma Z^\rho \Gamma_{\nu\rho,\sigma}^\mu]_p + o(\delta\tau)$$

The complete expansion thus is:

$$\begin{aligned} Z_r^\mu &= Z_q^\mu - [Y^\nu Z^\rho X^\sigma \Gamma_{\nu\rho,\sigma}^\mu]_p \delta\tau \delta\sigma - \frac{1}{2} [Y^\nu Y^\sigma Z^\rho \Gamma_{\nu\rho,\sigma}^\mu]_p \delta\sigma^2 + o(\delta^3) \\ &= Z_p^\mu - \frac{1}{2} \Gamma_{\nu\rho,\sigma}^\mu(p) [X^\nu X^\sigma Z^\rho \delta\tau^2 + 2Y^\nu Z^\rho X^\sigma \delta\tau \delta\sigma + Y^\nu Y^\sigma Z^\rho \delta\sigma^2] + o(\delta^3) \end{aligned}$$

Parallel transport along γ' leads to a similar expression (exchange of X and Y):

$$Z_r'^\mu = Z_p^\mu - \frac{1}{2} \Gamma_{\nu\rho,\sigma}^\mu(p) [Y^\nu Y^\sigma Z^\rho \delta\sigma^2 + 2X^\nu Z^\rho Y^\sigma \delta\sigma \delta\tau + X^\nu X^\sigma Z^\rho \delta\tau^2] + o(\delta^3)$$

The difference between the parallelly transported tangent vectors to leading order is:

$$\Delta Z_r^\mu = Z_r^\mu - Z_r'^\mu = - [\Gamma_{\nu\rho,\sigma}^\mu - \Gamma_{\sigma\rho,\nu}^\mu]_p [Y^\nu Z^\rho X^\sigma]_p \delta\tau \delta\sigma + o(\delta^3)$$

Recalling that $\Gamma_{\nu\rho}^\mu(p) = 0$, it is possible to write:

$$\Delta Z_r^\mu = - [R_{\rho\sigma\nu}^\mu Y^\nu Z^\rho X^\sigma]_p \delta\sigma \delta\tau + o(\delta^3) \quad (2.6.5)$$

It would be possible to evaluate the expression at r too, as it would differ only by higher order terms. Although the calculation was carried in a particular choice of coordinates, Eq. 2.6.5 is a tensor relation, therefore it must hold in all coordinate systems: the Riemann tensor thus determines the path dependence of parallel transport.

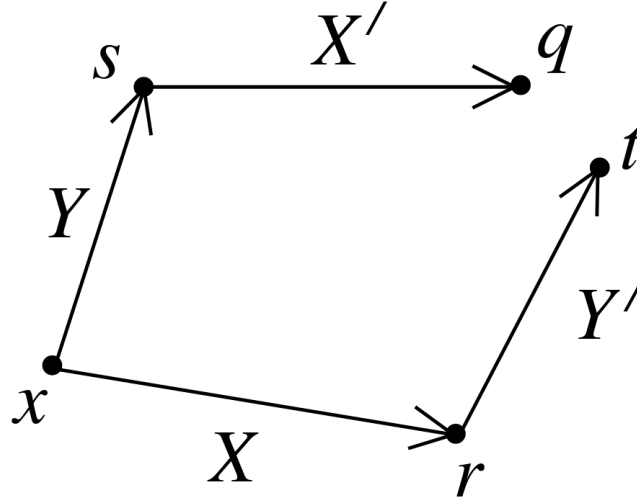


Figure 2.10: Visualization of torsion.

§2.6.2.1 Torsion

Consider two tangent vectors $X_p, Y_p \in T_p\mathcal{M}$ and a coordinate system x^μ such that $X_p = X^\mu \partial_\mu$ and $Y_p = Y^\mu \partial_\mu$. If $p : x^\mu$, as in Fig. 2.10 construct $r, s \in \mathcal{M}$ such that $r : x^\mu + \varepsilon X^\mu$ and $s : x^\mu + \varepsilon Y^\mu$, with ε an infinitesimal parameter. Now, parallel transport $X_p \in T_p\mathcal{M}$ along the direction of Y_p to $X'_s \in T_s\mathcal{M}$ and $Y_p \in T_p\mathcal{M}$ along X_p to $Y'_r \in T_r\mathcal{M}$; their components will be:

$$X'_s = (X^\mu - \varepsilon \Gamma_{\nu\rho}^\mu Y^\nu X^\rho) \quad Y'_r = (Y^\mu - \varepsilon \Gamma_{\nu\rho}^\mu X^\nu Y^\rho)$$

Repeating this process, starting from point s and moving along the direction of X'_s , a new point $q \in \mathcal{M}$ is determined, with coordinates:

$$q : x^\mu + \varepsilon (X^\mu + Y^\mu) - \varepsilon^2 \Gamma_{\nu\rho}^\mu Y^\nu X^\rho$$

Analogously, starting at point r and moving along the direction of Y'_r , a new point $t \in \mathcal{M}$ is determined, with coordinates:

$$t : x^\mu + \varepsilon (X^\mu + Y^\mu) - \varepsilon^2 \Gamma_{\nu\rho}^\mu X^\nu Y^\rho$$

If the connection is torsion-free, then $q \equiv t$. On the other hand, if $T_{\nu\rho}^\mu \neq 0$, the parallelogram fails to close, as in Fig. 2.10.

§2.6.3 Geodesic deviation

Definition 2.6.3 (Tangent and deviation vector fields)

Given a one-parameter family of geodesics $\{x^\mu(\tau; s)\}_{s \in \mathbb{R}}$ on a manifold $\mathcal{M}$, the **tangent vector field** and the **deviation vector field** are defined as:

$$X^\mu := \left. \frac{\partial x^\mu}{\partial \tau} \right|_s \quad S^\mu := \left. \frac{\partial x^\mu}{\partial s} \right|_\tau \quad (2.6.6)$$

The meaning of these vector fields is evident: the tangent vector field fixes a particular geodesics (i.e. a particular s) and assigns at each point of the geodesic its tangent vector, while the deviation vector field fixes a particular value of the affine parameter τ and assigns at each point with this value a vector which takes to a nearby geodesic (at the same τ).

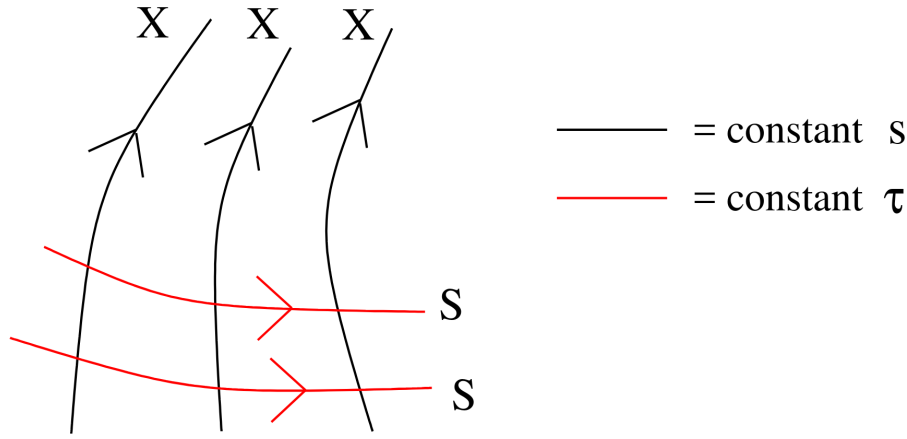


Figure 2.11: A one-parameter family of geodesics generated by X .

The family of geodesics sweeps a surface embedded in the manifold, so there is freedom in the choice of coordinates s and τ . In particular, it is always possible to pick them so that $X = \frac{\partial}{\partial \tau}$ and $S = \frac{\partial}{\partial s}$, in order for them to be linearly independent: $[X, S] = 0$, as in Fig. 2.11. Consider a torsion-free connection, so that:

$$\nabla_X S - \nabla_S X = [X, S] = 0 \quad \Rightarrow \quad \nabla_X \nabla_X S = \nabla_X \nabla_S X = \nabla_S \nabla_X X + R(X, S)X$$

But X is tangent to geodesics, so from Eq. 2.6.3 $\nabla_X X = 0$ and:

$$\nabla_X \nabla_X S = R(X, S)X \quad (2.6.7)$$

Restricting to an integral curve γ of the vector field X , i.e. $X^\mu|_\gamma = \frac{dx^\mu}{d\tau}$, the covariant derivative along γ becomes:

$$\nabla_X|_\gamma = X^\mu|_\gamma \nabla_\mu = \frac{dx^\mu}{d\tau} \nabla_\mu \equiv \frac{D}{D\tau}$$

Hence, in index notation, the change of the deviation vector along the geodesic is expressed as:

$$\frac{D^2 S^\mu}{D\tau^2} = R^\mu{}_{\nu\rho\sigma} X^\rho S^\sigma X^\nu \quad (2.6.8)$$

This can be interpreted as the relative acceleration of neighbouring geodesics, and it is determined by the Riemann tensor. Experimentally, such geodesic deviations are called **tidal forces**, which are the non-local manifestation of a gravitational field as a curvature of spacetime.

Advanced Topics in Differential Geometry

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§3.1 de Rham cohomology

Definition 3.1.1 (Closed form)

Let $\mathcal{M}$ be a smooth manifold. Then, $\omega \in \Omega^k(\mathcal{M})$ is **closed** if $d\omega = 0$.

Definition 3.1.2 (Exact form)

Let $\mathcal{M}$ be a smooth manifold. Then, $\omega \in \Omega^k(\mathcal{M})$ is **exact** if $\exists \eta \in \Omega^{k-1}(\mathcal{M}) : \omega = d\eta$, known as the *potential* of ω .

Since $d^2 = 0$, every exact form is closed. The converse, however, is in general not true: every closed form is exact only on manifolds with particular topological properties. By Stokes' theorem, it is clear that the integral of an exact form over any compact submanifold without boundary vanishes.

§3.1.1 Conservative covector fields

Given a smooth manifold $\mathcal{M}$, a smooth covector field $\omega \in \Omega^1(\mathcal{M})$ is **conservative** if its line integral over every piecewise smooth closed curve segment vanishes: it can be shown that this condition is equivalent to the requirement that the line integrals of γ are path-independent once the endpoints are fixed.

Theorem 3.1.1 (Conservative forms and exact forms)

Let $\mathcal{M}$ be a smooth manifold. Then, $\omega \in \Omega^1(\mathcal{M})$ is conservative if and only if it is exact.

Proof. If $\omega \in \Omega^1(\mathcal{M})$ is exact, then it is conservative by Eq. 2.3.17 it is conservative.

Suppose $\omega \in \Omega^1(\mathcal{M})$ is conservative, and assume $\mathcal{M}$ to be connected. Since the line integrals of ω are path-independent, denote by $\int_p^q \omega$ the line integral of ω along any piecewise smooth curve segment from $p \in \mathcal{M}$ to $q \in \mathcal{M}$. As a backward reparametrization of a path from p to q is a path to q to p , then Eq. 2.3.15 implies $\int_q^p \omega = -\int_p^q \omega$, while Eq. 2.3.13 implies:

$$\int_{p_1}^{p_2} \omega + \int_{p_2}^{p_3} \omega = \int_{p_1}^{p_3} \omega$$

Choose any $p_0 \in \mathcal{M}$ and define $f : \mathcal{M} \rightarrow \mathbb{R}$ as $f(q) = \int_{p_0}^q \omega$: WTS $df = \omega$. Let $q_0 \in \mathcal{M}$ arbitrary, let $(U, (x^i))$ be a smooth chart centered at q_0 and write the coordinate representation $\omega = \omega_i dx^i$ on U . $df_{q_0} = \omega_{q_0}$ is then equivalent to:

$$\frac{\partial f}{\partial x^j}(q_0) = \omega_j(q_0)$$

for $j = 1, \dots, n$. First, suppose $q_0 \in \text{Int } \mathcal{M}$, Fix j and let $\gamma : [-\varepsilon, \varepsilon] \rightarrow U$ be the smooth curve segment $\gamma(t) = (0, \dots, t, \dots, 0)$, with t in the j^{th} place, and with $\varepsilon \in \mathbb{R}^+$ small enough that $\gamma([-\varepsilon, \varepsilon]) \subseteq U$. Let $p_1 = \gamma(-\varepsilon)$ and define $\tilde{f} : \mathcal{M} \rightarrow \mathbb{R}$ as $\tilde{f}(q) = \int_{p_1}^q \omega$. Then, for all $q \in \mathcal{M}$:

$$f(q) - \tilde{f}(q) = \int_{p_0}^q \omega - \int_{p_1}^q \omega = \int_{p_0}^q \omega + \int_q^{p_1} \omega = \int_{p_0}^{p_1} \omega$$

which is independent of q , thus $\tilde{f}$ and f differ only by a constant, so it suffices to show that $\partial_j \tilde{f}(q_0) = \omega_j(q_0)$. Since $\gamma'(t) = \partial_j|_{\gamma(t)}$ by construction:

$$\omega_{\gamma(t)}(\gamma'(t)) = \omega_i(\gamma(t)) dx^i \left(\frac{\partial}{\partial x^j} \Big|_{\gamma(t)} \right) = \omega_j(\gamma(t))$$

Since the restriction of γ to $[-\varepsilon, t]$ is a smooth curve from p_1 to $\gamma(t)$:

$$\tilde{f} \circ \gamma(t) = \int_{p_1}^{\gamma(t)} \omega = \int_{-\varepsilon}^t ds \omega_{\gamma(s)}(\gamma'(s)) = \int_{-\varepsilon}^t ds \omega_j(\gamma(s))$$

Thus, by the fundamental theorem of calculus:

$$\frac{\partial \tilde{f}}{\partial x^j}(q_0) = \gamma'(0)(\tilde{f}) = \frac{d}{dt} \Big|_{t=0} \tilde{f} \circ \gamma(t) = \frac{d}{dt} \Big|_{t=0} \int_{-\varepsilon}^t ds \omega_j(\gamma(s)) = \omega_j(\gamma(0)) = \omega_j(q_0)$$

Therefore $df_{q_0} = \omega_{q_0}$ for $q_0 \in \text{Int } \mathcal{M}$. Now, consider $q_0 \in \partial \mathcal{M}$ with a boundary chart $(U, (x^i))$ centered at q_0 . The proof above still works, except for $j = n \equiv \dim_{\mathbb{R}} \mathcal{M}$, since x^n can only take non-negative values in this case. Let $p_1 = \gamma(0)$ (instead of $\gamma(-\varepsilon)$), and proceed as before: this shows that $df = \omega$ on $\mathcal{M}$ connected.

Finally, if $\mathcal{M}$ is not connected, let $\{\mathcal{M}_i\}$ be the connected components of $\mathcal{M}$. By the argument above, on each component there is a function $f_i \in \mathcal{C}^\infty(\mathcal{M}_i)$ such that $df_i = \omega$ on $\mathcal{M}_i$. Letting $f : \mathcal{M} \rightarrow \mathbb{R}$ be the function that is equal to f_i on $\mathcal{M}_i$ for each i , then $df = \omega$, completing the proof. $\square$

While not all closed forms are exact on a generic smooth manifold, since it depends on its shape, it is possible to show that *locally* this is always true. Given a finite-dimensional $\mathbb{R}$ -vector space V , then $U \subseteq V$ is said to be **star-shaped** if there is a point $x \in U$ such that for every $x \in U$ the line segment from c to x is entirely contained in U . For example, every convex subset is star-shaped.

Theorem 3.1.2 (Poincaré lemma)

If $U \in \mathcal{T}(\mathbb{R}^n)$ (or $\mathbb{U}^n$) is star-shaped, then every closed $\omega \in \Omega^1(U)$ is exact.

Proof. Suppose U is star-shaped with respect to $c \in U$, and let $\omega = \omega_i dx^i \in \Omega^1(U)$ closed. Since diffeomorphisms take closed forms to closed forms and exact forms to exact forms, translate U so that $c = 0$. Then, for any $x \in U$, let $\gamma_x : [0, 1] \rightarrow U$ denote the line segment from 0 to x parametrized as $\gamma_x(t) = tx$: by hypothesis $\gamma_x([0, 1]) \subseteq U \forall x \in U$. Define $f : U \rightarrow \mathbb{R}$ as:

$$f(x) = \int_{\gamma_x} \omega$$

WTS $df = \omega$, or equivalently $\partial_j f = \omega_j$ for $j = 1, \dots, n$. First, compute:

$$f(x) = \int_0^1 dt \omega_{\gamma_x(t)}(\gamma'_x(t)) = \int_0^1 dt \omega_i(tx) x^i$$

Since the integrand is smooth in all variables, it is permissible to differentiate under the integral sign:

$$\frac{\partial f}{\partial x^j}(x) = \int_0^1 dt \left[t \frac{\partial \omega_i}{\partial x^j}(tx) x^i + \omega_j(tx) \right]$$

Since ω is closed, $\partial_i \omega_j = \partial_j \omega_i$, hence:

$$\frac{\partial f}{\partial x^j}(x) = \int_0^1 dt \left[t \frac{\partial \omega_j}{\partial x^i}(tx) x^i + \omega_j(tx) \right] = \int_0^1 dt \frac{d}{dt} [t \omega_j(tx)] = [t \omega_j(tx)]_{t=0}^{t=1} = \omega_j(x)$$

which completes the proof. $\square$

Corollary 3.1.2.1 (Local exactness of closed forms)

Let $\mathcal{M}$ be a smooth manifold and $\omega \in \Omega^1(\mathcal{M})$ closed. Then, every point of $\mathcal{M}$ has a neighbourhood on which ω is exact.

Proof. Let $p \in \mathcal{M}$ be arbitrary. By hypothesis $\partial_i \omega_j = \partial_j \omega_i$ in some smooth coordinate ball or half-ball (U, φ) containing p . Balls and half-balls are convex and hence star-shaped, therefore applying Th. 3.1.4 to the coordinate representation of ω there exists $f \in \mathcal{C}^\infty(U) : \omega|_U = df$. $\square$

The key feature in constructing the potential function in Th. 3.1.4 is that every point of $x \in U$ can be reached by a definite path from $c \in U$ to x , chosen to vary smoothly as x varies. This fails, for example, on a punctured plane, i.e. a plane with a hole.

Example 3.1.1 (Swirl form on the plane)

Consider $\mathcal{M} = \mathbb{R}^2 - \{0\}$ and the closed $\omega \in \Omega^1(\mathcal{M})$ defined as:

$$\omega = \frac{x dy - y dx}{x^2 + y^2}$$

Choosing the smooth closed curve segment $\gamma : [0, 2\pi] \rightarrow \mathcal{M}$ defined as $\gamma(\vartheta) = (\cos \vartheta, \sin \vartheta)$:

$$\int_{\gamma} \omega = \int_{[0, 2\pi]} \frac{\cos \vartheta (\cos \vartheta d\vartheta) - \sin \vartheta (\sin \vartheta d\vartheta)}{\sin^2 \vartheta + \cos^2 \vartheta} = \int_0^{2\pi} d\vartheta = 2\pi$$

which shows, by Th. 3.1.1 that ω is not exact on $\mathbb{R}^2 - \{0\}$. However, restricting the domain to $U = \{(x, y) \in \mathbb{R}^2 : x > 0\}$, by direct computation $\omega = df$ with potential function $f : U \rightarrow \mathbb{R}$ given by:

$$f(x) = \arctan \frac{y}{x}$$

or, more clearly, in polar coordinates $\omega = d\vartheta$. The problem is then that there is no smooth (or even continuous) angle function on all of $\mathbb{R}^2 - \{0\}$, which is a consequence of the hole in the center.

§3.1.2 de Rham cohomology groups

The aim of de Rham cohomology is to study whether closed forms are exact on a smooth manifold by means of its global topological properties.

Let $\mathcal{M}$ be a smooth n -manifold. Since $d : \Omega^k(\mathcal{M}) \rightarrow \Omega^{k+1}(\mathcal{M})$ is defined for all $k = 0, \dots, n-1$, given $p \in \mathbb{N}_0$ denote $d_p : \Omega^p(\mathcal{M}) \rightarrow \Omega^{p+1}(\mathcal{M})$, with the convention $\Omega^p(\mathcal{M}) = \{0\} \equiv 0$ for $p < 0$ and $p > n$. Then, define:

$$\mathcal{Z}^p(\mathcal{M}) := \ker d_p \quad (3.1.1)$$

$$\mathcal{B}^p(\mathcal{M}) := \operatorname{Im} d_{p-1} \quad (3.1.2)$$

which are respectively the $\mathbb{R}$ -vector spaces of closed p -forms and exact p -forms on $\mathcal{M}$. By the above conventions, $\mathcal{B}^0(\mathcal{M}) = 0$ and $\mathcal{Z}^n(\mathcal{M}) = \Omega^n(\mathcal{M})$.

Definition 3.1.3 (de Rham cohomology groups)

Let $\mathcal{M}$ be a smooth manifold. The **de Rham cohomology group of degree p** , with $p \in \mathbb{N}_0$, is defined as:

$$\mathcal{H}^p(\mathcal{M}) := \frac{\mathcal{Z}^p(\mathcal{M})}{\mathcal{B}^p(\mathcal{M})} \quad (3.1.3)$$

This quotient vector space is well-defined for all $p \in \mathbb{N}_0$, since every exact form is closed, i.e. $\mathcal{B}^p(\mathcal{M}) \subseteq \mathcal{Z}^p(\mathcal{M})$, and it is a group under vector addition. Clearly, $\mathcal{H}^p(\mathcal{M}) = 0$ for $p < 0$ and $p > n$, while for $0 \leq p \leq n$ the condition $\mathcal{H}^p(\mathcal{M}) = 0$ holds if and only if every closed p -form on $\mathcal{M}$ is exact.

For any $\omega \in \Omega^p(\mathcal{M})$, let $[\omega] \in \mathcal{H}^p(\mathcal{M})$ denote the equivalence class of ω , known its *cohomology class*: if $[\omega] = [\omega']$, i.e. if ω and ω' differ by an exact form, then ω and ω' are said to be *cohomologous*.

Proposition 3.1.1 (Induced cohomology map)

For any smooth map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$, the pull-back $f^* : \Omega^p(\mathcal{M}_2) \rightarrow \Omega^p(\mathcal{M}_1)$ maps $\mathcal{Z}^p(\mathcal{M}_2) \mapsto \mathcal{Z}^p(\mathcal{M}_1)$ and $\mathcal{B}^p(\mathcal{M}_2) \mapsto \mathcal{B}^p(\mathcal{M}_1)$. It therefore descends to a linear map $f^* : \mathcal{H}^p(\mathcal{M}_2) \rightarrow \mathcal{H}^p(\mathcal{M}_1)$, called the **induced cohomology map**, such that $\mathbb{1}_{\mathcal{M}}^* = \mathbb{1}_{\mathcal{H}^p(\mathcal{M})}$ and, if $g : \mathcal{M}_2 \rightarrow \mathcal{M}_3$ is another smooth map, then:

$$(g \circ f)^* = f^* \circ g^* : \mathcal{H}^p(\mathcal{M}_3) \rightarrow \mathcal{H}^p(\mathcal{M}_1)$$

Proof. If ω is closed, then $d(f^*\omega) = f^*(d\omega) = 0$, so $f^*\omega$ is also closed. If $\omega = d\eta$ is exact, then $f^*\omega = f^*(d\eta) = d(f^*\eta)$, which is also exact. Therefore, f^* maps $\mathcal{Z}^p(\mathcal{M}_2) \mapsto \mathcal{Z}^p(\mathcal{M}_1)$ and $\mathcal{B}^p(\mathcal{M}_2) \mapsto \mathcal{B}^p(\mathcal{M}_1)$.

The induced cohomology map $f^* : \mathcal{H}^p(\mathcal{M}_2) \rightarrow \mathcal{H}^p(\mathcal{M}_1)$ is defined as follows: for $\omega \in \Omega^p(\mathcal{M})$ closed, let $f^*[\omega] = [f^*\omega]$, so if $\omega' = \omega + d\eta$, then $[f^*\omega'] = [f^*\omega + d(f^*\eta)] = [f^*\omega] = f^*[\omega]$, hence this map is well-defined. Its properties follow immediately from the analogous properties of pull-back maps on forms. $\square$

Corollary 3.1.1.1 (Diffeomorphism-invariance of de Rham cohomology)

Diffeomorphic smooth manifolds have isomorphic de Rham cohomology groups.

Although in general the direct computation of de Rham groups is not easy, this process can be simplified with the following result.

Proposition 3.1.2 (Cohomology of disjoint unions)

Let $\{\mathcal{M}_j\}$ be a countable collection of smooth n -manifolds, and let $\mathcal{M} = \bigsqcup_j \mathcal{M}_j$. The inclusion maps $\iota_j : \mathcal{M}_j \hookrightarrow \mathcal{M}$ induce an isomorphism:

$$\mathcal{H}^p(\mathcal{M}) \cong \prod_j \mathcal{H}^p(\mathcal{M}_j)$$

Proof. The pullback maps $\iota_j^* : \Omega^p(\mathcal{M}) \rightarrow \Omega^p(\mathcal{M}_j)$ already induce an isomorphism $\Omega^p(\mathcal{M}) \cong \prod_j \Omega^p(\mathcal{M}_j)$ as:

$$\omega \mapsto (\iota_1^* \omega, \iota_2^* \omega, \dots) = (\omega|_{\mathcal{M}_1}, \omega|_{\mathcal{M}_2}, \dots)$$

since it is injective, as any $\omega \in \Omega^p(\mathcal{M})$ whose restriction to each $\mathcal{M}_j$ is zero must itself be zero, and surjective, since defining an arbitrary $\omega_j \in \Omega^p(\mathcal{M}_j)$ on each $\mathcal{M}_j$ defines a global $\omega \in \Omega^p(\mathcal{M})$. $\square$

Therefore, each de Rham group of a disconnected manifold is just the direct product of the corresponding groups of its components. The main objective is thus to compute the de Rham groups of connected manifolds.

Example 3.1.2 (Cohomology of degree zero)

Let $\mathcal{M}$ be a connected smooth manifolds. First, note that $\mathcal{B}^0(\mathcal{M}) = 0$, since there are no (-1) -forms. Then, observe that a closed 0-form is $f \in \mathcal{C}^\infty(\mathcal{M}) : df = 0$, and since $\mathcal{M}$ is connected, this is true if and only if f is constant: therefore, $\mathcal{H}^0(\mathcal{M}) = \mathcal{Z}^0(\mathcal{M})$ is just the space of constant functions, which is 1-dimensional.

If $\mathcal{M}$ is a 0-manifold, then $\mathcal{H}^0(\mathcal{M})$ is a direct product of 1-dimensional vector spaces, one for each point of $\mathcal{M}$, and the cohomology groups with non-zero degree vanish for dimensional reasons.

§3.1.3 Homotopy invariance

Cor. 3.1.1.1 has a profound generalization: de Rham cohomology groups are topological invariants (actually, they are homotopy invariants, see §A.1 for homotopy equivalence).

Definition 3.1.4 (Cochain homotopy)

Let $f, g : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be smooth maps. A collection of linear maps $h_p : \Omega^p(\mathcal{M}_2) \rightarrow \Omega^{p-1}(\mathcal{M}_1)$ is a **cochain homotopy** between f^* and g^* if, for all $\omega \in \Omega^p(\mathcal{M}_2)$:

$$(d_{p-1} \circ h_p \omega + h_{p+1} \circ d_p) \omega = g^* \omega - f^* \omega \quad (3.1.4)$$

Proposition 3.1.3 (Cochain homotopy and induced cohomology maps)

Let $f, g : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be smooth maps. If there exists a cochain homotopy between f^* and g^* , then the induced cohomology maps $f^*, g^* : \mathcal{H}^p(\mathcal{M}_2) \rightarrow \mathcal{H}^p(\mathcal{M}_1)$ are equal.

Proof. Consider $\omega \in \Omega^p(\mathcal{M})(\mathcal{M}_2)$ closed. Then, by Eq. 3.1.4:

$$d_{p-1}(h_p \omega) = g^* \omega - f^* \omega \implies g^*[\omega] - f^*[\omega] = [d\eta] = 0$$

where $\eta \equiv h_p \omega \in \Omega^{p-1}(\mathcal{M}_1)$. $\square$

Consider a smooth manifold $\mathcal{M}$ and, for each $t \in [0, 1]$, let $\iota_t : \mathcal{M} \rightarrow \mathcal{M} \times [0, 1]$ be the map $\iota_t(x) = (x, t)$. If $\partial\mathcal{M} = \emptyset$, then $\mathcal{M} \times I$ is a smooth manifold with boundary, while if $\partial\mathcal{M} \neq \emptyset$, then $\mathcal{M} \times I$ is a smooth manifold with corners: it is straightforward to check that the definitions of de Rham groups and induced cohomology maps remain unchanged in the context of smooth manifolds with corners.

Lemma 3.1.1 (Existence of cochain homotopy)

For any smooth manifold $\mathcal{M}$, there exists a cochain homotopy between the two maps $\iota_0^*, \iota_1^* : \Omega(\mathcal{M} \times [0, 1]) \rightarrow \Omega(\mathcal{M})$.

Proof. For each $p \in \mathbb{N}_0$, WTS there exists a linear map $h_p : \Omega^p(\mathcal{M} \times I) \rightarrow \Omega^{p-1}(\mathcal{M})$ such that, for all $\omega \in \Omega^p(\mathcal{M} \times I)$:

$$h_{p+1}(\mathrm{d}_p \omega) + \mathrm{d}_{p-1}(h_p \omega) = \iota_1^* \omega - \iota_0^* \omega$$

Let s denote the standard coordinate on $\mathbb{R}$, and let S be the vector field on $\mathcal{M} \times \mathbb{R}$ defined as $S_{q,s} = (0, \partial/\partial s|_s)$ under the usual identification $T_{(q,s)}\mathcal{M} \cong T_q\mathcal{M} \times T_s\mathbb{R}$. Given $\omega \in \Omega^p(\mathcal{M} \times [0, 1])$, define $h_p \omega \in \Omega^{p-1}(\mathcal{M})$ as:

$$h_p \omega = \int_0^1 \mathrm{d}t \iota_t^*(\iota_S \omega)$$

which, for any $q \in \mathcal{M}$, is expressed as:

$$(h_p \omega)_p = \int_0^1 \mathrm{d}t \iota_t^*((\iota_S \omega)_{(q,t)})$$

where the integrand is a map $t \mapsto \bigwedge^{p-1}(T_q\mathcal{M})$. On any smooth coordinate domain $U \subseteq \mathcal{M}$, the components of the integrand are smooth function of $(q, t) \in U \times I$, so correctly $h_p \omega \in \Omega^{p-1}(\mathcal{M})$. Then, $\mathrm{d}_{p-1}(h_p \omega)$ can be computed at any point by differentiating under the integral sign in local coordinates:

$$\mathrm{d}_{p-1}(h_p \omega) = \int_0^1 \mathrm{d}t \mathrm{d}(\iota_t^*(\iota_S \omega))$$

Therefore, using Th. 2.2.2:

$$\begin{aligned} h_{p+1}(\mathrm{d}_p \omega) + \mathrm{d}_{p-1}(h_p \omega) &= \int_0^1 \mathrm{d}t [\iota_t^*(\iota_S \mathrm{d}\omega) + \mathrm{d}(\iota_t^*(\iota_S \omega))] \\ &= \int_0^1 \mathrm{d}t \iota_t^*(\iota_S(\mathrm{d}\omega) + \mathrm{d}(\iota_S \omega)) = \int_0^1 \mathrm{d}t \iota_t^* \mathcal{L}_S \omega \end{aligned}$$

To simplify this expression, use the flow of S over $\mathcal{M} \times \mathbb{R}$, which is given explicitly by $\theta_t(q, s) = (q, t + s)$. Then, using Eq. 1.6.24:

$$\iota_t^*(\mathcal{L}_S \omega) = \iota_0^*(\theta_t^*(\mathcal{L}_S \omega)) = \iota_0^*\left(\frac{\mathrm{d}}{\mathrm{d}t}(\theta_t^* \omega)\right) = \frac{\mathrm{d}}{\mathrm{d}t} \iota_0^*(\theta_t^* \omega) = \frac{\mathrm{d}}{\mathrm{d}t} \iota_t^* \omega$$

Inserting this into the integral and applying the fundamental theorem of calculus yields the thesis. $\square$

This lemma allows to prove the homotopy invariance of de Rham cohomology.

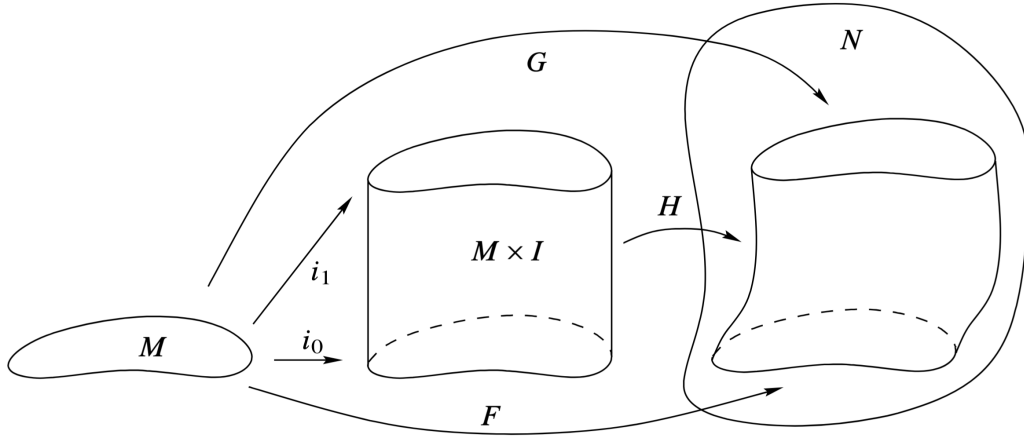


Figure 3.1: Homotopic maps.

Theorem 3.1.3 (Homotopy invariance of de Rham cohomology)

Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be homotopy equivalent smooth manifolds. Then, any smooth homotopy equivalence $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ induces an isomorphism $\mathcal{H}^p(\mathcal{M}_1) \cong \mathcal{H}^p(\mathcal{M}_2)$ for each $p \in \mathbb{N}_0$.

Proof. Let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a homotopy equivalence, with homotopy inverse $g : \mathcal{M}_2 \rightarrow \mathcal{M}_1$: by the Whitney approximation theorem Th. 1.3.13, there are smooth maps $\tilde{f} : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ homotopic to f and $\tilde{g} : \mathcal{M}_2 \rightarrow \mathcal{M}_1$ homotopic to g . Since homotopy is preserved by composition, it follows that $\tilde{f} \circ \tilde{g} \simeq f \circ g \simeq \mathbb{1}_{\mathcal{M}_2}$ and $\tilde{g} \circ \tilde{f} \simeq g \circ f \simeq \mathbb{1}_{\mathcal{M}_1}$, hence $\tilde{f}$ and $\tilde{g}$ are homotopy inverses of each other.

Lemma 3.1.2

If $\mathcal{M}_1, \mathcal{M}_2$ are smooth manifolds and $f, g : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ are homotopic smooth maps, then the induced cohomology maps $f^*, g^* : \mathcal{H}^p(\mathcal{M}_2) \rightarrow \mathcal{H}^p(\mathcal{M}_1)$ are equal for every $p \in \mathbb{N}_0$.

Proof. By Lemma 3.1.1 the two cohomology maps $\iota_0^*, \iota_1^* : \mathcal{H}^p(\mathcal{M}_1 \times [0, 1]) \rightarrow \mathcal{H}^p(\mathcal{M}_1)$ are equal. Moreover, it can be shown that if $f, g : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ are homotopic smooth maps, then they are smoothly homotopic, i.e. there is a smooth homotopy $H : \mathcal{M}_1 \times [0, 1] \rightarrow \mathcal{M}_2$ from f to g . Since $f = H \circ \iota_0$ and $g = H \circ \iota_1$ (see Fig. 3.1), by Prop. 3.1.1:

$$f^* = (H \circ \iota_0)^* = \iota_0^* \circ H^* = \iota_1^* \circ H^* = (H \circ \iota_1)^* = g^*$$

which is the thesis. □

Using Lemma 3.1.2, then, on the cohomology:

$$\tilde{f}^* \circ \tilde{g}^* = (\tilde{g} \circ \tilde{f})^* = \mathbb{1}_{\mathcal{M}_1}^* = \mathbb{1}_{\mathcal{H}^p(\mathcal{M}_1)}$$

and equivalently for $\tilde{g}^* \circ \tilde{f}^* = \mathbb{1}_{\mathcal{H}^p(\mathcal{M}_2)}$. Therefore, $\tilde{f}^* : \mathcal{H}^p(\mathcal{M}_2) \rightarrow \mathcal{H}^p(\mathcal{M}_1)$ is an isomorphism. □

Corollary 3.1.3.1 (Topological invariance of de Rham cohomology)

Let $\mathcal{M}_1$ and $\mathcal{M}_2$ are homeomorphic smooth manifolds, then their de Rham cohomology groups are isomorphic.

Proof. Every homeomorphism is a homotopy equivalence, thus use Th. 3.1.3. $\square$

Homotopy invariance can be used to compute some de Rham groups.

Proposition 3.1.4 (Cohomology of contractible manifolds)

Let $\mathcal{M}$ be a contractible¹ smooth manifold. Then, $\mathcal{H}^p(\mathcal{M}) = 0$ for $p \geq 1$.

Proof. By the hypothesis, there is some $q \in \mathcal{M}$ such that $\mathbb{1}_{\mathcal{M}} \simeq c_q : \mathcal{M} \rightarrow \mathcal{M}$ which sends all of $\mathcal{M}$ to q . Denoting the inclusion map as $\iota_q : \{q\} \hookrightarrow \mathcal{M}$, it follows that $c_q \circ \iota_q = \mathbb{1}_{\{q\}}$ and $\iota_q \circ c_q \simeq \mathbb{1}_{\mathcal{M}}$, so ι_q is a homotopy equivalence between $\{q\}$ and $\mathcal{M}$. Then, by homotopy invariance, the de Rham groups of $\mathcal{M}$ are isomorphic to those of a 0-manifold, which vanish for $p \geq 1$ as per Ex. 3.1.2. $\square$

The Poincaré lemma Th. 3.1.2 can be generalized to forms of all degrees.

Theorem 3.1.4 (Poincaré lemma)

If $U \in \mathcal{T}(\mathbb{R}^n)$ (or $\mathbb{U}^n$) is star-shaped, then $\mathcal{H}^p(U) = 0$ for $p \geq 1$.

Proof. If U is star-shaped with respect to $c \in \mathbb{R}^n$, then it is contractible by the following straight-line homotopy:

$$H(x, t) = c + t(x - c)$$

The thesis follows from Prop. 3.1.4. $\square$

Corollary 3.1.4.1 (Local exactness of closed forms)

Each point of a smooth manifold has a neighbourhood on which every closed form is exact.

Proof. Every point of a smooth manifold has a neighbourhood diffeomorphic to an open ball in $\mathbb{R}^n$ or half-ball in $\mathbb{U}^n$, each of which are star-shaped. The result follows from Th. 3.1.4. $\square$

Since $\mathbb{R}^n$ and $\mathbb{U}^n$ are both star-shaped for $n \in \mathbb{N}_0$, $\mathcal{H}^p(\mathbb{R}^n) = 0$ and $\mathcal{H}^p(\mathbb{U}^n) = 0$ for $p \geq 1$.

Another case in which de Rham cohomology can be studied quite extensively is in degree 1. Let $\mathcal{M}$ be a connected smooth manifold and $q \in \mathcal{M}$. Let $\text{Hom}(\pi_1(\mathcal{M}, q), \mathbb{R})$ denote the set of group homomorphisms from $\pi_1(\mathcal{M}, q)$ to the additive group $\mathbb{R}(+)$: this is a $\mathbb{R}$ -vector space under pointwise addition of homomorphisms and multiplication by constants. Define a linear map $\Phi : \mathcal{H}^1(\mathcal{M}) \rightarrow \text{Hom}(\pi_1(\mathcal{M}, q), \mathbb{R})$ as follows: given a cohomology class $[\omega] \in \mathcal{H}^1(\mathcal{M})$, define $\Phi[\omega] : \pi_1(\mathcal{M}, q) \rightarrow \mathbb{R}$ as:

$$\Phi[\omega][\gamma] \equiv \int_{\tilde{\gamma}} \omega$$

where γ is any piecewise smooth curve representing the path homotopy class $[\gamma] \in \pi_1(\mathcal{M}, q)$.

Theorem 3.1.5 (First cohomology and fundamental group)

Let $\mathcal{M}$ be a connected smooth manifold. For each $q \in \mathcal{M}$, the linear map $\Phi : \mathcal{H}^1(\mathcal{M}) \rightarrow \text{Hom}(\pi_1(\mathcal{M}, q), \mathbb{R})$ is an isomorphism.

Proof. The proof that Φ is surjective requires more advanced instruments, so WTS Φ is well-defined and injective. Given $[\gamma] \in \pi_1(\mathcal{M}, q)$, by the Whitney approximation theorem Th. 1.3.13

¹A topological space X is *contractible* if $\mathbb{1}_X$ is homotopic to a constant map.

there is a smooth closed curve segment $\tilde{\gamma}$ in the same path class of γ , and by Th. 2.3.3, $\int_{\tilde{\gamma}} \omega$ gives the same result for every piecewise smooth curve $\tilde{\gamma}$ in the given class. Moreover, if $\tilde{\omega} \in \Omega^1(\mathcal{M})$ is in the same cohomology class of ω , then $\tilde{\omega} - \omega = df$ for some $f \in C^\infty(\mathcal{M})$, so:

$$\int_{\tilde{\gamma}} \tilde{\omega} - \int_{\tilde{\gamma}} \omega = \int_{\tilde{\gamma}} df = f(q) - f(q) = 0$$

Thus Φ is well-defined. By Eq. 2.3.13, $\Phi[\omega]$ is a group homomorphism from $\pi_1(\mathcal{M}, q)$ to $\mathbb{R}$, and from linearity of the line integral Φ is itself a linear map.

To see that Φ is injective, suppose $\Phi[\omega] \equiv 0$, which means that $\int_{\tilde{\gamma}} \omega = 0$ for every piecewise smooth closed curve starting at q . If σ is a piecewise smooth closed curve starting at some other point $q' \in \mathcal{M}$, then a piecewise smooth curve α from q to q' can be chosen so that the path product $\alpha \cdot \sigma \cdot \bar{\alpha}$ is a closed curve based at q , where $\bar{\alpha}$ is a backward reparametrization of α . Then:

$$0 = \int_{\alpha \cdot \sigma \cdot \bar{\alpha}} \omega = \int_{\alpha} \omega + \int_{\sigma} \omega - \int_{\alpha} \omega = \int_{\sigma} \omega$$

Therefore, ω is conservative and thus exact. $\square$

By Th. 2.3.3, on simply connected smooth manifolds every closed 1-form is exact, so $\mathcal{H}^1(\mathcal{M}) = 0$. This result can be generalized.

Corollary 3.1.5.1

If $\mathcal{M}$ is a connected smooth manifold with finite fundamental group, then $\mathcal{H}^1(\mathcal{M}) = 0$.

Proof. There are no non-trivial homomorphisms from a finite group to $\mathbb{R}$. $\square$

The same result holds if the fundamental group of a connected smooth manifold is a torsion group².

§3.1.4 Mayer–Vietoris theorem

§3.1.4.1 Concepts of algebraic topology

Let $\mathcal{R}$ be a commutative ring³, and consider the following sequence of $\mathcal{R}$ -modules and $\mathcal{R}$ -linear maps:

$$\dots \longrightarrow A^{p-1} \xrightarrow{d_{p-1}} A^p \xrightarrow{d_p} A^{p+1} \longrightarrow \dots \quad (3.1.5)$$

²A group Γ is a *torsion group* if $\forall g \in \Gamma \exists k \in \mathbb{N} : g^k = e$.

³A *ring* is a set $\mathcal{R}$ equipped with two binary operations $+, \cdot : \mathcal{R} \times \mathcal{R} \rightarrow \mathcal{R}$ (addition and multiplication) which satisfy the following ring axioms:

- (1) $(\mathcal{R}, +)$ is an abelian group;
- (2) $(\mathcal{R}, \cdot)$ is a monoid (i.e. multiplication is associative and has an identity);
- (3) multiplication is distributive (from left and right) with respect to addition.

The identities of addition and multiplication are usually denoted by 0 and 1 , respectively. If multiplication is commutative, $\mathcal{R}$ is a commutative ring.

A *left-module* over a ring is an abelian group $(M, +)$ with an operation $\odot : \mathcal{R} \times M \rightarrow M$ (scalar multiplication) such that, for all $r, s \in \mathcal{R}$ and $x, y \in M$:

- (1) $r \odot (x + y) = r \odot x + r \odot y$ and $(r + s) \odot x = r \odot x + s \odot x$;
- (2) $(r \cdot s) \odot x = r \odot (s \odot x)$ and $1 \odot x = x$.

denoted by A^* . Such a sequence is called a **(cochain) complex**⁴ if the composition of any successive applications is the zero map, i.e. if for each $p \in \mathbb{N}_0$:

$$d_p \circ d_{p+1} \equiv 0 : A^p \rightarrow A^{p+2} \quad (3.1.6)$$

If $\text{Im } d_p = \ker d_{p+1}$, the sequence is said to be an **exact sequence**. Clearly, every exact sequence is a complex, while the converse is not true in general.

If A^* is a cochain complex, it is possible to define the p^{th} **cohomology group** of A^* as the quotient module which gives a quantitative measure of the failure of exactness at A^p :

$$\mathcal{H}^p(A^*) := \frac{\ker d_p}{\text{Im } d_{p-1}} \quad (3.1.7)$$

If A^* and B^* are cochain complexes, a **cochain map** from A^* to B^* , denoted by $f : A^* \rightarrow B^*$, is a collection of linear maps $f_p : A^p \rightarrow B^p$ such that the following diagram commutes for each $p \in \mathbb{N}_0$:

$$\begin{array}{ccccccc} \dots & \longrightarrow & A^p & \xrightarrow{d_p} & A^{p+1} & \longrightarrow & \dots \\ & & \downarrow f_p & & \downarrow f_{p+1} & & \\ \dots & \longrightarrow & B^p & \xrightarrow{d_p} & B^{p+1} & \longrightarrow & \dots \end{array}$$

Suppressing indices, this condition can be written as $f \circ d = d \circ f$, which means that any cochain map $f : A^* \rightarrow B^*$ induces a linear cohomology map $f_p : \mathcal{H}^p(A^*) \rightarrow \mathcal{H}^p(B^*)$ for each $p \in \mathbb{N}_0$. A **short exact sequence** of complexes consists of three complexes A^*, B^*, C^* together with cochain maps:

$$0 \longrightarrow A^* \xrightarrow{f} B^* \xrightarrow{g} C^* \longrightarrow 0 \quad (3.1.8)$$

such that each sequence:

$$0 \longrightarrow A^p \xrightarrow{f_p} B^p \xrightarrow{g_p} C^p \longrightarrow 0$$

is exact. This means that f_p is injective, g_p is surjective and $\text{Im } f_p = \ker g_p$ for each $p \in \mathbb{N}_0$.

Lemma 3.1.3 (Zigzag lemma)

Given a short exact sequence of complexes as in Eq. 3.1.8, for each $p \in \mathbb{N}_0$ there is a linear map $\delta_p : \mathcal{H}^p(C^*) \rightarrow \mathcal{H}^{p+1}(A^*)$, called the **connecting homomorphism**, such that the following sequence is exact:

$$\dots \xrightarrow{\delta_{p-1}} \mathcal{H}^p(A^*) \xrightarrow{f_p^*} \mathcal{H}^p(B^*) \xrightarrow{g_p^*} \mathcal{H}^p(C^*) \xrightarrow{\delta_p} \mathcal{H}^{p+1}(A^*) \xrightarrow{f_{p+1}^*} \dots$$

§3.1.4.2 Proof of the Mayer–Vietoris theorem

The discussion of cochain complexes in §3.1.4.1 applies equally well e.g. to $\mathcal{R} = \mathbb{Z}$, in which case the sequence is composed of abelian groups and homomorphisms, and to $\mathcal{R} = \mathbb{R}$, in which case it is composed of vector spaces and linear maps. The case of interest here is $\mathcal{R} = \mathbb{R}$.

A *right-module* is defined analogously with $\odot : M \times \mathcal{R} \rightarrow M$. If $\mathcal{R}$ is a commutative ring, the two definitions coincide and $(M, +)$ is just a *module* over $\mathcal{R}$. In practice, $\cdot$ is omitted for multiplication and used in place of $\odot$.

⁴A *chain complex* is defined analogously, but with maps $\partial_p : A^p \rightarrow A^{p-1}$ going in the opposite direction. In this case, the term *homology* is used in place of cohomology. The conclusions of this section hold for chain complexes too: in particular, chain maps induce linear maps on homologies.

Let $\mathcal{M}$ be a smooth manifold, and let $U, V \in \mathcal{T}(\mathcal{M}) : \mathcal{M} = U \cup V$. There are then four inclusion maps and four induced pull-back maps:

$$\begin{array}{ccc}
 & U & \\
 i \nearrow & & \searrow k \\
 U \cup V & & \mathcal{M} \\
 j \searrow & & \nearrow l \\
 & V &
 \end{array}
 \qquad
 \begin{array}{ccc}
 & \Omega^p(U) & \\
 i^* \nearrow & & \nwarrow k^* \\
 \Omega^p(U \cup V) & & \Omega^p(\mathcal{M}) \\
 j^* \searrow & & \nwarrow l^* \\
 & \Omega^p(V) &
 \end{array}$$

There are also corresponding cohomology maps. Note that these pull-back maps are just restrictions, e.g. $k^*\omega = \omega|_U$. Consider the following sequence of maps:

$$0 \longrightarrow \Omega^p(\mathcal{M}) \xrightarrow{k^* \oplus l^*} \Omega^p(U) \oplus \Omega^p(V) \xrightarrow{i^* - j^*} \Omega^p(U \cap V) \longrightarrow 0 \quad (3.1.9)$$

with:

$$(k^* \oplus l^*)\omega \equiv (k^*\omega, l^*\omega) \qquad (i^* - j^*)(\omega, \eta) \equiv i^*\omega - j^*\eta \quad (3.1.10)$$

Since pull-backs commute with exterior derivatives, these maps descend to linear maps on the corresponding de Rham cohomology groups.

Theorem 3.1.6 (Mayer–Vietoris theorem)

Let $\mathcal{M}$ be a smooth manifold and $U, V \in \mathcal{T}(\mathcal{M}) : \mathcal{M} = U \cup V$. For each $p \in \mathbb{N}_0$, there is a linear map $\delta_p : \mathcal{H}^p(U \cap V) \rightarrow \mathcal{H}^{p+1}(\mathcal{M})$ such that the following sequence, called the **Mayer–Vietoris sequence** for the open cover $\{U, V\}$, is exact:

$$\dots \xrightarrow{\delta_p} \mathcal{H}^p(\mathcal{M}) \xrightarrow{k^* \oplus l^*} \mathcal{H}^p(U) \oplus \mathcal{H}^p(V) \xrightarrow{i^* - j^*} \mathcal{H}^p(U \cap V) \xrightarrow{\delta_{p+1}} \mathcal{H}^{p+1}(\mathcal{M}) \xrightarrow{k^* \oplus l^*} \dots$$

Proof. It suffices to show that the sequence Eq. 3.1.9 is exact for each p , as then, because then, since pull-back maps commute with exterior derivatives, it defines a short exact sequence of cochain maps, and the Mayer–Vietoris sequence is exact by Lemma 3.1.3.

First, prove exactness at $\Omega^p(\mathcal{M})$: WTS $k^* \oplus l^*$ is injective. Let $\sigma \in \Omega^p(\mathcal{M}) : (k^* \oplus l^*)\sigma = (\sigma|_U, \sigma|_V) = (0, 0)$: since $\{U, V\}$ is an open cover of $\mathcal{M}$, this implies $\sigma = 0$, hence $k^* \oplus l^*$ is injective.

Now, prove exactness at $\Omega^p(U) \oplus \Omega^p(V)$. Observe that:

$$(i^* - j^*) \circ (k^* \oplus l^*)(\sigma) = (i^* - j^*)(\sigma|_U, \sigma|_V) = \sigma|_{U \cap V} - \sigma|_{U \cap V} = 0$$

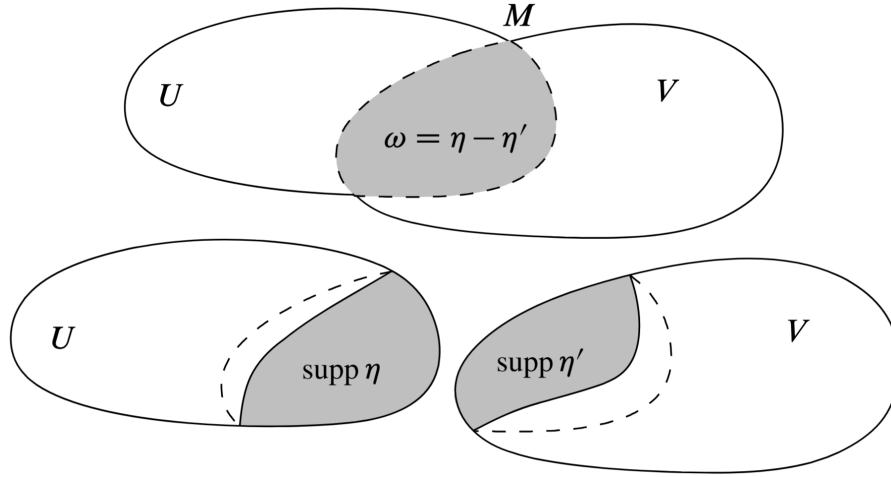
which shows $\text{Im}(k^* \oplus l^*) \subseteq \ker(i^* - j^*)$. Conversely, let $(\eta, \eta') \in \Omega^p(U) \oplus \Omega^p(V) : (i^* - j^*)(\eta, \eta') = 0$, which means that $\eta|_{U \cap V} = \eta'|_{U \cap V}$, so there is a global $\sigma \in \Omega^p(\mathcal{M})$ defined by:

$$\sigma = \begin{cases} \eta & \text{on } U \\ \eta' & \text{on } V \end{cases}$$

Clearly $(\eta, \eta') = (k^* \oplus l^*)\sigma$, hence $\ker(i^* - j^*) \subseteq \text{Im}(k^* \oplus l^*)$, i.e. $\ker(i^* - j^*) = \text{Im}(k^* \oplus l^*)$.

Finally, prove exactness at $\Omega^p(U \cap V)$. Let $\omega \in \Omega^p(U \cap V)$ arbitrary: WTS there exist $\eta \in \Omega^p(U)$ and $\eta' \in \Omega^p(V)$ such that (see Fig. 3.2):

$$\omega = (i^* - j^*)(\eta, \eta') = i^*\eta - j^*\eta' = \eta|_{U \cap V} - \eta'|_{U \cap V}$$

Figure 3.2: Surjectivity of $i^* - j^*$.

Let $\{\varphi, \psi\}$ be a smooth partition of unity subordinate to the open cover $\{U, V\}$, and define $\eta \in \Omega^p(U)$ and $\eta' \in \Omega^p(V)$ as:

$$\eta = \begin{cases} \psi\omega & \text{on } U \cap V \\ 0 & \text{on } U - \text{supp } \psi \end{cases} \quad \eta' = \begin{cases} -\varphi\omega & \text{on } U \cap V \\ 0 & \text{on } V - \text{supp } \varphi \end{cases}$$

In each form, on the sets where the two definitions overlap, they both vanish, so $\eta \in \Omega^p(U)$ and $\eta' \in \Omega^p(V)$ are well-defined. Then:

$$\eta|_{U \cap V} - \eta'|_{U \cap V} = \psi\omega - (-\varphi\omega) = (\psi + \varphi)\omega = \omega$$

which concludes the proof. $\square$

Corollary 3.1.6.1 (Characterization of connecting homomorphism)

The **connecting homomorphism** $\delta_p : \mathcal{H}^p(U \cap V) \rightarrow \mathcal{H}^{p+1}(\mathcal{M})$ of the Mayer–Veitoris sequence is defined as follows. For each $\omega \in \mathcal{Z}^p(U \cap V)$, there are $\eta \in \Omega^p(U), \eta' \in \Omega^p(V)$ such that $\omega = \eta|_{U \cap V} - \eta'|_{U \cap V}$; then, $\delta_p[\omega] = [\sigma]$, where $\sigma \in \Omega^{p+1}(\mathcal{M})$ such that $\sigma|_U = d\eta$ and $\sigma|_V = d\eta'$. If $\{\varphi, \psi\}$ is a smooth partition of unity subordinate to $\{U, V\}$, then one can take $\eta = \psi\omega$ and $\eta' = -\varphi\omega$, both extended to zero outside of the supports of ψ and φ .

§3.1.4.3 Applications of the Mayer–Vietoris theorem

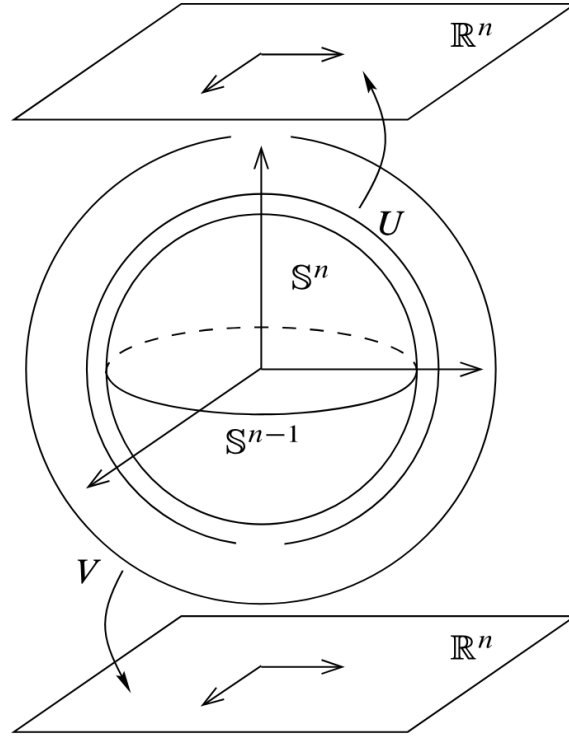
The Mayer–Vietoris theorem allows to compute all of the de Rham cohomology groups of spheres.

Proposition 3.1.5 (Cohomology of spheres)

For $n \in \mathbb{N}$, the de Rham cohomology groups of $\mathbb{S}^n$ are:

$$\mathcal{H}^p(\mathbb{S}^n) \cong \begin{cases} \mathbb{R} & p = 0, n \\ 0 & 0 < p < n \end{cases} \quad (3.1.11)$$

The cohomology class of any smooth orientation form is a basis for $\mathcal{H}^p(\mathbb{S}^n)$.

Figure 3.3: Stereographic projections on $\mathbb{S}^n$.

Proof. By Ex. 3.1.2, $\mathcal{H}^0(\mathbb{S}^n) \cong \mathbb{R}$, so consider $p \geq 1$. The proof is by induction on $n \in \mathbb{N}$.

($n = 1$) Any orientation for on $\mathbb{S}^1$ has non-zero integral, so it is not exact by Stokes' theorem, hence $\dim_{\mathbb{R}} \mathcal{H}^1(\mathbb{S}^1) \geq 1$. On the other hand, by Th. 3.1.5 there is an injective linear map $\mathcal{H}^1(\mathbb{S}^1) \rightarrow \text{Hom}(\pi_1(\mathbb{S}^1, 1), \mathbb{R})$, but $\dim_{\mathbb{R}} \text{Hom}(\pi_1(\mathbb{S}^1, 1), \mathbb{R}) = 1$, thus $\dim_{\mathbb{R}} \mathcal{H}^1(\mathbb{S}^1) = 1$, and is spanned by the cohomology class of any smooth orientation form.

($n \geq 2$) Assume that the theorem is true for $\mathbb{S}^{n-1}$. Since $\mathbb{S}^n$ is simply connected (for $n \geq 1$), $\mathcal{H}^1(\mathbb{S}^n) = 0$ by Cor. 3.1.5.1. For $p > 1$, use Th. 3.1.6 as follows: let N and S be the north and south poles in $\mathbb{S}^n$, respectively, and let $U = \mathbb{S}^n - \{S\}$ and $V = \mathbb{S}^n - \{N\}$, which are both diffeomorphic to $\mathbb{R}^n$ (through the stereographic projection, see Fig. 3.3), thus $U \cap V$ is diffeomorphic to $\mathbb{R}^n - \{0\}$. Part of the Mayer–Vietoris sequence for $\{U, V\}$ reads:

$$\mathcal{H}^{p-1}(U) \oplus \mathcal{H}^{p-1}(V) \rightarrow \mathcal{H}^{p-1}(U \cap V) \rightarrow \mathcal{H}^p(\mathbb{S}^n) \rightarrow \mathcal{H}^p(U) \oplus \mathcal{H}^p(V)$$

Since U, V are diffeomorphic to $\mathbb{R}^n$, the groups at both ends are trivial when $p > 1$, hence $\mathcal{H}^p(\mathbb{S}^n) \cong \mathcal{H}^{p-1}(U \cap V)$. Moreover, $U \cap V$ is diffeomorphic to $\mathbb{R}^n - \{0\}$ and thus homotopy equivalent to $\mathbb{S}^{n-1}$ by Ex. A.1.2, so in the end $\mathcal{H}^p(\mathbb{S}^n) \cong \mathcal{H}^{p-1}(\mathbb{S}^{n-1})$ for $p > 1$. The thesis then follows by induction. As in the $n = 1$ case, any smooth orientation on $\mathbb{S}^n$ determines a non-zero cohomology class, which therefore spans $\mathcal{H}^n(\mathbb{S}^n)$. $\square$

It can be shown that $\eta \in \Omega^n(\mathbb{S}^n)$ is exact if and only if $\int_{\mathbb{S}^n} \eta = 0$.

Corollary 3.1.5.1 (Cohomology of punctured Euclidean spaces)

Let $x \in \mathbb{R}^n$ and $\mathcal{M} = \mathbb{R}^n - \{x\}$, with $n \geq 2$. The only non-trivial de Rham groups of $\mathcal{M}$ are $\mathcal{H}^0(\mathcal{M}) \cong \mathbb{R}$ and $\mathcal{H}^{n-1}(\mathcal{M}) \cong \mathbb{R}$. A closed $\eta \in \Omega^{n-1}(\mathcal{M})$ is exact if and only if $\int_S \eta = 0$ for some (and hence every) $(n-1)$ -dimensional sphere $S \subseteq \mathcal{M}$ centered at x .

Proof. Let $S \subseteq \mathcal{M}$ be any $(n-1)$ -dimensional sphere centered at x . The inclusion map $\iota : S \hookrightarrow \mathcal{M}$ is a homotopy equivalence, thus $\iota^* : \mathcal{H}^p(\mathcal{M}) \rightarrow \mathcal{H}^p(S)$ is an isomorphism for each $p \in \mathbb{N}_0$ and the result on de Rham groups of $\mathcal{M}$ follows from Prop. 3.1.5.

If $\eta \in \mathcal{Z}^{n-1}(\mathcal{M})$, then $\eta \in \mathcal{B}^{n-1}(\mathcal{M})$ if and only if $\iota^*\eta \in \mathcal{B}^{n-1}(S)$, which is true if and only if $\int_S \eta = \int_S \iota^*\eta = 0$ by the observation above. $\square$

This result remains true even when $\mathcal{M} = \mathbb{R}^n - \bar{B}$, for some closed ball $\bar{B} \subseteq \mathbb{R}^n$.

Corollary 3.1.5.2

Let $U \in \mathcal{T}(\mathbb{R}^n)$, with $n \geq 2$, and $x \in U$. Then $\mathcal{H}^{n-1}(U - \{x\}) \neq 0$.

Proof. Since U is open, there is an $(n-1)$ -dimensional sphere S centered at x such that $S \subseteq U - \{x\}$. Let $\iota : S \hookrightarrow U - \{x\}$ be the inclusion map and $r : U - \{x\} \rightarrow S$ be the radial projection onto S . Then, r and ι are smooth with $r \circ \iota = \mathbb{1}_S$, which implies $\iota^* \circ r^* = \mathbb{1}_{\mathcal{H}^{n-1}(S)}$, and therefore $r^* : \mathcal{H}^{n-1}(S) \rightarrow \mathcal{H}^{n-1}(U - \{x\})$ is injective: since $\mathcal{H}^{n-1}(S) \neq 0$ by Prop. 3.1.5, the thesis follows. $\square$

An important application of the topological invariance of the de Rham cohomology groups is the proof of the topological invariance of dimension.

Theorem 3.1.7 (Topological invariance of dimension)

A non-empty n -dimensional topological manifold cannot be homeomorphic to an m -dimensional topological manifold, unless $m = n$.

Proof. If $\mathcal{M}$ is a topological n -manifold that is homeomorphic to an m -manifold, then $\mathcal{M}$ is itself both an n -manifold and an m -manifold. The case in which $m = 0$ or $n = 0$ is trivial, hence WLOG $m > n \geq 1$. Since $\mathcal{M}$ is an m -manifold, there is $V \in \mathcal{T}(\mathcal{M})$ that is homeomorphic to $\mathbb{R}^m$, but an open subset of an n -manifold is itself an n -manifold, hence any point $x \in V$ has a neighbourhood $U \subseteq V$ that is homeomorphic to $\mathbb{R}^n$. Since U is homeomorphic to $\mathbb{R}^n$, the homeomorphism can be used to define a smooth structure on U , and then $\mathcal{H}^{m-1}(U - \{x\}) = 0$ by Cor. 3.1.5.1. On the other hand, since U is homeomorphic to an open set of $\mathbb{R}^m$, this homeomorphism can be used to define another smooth structure on U , and Cor. 3.1.5.2 implies $\mathcal{H}^{m-1}(U - \{x\}) \neq 0$, which contradicts the topological invariance of de Rham cohomology $\rightarrow \times$ $\square$

§3.1.5 Degree theory

Top-degree cohomology can be used to prove some properties of smooth functions between certain compact manifolds of the same dimension.

It is useful to introduce a generalization of the de Rham cohomology groups using compactly supported forms. Let $\mathcal{M}$ be a smooth manifold: the $\mathbb{R}$ -vector space of compactly supported smooth p -forms on $\mathcal{M}$ is denoted by $\Omega_c^p(\mathcal{M}) \subseteq \Omega^p(\mathcal{M})$. Then, restricting to compactly supported smooth p -forms the spaces $\mathcal{Z}_c^p(\mathcal{M}) \subseteq \mathcal{Z}^p(\mathcal{M})$ and $\mathcal{B}_c^p(\mathcal{M}) \subseteq \mathcal{B}^p(\mathcal{M})$, the p^{th} compactly supported de Rham cohomology group of $\mathcal{M}$ is the quotient space:

$$\mathcal{H}_c^p(\mathcal{M}) := \frac{\mathcal{Z}_c^p(\mathcal{M})}{\mathcal{B}_c^p(\mathcal{M})} \quad (3.1.12)$$

This reduces to the ordinary de Rham cohomology when $\mathcal{M}$ is compact. For non-compact manifolds,

however, these are different groups: for example, when $n \geq 1$:

$$\mathcal{H}_c^p(\mathbb{R}^n) \cong \begin{cases} \mathbb{R} & p = n \\ 0 & 0 \leq p < n \end{cases} \quad (3.1.13)$$

contrary to $\mathcal{H}^p(\mathbb{R}^n) = 0$ for $p \geq 1$. In general, smooth maps need not pull-back compactly supported forms to compactly supported forms, so they don't induce maps on compactly supported cohomology. However, a proper map does pull-back compactly supported forms to compactly supported forms, so a proper map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ induces a cohomology map $f^* : \mathcal{H}_c^p(\mathcal{M}_2) \rightarrow \mathcal{H}_c^p(\mathcal{M}_1)$ for each $p \in \mathbb{N}_0$.

§3.1.5.1 Top-degree cohomology

Let $\mathcal{M}$ be an oriented smooth n -manifold. There is a natural linear map $I : \Omega_c^n(\mathcal{M}) \rightarrow \mathbb{R}$ given by integration over $\mathcal{M}$:

$$I(\omega) := \int_{\mathcal{M}} \omega$$

Since the integral of the exterior derivative of a compactly supported $(n-1)$ -form vanishes, I descends to a linear map $I : \mathcal{H}_c^n(\mathcal{M}) \rightarrow \mathbb{R}$ (every smooth n -form on an n -manifold is closed).

Proposition 3.1.6 (Top cohomology, orientable compact support case)

If $\mathcal{M}$ is a connected oriented n -manifold, then the integration map $I : \mathcal{H}_c^n(\mathcal{M}) \rightarrow \mathbb{R}$ is an isomorphism, so $\dim_{\mathbb{R}} \mathcal{H}_c^n(\mathcal{M}) = 1$.

Proof. If $n = 0$, the thesis is trivial by Ex. 3.1.2, so assume $n \geq 1$. Let $(U, (x^i))$ be an oriented smooth coordinate chart on $\mathcal{M}$, and let f be a smooth bump function with compact support in U : then, the n -form defined as $\theta_0 = f dx^1 \wedge \cdots \wedge dx^n$ in U and 0 outside U is smooth and compactly supported on $\mathcal{M}$, and satisfies $I(\theta_0) > 0$. Therefore, I is surjective, so WTS I is injective, which is equivalent to showing that if $\omega \in \Omega_c^n(\mathcal{M})$ such that $\int_{\mathcal{M}} \omega = 0$, then $\exists \eta \in \Omega_c^{n-1}(\mathcal{M}) : \omega = d\eta$. Let $\{U_i\}_{i \in \mathbb{N}}$ be a countable open cover of $\mathcal{M}$ such that each U_i is diffeomorphic to $\mathbb{R}^n$, and let $\mathcal{M}_k \equiv \bigcup_{i=1}^k U_i$ for each $k \in \mathbb{N}$. Since $\mathcal{M}$ is connected, it is possible to arrange the series so that $\mathcal{M}_k \cap U_{k+1} \neq \emptyset \forall k \in \mathbb{N}$. Since every compactly supported n -form is supported in $\mathcal{M}_k$ for some $k \in \mathbb{N}$, it suffices to prove that if $\omega \in \Omega_c^n(\mathcal{M}_k)$ has $\int_{\mathcal{M}_k} \omega = 0$, then $\omega = d\eta$ for some $\eta \in \Omega_c^{n-1}(\mathcal{M}_k)$, which can then be extended by zero to a compactly supported form on all of $\mathcal{M}$. The proof is by induction on $k \in \mathbb{N}$.

($k = 1$) Since $\mathcal{M}_1 = U_1$ is diffeomorphic to $\mathbb{R}^n$, the claim is proved by the following lemma.

Lemma 3.1.4 (Poincaré lemma with compact support)

Let $n \geq p \geq 1$ and let ω be a compactly supported p -form on $\mathbb{R}^n$. If $p = n$, suppose $\int_{\mathbb{R}^n} \omega = 0$. Then, there exists $\eta \in \Omega_c^{p-1}(\mathbb{R}^n)$ such that $\omega = d\eta$.

($k \geq 1$) Assume the claim to be true for some $k \geq 1$, and suppose $\omega \in \Omega_c^n(\mathcal{M}_{k+1})$, with $\mathcal{M}_{k+1} = \mathcal{M}_k \cup U_{k+1}$ that satisfies $\int_{\mathcal{M}_{k+1}} \omega = 0$. Let $\theta \in \Omega_c^n(\mathcal{M}_{k+1})$ be supported in $\mathcal{M}_k \cap U_{k+1}$ and usch that $\int_{\mathcal{M}_{k+1}} \theta = 1$. Let $\{\varphi, \psi\}$ be a smooth partition of unity for $\mathcal{M}_{k+1}$ subordinate to the cover $\{\mathcal{M}_k, U_{k+1}\}$, as in Fig. 3.4, and let $c \equiv \int_{\mathcal{M}_{k+1}} \varphi \omega$. Since $\varphi \omega - c\theta$ is compactly supported in $\mathcal{M}_k$ and its integral vanished by definition of c , by the inductive hypothesis there is $\alpha \in \Omega_c^{n-1}(\mathcal{M}_k)$

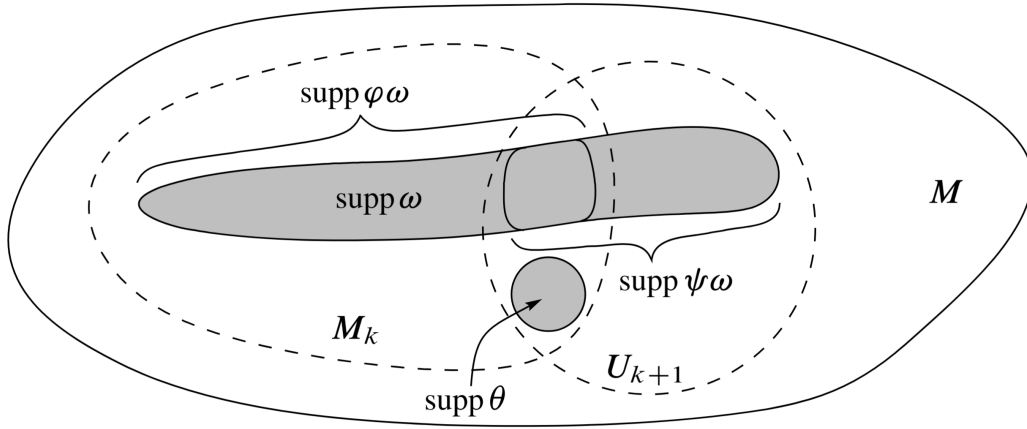


Figure 3.4: Computation of top-degree cohomology.

such that $d\alpha = \varphi\omega - c\theta$. Similarly, $\psi\omega + c\theta$ is compactly supported in U_{k+1} and:

$$\int_{U_{k+1}} (\psi\omega + c\theta) = \int_{\mathcal{M}_{k+1}} (1 - \vartheta)\omega + c \int_{\mathcal{M}_{k+1}} \theta = \int_{\mathcal{M}_{k+1}} \omega - \int_{\mathcal{M}_{k+1}} \varphi\omega + c = 0$$

Thus, by Lemma 3.1.4, there exists $\beta \in \Omega_c^{n-1}(U_{k+1})$ such that $d\beta = \psi\omega + c\theta$. Both α and β can be extended by zero to smooth compactly supported forms on $\mathcal{M}_{k+1}$, and:

$$d(\theta + \beta) = (\varphi\omega - c\theta) + (\psi\omega + c\theta) = (\varphi + \psi)\omega = \omega$$

which completes the proof. $\square$

Corollary 3.1.6.1 (Top cohomology, orientable compact case)

If $\mathcal{M}$ is a compact connected orientable smooth n -manifold, then $\dim_{\mathbb{R}} \mathcal{H}^n(\mathcal{M}) = 1$ and is spanned by the cohomology class of any smooth orientation form.

Proof. For compact manifolds $\mathcal{H}_c^p(\mathcal{M}) = \mathcal{H}^p(\mathcal{M})$, hence use Prop. 3.1.6 and the fact that any orientation form has non-vanishing integral. $\square$

Proposition 3.1.7 (Top cohomology, orientable non-compact case)

If $\mathcal{M}$ is a non-compact connected orientable smooth n -manifold, then $\mathcal{H}^n(\mathcal{M}) = 0$.

Proof. First, choose an orientation on $\mathcal{M}$. Let $f \in C^\infty(\mathcal{M})$ be a smooth exhaustion function: by adding a constant, it is possible to set $\inf_{\mathcal{M}} f = 0$, and then connectedness and non-compactness of $\mathcal{M}$ imply $f(\mathcal{M}) = [0, \infty)$. For each $i \in \mathbb{N}$, let $V_i = f^{-1}((i-1, i))$: clearly, $\{V_i\}_{i \in \mathbb{N}}$ is a cover of $\mathcal{M}$ by non-empty precompact⁵ open sets, with $V_i \cap V_j \neq \emptyset$ if and only if $j = i-1, i, i+1$. Let $\{\psi_i\}$ be a smooth partition of unity subordinate to this cover and, for each $i \in \mathbb{N}$, let $\theta_i \in \Omega_c^n(\mathcal{M})$ be compactly supported in $V_i \cap V_{i+1}$ with $\int_{\mathcal{M}} \theta_i = 1$.

Let $\omega \in \Omega^n(\mathcal{M})$ arbitrary, and let $\omega_i = \psi_i\omega \in \Omega_c^n(V_i)$ for each $i \in \mathbb{N}$. Let $c_1 = \int_{V_1} \omega_1$, so that $\omega_1 - c_1\theta_1$ is compactly supported in V_1 and has non-zero integral. It follows from Prop. 3.1.6 that there exists $\eta_1 \in \Omega_c^n(V_1)$ such that $d\eta_1 = \omega_1 - c_1\theta_1$. Next, choose $c_2 \in \mathbb{R}$ such that $\int_{V_2} (\omega_2 + c_1\theta_1 - c_2\theta_2) = 0$, so there exists $\eta_2 \in \Omega_c^n(V_2)$ such that $d\eta_2 = \omega_2 + c_1\theta_1 - c_2\theta_2$. Continuing by induction, choose $c_j \in \mathbb{R}$ and $\eta_j \in \Omega_c^n(V_j)$ such that $d\eta_j = \omega_j + c_{j-1}\theta_{j-1} - c_j\theta_j$.

Now, set $\eta \equiv \sum_{j \in \mathbb{N}} \eta_j$, with each η_j extended by zero on $\mathcal{M} - V_j$. Since at most three terms in this sum are non-zero on each V_i , this is $\eta \in \Omega^n(\mathcal{M})$. When taking the exterior derivative, the $c_j \theta_j$ terms cancel, so $d\eta = \sum_{j \in \mathbb{N}} \omega_j = \omega$. $\square$

An analogous result for non-orientable manifolds can be found in a similar way.

Proposition 3.1.8 (Top cohomology, non-orientable case)

If $\mathcal{M}$ is a connected non-orientable smooth n -manifold, then $\mathcal{H}_c^n(\mathcal{M}) = 0$ and $\mathcal{H}^n(\mathcal{M}) = 0$.

§3.1.5.2 Smooth maps

Once the top-degree cohomology groups of all compact manifolds are found, smooth maps between them can be studied by assigning an integer to them in such a way that homotopic maps have the same integer.

Theorem 3.1.8 (Degree of smooth map)

Let $\mathcal{M}_1, \mathcal{M}_2$ be compact oriented smooth n -manifolds, and let $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ be a smooth map. There exists a unique $k_f \in \mathbb{Z}$, called the **degree** of f , that satisfies:

- (1) for every $\omega \in \Omega^n(\mathcal{M}_2)$:

$$\int_{\mathcal{M}_1} f^* \omega = k_f \int_{\mathcal{M}_2} \omega$$

- (2) if $q \in \mathcal{M}_2$ is a regular value of f , then:

$$k_f = \sum_{x \in f^{-1}(q)} \text{sgn}(x)$$

where $\text{sgn}(x) = +1$ if df_x is orientation-preserving, and -1 otherwise.

Proof. By Cor. 3.1.6.1, two smooth n -forms on either $\mathcal{M}_1$ or $\mathcal{M}_2$ are cohomologous if and only if they have the same integral. Let $\theta \in \Omega^n(\mathcal{M}_2) : \int_{\mathcal{M}_2} \theta = 1$, and let $k_f = \int_{\mathcal{M}_1} f^* \theta$. If $\omega \in \Omega^n(\mathcal{M}_2)$ is arbitrary, then ω is cohomologous to $a\theta$, where $a \in \mathbb{R}$, and therefore $f^* \omega$ is cohomologous to $a f^* \theta$. Then:

$$\int_{\mathcal{M}_1} f^* \omega = a \int_{\mathcal{M}_1} f^* \theta = a k_f = k_f \int_{\mathcal{M}_2} \omega$$

Thus k_f satisfies (1) and is clearly the only number that does so.

Next WTS (2), from which it also follows that $k_f \in \mathbb{Z}$. Let $q \in \mathcal{M}_2$ be an arbitrary regular value of f : since $f^{-1}(q)$ is a properly embedded 0-submanifold of $\mathcal{M}$, it is finite. Suppose first that $f^{-1}(q)$ is not empty, i.e. $f^{-1}(q) = \{x_1, \dots, x_m\}$. By the inverse function theorem, for each $i = 1, \dots, m$ there is a neighbourhood U_i of x_i such that f is a diffeomorphism from U_i to a neighbourhood W_i of q , and, by shrinking the U_i sets if necessary, they can be assumed to be disjoint. Then, $K = \mathcal{M} - (U_1 \cup \dots \cup U_m)$ is closed in $\mathcal{M}$ and thus compact, so $f(K)$ is closed in $\mathcal{M}_2$ and disjoint from q . Let W be the connected component of $W_1 \cap \dots \cap W_m \cap (\mathcal{M}_2 - f(K))$ containing q , and let $V_i = f^{-1}(W) \cap U_i$: it follows that W is a connected neighbourhood of q such that $f^{-1}(W) = V_1 \sqcup \dots \sqcup V_m$, and f restricts to a diffeomorphism from each V_i to W . Since each V_i is connected, the restriction of f to V_i must be either orientation-preserving or reversing.

⁵Given a topological space X , $U \subseteq X$ is *precompact* if $\overline{U}$ is compact.

Let $\omega \in \Omega^n(\mathcal{M}_2)$ be compactly supported in W and satisfying $\int_{\mathcal{M}_2} \omega = \int_W \omega = 1$. From (1) it follows that $\int_{\mathcal{M}_1} f^* \omega = k_f$ and, since $f^* \omega$ is compactly supported in $f^{-1}(W)$, $\int_{\mathcal{M}_1} f^* \omega = \sum_{i=1}^m \int_{V_i} f^* \omega$. From (4) of Lemma 2.3.5, $\int_{V_i} f^* \omega = \pm \int_W \omega = \pm 1$ for each $i = 1, \dots, m$, with positive sign if f is orientation-preserving on V_i and negative otherwise. This proves (2) when $f^{-1}(q) \neq \emptyset$.

If $f^{-1}(q) = \emptyset$, then q has a neighbourhood $W \subseteq \mathcal{M}_2 - f(\mathcal{M}_1)$ since $f(\mathcal{M}_1)$ is compact and thus closed. Given $\omega \in \Omega^n(\mathcal{M}_2)$ compactly supported in W , then $\int_{\mathcal{M}_1} f^* \omega = 0$, so $k_f = 0$. This proves (2). $\square$

The importance of degree theory comes from the fact that these two characterizations are independent of each other: for example, it is often easy to compute the degree of a map by simply counting the points in the preimage of a regular value, and then use the characterization in terms of differential forms to prove important properties.

Lemma 3.1.5 (Properties of degree)

Let $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$ be compact connected oriented smooth n -manifolds. Then:

- (1) if $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ and $g : \mathcal{M}_2 \rightarrow \mathcal{M}_3$ are smooth maps, then $k_{g \circ f} = k_g k_f$;
- (2) if $f : \mathcal{M}_1 \rightarrow \mathcal{M} - 2$ is a diffeomorphism, the $k_f = +1$ if f is orientation-preserving and -1 otherwise;
- (3) if two smooth maps $f_0, f_1 : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ are homotopic, then $k_{f_0} = k_{f_1}$.

By Lemma 3.1.5, it is possible to define the degree of a continuous map $f : \mathcal{M}_1 \rightarrow \mathcal{M}_2$ between compact connected oriented smooth n -manifolds as the degree of any smooth map that is homotopic to f : the Whitney approximation theorem Th. 1.3.13 guarantees the existence of such a map.

Proposition 3.1.9

Let $\mathcal{M}$ be a compact connected oriented smooth n -manifold, and X a compact oriented smooth $(n+1)$ -manifold with boundary. If $f : \partial X \rightarrow \mathcal{M}$ is a continuous map that has a continuous extension to X , then $k_f = 0$.

Proof. Let $F : X \rightarrow \mathcal{M}$ be the continuous extension of f on X . By Th. 1.3.13, there is a smooth map $\tilde{f} : X \rightarrow \mathcal{M}$ that is homotopic to F . Replacing F by $\tilde{F}$ and f by $\tilde{f}|_{\partial X}$, it is possible to assume that both f and F are smooth.

Let $\omega \in \Omega^n(\mathcal{M})$: then, $d\omega = 0$ since ω is a top form. From Stokes' theorem:

$$\int_{\partial X} f^* \omega = \int_{\partial X} F^* \omega = \int_X d(F^* \omega) = \int_X F^* d\omega = 0$$

By (1) of Th. 3.1.8, then, $k_f = 0$. $\square$

Theorem 3.1.9 (Brouwer fixed-point theorem)

Every continuous map $\mathbb{D}^n \rightarrow \mathbb{D}^n$ has a fixed point.

Proof. Suppose that $F : \mathbb{D}^n \rightarrow \mathbb{D}^n$ is continuous and has no fixed points. It is then possible to

define a continuous map $G : \mathbb{D}^n \rightarrow \mathbb{S}^{n-1}$ as:

$$G(x) = \frac{x - F(x)}{\|x - F(x)\|}$$

and let $g = G|_{\mathbb{S}^{n-1}} : \mathbb{S}^{n-1} \rightarrow \mathbb{S}^{n-1}$. By Prop. 3.1.9 $k_g = 0$. On the other hand, consider $H : \mathbb{S}^{n-1} \times [0, 1] \rightarrow \mathbb{S}^{n-1}$ defined as:

$$H(x, t) = \frac{x - tF(x)}{\|x - tF(x)\|}$$

which is well-defined since the denominator never vanishes when $t = 1$, as F has no fixed points, and when $t < 1$, as $\|x\| = 1$ while $\|tF(x)\| \leq t < 1$. Therefore, H is continuous and a homotopy from $\mathbb{1}_{\mathbb{S}^{n-1}}$ to g : this implies $k_g = 1$ by (3) of Lemma 3.1.5 $\dashv$ $\square$

§3.2 Hodge theory

The study of the de Rham cohomology of a metric manifold makes use of the additional structure given by the metric.

§3.2.1 Inner product on differential forms

Recall §2.1.3: on a metric manifold $(\mathcal{M}, g)$, the canonical smooth bundle isomorphism $T\mathcal{M} \cong T^*\mathcal{M}$ allows the metric to define an inner product on $T_p^*\mathcal{M}$ as:

$$\langle \omega, \eta \rangle_g \equiv \langle \omega^\sharp, \eta^\sharp \rangle_g$$

where the inner product on the right-hand side is that on $T_p\mathcal{M}$, i.e. $\langle v, w \rangle_g := g_p(v, w)$, which is positive-definite on Riemannian manifolds. Clearly, then, $\langle \omega, \eta \rangle_g = \omega(\eta^\sharp) = \eta(\omega^\sharp)$ by definition of the sharp isomorphism.

This construction is easily generalized to decomposable differential forms. Indeed, given two decomposable p -forms $\alpha = \alpha_1 \wedge \dots \wedge \alpha_p$ and $\beta = \beta_1 \wedge \dots \wedge \beta_p$, their metric-induced inner product can be defined through the **Gram determinant**:

$$\langle \alpha, \beta \rangle_g := \det[\langle \alpha_i, \beta_j \rangle_g] \quad (3.2.1)$$

This is a natural choice since the determinant is alternating under the exchange of columns, thus preserving the alternating nature of the exterior product, and it reduces to the inner product on 1-forms defined above. This definition clearly generalizes bilinearly to arbitrary p -forms, since any p -form can be expressed as a linear combination of decomposable forms (Eq. 2.2.19).

Proposition 3.2.1 (Inner product of p -forms)

Let $(\mathcal{M}, g)$ be a metric manifold, and let ω, η be two p -forms on $\mathcal{M}$. Then, their inner product is:

$$\langle \omega, \eta \rangle_g = \frac{1}{p!} \omega_{i_1 \dots i_p} \eta_{j_1 \dots j_p} g^{i_1 j_1} \dots g^{i_p j_p} \quad (3.2.2)$$

on every local oriented orthonormal frame on $\mathcal{M}$.

Proof. Let (x^i) be a smooth chart on $\mathcal{M}$, so that ω and η can be expressed as in Eq. 2.2.19. Then, extending bilinearly Eq. 3.2.1:

$$\begin{aligned}\langle \omega, \eta \rangle_g &= \frac{1}{(p!)^2} \omega_{i_1 \dots i_p} \eta_{j_1 \dots j_p} \det[\langle dx^{i_a}, dx^{j_b} \rangle_g] = \frac{1}{(p!)^2} \omega_{i_1 \dots i_p} \eta_{j_1 \dots j_p} \det[g^{i_a j_b}] \\ &= \frac{1}{(p!)^2} \omega_{i_1 \dots i_p} \eta_{j_1 \dots j_p} \sum_{\sigma \in S^p} \operatorname{sgn}(\sigma) g^{i_1 j_{\sigma(1)}} \dots g^{i_p j_{\sigma(p)}} = \frac{1}{p!} \omega_{i_1 \dots i_p} \eta_{j_1 \dots j_p} g^{i_1 j_1} \dots g^{i_p j_p}\end{aligned}$$

where in the last step the total antisymmetry of $\omega_{i_1 \dots i_p}$ was used. To show that this expression is basis-independent, consider two oriented orthonormal frames $(E_1, \dots, E_n)$ and $(\tilde{E}_1, \dots, \tilde{E}_n)$ on $\mathcal{M}$ such that:

$$\tilde{E}_i = A_i^j E_j$$

for some smooth functions $[A_i^j]$. Since these frames are consistently oriented, $[A_i^j](p) \in \operatorname{SO}(n)$ for each $p \in \mathcal{M}$. Then, the components of ω in the two bases are linked by (Eq. 1.5.11):

$$\tilde{\omega}_{i_1 \dots i_p} = A_{i_1}^{j_1} \dots A_{i_p}^{j_p} \omega_{j_1 \dots j_p}$$

and similarly for η . Then, given the expression for the metric $g^{ij} = \langle \varepsilon^i, \varepsilon^j \rangle_g$ and the transformation law $\tilde{\varepsilon}^i = [A^{-1}]_j^i \varepsilon^j$ for the dual coframes, the inner product in the transformed frame is:

$$\begin{aligned}\langle \omega, \eta \rangle_g &= \tilde{\omega}_{i_1 \dots i_p} \tilde{\eta}_{j_1 \dots j_p} \tilde{g}^{i_1 j_1} \dots \tilde{g}^{i_p j_p} \\ &= A_{i_1}^{\ell_1} \dots A_{i_p}^{\ell_p} \omega_{\ell_1 \dots \ell_p} A_{j_1}^{m_1} \dots A_{j_p}^{m_p} \eta_{m_1 \dots m_p} [A^{-1}]_{a_1}^{i_1} [A^{-1}]_{b_1}^{j_1} \dots [A^{-1}]_{a_p}^{i_p} [A^{-1}]_{b_p}^{j_p} g^{a_1 b_1} \dots g^{a_p b_p} \\ &= \delta_{a_1}^{\ell_1} \delta_{b_1}^{m_1} \dots \delta_{a_p}^{\ell_p} \delta_{b_p}^{m_p} \omega_{\ell_1 \dots \ell_p} \eta_{m_1 \dots m_p} g^{a_1 b_1} \dots g^{a_p b_p} = \omega_{i_1 \dots i_p} \eta_{j_1 \dots j_p} g^{i_1 j_1} \dots g^{i_p j_p}\end{aligned}$$

which indeed shows that the above expression for the inner product is independent of the choice of oriented orthonormal frame. $\square$

§3.2.1.1 L^2 -inner product

Another useful inner product is the L^2 -inner product of compactly supported p -forms, which can be defined on orientable metric manifolds:

$$(\omega, \eta) := \int_{\mathcal{M}} \langle \omega, \eta \rangle_g dV_g \quad (3.2.3)$$

where ω_g is the volume form on $\mathcal{M}$ induced by g (Eq. 2.3.3). To be precise, this inner product is well-defined on *square-integrable* p -forms:

$$\Omega_{L^2}^p(\mathcal{M}) := \{\omega \in \Omega^p(\mathcal{M}) : \|\omega\|_{L^2} < \infty\} \quad (3.2.4)$$

where $\|\omega\|_{L^2} = (\omega, \omega)$. Clearly, $\Omega_c^p(\mathcal{M}) \subsetneq \Omega_{L^2}^p(\mathcal{M}) \subsetneq \Omega^p(\mathcal{M})$ in general (if $\mathcal{M}$ is compact, they are all equal), and it can be shown that $\Omega_c^p(\mathcal{M})$ is dense⁶ in $\Omega_{L^2}^p(\mathcal{M})$. Note that $\Omega_{L^2}^p(\mathcal{M})$ is a pre-Hilbert space, since it is not complete: it is possible to construct a Cauchy sequence of forms in $\Omega_{L^2}^p(\mathcal{M})$ whose limit exists in a L^2 sense, but is not smooth.

⁶Given a topological space X and a subset $A \subseteq X$, a *limit point* of A in X is a point $p \in X$ such that every neighbourhood of x contains at least one point of A different from x itself. Then, A is said to be *dense* in X if every point of X either is in A or is a limit point of A .

§3.2.2 Hodge dual

Definition 3.2.1 (Hodge operator)

Let $(\mathcal{M}, g)$ be a metric manifold. The **Hodge operator** $\star : \Omega^p(\mathcal{M}) \rightarrow \Omega^{n-p}(\mathcal{M})$ is defined such that, given $\omega, \eta \in \Omega^p(\mathcal{M})$:

$$\omega \wedge \star \eta = \langle \omega, \eta \rangle_g dV_g \quad (3.2.5)$$

The image $\star \eta \in \Omega^{n-p}(\mathcal{M})$ is often called the *Hodge dual* of $\eta \in \Omega^p(\mathcal{M})$. It is trivial to show that the Hodge operator is a linear operator, hence it is completely determined by its action on the basis p -forms (in each smooth map on $\mathcal{M}$):

$$\star(dx^{i_1} \wedge \cdots \wedge dx^{i_p}) = \frac{\sqrt{g}}{(n-p)!} g^{i_1 j_1} \cdots g^{i_p j_p} \epsilon_{j_1 \dots j_n} dx^{j_{p+1}} \wedge \cdots \wedge dx^{j_n} \quad (3.2.6)$$

where the factorial accounts for the contraction of $n-p$ indices between two totally antisymmetric objects. This expression makes it clear that $dV_g = \star 1$. Equivalently, the Hodge dual of $\omega \in \Omega^p(\mathcal{M})$ can be expressed as:

$$\star \omega = \frac{1}{(n-p)!} (\star \omega)_{i_1 \dots i_{n-p}} dx^{i_1} \wedge \cdots \wedge dx^{i_{n-p}}$$

with components:

$$(\star \omega)_{i_1 \dots i_{n-p}} = \frac{\sqrt{g}}{n!} \omega_{j_1 \dots j_p} g^{j_1 \ell_1} \cdots g^{j_p \ell_p} \epsilon_{\ell_1 \dots \ell_k i_1 \dots i_{n-p}} \quad (3.2.7)$$

Lemma 3.2.1 (Invariance of the Hodge dual)

The Hodge dual is independent of the choice of oriented orthonormal frame.

Proof. This invariance is trivially derived from Eq. 3.2.5, since both the inner product and the volume form are independent of the choice of oriented orthonormal frame, respectively by Prop. 3.2.1 and Prop. 2.3.10.

This can also be seen directly from Eq. 3.2.6 (alibi interpretation) or Eq. 3.2.7 (alias interpretation). For example, given two smooth charts $(U, x^i), (V, y^i) : U \cap V \neq \emptyset$, then on $U \cap V$:

$$\begin{aligned} \star(dy^{i_1} \wedge \cdots \wedge dy^{i_p}) &= \frac{1}{(n-p)!} \sqrt{|\det[\tilde{g}_{ij}]|} \tilde{g}^{i_1 j_1} \cdots \tilde{g}^{i_p j_p} \epsilon_{j_1 \dots j_n} dy^{j_{p+1}} \wedge \cdots \wedge dy^{j_n} \\ &= \frac{1}{(n-p)!} \sqrt{\left| \det \left[\frac{\partial x^s}{\partial y^i} \right] \det \left[\frac{\partial x^t}{\partial y^j} \right] \det[g_{st}] \right|} \times \\ &\quad \times \frac{\partial y^{i_1}}{\partial x^{a_1}} \frac{\partial y^{j_1}}{\partial x^{b_1}} g^{a_1 b_1} \cdots \frac{\partial y^{i_p}}{\partial x^{a_p}} \frac{\partial y^{j_p}}{\partial x^{b_p}} g^{a_p b_p} \epsilon_{j_1 \dots j_n} \frac{\partial y^{j_{p+1}}}{\partial x^{b_{p+1}}} dx^{b_{p+1}} \wedge \cdots \wedge \frac{\partial y^{j_n}}{\partial x^{b_n}} dx^{b_n} \\ &= \left| \det \left[\frac{\partial x}{\partial y} \right] \right| \frac{\partial y^{i_1}}{\partial x^{a_1}} \frac{\partial y^{j_1}}{\partial x^{b_1}} \cdots \frac{\partial y^{i_p}}{\partial x^{a_p}} \frac{\partial y^{j_p}}{\partial x^{b_p}} \frac{\partial y^{j_{p+1}}}{\partial x^{b_{p+1}}} \cdots \frac{\partial y^{j_n}}{\partial x^{b_n}} \epsilon_{j_1 \dots j_n} \times \\ &\quad \times \frac{\sqrt{g}}{(n-p)!} g^{a_1 b_1} \cdots g^{a_p b_p} dx^{b_{p+1}} \wedge \cdots \wedge dx^{b_n} \end{aligned}$$

Now, recall the following identity for $M \in \mathbb{R}^{n \times n}$ has been used:

$$\epsilon^{i_1 \dots i_n} M_{i_1 j_1} \cdots M_{i_n j_n} = \epsilon_{j_1 \dots j_n} \det M \quad (3.2.8)$$

so that the above expression becomes:

$$\begin{aligned} \star(\mathrm{d}y^{i_1} \wedge \cdots \wedge \mathrm{d}y^{i_p}) &= \left| \det \left[\frac{\partial x}{\partial y} \right] \right| \frac{\partial y^{i_1}}{\partial x^{a_1}} \cdots \frac{\partial y^{i_p}}{\partial x^{a_p}} \epsilon_{b_1 \dots b_n} \det \left[\frac{\partial y}{\partial x} \right] \\ &\quad \times \frac{\sqrt{g}}{(n-p)!} g^{a_1 b_1} \cdots g^{a_p b_p} \mathrm{d}x^{b_{p+1}} \wedge \cdots \wedge \mathrm{d}x^{b_n} \\ &= \operatorname{sgn} \left(\det \left[\frac{\partial y}{\partial x} \right] \right) \frac{\partial y^{i_1}}{\partial x^{j_1}} \cdots \frac{\partial y^{i_p}}{\partial x^{j_p}} \star(\mathrm{d}x^{j_1} \wedge \cdots \wedge \mathrm{d}x^{j_p}) \end{aligned}$$

Since the two charts are assumed to have the same orientation, the sign function returns +1, and the relation found clearly shows that the factors coming from the transformation of $\star(\mathrm{d}y^{i_1} \wedge \cdots \wedge \mathrm{d}y^{i_p})$ exactly cancel those in the transformation of $\omega_{i_1 \dots i_p}$, leaving $\star\omega$ unchanged. $\square$

Therefore, the Hodge operator is a well-defined operator.

Lemma 3.2.2 (Nilpotence of the Hodge operator)

Let $(\mathcal{M}, g)$ be a metric manifold, and let $\omega \in \Omega^p(\mathcal{M})$. Then:

$$\star(\star\omega) = \sigma(-1)^{k(n-p)}\omega \quad (3.2.9)$$

where $\sigma := \operatorname{sgn} \det[g_{ij}]$ is the **parity** of the signature of g (basis-independent).

Proof. By linearity, it suffices to show the result only for basis p -forms:

$$\begin{aligned} \star(\star(\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_p})) &= \frac{\sqrt{g}}{(n-p)!} g^{i_1 j_1} \cdots g^{i_p j_p} \epsilon_{j_1 \dots j_n} \star(\mathrm{d}x^{j_{p+1}} \wedge \cdots \wedge \mathrm{d}x^{j_n}) \\ &= \frac{|\det[g_{ij}]|}{(n-p)!p!} g^{i_1 j_1} \cdots g^{i_p j_p} g^{j_{p+1} a_1} \cdots g^{j_n a_{n-p}} \epsilon_{j_1 \dots j_n} \epsilon_{a_1 \dots a_n} \times \\ &\quad \times \mathrm{d}x^{a_{n-p+1}} \wedge \cdots \wedge \mathrm{d}x^{a_n} \\ &= \frac{|\det[g_{ij}]|}{(n-p)!p!} \det[g^{ij}] \epsilon^{i_1 \dots i_p a_1 \dots a_{n-p}} \epsilon_{a_1 \dots a_n} \mathrm{d}x^{a_{n-p+1}} \wedge \cdots \wedge \mathrm{d}x^{a_n} \\ &= \frac{\sigma}{(n-p)!p!} (-1)^{p(n-p)} \epsilon^{a_1 \dots a_{n-p} i_1 \dots i_p} \epsilon_{a_1 \dots a_{n-p} j_1 \dots j_p} \mathrm{d}x^{j_1} \wedge \cdots \wedge \mathrm{d}x^{j_p} \end{aligned}$$

where Eq. 3.2.8 has been used. Now, use the following identity for the generalized Kroncker delta:

$$m! \delta_{j_1 \dots j_p}^{i_1 \dots i_p} = \epsilon^{a_1 \dots a_m i_1 \dots i_p} \epsilon_{a_1 \dots a_m j_1 \dots j_p} \quad (3.2.10)$$

so that:

$$\begin{aligned} \star(\star(\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_p})) &= \frac{\sigma}{p!} (-1)^{p(n-p)} \delta_{j_1 \dots j_p}^{i_1 \dots i_p} \mathrm{d}x^{j_1} \wedge \cdots \wedge \mathrm{d}x^{j_p} \\ &= \sigma (-1)^{p(n-p)} \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_p} \end{aligned}$$

which is the thesis. $\square$

The nilpotence of the Hodge operator allows to define a simple expression for its inverse $\star^{-1} : \Omega^p(\mathcal{M}) \rightarrow \Omega^{n-p}(\mathcal{M})$:

$$\star^{-1} = \sigma(-1)^{p(n-p)}\star \quad (3.2.11)$$

§3.2.2.1 Codifferential

Using the Hodge operator, the L^2 -inner product of $\omega, \eta \in \Omega_{L^2}^p(\mathcal{M})$ can be written as:

$$(\omega, \eta) = \int_{\mathcal{M}} \omega \wedge \star \eta \quad (3.2.12)$$

Proposition 3.2.2 (Codifferential)

Let $(\mathcal{M}, g)$ be a metric manifold without boundary. Then, the **codifferential** is the unique operator $d^\dagger : \Omega^p(\mathcal{M}) \rightarrow \Omega^{p-1}(\mathcal{M})$ such that, for any $\omega \in \Omega_{L^2}^{p-1}(\mathcal{M}), \eta \in \Omega_{L^2}^p(\mathcal{M})$:

$$(d\omega, \eta) = (\omega, d^\dagger \eta) \quad (3.2.13)$$

Its expression is:

$$d^\dagger = \sigma(-1)^{n(p+1)+1} \star d \star \quad (3.2.14)$$

Proof. It suffices to consider $\omega \in \Omega_{L^2}^{p-1}(\mathcal{M}), \eta \in \Omega_{L^2}^p(\mathcal{M})$. Then, $\omega \wedge \star \eta \in \Omega^{n-1}(\mathcal{M})$, and by Eq. 2.2.23:

$$d(\omega \wedge \star \eta) = d\omega \wedge \star \eta + (-1)^{p-1} \omega \wedge d \star \eta \in \Omega^n(\mathcal{M})$$

By Stokes' theorem, then, since $\partial \mathcal{M} = \emptyset$:

$$0 = \int_{\mathcal{M}} d(\omega \wedge \star \eta) = \int_{\mathcal{M}} d\omega \wedge \star \eta + (-1)^{p-1} \int_{\mathcal{M}} \omega \wedge d \star \eta$$

The first term is just $(d\omega, \eta)$, while the second term is proportional to $(\omega, \star d \star \eta)$. To determine the coefficient, note that $d \star \eta \in \Omega^{n-p+1}(\mathcal{M})$, so using Eq. 3.2.9:

$$(\omega, \star d \star \eta) = \int_{\mathcal{M}} \omega \wedge \star(\star d \star \eta) = \sigma(-1)^{(n-p+1)(p-1)} \int_{\mathcal{M}} \omega \wedge d \star \eta$$

Thus, combining the two relations:

$$(\omega, \star d \star \eta) = -\sigma(-1)^{(n-p)(p-1)} (d\omega, \eta)$$

Noting that $\sigma^2 = 1$ and:

$$(n-p)(p-1) + 1 = n(p-1) - p(p-1) + 1 \equiv n(p+1) + 1 \pmod{2}$$

yields the expression for $d^\dagger$. □

For metric manifolds with boundary, the $d^\dagger$ is not the adjoint of d in the operator-theoretic sense, since $(d\omega, \eta)$ and $(\omega, d^\dagger \eta)$ differ by a boundary term proportional to $\int_{\partial \mathcal{M}} \omega \wedge \star \eta$.

Definition 3.2.2 (Laplace–de Rham operator)

Let $(\mathcal{M}, g)$ be a metric manifold. The **Laplace–de Rham operator** $\Delta_d \in \text{End } \Omega^p(\mathcal{M})$ is the operator defined as:

$$\Delta_d := (d + d^\dagger)^2 \quad (3.2.15)$$

To simplify the expression for the Laplace–de Rham operator, it is crucial to extend the nilpotence $d^2 = 0$ to the codifferential.

Lemma 3.2.3 (Nilpotence of the codifferential)

$$d^\dagger \circ d^\dagger \equiv 0 \quad (3.2.16)$$

Proof. By direct computation, given $\omega \in \Omega^p(\mathcal{M})$, then $d \star \omega \in \Omega^{n-p+1}(\mathcal{M})$:

$$d^\dagger(d^\dagger \omega) = \star d \star \star d \star \omega = \sigma(-1)^{(n-p+1)(p-1)} \star d \circ d \star \omega$$

Using $d \circ d \equiv 0$ completes the proof. $\square$

By the nilpotence of both d and $d^\dagger$, the Laplace–de Rham operator can be expressed as:

$$\Delta_d = dd^\dagger + d^\dagger d \quad (3.2.17)$$

Example 3.2.1 (Laplacian of functions)

Let $(\mathcal{M}, g)$ be an oriented metric manifold. Trivially, $d^\dagger f = 0$ for any $f \in \mathcal{C}^\infty(\mathcal{M}) = d0$, since $\star f \in \Omega^n(\mathcal{M})$ is a top form, hence in a smooth chart (U, x^i) on $\mathcal{M}$:

$$\begin{aligned} \Delta_d f &= d^\dagger d f = \sigma(-1)^{2n+1} \star d \star (\partial_i f dx^i) = -\sigma \star d(\partial_i f \star dx^i) \\ &= -\frac{\sigma}{(n-1)!} \star d(\partial_i f \sqrt{g} g^{ij_1} \epsilon_{j_1 \dots j_n} dx^{j_2} \wedge \dots \wedge dx^{j_n}) \\ &= -\frac{\sigma}{(n-1)!} \star [\partial_\ell (\sqrt{g} g^{ij_1} \partial_i f) \epsilon_{j_1 \dots j_n} dx^\ell \wedge dx^{j_2} \wedge \dots \wedge dx^{j_n}] = -\sigma \partial_\ell (\sqrt{g} g^{i\ell} \partial_i f) \star dV \end{aligned}$$

Now, since $dV_g = \star 1$, by Eq. 3.2.9 it follows that $\star dV_g = \sigma$, so inserting $dV_g = \sqrt{g} dV$ the final expression for the Laplacian of a function is found:

$$\Delta_d f = -\frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i} \left(\sqrt{g} g^{ij} \frac{\partial f}{\partial x^j} \right) \quad (3.2.18)$$

As hinted by Eq. 3.2.18, the Laplace–de Rham operator is the opposite of the ordinary Laplace–Beltrami operator: $\Delta_d = -\nabla^2$.

§3.2.2.2 Harmonic forms and de Rham cohomology

The Laplace–de Rham operator is the link connecting Hodge theory to the de Rham cohomology. In particular, it is possible to define a new space of differential forms.

Let $(\mathcal{M}, g)$ be a metric manifold: $\omega \in \Omega^p(\mathcal{M})$ is said to be **harmonic** if $\Delta_d \omega = 0$. The space of such harmonic p -forms is denoted by $\mathcal{H}^p(\mathcal{M}) \subseteq \Omega^p(\mathcal{M})$. Note that, on oriented Riemannian manifolds, harmonic forms are both closed and co-closed, since:

$$0 = (\omega, \Delta_d \omega) = (d\omega, d\omega) + (d^\dagger \omega, d^\dagger \omega)$$

As the L^2 -inner product is positive-definite in the Riemannian case, this relation holds if and only if both $d\omega = 0$ and $d^\dagger \omega = 0$. The Riemannian case also has additional properties.

Theorem 3.2.1 (Harmonic decomposition of forms)

Let $(\mathcal{M}, g)$ be a compact Riemannian manifold. Then, for all $p \in \mathbb{N}_0$:

$$\Omega^k(\mathcal{M}) = \text{Im } d_{p-1} \oplus \text{Im } d_{p+1}^\dagger \oplus \mathcal{H}^p(\mathcal{M}) \quad (3.2.19)$$

| *Proof.* See §5 of [19]. □

As in §3.1.2, the form-degree of $d_p : \Omega^p(\mathcal{M}) \rightarrow \Omega^{p+1}(\mathcal{M})$ and $d_p^\dagger : \Omega^p(\mathcal{M}) \rightarrow \Omega^{p-1}(\mathcal{M})$ has been made explicit for convenience. With this direct sum, on a compact Riemannian manifold $\mathcal{M}$, any $\omega \in \Omega^p(\mathcal{M})$ can be uniquely decomposed as $\omega = d\alpha + d^\dagger\beta + \gamma$, with $\alpha \in \Omega^{p-1}(\mathcal{M})$, $\beta \in \Omega^{p+1}(\mathcal{M})$ and $\gamma \in \Omega^p(\mathcal{M})$.

Theorem 3.2.2 (Hodge's theorem)

Let $(\mathcal{M}, g)$ be a compact Riemannian manifold. Then, for all $p \in \mathbb{N}_0$:

$$\mathcal{H}^p(\mathcal{M}) \cong \mathcal{H}^p(\mathcal{M}) \quad (3.2.20)$$

Proof. Since harmonic p -forms are closed, $\mathcal{H}^p(\mathcal{M}) \subseteq \mathcal{Z}^p(\mathcal{M})$. Moreover, Th. 3.2.1 implies that $\mathcal{H}^p(\mathcal{M}) \cap \text{Im } d_{p-1} = \emptyset$, hence $\mathcal{H}^p(\mathcal{M}) \subseteq \mathcal{H}^p(\mathcal{M})$.

WTS any equivalence class $[\omega] \in \mathcal{H}^p(\mathcal{M})$ can be represented by a harmonic p -form. By Th. 3.2.1, let $\omega = d\alpha + d^\dagger\beta + \gamma$ be the unique decomposition of $\omega \in \mathcal{Z}^p(\mathcal{M})$, with $\alpha \in \Omega^{p-1}(\mathcal{M})$, $\beta \in \Omega^{p+1}(\mathcal{M})$ and $\gamma \in \mathcal{H}^p(\mathcal{M})$. By definition $d\omega = 0$, but $d^2 = 0$ and $d\gamma = 0$ since harmonic forms are closed, hence $d\omega = dd^\dagger\beta = 0$ and:

$$0 = (d\omega, \beta) = (dd^\dagger\beta, \beta) = (d^\dagger\beta, d^\dagger\beta)$$

Since the L^2 -inner product is positive-definite on Riemannian manifolds, this implies $d^\dagger\beta = 0$, therefore $\omega = d\alpha + \gamma$. By definition $\mathcal{H}^p(\mathcal{M}) := \mathcal{Z}^p(\mathcal{M})/\mathcal{B}^p(\mathcal{M})$, so $[\omega] = \gamma$ as sought. □

This is the fundamental result of Hodge theory: each de Rham cohomology class contains exactly one harmonic form.

§3.3 Cartan formalism

Given a metric n -manifold $\mathcal{M}$ and a smooth chart $(U, (x^\mu))$, consider a local smooth frame on U defined as a linear combination of the coordinate basis:

$$E_a = e_a^\mu \partial_\mu \quad (3.3.1)$$

where both $a, \mu = 1, \dots, n$. A particularly useful choice of a non-coordinate frame is one in which the metric looks flat:

$$g(E_a, E_b) = g_{\mu\nu} e_a^\mu e_b^\nu \equiv \eta_{ab} \quad (3.3.2)$$

where $\eta_{ab} \equiv \text{diag}(-1, \dots, -1, 1, \dots, 1)$ is the flat metric with signature (r, s) (which is assumed to exist on $\mathcal{M}$). Such a frame (E_a) is called *vielbein* or **tetrad frame**. In the following computations, Greek indices denote quantities computed in the (∂_μ) coordinate frame, which are raised/lowered by the metric $g_{\mu\nu}$, while Latin indices denote quantities computed in the (E_a) tetrad frame, which are raised/lowered by the metric η_{ab} .

The dual tetrad frame (θ^a) is defined by $\theta^a(E_b) = \delta_b^a$. The relation to the dual coordinate frame is:

$$\theta^a = e_a^\mu dx^\mu \quad (3.3.3)$$

where the coefficients satisfy $e_a^\mu e_b^\mu = \delta_b^a$ and $e_a^\mu e_a^\nu = \delta_\mu^\nu$. The metric tensor can be written in both frames as $g = g_{\mu\nu} dx^\mu \otimes dx^\nu = \eta_{ab} \theta^a \otimes \theta^b$, thus the inverse relation of Eq. 3.3.2 is found:

$$g_{\mu\nu} = e_a^\mu e_b^\nu \eta_{ab} \quad (3.3.4)$$

Observation 3.3.1 (Vielbeins and Lorentz transformations)

Given a smooth chart $(U, (x^\mu))$ on $\mathcal{M}$, vielbeins are not uniquely-defined. Indeed, if (E_a) is a tetrad frame, it is always possible to find a new tetrad frame like:

$$\tilde{E}_a = L_a^b E_b \quad \Longleftrightarrow \quad \tilde{e}_a^\mu = L_a^b e_b^\mu$$

The transformation matrix must then satisfy the condition imposed by Eq. 3.3.2:

$$L_a^c L_b^d \eta_{cd} = \eta_{ab}$$

which must hold locally at each $p \in U$. Rewriting in standard contravariant notation $\Lambda \equiv L^{-1}$, then:

$$\tilde{e}_a^\mu = e_b^\mu (\Lambda^{-1})^b_a \quad \Lambda_a^c \Lambda_b^d \eta_{cd} = \eta_{ab} \quad (3.3.5)$$

This is the definition of a local $O(r, s)$ at each $p \in U$. In the Lorentzian case, then, $\Lambda \in O(1, n-1)$: this group has four disconnected components, defined by the four combinations of $\det \Lambda = \pm 1$ and $\Lambda^0_0 \geq 1$ or $\Lambda^0_0 \leq -1$; the special orthochronous component $SO^+(1, n-1)$, defined by $\det \Lambda = 1$ and $\Lambda^0_0 \geq 1$, is the group of Lorentz transformations on $T_p U$ (for a more detailed account on the Lorentz group, see §1.2 of [4]). Transformations between tetrad frames therefore are **local Lorentz transformations**, since, contrary to usual Lorentz transformations, they can vary from point to point of U .

§3.3.1 Connection 1-form

In a vielbein frame (E_a) , connection components are computed in the usual way:

$$\nabla_{E_c} E_b \equiv \omega_{cb}^a E_a \quad (3.3.6)$$

where the n^3 functions ω_{cb}^a are just the connection coefficients Γ_{cb}^a on the vielbein basis.

Definition 3.3.1 (Connection 1-form)

Let $(\mathcal{M}, g)$ be a metric manifold, and let (E_a) be a vielbein frame on $\mathcal{M}$ with dual coframe (θ^a) . Then, the **connection 1-form** $\omega_b^a \in \Omega^1(\mathcal{M})$ is defined as:

$$\omega_b^a := \omega_{cb}^a \theta^c \quad (3.3.7)$$

Note that these are truly n^2 1-forms, according to values of $a, b = 1, \dots, n$. This is also known as the *spin connection*, due to its relationship to spinors in curved spacetime. Then, the connection in the vielbein basis is easily defined:

$$\nabla_X Y = X(Y^a) E_a + Y^a X^b \nabla_{E_b} E_a = X^b [E_b(Y^a) + \omega_{bc}^a Y^c] E_a$$

Hence, setting the notation $\nabla_b Y^a \equiv (\nabla_{E_b} Y)^a$, it is clear that the connection takes the form:

$$\nabla_b Y^a = E_b(Y^a) + \omega_{bc}^a Y^c \quad (3.3.8)$$

In general, $E_b(Y^a) = e_b^\mu \partial_\mu Y^a$. Using Eq. 2.4.16, the covariant derivative of forms is found:

$$\nabla_b \zeta_a = E_b(\zeta_a) - \omega_{ba}^c \zeta_c \quad (3.3.9)$$

Finally, for tensors of general rank:

$$\nabla_c T^{a_1 \dots a_r}_{b_1 \dots b_s} = E_c(T^{a_1 \dots a_r}_{b_1 \dots b_s}) + \sum_{j=1}^r \omega_{cd_j}^{a_j} T^{a_1 \dots d_j \dots a_r}_{b_1 \dots b_s} - \sum_{j=1}^s \omega_{cb_j}^{d_j} T^{a_1 \dots a_r}_{b_1 \dots d_j \dots b_s} \quad (3.3.10)$$

Lemma 3.3.1 (Transformation law)

Given a local Lorentz transformation Λ acting as in Eq. 3.3.5:

$$\tilde{\omega}^a_b = \Lambda^a_c \omega^c_d (\Lambda^{-1})^d_b + \Lambda^a_c d(\Lambda^{-1})^c_b \quad (3.3.11)$$

Proof. For ease of notation, rename the transformation in Eq. 3.3.5 as $\Lambda^{-1} \equiv L$. Then, from the orthogonality condition $\theta^a(e_b) = \delta^a_b$, it is clear that:

$$\tilde{E}_a = L^b_a E_b \quad \tilde{\theta}^a = (L^{-1})^a_b \theta^b$$

Then, the connection coefficients can be computed as:

$$\begin{aligned} \tilde{\omega}^a_{cb} \tilde{E}_a &= \nabla_{\tilde{E}_c} \tilde{E}_b = L^d_c \nabla_{E_d} (L^f_b E_f) = L^d_c \left[\nabla_{E_d} L^f_b E_f + L^f_b \nabla_{E_d} E_f \right] \\ &= L^d_c \left[\nabla_{E_d} L^f_b E_f + L^f_b \omega^g_{df} E_g \right] = L^d_c \left[\nabla_{E_d} L^f_b + L^g_b \omega^f_{dg} \right] (L^{-1})^a_f \tilde{E}_a \end{aligned}$$

Finally, the transformed connection 1-form is:

$$\begin{aligned} \tilde{\omega}^a_b &= \tilde{\omega}^a_{cb} \tilde{\theta}^c = L^d_c \left[\nabla_{E_d} L^f_b + L^g_b \omega^f_{dg} \right] (L^{-1})^a_f (L^{-1})^c_h \theta^h \\ &= \left[\nabla_{E_d} L^f_b + L^g_b \omega^f_{dg} \right] (L^{-1})^a_f \theta^d = (L^{-1})^a_c \left[dx^\mu \nabla_\mu L^c_b + \omega^c_d L^d_b \right] \end{aligned}$$

Recalling that $df = dx^\mu \nabla_\mu f$, with $f \in \mathcal{C}^\infty(\mathcal{M})$, and that $\Lambda = L^{-1}$ yields the thesis. $\square$

The second term reflects the second term in Prop. 2.4.3, which involves the derivative of the coordinate transformation.

As for the connection coefficients, it is possible to compute the torsion tensor on a vielbein basis as $\mathcal{T}^a_{bc} = \mathcal{T}(\theta^a, E_b, E_c)$, which retains the antisymmetry $\mathcal{T}^a_{bc} = -\mathcal{T}^a_{cb}$.

Definition 3.3.2 (Torsion 2-form)

Let $(\mathcal{M}, g)$ be a metric manifold, and let (E_a) be a vielbein frame on $\mathcal{M}$ with dual coframe (θ^a) . Then, the **torsion 2-form** $T^a \in \Omega^2(\mathcal{M})$ is defined as:

$$T^a := \frac{1}{2} \mathcal{T}^a_{bc} \theta^b \wedge \theta^c \quad (3.3.12)$$

This is a vector-valued 2-form, thus T^a has n independent 2-forms. Like Eq. 2.5.5, the torsion 2-form is linked to the connection 1-form by the first Cartan structure equation.

Theorem 3.3.1 (First Cartan structure equation)

$$T^a = d\theta^a + \omega^a_b \wedge \theta^b \quad (3.3.13)$$

Proof. From Eq. 3.3.6 $\omega_{cb}^a = e^a_\rho e_c^\mu \nabla_\mu e_b^\rho$, thus inserting Eq. 2.4.6 and Eq. 3.3.7:

$$\begin{aligned}\omega_b^a \wedge \theta^b &= \omega_{cb}^a (e_c^\mu dx^\mu) \wedge (e_b^\nu dx^\nu) = e^a_\rho e_c^\lambda (\partial_\lambda e_b^\rho + e_b^\sigma \Gamma_{\lambda\sigma}^\rho) (e_c^\mu dx^\mu) \wedge (e_b^\nu dx^\nu) \\ &= e^a_\rho e_c^\lambda e_c^\mu e_b^\nu (\partial_\lambda e_b^\rho + e_b^\sigma \Gamma_{\lambda\sigma}^\rho) dx^\mu \wedge dx^\nu = e^a_\rho e_b^\nu (\partial_\mu e_b^\rho + e_b^\sigma \Gamma_{\mu\sigma}^\rho) dx^\mu \wedge dx^\nu\end{aligned}$$

Now, $e_b^\nu e_b^\rho = \delta_\nu^\rho$ implies $e_b^\nu \partial_\mu e_b^\rho = -e_b^\rho \partial_\mu e_b^\nu$, hence:

$$\begin{aligned}\omega_b^a \wedge \theta^b &= [-e^a_\rho e_b^\rho \partial_\mu e_b^\nu + e^a_\rho e_b^\nu e_b^\sigma \Gamma_{\mu\sigma}^\rho] dx^\mu \wedge dx^\nu \\ &= -\partial_\mu e^a_\nu dx^\mu \wedge dx^\nu + e^a_\rho \Gamma_{\mu\nu}^\rho dx^\mu \wedge dx^\nu = -d\theta^a + e^a_\rho \Gamma_{[\mu\nu]}^\rho dx^\mu \wedge dx^\nu\end{aligned}$$

Now, from the multilinearity of the torsion tensor:

$$T^a = \frac{1}{2} \mathcal{T}^a_{bc} \theta^b \wedge \theta^c = \frac{1}{2} e^a_\rho e_b^\alpha e_c^\beta e_b^\mu e_c^\nu T^\rho_{\beta\gamma} dx^\mu \wedge dx^\nu = \frac{1}{2} e^a_\rho \mathcal{T}^\rho_{\mu\nu} dx^\mu \wedge dx^\nu$$

Writing $\mathcal{T}^\rho_{\mu\nu} = 2\Gamma_{[\mu\nu]}^\rho$ completes the proof. $\square$

This construction can be generalized to arbitrary vector-valued 1-forms, defining the so-called **covariant exterior derivative**:

$$D\zeta^a := d\zeta^a + \omega_b^a \wedge \zeta^b \quad (3.3.14)$$

This is a map $D : \Omega^p(\mathcal{M}) \rightarrow \Omega^{p+1}(\mathcal{M})$, like the exterior derivative, but it has an important invariance property.

Lemma 3.3.2 (Lorentz covariance of covariant exterior derivative)

The covariant exterior derivative transforms under local Lorentz transformations as:

$$\zeta^a \mapsto \Lambda^a_b \zeta^b \implies D\zeta^a \mapsto \Lambda^a_b D\zeta^b \quad (3.3.15)$$

Proof. Denoting transformed quantities with a tilde and recalling Eq. 3.3.11:

$$\begin{aligned}\tilde{D}\tilde{\zeta}^a &= d\tilde{\zeta}^a + \tilde{\omega}_b^a \wedge \tilde{\zeta}^b \\ &= d\Lambda^a_b \wedge \zeta^b + \Lambda^a_b d\zeta^b + \Lambda^a_c (\Lambda^{-1})^d_b \Lambda^b_f \omega^c_d \wedge \zeta^f + \Lambda^a_c \Lambda^b_d d(\Lambda^{-1})^c_b \wedge \zeta^d \\ &= d\Lambda^a_b \wedge \zeta^b + \Lambda^a_b d\zeta^b + \Lambda^a_c \omega^c_d \wedge \zeta^d - \Lambda^a_c (\Lambda^{-1})^c_b d\Lambda^b_d \wedge \zeta^d = \Lambda^a_b [d\zeta^b + \omega^b_c \zeta^c]\end{aligned}$$

where $(\Lambda^{-1})^c_b \Lambda^b_d = \delta^c_d \implies \Lambda^b_d d(\Lambda^{-1})^c_b = -(\Lambda^{-1})^c_b d\Lambda^b_d$ has been used. $\square$

The exterior covariant derivative can be used to extend the covariant derivative to vector-valued 1-forms (using Eq. 3.3.8):

$$D\zeta^a = \partial_\mu \zeta^a e_c^\mu \wedge \theta^c + \omega_{cb}^a \zeta^b \wedge \theta^c = [E_c(\zeta^a) + \omega_{cb}^a \zeta^b] \wedge \theta^c \equiv \zeta^a_{;c} \wedge \theta^c \quad (3.3.16)$$

The same construction for the exterior covariant derivative can be generalized to tensor-valued p -forms as:

$$D\zeta^{a_1 \dots a_r}_{b_1 \dots b_s} = d\zeta^{a_1 \dots a_r}_{b_1 \dots b_s} + \sum_{j=1}^r \omega^{a_j}_c \wedge \zeta^{a_1 \dots c \dots a_r}_{b_1 \dots b_s} - \sum_{j=1}^s \omega^c_{b_j} \wedge \zeta^{a_1 \dots a_r}_{b_1 \dots c \dots b_s} \quad (3.3.17)$$

It is then trivial to show, using Lemma 3.3.2, that a local Lorentz transformation acts as:

$$\zeta^{a_1 \dots a_r}_{b_1 \dots b_s} \mapsto \Lambda^{a_1}_{c_1} \dots \Lambda^{a_r}_{c_r} \zeta^{c_1 \dots c_r}_{d_1 \dots d_s} (\Lambda^{-1})^{d_1}_{b_1} \dots (\Lambda^{-1})^{d_s}_{b_s} \quad (3.3.18)$$

$$D\zeta^{a_1 \dots a_r}_{b_1 \dots b_s} \mapsto \Lambda^{a_1}_{c_1} \dots \Lambda^{a_r}_{c_r} D\zeta^{c_1 \dots c_r}_{d_1 \dots d_s} (\Lambda^{-1})^{d_1}_{b_1} \dots (\Lambda^{-1})^{d_s}_{b_s} \quad (3.3.19)$$

where the transformation rule $\zeta_a \mapsto \zeta_b (\Lambda^{-1})^b_a$ follows from the requirement of $\zeta^a \xi_a$ being Lorentz-invariant (as for usual Lorentz transformation).

§3.3.1.1 Levi–Civita spin connection

If the connection is torsionless, the first structure equation reduces to:

$$D\theta^a = 0 \quad (3.3.20)$$

For a Levi–Civita connection, a stronger result holds.

Proposition 3.3.1

On a metric manifold with a Levi–Civita connection:

$$\omega_{ab} = -\omega_{ba} \quad (3.3.21)$$

Proof. Consider the vielbein condition Eq. 3.3.2. For a Levi–Civita connection, compatibility with the metric implies:

$$\begin{aligned} 0 &= \nabla_X (g_{\mu\nu} e_a^\mu e_b^\nu) = X^c g_{\mu\nu} (\nabla_{E_c} e_a^\mu e_b^\nu + e_a^\mu \nabla_{E_c} e_b^\nu) \\ &= X^c g_{\mu\nu} (\omega_{ca}^d e_d^\mu e_b^\nu + \omega_{cb}^d e_a^\mu e_d^\nu) = X^c (\omega_{ca}^d \eta_{db} + \omega_{cb}^d \eta_{ad}) \end{aligned}$$

Note that $\omega_b^a(X) = \omega_{cb}^a \theta^c(X) = \omega_{cb}^a X^c$, so that:

$$0 = \omega_a^d(X) \eta_{db} + \omega_b^d(X) \eta_{ad} = \omega_{ba}(X) + \omega_{ab}(X)$$

By the arbitrariness of X , it follows the thesis. $\square$

Eq. 3.3.20–3.3.21 allow to quickly compute the spin connection of a Levi–Civita connection, as they uniquely define it. Indeed, by antisymmetry ω_{ab} has $\frac{1}{2}n(n-1)$ independent 1-forms, i.e. $\frac{1}{2}n^2(n-1)$ independent components. On the other hand, the first structure equation constraints a vector-valued 2-form, which has n 2-forms with $\frac{1}{2}n(n-1)$ independent components each, thus posing $\frac{1}{2}n^2(n-1)$ constraints: this is the same number as the independent components of ω_{ab} , therefore the spin connection is uniquely determined.

The Levi–Civita connection is the unique connection such that Eq. 3.3.20–3.3.21 both hold. Moreover, for a Levi–Civita connection $D(\zeta^a \xi_a) = d(\zeta^a \xi_a)$ and $D\eta_{ab} = 0$, since $\omega_a^b = -\omega^b_a$.

§3.3.2 Curvature 2-form

As for the torsion tensor, it is possible to compute the curvature tensor on a vielbein basis as $\mathcal{R}^a_{bcd} = R(\theta^a, E_b, E_c, E_d)$, which retains the symmetries Prop. 2.5.4.

Definition 3.3.3 (Curvature 2-form)

Let $(\mathcal{M}, g)$ be a metric manifold, and let (E_a) be a vielbein frame on $\mathcal{M}$ with dual coframe (θ^a) . Then, the **curvature 2-form** $\Omega^a_b \in \Omega^2(\mathcal{M})$ is defined as:

$$\Omega^a_b := \frac{1}{2} \mathcal{R}^a_{bcd} \theta^c \wedge \theta^d \quad (3.3.22)$$

The curvature 2-form is a tensor-valued 2-form, hence Ω^a_b has n^2 2-forms, i.e. $\frac{1}{2}n^3(n-1)$ independent components (the same number as the curvature tensor, as expected). It is possible to derive a structure equation for the curvature 2-form too.

Theorem 3.3.2 (Second Cartan structure equation)

$$\Omega^a_b = d\omega^a_b + \omega^a_c \wedge \omega^c_b \quad (3.3.23)$$

Proof. Recalling Eq. 2.5.4, it is possible to write:

$$\mathcal{R}^a_{bcd}E_a = R(E_c, E_d)E_b = \nabla_{E_c}\nabla_{E_d}E_b - \nabla_{E_d}\nabla_{E_c}E_b - \nabla_{[E_c, E_d]}E_b$$

In general, by linearity of the connection, $\nabla_X E_b = \omega^c_b(X)E_c$, hence (using $\nabla_X f = X(f)$):

$$\begin{aligned} \mathcal{R}^a_{bcd}E_a &= \nabla_{E_c}[\omega^a_{db}E_a] - \nabla_{E_d}[\omega^a_{cb}E_a] - \omega^a_b([E_c, E_d])E_a \\ &= E_c(\omega^a_{db})E_a + \omega^f_{db}\omega^a_{cf}E_a - E_d(\omega^a_{cb})E_a - \omega^f_{cb}\omega^a_{df}E_a - \omega^a_b([E_c, E_d])E_a \end{aligned}$$

Lemma 3.3.3

$$d\omega^m_n(E_a, E_b) = E_a(\omega^m_{bn}) - E_b(\omega^m_{an}) - \omega^m_n([E_a, E_b]) \quad (3.3.24)$$

Proof. Writing $\partial_\mu = e^a_\mu E_a$ and $dx^\mu = e^\mu_a \theta^a$, by linearity of the connection:

$$d\omega^m_n = \nabla_{E_a}(\omega^m_{cn}e^c_\mu)e_b^\mu \theta^a \wedge \theta^b = \frac{1}{2} [\nabla_{E_a}(\omega^m_{cn}e^c_\mu)e_b^\mu - \nabla_{E_b}(\omega^m_{cn}e^c_\mu)e_a^\mu] \theta^a \wedge \theta^b$$

Since $\omega^m_{cn}e^c_\mu$ is a scalar function, using the Leibniz rule:

$$d\omega^m_n(E_a, E_b) = E_a(\omega^m_{bn}) - E_b(\omega^m_{an}) + \omega^m_{cn} [E_a(e^c_\mu)e_b^\mu - E_b(e^c_\mu)e_a^\mu]$$

Now, consider the action of the commutator of vielbeins on $\mathcal{C}^\infty(\mathcal{M})$:

$$\begin{aligned} [E_a, E_b](f) &:= E_a(E_b(f)) - E_b(E_a(f)) = e_a^\mu \partial_\mu (e_b^\nu \partial_\nu f) - e_b^\nu \partial_\nu (e_a^\mu \partial_\mu f) \\ &= e_a^\mu \partial_\mu e_b^\nu \partial_\nu f - e_b^\nu \partial_\nu e_a^\mu \partial_\mu f = [E_a(e_b^\mu)e^c_\mu - E_b(e_a^\mu)e^c_\mu] E_c(f) \end{aligned}$$

As $e_b^\mu e^c_\mu = \delta^c_b$, by the Leibniz rule $E_a(e_b^\mu)e^c_\mu + e_b^\mu E_a(e^c_\mu) = 0$, hence:

$$([E_a, E_b])^c = e_a^\mu E_b(e^c_\mu) - e_b^\mu E_a(e^c_\mu) \quad (3.3.25)$$

which, inserted in the expression for $d\omega^m_n(E_a, E_b)$, yields the thesis. $\square$

Using Lemma 3.3.3, the above expression becomes:

$$\mathcal{R}^a_{bcd} = d\omega^a_b(E_c, E_d) + \omega^f_{db}\omega^a_{cf} - \omega^f_{cb}\omega^a_{df}$$

But $\mathcal{R}^a_{bcd} = \Omega^a_b(E_c, E_d)$, while:

$$\omega^a_f \wedge \omega^f_b = \omega^a_{cf}\omega^f_{db}\theta^c \wedge \theta^d = \frac{1}{2} [\omega^f_{db}\omega^a_{cf} - \omega^f_{cb}\omega^a_{df}] \theta^c \wedge \theta^d$$

thus the proof is complete. $\square$

Combining the two structure equations, it is possible to find simple expressions for the exterior covariant derivative of the torsion and curvature 2-forms.

Proposition 3.3.2

$$DT^a = \Omega^a_b \wedge E^b \quad D\Omega^a_b = 0 \quad (3.3.26)$$

Proof. From Eq. 3.3.20, taking the exterior derivative and using Eq. 2.2.23:

$$dT^a = d\omega^a_b \wedge \theta^b - \omega^a_b \wedge d\theta^b = d\omega^a_b \wedge \theta^b - \omega^a_b \wedge (T^b - \omega^b_c \wedge \theta^c)$$

Rearranging the terms and inserting Eq. 3.3.23:

$$dT^a + \omega^a_b \wedge T^b = (d\omega^a_b + \omega^a_c \wedge \omega^c_b) \wedge \theta^b \implies DT^a = \Omega^a_b \wedge E^b$$

Now, take the exterior derivative of Eq. 3.3.23 and apply Eq. 2.2.23:

$$d\Omega^a_b = d\omega^a_c \wedge \omega^c_b - \omega^a_c \wedge d\omega^c_b = (\Omega^a_c - \omega^a_d \wedge \omega^d_c) \wedge \omega^c_b - \omega^a_c \wedge (\Omega^c_b - \omega^c_d \wedge \omega^d_b)$$

The second and fourth terms cancel by renaming the mute indices, thus:

$$0 = d\Omega^a_b + \omega^a_c \wedge \Omega^c_b - \Omega^a_c \wedge \omega^c_b = D\Omega^a_b$$

since $\Omega^a_c \wedge \omega^c_b = \omega^c_b \wedge \Omega^a_c$ by Eq. 2.2.9. □

§3.3.2.1 Bianchi identities

The Bianchi identities are useful identities which hold for a torsionless connection and put symmetry constraints on the curvature tensor.

Theorem 3.3.3 (Bianchi identities)

For a torsionless connection:

$$\mathcal{R}^a_{[bcd]} = 0 \quad \mathcal{R}^a_{b[cd;k]} = 0 \quad (3.3.27)$$

Proof. Setting $T^a \equiv 0$ in the first Eq. 3.3.26:

$$0 = \Omega^a_b \wedge \theta^b = \frac{1}{2} \mathcal{R}^a_{bcd} \theta^c \wedge \theta^d \wedge \theta^b = \frac{1}{2} \mathcal{R}^a_{[bcd]} \theta^b \wedge \theta^c \wedge \theta^d$$

which is the first Bianchi identity. Then, from the second Eq. 3.3.26:

$$0 = D\Omega^a_b = \frac{1}{2} D [\mathcal{R}^a_{bcd} \theta^c \wedge \theta^d] = \frac{1}{2} [\mathcal{R}^a_{bcd;k} \theta^k \wedge \theta^c \wedge \theta^d + \mathcal{R}^a_{bcd} (D\theta^c \wedge \theta^d - \theta^c \wedge D\theta^d)]$$

But $D\theta^a = T^a \equiv 0$, hence this equation is equivalent to the second Bianchi identity. □

In the vielbein basis, the Ricci tensor is defined as the contraction $\mathcal{R}_{bd} \equiv \mathcal{R}^a_{bad}$, while the Ricci scalar is the trace $\mathcal{R} \equiv \eta^{ab} \mathcal{R}_{ab}$. In the same way, the Einstein tensor is defined as $G_{ab} \equiv \mathcal{R}_{ab} - \frac{1}{2} \eta_{ab} \mathcal{R}$.

Corollary 3.3.3.1 (Einstein–Bianchi identity)

For a Levi-Civita connection:

$$G_{ab}{}^{;b} = 0 \quad (3.3.28)$$

Proof. From the second Bianchi identity Eq. 3.3.27, using the antisymmetry $\mathcal{R}^a{}_{bcd} = \mathcal{R}^a{}_{b[cd]}$:

$$\mathcal{R}_{bd;k} + \mathcal{R}^a{}_{bka;d} + \mathcal{R}^a{}_{bdk;a} = 0 \quad \implies \quad \mathcal{R}_{bd;k} - \mathcal{R}_{bk;d} + \mathcal{R}^a{}_{bdk;a} = 0$$

Now, contract with η^{bd} :

$$\mathcal{R}_{;k} - \mathcal{R}^b{}_{k;b} + \mathcal{R}^{ad}{}_{dk;a} = 0$$

Lemma 3.3.4 (Symmetries of Riemann tensor)

For a Levi-Civita connection:

$$\mathcal{R}_{abcd} = \mathcal{R}_{[ab][cd]} \quad (3.3.29)$$

Proof. By Eq. 3.3.22 $\mathcal{R}^a{}_{bcd} = \Omega^a{}_b(E_c, E_d)$, thus to compute this expression first consider:

$$d\omega^a{}_b = d\omega^a{}_{mb} \wedge \theta^m + \omega^a{}_{mb} d\theta^m = d\omega^a{}_{mb} \wedge \theta^m - \omega^a{}_{mb} \omega^m{}_k \wedge \theta^k$$

where Eq. 3.3.20 was used. Then, since $df = \partial_\mu f e_a{}^\mu \theta^a = E_a(f) \theta^a \forall f \in \mathcal{C}^\infty(\mathcal{M})$:

$$d\omega^a{}_b = E_k(\omega^a{}_{mb}) \theta^k \wedge \theta^m - \omega^a{}_{mb} \omega^m{}_k \wedge \theta^k = [E_m(\omega^a{}_{kb}) - \omega^a{}_{cb} \omega^c{}_{mk}] \theta^m \wedge \theta^k$$

Therefore, from Eq. 3.3.23:

$$\Omega^a{}_b = [E_m(\omega^a{}_{kb}) - \omega^a{}_{cb} \omega^c{}_{mk} + \omega^a{}_{mc} \omega^c{}_{kb}] \theta^m \wedge \theta^k$$

Now, to evaluate $\Omega^a{}_b(E_c, E_d)$, use:

$$\theta^m \wedge \theta^k(E_c, E_d) \equiv \begin{vmatrix} \theta^m(E_c) & \theta^m(E_d) \\ \theta^k(E_c) & \theta^k(E_d) \end{vmatrix} = \delta_c^m \delta_d^k - \delta_d^m \delta_c^k =: \delta_{cd}^{mk}$$

Hence, defining $\omega_{abc} \equiv \eta_{ak} \omega^k{}_{bc}$, the curvature tensor is found to be:

$$\mathcal{R}_{abcd} = [E_m(\omega_{akb}) - \omega_{agb} \omega^g{}_{mk} + \omega_{amg} \omega^g{}_{kb}] \delta_{cd}^{mk}$$

Then, since $\delta_{cd}^{mk} = \delta_{[cd]}^{[mk]}$ and, recalling Eq. 3.3.21 for a Levi-Civita connection, $\omega_{ab} = -\omega_{ba} \implies \omega_{acb} = \omega_{[a|c|b]}$, this is equivalent to the thesis. $\square$

Using Lemma 3.3.4, the above expression reduces to:

$$\mathcal{R}_{;k} = 2\mathcal{R}^a{}_{k;a}$$

Then, the thesis is trivial from the metric compatibility of the connection. $\square$

Note that Lemma 3.3.4 allows to explicitly compute the curvature tensor from connection coefficients:

$$\mathcal{R}_{abcd} = E_c(\omega_{adb}) - E_d(\omega_{acb}) - \omega_{agb}(\omega_{cd}^g - \omega_{dc}^g) + \omega_{acg}\omega_{db}^g - \omega_{adg}\omega_{cb}^g \quad (3.3.30)$$

Moreover, by multilinearity, the curvature tensor in the vielbein basis retains the symmetry $\mathcal{R}_{abcd} = \mathcal{R}_{cdab}$. This, together with $\mathcal{R}_{abcd} = \mathcal{R}_{[ab][cd]}$, means that the Riemann tensor can be thought as having two symmetrized multi-indices $[ab]$ and $[cd]$, hence the number of independent components is

$\frac{1}{2}N(N+1)$, where N is the number of possible values for a single multi-index, i.e. $N = \frac{1}{2}n(n-1)$. However, one has to account for the constraint posed by the first Bianchi identity: since there is one such constraint for each choice of four distinct indices, the total number of independent components of the Riemann tensor is:

$$f_{\text{Riemann}} = \frac{1}{2} \frac{n(n-1)}{2} \left(\frac{n(n-1)}{2} + 1 \right) - \binom{n}{4} = \frac{1}{8}(n-1)(n(n-1)+2) - \frac{1}{24}n(n-1)(n-2)(n-3)$$

Simplifying this expression yields the classical result:

$$f_{\text{Riemann}} = \frac{n^2(n^2-1)}{12} \quad (3.3.31)$$

In particular, on 4-dimensional manifolds this analysis yields the separation of the 20 degrees of freedom of the Riemann tensor into the 10 degrees of freedom of the Ricci tensor (“symmetric” part) and the 10 of the Weyl tensor (“antisymmetric” part).

Note that, if the curvature tensor is computed on a non-coordinate and non-tetrad basis, then it is not possible to impose the symmetry $\mathcal{R}_{abcd} = \mathcal{R}_{cdab}$ and the unconstrained components rise to 36 (i.e. N^2): the spurious 16 degrees of freedom thus need to be eliminated with additional constraints, depending on the formalism employed.

Part II

General Relativity

Notation

In the General Relativity part, Lorentzian metrics are always taken with mostly-positive signature $(1, n - 1)$, i.e. $(-, +, \dots, +)$, therefore a vector $X_p \in T_p\mathcal{M}$ is:

- timelike if $X_p^2 \equiv g_p(X_p, X_p) < 0$;
- null if $X_p^2 \equiv g_p(X_p, X_p) = 0$;
- spacelike if $X_p^2 \equiv g_p(X_p, X_p) > 0$;

Greek indices are Lorentzian indices, while Latin indices are Euclidean indices (or generic $\mathbb{N}_0$ - or $\mathbb{Z}$ -indices), and the Einstein summation convention on repeated indices is used unless otherwise specified.

Moreover, natural units are used, with $c = \hbar = k_B \equiv 1$, so that mass, energy, temperature, inverse length and inverse time all have the same units.

External Gravitational Fields

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In classical field theories, two distinct classes of objects are considered: particles and fields. Fields determine the motion of particles, while particles induce oscillations in fields.

§4.1 Geodetic motion

Free particles moving on an n -dimensional metric manifold $(\mathcal{M}, g)$ have geodesics as trajectories, which are found as extrema of the action functional.

§4.1.1 Non-relativistic particles

The action of a free particle is composed of the sole kinetic term, which, for a non-relativistic particle, is $\frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}}$, hence:

$$L = \frac{m}{2} g_{ij}(\mathbf{x}) \dot{x}^i \dot{x}^j \quad (4.1.1)$$

where $\{x^i\}_{i=1,\dots,n}$ are the coordinates of the chart(s) which cover the portion of $\mathcal{M}$ spanned by the trajectory.

Proposition 4.1.1 (Non-relativistic motion)

The equations of motion of the Lagrangian Eq. 4.1.1 are:

$$\ddot{x}^i + \Gamma_{jk}^i \dot{x}^j \dot{x}^k = 0 \quad (4.1.2)$$

where Γ_{jk}^i are the Christoffel symbols of the Levi-Civita connection on $\mathcal{M}$.

Proof. Varying the action of Eq. 4.1.1 on trajectories between $\mathbf{x}_1 \equiv \mathbf{x}(t_1)$, $\mathbf{x}_2 \equiv \mathbf{x}(t_2) \in \mathcal{M}$:

$$\delta S = \frac{m}{2} \int_{\mathbf{x}_1}^{\mathbf{x}_2} d^n x \sqrt{g} \delta [g_{ij} \dot{x}^i \dot{x}^j] = \frac{m}{2} \int_{\mathbf{x}_1}^{\mathbf{x}_2} d^n x \sqrt{g} \delta x^k \left[g_{ij,k} \dot{x}^i \dot{x}^j - 2 \frac{d}{dt} (g_{ik} \dot{x}^i) \right]$$

Setting the term in parentheses to zero yields:

$$\frac{1}{2} g_{ij,k} \dot{x}^i \dot{x}^j - g_{ik,j} \dot{x}^i \dot{x}^j - g_{ik} \ddot{x}^i = 0$$

Symmetrizing $\frac{1}{2} g_{ij,k} - g_{ik,j}$ results in $-g_{il} \Gamma_{jk}^l$, hence the geodesic equation is recovered. $\square$

This is precisely the geodesic equation on $\mathcal{M}$.

§4.1.2 Relativistic particles

Consider now a relativistic particle moving on an n -dimensional Lorentzian manifold $(\mathcal{M}, g)$. In this case, the free action is the distance between the two endpoints $x_1, x_2 \in \mathcal{M}$ of the trajectory:

$$\mathcal{S} = -mc \int_{x_1}^{x_2} \sqrt{-ds^2} = -mc \int_{\sigma_1}^{\sigma_2} d\sigma \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}} \quad (4.1.3)$$

where the negative sign clarifies that particles can only move on timelike and null geodesics, and $\sigma \in [\sigma_1, \sigma_2] \subseteq \mathbb{R}$ parametrizes the trajectory. Note that $g_{\mu\nu} = g_{\mu\nu}(x(\sigma))$, and the Lagrangian of the relativistic free particle is:

$$L = -mc \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}} \quad (4.1.4)$$

This action is clearly reparametrization-invariant, since:

$$\tilde{\mathcal{S}} = -mc \int_{\tilde{\sigma}_1}^{\tilde{\sigma}_2} d\tilde{\sigma} \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\tilde{\sigma}} \frac{dx^\nu}{d\tilde{\sigma}}} = -mc \int_{\sigma_1}^{\sigma_2} d\sigma \frac{d\tilde{\sigma}}{d\sigma} \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma} \left(\frac{d\sigma}{d\tilde{\sigma}}\right)^2} = \mathcal{S}$$

By this invariance, the value of the action between two points on a trajectory has a well-defined meaning: it is the elapsed **proper time** measured by the particle along the motion:

$$\tau(\sigma) \equiv \frac{1}{c} \int_0^\sigma d\sigma' \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma'} \frac{dx^\nu}{d\sigma'}} \quad (4.1.5)$$

Being this a monotone function on the trajectory, it is possible to parametrize the action in terms of proper time:

$$\mathcal{S} = -mc \int_{\tau_1}^{\tau_2} d\tau \sqrt{-g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} \quad (4.1.6)$$

where it is conventional to set $\dot{x} \equiv \frac{dx}{d\tau}$.

Proposition 4.1.2 (Relativistic motion)

The equations of motion of the action Eq. 4.1.3 are:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0 \quad (4.1.7)$$

where $\Gamma_{\nu\rho}^\mu$ are the Christoffel symbols of the Levi-Civita connection on $\mathcal{M}$.

Proof. Varying the action yields the following equations of motion:

$$-\frac{1}{2L} g_{\mu\nu, \rho} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma} - \frac{d}{d\sigma} \left(-\frac{1}{L} g_{\rho\nu} \frac{dx^\nu}{d\sigma} \right) = 0$$

Performing the differentiation and using the fact that $L \neq 0$ (for a massive particle):

$$g_{\mu\rho} \frac{d^2 x^\rho}{d\sigma^2} + \frac{1}{2} (g_{\mu\rho, \nu} + g_{\mu\nu, \rho} - g_{\nu\rho, \mu}) \frac{dx^\mu}{d\sigma} \frac{dx^\rho}{d\sigma} = \frac{1}{L} \frac{dL}{d\sigma} g_{\mu\rho} \frac{dx^\rho}{d\sigma}$$

This is the equation for a non-affinely-parametrized geodesic. To see this, note that from Eq. 4.1.5:

$$c \frac{d\tau}{d\sigma} = L(\sigma) \quad \Rightarrow \quad L(\tau) = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}} = \frac{d\sigma}{d\tau} L(\sigma) = c$$

Therefore, the Lagrangian can be made constant with any affine parametrization $\sigma = a\tau + b$, with $a \in \mathbb{R} - \{0\}$, $b \in \mathbb{R}$. This completes the proof. $\square$

It is straightforward to compute the conjugate momentum from Eq. 4.1.4:

$$p_\mu = \frac{\partial L}{\partial \dot{x}^\mu} = -mc \frac{\partial}{\partial \dot{x}^\mu} \sqrt{-g_{\nu\rho} \dot{x}^\nu \dot{x}^\rho} = -\frac{m^2 c^2}{L} \dot{x}_\mu$$

which results into the **mass-shell condition**:

$$p^2 = -m^2 c^2 \quad (4.1.8)$$

Therefore, the time component of p^μ is found to be $(p^0)^2 = m^2 c^2 + \mathbf{p}^2$, which has two important consequences: first of all, $p^0 \neq 0$, which means that the particle is never at rest in the temporal direction (which was to be expected); secondly, p^μ only has $n - 1$ independent components, which is a consequence of the reparametrization invariance of the action¹.

§4.1.3 Interactions

Once the relativistic motion of a (massive) free particle on a general n -dimensional Lorentzian manifold has been clarified, focus now on Minkowski spacetime $\mathbb{R}^{1,3}$ with the Lorentz–Minkowski metric $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$. The action then takes the form:

$$\mathcal{S} = -mc \int_{\sigma_1}^{\sigma_2} d\sigma \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}}$$

Clearly, there are two possibilities to introduce an interaction term: either inside or outside the square root.

§4.1.3.1 Electromagnetism

The addition of a term $+d\sigma V(x)$ outside the square root is not reparametrization-invariant. Indeed, the simplest interaction term which satisfies both reparametrization invariance and Lorentz invariance is:

$$\mathcal{S} = \int_{\sigma_1}^{\sigma_2} d\sigma \left[-mc \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}} - q A_\mu(x) \frac{dx^\mu}{d\sigma} \right] \quad (4.1.9)$$

where $A_\mu(x)$ is the potential of the field and $q \in \mathbb{R}$ is the charge of the coupling. To determine the potential A_μ for the electromagnetic interaction, compare the classical limit of Eq. 4.1.9 to the action of a non-relativistic particle in an external electromagnetic field:

$$\mathcal{S} \xrightarrow{c \rightarrow \infty} \int_{t_1}^{t_2} dt \left[-mc^2 + \frac{m}{2} \dot{\mathbf{x}}^2 - q A_0(x) c - q \mathbf{A}(x) \cdot \dot{\mathbf{x}} \right]$$

¹The n equations of motion are $x^\mu = x^\mu(\sigma)$, but σ cannot contain any information about the system, thus one equation must be used to eliminate the σ dependence, resulting in $n - 1$ degrees of freedom. This was to be expected from the non-relativistic limit, which has $n - 1$ degrees of freedom too (since time is absolute in this limit).

$$\mathcal{S}_{\text{em}} = \int_{t_1}^{t_2} dt \left[\frac{m}{2} \dot{\mathbf{x}}^2 - q\phi(x) - q\mathbf{A}(x) \cdot \dot{\mathbf{x}} \right]$$

Comparing the individual terms, the electromagnetic interaction is described by the vector potential $A_\mu(x) = (\phi(x)/c, \mathbf{A}(x))$.

§4.1.3.2 Gravity

While the addition of an interaction outside the square root forces it to be described by a vector field (at least in its simplest form), doing so inside the square root requires a tensor field, in order to satisfy both reparametrization invariance and Lorentz invariance. The resulting action then is:

$$\mathcal{S} = -mc \int_{\sigma_1}^{\sigma_2} d\sigma \sqrt{-(\eta_{\mu\nu} + V_{\mu\nu}(x)) \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}}$$

However, this can be interpreted² as the action of a relativistic free particle on a curved 4-dimensional Lorentzian manifold with metric $g_{\mu\nu}(x) \equiv \eta_{\mu\nu} + V_{\mu\nu}(x)$. To compute the non-relativistic limit of this action, expand the term in the square root:

$$\mathcal{S} = -mc^2 \int_{t_1}^{t_2} dt \sqrt{-g_{00}(x) - \frac{1}{c} \left(2g_{0i}\dot{x}^i + \frac{1}{c} g_{ij}\dot{x}^i\dot{x}^j \right)}$$

The only interesting case is $V_{\mu\nu}(x) \equiv -c^{-2}V(x)\delta_{\mu,0}\delta_{\nu,0}$, in which case:

$$\mathcal{S} \xrightarrow{c \rightarrow \infty} \int_{t_1}^{t_2} dt \left[-mc^2 - m \frac{V(x)}{2} + \frac{m}{2} \dot{\mathbf{x}}^2 \right]$$

Now, compare this to the action of a non-relativistic particle in an external gravitational field:

$$\mathcal{S}_g = \int_{t_1}^{t_2} dt \left[-mc^2 + \frac{m}{2} \dot{\mathbf{x}}^2 - m\Phi(x) \right]$$

The comparison gives $V(x) = 2\Phi(x)$: therefore, the gravitational interaction naturally induces the notion of a curved spacetime. Moreover, this is the condition that a metric describing the gravitational interaction must satisfy in the weak-field limit (i.e. the non-relativistic limit):

$$g_{00}(x) \approx - \left[1 + \frac{2\Phi(x)}{c^2} \right] \quad (4.1.10)$$

with $\Phi(x)$ the Newtonian potential (indeed, this is seldom called the “Newtonian limit”).

§4.2 Equivalence principle

As per the discussion in §4.1.3.2, the particle’s mass m can be regarded as the “charge” which couples it to the gravitational field. This fact is known as the **weak equivalence principle** (WEP), which states the equivalence between inertial mass $m_i : \mathbf{F}_{\text{tot}} = m_i \ddot{\mathbf{x}}$ and gravitational mass m_g :

$$m_i = m_g \quad (4.2.1)$$

This equivalence has been experimentally verified with relative error $\sim 10^{-13}$.

²To be precise, this is true assuming that the tensor $g_{\mu\nu}(x)$ satisfies the conditions in Def. 2.1.1.

§4.2.1 Kottler–Möller metric

A direct consequence of the WEP is the indistinguishability of a uniform gravitational field from a constant acceleration.

Consider a massive particle with a constant acceleration $\mathbf{a} = a\hat{\mathbf{e}}_x$ with respect to an inertial reference frame $\mathcal{O}$ in Minkowski spacetime. A constant acceleration means that the rapidity grows linearly in proper time as $\varphi(\tau) = \frac{a\tau}{c}$, so that the particle's velocity (along $\hat{\mathbf{e}}_x$) is:

$$v(\tau) = c \operatorname{sech} \frac{a\tau}{c} \quad (4.2.2)$$

Since $\frac{dt}{d\tau} = \gamma(\tau)$ and $\frac{dx}{d\tau} = v(\tau)$, inserting the expression for $v(\tau)$ in the Lorentz factor yields:

$$t(\tau) = \frac{c}{a} \sinh \frac{a\tau}{c} \quad x(\tau) = \frac{c^2}{a} \cosh \frac{a\tau}{c} - \frac{c^2}{a} \quad (4.2.3)$$

where the integration constants have been chosen so that $t(0) = 0$ and $x(0) = 0$. The worldline (i.e. trajectory) of the particle is thus described by a hyperbola in $\mathcal{O}$:

$$\left(x + \frac{c^2}{a}\right)^2 - c^2 t^2 = \frac{c^4}{a^2} \quad (4.2.4)$$

The coordinate transformation from the inertial reference frame $\mathcal{O}$, with coordinates (t, x) , to the non-inertial comoving reference frame of the particle, with coordinates (τ, ρ) , is given by:

$$ct = \left(\rho + \frac{c^2}{a}\right) \sinh \frac{a\tau}{c} \quad x = \left(\rho + \frac{c^2}{a}\right) \cosh \frac{a\tau}{c} - \frac{c^2}{a} \quad (4.2.5)$$

Indeed, the particle's spatial trajectory is correctly described by $\rho = 0$. Note that the comoving coordinates (τ, ρ) do not cover the whole Minkowski spacetime: this shows that there are regions of spacetime which are causally disconnected from the particle. The pull-back of the metric in the comoving frame is:

$$ds^2 = -\left(1 + \frac{a\rho}{c^2}\right)^2 c^2 d\tau^2 + d\rho^2 + dy^2 + dz^2 \quad (4.2.6)$$

This is known as the **Kottler–Möller metric**. This metric tensor has precisely the form discussed in §4.1.3.2, since in the sub-relativistic limit its temporal component is $g_{00} \approx -\left(1 + \frac{2a\rho}{c^2}\right)$, and it describes a gravitational field with potential $\Phi(\rho) = a\rho$, which is a uniform gravitational field.

§4.2.2 Einstein's equivalence principle

A corollary of the WEP is that a uniform gravitational field can be nullified by choosing a particular non-inertial reference frame, the *free-fall reference frame*, which is comoving with the considered particle.

Einstein generalized the WEP to a stronger principle, **Einstein's equivalence principle**, which states that *any* gravitational field can be locally nullified by a suitable choice of (non-inertial) reference frame. Formally, this is equivalent to Th. 2.6.1: one can always choose normal coordinates in a given point so that spacetime locally looks flat, i.e. $g_{\mu\nu}(p) = \eta_{\mu\nu}$.

Although locally nullifiable, the effects of a non-uniform gravitational field become measurable when it is possible to conduct non-local measurements, due to the presence of **tidal forces**.

Example 4.2.1 (Einstein's elevator Gedankenexperiment)

Consider an observer in a closed box which is falling towards the Earth. Although in the free-fall reference frame the observer cannot establish whether it is fluctuating in space or falling towards the Earth, they can do so with a non-local experiment: taking two test masses (i.e. not affected by their mutual gravitational attraction), if the box is fluctuating in space they will remain in their initial position due to inertia, meanwhile if the box is falling towards the Earth they will have a horizontal displacement, due to the fact that the Earth's gravitational field is radial (and thus non-uniform). This is the case of a tidal force.

§4.2.3 Gravitational time dilation

Consider a spherically-symmetric body of uniform mass M . In the weak-field approximation, the gravitational field generated by the body is given by a metric such that $g_{00}(x) = -1 - \frac{2\Phi(x)}{c^2}$, with $\Phi(x) = -\frac{GM}{r}$ and $r \equiv \|\mathbf{x}\|$. Therefore, a stationary observer at a distance r from the body will measure time intervals given by (using $d\tau^2 := -ds^2$):

$$d\tau^2 = \left(1 - \frac{2GM}{rc^2}\right) dt^2$$

Defining the Schwarzschild radius of a mass M as $R_s \equiv 2GM/c^2$ and integrating, the **gravitational time dilation** is found:

$$\Delta\tau(r) = \Delta t \sqrt{1 - \frac{R_s}{r}} \quad (4.2.7)$$

which means that time flows slower near massive objects.

Proposition 4.2.1 (Relative gravitational time dilation)

Given two stationary observers at distances r and $r + \Delta r$ from a mass M , with $\Delta r \ll r$ and $R_s \ll r$, then:

$$\frac{\Delta\tau(r + \Delta r)}{\Delta\tau(r)} = 1 + \frac{R_s \Delta r}{2r^2} \quad (4.2.8)$$

Proof. Using Eq. 4.2.7:

$$\begin{aligned} \Delta\tau(r + \Delta r) &= \Delta t \sqrt{1 - \frac{R_s}{r + \Delta r}} \approx \Delta t \sqrt{1 - \frac{R_s}{r} + \frac{R_s \Delta r}{r^2}} \\ &\approx \Delta t \sqrt{1 - \frac{R_s}{r}} \left(1 + \frac{R_s \Delta r}{2r^2}\right) = \Delta\tau(r) \left(1 + \frac{R_s \Delta r}{2r^2}\right) \end{aligned}$$

which is the thesis. $\square$

Accounting for the relative gravitational time dilation is extremely important in precision applications (like the GPS): for example, with an elevation difference of $\Delta r \sim 10^3$ m with respect to sea level $r \sim 6 \cdot 10^6$ m, the measured time difference is $\sim 10^{-12}$ s.

Proposition 4.2.2 (Gravitational redshift)

Consider a photon emitted with frequency ω_1 at a distance r_1 from the center of a weak gravitational potential $\Phi(r)$. An observer at distance r_2 will measure a shifted frequency ω_2 given

by:

$$\frac{\omega_2}{\omega_1} = \left[1 + \frac{\Phi(r_2) - \Phi(r_1)}{c^2} \right]^{-1} \quad (4.2.9)$$

Proof. Since $\omega = \frac{2\pi}{\Delta\tau}$, where $\Delta\tau$ is the time elapsed during an oscillation of a full wavelength, by Eq. 4.1.10:

$$\frac{\Delta\tau_2}{\Delta\tau_1} = \sqrt{\frac{1 + 2\Phi(r_2)/c^2}{1 + 2\Phi(r_1)/c^2}} \approx 1 + \frac{\Phi(r_2) - \Phi(r_1)}{c^2}$$

This is equivalent to the thesis. □

In the case of the (weak) gravitational field generated by a mass M , since $\Phi(r) \propto -r^{-1}$, then $r_2 > r_1 \implies \omega_2 < \omega_1$ (redshift) and vice versa (blueshift).

Geometrodynamics

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The force of gravity is mediated by a gravitational field, which is identified with a metric $g(x)$ on a n -dimensional Lorentzian manifold. This metric is a dynamical object, as all other fields in Nature, thus the laws governing its dynamics must be studied.

§5.1 Geometrodynamics in vacuum

Differential Geometry poses strict limits to the possible actions that can be formulated, ensuring it is something intrinsic to the metric and independent on the choice of coordinates.

Definition 5.1.1 (Spacetime)

An orientable Lorentzian d -manifold $(\mathcal{M}, g)$ with $d \geq 3$ and equipped with a torsion-free Levi-Civita connection is called a **spacetime**.

To define the action for the metric of a spacetime $(\mathcal{M}, g)$, recalling the canonical volume form in Eq. 2.3.3, there needs to be a metric-dependent scalar function on $\mathcal{M}$: the obvious non-trivial choice is the Ricci scalar $\mathcal{R}$. The resulting action is the **Einstein–Hilbert action**:

$$\mathcal{S}_{\text{EH}}[g] = \int_{\mathcal{M}} d^d x \sqrt{g} \mathcal{R} \quad (5.1.1)$$

This is the simplest non-trivial action for the gravitational field allowed by Differential Geometry. Schematically, $\mathcal{R} \sim \partial\Gamma + \Gamma\Gamma$ and $\Gamma \sim \partial g$, so the Einstein–Hilbert Lagrangian appears to be dependent on second derivatives of the metric: however, the terms proportional to $\partial^2 g$ can be shown to form a total divergence term, thus making $\mathcal{L}_{\text{EH}}$ “effectively” first order, like most other actions in physics. The total divergence term needs to be treated carefully on manifolds with boundary, since a well-posed variational problem with Dirichlet boundary conditions (fixing $g|_{\partial\mathcal{M}}$) requires vanishing normal derivatives of δg at the boundary $\partial\mathcal{M}$: this is not satisfied by $\mathcal{S}_{\text{EH}}$ alone, and the standard fix is to add a boundary term (the so-called *Gibbons–Hawking–York boundary term* $\mathcal{S}_{\text{GHY}}$) which ensured $\delta g|_{\partial\mathcal{M}} = 0$. In the following, the inclusion of a GHY term in the action $\mathcal{S}_{\text{grav}} = \mathcal{S}_{\text{EH}} + \mathcal{S}_{\text{GHY}}$ is made implicit by setting $\delta g|_{\partial\mathcal{M}} = 0$.

The Einstein–Hilbert action justifies the restriction $d \geq 3$ in the dimensionality of spacetime: in $d = 1$ the Riemann tensor, and so the Ricci scalar curvature, identically vanishes, while in $d = 2$ the action $\mathcal{S}_{\text{EH}}$ is proportional to the Euler characteristic of $\mathcal{M}$ (by the Gauss–Bonnet theorem), which is a topological invariant of $\mathcal{M}$ independent of g . Also, $G \equiv 0$ in $d = 2$.

Observation 5.1.1 (Dimensionale analysis)

The Einstein–Hilbert action in Eq. 5.1.1 does not have the right dimensions, which will be necessary when considering the presence of matter. If x has dimension of length, then g is dimensionless and the Ricci scalar is $[\mathcal{R}] = L^{-2}$. Including the integration measure, $[\mathcal{S}_{\text{EH}}] = L^{d-2}$, but it must have the same dimension of $[\hbar] = ML^2T^{-1}$ (i.e. energy $\times$ time). Thus, the action with the right dimension is:

$$\mathcal{S}_{\text{EH}}[g] = \frac{c^3}{16\pi G_N^{(d)}} \int_{\mathcal{M}} d^d x \sqrt{g} \mathcal{R} \quad (5.1.2)$$

where the d -dimensional Newton constant has dimension $[G_N^{(d)}] = M^{-1}L^{d-1}T^{-2}$. In the following, natural units $c = \hbar = k_B \equiv 1$ are adopted, so that $L \propto T \propto M^{-1}$: in particular, this means that the Newton constant has mass dimension $d - 2$, which makes any quantum gravitational theory non-renormalizable for $d > 2$.

To determine the Euler–Lagrange equations of the Einstein–Hilbert action, consider a fixed initial metric $g(x)$ and a perturbation $g(x) \mapsto g(x) + \delta g(x)$.

Lemma 5.1.1 (Metric variations)

Let $(\mathcal{M}, g)$ be a metric manifold. Under a variation $g(x) \mapsto g(x) + \delta g(x)$:

$$\delta g_{\mu\nu} \equiv (\delta g)_{\mu\nu} \quad \delta g^{\mu\nu} = -g^{\mu\alpha} g^{\nu\beta} \delta g_{\alpha\beta} \quad \delta \sqrt{g} = \frac{\sigma}{2} \sqrt{g} g_{\mu\nu} \delta g^{\mu\nu} \quad (5.1.3)$$

$$\delta \Gamma_{\mu\nu}^{\rho [1]} = \frac{1}{2} g^{\rho\sigma} (\nabla_{\mu} \delta g_{\sigma\nu} + \nabla_{\nu} \delta g_{\sigma\mu} - \nabla_{\sigma} \delta g_{\mu\nu}) \quad (5.1.4)$$

$$\delta \mathcal{R}_{\mu\nu} = \nabla_{\rho} \delta \Gamma_{\mu\nu}^{\rho} - \nabla_{\nu} \delta \Gamma_{\rho\mu}^{\rho} \quad (5.1.5)$$

where σ is the parity of the signature of g .

Proof. $\delta g_{\mu\nu} \equiv (\delta g)_{\mu\nu}$ is a definition. For the inverse metric:

$$g_{\mu\nu} g^{\nu\rho} = \delta_{\mu}^{\rho} \quad \Rightarrow \quad \delta g_{\mu\nu} g^{\nu\rho} + g_{\mu\nu} \delta g^{\nu\rho} = 0 \quad \Rightarrow \quad \delta g^{\mu\nu} = -g^{\mu\rho} g^{\nu\sigma} \delta g_{\rho\sigma}$$

The variation of $\sqrt{g}$ requires a classical result for invertible matrices: $\log \det A = \text{tr} \log A$, i.e. $(\det A)^{-1} \delta(\det A) = \text{tr}(A^{-1} \delta A)$. Applying this to the metric:

$$\delta \sqrt{g} = \frac{1}{2} \frac{1}{\sqrt{g}} \delta g = \frac{1}{2} \frac{1}{\sqrt{g}} (\sigma g) g^{\mu\nu} \delta g_{\mu\nu} = \frac{\sigma}{2} \sqrt{g} g^{\mu\nu} \delta g_{\mu\nu} = \frac{\sigma}{2} \sqrt{g} g_{\mu\nu} \delta g^{\mu\nu}$$

Now, consider the Christoffel symbols, and in particular note that, although $\Gamma_{\mu\nu}^{\rho}$ is not a tensor, $\delta \Gamma_{\mu\nu}^{\rho}$ is, since it is the difference of two Christoffel symbols, one computed with $g_{\mu\nu}$ and the other with $g_{\mu\nu} + \delta g_{\mu\nu}$, and the extra term in the transformation law of $\Gamma_{\mu\nu}^{\rho}$ is independent of the metric, thus it cancels out. This observation allows to work in normal coordinates at $p \in \mathcal{M}$, so that $\partial_{\rho} g_{\mu\nu}|_p = 0$ and $\Gamma_{\mu\nu}^{\rho}|_p = 0$. Then, to linear order in the variation:

$$\delta \Gamma_{\mu\nu}^{\rho [1]}|_p = \frac{1}{2} g^{\rho\sigma} (\partial_{\mu} \delta g_{\sigma\nu} + \partial_{\nu} \delta g_{\sigma\mu} - \partial_{\sigma} \delta g_{\mu\nu})|_p = \frac{1}{2} g^{\rho\sigma} (\nabla_{\mu} \delta g_{\sigma\nu} + \nabla_{\nu} \delta g_{\sigma\mu} - \nabla_{\sigma} \delta g_{\mu\nu})$$

This is a tensor equation, hence valid in all coordinate system and, by the arbitrariness of p , on all $\mathcal{M}$. In normal coordinates, the Riemann tensor becomes $\mathcal{R}^{\sigma}_{\mu\rho\nu}|_p = \partial_{\rho} \Gamma_{\nu\mu}^{\sigma}|_p - \partial_{\nu} \Gamma_{\rho\mu}^{\sigma}|_p$, so:

$$\delta \mathcal{R}^{\sigma}_{\mu\rho\nu}|_p = \partial_{\rho} \delta \Gamma_{\nu\mu}^{\sigma}|_p - \partial_{\nu} \delta \Gamma_{\rho\mu}^{\sigma}|_p = \nabla_{\rho} \delta \Gamma_{\nu\mu}^{\sigma}|_p - \nabla_{\nu} \delta \Gamma_{\rho\mu}^{\sigma}|_p$$

This is a tensor equation, hence again valid in all coordinates systems and on all $\mathcal{M}$. Contracting indices σ, ρ yields the thesis. $\square$

With these expressions it is now possible to compute the variations of the Einstein–Hilbert action to find the equations of motion of the gravitational field.

Proposition 5.1.1 (Einstein field equations in vacuum)

The Euler–Lagrange equations of the Einstein–Hilbert action are:

$$G_{\mu\nu} = 0 \quad (5.1.6)$$

Proof. Varying the Einstein–Hilbert action:

$$\delta \mathcal{S}_{\text{EH}}[g] = \delta \int_{\mathcal{M}} d^d x \sqrt{g} g^{\mu\nu} \mathcal{R}_{\mu\nu} = \int_{\mathcal{M}} d^d x [(\delta \sqrt{g}) g^{\mu\nu} \mathcal{R}_{\mu\nu} + \sqrt{g} (\delta g^{\mu\nu}) \mathcal{R}_{\mu\nu} + \sqrt{g} g^{\mu\nu} (\delta \mathcal{R}_{\mu\nu})]$$

Inserting the expressions in Lemma 5.1.1:

$$\delta \mathcal{S}_{\text{EH}}[g] = \int_{\mathcal{M}} d^d x \sqrt{g} \left[\left(-\frac{1}{2} \mathcal{R} g_{\mu\nu} + \mathcal{R}_{\mu\nu} \right) \delta g^{\mu\nu} + g^{\mu\nu} (\nabla_{\rho} \delta \Gamma_{\mu\nu}^{\rho} - \nabla_{\nu} \Gamma_{\rho\mu}^{\rho}) \right]$$

Then, using the metric compatibility of the Levi–Civita connection:

$$\mathcal{S}_{\text{EH}}[g] = \int_{\mathcal{M}} d^d x \sqrt{g} \left[\left(\mathcal{R}_{\mu\nu} - \frac{1}{2} \mathcal{R} g_{\mu\nu} \right) \delta g^{\mu\nu} + \nabla_{\mu} (g^{\rho\nu} \delta \Gamma_{\rho\nu}^{\mu} - g^{\mu\nu} \delta \Gamma_{\rho\nu}^{\rho}) \right]$$

The last term is a total derivative, hence by the divergence theorem Eq. 2.5.19 it yields a boundary term which vanishes as $\delta g|_{\partial \mathcal{M}} = 0$. The Euler–Lagrange equations are then found imposing $\delta \mathcal{S}_{\text{EH}} = 0$ for arbitrary δg , so recalling the definition of the Einstein tensor Eq. 2.5.16 the proof is completed. $\square$

These equations are the Einstein field equations in vacuum, which govern the geometrodynamics of spacetime in the absence of any matter. For this reason, they can be further simplified with a contraction: in general, the trace of the Einstein tensor is:

$$G = \frac{2-d}{2} \mathcal{R} \quad (5.1.7)$$

This means that vacuum spacetimes of dimension $d \neq 2$ are Ricci-flat. Metric 2-manifolds are instead trivialized by the existence of isothermal coordinates (see Ex. 7.9 in §7.6.2 of [11]).

While a constant gravitational Lagrangian has no physical relevance, as it uniquely implies $g \equiv 0$, Eq. 5.1.2 allows for the addition of a constant term to the Ricci scalar. An interesting expression is:

$$\mathcal{S}_{\text{grav}}[g] = \frac{1}{16\pi G_{\text{N}}^{(d)}} \int_{\mathcal{M}} d^d x \sqrt{g} [\mathcal{R} - (d-2)\Lambda] \quad (5.1.8)$$

$\Lambda \in \mathbb{R}$ is known as a *cosmological constant* and has dimension $[\Lambda] = L^{-2}$. The resulting field equations are:

$$\mathcal{R}_{\mu\nu} - \frac{1}{2} \mathcal{R} g_{\mu\nu} = -\frac{d-2}{2} \Lambda g_{\mu\nu} \quad (5.1.9)$$

Contracting with $g^{\mu\nu}$ yields $\mathcal{R} = \frac{2d}{d-2} \Lambda$, thus in the absence of matter:

$$\mathcal{R}_{\mu\nu} = \Lambda g_{\mu\nu} \quad (5.1.10)$$

Manifolds on which $\text{Ric} = \kappa g$ for a constant $\kappa \in \mathbb{R}$ are known as **Einstein manifolds**: clearly, Ricci-flat manifolds are a special class of Einstein manifolds with $\kappa = 0$. Notice that Einstein manifolds have constant Ricci scalar curvature $\mathcal{R} = \kappa d$.

Note that the Einstein field equations are not linear in the metric, hence the superposition principle is not valid for gravitational fields, except at NLO in the approximation of weak gravity (see §6.1.1).

§5.1.1 Symmetries

The Einstein equations are highly non-linear, hence they are extremely hard to solve. In the study of their solutions, a fundamental role is played by symmetries. For example, consider the simplest solution, i.e. Minkowski spacetime: its symmetries are familiar, namely translations and rotations of spacetime (with the latter splitting between proper rotations and Lorentz boosts), whose existence allows to define conserved energy, momentum and angular momentum on a fixed Minkowski background. It is then important to characterize the symmetries of a general metric.

Being the metric a symmetric $\mathbb{R}^{d \times d}$ matrix, it should have $\frac{1}{2}d(d+1)$ degrees of freedom. However, not all the components of $g_{\mu\nu}$ are physical. Indeed, two metrics related by a change of coordinates $x^\mu \mapsto \tilde{x}^\mu(x)$ describe the same physical spacetime, thus there is a redundancy in any given representation of the metric which removes precisely d degrees of freedom, leaving only $\frac{1}{2}d(d-1)$ physical degrees of freedom.

Formally, a diffeomorphism $\phi : \mathcal{M} \rightarrow \mathcal{M}$ maps all fields on $\mathcal{M}$ to a new set of fields on $\mathcal{M}$: the result is physically indistinguishable from the original, describing the same spacetime but in different coordinates. Such diffeomorphisms are analogous to gauge symmetries in Yang-Mills theories.

Consider a diffeomorphism $x^\mu \mapsto \tilde{x}^\mu = x^\mu + \delta x^\mu$: this can be viewed as an active diffeomorphism, where points in spacetime are mapped from one another, or as a passive diffeomorphism, in which only the coordinates of each point are affected, but the two interpretations are equivalent.

Lemma 5.1.2 (Transformation of tensors under infinitesimal diffeomorphisms)

Let $(\mathcal{M}, g)$ be a metric manifold, $Z \in \mathfrak{X}(\mathcal{M})$ and $T \in \mathfrak{T}_s^r(\mathcal{M})$. Given a smooth chart $(U, (x^\mu))$ on $\mathcal{M}$, under a coordinate change $x^\mu \mapsto \tilde{x}^\mu$, define the **pointwise variation** of T as:

$$\delta T(x) := \tilde{T}(x) - T(x) \quad (5.1.11)$$

Then, if $\tilde{x}^\mu(x) = x^\mu - Z^\mu(x)$, the pointwise variation of T is:

$$\delta T(x) = \mathcal{L}_Z T(x) + o(Z^2) \quad (5.1.12)$$

Proof. By Eq. 1.6.13:

$$\tilde{T}_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}(\tilde{x}) = \frac{\partial \tilde{x}^{\mu_1}}{\partial x^{\alpha_1}} \dots \frac{\partial x^{\mu_r}}{\partial x^{\alpha_r}} \frac{\partial x^{\beta_1}}{\partial \tilde{x}^{\nu_1}} \dots \frac{\partial x^{\beta_s}}{\partial \tilde{x}^{\nu_s}} T_{\beta_1 \dots \beta_s}^{\alpha_1 \dots \alpha_r}(x)$$

Now, note that^a:

$$\frac{\partial \tilde{x}^\mu}{\partial x^\alpha} = \delta_\alpha^\mu - \partial_\alpha Z^\mu \quad \frac{\partial x^\beta}{\partial \tilde{x}^\nu} = \delta_\nu^\beta + \partial_\nu Z^\beta + o(Z^2)$$

so that:

$$\tilde{T}_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}(\tilde{x}) = T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}(x) - \sum_{i=1}^r T_{\nu_1 \dots \nu_i \dots \nu_s}^{\mu_1 \dots \rho \dots \mu_r}(x) \partial_\rho Z^{\mu_i}(x) + \sum_{j=1}^s T_{\nu_1 \dots \rho \dots \nu_s}^{\mu_1 \dots \mu_r}(x) \partial_{\nu_j} Z^\rho(x) + o(Z^2)$$

On the other hand:

$$\tilde{T}_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}(\tilde{x}) = \tilde{T}_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}(x) - Z^\rho \partial_\rho \tilde{T}_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}(x) = \tilde{T}_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}(x) - Z^\rho \partial_\rho T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}(x) + o(Z^2)$$

Therefore, by definition:

$$\begin{aligned} \delta T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}(x) &= Z^\rho \partial_\rho T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}(x) - \sum_{i=1}^r T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r} \rho \dots \mu_r \nu_1 \dots \nu_l(x) \partial_\rho Z^{\mu_i}(x) \\ &\quad + \sum_{j=1}^s T_{\nu_1 \dots \rho \dots \nu_s}^{\mu_1 \dots \mu_r}(x) \partial_{\nu_j} Z^\rho(x) + o(Z^2) \end{aligned}$$

which is precisely Eq. 1.6.19. □

^aThe second relation needs proof.

By Lemma 5.1.2, at first order in Z the metric transforms as:

$$\delta g_{\mu\nu} = (\mathcal{L}_Z g)_{\mu\nu} \quad (5.1.13)$$

Thus, an infinitesimal diffeomorphism along $Z \in \mathfrak{X}(\mathcal{M})$ makes the metric change by an infinitesimal amount given by its Lie derivative along Z , which can be viewed as the leading term in the Taylor expansion along Z . These equations can be rewritten with Eq. 1.6.19:

$$\delta g_{\mu\nu} = Z^\rho \partial_\rho g_{\mu\nu} + g_{\rho\nu} \partial_\mu Z^\rho + g_{\mu\rho} \partial_\nu Z^\rho$$

Using the metric compatibility $\nabla_\rho g_{\mu\nu} = 0$ of the Levi-Civita connection, from which it follows that $\partial_\rho g_{\mu\nu} = \Gamma_{\rho\mu}^\lambda g_{\lambda\nu} + \Gamma_{\rho\nu}^\lambda g_{\mu\lambda}$, this expression becomes:

$$\delta g_{\mu\nu} = g_{\mu\rho} (\partial_\nu Z^\rho + \Gamma_{\nu\lambda}^\rho Z^\lambda) + g_{\rho\nu} (\partial_\mu Z^\rho + \Gamma_{\mu\lambda}^\rho Z^\lambda) = g_{\mu\rho} \nabla_\nu Z^\rho + g_{\rho\nu} \nabla_\mu Z^\rho$$

Using again the metric compatibility, the metric factors can be brought inside the covariant derivatives, yielding a simple expression:

$$\delta g_{\mu\nu} = \nabla_\mu Z_\nu + \nabla_\nu Z_\mu \quad (5.1.14)$$

This equation can be used in the path integral. In fact, insisting that $\delta\mathcal{S} = 0$ for any $\delta g_{\mu\nu}$ gives the equations of motion; on the other hand, those variations $\delta g_{\mu\nu}$ for which $\delta\mathcal{S} = 0$ for any metric are the *symmetries* of the action. From Prop. 5.1.1:

$$\delta\mathcal{S}_{\text{EH}}[g] = \int_{\mathcal{M}} d^d x \sqrt{g} G^{\mu\nu} \delta g_{\mu\nu} = 2 \int_{\mathcal{M}} d^d x \sqrt{g} G^{\mu\nu} \nabla_\mu Z_\nu$$

Invariance for $x^\mu \mapsto x^\mu - Z^\mu$ means that $\delta\mathcal{S} = 0$ for all $Z \in \mathfrak{X}(\mathcal{M})$, hence, integrating by parts:

$$\nabla_\mu G^{\mu\nu} = 0 \quad (5.1.15)$$

This is the Bianchi identity: although it is a result from Differential Geometry, it follows from diffeomorphism invariance of the Einstein-Hilbert action. This identity contains 4 equations, which make the 10 Einstein equations not completely independent: in fact, only 6 of them are independent, the same number of independent components of the metric, thus ensuring that the field equations are not over-determined.

§5.1.1.1 Isometries

With Eq. 5.1.14, it is possible to find those vector fields which identically have $\delta g_{\mu\nu} \equiv 0$, i.e. those under whose flow the metric tensor remains unchanged.

Definition 5.1.2 (Isometry)

Let $(\mathcal{M}, g)$ be a metric manifold. The flow associated to $\xi \in \mathfrak{X}(\mathcal{M})$ is an **isometry** if:

$$\mathcal{L}_\xi g = 0 \quad (5.1.16)$$

By Eq. 5.1.13-5.1.14, in a spacetime this condition can be specialized to a local frame as:

$$\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = 0 \quad (5.1.17)$$

Def. 5.1.2 means that the metric does not change along flow lines: this is called *Killing equation*, and a vector field which satisfied it is a **Killing vector field**. It can be shown that the Killing vector fields of a manifold form a Lie algebra: this is the Lie algebra of the isometry group of the manifold, which is thus generated by its Killing vector fields.

Given a Killing vector field $\xi \in \mathfrak{X}(\mathcal{M})$, it is always possible to choose coordinates $(x^0, \dots, x^{d-2}, y)$ on $\mathcal{M}$ such that $\xi = \frac{\partial}{\partial y}$, known as *adapted coordinates*, in which the Killing equation simply reads $\partial_y g_{\mu\nu} = 0$, as in the case of an ignorable degree of freedom. It is possible to reframe Noether's theorem in the context of Killing vectors. Indeed, the presence of a Killing vector field gives rise to a conserved quantity.

Theorem 5.1.1 (Noether's theorem)

Let $(\mathcal{M}, g)$ be a spacetime, and suppose that it admits a Killing vector field $\xi \in \mathfrak{X}(\mathcal{M})$. Then, given an affinely-parametrized geodesic with tangent vector field U , it is possible to define a **Noether charge** associated to ξ as:

$$Q_\xi := \langle U, \xi \rangle_g \quad (5.1.18)$$

which is conserved along the geodesic, i.e. $\nabla_U Q_\xi = 0$.

Proof. Since the geodesic is affinely-parametrized, $\nabla_U U = 0$, hence by direct computation:

$$\begin{aligned} \nabla_U Q_\xi &= \nabla_U \langle U, \xi \rangle_g = \langle \nabla_U U, \xi \rangle_g + \langle U, \nabla_U \xi \rangle_g \\ &= \langle U, \nabla_U \xi \rangle_g = U^\mu U^\nu \nabla_\nu \xi_\mu = U^\mu U^\nu \nabla_{(\nu} \xi_{\mu)} = 0 \end{aligned}$$

which vanishes by the Killing equation $\nabla_{(\nu} \xi_{\mu)} = 0$. □

The interpretation of the Noether charge depends on the nature of the associated Killing vector field, e.g. if ξ is globally timelike then Q_ξ can be interpreted as an energy.

Example 5.1.1 (Static spherically-symmetric spacetime)

Consider a static spherically-symmetric spacetime with metric:

$$ds^2 = -h(r)dt^2 + h(r)^{-1}dr^2 + r^2 d\Omega_2^2$$

There are 4 Killing vector fields: $\frac{\partial}{\partial t}$, since t is a cyclic variable, and the 3 Killing vector fields, among which is $\frac{\partial}{\partial \varphi}$, associated to the spatial isometry subgroup $SO(3)$.

Consider a generic geodesic γ with tangent vector field u , with $u^2 = -1$ (timelike) or $u^2 = 0$ (null). By Noether's theorem Th. 5.1.1, then, $\langle u, \frac{\partial}{\partial t} \rangle_g$ is constant along γ :

$$\left\langle u, \frac{\partial}{\partial t} \right\rangle_g = u^\mu g_{\mu 0} = -u^0 h$$

This expression resembles that for the energy of a particle with 4-momentum p measured by an

observer with 4-velocity v :

$$E_{\text{local}} = -\langle p, v \rangle_g \quad (5.1.19)$$

Indeed, the 4-velocity of a stationary ($r, \vartheta, \varphi = \text{const.}$) observer at infinity is just $v = \frac{\partial}{\partial t}$, so assuming a timelike geodesic with $u^\mu = \frac{1}{m}p^\mu$:

$$E_\infty = -\langle p, v \rangle = -m \left\langle u, \frac{\partial}{\partial t} \right\rangle_g$$

Therefore, the Noether charge associated to $\frac{\partial}{\partial t}$ is the energy per unit mass measured at infinity, which is constant along timelike geodesics. Equivalently, $\langle u, \frac{\partial}{\partial \varphi} \rangle_g$ turns out to be the particle's angular momentum per unit mass measured at infinity.

§5.1.1.2 Maximally symmetric spacetimes

Starting from the Killing equation, it is possible to further characterize Killing vector fields with the curvature tensor.

Lemma 5.1.3 (Killing vectors and Riemann tensor)

Let $(\mathcal{M}, g)$ be a metric manifold with a torsion-free Levi-Civita connection, and let $\xi \in \mathfrak{X}(\mathcal{M})$ be a Killing vector field. Then:

$$\nabla_\nu \nabla_\alpha \xi_\mu = -\mathcal{R}_{\alpha\mu\nu}{}^\beta \xi_\beta \quad (5.1.20)$$

Proof. By the Ricci identity Eq. 2.5.8 for a torsion-free connection:

$$\nabla_\mu \nabla_\nu \xi_\alpha - \nabla_\nu \nabla_\mu \xi_\alpha = \mathcal{R}_{\alpha\beta\mu\nu} \xi^\beta = \mathcal{R}_{\mu\nu\alpha}{}^\beta \xi_\beta$$

where the first symmetry of Eq. 2.5.11 has been used. By the Killing equation $\nabla_\nu \nabla_\mu \xi_\alpha = -\nabla_\nu \nabla_\alpha \xi_\mu$, so:

$$\nabla_\mu \nabla_\nu \xi_\alpha + \nabla_\nu \nabla_\alpha \xi_\mu = \mathcal{R}_{\mu\nu\alpha}{}^\beta \xi_\beta$$

Performing cyclic permutations of indices, a set of three equations is found (including the previous one):

$$\nabla_\mu \nabla_\nu \xi_\alpha + \nabla_\nu \nabla_\alpha \xi_\mu = \mathcal{R}_{\mu\nu\alpha}{}^\beta \xi_\beta$$

$$\nabla_\nu \nabla_\alpha \xi_\mu + \nabla_\alpha \nabla_\mu \xi_\nu = \mathcal{R}_{\nu\alpha\mu}{}^\beta \xi_\beta$$

$$\nabla_\alpha \nabla_\mu \xi_\nu + \nabla_\mu \nabla_\nu \xi_\alpha = \mathcal{R}_{\alpha\mu\nu}{}^\beta \xi_\beta$$

Summing the first two and subtracting the last one:

$$2\nabla_\nu \nabla_\alpha \xi_\mu = (\mathcal{R}_{\mu\nu\alpha}{}^\beta + \mathcal{R}_{\nu\alpha\mu}{}^\beta - \mathcal{R}_{\alpha\mu\nu}{}^\beta) \xi_\beta$$

The Bianchi identity $\mathcal{R}_{[\mu\nu\alpha]}{}^\beta = 0 \implies \mathcal{R}_{\mu\nu\alpha}{}^\beta + \mathcal{R}_{\nu\alpha\mu}{}^\beta = -\mathcal{R}_{\alpha\mu\nu}{}^\beta$ completes the proof. $\square$

Eq. 5.1.17, 5.1.20 can be used to determine Killing vector fields: starting from a point $p \in \mathcal{M}$ and moving along a curve, initial conditions are imposed on $\xi^\mu|_p$ and $\nabla_\mu \xi_\nu|_p$, however the Killing equation reduces the number of independent initial conditions, since only those on $\nabla_{[\mu} \xi_{\nu]}$ are relevant. On an n -dimensional Lorentzian manifold $\mathcal{M}$, then, the maximum number of independent Killing

vector fields (i.e. of independent initial conditions) is:

$$N_{\max} = d + \frac{d(d-1)}{2} = \frac{d(d+1)}{2} \quad (5.1.21)$$

A spacetime which admits $N_{\max}$ Killing vector fields is said to be **maximally symmetric**. These $N_{\max}$ Killing vector fields split between generators of translations and rotations: indeed, in an d -dimensional space there are d axes along which to define translations and $\frac{1}{2}d(d-1)$ planes in which to define rotations, thus reproducing the above split between initial conditions.

Proposition 5.1.2 (Riemann tensor of maximally symmetric spacetime)

The curvature tensor of a maximally symmetric spacetime can be written as:

$$\mathcal{R}_{\mu\nu\alpha\beta} = \lambda(g_{\mu\alpha}g_{\nu\beta} - g_{\nu\alpha}g_{\mu\beta}) \quad \lambda \in \mathbb{R} \quad (5.1.22)$$

Proof. In a maximally symmetric spacetime the curvature is independent under all possible translations and rotations, hence choose normal coordinates at an arbitrary $p \in \mathcal{M}$, so that $g_{\mu\nu}|_p = \eta_{\mu\nu}$. These coordinates are not unique, since any Lorentz transformation leaves the metric components unchanged and equal to $\eta_{\mu\nu}$ (by definition of Lorentz transformation, recall Obs. 3.3.1). Lorentz invariance must then be imposed on the Riemann tensor: this means that it can only be expressed in terms of Lorentz invariant tensors, which uniquely are the Lorentz–Minkowski metric, the Kronecker delta and the Levi–Civita tensor. The only combination of these tensor which matches the symmetries of the Riemann tensor (Prop. 2.5.4) is:

$$\mathcal{R}_{\mu\nu\alpha\beta}|_p \propto \eta_{\mu\alpha}\eta_{\nu\beta} - \eta_{\mu\beta}\eta_{\nu\alpha}$$

Replacing $\eta_{\mu\nu}$ with $g_{\mu\nu}|_p = \eta_{\mu\nu}$ turns this into a tensorial equation, valid on all of $\mathcal{M}$ by the arbitrariness of p , hence the thesis. $\square$

The constant $\lambda \in \mathbb{R}$ has dimensions of $[\lambda] = L^{-2}$. The Ricci tensor and scalar are consequently expressed by:

$$\mathcal{R}_{\mu\nu} = (d-1)\lambda g_{\mu\nu} \quad \mathcal{R} = d(d-1)\lambda \quad (5.1.23)$$

Therefore, maximally symmetric spacetimes are Einstein manifolds, and as such have constant Ricci scalar curvature. An explicit construction of maximally symmetric solutions to Eq. 2.5.16 is given in §5.1.2.

§5.1.1.3 Komar integrals

Noether’s theorem is not the only way to associate conserved quantities to Killing vector fields in matter-less spacetimes.

Definition 5.1.3 (Komar form)

Let $(\mathcal{M}, g)$ be a spacetime, and let $\xi \in \mathfrak{X}(\mathcal{M})$ be a Killing vector field. The **Komar charge** $\mathbf{K}[\xi] \in \Omega^{d-2}(\mathcal{M})$ associated to ξ is defined as:

$$\mathbf{K}[\xi] := -\frac{1}{16\pi G_N^{(d)}} \star d\xi^b \quad (5.1.24)$$

The Killing 2-form $d\xi^b$ is a gravitational generalization of the field-strength tensor of a Yang–Mills theory (see §B.1 for the $U(1)$ case). Indeed, $\Gamma_{[\mu\nu]}^\rho = 0$ for torsion-free connections, hence:

$$d\xi^b = \partial_\mu \xi_\nu dx^\mu \wedge dx^\nu = \nabla_{[\mu} \xi_{\nu]} dx^\mu \wedge dx^\nu$$

Theorem 5.1.2 (Gravitational Maxwell equations)

Let $(\mathcal{M}, g)$ be a Ricci-flat spacetime, and let $\xi \in \mathfrak{X}(\mathcal{M})$ be a Killing vector field. Then:

$$d\mathbf{K}[\xi] = 0 \quad (5.1.25)$$

Proof. To prove this result, it is convenient to work in a local frame, so that, by Prop. B.1.2, the thesis is equivalent to $\nabla^\mu X_{\mu\nu} = 0$, where $d\xi^b = \frac{1}{2} X_{\mu\nu} dx^\mu \wedge dx^\nu$.

Contracting μ, ν in Eq. 5.1.20 yields $\nabla_\mu \nabla_\alpha \xi^\mu = \mathcal{R}_{\alpha\beta} \xi^\beta$, so using the Killing equation $\nabla_\mu \xi_\nu = -\nabla_\nu \xi_\mu$:

$$\nabla^\mu X_{\mu\nu} = \nabla^\mu (\nabla_\mu \xi_\nu - \nabla_\nu \xi_\mu) = -2\nabla^\mu \nabla_\nu \xi_\mu = -2\mathcal{R}_{\nu\beta} \xi^\beta$$

For a Ricci-flat spacetime $\mathcal{R}_{\mu\nu} \equiv 0$, hence the thesis. $\square$

The fact that Komar forms are conserved (i.e. closed) in Ricci-flat spacetimes is often frased as the fact that they are “on-shell closed”, as they require the vacuum field equations to be satisfied in order to be closed. Like other on-shell closed $(d-2)$ -forms, the integral of the Komar form $\mathbf{K}[\xi]$ over a suitable spatial $(d-2)$ -hypersurface gives the (gravitational) charge associated with the Killing vector field ξ .

Definition 5.1.4 (Komar charge)

Let $(\mathcal{M}, g)$ be a spacetime, and let $\xi \in \mathfrak{X}(\mathcal{M})$ be a Killing vector field. Given a closed oriented spacelike $(d-2)$ -submanifold $S \subseteq \mathcal{M}$, the **Komar charge** associated to ξ in S is define as:

$$Q_K[\xi; S] := \int_S \mathbf{K}[\xi] \quad (5.1.26)$$

For asymptotically-flat spacetimes, a $(d-2)$ -sphere at spatial infinity $\mathbb{S}_\infty^{d-2}$ is usually chosen, resulting in the so-called *Komar integral* of ξ :

$$Q_K[\xi] \equiv \int_{\mathbb{S}_\infty^{d-2}} \mathbf{K}[\xi] = -\frac{1}{16\pi G_N^{(d)}} \int_{\mathbb{S}_\infty^{d-2}} \star d\xi^b \quad (5.1.27)$$

As for Noether integrals of a particle, Komar integrals of spacetime are interpreted based on the defining Killing vector. For example, if ξ is everywhere timelike, then its Komar integral can be identified with the energy (or the mass) of spacetime; if the Killing vector is related to rotations, instead, the conserved charge is the angular momentum of spacetime.

Note that the sign of Eq. 5.1.24 and Eq. 5.1.26 is dictated by the convention used for the Hodge dual Eq. 3.2.6: if $\varepsilon_{j_{p+1}\dots j_n j_1 \dots j_p}$ is used in place of $\varepsilon_{j_1 \dots j_p}$, as in [1], then the correct sign is $(-1)^{d-1}$.

§5.1.2 Maximally symmetric solutions

The Einstein equations in the absence of matter reduce to $\mathcal{R}_{\mu\nu} = \Lambda g_{\mu\nu}$, with $\Lambda \in \mathbb{R}$. In the case of maximally symmetric solutions, $\Lambda = (n-1)R$, and the possible solutions split into three classes: $\Lambda = 0$ (Minkowski), $\Lambda > 0$ (de Sitter) and $\Lambda < 0$ (anti-de Sitter).

§5.1.2.1 Minkowski space

The simplest case is $\Lambda = 0$. The Einstein equations then reduce to $\mathcal{R}_{\mu\nu} = 0$, with the condition of $g_{\mu\nu}$ being non-degenerate, since it is a metric and the field equations require the existence of the inverse metric $g^{\mu\nu}$. This restriction is physically unusual: it is not a holonomic constraint on the physical degrees of freedom, but an inequality $\det g < 0$ (with required signature $(-, +, +, +)$) which is not found for other fields of the Standard Model.

The simplest Ricci flat metric is the Lorentz–Minkowski metric $ds^2 = -dt^2 + d\mathbf{x}^2$ on $\mathbb{R}^{1,d-1}$, but it is not the only one: another example is the Schwarzschild metric (§5.3).

§5.1.2.2 de Sitter space

Consider now $\Lambda > 0$ and $d = 4$. A static spherically-symmetric ansatz is:

$$ds^2 = -f(r)^2 dt^2 + f(r)^{-2} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2)$$

Riemann tensor First, one needs to compute the Ricci tensor for this metric: a simple way is using the curvature form. The non-coordinate 1-forms satisfy $ds^2 = \eta_{ab} \theta^a \otimes \theta^b$, thus:

$$\theta^0 = f dt \quad \theta^1 = f^{-1} dr \quad \theta^2 = r d\vartheta \quad \theta^3 = r \sin \vartheta d\varphi$$

$$d\theta^0 = f' dr \wedge dt \quad d\theta^1 = 0 \quad d\theta^2 = dr \wedge d\vartheta \quad d\theta^3 = \sin \vartheta dr \wedge d\varphi + r \cos \vartheta d\vartheta \wedge d\varphi$$

The first Cartan structure relation Eq. 3.3.20, together with Eq. 3.3.21, allow to determine the connection 1-form. For example, the first equation is $\omega^0_1 = f' f dt = f' d\theta^0$, and then $\omega^1_0 = \omega_{10} = -\omega_{01} = \omega^0_1$. Using the other structure equation, the only non-vanishing components of the connection 1-form are:

$$\omega^0_1 = \omega^1_0 = f' \hat{\theta}^0 \quad \omega^2_1 = -\omega^1_2 = \frac{f}{r} \hat{\theta}^2 \quad \omega^3_1 = -\omega^1_3 = \frac{f}{r} \hat{\theta}^3 \quad \omega^3_2 = -\omega^2_3 = \frac{\cot \vartheta}{r} \hat{\theta}^3$$

The curvature 2-form can be computed from the second Cartan structure relation Eq. 3.3.23. For example, $\Omega^0_1 = d\omega^0_1 + \omega^0_c \wedge \omega^c_1$, but $\omega^0_c \wedge \omega^c_1 = \omega^0_1 \wedge \omega^1_1 = 0$, thus $\Omega^0_1 = d\omega^0_1 = ((f')^2 + f'' f) dr \wedge dt$. From the curvature 2-form, the Riemann tensor can be computed via Eq. 3.3.22 (recall the anti-symmetries of the Riemann tensor), finding the only non-vanishing components:

$$\mathcal{R}_{0101} = f f'' + (f')^2 \quad \mathcal{R}_{0202} = \mathcal{R}_{0303} = \frac{f f'}{r} \quad \mathcal{R}_{1212} = \mathcal{R}_{1313} = -\frac{f f'}{r} \quad \mathcal{R}_{2323} = \frac{1 - f^2}{r^2}$$

To convert back to $x^\mu = (t, r, \vartheta, \varphi)$, use $\mathcal{R}_{\mu\nu\rho\sigma} = e^a_\mu e^b_\nu e^c_\rho e^d_\sigma \mathcal{R}_{abcd}$, which is particularly easy given that the matrices e_a^μ which define the non-coordinate 1-forms are diagonal:

$$\begin{aligned} \mathcal{R}_{trtr} &= f(r) f''(r) + f'(r)^2 & \mathcal{R}_{t\vartheta t\vartheta} &= f(r)^3 f'(r) r & \mathcal{R}_{t\varphi t\varphi} &= f(r)^3 f'(r) r \sin^2 \vartheta \\ \mathcal{R}_{r\vartheta r\vartheta} &= -\frac{f'(r) r}{f(r)} & \mathcal{R}_{r\varphi r\varphi} &= -\frac{f'(r) r}{f(r)} \sin^2 \vartheta & \mathcal{R}_{\vartheta\varphi\vartheta\varphi} &= (1 - f(r)^2) r^2 \sin^2 \vartheta \end{aligned}$$

Ricci tensor Given the Riemann tensor, it is easy to check that the Ricci tensor is diagonal with components:

$$\begin{aligned} \mathcal{R}_{tt} &= -f(r)^4 R_{rr} = f(r)^3 \left[f''(r) + \frac{2f'(r)}{r} + \frac{f'(r)^2}{f(r)} \right] \\ \mathcal{R}_{\varphi\varphi} &= \sin^2 \vartheta \mathcal{R}_{\vartheta\vartheta} = (1 - f(r)^2 - 2f(r) f'(r) r) \sin^2 \vartheta \end{aligned}$$

Imposing $\mathcal{R}_{\mu\nu} = \Lambda g_{\mu\nu}$ determines two constraints on $f(r)$:

$$f''(r) + \frac{2f'(r)}{r} + \frac{f'(r)^2}{f(r)} = -\frac{\Lambda}{f(r)} \quad 1 - 2f(r)f'(r)r - f(r)^2 = \Lambda r^2$$

A possible solution is:

$$f(r) = \sqrt{1 - \frac{r^2}{R^2}} \quad R^2 \equiv \frac{3}{\Lambda}$$

This determines the metric of **de Sitter space**:

$$ds^2 = -\left(1 - \frac{r^2}{R^2}\right) dt^2 + \left(1 - \frac{r^2}{R^2}\right)^{-1} dr^2 + r^2(d\vartheta^2 + \sin^2 \vartheta d\varphi^2) \quad (5.1.28)$$

More precisely, this is the *static patch* of de Sitter space.

de Sitter geodesics To interpret a metric, it is useful to study its geodesics. First, note that a non-trivial $g_{tt}(r)$ term means that a particle won't remain at rest at $r \neq 0$, but it will be pushed to smaller values of $g_{tt}(r)$, i.e. larger values of r . The action of a particle in the general $f(r)$ metric, parametrized by its proper time σ , is:

$$\mathcal{S} = \int d\sigma \left[-f(r)^2 \dot{t}^2 + f(r)^{-2} \dot{r}^2 + r^2(\dot{\vartheta}^2 + \sin^2 \vartheta \dot{\varphi}^2) \right] \quad (5.1.29)$$

where $\dot{x}^\mu \equiv \frac{dx^\mu}{d\sigma}$. This Lagrangian has two ignorable degrees of freedom which lead to conserved quantities: $t(\sigma)$ and $\varphi(\sigma)$, as they appear only with time derivatives. The first one has energy as the conserved quantity, while the second has angular momentum:

$$E = -\frac{1}{2} \frac{\partial L}{\partial \dot{t}} = f(r)^2 \dot{t} \quad (5.1.30)$$

$$\ell = \frac{1}{2} \frac{\partial L}{\partial \dot{\varphi}} = r^2 \sin^2 \vartheta \dot{\varphi} \quad (5.1.31)$$

The $\frac{1}{2}$ are due to the absence of the usual factor in the kinetic term. The equations of motion from the action Eq. 5.1.29 should be supplemented with a constraint to distinguish whether a particle is massive or massless. For a massive particle, the trajectory must be timelike, so:

$$-f(r)^2 \dot{t}^2 + f(r)^{-2} \dot{r}^2 + r^2(\dot{\vartheta}^2 + \sin^2 \vartheta \dot{\varphi}^2) = -1$$

WLOG consider geodesics that lie in the $\vartheta = \frac{\pi}{2}$ plane, so that $\dot{\vartheta} = 0$ and $\sin^2 \vartheta = 1$. Replacing (t, φ) with (E, ℓ) , the constraint becomes:

$$\dot{r}^2 + V_{\text{eff}}(r) = E^2 \quad V_{\text{eff}}(r) = \left(1 + \frac{\ell^2}{r^2}\right) f(r)^2 \quad (5.1.32)$$

For de Sitter space:

$$V_{\text{eff}}(r) = \left(1 + \frac{\ell^2}{r^2}\right) \left(1 - \frac{r^2}{R^2}\right) \quad (5.1.33)$$

This effective potential is plotted in Fig. 5.1: immediately, one sees that the potential pushes the particle to larger values of r . Focusing on geodesics with $\ell = 0$, the potential is a harmonic repulsor:

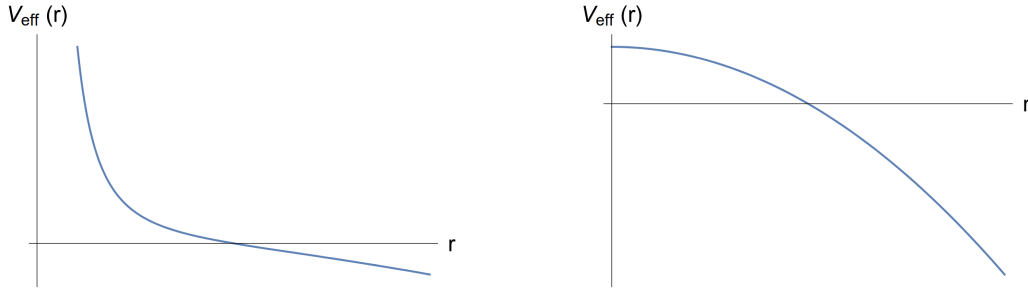


Figure 5.1: Effective potential for de Sitter geodesics, with $\ell \neq 0$ and $\ell = 0$ respectively.

a particle stationary at $r = 0$ is an unstable geodesic, since if it has non-zero initial velocity it will follow the trajectory:

$$r(\sigma) = R\sqrt{E^2 - 1} \sinh \frac{\sigma}{R} \quad (5.1.34)$$

Note that the metric is singular at $r = R$, a fact not manifest in the geodesic Eq. 5.1.34 which shows that any observer reaches $r = R$ in finite proper time. Problems arise when studying the coordinate time t , which has the interpretation of the time experienced by some stationary at $r = 0$; from Eq. 5.1.30:

$$\frac{dt}{d\sigma} = E \left(1 - \frac{r^2}{R^2} \right)^{-1}$$

This shows that $t(\sigma) \rightarrow \infty$ as $r(\sigma) \rightarrow R$: in fact, if $r(\sigma^*) = R$, then for $\sigma = \sigma^* - \varepsilon$ one has $\frac{dt}{d\sigma} = -\frac{\alpha}{\varepsilon}$, i.e. $t(\varepsilon) = -\log(\varepsilon/R)$, so $t(\varepsilon) \rightarrow \infty$ as $\varepsilon \rightarrow 0$. This means that a particle moving along the geodesic Eq. 5.1.34 will reach $r = R$ in finite proper time, but a stationary observer at $r = 0$ will measure an infinite amount of time.

This strange behaviour is similar to what happens at the horizon of a black hole ($r = 2GM$): however, the Schwarzschild metric has a singularity at $r = 0$, while de Sitter metric looks flat at $r = 0$. de Sitter space seems like an inverted black hole in which particles are pushed outwards to $r = R$.

de Sitter embeddings de Sitter space can be embedded as a submanifold of $\mathbb{R}^{1,4}$ with metric:

$$ds^2 = -(dX^0)^2 + \sum_{i=1}^4 (dX^i)^2 \quad (5.1.35)$$

In particular, de Sitter metric Eq. 5.1.28 is a metric on the submanifold of $\mathbb{R}^{1,4}$ defined by the timelike hyperboloid:

$$-(X^0)^2 + \sum_{i=1}^4 (X^i)^2 = R^2 \quad (5.1.36)$$

A way of parametrizing this constraint is by imposing that $r^2 = (X^1)^2 + (X^2)^2 + (X^3)^2$, so that:

$$R^2 - r^2 = (X^4)^2 - (X^0)^2$$

The solutions are parametrized as:

$$X^0 = \sqrt{R^2 - r^2} \sinh \frac{t}{R} \quad X^4 = \sqrt{R^2 - r^2} \cosh \frac{t}{R}$$

Computing the respective variations, along with $\sum_{i=1}^3 (dX^i)^2 = dr^2 + r^2 d\Omega_n^2$, where $d\Omega_n^2$ is the metric element on $\mathbb{S}^n$, allows to show that the pull-back of the Minkowski metric Eq. 5.1.35 on

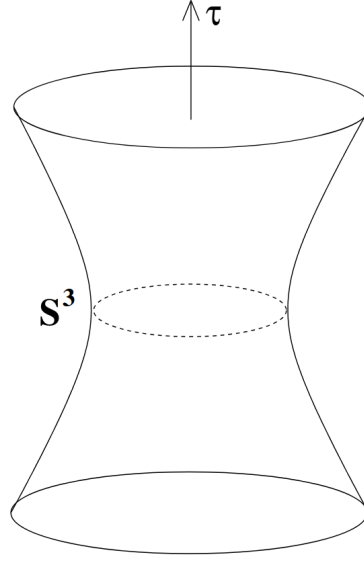


Figure 5.2: de Sitter space visualization with global coordinates.

the hyperboloid Eq. 5.1.36 so parametrized gives the de Sitter metric Eq. 5.1.28 in the static patch coordinates.

The coordinates $\{X^i\}_{i=0,\dots,4}$ so defined are not the most intuitive: they single out X^4 as special, when the constraint does no such thing, and they do not cover the whole hyperboloid, as they are limited to $X^4 \geq 0$. A better choice of parametrization is:

$$X^0 = R \sinh \frac{\tau}{R} \quad X^i = y^i R \cosh \frac{\tau}{R} : \sum_{i=1}^4 (y^i)^2 = 1$$

Given this constraint, $\{y^i\}_{i=1,2,3,4}$ parametrize $\mathbb{S}^3$. These coordinates retain more of the symmetry of de Sitter space and cover the whole space, thus are a better parametrization. The pull-back of Minkowski metric, however, gives a rather different metric on de Sitter space:

$$ds^2 = -d\tau^2 + R^2 \cosh^2 \frac{\tau}{R} d\Omega_3^2 \quad (5.1.37)$$

These are known as *global coordinates*, as they cover the whole space (except for usual singularities at the poles for any choice of coordinates on $\mathbb{S}^3$). Since this metric is related to that in Eq. 5.1.28 by a change of coordinates, it too must obey the Einstein equations. Global coordinates also show that the singularity which happens at $X^4 = 0$, i.e. at $r = R$, is nothing but a coordinate singularity.

These coordinates provide a clearer intuition for the physics of de Sitter space: it is a time-dependent solution of the field equations in which a spatial $\mathbb{S}^3$ first shrinks to a minimal radius R and then expands, as shown in Fig. 5.2. The expansionary phase is a good approximation to our current universe on large scales.

§5.1.2.3 Anti-de Sitter space

Finally, consider $\Lambda < 0$. With the same ansatz as for de Sitter space, it is easy to find the metric for *anti-de Sitter space* (AdS):

$$ds^2 = -\left(1 + \frac{r^2}{R^2}\right) dt^2 + \left(1 + \frac{r^2}{R^2}\right)^{-1} dr^2 + r^2 d\Omega_2^2 \quad R^2 \equiv -\frac{3}{\Lambda} \quad (5.1.38)$$

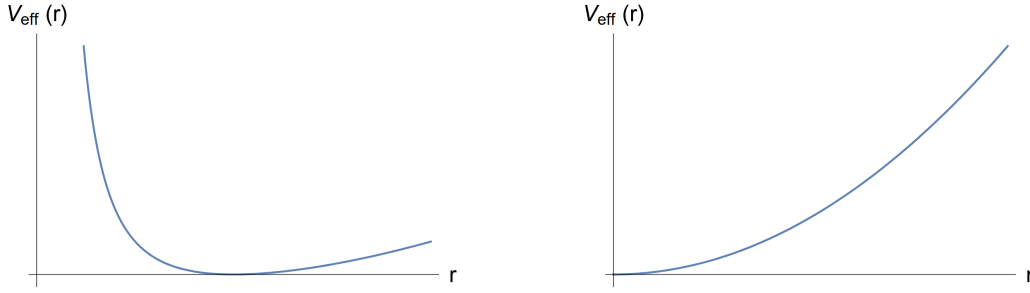


Figure 5.3: Effective potential for anti-de Sitter geodesics, with $\ell \neq 0$ and $\ell = 0$ respectively.

Equivalent coordinates are found setting $r = R \sinh \rho$:

$$ds^2 = -\cosh^2 \rho dt^2 + R^2 d\rho^2 + R^2 \sinh^2 \rho d\Omega_2^2 \quad (5.1.39)$$

In AdS there is no coordinate singularity in r , and indeed coordinates cover the whole space.

Anti-de Sitter geodesics AdS has the same action as in Eq. 5.1.29, thus the radial trajectory of a massive particle moving along a geodesic in the $\vartheta = \frac{\pi}{2}$ plane is $\dot{r}^2 + V_{\text{eff}}(r) = E^2$, with:

$$V_{\text{eff}}(r) = \left(1 + \frac{\ell^2}{r^2}\right) \left(1 + \frac{r^2}{R^2}\right) \quad (5.1.40)$$

Plotting it in Fig. 5.3, the geodesics' behaviour is clear: with no angular momentum, anti-de Sitter space acts like a harmonic oscillator, pulling the particle towards the origin and making it oscillate around $r = 0$. If the particle has non-zero angular momentum, then the potential has a minimum at $r_*^2 = R\ell$, thus particles oscillate around r_* : importantly, particles with finite energy cannot escape to $r \rightarrow \infty$, but are constrained by the spacetime within some finite distance from the origin.

This picture of AdS as a harmonic trap which pulls particle to the origin clashes with the fact that AdS is a homogeneous space (roughly, all points are the same). To understand how these two facts are compatible, consider a stationary observer at $r = 0$: this is a geodesic and, from its perspective, other observers (with $\ell = 0$) will oscillate around $r = 0$ along geodesics. However, since these observers move along geodesics, in their reference frame they are stationary at $r = 0$, with all other observers oscillating. Thus, while in dS each observer views themselves at the center of the universe, with other observers moving away from them, in AdS each observer views themselves as the center of the universe, with other observer oscillating around them.

Its possible to study geodesics for a massless particle too. This time, the constraint to the action is:

$$-f(r)^2 \dot{t}^2 + f(r)^{-2} \dot{r}^2 + r^2 (\dot{\vartheta}^2 + \sin^2 \vartheta \dot{\phi}^2) = 0$$

This means that the particle follows a null geodesic. Its equation of motion is:

$$\dot{r}^2 + V_{\text{null}}(r) = E^2 \quad V_{\text{null}}(r) = \frac{\ell^2}{r^2} \left(1 + \frac{r^2}{R^2}\right)$$

Its plot in Fig. 5.4 makes it clear that any massless particle can escape to $r \rightarrow \infty$, as the null potential is asymptotically constant, and it will experience the usual gravitational redshift: AdS only confines massive particles. To solve for the trajectory, it is easier to work with $r = R \sinh \rho$ and $\ell = 0$, so that:

$$R\dot{\rho} = \pm \frac{E}{\cosh \rho} \quad \Rightarrow \quad R \sinh \rho(\sigma) = E(\sigma - \sigma_0)$$

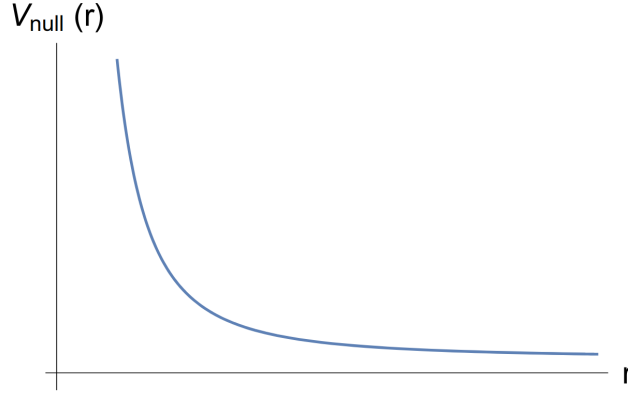


Figure 5.4: Potential experienced by massless particles with $\ell \neq 0$ in AdS.

One sees that $\rho \rightarrow \infty$ only in infinite affine time (i.e. $\sigma \rightarrow \infty$). However, in coordinate time, recalling Eq. 5.1.30, $E = \cosh^2 \rho \dot{t}$, so:

$$R \tan \frac{t}{R} = E(\sigma - \sigma_0)$$

Hence, as affine time $\sigma \rightarrow \infty$, coordinate time $t \rightarrow \frac{\pi}{2} R$. This means that light rays escape to infinity in a finite amount of coordinate time: to make sense of dynamics in AdS, one must specify some boundary conditions at infinity to dictate how massless particles and field behave. AdS does not appear to be of any cosmological interest.

Anti-de Sitter embeddings AdS too has a natural embedding in a 5-dimensional spacetime: it is a submanifold of $\mathbb{R}^{2,3}$ with metric:

$$ds^2 = -(dX^0)^2 - (dX^4)^2 + \sum_{i=1}^3 (dX^i)^2 \quad (5.1.41)$$

In particular, anti-de Sitter metric Eq. 5.1.38 is a metric on the hyperboloid:

$$-(X^0)^2 - (X^4)^2 + \sum_{i=1}^3 (X^i)^2 = -R^2 \quad (5.1.42)$$

This constraint can be solved via the parametrization:

$$X^0 = R \cosh \rho \sin \frac{t}{R} \quad X^4 = R \cosh \rho \cos \frac{t}{R} \quad X^i = y^i R \sinh \rho : \sum_{i=1}^3 (y^i)^2 = 1$$

Given this constraint, $\{y^i\}_{i=1,2,3}$ parametrize $\mathbb{S}^2$. The pull-back of the metric Eq. 5.1.41 on the hyperboloid so parametrized gives anti-de Sitter metric Eq. 5.1.38.

Note a small subtlety: the embedding hyperboloid has topology $\mathbb{S}^1 \times \mathbb{R}^3$, not $\mathbb{R}^4$: this corresponds to a compact time direction, as $t \in [0, 2\pi R)$. However, AdS metric does not have such restriction, with $t \in \mathbb{R}$: this is a universal covering of the hyperboloid.

Another useful parametrization of the hyperboloid is the following:

$$X^4 - X^3 = r \quad X^4 + X^3 = \frac{R^2}{r} + \frac{r}{R^2} \eta_{ij} x^i x^j \quad X^i = \frac{r}{R} x^i, \quad i = 0, 1, 2$$

with $r \in [0, \infty)$. With these coordinates, the metric takes the form:

$$ds^2 = R^2 \frac{dr^2}{r^2} + \frac{r^2}{R^2} \eta_{ij} dx^i dx^j$$

These coordinates do not cover the whole AdS, but only one-half of the hyperboloid, restricted to $X^4 - X^3 > 0$: this is known as the *Poincaré patch* of AdS. Moreover, x^0 cannot be further extended beyond $x^0 \in (-\infty, +\infty)$, thus in global coordinates the restriction $t \in [0, 2\pi R)$ remains. In this case, massive particles fall towards $r = 0$.

Finally, there are two more possible coordinate systems on the Poincaré patch, obtained by setting $z = \frac{R^2}{r}$ and $r = Re^\rho$:

$$ds^2 = \frac{R^2}{z^2} (dz^2 + \eta_{ij} dx^i dx^j) \quad ds^2 = R^2 d\rho^2 + e^{2\rho} \eta_{ij} dx^i dx^j$$

In each case, massive particles fall towards $z = \infty$ or $\rho = -\infty$.

§5.1.2.4 Isometry groups

Minkowski spacetime Consider Minkowski spacetime $\mathbb{R}^{1,3}$ with $g_{\mu\nu} = \eta_{\mu\nu}$. The Killing equation is:

$$\partial_\mu K_\nu + \partial_\nu K_\mu = 0$$

There are two forms of solution. The first one is:

$$K_\mu = c_\mu$$

for any constant vector c_μ . These generate translations. Alternatively:

$$K_\mu = \omega_{\mu\nu} x^\nu : \omega_{\mu\nu} = -\omega_{\nu\mu}$$

These generate rotations and Lorentz boosts (see §1.2 of [4]). The emergence of the Lie algebra structure can be elucidated defining the Killing vectors:

$$P_\mu := \partial_\mu \quad \mathcal{J}_{\mu\nu} := \eta_{\mu\rho} x^\rho \partial_\nu - \eta_{\nu\rho} x^\rho \partial_\mu$$

These are 10 Killing vectors, 4 for translations and 6 for rotations and boosts, which obey the Poincaré algebra:

$$\begin{aligned} [P_\mu, P_\nu] &= 0 & [\mathcal{J}_{\mu\nu}, P_\sigma] &= -\eta_{\mu\sigma} P_\nu + \eta_{\nu\sigma} P_\mu \\ [\mathcal{J}_{\mu\nu}, \mathcal{J}_{\rho\sigma}] &= \eta_{\mu\sigma} \mathcal{J}_{\nu\rho} + \eta_{\nu\rho} \mathcal{J}_{\mu\sigma} - \eta_{\mu\rho} \mathcal{J}_{\nu\sigma} - \eta_{\nu\sigma} \mathcal{J}_{\mu\rho} \end{aligned}$$

This is precisely the Lie algebra of the Poincaré group $\text{ISO}(1, 3) := \mathbb{R}^4 \rtimes \text{SO}(1, 3)$, i.e. the isometry group of Minkowski spacetime.

dS and AdS spacetime The isometries of dS and AdS are best seen from their embeddings. The constraint Eq. 5.1.36 which defined de Sitter space is invariant under the rotations of $\mathbb{R}^{1,4}$, thus dS inherits the $\text{SO}(1, 4)$ isometry group. Similarly, the constraint Eq. 5.1.42 which defines anti-de Sitter space is invariant under rotations of $\mathbb{R}^{2,3}$, so AdS inherits the $\text{SO}(2, 3)$ isometry group. Note that both these groups are 10-dimensional, as $\text{ISO}(1, 3)$.

It is simple to determine the 10 Killing spinors of 5-dimensional spacetime:

$$M_{AB} = \eta_{AC} X^C \partial_B - \eta_{BC} X^C \partial_A$$

where X^A , $A = 0, 1, 2, 3, 4$ are 5-dimensional coordinates and η_{AB} is the appropriate Minkowski metric, with signature $(-, +, +, +, +)$ for dS and $(-, -, +, +, +)$ for AdS. In either case, the Lie algebra is that of the appropriate Lorentz group:

$$[M_{AB}, M_{CD}] = \eta_{AD}M_{BC} + \eta_{BC}M_{AD} - \eta_{AC}M_{BD} - \eta_{BD}M_{AC}$$

The embedding hyperbolae are both invariant under these Killing vectors: flows generated by them map points on the hyperbolae to other points on the hyperbolae. Therefore, these Killing vectors are inherited by dS and AdS respectively.

Energy is defined as the Noether charge of a timelike Killing vector. In anti-de Sitter space there is no problem finding a timelike Killing vector, as the metric in global coordinates Eq. 5.1.38 is time-independent, so $K = \partial_t$. However, this is not the case in de Sitter space.

Considering the static patch of dS space, the metric Eq. 5.1.28 is time-independent, thus $K = \partial_t$ is a Killing vector; pushing it forward to the 5-dimensional space:

$$\partial_t = \frac{\partial X^A}{\partial t} \partial_A = \frac{1}{R}(X^4 \partial_0 + X^0 \partial_4)$$

On the static patch, this Killing vector is timelike and the energy Eq. 5.1.30 follows from it. The problem is that the static patch is only one-half of the hyperboloid: when considering the whole AdS, one must account for the case $X^0 = 0, X^4 < 0$, in which the Killing vector points in the past direction, or $X^0 \neq 0, X^4 = 0$, in which it is spacelike. Therefore, the Killing vector can be both timelike (future-directed and past-directed) or spacelike when spanning the whole manifold: for this reason, it is not possible to define a global positive conserved energy on the total de Sitter space. The same conclusion follows by noting that the metric in global coordinates Eq. 5.1.37 is time-dependent.

§5.2 Matter coupling

Spacetime is not merely the background on which matter exists, but it is dynamically influenced by the matter distribution on it. It is therefore necessary to study how matter couples to the spacetime metric.

Observation 5.2.1 (Field theories in curved spacetime)

The simplest way to describe matter is by fields governed by a Lagrangian. Consider a scalar field $\phi(x)$ in a flat Minkowski spacetime $\mathbb{R}^{1,d-1}$, whose action is:

$$\mathcal{S}[\phi] := \int_{\mathcal{M}} d^d x \left[-\frac{1}{2} \eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right] \quad (5.2.1)$$

The negative sign of the kinetic term follows from the signature choice $(-, +, +, +)$. The generalization to curved spacetime is straightforward with the minimal-coupling prescription: $\partial_\mu \mapsto \nabla_\mu$ and $d^d x \mapsto \omega_g \equiv d^d x \sqrt{g}$. Curved spacetime also introduces the possibility to add new terms to the Lagrangian, for example:

$$\mathcal{S}[\phi] = \int_{\mathcal{M}} d^d x \sqrt{g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) - \frac{1}{2} \kappa \mathcal{R} \phi^2 \right] \quad (5.2.2)$$

for some $\kappa \in \mathbb{R}$ (renormalizable $\forall d \in \mathbb{N}$). This theory correctly reduces to Eq. 5.2.1 in flat spacetime, as $\mathcal{R} = 0$. To derive the equation of motion, vary $\mathcal{S}$ with respect to ϕ :

$$\begin{aligned}\delta\mathcal{S}[\phi] &= \int_{\mathcal{M}} d^d x \sqrt{g} \left[-g^{\mu\nu} \nabla_\mu \delta\phi \nabla_\nu \phi - \frac{\partial V}{\partial \phi} \delta\phi - \kappa \mathcal{R} \phi \delta\phi \right] \\ &= \int_{\mathcal{M}} d^d x \sqrt{g} \left[\left(g^{\mu\nu} \nabla_\mu \nabla_\nu \phi - \frac{\partial V}{\partial \phi} - \kappa \mathcal{R} \phi \right) \delta\phi - \nabla_\mu (\delta\phi \nabla^\mu \phi) \right]\end{aligned}$$

where integration by parts was possible by the metric compatibility of the Levi-Civita connection. The last term is, by divergence theorem, a boundary term, thus the equation of motion is:

$$g^{\mu\nu} \nabla_\mu \nabla_\nu \phi - \frac{\partial V}{\partial \phi} - \kappa \mathcal{R} \phi = 0$$

Now covariant derivatives are necessary, as $\nabla_\mu \nabla_\nu \neq \partial_\mu \partial_\nu$. It is useful to introduce the **d'Alembert operator** $\square := \nabla^\mu \nabla_\mu$, so that the equation of motion for the scalar field ϕ becomes:

$$\square\phi - \frac{\partial V}{\partial \phi} - \kappa \mathcal{R} \phi = 0 \quad (5.2.3)$$

§5.2.1 Energy-momentum tensor

In classical (and quantum) field theory in a Minkowski background, given a Poincaré-invariant Lagrangian $\mathcal{L} = \mathcal{L}(\Phi, \partial\Phi)$, i.e. a Lagrangian invariant under $\text{ISO}(1, d-1)$, it is possible to define the *canonical energy-momentum tensor* (C-EMT) as the Noether current associated to spacetime translations (see §1.3.1 of [4] for additional details):

$$\mathcal{T}^\mu_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi_a)} \partial_\nu \Phi_a - \delta^\mu_\nu \mathcal{L} \quad (5.2.4)$$

Like all Noether currents, the C-EMT is on-shell conserved, i.e. $\partial_\mu \mathcal{T}^\mu_\nu = 0$ if the equations of motion are satisfied. The main problem of the C-EMT, however, is that it is not symmetric; moreover, its generalization to a generic curved spacetime is not obvious. For this reason, it is convenient to define another energy-momentum tensor starting from the action $\mathcal{S}[\Phi] = \int_{\mathcal{M}} d^d x \sqrt{g} \mathcal{L}(\Phi, \partial\Phi, g)$:

$$T_{\mu\nu} := -\frac{2}{\sqrt{g}} \frac{\delta(\sqrt{g} \mathcal{L})}{\delta g^{\mu\nu}} \quad (5.2.5)$$

This is the so-called **Belinfante–Rosenfeld energy-momentum tensor** (BR-EMT), which is manifestly a symmetric $(0, 2)$ -tensor, as it inherits the symmetries of the metric. It can be shown that, in a flat background, the C-EMT and the BR-EMT only differ by a total derivative:

$$T^{\mu\nu}|_{g=\eta} = \mathcal{T}^{\mu\nu} + \partial_\alpha B^{\alpha\mu\nu} \quad (5.2.6)$$

where the form of the potential $B^{\alpha\mu\nu}$ depends on the nature of the field ϕ , and in particular on its spin. Indeed, if the field Φ transforms under a finite-dimensional representation of the Lorentz group $\text{SO}^+(1, d-1)$, with generators $[\mathcal{J}^{\mu\nu}]^a_b$ obeying the Lorentz algebra¹, then the Belinfante potential has a universal expression in terms of the *spin current* (see §7.4 of [21]):

$$B^{\alpha\mu\nu} = \frac{i}{2} (S^{\alpha\mu\nu} + S^{\mu\nu\alpha} + S^{\nu\mu\alpha}) \quad S^{\alpha\mu\nu} \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\alpha \Phi^a)} [\mathcal{J}^{\mu\nu}]^a_b \Phi^b \quad (5.2.7)$$

It can be shown that the combination $\mathcal{T}^{\mu\nu} + \partial_\alpha B^{\alpha\mu\nu}$ is indeed symmetric, and the manifest antisymmetry $B^{\alpha\mu\nu} = -B^{\mu\alpha\nu}$ ensures that $\partial_\mu T^{\mu\nu} = \partial_\mu \mathcal{T}^{\mu\nu}$, i.e. that the BR-EMT is on-shell conserved too. Moreover, the addition of a total derivative does not affect the Noether charge $P^\mu \equiv \int_\Sigma d^{d-1}x \mathcal{T}^{0\mu}$, since it only determines a vanishing spatial boundary term, hence the C-EMT and the BR-EMT encode the same dynamical information: for this reason, in the following $T^{\mu\nu}$ is simply called the energy-momentum tensor (EMT).

§5.2.1.1 Geometrodynamics sourced by matter

To understand how matter fields back-react on the spacetime metric, consider the combined action:

$$\mathcal{S}[g, \Phi] = \frac{1}{16\pi G_N^{(d)}} \int_{\mathcal{M}} d^d x \sqrt{g} (\mathcal{R} - (d-2)\Lambda) + \mathcal{S}_{\text{mat}}[g, \Phi] \quad (5.2.9)$$

where $\mathcal{S}_m$ is the combined action for all matter fields (here collectively denoted by Φ), which in general depends on both the fields and the metric.

Proposition 5.2.1 (Einstein field equations)

The equations of motion determined by the action Eq. 5.2.9 are:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N^{(d)} T_{\mu\nu} \quad \frac{\delta \mathcal{S}_{\text{mat}}}{\delta \Phi} = 0 \quad (5.2.10)$$

Proof. Varying the full action and inserting Eq. 5.2.5:

$$\delta \mathcal{S} = \frac{1}{16\pi G_N^{(d)}} \int_{\mathcal{M}} d^d x \sqrt{g} [G_{\mu\nu} + \Lambda g_{\mu\nu}] \delta g^{\mu\nu} + \int_{\mathcal{M}} d^d x \sqrt{g} \left[-\frac{1}{2} T_{\mu\nu} \delta g^{\mu\nu} + \frac{\delta \mathcal{L}_{\text{mat}}}{\delta \Phi} \delta \Phi \right] = 0$$

where matter fields are assumed to vanish asymptotically. $\square$

These are the full Einstein field equations, describing gravity coupled to matter. It is possible to rewrite them by observing that the cosmological constant can be absorbed in the EMT as an additive component:

$$(T_\Lambda)_{\mu\nu} = -\frac{\Lambda}{8\pi G} g_{\mu\nu} \quad (5.2.11)$$

This is justified by the fact that matter fields often mimic a cosmological constant. Taking the trace then gives $\frac{2-d}{2} \mathcal{R} = 8\pi G_N^{(d)} T$, where $T \equiv g^{\mu\nu} T_{\mu\nu}$, hence:

$$\mathcal{R}_{\mu\nu} = 8\pi G_N^{(d)} \left(T_{\mu\nu} - \frac{1}{d-2} T g_{\mu\nu} \right) \quad (5.2.12)$$

with cosmological constant absorbed in the EMT.

¹The Lorentz algebra $\mathfrak{so}^+(1, d-1)$ is composed of the generators of $d-1$ boosts (hyperbolic rotations) and $\frac{1}{2}(d-1)(d-2)$ spatial rotations, hence it has $\frac{1}{2}d(d-1)$ generators $\mathcal{J}^{\mu\nu} = -\mathcal{J}^{\nu\mu}$ satisfying:

$$[\mathcal{J}^{\mu\nu}, \mathcal{J}^{\rho\sigma}] = i(\eta^{\mu\sigma} \mathcal{J}^{\nu\rho} - \eta^{\mu\rho} \mathcal{J}^{\nu\sigma} - \eta^{\nu\sigma} \mathcal{J}^{\mu\rho} + \eta^{\nu\rho} \mathcal{J}^{\mu\sigma}) \quad (5.2.8)$$

§5.2.1.2 Symmetries

The combined action Eq. 5.2.9 must be diffeomorphism-invariant, thus, given $Z \in \mathfrak{X}(\mathcal{M}) : \delta g = \mathcal{L}_Z g$ and following the same argument which lead to the Bianchi identity Eq. 5.1.15:

$$\delta \mathcal{S} = \int_{\mathcal{M}} d^d x \sqrt{g} \left[\frac{1}{2} \left(\frac{1}{8\pi G_N^{(d)}} G_{\mu\nu} - T^{\mu\nu} \right) \nabla^{(\mu} Z^{\nu)} + \frac{\delta \mathcal{L}_{\text{mat}}}{\delta \Phi} \delta_Z \Phi \right] = 0$$

Using the Bianchi identity Eq. 5.1.15 to rewrite the first term:

$$G_{\mu\nu} \nabla^{(\mu} Z^{\nu)} = G_{\mu\nu} \nabla^\nu Z^\mu = \nabla^\nu (G_{\mu\nu} Z^\mu) \equiv \nabla^\nu q_\nu$$

So, by Gauss's theorem Th. 2.5.4 applied to $q \in \mathfrak{X}(\mathcal{M})$:

$$\int_{\mathcal{M}} d^d x \sqrt{g} \nabla^\nu q_\nu = \int_{\partial \mathcal{M}} d^{d-1} x \sqrt{\gamma} n_\nu q^\nu \equiv 0$$

which can be made vanishing by a suitable choice of boundary conditions. Then, using the same argument for the EMT:

$$T_{\mu\nu} \nabla^{(\mu} Z^{\nu)} = \nabla^\nu (T_{\mu\nu} Z^\mu) - Z^\mu \nabla^\nu T_{\mu\nu}$$

The first term produces a neglectable boundary term, as before, so the remaining variation of the action is:

$$\delta \mathcal{S} = \int_{\mathcal{M}} d^d x \sqrt{g} \left[Z^\mu \nabla^\nu T_{\mu\nu} + \frac{\delta \mathcal{L}_{\text{mat}}}{\delta \Phi} \delta_Z \Phi \right] = 0$$

For on-shell matter fields the second term vanishes by the Euler–Lagrange equations, hence the EMT is on-shell covariantly conserved:

$$\nabla^\nu T_{\mu\nu} = 0 \quad (5.2.13)$$

§5.2.1.3 Examples of EMTs

Scalar field theory It is straightforward to compute $T_{\mu\nu}$ for a scalar field $\phi(x)$. Recall Eq. 5.2.2 (with $\xi = 0$) and Lemma 5.1.1:

$$\delta \mathcal{S}_{\text{scalar}} = \int d^4 x \sqrt{g} \left[\frac{1}{4} g_{\mu\nu} \nabla^\rho \phi \nabla_\rho \phi + \frac{1}{2} g_{\mu\nu} V(\phi) - \frac{1}{2} \nabla_\mu \phi \nabla_\nu \phi \right] \delta g^{\mu\nu}$$

This gives the energy-momentum tensor:

$$T_{\mu\nu} = \nabla_\mu \phi \nabla_\nu \phi - g_{\mu\nu} \left(\frac{1}{2} \nabla^\rho \phi \nabla_\rho \phi + V(\phi) \right) \quad (5.2.14)$$

Restricting to flat Minkowski spacetime:

$$T_{00} = \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} (\nabla \phi)^2 + V(\phi)$$

which is the energy density of a scalar field.

Maxwell theory Varying Maxwell action Eq. B.1.3:

$$\delta S_{\text{Maxwell}} = -\frac{1}{4} \int d^4x \sqrt{g} \left[-\frac{1}{2} g_{\mu\nu} F^{\rho\sigma} F_{\rho\sigma} + 2g^{\rho\sigma} F_{\mu\rho} F_{\nu\sigma} \right] \delta g^{\mu\nu}$$

So the energy-momentum tensor for Maxwell theory is:

$$T_{\mu\nu} = g^{\rho\sigma} F_{\mu\rho} F_{\nu\sigma} - \frac{1}{4} g_{\mu\nu} F^{\rho\sigma} F_{\rho\sigma} \quad (5.2.15)$$

In flat Minkowski spacetime:

$$T_{00} = \frac{1}{2} \mathbf{E}^2 + \frac{1}{2} \mathbf{B}^2$$

which is the energy density of the electromagnetic field.

Perfect fluids A perfect fluid is described by its energy density $\rho(\mathbf{x}, t)$, pressure $p(\mathbf{x}, t)$ and velocity 4-vector field $u^\mu(\mathbf{x}, t) : u^\mu u_\mu = -1$. Pressure and energy density are related by an *equation of state* $p = p(\rho)$: cosmological fluids have a simple equation of state of the form $p = w\rho$, e.g. $w = \frac{1}{3}$ for radiation and $w = 0$ for dust. The energy-momentum tensor of a perfect fluid is:

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu + pg^{\mu\nu} \quad (5.2.16)$$

A fluid at rest ($u^\mu = \delta^{\mu,0}$) in flat Minkowski spacetime has $T^{\mu\nu} = \text{diag}(\rho, p, p, p)$, thus T_{00} is yet again the energy density, as expected. Generally, $\rho = T_{\mu\nu} u^\mu u^\nu$ is the energy density measured by an observer co-moving with the fluid.

Bianchi identity $\nabla_\mu T^{\mu\nu} = 0$ determines two constraints. The first is:

$$u^\mu \nabla_\mu \rho + (\rho + p) \nabla_\mu u^\mu = 0 \quad (5.2.17)$$

which is a generalization of mass conservation (where mass is identified with ρ). The first term calculates how fast ρ changes along u^μ , while the second expresses it depending on the rate of flow out of the region $\nabla_\mu u^\mu$. The second constraint is:

$$(\rho + p)u^\mu \nabla_\mu u^\nu = -(g^{\mu\nu} + u^\mu u^\nu) \nabla_\mu p \quad (5.2.18)$$

which is a generalization of Euler equation, i.e. the fluid equivalent of Newton's second law.

§5.2.2 Energy conservation

Being the EMT a tensor current, its covariant conservation has more subtle implications than the covariant conservation of vector currents.

Observation 5.2.2 (Covariant conservation of vector currents)

Consider a conserved vector current $J \in \mathfrak{X}(\mathcal{M}) : \nabla_\mu J^\mu = 0$. By Gauss's theorem Th. 2.5.4, given a d -dimensional submanifold $\mathcal{V}$:

$$0 = \int_{\mathcal{V}} d^d x \sqrt{g} \nabla_\mu J^\mu = \int_{\partial\mathcal{V}} d^{d-1} x \sqrt{\gamma} n_\mu J^\mu$$

Now, assume that the boundary has a tubular shape like $\partial\mathcal{V} = \Sigma_1 \cup \Sigma_2 \cup S$, where $\Sigma_{1,2}$ are past and future spacelike hypersurfaces and S is a (lateral) timelike hypersurface. If no current flows in/out of $\mathcal{V}$, i.e. if $n_\mu J^\mu|_S = 0$, then the above equation reduces to charge conservation

$Q_J(\Sigma_1) = Q_J(\Sigma_2)$, where the charge associated to J on a spacelike hypersurface Σ is defined as:

$$Q_J(\Sigma) := \int_{\Sigma} d^{d-1}x \sqrt{\gamma} n_{\mu} J^{\mu} \quad (5.2.19)$$

Therefore, for a vector current, covariant conservation is equivalent to charge conservation.

The same argument as in Obs. 5.2.2 does not apply to the energy-momentum tensor: the problem arises from generalizing $\nabla_{\mu} J^{\mu} = \frac{1}{\sqrt{g}} \partial_{\mu} (\sqrt{g} J^{\mu})$ to higher-order tensor fields, which is necessary in order to have a divergence theorem like Eq. 2.5.19. Indeed, for a $(0, 2)$ -tensor field:

$$\nabla_{\mu} T^{\mu\nu} = \partial_{\mu} T^{\mu\nu} + \Gamma_{\mu\rho}^{\mu} T^{\rho\nu} + \Gamma_{\mu\rho}^{\nu} T^{\mu\rho} = \frac{1}{\sqrt{g}} \partial_{\mu} (\sqrt{g} T^{\mu\nu}) + \Gamma_{\mu\rho}^{\nu} T^{\mu\rho}$$

The last term does not allow to convert the integral of $\nabla_{\mu} T^{\mu\nu} = 0$ to a boundary term. Instead:

$$\partial_{\mu} (\sqrt{g} T^{\mu\nu}) = -\sqrt{g} \Gamma_{\mu\rho}^{\nu} T^{\mu\rho} \quad (5.2.20)$$

Thus, for higher-order tensors, covariant conservation is not equivalent to charge conservation. For the EMT, however, it is possible to define a conserved vector current associated to a Killing vector field $\xi \in \mathfrak{X}(\mathcal{M})$:

$$J_T^{\nu}[\xi] := \xi_{\mu} T^{\mu\nu} \quad (5.2.21)$$

This current is covariantly conserved, as:

$$\nabla_{\nu} J_T^{\nu}[\xi] = T^{\mu\nu} \nabla_{\nu} \xi_{\mu} + K_{\mu} \nabla_{\nu} \xi^{\mu\nu} = T^{\mu\nu} \nabla_{(\nu} \xi_{\mu)} = 0$$

Its associated conserved charge on a spacelike hypersurface Σ is defined as:

$$Q_T[\xi; \Sigma] := \int_{\Sigma} d^{d-1}x \sqrt{\gamma} n_{\mu} J_T^{\mu} \quad (5.2.22)$$

The interpretation of this conserved charge depends on the nature of the associated Killing vector field, e.g. if ξ is globally timelike, then the charge can be interpreted as the energy of matter.

The problem of energy conservation becomes subtle when dealing with spacetimes which do not admit any globally timelike Killing vector field. For example, a system comprised of two orbiting stars is modelled by a spacetime which does not have such a Killing vector field: however, the problem of energy conservation does not arise in this case, as the stars, while orbiting each other, emit gravitational waves, thus losing energy and eventually spiraling towards each other. Nonetheless, a meaningful question is that of energy conservation of the total system, i.e. the two stars and the gravitational field. A guess would be to consider a total energy-momentum tensor defined similarly to Eq. 5.2.5, but:

$$T_{\mu\nu}^{\text{total}} = -\frac{2}{\sqrt{g}} \left[\frac{1}{16\pi G} \frac{\delta \mathcal{S}_{\text{EH}}}{\delta g^{\mu\nu}} + \frac{\delta \mathcal{S}_{\text{m}}}{\delta g^{\mu\nu}} \right] = -\frac{1}{8\pi G} G_{\mu\nu} + T_{\mu\nu} = 0$$

by the Einstein field equations. This equation has no physical significance other than expressing the subtlety of energy conservation in General Relativity.

Clearly, one could try to understand the energy carried by the gravitational field alone. Unfortunately, there are compelling arguments that there exists no tensor which can be thought as the local energy density of the gravitational field: roughly speaking, the energy density of the Newtonian gravitational field Φ is proportional to $(\nabla \Phi)^2$, so the relativistic equivalent should be proportional to the first derivatives of the metric, which can be made locally vanishing by normal coordinates due to the equivalence principle, and a tensor which vanishes in one coordinate frame does so in all of them.

§5.2.3 Energy conditions

To study the general properties of a spacetime without explicitly referencing the specific characteristics of matter, one needs to place certain restrictions on the kinds of EMTs which are considered physical: these are the *energy conditions* and express the idea that energy should be positive.

Weak energy condition The weak energy condition states that, for any timelike vector field $X \in \mathfrak{X}(\mathcal{M})$:

$$T_{\mu\nu}X^\mu X^\nu \geq 0$$

This quantity is the energy measured by an observer moving along the timelike integral curves of X , and it should be non-negative. A timelike curve can be arbitrarily close to a null curve, thus by continuity:

$$T_{\mu\nu}X^\mu X^\nu \geq 0 \quad \forall X \in \mathfrak{X}(\mathcal{M}) : X_\mu X^\mu \leq 0 \quad (5.2.23)$$

Strong energy condition The strong energy condition states that:

$$\mathcal{R}_{\mu\nu}X^\mu X^\nu \geq 0 \quad \forall X \in \mathfrak{X}(\mathcal{M}) : X_\mu X^\mu \leq 0 \quad (5.2.24)$$

This condition ensures that timelike and null geodesics eventually converge, which means that gravity is attractive. Using Eq. 5.2.12, this condition can be rewritten as:

$$\left(T_{\mu\nu} - \frac{1}{d-2} T g_{\mu\nu} \right) X^\mu X^\nu \geq 0$$

Null energy condition The null energy condition states that:

$$T_{\mu\nu}X^\mu X^\nu \geq 0 \quad \forall X \in \mathfrak{X}(\mathcal{M}) : X_\mu X^\mu = 0 \quad (5.2.25)$$

This condition is implied by both the weak and strong energy condition, but the converse is not true: it is weaker than both conditions.

Dominant energy condition Given a future-directed timelike vector field $X \in \mathfrak{X}(\mathcal{M})$, it is possible to define the energy density current measured by an observer moving along integral curves of X as:

$$J^\mu \equiv -T^{\mu\nu} X_\nu$$

The dominant energy condition requires that, in addition to the WEC Eq. 5.2.23:

$$J_\mu J^\mu \leq 0 \quad (5.2.26)$$

This means that the energy density current is either timelike or null, so requiring that energy does not flow faster than light.

Example 5.2.1 (Fluids)

Recall the energy-momentum tensor for a fluid Eq. 5.2.16 and impose the weak energy condition (WLOG $\langle X, X \rangle_g = -1$):

$$(\rho + p)(u \cdot X)^2 - p \geq 0$$

In the rest-frame $u^\mu = (1, 0, 0, 0)$, so considering a constant $X^\mu = (\cosh \varphi, \sinh \varphi, 0, 0)$, whose integral curves are world-lines of observers boosted with rapidity φ with respect to the fluid, then:

$$(\rho + p) \cosh^2 \varphi - p \geq 0 \quad \Rightarrow \quad \begin{cases} \rho \geq 0 & \varphi = 0 \\ p \geq -\rho & \varphi \rightarrow \infty \end{cases}$$

The first condition ensures that the energy density is positive, the second allows for negative pressure limited from below. Note that there are situations in which negative energy density makes physical sense: viewing the cosmological constant as part of the energy-momentum density, then any $\Lambda < 0$ violates the weak energy condition. In this sense, AdS space violates the weak energy condition.

On the other hand, the strong energy condition for the energy-momentum tensor of a fluid Eq. 5.2.16 in the rest-frame reads:

$$(\rho + p)(u \cdot X)^2 - p + \frac{1}{2}(3p - \rho) \geq 0 \quad \Rightarrow \quad (\rho + p) \cosh^2 \varphi - p + \frac{1}{2}(3p - \rho) \geq 0$$

whose solution is:

$$\begin{cases} p \geq -\rho/3 & \varphi = 0 \\ p \geq -\rho & \varphi \rightarrow \infty \end{cases}$$

It is possible to show that $\Lambda > 0$ violates the strong energy condition: dS space is incompatible with it, as timelike geodesics are pulled apart by the expansion of space.

Furthermore, it can be shown that the null energy condition implies $\rho + p \geq 0$, while the dominant energy condition imposes $\rho^2 \geq p^2$.

Example 5.2.2 (Scalar field theories)

The weak energy condition for the energy-momentum tensor of a scalar field theory Eq. 5.2.14 reads:

$$(X^\mu \partial_\mu \phi)^2 + \frac{1}{2} \partial_\mu \phi \partial^\mu \phi + V(\phi) \geq 0$$

The sum of the first two terms is always positive: define $Y_\mu = \partial_\mu \phi + X_\mu X^\nu \partial_\nu \phi : X_\mu Y^\mu = 0$, i.e. orthogonal to X_μ , so it must be spacelike or null, hence $Y_\mu Y^\mu \geq 0$. Rewriting the above condition:

$$\frac{1}{2}(X^\mu \partial_\mu \phi)^2 + \frac{1}{2} Y_\mu Y^\mu + V(\phi) \geq 0 \quad \Rightarrow \quad V(\phi) \geq 0$$

The weak energy condition is clearly violated by any classical field theory with $V(\phi) \leq 0$. On the contrary, any classical scalar field theory with $V(\phi) \geq 0$ violates the strong energy condition, while the null is satisfied by any classical scalar field theory.

§5.3 Schwarzschild solution

The Schwarzschild solution is the unique static and spherically-symmetric vacuum solution to the Einstein field equations Eq. 5.1.6: it describes spacetime outside a spherically-symmetric body, and it is the most important exact solution of the field equations.

§5.3.1 Birkhoff theorem

To prove the uniqueness of the Schwarzschild solution, first a definition of “static” spacetime is required. Given that symmetries are defined through Killing vector fields (§5.1.1.1), it is straightforward to formulate precise definitions of “stationary” and “static” using them.

Definition 5.3.1 (Stationary spacetime)

A Lorentzian manifold $(\mathcal{M}, g)$ is **stationary** if there exists a globally-timelike Killing vector field $K \in \mathfrak{X}(\mathcal{M})$.

This means that an observer moving along the flow of K , which is timelike, perceives no changes in spacetime, since the flow of K is an isometry group of $\mathcal{M}$. It may happen that K is timelike only in some open region of $\mathcal{M}$: then, only this part of spacetime is said stationary.

Proposition 5.3.1 (Stationary metric)

If $(\mathcal{M}, g)$ is a stationary Lorentzian manifold, then $g_{\mu\nu,0} = 0 \ \forall \mu, \nu = 0, 1, 2, 3$.

Proof. Consider a spacelike hypersurface $\Sigma \subseteq \mathcal{M}$ and be $\phi : \mathbb{R} \times \mathcal{M} \rightarrow \mathcal{M}$ the flow of K on $\mathcal{M}$. Then, define the following local coordinates on $\mathcal{M}$: if $p = \phi_t(p_0)$, with $p_0 \in \Sigma$ and $t \in \mathbb{R}$, then the coordinates of p are $(t, x^1(p_0), x^2(p_0), x^3(p_0))$, where x^μ are local coordinates on Σ . In terms of these coordinates, clearly $K = \partial_0$, hence using Eq. 1.6.19:

$$(\mathcal{L}_K g)_{\mu\nu} = g_{\mu\nu,\rho} K^\rho + g_{\rho\nu} K^\rho_{,\mu} + g_{\mu\rho} K^\rho_{,\nu} = g_{\mu\nu,0} + 0 + 0 = g_{\mu\nu,0}$$

Therefore, Killing equation $\mathcal{L}_K g = 0$ implies the thesis. $\square$

These coordinates are called *adapted coordinates* to the Killing field. To deduce a definition of static spacetime, which is a special case of stationary spacetime, consider adapted coordinates such that $g_{0i} = 0 \ \forall i = 1, 2, 3$, so that the Killing field is orthogonal to all spatial sections $\{t = \text{const.}\}$. Then, the 1-form $\omega = K^\flat$ associated to K is:

$$\omega = g_{\mu\nu} K^\mu dx^\nu = g_{00} dt = g(K, K) dt$$

since $g_{00} = g(\partial_0, \partial_0) = g(K, K)$. This trivially implies the **Frobenius condition** for ω :

$$\omega \wedge d\omega = 0 \tag{5.3.1}$$

Conversely, assume that the Frobenius condition holds for the Killing field K of a stationary Lorentzian manifold $(\mathcal{M}, g)$. Then, applying ι_K to Eq. 5.3.1 and using Th. 2.2.2 yields:

$$0 = \iota_K(\omega \wedge d\omega) = (\iota_K \omega) \wedge d\omega - \omega \wedge (\iota_K d\omega) = g(K, K) d\omega - \omega \wedge (\mathcal{L}_K \omega - dg(K, K))$$

To show that $\mathcal{L}_K \omega = 0$, expand both $\mathcal{L}_K \omega$ and $\mathcal{L}_K g$:

$$(\mathcal{L}_K \omega)(X) = K(\omega(X)) - \omega([K, X]) = K(g(K, X)) - g(K, [K, X])$$

$$0 = (\mathcal{L}_K g)(K, X) = K(g(K, X)) - g([K, K], X) - g([K, X], K) = K(g(K, X)) - g(K, [K, X])$$

Hence, setting $V \equiv g(K, K) \neq 0$, the above equation becomes:

$$0 = V d\omega + \omega \wedge dV = d(V^{-1} \omega)$$

Assuming $\mathcal{M}$ to be simply connected, by the Poincaré Lemma $\omega = Vdf$, with $f \in \mathcal{C}^\infty(\mathcal{M})$. Using this function as a time coordinate, then, the expression $\omega = g(K, K)dt$ is recovered, which shows that K is orthogonal to all spacelike sections $\{t = \text{const.}\}$, since taken $X \in \mathfrak{X}(\mathcal{M})$ tangential to one of these sections $g(K, X) = \omega(X) = Vdt(X) = VX(dt) = 0$. Switching to adapted coordinates, $K = \partial_t$, so $g_{0i} = g(\partial_t, \partial_i) = g(K, \partial_i) = 0$. Therefore, this reasoning shows that, if the globally-timelike Killing vector field K satisfies the Froebenius condition, then the metric locally splits as:

$$ds^2 = g_{00}(\mathbf{x})dt^2 + g_{ij}(\mathbf{x})dx^i dx^j \quad (5.3.2)$$

This justifies the following definition.

Definition 5.3.2 (Static spacetime)

Given a stationary Lorentzian manifold $(\mathcal{M}, g)$ with globally-timelike Killing vector field K , then it is said to be **static** if, setting $\omega \equiv K^\flat$, then $\omega \wedge d\omega = 0$ and locally $\omega = g(K, K)dt$ for an adapted time coordinate t , which is unique up to an additive constant.

In a static spacetime, the flow of K isometrically maps spacelike sections $\{t = \text{const.}\}$ onto each other, and an observer at rest moves along integral curves of K . In the following, K is assumed to be the only globally-timelike Killing vector field, so that there is a *distinguished time*.

Theorem 5.3.1 (Birkhoff's theorem)

A spherically-symmetric vacuum solution to the Einstein field equations must be static, and it is uniquely given by the **Schwarzschild solution**:

$$ds^2 = -\left(1 - \frac{2m}{r}\right) dt^2 + \left(1 - \frac{2m}{r}\right)^{-1} dr^2 + r^2 [d\vartheta^2 + \sin^2 \vartheta d\varphi^2] \quad (5.3.3)$$

on the manifold $\mathcal{M} = \mathbb{R} \times \mathbb{R}_{>2m}^+ \times \mathbb{S}^2$, with $m \in \mathbb{R}$.

Proof. First of all, spherical symmetry requires that the Lorentzian manifold considered has $\text{SO}(3)$ as an isometry group, with the conditions that the group orbits^a are 2-dimensional spacelike hypersurfaces. Thus, the manifold splits into $\mathcal{M} = \Sigma \times \mathbb{S}^2$, where Σ is a 2-dimensional Lorentzian submanifold. Since the metric on a 2-dimensional pseudo-Riemannian manifold can be always diagonalized^b, the most general spherically-symmetric metric takes the form:

$$ds^2 = -e^{2a(t,r)} dt^2 + e^{2b(t,r)} dr^2 + r^2 [d\vartheta^2 + \sin^2 \vartheta d\varphi^2]$$

The ϑ, φ are the standard coordinates on $\mathbb{S}^2$, $t \in \mathbb{R}$ and the radial coordinate r is determined by the foliation of the spatial part of the manifold by invariant 2-spheres: in particular, each 2-sphere is characterized by a surface area A , and r is defined by $A = 4\pi r^2$.

With notation $f' \equiv \frac{df}{dr}$ and $\dot{f} \equiv \frac{df}{dt}$, the only non-vanishing components of the Einstein tensor are computed to be:

$$\begin{aligned} G^0_0 &= -\frac{1}{r^2} + e^{-2b} \left(\frac{1}{r^2} - \frac{2b'}{r} \right) & G^1_1 &= -\frac{1}{r^2} + e^{-2b} \left(\frac{1}{r^2} + \frac{2a'}{r} \right) & G^1_0 &= \frac{2\dot{b}}{r} e^{-a-b} \\ G^2_2 &= G^3_3 = e^{-2b} \left(a'^2 - a'b' + a'' + \frac{a' - b'}{r} \right) & & & & + e^{-2a} \left(-\dot{b}^2 + \dot{a}\dot{b} - \ddot{b} \right) \end{aligned}$$

Now, the vacuum field equations read $G^\mu_\nu = 0$. From $G^1_0 = 0$ it is clear that $b = b(r)$; then, $G_{11} + G_{00} = 0$ implies the partial differential equation $a' + b' = 0$, which has the general solution

$a(t, r) = -b(r) + f(t)$. To determine $b(r)$, consider $G^0_0 = 0$:

$$-\frac{1}{r^2} + e^{-2b} \left(\frac{1}{r^2} - \frac{2b'}{r} \right) = 0 \quad \Rightarrow \quad (re^{-2b})' = 1 \quad \Rightarrow \quad e^{-2b} = 1 - \frac{2m}{r}$$

where $m \in \mathbb{R}$ is an integration constant. Then, the other vacuum field equations are satisfied, and the metric takes the form:

$$ds^2 = -e^{2f(t)} \left(1 - \frac{2m}{r} \right) dt^2 + \left(1 - \frac{2m}{r} \right)^{-1} + r^2 [d\vartheta^2 + \sin^2 \vartheta d\varphi^2]$$

Introducing a new time coordinate $\tilde{t} = \int dt e^{f(t)}$, this metric reduces to the Schwarzschild solution, which is manifestly static with the timelike Killing vector field $K = \partial_t$. $\square$

^aGiven a group G which acts on a set X , the orbit of $x \in X$ is defined as $G(x) := \{gx \in X : g \in G\} \subseteq X$.

^bThis is achieved with the so-called *isothermal coordinates*.

The converse of Birkhoff's theorem is also true, and it is called Israel's theorem. The physical meaning of the parameter $m \in \mathbb{R}$ is clarified by studying the Newtonian limit for g_{00} at large distances:

$$-\left(1 - \frac{2m}{r} \right) \approx -\left(1 + \frac{2\Phi(r)}{c^2} \right)$$

Since $\Phi(r) = -GM/r$, where M is the total mass generating the gravitational field, then:

$$m = \frac{GM}{c^2} \quad (5.3.4)$$

Hence, m can be seen as the (normalized) mass of the spherically-symmetric object which generates the Schwarzschild field: this restricts the physical significance of m to $m \in \mathbb{R}^+$, as $m = 0$ corresponds to flat Minkowski space. As a consequence, the Schwarzschild metric describes the spacetime outside any non-rotating spherically-symmetric object, even in presence of time dependence in the internal distribution of matter (e.g. a collapsing star).

Note that the solution is singular at $r = 2m \equiv R_s$: indeed, Birkhoff's theorem only applies in the $r > R_s$ region, while the “interior” region, as well as the nature of the singularity at $r = R_s$ and $r = 0$, remains to be studied (in §7.4.1). It is important to notice that r is not the “distance from the center”: indeed, given the foliated structure of the Schwarzschild spacetime, dr is not the distance between two 2-spheres with r and $r+dr$ (defined by their surface area), which is instead $dl = dr/\sqrt{1 - 2m/r}$. Therefore, it is clear that the $\{t = \text{const.}\}$ sections of spacetime are non-Euclidean.

§5.3.2 Motion in a Schwarzschild field

The Lagrangian for a test body in a Schwarzschild field, parametrized with proper time, is:

$$2L = -\left(1 - \frac{2m}{r} \right) \dot{t}^2 + \left(1 - \frac{2m}{r} \right)^{-1} \dot{r}^2 + r^2 [\dot{\vartheta}^2 + \sin^2 \vartheta \dot{\varphi}^2]$$

Note that, along the worldline $x^\mu(\tau)$, which is a geodesic on $\mathcal{M}$, the Lagrangian takes the value $2L = -1$, as $ds^2 = -d\tau^2$. The equation of motion for ϑ reads:

$$\frac{d}{d\tau}(r^2 \dot{\vartheta}) = r^2 \sin \vartheta \cos \vartheta \dot{\varphi}^2$$

Therefore, if the motion is initially in the equatorial plane, i.e. $\vartheta = \frac{\pi}{2}$ and $\dot{\vartheta} = 0$, it will remain so, hence WLOG $\vartheta \equiv \frac{\pi}{2}$ and the Lagrangian reduces to:

$$2L = - \left(1 - \frac{2m}{r}\right) \dot{t}^2 + \left(1 - \frac{2m}{r}\right)^{-1} \dot{r}^2 + r^2 \dot{\varphi}^2 \quad (5.3.5)$$

The variables t and φ are cyclic, and they correspond to two conserved quantities:

$$\frac{\partial L}{\partial \dot{\varphi}} = r^2 \dot{\varphi} \equiv \ell \quad - \frac{\partial L}{\partial \dot{t}} = \left(1 - \frac{2m}{r}\right) \dot{t} \equiv E \quad (5.3.6)$$

which are identified respectively with the angular momentum and the energy of the test body. Inserting these definitions into $2L = -1$, the equation of motion becomes:

$$\dot{r}^2 + V_{\text{eff}}(r) = E^2 \quad (5.3.7)$$

with the effective potential:

$$V_{\text{eff}}(r) = \left(1 - \frac{2m}{r}\right) \left(1 + \frac{\ell^2}{r^2}\right) \quad (5.3.8)$$

This potential always has asymptotics $V_{\text{eff}}(r) \rightarrow -\infty$ for $r \rightarrow 0$ and $V_{\text{eff}}(r) \rightarrow 1$ for $r \rightarrow +\infty$, but the intermediate behaviour depends on the ratio ℓ/m : in particular, for $\ell/m < 2\sqrt{3}$ the potential has no critical points, hence any incoming particle falls towards the singularity at $r = R_s$.

The main physically-interesting quantity is the orbit $r = r(\varphi)$, which can be determined from this equation of motion.

Proposition 5.3.2 (Quasi-Keplerian motion)

Defining $u(\varphi) \equiv \frac{1}{r(\varphi)}$ and $u' \equiv \frac{du}{d\varphi}$, then the orbital equation becomes:

$$u'' + u = \frac{m}{\ell^2} + 3mu^2 \quad (5.3.9)$$

Proof. First, note that by the chain rule:

$$\dot{r} = r' \dot{\varphi} = r' \frac{\ell}{r^2} \quad \Rightarrow \quad r'^2 \frac{\ell^2}{r^4} = E^2 - V_{\text{eff}}(r)$$

Then, introducing $r' = -u'/u^2$:

$$\ell^2 u'^2 = E^2 - (1 - 2mu) (1 + \ell^2 u^2)$$

Differentiating with respect to φ results in:

$$2u'u'' + 2uu' = \frac{2m}{\ell^2} u' + 6mu^2 u'$$

Hence, either $u' = 0$, which is the equation for circular motion $r = \text{const.}$, or it solves the ODE Eq. 5.3.9. $\square$

In the Newtonian case $L = \frac{1}{2} \dot{\mathbf{x}}^2 + \frac{GM}{r^2}$, which results in the Keplerian orbital equation:

$$u'' + u = \frac{GM}{\ell^2} \quad (5.3.10)$$

Therefore, the motion in a Schwarzschild field is quasi-Keplerian, since the orbital equation contains an additional perturbative term $3mu^2$. For our planet system, this perturbation is relatively small, e.g. for Mercury:

$$\frac{3mu^2}{m/\ell^2} = 3u^2\ell^2 = \frac{3}{r^2} (r^2\dot{\varphi})^2 \simeq \frac{3}{c^2} \left(r \frac{d\varphi}{dt} \right)^2 \simeq \frac{3v_{\perp}^2}{c^2} \simeq 7.7 \cdot 10^{-8}$$

where $v_{\perp}$ is the velocity component perpendicular to the radius vector.

§5.3.2.1 Freely-falling emitter

Consider a point-like source, emitting light in a constant and isotropic way in its comoving reference frame, which is radially falling from infinity. The emitted light is detected by an observer in a fixed position, along the same radial direction, at $r = R \gg R_s$.

Since light propagates on null curves with $ds^2 = 0$, from Eq. 5.3.3:

$$dt = \pm \frac{r}{r - R_s} dr \quad \Rightarrow \quad t = r + R_s \ln(r - R_s) + \text{const.}$$

where the positive solution has been chosen and the $\mathbb{S}^2$ component has been ignored, since the motion is radial (i.e. with fixed ϑ and φ). Now, fix an emission event (t_e, r_e) , which is detected at (t_r, R) . Clearly, by the above relation:

$$t_r - R - R_s \ln(R - R_s) = t_e - r_e - R_s \ln(r_e - R_s) \quad (5.3.11)$$

It is possible to make the relation between t_e and r_e explicit. Recall Eq. 5.3.8 and set $\ell = 0$ (radial motion), so that:

$$\dot{r}^2 = E^2 - 1 + \frac{R_s}{r} \equiv \frac{R_s}{r} - \frac{R_s}{r_0} \quad r_0 \equiv \frac{R_s}{1 - E^2}$$

where r_0 is the initial radial coordinate of the geodesic. Note that this is the same equation which can be derived from energy conservation in Newtonian mechanics, with $\Phi(r) = -GMm_s/r$. Since the source is falling from infinity, $E = 1$ (which justifies the choice of the positive solution for $t = t(r)$), hence the geodesic is described by:

$$r = R_s s^2 \quad \tau = -\frac{2}{3} R_s s^3 \quad (5.3.12)$$

where $s \in \mathbb{R}_{>1}^+$ is an auxiliary variable and the choice of the sign of τ is dictated by the fact that the source is falling, so s decreases along the trajectory while τ must always increase. Then, by Eq. 5.3.6:

$$\frac{r - R_s}{r} \frac{dt}{d\tau} = 1 \quad \Rightarrow \quad t = -\frac{2}{3} R_s s^3 - 2R_s s + R_s \ln \frac{s+1}{s-1}$$

Inserting this into Eq. 5.3.11 in place of (t_e, r_e) :

$$t_r = -\frac{2}{3} R_s s^3 - R_s s^2 - 2R_s s - 2R_s \ln(s-1) + R + R_s \ln \left(\frac{R}{R_s} - 1 \right)$$

In the limit $s \rightarrow 1$, i.e. $r \rightarrow R_s$, this equation becomes:

$$t_r \rightarrow -2R_s \ln(s-1) + \text{const.} \quad \Rightarrow \quad s \rightarrow 1 + ae^{-\frac{t_r}{2R_s}}$$

with $a \equiv \exp(\text{const.}/2R_s)$. Differentiating this equation and Eq. 5.3.12, then:

$$\frac{dt_r}{d\tau} \rightarrow a^{-1} e^{\frac{t_r}{2R_s}} \quad (5.3.13)$$

Note that τ is the proper time of the source, while t_r can be identified with the proper time of the observer, since $R \gg R_s$. This means that the rate of reception of photons by the observer is exponentially damped, resulting in an exponential decrease of the observed intensity of the source as it approaches $r = R_s$. Additionally, the light emitted by the source is exponentially redshifted², near the horizon: combining these effects, the source eventually becomes undetectable, but the radiation intensity never drops to zero, although the source crosses the horizon in a finite proper time (as per Eq. 5.3.12).

§5.3.3 Classical tests of General Relativity

§5.3.3.1 Gravitational redshift

Consider a stationary clock, i.e. a clock on a spacetime section $\{(r, \vartheta, \varphi) = \text{const.}\}$. Then, the time measured by it is:

$$d\tau = dt \sqrt{1 - \frac{2m}{r}} \quad (5.3.14)$$

Then, if two stationary clocks at $r_1 = r$ and $r_2 = r + \Delta r$ exchange light signals, the non-Euclidean nature of spacetime determines a gravitational redshift:

$$\frac{\nu_2}{\nu_1} = \sqrt{\frac{1 - R_s/r_1}{1 - R_s/r_2}} \simeq 1 - \frac{R_s \Delta r}{2r^2} \quad (5.3.15)$$

where the approximation holds for $\Delta r \ll r$ and $R_s \ll r$.

§5.3.3.2 Periastron precession

The perihelion shift of Mercury was, at the start of the 20th century, one of the main unsolved problem of celestial mechanics: even after the perturbing effects of other planets on Mercury's orbit had been accounted for, there remained in the data an unexplained advance in its perihelion. This precession is explained by General Relativity.

To study in general the precession of the periastron of a generic planet, Eq. 5.3.9 is treated perturbatively. In the Newtonian limit, the Keplerian orbit is an ellipse:

$$u(\varphi) = \frac{1}{p} (1 + e \cos \varphi) \quad p = a (1 + e^2) = \frac{\ell^2}{m} \quad (5.3.16)$$

where e is the eccentricity of the orbit and a its greater semiaxis. Then, at first order of approximation, this solution is inserted in the perturbative term, yielding:

$$u'' + u = \frac{m}{\ell^2} + \frac{3m^3}{\ell^4} (1 + e \cos \varphi)^2$$

²Requires proof.

Lemma 5.3.1

For a constant $A \in \mathbb{R}$, the following ODEs have particular solutions:

$$v'' + v = \begin{cases} A \\ A \cos \varphi \\ A \cos^2 \varphi \end{cases} \quad \Rightarrow \quad v(\varphi) = \begin{cases} A \\ \frac{1}{2}A\varphi \sin \varphi \\ \frac{1}{2}A - \frac{1}{6}A \cos 2\varphi \end{cases} \quad (5.3.17)$$

Using Lemma 5.3.1, it is trivial to see that the perturbative term leads to the following relativistic correction to the Keplerian elliptic orbit:

$$u(\varphi) = \frac{m}{\ell^2} (1 + e \cos \varphi) + \frac{3m^3}{\ell^4} \left[1 + \frac{e^2}{2} - \frac{e^2}{6} \cos 2\varphi + e\varphi \sin \varphi \right] \quad (5.3.18)$$

with an initial periastron at $\varphi = 0$. The important term, in this perturbation, is the *secular term* proportional to $\varphi \sin \varphi$, which is responsible (at first-order approximation) for the periastron precession.

Proposition 5.3.3 (Periastron precession)

If $\varphi = 0$ is a periastron, then the next periastron is at $\varphi = 2\pi + \delta\varphi$, with:

$$\delta\varphi \simeq 3\pi \frac{R_s}{a(1 - e^2)} \quad (5.3.19)$$

Proof. The periastron is determined by the condition $u' = 0$, i.e.:

$$\sin \varphi = \frac{3m^3}{\ell^2} \left[\frac{e}{3} \sin 2\varphi + \sin \varphi + \varphi \cos \varphi \right]$$

Introducing the periastron anomaly $\varphi = 2\pi + \delta\varphi$, this becomes:

$$\sin \delta\varphi = \frac{3m^3}{\ell^2} \left[\frac{e}{3} \sin 2\delta\varphi + \sin \delta\varphi + \delta\varphi \cos \delta\varphi + 2\pi \cos \delta\varphi \right]$$

Now, assume that $e \ll 1$ and $|\delta\varphi| \ll 1$: then, $|\sin \delta\varphi| \ll 1$ and $\cos \delta\varphi \simeq 1$, so that the only term which determines the anomaly is, approximately, the last one:

$$\sin \delta\varphi \simeq 6\pi \frac{m^3}{\ell^2} \cos \delta\varphi$$

Then, using $\tan \delta\varphi \simeq \delta\varphi$, $2m \equiv R_s$ and Eq. 5.3.16, the thesis is obtained. $\square$

Although an approximation valid for $e \ll 1$ and $|\delta\varphi| \ll 1$, this expression is incredibly accurate for predictions in the solar system. Defining the rate of periastron shift:

$$\dot{\omega} \equiv \frac{\delta\varphi}{T} \quad T = 2\pi \sqrt{\frac{a^3}{GM}} \quad (5.3.20)$$

where the orbital period is computed from Kepler's third law, the prediction for Mercury is $\dot{\omega} \simeq 42.98''$ per century, which is accurate at 0.1% error with respect to the observed data.

§5.3.3.3 Deflection of light

Recalling Eq. 5.3.5, the motion of lightrays is determined by $L = 0$, since for null curves $ds^2 = 0$:

$$\left(1 - \frac{2m}{r}\right)^{-1} E^2 - \left(1 - \frac{2m}{r}\right)^{-1} \dot{r}^2 - \frac{\ell^2}{r^2} = 0 \quad (5.3.21)$$

This equation results in a simpler orbital equation.

Proposition 5.3.4 (Motion of lightrays)

$$u'' + u = 3mu^2 \quad (5.3.22)$$

Proof. Eq. 5.3.21 can be rewritten, with $\dot{r} = -u'\ell$, as:

$$E^2 - (1 - 2mu)\ell^2 u^2 = \ell^2 u'^2 \implies u'^2 + u^2 = \frac{E^2}{\ell^2} + 2mu^3$$

Differentiating yields the thesis. $\square$

In the solar system, the perturbative term is relatively small, since:

$$\frac{3mu^2}{u} = \frac{3R_s}{2r} \leq \frac{R_s}{R_\odot} \simeq 4.24 \cdot 10^{-6}$$

The unperturbed solution, found neglecting the $3mu^2$ term, is a straight light-path:

$$u(\varphi) = \frac{1}{b} \sin \varphi \quad (5.3.23)$$

where b is the impact parameter, i.e. the distance between the asymptotic initial direction of the lightray and the center of the mass distribution which determines the curvature of spacetime. At first order in perturbation theory, then:

$$u'' + u = \frac{3m}{b^2} (1 - \cos^2 \varphi)$$

Solving this with unperturbed solution + particular solution yields the first-order solution:

$$u(\varphi) = \frac{1}{b} \sin \varphi + \frac{3m}{2b^2} \left(1 + \frac{1}{3} \cos 2\varphi\right) \quad (5.3.24)$$

For large r (small u), $|\varphi|$ is either small or close to π . For small φ , take $\sin \varphi \simeq \varphi$ and $\cos \varphi \simeq 1$, and define the asymptotic initial direction $\varphi_\infty \equiv \lim_{u \rightarrow 0} \varphi$, so that by Eq. 5.3.24:

$$\varphi_\infty = -\frac{2m}{b} \quad (5.3.25)$$

For an observer at large distance, then, the **total deflection** is $\delta = 2|\varphi_\infty|$, that is:

$$\delta = \frac{2R_s}{b} \quad (5.3.26)$$

For the Sun, this results in $\delta \simeq 1.75'' R_s/b$, which as of today is accurate within 0.02% from radioastronomical measurements of quasars: the experimental confirmation of this deviation was the first historical test of General Relativity.

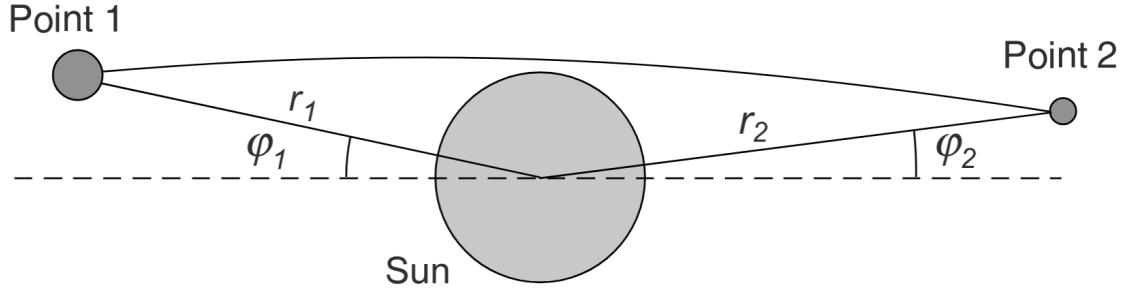


Figure 5.5: Path of a radar signal between two points in Schwarzschild spacetime.

§5.3.3.4 Gravitational time delay

Suppose that a signal is transmitted from a point (r_1, φ_1) to another point (r_2, φ_2) on the equatorial plane $\vartheta = \frac{\pi}{2}$. It is possible to compute the coordinate time required for this transmission.

Proposition 5.3.5 (Elapsed coordinate time)

Let $r_0 \in \mathbb{R}_{\geq R_s}^+$ be the distance of closest approach of a lightray. Then, the coordinate time elapsed during a transmission between r_0 and $r \geq r_0$ is:

$$\Delta t(r, r_0) = \int_{r_0}^r dr \left(1 - \frac{R_s}{r}\right)^{-1} \left[1 - \frac{1 - R_s/r}{1 - R_s/r_0} \left(\frac{r_0}{r}\right)^2\right]^{-\frac{1}{2}} \quad (5.3.27)$$

Assuming that $r, r_0 \gg R_s$, then:

$$\Delta t(r, r_0) \simeq \sqrt{r^2 - r_0^2} + R_s \ln \left[\frac{r + \sqrt{r^2 - r_0^2}}{r_0} \right] + \frac{R_s}{2} \sqrt{\frac{r - r_0}{r + r_0}} \quad (5.3.28)$$

Proof. First, express $\dot{r}$ in terms of E :

$$\dot{r} = \frac{dr}{dt} \dot{t} = \frac{dr}{dt} \frac{E}{1 - R_s/r}$$

Then, recalling Eq. 5.3.21:

$$\left(1 - \frac{R_s}{r}\right)^{-3} \left(\frac{dr}{dt}\right)^2 = \left(1 - \frac{R_s}{r}\right)^{-1} - \frac{\ell^2}{E^2} \frac{1}{r^2}$$

If r_0 is the distance of closest approach, then $\frac{dr}{dt} \Big|_{r=r_0} = 0$, so that:

$$\frac{\ell^2}{E^2} = \frac{r_0^2}{1 - R_s/r_0}$$

Substituting this into the above equation and integrating yields the thesis. Then, if $r, r_0 \gg R_s$:

$$\Delta t(r, r_0) \simeq \int_{r_0}^r dr \left(1 + \frac{R_s}{r}\right) \left[1 - \left(\frac{R_s}{r_0} - \frac{R_s}{r}\right) \left(\frac{r_0}{r}\right)^2\right]^{-\frac{1}{2}}$$

This expression can be rearranged as:

$$\Delta t(r, r_0) \simeq \int_{r_0}^r dr \left(1 - \frac{r_0^2}{r^2}\right)^{-\frac{1}{2}} \left[1 + \frac{R_s}{r} + \frac{R_s r_0}{2r(r + r_0)}\right]$$

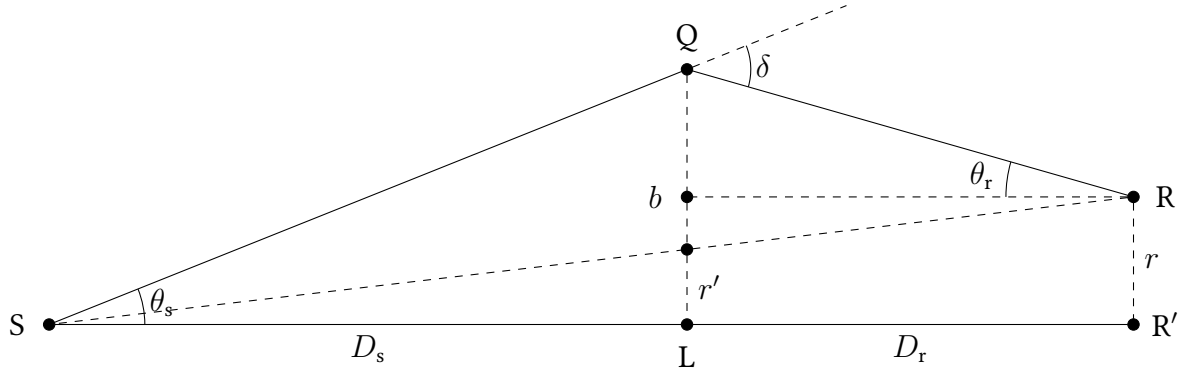


Figure 5.6: Schematics of the lens geometry.

Integrating gives the thesis. □

Note that the first term in the approximate expression is the non-relativistic term, while the other terms give the correction determined by the non-Euclidean nature of the Schwarzschild spacetime. Now, if $|\varphi_1 - \varphi_2| > \frac{\pi}{2}$ as in Fig. 5.5, the total elapsed coordinate time is $\Delta t_{1,2} = \Delta t(r_1, r_0) + \Delta t(r_2, r_0)$, hence the time delay due to non-Euclideanity is given by the **Shapiro delay in coordinate time**:

$$\begin{aligned} \Delta t &:= 2 \left[\Delta t(r_1, r_0) + \Delta t(r_2, r_0) - \sqrt{r_1^2 - r_0^2} - \sqrt{r_2^2 - r_0^2} \right] \\ &\simeq 2R_s \ln \left[\frac{(r_1 + \sqrt{r_1^2 - r_0^2})(r_2 + \sqrt{r_2^2 - r_0^2})}{r_0^2} \right] + R_s \left[\sqrt{\frac{r_1 - r_0}{r_1 + r_0}} + \sqrt{\frac{r_2 - r_0}{r_2 + r_0}} \right] \end{aligned} \quad (5.3.29)$$

Although the coordinate time is not observable, since clocks measure their proper time, the Shapiro delay in coordinate time still gives a general idea of the magnitude of the general relativistic time delay.

Example 5.3.1 (Earth–Mars time delay)

Consider a radar transmission from Earth to Mars and then back to Earth. The Shapiro delay in coordinate time is (with $r_0 \ll r_1, r_2$):

$$\Delta t \simeq 2R_s \left[\ln \frac{4r_1 r_2}{r_0^2} + 1 \right] \lesssim 240 \mu s$$

§5.3.3.5 Gravitational microlensing

A direct consequence of the deflection of light due to a mass density is the lensing effect of the latter on the light emitted from objects further along the line of sight.

It is important to distinguish between two kinds of gravitational lensing: the cosmological lensing, where the source is a distant quasar and the lens is galaxy, and the gravitational microlensing, where the source is a distant star and the lens is a concentrated and relatively small mass density. The former has a much more complex theory, due to the extended nature of the lens, hence only microlensing is considered here: the main experimental application of microlensing is the search for massive galactic halo objects (MACHOs, ~ 10 kpc), which act as lenses for stars in the Magellanic clouds (~ 50 kpc) and whose main candidates are brown dwarfs, inactive white dwarfs, neutron stars and black holes.

Optical schematics The optical scheme of a gravitational lens is composed of three points: the source S, the lens L and the observer R. Then, set R' the projection of R on the line SL and define the distances $D_s \equiv \overline{SL}$, $D_r \equiv \overline{LR'}$ and $r \equiv \overline{RR'}$, as in Fig. Fig. 5.6.

At first approximation, it is possible to identify the lightray emitted from the source and deflected by the lens with its two asymptotes, which are lines passing through a common point Q at a distance b from the lens and forming an angle δ given by Eq. 5.3.26, with $\overline{QL} \perp \overline{SL}$: these assumptions only induce errors of order $o(b^{-2})$, where $b, r \ll D_s, D_r$ is supposed.

Proposition 5.3.6 (Lensed image)

The image produced by the lens has dimension:

$$b_{\pm} = \frac{1}{2} \left[r' \pm \sqrt{r'^2 + 4b_E^2} \right] \quad (5.3.30)$$

with:

$$b_E \equiv \sqrt{2R_s \frac{D_s D_r}{D_s + D_r}} \quad r' \equiv \frac{D_s}{D_s + D_r} r \quad (5.3.31)$$

Proof. Geometrically, the following relations are clear (with $b \ll R_s$):

$$\theta_s + \theta_r = \delta \quad D_s \theta_s = b \quad D_r \theta_r = b - r$$

Substituting Eq. 5.3.26 and combining these equations, then:

$$\frac{b}{D_s} + \frac{b-r}{D_r} = \frac{2R_s}{b} \quad \Rightarrow \quad b^2 - \frac{D_s r}{D_s + D_r} b = 2R_s \frac{D_s D_r}{D_s + D_r}$$

Solving this equation yields the thesis. $\square$

Note that the two solutions correspond to the lightray passing either “above” (like in Fig. 5.6) or “below” the lens: this does not mean that the lens forms two images, since, for an image to form, it is necessary that a small beam of light passes through the same point after being lensed, but this does not happen in gravitational lenses, which are highly astigmatic.

While r' has a clear geometrical meaning, b_E corresponds to the case of a perfect alignment, i.e. $r = 0$. In this case, the lightrays which get lensed are not only the two already analyzed, which are confined on the plane determined by S, L and R, but are infinitely many, and they lie on a cone with axis $\overline{SR}$ and semiangle θ_r . In this case, the lens does not produce two images, but a whole circumference of radius b_E : an **Einstein ring**.

Note that these Einstein rings are nearly impossible to observe for relatively close sources: with $D_s = D_r = 1$ kpc and $R_s = 3$ km, the angular semiopening of the ring is $\theta_r = b_E/D_r \simeq 0.002''$. On the other hand, these rings are fairly common on cosmological scales, e.g. with a galaxy as lens, but on these scales it is not possible to apply a Euclidean analysis.

Microlensing In the conditions of gravitational microlensing, observing lensed images or Einstein rings is impossible, as already seen. However, an observable effect is the magnification of the source. Assuming that the source emits isotropically a luminosity L , the power irradiated on an area $d\sigma$ in the absence of any lens is:

$$P_0 = \frac{L}{4\pi} \frac{d\sigma}{(D_s + D_r)^2}$$

On the other hand, if a lens is present, the power irradiated on $d\sigma$ is given by the luminosity emitted in the solid angle $d\Omega = \sin \theta_s d\theta_s d\varphi$, while $d\sigma = r dr d\varphi$ by axial symmetry, so that:

$$P = L \frac{d\Omega}{4\pi}$$

The magnification A is the ratio between P and P_0 , that is:

$$A = \frac{P}{P_0} = (D_s + D_r)^2 \frac{d\Omega}{d\sigma} \simeq (D_s + D_r)^2 \frac{\theta_s d\theta_s}{r dr} = \left(\frac{D_s + D_r}{D_s} \right)^2 \frac{b db}{r dr} = \frac{b db}{r' dr'}$$

Recalling Eq. 5.3.30, one must add the two contributions from each image, thus finding:

$$A = \frac{b_+^2}{r' |2b_+ - r'|} + \frac{b_-^2}{r' |2b_- - r'|}$$

where the absolute value is necessary in order to keep A positive. A trivial manipulation yields the final expression:

$$A = \frac{r'^2 + 2b_E^2}{2r' \sqrt{r'^2 + 4b_E^2}} \quad (5.3.32)$$

This expression seems to have a pole for $r' \rightarrow 0$: this is easily avoided by considering the source as an extended object, rather than a point.

Searching for these kinds of magnifications in the light of distant stars allows to study MACHOs: to distinguish the resulting luminous variability from variable stars, it is important to note that the microlensing effect is exactly symmetrical in time and non-dispersive, i.e. equal for all wavelengths.

Gravitational waves

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Although the Einstein field equations, being second-order non-linear inhomogeneous PDEs, are highly non-trivial to solve, they admit wave-like solutions, which produce what are known as *gravitational waves* (GWs). To study GWs, three approaches are possible: perturbative solutions, exact solutions and asymptotic solutions of radiating systems.

§6.1 Perturbations of spacetime

The basic perturbative approach for the study of GWs is to consider a background metric $\bar{g}$ and introduce a perturbation h describing the GWs, so to obtain a spacetime metric like:

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu} \quad (6.1.1)$$

A key property of GWs is the time-domain splitting between the background metric and the perturbations: indeed, observable GWs have frequencies ~ 1 Hz, so the variations of h have timescale of ~ 1 sec, while those of $\bar{g}$ have cosmological timescale $\sim H_0^{-1}$, i.e. in the order of billions of years. Note that this time-domain splitting is not gauge invariant, since it depends on the explicit coordinates used.

§6.1.1 Weak gravity

To generalize the perturbative expansion in Eq. 6.1.1, consider a 1-parameter family of metrics $q(\lambda)$ analytic in $\lambda \in \mathbb{R}$:

$$q(0) = \bar{g} \quad \left. \frac{dq}{d\lambda} \right|_{\lambda=0} = h$$

Then, as $\lambda \rightarrow 0$ the metric reduces to $q(\lambda) = \bar{g} + \lambda h + o(\lambda^2)$, which is just the above expression. Now, consider the two Levi-Civita connections associated to $q(\lambda)$ and $\bar{g}$ respectively:

$$\bar{\nabla}_\alpha \bar{g}_{\mu\nu} = 0 \quad \tilde{\nabla}_\alpha q_{\mu\nu} = 0 \quad (6.1.2)$$

Lemma 6.1.1 (Difference tensor)

Let $\mathcal{M}$ be a metric manifold, and let $\tilde{\nabla}, \bar{\nabla}$ be two Koszul connections in $\mathcal{M}$. Then, given $\omega \in$

$\Omega^1(\mathcal{M})$:

$$\tilde{\nabla}_\mu \omega_\alpha - \bar{\nabla}_\mu \omega_\alpha = -C_{\mu\alpha}^\rho \omega_\rho \quad (6.1.3)$$

with $C \in \mathfrak{T}_2^1(\mathcal{M})$. If both families of connections are torsion-free, $C_{\mu\alpha}^\rho = C_{\alpha\mu}^\rho$.

Proof. To show that $C \in \mathfrak{T}_2^1(\mathcal{M})$, recall that any connection acts on $T_0^0 \mathcal{M} = \mathcal{C}^\infty(\mathcal{M})$ as $\nabla_\mu f = \partial_\mu f$, hence all connections in $\mathcal{M}$ agree on $T_0^0 \mathcal{M}$. Now, let $f \in \mathcal{C}^\infty(\mathcal{M})$, $\omega \in \Omega^1(\mathcal{M})$:

$$\tilde{\nabla}_\mu (f\omega_\alpha) - \bar{\nabla}_\mu (f\omega_\alpha) = (\tilde{\nabla}_\mu f)\omega_\alpha + f(\tilde{\nabla}_\mu \omega_\alpha) - (\bar{\nabla}_\mu f)\omega_\alpha - f(\bar{\nabla}_\mu \omega_\alpha) = f(\tilde{\nabla}_\mu \omega_\alpha - \bar{\nabla}_\mu \omega_\alpha)$$

Now, consider $\eta \in \Omega^1(\mathcal{M}) : \eta|_p = \omega|_p$: it can be shown that there exist $\{f_a\} \subseteq \mathcal{C}^\infty(\mathcal{M}) : f_a(p) = 0$ and $\{\zeta^a\} \subseteq \Omega^1(\mathcal{M})$ such that:

$$\eta - \omega = \sum_a f_a \zeta^a$$

Then:

$$\tilde{\nabla}_\mu (\eta_\alpha - \omega_\alpha) - \bar{\nabla}_\mu (\eta_\alpha - \omega_\alpha) = \sum_a \left[\tilde{\nabla}_\mu (f_a \zeta_\alpha^a) - \bar{\nabla}_\mu (f_a \zeta_\alpha^a) \right] = \sum_a f_a \left[\tilde{\nabla}_\mu \zeta_\alpha^a - \bar{\nabla}_\mu \zeta_\alpha^a \right]$$

which vanishes at p since $f_a(p) = 0$. Therefore:

$$\left[\tilde{\nabla}_\mu \eta_\alpha - \bar{\nabla}_\mu \eta_\alpha \right]_p = \left[\tilde{\nabla}_\mu \omega_\alpha - \bar{\nabla}_\mu \omega_\alpha \right]_p$$

which shows that the difference $\tilde{\nabla}_\mu \omega_\alpha - \bar{\nabla}_\mu \omega_\alpha$ only depends on the value of $\omega_\alpha|_p$. This means that $\tilde{\nabla}_\mu - \bar{\nabla}_\mu$ defines a linear map $T_p^* \mathcal{M} \rightarrow T_2^0(T_p \mathcal{M})$, and so a $(1, 2)$ -tensor field C with components Eq. 6.1.3.

Now, consider $\omega_\alpha = \nabla_\alpha f$ with $f \in \mathcal{C}^\infty(\mathcal{M})$:

$$\tilde{\nabla}_\mu \tilde{\nabla}_\nu f - \bar{\nabla}_\mu \bar{\nabla}_\nu f = -C_{\mu\nu}^\rho f$$

However, for a torsion-free connection ∇ :

$$\nabla_\mu \nabla_\nu f = \nabla_\mu (\partial_\nu f) = \partial_\mu \partial_\nu f - \Gamma_{\mu\nu}^\rho \partial_\rho f = \partial_\nu \partial_\mu f - \Gamma_{\nu\mu}^\rho \partial_\rho f = \nabla_\nu \nabla_\mu f$$

Therefore, if both $\tilde{\nabla}$ and $\bar{\nabla}$ are torsion-free, $C_{\mu\nu}^\rho = C_{\nu\mu}^\rho$. □

The difference tensor can be used to characterize the curvature tensors of the perturbed metric.

Proposition 6.1.1 (Difference tensor of perturbed metric)

Given the two Levi-Civita connections defined in Eq. 6.1.2, the difference tensor defined by ($\omega \in \Omega^1(\mathcal{M})$):

$$\tilde{\nabla}_\mu \omega_\alpha - \bar{\nabla}_\mu \omega_\alpha = -C_{\mu\alpha}^\rho \omega_\rho$$

is expressed as:

$$C_{\mu\nu}^\rho(\lambda) = \frac{1}{2} q^{\rho\alpha} (\bar{\nabla}_\mu q_{\nu\alpha} - \bar{\nabla}_\nu q_{\mu\alpha} - \bar{\nabla}_\alpha q_{\mu\nu}) \quad (6.1.4)$$

Proof. By the metric-compatibility of the Levi-Civita connection:

$$0 = \tilde{\nabla}_\alpha q_{\mu\nu} = \bar{\nabla}_\alpha q_{\mu\nu} - C_{\alpha\mu}^\rho q_{\rho\nu} - C_{\alpha\nu}^\rho q_{\mu\rho}$$

Exchanging $\alpha \leftrightarrow \mu$ and then $\mu \leftrightarrow \nu$, a set of three equations is found:

$$\bar{\nabla}_\alpha q_{\mu\nu} - C_{\alpha\mu}^\rho q_{\rho\nu} - C_{\alpha\nu}^\rho q_{\mu\rho} = 0$$

$$\bar{\nabla}_\mu q_{\alpha\nu} - C_{\mu\alpha}^\rho q_{\rho\nu} - C_{\mu\nu}^\rho q_{\alpha\rho} = 0$$

$$\bar{\nabla}_\nu q_{\alpha\mu} - C_{\nu\alpha}^\rho q_{\rho\mu} - C_{\nu\mu}^\rho q_{\alpha\rho} = 0$$

Summing the last two equations and subtracting the first one:

$$\bar{\nabla}_\mu q_{\alpha\nu} + \bar{\nabla}_\nu q_{\alpha\mu} - \bar{\nabla}_\alpha q_{\mu\nu} - 2C_{\mu\nu}^\rho q_{\alpha\rho} = 0$$

which yields the thesis. $\square$

The Riemann curvature tensor can now be computed directly from definition (Eq. 2.5.4).

Lemma 6.1.2 (Difference of Riemann tensors)

Given the two Levi–Civita connections defined in Eq. 6.1.2, the difference of the associated Riemann tensors is:

$$\tilde{\mathcal{R}}_{\mu\nu\alpha}{}^\rho - \bar{\mathcal{R}}_{\mu\nu\alpha}{}^\rho = 2\bar{\nabla}_{[\nu} C_{\mu]}^\rho{}_\alpha + C_{\nu\beta}^\rho C_{\mu\alpha}^\beta - C_{\nu\alpha}^\rho C_{\mu\beta}^\rho \quad (6.1.5)$$

Proof. By Eq. 2.5.4:

$$\begin{aligned} \tilde{\mathcal{R}}_{\mu\nu\alpha}{}^\rho \omega_\rho &= \tilde{\nabla}_\mu \tilde{\nabla}_\nu \omega_\alpha - \tilde{\nabla}_\nu \tilde{\nabla}_\mu \omega_\alpha \\ &= \tilde{\nabla}_\mu (\bar{\nabla}_\nu \omega_\alpha - C_{\nu\alpha}^\rho \omega_\rho) - \tilde{\nabla}_\nu (\bar{\nabla}_\mu \omega_\alpha - C_{\mu\alpha}^\rho \omega_\rho) \\ &= \bar{\nabla}_\mu \bar{\nabla}_\nu \omega_\alpha - C_{\mu\nu}^\rho \bar{\nabla}_\rho \omega_\alpha - C_{\mu\alpha}^\rho \bar{\nabla}_\nu \omega_\rho - \tilde{\nabla}_\mu (C_{\nu\alpha}^\rho \omega_\rho) \\ &\quad - \bar{\nabla}_\nu \bar{\nabla}_\mu \omega_\alpha + C_{\nu\mu}^\rho \bar{\nabla}_\rho \omega_\alpha + C_{\nu\alpha}^\rho \bar{\nabla}_\mu \omega_\rho + \tilde{\nabla}_\nu (C_{\mu\alpha}^\rho \omega_\rho) \\ &= \bar{\mathcal{R}}_{\mu\nu\alpha}{}^\rho \omega_\rho - C_{\mu\alpha}^\rho \bar{\nabla}_\nu \omega_\rho + C_{\nu\alpha}^\rho \bar{\nabla}_\mu \omega_\rho - \tilde{\nabla}_\mu (C_{\nu\alpha}^\rho \omega_\rho) + \tilde{\nabla}_\nu (C_{\mu\alpha}^\rho \omega_\rho) \end{aligned}$$

Now, note that:

$$\begin{aligned} \tilde{\nabla}_\mu (C_{\nu\alpha}^\rho \omega_\rho) &= \nabla_\mu (C_{\nu\alpha}^\rho) \omega_\rho + C_{\nu\alpha}^\rho \tilde{\nabla}_\mu \omega_\rho \\ &= \bar{\nabla}_\mu C_{\nu\alpha}^\rho \omega_\rho + C_{\mu\beta}^\rho C_{\nu\alpha}^\beta \omega_\rho - C_{\mu\nu}^\beta C_{\beta\alpha}^\rho \omega_\rho - C_{\mu\alpha}^\beta C_{\nu\beta}^\rho \omega_\rho + C_{\nu\alpha}^\rho \bar{\nabla}_\mu \omega_\rho - C_{\nu\alpha}^\rho C_{\mu\rho}^\beta \omega_\beta \end{aligned}$$

Thus:

$$\begin{aligned} \tilde{\mathcal{R}}_{\mu\nu\alpha}{}^\rho \omega_\rho &= \bar{\mathcal{R}}_{\mu\nu\alpha}{}^\rho \omega_\rho - C_{\mu\alpha}^\rho \bar{\nabla}_\nu \omega_\rho + C_{\nu\alpha}^\rho \bar{\nabla}_\mu \omega_\rho - \tilde{\nabla}_\mu (C_{\nu\alpha}^\rho \omega_\rho) + \tilde{\nabla}_\nu (C_{\mu\alpha}^\rho \omega_\rho) \\ &\quad - \bar{\nabla}_\mu C_{\nu\alpha}^\rho \omega_\rho - C_{\mu\beta}^\rho C_{\nu\alpha}^\beta \omega_\rho + C_{\mu\nu}^\beta C_{\beta\alpha}^\rho \omega_\rho + C_{\mu\alpha}^\beta C_{\nu\beta}^\rho \omega_\rho - C_{\nu\alpha}^\rho \bar{\nabla}_\mu \omega_\rho + C_{\nu\alpha}^\rho C_{\mu\rho}^\beta \omega_\beta \\ &\quad + \bar{\nabla}_\nu C_{\mu\alpha}^\rho \omega_\rho + C_{\nu\beta}^\rho C_{\mu\alpha}^\beta \omega_\rho - C_{\nu\mu}^\beta C_{\beta\alpha}^\rho \omega_\rho - C_{\nu\alpha}^\beta C_{\mu\beta}^\rho \omega_\rho + C_{\mu\alpha}^\rho \bar{\nabla}_\nu \omega_\rho - C_{\mu\alpha}^\rho C_{\nu\rho}^\beta \omega_\beta \\ &= \bar{\mathcal{R}}_{\mu\nu\alpha}{}^\rho \omega_\rho + 2\bar{\nabla}_{[\nu} C_{\mu]}^\rho{}_\alpha \omega_\rho - 2C_{\nu\alpha}^\beta C_{\mu\beta}^\rho \omega_\rho + 2C_{\nu\beta}^\rho C_{\mu\alpha}^\beta \omega_\rho + C_{\nu\alpha}^\rho C_{\mu\rho}^\beta \omega_\beta - C_{\mu\alpha}^\rho C_{\nu\rho}^\beta \omega_\beta \end{aligned}$$

Renaming dummy indices yields the thesis. $\square$

Eq. 6.1.4 and Eq. 6.1.5 are valid at all orders of the analytic parameter $\lambda \in \mathbb{R}$, hence they are well suited for a perturbative expansion. At NLO, the inverse metric tensor reads:

$$q^{\mu\nu} = \bar{g}^{\mu\nu} - \lambda \bar{g}^{\mu\alpha} \bar{g}^{\nu\beta} h_{\alpha\beta} + o(\lambda^2) \quad (6.1.6)$$

hence indices of quantities proportional to h are raised/lowered by $\bar{g}$ only: $h^{\mu\nu} = \bar{g}^{\mu\alpha} \bar{g}^{\nu\beta} h_{\alpha\beta} + o(\lambda^2)$.

Then, the difference tensor and the Riemann tensor at linear order become:

$$C_{\mu\nu}^\rho(\lambda) = \frac{1}{2}\bar{g}^{\rho\alpha}(\bar{\nabla}_\mu h_{\nu\alpha} + \bar{\nabla}_\nu h_{\mu\alpha} - \bar{\nabla}_\alpha h_{\mu\nu})\lambda + o(\lambda^2) \quad (6.1.7)$$

$$\tilde{\mathcal{R}}_{\mu\nu\alpha}{}^\rho = \bar{\mathcal{R}}_{\mu\nu\alpha}{}^\rho + 2\bar{\nabla}_{[\nu}[C^{[1]}]_{\mu]\alpha}^\rho\lambda + o(\lambda^2) \quad (6.1.8)$$

Therefore, $[\tilde{\mathcal{R}}^{[1]}]_{\mu\nu\alpha}{}^\rho = 2\bar{\nabla}_{[\nu}[C^{[1]}]_{\mu]\alpha}^\rho$. With this formalism, it is possible to analytically expand the Einstein field equations in powers of $\lambda \in \mathbb{R}$:

$$\bar{G}_{\mu\nu} + [G^{[1]}]_{\mu\nu}\lambda + o(\lambda^2) = 8\pi G_N^{(d)} (\bar{T}_{\mu\nu} + [T^{[1]}]_{\mu\nu}\lambda + o(\lambda^2))$$

There is an order-by-order split of the field equations into background and NⁿLO equations:

$$\bar{G}_{\mu\nu} = 8\pi G_N^{(d)} \bar{T}_{\mu\nu} \quad [G^{[n]}]_{\mu\nu} = 8\pi G_N^{(d)} [T^{[n]}]_{\mu\nu} \quad (6.1.9)$$

Observation 6.1.1 (Gauge invariance of analytic expansion)

Under an infinitesimal diffeomorphism generated by $Z \in \mathfrak{X}(\mathcal{M}) : x^\mu \mapsto x^\mu + Z^\mu$, the metric changes as $q_{\mu\nu} \mapsto q_{\mu\nu} + \delta q_{\mu\nu}$ with:

$$\delta q = \mathcal{L}_Z q = \sum_{n \in \mathbb{N}_0} \mathcal{L}_Z q^{[n]} \lambda^n = \mathcal{L}_Z \bar{g} + \lambda \mathcal{L}_Z h + o(\lambda^2)$$

However, the perturbative expansion requires $\lambda \rightarrow 0$, hence the second term is a second order one (as both Z and λ are infinitesimal) and the variation δq only has beyond-leading-order terms. Therefore, under an infinitesimal diffeomorphism, the perturbed metric at first-order transforms as:

$$\bar{g} \mapsto \bar{g} \quad h \mapsto h + \mathcal{L}_Z \bar{g} \quad (6.1.10)$$

i.e. $\delta h_{\mu\nu} = 2\bar{\nabla}_{(\mu} Z_{\nu)}$, with $Z_\mu = \bar{g}_{\mu\nu} Z^\nu$. In the same way, $\delta[\tilde{\mathcal{R}}^{[1]}]_{\mu\nu\alpha}{}^\rho = \mathcal{L}_Z \bar{\mathcal{R}}_{\mu\nu\alpha}{}^\rho$.

The perturbed Einstein field equations Eq. 6.1.9 are most conveniently expressed in terms of a modified perturbation metric:

$$\bar{h}_{\mu\nu} := h_{\mu\nu} - \frac{1}{2}h\bar{g}_{\mu\nu} \quad h \equiv \bar{g}^{\mu\nu} h_{\mu\nu} \quad (6.1.11)$$

The bar notation in $\bar{h}_{\mu\nu}$ has nothing to do with the background metric. Then, setting $\bar{\square} \equiv \bar{g}^{\mu\nu} \bar{\nabla}_\mu \bar{\nabla}_\nu$, the Einstein tensor at NLO can be shown to be:

$$[G^{[1]}]_{\mu\nu} = -\frac{1}{2}\bar{\square}\bar{h}_{\mu\nu} + \bar{h}_{\mu\nu}\bar{\mathcal{R}} + \bar{\nabla}_{(\nu}\bar{\nabla}^\alpha\bar{h}_{\mu)\alpha} + \bar{\mathcal{R}}_{(\mu}{}^\alpha\bar{h}_{\nu)\alpha} + \bar{\mathcal{R}}_{\alpha\nu\mu\beta}\bar{h}^{\alpha\beta} - \frac{1}{2}\bar{g}_{\mu\nu} [\bar{\mathcal{R}}_{\alpha\beta}\bar{h}^{\alpha\beta} + \bar{\nabla}^\alpha\bar{\nabla}^\beta\bar{h}_{\alpha\beta}] \quad (6.1.12)$$

This expression is greatly simplified by imposing a gauge-fixing condition (by Obs. 6.1.1). In particular, a convenient choice is the **Lorentz gauge**:

$$\bar{\nabla}^\alpha \bar{h}_{\alpha\mu} = 0 \quad (6.1.13)$$

In this gauge, the NLO Einstein tensor reduces to:

$$[G^{[1]}]_{\mu\nu} = -\frac{1}{2}\bar{\square}\bar{h}_{\mu\nu} + \bar{h}_{\mu\nu}\bar{\mathcal{R}} + \bar{\mathcal{R}}_{(\mu}{}^\alpha\bar{h}_{\nu)\alpha} + \bar{\mathcal{R}}_{\alpha\nu\mu\beta}\bar{h}^{\alpha\beta} - \frac{1}{2}\bar{g}_{\mu\nu}\bar{\mathcal{R}}_{\alpha\beta}\bar{h}^{\alpha\beta}$$

The time-domain splitting between the background metric and the perturbation in Eq. 6.1.1 can be generally represented by an inequality between the spacetime scale L_b of the background and that λ of the perturbation: indeed, by dimensional analysis, the background curvature scales like $\bar{\mathcal{R}} \sim L_b^{-2}$, while the perturbation like $\partial h \sim \lambda^{-1}$, hence by the split $\lambda \ll L_b$ it is clear that the first perturbation term dominates over the other curvature terms. The NLO field equations thus have the schematic form of a wave equation:

$$\square \bar{h}_{\mu\nu} = -16\pi G_N [T^{[1]}]_{\mu\nu} + (\text{background curvature terms})$$

Observation 6.1.2 (Quantum gravity)

The NLO Einstein tensor can be found by the variation of the linearized Einstein–Hilber action, also known as the **Fierz–Pauli action**, which turns out to be:

$$\mathcal{S}_{\text{FP}}^{[1]} = \frac{1}{8\pi G_N^{(d)}} \int d^4x \left[-\frac{1}{4} \partial_\rho h_{\mu\nu} \partial^\rho h^{\mu\nu} + \frac{1}{2} \partial_\rho h_{\mu\nu} \partial^\nu h^{\rho\mu} + \frac{1}{4} \partial_\mu h \partial^\mu h - \frac{1}{2} \partial_\nu h^{\mu\nu} \partial_\mu h \right] \quad (6.1.14)$$

An effective free theory of the graviton is given by the Fierz–Pauli lagrangian, which can then be generalized to an interacting theory introducing an interaction Lagrangian $\mathcal{L}_{\text{int}}$. Note that $h_{\mu\nu}$ is dimensionless, hence to describe the graviton it is customary to define a canonical field $\hat{h}_{\mu\nu} \equiv (G_N^{(d)})^{-1/2} h_{\mu\nu}$: recall by Obs. 5.1.1 that $[G_N^{(d)}] = d-2$, hence $[\hat{h}] = \frac{2-d}{2}$ and correctly $[\mathcal{L}_{\text{FP}}] = -d$. Since the Fierz–Pauli action must be at least of order $(\partial h)^2$, the lowest-order interacting term has schematic structure:

$$\mathcal{L}_{\text{int}} \sim \frac{1}{G_N^{(d)}} (\partial h)^2 h \sim \sqrt{G_N^{(d)}} (\partial \hat{h})^2 \hat{h} \sim \frac{1}{M_p^2} (\partial \hat{h}) \hat{h}$$

where the Plank mass is define (in natural units) as $M_p^2 \equiv 1/8\pi G_N^{(d)}$. The coupling has mass dimension $\frac{d-2}{2}$, hence this quantum field theory of the graviton is non-renormalizable for $d > 2$. This was to be expected from an effective theory: indeed, the inverse metric $q^{\mu\nu}$ truly has infinite powers of h , hence the radiative corrections given by virtual gravitons are extremely small, and they only become non-negligible when the perturbative expansion diverges (for $E \gtrsim M_p$), with consequent loss of meaning of the effective theory.

§6.1.2 Linear theory of GWs

Consider a $d = 4$ background metric $\bar{g}_{\mu\nu} = \eta_{\mu\nu}$: this is, of course, only an approximation, since the metric perturbation would have a back-propagation effect which curves the background, nonetheless it can be seen as restricting to a spacetime region which is small when compared to the curvature scale of the background. The Lorentz gauge-fixing condition then reduces to:

$$\partial^\mu \bar{h}_{\mu\nu} = 0$$

Under an infinitesimal diffeomorphism generated by $Z \in \mathfrak{X}(\mathcal{M})$:

$$\delta h_{\mu\nu} = (\mathcal{L}_Z \eta)_{\mu\nu} = \partial_\mu Z_\nu + \partial_\nu Z_\mu \quad \delta h = 2\partial^\mu Z_\mu$$

Then, setting $\square = \eta^{\mu\nu} \partial_\mu \partial_\nu$ (the bar notation is suppressed for objects of the flat background):

$$\delta \bar{h}_{\mu\nu} = \partial_\mu Z_\nu + \partial_\nu Z_\mu - \eta_{\mu\nu} \partial^\alpha Z_\alpha \quad \implies \quad \delta(\partial^\mu \bar{h}_{\mu\nu}) = \square Z_\nu$$

The Lorentz gauge thus presents a residual gauge freedom for $Z \in \mathfrak{X}(\mathcal{M})$, such that:

$$\square Z^\mu = 0 \quad (6.1.15)$$

For this reason, the Lorentz gauge is also known as the *harmonic gauge* (or *de Donder gauge*), since it preserves the condition $\square x^\mu = 0$ (as $x^\mu \mapsto x^\mu + Z^\mu$).

In $d \geq 4$ it makes physical sense¹ to consider the propagation of GWs in vacuum $[T^{[1]}]_{\mu\nu} = 0$:

$$\square \bar{h}_{\mu\nu} = 0 \quad (6.1.16)$$

$\bar{h}_{\mu\nu}$ can then be seen as a massless spin-2 field propagating in Minkowski spacetime: by symmetry, it would have 10 independent components, but the Lorentz gauge eliminates 4 degrees of freedom, and the residual gauge eliminates additional 4 of them, leaving only 2 degrees of freedom for the propagating GW.

To explicitly see this, consider the general ansatz for the vacuum wave equation in the homogeneous and isotropic spacetime $\mathbb{R}^{1,3}$:

$$\bar{h}_{\mu\nu}(x) = \int_{\mathbb{R}^{1,3}} \frac{d^4 k}{(2\pi)^4} e^{ik \cdot x} \bar{\varepsilon}_{\mu\nu}(k)$$

for some polarization matrix $\bar{\varepsilon}_{\mu\nu}(k) \in \mathbb{C}^{4 \times 4}$ (the physical perturbation is the real part of this Fourier transform). In order to be a solution, the on-shell condition given by the equations of motion Eq. 6.1.16 must be satisfied:

$$\eta^{\alpha\beta} k_\alpha k_\beta \bar{\varepsilon}_{\mu\nu}(k) = 0 \quad \implies \quad k^2 = 0$$

confirming that $\bar{h}_{\mu\nu}$ is a massless field². Then, the solution to the linearized field equations reads:

$$\bar{h}_{\mu\nu}(x) = \int_{\mathbb{R}^{1,3}} \frac{d^4 k}{(2\pi)^4} \delta^{(4)}(k^2) e^{ik \cdot x} \bar{\varepsilon}_{\mu\nu}(k) \quad (6.1.17)$$

Imposing the Lorentz gauge-fixing condition:

$$\partial^\mu \bar{h}_{\mu\nu} = 0 \quad \implies \quad k^\mu \bar{\varepsilon}_{\mu\nu} = 0$$

which means that the polarization matrix must be transverse to the direction of propagation of the GW. The residual gauge freedom can be exploited writing $Z \in \mathfrak{X}(\mathcal{M})$ as:

$$Z^\mu(x) = \int_{\mathbb{R}^{1,3}} \frac{d^4 k}{(2\pi)^4} \hat{Z}^\mu(k) e^{ik \cdot x}$$

Then, the gauge condition $\square Z^\mu = 0$ simply becomes $k^2 \hat{Z}^\mu = 0$, i.e. $k^2 = 0$. This means that on-shell it is possible to freely choose the component functions Z^μ . First, observe that the infinitesimal gauge transformation $x^\mu \mapsto x^\mu + Z^\mu$ acts as:

$$h_{\mu\nu}(x) \mapsto \bar{h}_{\mu\nu}(x) + 2\partial_{(\mu} Z_{\nu)}(x) = \int_{\mathbb{R}^{1,3}} \frac{d^4 k}{(2\pi)^4} \delta^{(4)}(k^2) [\varepsilon_{\mu\nu} + 2ik_{(\mu} \hat{Z}_{\nu)}] e^{ik \cdot x}$$

¹In $d = 2, 3$ the Riemann tensor has respectively 1 and 6 independent components (recall Eq. 3.3.31), so the vacuum field equations (which are 10 independent equations) completely nullify the Riemann tensor, making it identically vanish.

²For a massless field there is no mass gap, since the energy $k^0 = \pm\sqrt{\mathbf{k}^2} = \pm\omega$ can be made arbitrarily small.

which corresponds to a transformation of the polarization matrix:

$$\varepsilon_{\mu\nu} \mapsto \varepsilon_{\mu\nu} + i(k_\mu \hat{Z}_\nu + k_\nu \hat{Z}_\mu) \quad (6.1.18)$$

The component $\hat{Z}^0$ is chosen so that $h = \eta^{\mu\nu} h_{\mu\nu} = 0$, which means that:

$$\eta^{\mu\nu} \varepsilon_{\mu\nu} + 2ik_\mu \hat{Z}^\mu = 0 \quad \Rightarrow \quad \hat{Z}^0 = \frac{1}{k^0} (\eta^{\mu\nu} \varepsilon_{\mu\nu} + \mathbf{k} \cdot \mathbf{Z})$$

Once this condition is imposed, $\bar{h}_{\mu\nu} = h_{\mu\nu}$ and $\bar{\varepsilon}_{\mu\nu} = \varepsilon_{\mu\nu}$. Then, the remaining components $\hat{Z}^i$ are chosen so that $\bar{h}_{0i} = h_{0i} = 0$, which means that:

$$\varepsilon_{0i} + i(k_0 \hat{Z}_i + k_i \hat{Z}_0) = 0$$

The complete gauge fixing then results into:

$$k^\mu \varepsilon_{\mu\nu} = 0 \quad \eta^{\mu\nu} \varepsilon_{\mu\nu} = 0 \quad \varepsilon_{0i} = 0$$

This are indeed 8 conditions on the polarization matrix, corresponding to the 4 Lorentz gauge-fixing conditions and the 4 residual gauge-fixing conditions. It is trivial to check that these conditions are equivalent to:

$$\varepsilon_{0\mu} = 0 \quad k^i \varepsilon_{ij} = 0 \quad \delta^{ij} \varepsilon_{ij} = 0 \quad (6.1.19)$$

For this reason, this is known as the **transverse-traceless gauge** (TT gauge). In the TT gauge, then:

$$\varepsilon_{\mu\nu}^{\text{TT}} = \begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{0} & \varepsilon_{ij} \end{bmatrix}$$

where ε_{ij} is a transverse-traceless spatial polarization matrix.

Observation 6.1.3 (Transverse and traceless matrices)

In general, any $\mathbb{R}^{d \times d}$ matrix can be made traceless and transverse with respect to some versor $\mathbf{n} \in \mathbb{R}^d$. Indeed, defining the transverse projector:

$$\mathcal{P}_{ij} = \delta_{ij} - n_i n_j \quad (6.1.20)$$

it is clear that $n_i \mathcal{P}_{ij} = n_j - n_j = 0$. On the other hand, any $q_{ij} \in \mathbb{R}^{d \times d}$ can be made traceless by:

$$q_{ij} \mapsto q_{ij} - \frac{1}{d} q_{mn} \delta_{mn} \delta_{ij}$$

Hence, it is possible to define a transverse-traceless projector as:

$$q_{ij}^{\text{TT}} = \Lambda_{imjn} q_{mn} \quad \Lambda_{imjn} = \mathcal{P}_{im} \mathcal{P}_{jn} - \frac{1}{d-1} \mathcal{P}_{ij} \mathcal{P}_{mn} \quad (6.1.21)$$

which trivially satisfies $n_i \Lambda_{imjn} = n_j \Lambda_{imjn} = \delta_{ij} \Lambda_{imjn} = 0$.

By the construction outlined in Obs. 6.1.3, assuming WLOG that the GW propagates along $k^\mu = (\omega, 0, 0, \omega)$, the polarization matrix is found to be:

$$\varepsilon_{\mu\nu}^{\text{TT}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & H_+ & H_\times & 0 \\ 0 & H_\times & -H_+ & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (6.1.22)$$

which shows the two linearly-independent physical polarizations:

$$\varepsilon_{ij}^{\text{TT}} = H_+ \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + H_\times \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

§6.1.3 Geodesic deviation

To understand how GWs can be detected, it is necessary to study how the passage of a GW affects the distance between neighbouring geodesics.

Definition 6.1.1 (Geodesic congruence)

Let $(\mathcal{M}, g)$ be a spacetime. Given a nowhere-vanishing $X \in \mathfrak{X}(\mathcal{M})$, the set of its integral curves $\{\gamma_s(\tau)\}$ is called a **congruence**. If it is possible to define a tangent vector field $u = \frac{\partial \gamma_s}{\partial \tau}$ such that $\nabla_u u = 0$, it is called a **geodesic congruence**.

A congruence is characterized by its nowhere-vanishing vector field $X \in \mathfrak{X}(\mathcal{M})$: if X is timelike/spacelike/null, the congruence is called a timelike/spacelike/null congruence. For example, a timelike geodesic congruence can be interpreted as a family of free-falling test particles (i.e. not affected by mutual interactions), while a null geodesic congruence can be seen as a family of freely propagating light rays.

Consider a timelike geodesic congruence $\{\gamma_s(\tau)\}$, and define the tangent vector field and the deviation vector field:

$$u = \frac{\partial}{\partial \tau} \gamma_s(\tau) \quad \xi = \frac{\partial}{\partial s} \gamma_s(\tau) \quad (6.1.23)$$

In particular, ξ is the vector which, at fixed affine parameter, points from a given geodesic to the next one in the congruence. In general, u and ξ need not be orthogonal, but they can be chosen so. Indeed, the angle between them is constant:

$$\frac{d}{d\tau} \langle \xi, u \rangle_g = \nabla_u \langle \xi, u \rangle_g = \langle \nabla_u \xi, u \rangle_g = u_\alpha u^\beta \nabla_\beta \xi^\alpha$$

Choosing u and ξ such that $[u, \xi] = 0$ (e.g. by choosing them among local coordinates) yields $\xi^\alpha \nabla_\alpha u^\mu - u^\alpha \nabla_\alpha \xi^\mu = 0$, hence:

$$\frac{d}{d\tau} \langle \xi, u \rangle_g = u_\alpha \xi^\beta \nabla_\beta u^\alpha = \frac{1}{2} \nabla_\xi u^2 = 0$$

since for a timelike congruence u can be chosen so that $u^2 = -1$. This allows to fix $\langle u, \xi \rangle_g = 0$ for a particular value of the affine parameter.

The relative deviation of the geodesics is encoded in the variation of ξ . One defines the relative velocity and acceleration of the geodesic deviation along the congruence as:

$$v = \nabla_u \xi \quad a = \nabla_u v$$

Lemma 6.1.3 (Geodesic deviation acceleration)

Let $(\mathcal{M}, g)$ be a spacetime. Given a geodesic congruence, the geodesic deviation acceleration is:

$$a^\mu = \mathcal{R}^\mu_{\rho\alpha\beta} u^\rho u^\alpha \xi^\beta \quad (6.1.24)$$

Proof. By definition:

$$a^\mu = \nabla_u v^\mu = u^\alpha \nabla_\alpha (u^\beta \nabla_\beta \xi^\mu) = u^\alpha \nabla_\alpha (\xi^\beta \nabla_\beta u^\mu) = u^\alpha \xi^\beta \nabla_\alpha \nabla_\beta u^\mu + u^\alpha \nabla_\alpha \xi^\beta \nabla_\beta u^\mu$$

where in the second equality $[\xi, u] = 0$ was used. Inserting the Ricci identity Eq. 2.5.8 for a torsion-free connection and using $[\xi, u] = 0$ again:

$$\begin{aligned} a^\mu &= u^\alpha \xi^\beta (\nabla_\beta \nabla_\alpha u^\mu + \mathcal{R}^\mu_{\rho\alpha\beta} u^\rho) + \xi^\alpha \nabla_\alpha u^\beta \nabla_\beta u^\mu \\ &= u^\alpha \xi^\beta \mathcal{R}^\mu_{\rho\alpha\beta} u^\rho + \xi^\rho (u^\alpha \nabla_\beta \nabla_\alpha u^\mu + \nabla_\beta u^\alpha \nabla_\alpha u^\mu) \\ &= u^\alpha \xi^\beta \mathcal{R}^\mu_{\rho\alpha\beta} u^\rho + \xi^\beta \nabla_\beta (\nabla_u u^\mu) = u^\alpha \xi^\beta \mathcal{R}^\mu_{\rho\alpha\beta} u^\rho \end{aligned}$$

which is the thesis. $\square$

Eq. 6.1.24 shows that in flat spacetime the elements of a congruence have no relative non-uniform motion, and the only relative acceleration is provided by the curvature of spacetime: this makes the geodesic deviation a primary experimental probe of the passage of GWs.

Consider two test masses displaced (at constant time) by a spatial vector ξ . Both test masses are assumed to be in free fall in a flat region of spacetime, so that the background metric is $\bar{g}_{\mu\nu} = \eta_{\mu\nu}$. Then, upon the passage of a GW, the metric is perturbed (in the TT frame) as:

$$ds^2 = -dt^2 + (\delta_{ij} + h_{ij}^{\text{TT}}) dx^i dx^j$$

Since the two test masses move along timelike geodesics, their trajectories are described by the geodesic equation:

$$\frac{d^2 x^i}{d\tau^2} + \Gamma^i_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0 \quad \left. \frac{dx^i}{d\tau} \right|_{\tau=0} = 0$$

where the initial condition ensures that they are initially at rest. Inserting the initial condition in the geodesic equation:

$$\left. \frac{d^2 x^i}{d\tau^2} \right|_{\tau=0} = -\Gamma^i_{00} \left(\frac{dx^0}{d\tau} \right)^2 \Big|_{\tau=0} = -\frac{1}{2} g^{i\alpha} (2\partial_0 g_{0\alpha} - \partial_\alpha g_{00}) \left(\frac{dx^0}{d\tau} \right)^2 \Big|_{\tau=0} = 0$$

This means that the test masses remain at rest if they are initially at rest. However, their proper distance is indeed affected by the passage of the GW:

$$\xi^2 = g_{\mu\nu} \xi^\mu \xi^\nu = (\delta_{ij} + h_{ij}^{\text{TT}}) \xi^i \xi^j = d_0^2 + h_{ij}^{\text{TT}} \xi^i \xi^j$$

So, the proper distance between the two test masses oscillates with the GW:

$$d(t) = \|\xi\| = d_0 \sqrt{1 + \frac{1}{d_0^2} h_{ij}^{\text{TT}} \xi^i \xi^j} \simeq d_0 + \frac{1}{2d_0} h_{ij}^{\text{TT}} \xi^i \xi^j$$

Observation 6.1.4 (Free-falling frame and detector frame)

Regardless of the motion of an observer along a curve $\gamma(\tau)$, it is always possible to choose suitable local coordinates, known as *Fermi normal coordinates*, such that:

$$g_{\mu\nu}|_{\gamma(\tau)} = \eta_{\mu\nu} \quad \Gamma^\mu_{\alpha\beta}|_{\gamma(\tau)} = 0$$

The metric then differs from the Lorentz–Minkowski metric only by terms proportional to the spacetime curvature and to the acceleration of the observer. In the case of a free-falling observer

$\nabla_u u = 0$, and the metric can be shown to be:

$$ds^2 = -dt^2 [1 + \mathcal{R}_{i0j0} x^i x^j] - 2dt dx^i \frac{2}{3} \mathcal{R}_{0jik} x^j x^k + dx^i dx^j \left[\delta_{ij} - \frac{1}{3} \mathcal{R}_{ijkl} x^k x^l \right] + \dots$$

On Earth, observers comoving with the detector are not in free fall, hence acceleration effects should be taken into consideration.

Now, apply the geodesic deviation equation Eq. 6.1.24 to the geodesics of the two test masses. Clearly, $u = \frac{d}{d\tau}$, hence it becomes:

$$\frac{d^2 \xi^\mu}{d\tau^2} = -\mathcal{R}_{\beta\alpha}{}^\mu{}_\rho u^\rho u^\alpha \xi^\beta$$

In the free-fall frame $u^\mu = (1, \mathbf{0})$, so $\langle u, \xi \rangle_g = 0$ implies that $\xi^\mu = (0, \xi)$ and:

$$\frac{d^2 \xi^i}{d\tau^2} = -\mathcal{R}_{j0}{}^i{}_0 \xi^j$$

For non-relativistic motion the proper time τ can be replaced with the coordinate time t . Moreover, at linear order the Riemann tensor is gauge-invariant, hence it can be computed in the preferred gauge. In the TT gauge at linear order:

$$\mathcal{R}_{j0}{}^i{}_0 = [\mathcal{R}^{[1]}]_{j0}{}^i{}_0 = \mathcal{R}_{i0j0}^{\text{TT}} = \frac{1}{2} \partial_0^2 h_{ij}^{\text{TT}}$$

The geodesic deviation is then described at linear order by a simple ODE:

$$\frac{d^2 \xi^i}{dt^2} = \frac{1}{2} \partial_0^2 h_{ij}^{\text{TT}} \xi^j \quad (6.1.25)$$

Note that this approximation is only valid if the detector frame can be taken to be in free fall: this is valid if the characteristic dimension of the detector x is much smaller than the scale of spacetime curvature L , i.e. $x \ll L$. In this case $L^{-2} \sim \mathcal{R} \sim \partial_0^2 h^{\text{TT}} \sim \lambda_{\text{GW}}^{-2}$, hence $x \ll \lambda_{\text{GW}}$, where λ_{GW} is the wavelength of the GW: taking $\lambda_{\text{GW}} \sim 10^5$ km as a reference scale³, Earth-based interferometers (e.g. LIGO, VIRGO, KAGRA) with $x \sim 10$ km are well described by this approximation, while space-based detectors (e.g. LISA) with $x \sim 10^6$ km require a more careful analysis.

§6.2 Gravitational energy

The equivalence principle forbids the localization of gravitational energy and momentum, as already noted in §5.2.2. Nonetheless, non-local objects describing these quantities do exist.

In classical Electromagnetism, given a system of electric charges described by a charge density $\varrho = \varrho(\mathbf{x})$, it is possible to define the total electric charge inside a spatial region V as:

$$q \equiv \int_V d^3x \varrho(\mathbf{x})$$

Using the first Maxwell equation $\nabla \cdot \mathbf{E} = 4\pi\varrho$, it is possible to express q in terms of $\mathbf{E}$:

$$q = \frac{1}{4\pi} \int_V d^3x \nabla \cdot \mathbf{E} = \frac{1}{4\pi} \int_{\partial V} d\mathbf{S} \cdot \mathbf{E}$$

³ $\lambda_{\text{GW}} \sim 10^5$ km corresponds to $\nu_{\text{GW}} \sim 1$ Hz, which is the typical frequency scale of GWs generated by compact binaries.

Therefore, only the flux of the electric field through ∂V modifies the total charge inside V . The same construction can be applied to Newtonian gravity, where a generic gravitational system is described by a mass density $\varrho = \varrho(\mathbf{x})$, so that the total mass (gravitational charge) inside a spatial region V is:

$$M \equiv \int_V d^3x \varrho(\mathbf{x})$$

In this case, use the Poisson equation $\nabla^2 \Phi = 4\pi G_N \varrho$ to rewrite:

$$M = \frac{1}{4\pi G_N} \int_V d^3x \nabla \cdot \nabla \Phi = \frac{1}{4\pi G_N} \int_{\partial V} d\mathbf{S} \cdot \nabla \Phi$$

Again, only the flux of the gravitational field through ∂V modifies the total mass inside V . A similar argument can be extended to the general relativistic case.

§6.2.1 Energy-momentum pseudotensor

Since any gravitational field can be locally nullified with the choice of a local inertial frame (through Fermi normal coordinates), the energy of a gravitational field is not suitably described by a tensor, since a tensor which vanishes in a particular frame does so in all of them.

For this reason, it is necessary to find a non-tensorial object $t^{\mu\nu}$, constructed from the metric, which is symmetric and such that $\partial_\mu [g(T^{\mu\nu} + t^{\mu\nu})] = 0$, so that the total energy-momentum distribution of the gravitational field together with matter fields is conserved in the usual sense⁴.

§6.2.1.1 Weak gravity

Consider first the simplified case of a perturbation $h_{\mu\nu}$ propagating on a $d = 4$ Minkowski background. In this case, the background Einstein equations are identically satisfied (since the EMT of matter must vanish at leading order, for gravity to be weak), hence the energy of the gravitational field needs to be studied at NLO.

Proposition 6.2.1 (Linearized Einstein equations)

The NLO field equation in Eq. 6.1.9 can be written as:

$$\partial^\alpha \partial^\beta H_{\mu\alpha\nu\beta} = 16\pi G_N [T^{[1]}]_{\mu\nu} \quad (6.2.1)$$

where the **Lorentz tensor** is defined as:

$$H_{\mu\alpha\nu\beta} \equiv -\bar{h}_{\mu\nu}\eta_{\alpha\beta} - \bar{h}_{\alpha\beta}\eta_{\mu\nu} + \bar{h}_{\alpha\nu}\eta_{\mu\beta} + \bar{h}_{\mu\beta}\eta_{\alpha\nu} \quad (6.2.2)$$

Proof. Given the metric tensor $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, at NLO the inverse metric is $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$, thus:

$$[\Gamma^{[1]}]_{\nu\rho}^\sigma = \frac{1}{2}\eta^{\sigma\lambda} (\partial_\nu h_{\lambda\rho} + \partial_\rho h_{\nu\lambda} - \partial_\lambda h_{\nu\rho})$$

Recalling Eq. 2.5.6, the $\Gamma^{[1]}\Gamma^{[1]} \sim h^2$ terms of the Riemann tensor are NNLO and hence negligible, so:

$$[\mathcal{R}^{[1]}]_{\rho\mu\nu}^\sigma = \frac{1}{2}\eta^{\sigma\lambda} (\partial_\mu \partial_\rho h_{\nu\lambda} - \partial_\mu \partial_\lambda h_{\nu\rho} - \partial_\nu \partial_\rho h_{\mu\lambda} + \partial_\nu \partial_\lambda h_{\mu\rho})$$

⁴Add clarification on conservation of currents.

Contracting (σ, ρ) , the Ricci tensor is:

$$[\mathcal{R}^{[1]}]_{\mu\nu} = \frac{1}{2} (\partial^\rho \partial_\mu h_{\nu\rho} + \partial^\rho \partial_\nu h_{\mu\rho} - \square h_{\mu\nu} - \partial_\mu \partial_\nu h)$$

where $\square \equiv \partial^\mu \partial_\mu$. The Ricci scalar then is:

$$\mathcal{R}^{[1]} = \partial^\mu \partial^\nu h_{\mu\nu} - \square h$$

Finally, the Einstein tensor can be expressed as:

$$[G^{[1]}]_{\mu\nu} = \frac{1}{2} [\partial^\rho \partial_\mu h_{\nu\rho} + \partial^\rho \partial_\nu h_{\mu\rho} - \square h_{\mu\nu} - \partial_\mu \partial_\nu h - (\partial^\rho \partial^\sigma h_{\rho\sigma} - \square h) \eta_{\mu\nu}]$$

It suffices to show that $[G^{[1]}]_{\mu\nu} = \frac{1}{2} \partial^\alpha \partial^\beta H_{\mu\alpha\nu\beta}$. By direct computation:

$$\begin{aligned} \partial^\alpha \partial^\beta H_{\mu\alpha\nu\beta} &= -\square \bar{h}_{\mu\nu} - \eta^{\mu\nu} \partial^\alpha \partial^\beta \bar{h}_{\alpha\beta} + \partial_\mu \partial^\alpha \bar{h}_{\alpha\nu} + \partial_\nu \partial^\alpha \bar{h}_{\mu\alpha} \\ &= -\square \left(h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \right) - \eta_{\mu\nu} \left(\partial^\alpha \partial^\beta h_{\alpha\beta} - \frac{1}{2} \square h \right) \\ &\quad + \partial_\mu \partial^\alpha \left(h_{\alpha\nu} - \frac{1}{2} \eta_{\alpha\nu} h \right) + \partial_\nu \partial^\alpha \left(h_{\mu\alpha} - \frac{1}{2} \eta_{\mu\alpha} h \right) \\ &= -\square h_{\mu\nu} + \partial_\mu \partial^\alpha h_{\alpha\nu} + \partial_\nu \partial^\alpha h_{\mu\alpha} - \partial_\mu \partial_\nu h - \eta_{\mu\nu} (\partial^\alpha \partial^\beta h_{\alpha\beta} - \square h) \end{aligned}$$

Renaming dummy indices yields the thesis. $\square$

The Lorentz tensor has the same symmetries of the Riemann tensor, namely:

$$H_{\mu\alpha\nu\beta} = H_{\nu\beta\mu\alpha} = H_{[\mu\alpha][\nu\beta]} \quad H_{\mu[\alpha\nu\beta]} = 0 \quad (6.2.3)$$

With Eq. 6.2.1, it is possible to define the total 4-momentum in a spatial region V in the usual way:

$$\begin{aligned} P^\mu &\equiv \int_V d^3x T^{\mu 0} = \frac{1}{16\pi G_N} \int_V d^3x \partial_\alpha \partial_\beta H^{\mu\alpha 0\beta} \\ &= \frac{1}{16\pi G_N} \int_V d^3x \partial_\alpha \partial_i H^{\mu\alpha 0i} = \frac{1}{16\pi G_N} \int_{\partial V} d^2S n_i \partial_\alpha H^{\mu\alpha 0i} \end{aligned}$$

where the index β is restricted to be a spatial index by the total antisymmetry of $H^{\mu\alpha\nu\beta}$ in the last three indices. Similarly, for the total angular momentum:

$$\begin{aligned} J^{\mu\nu} &\equiv \int_V d^3x (x^\mu T^{\nu 0} - x^\nu T^{\mu 0}) = \frac{1}{16\pi G_N} \int_V d^3x (x^\mu \partial_\alpha \partial_\beta H^{\nu\alpha 0\beta} - x^\nu \partial_\alpha \partial_\beta H^{\mu\alpha 0\beta}) \\ &= \frac{1}{8\pi G_N} \int_V d^3x (\partial_\beta (x^{[\mu} \partial_\alpha H^{\nu]\alpha 0\beta}) - \delta_\beta^{[\mu} \partial_\alpha H^{\nu]\alpha 0\beta}) \\ &= \frac{1}{8\pi G_N} \int_V d^3x (\partial_i (x^{[\mu} \partial_\alpha H^{\nu]\alpha 0i}) - \partial_\alpha H^{[\nu\alpha 0|\mu]}) \\ &= \frac{1}{8\pi G_N} \int_V d^3x \partial_i (x^{[\mu} \partial_\alpha H^{\nu]\alpha 0i} + H^{[\mu|i 0|\nu]}) + \frac{1}{8\pi G_N} \int_V d^3x \partial_0 H^{[\mu|00|\nu]} \end{aligned}$$

The second term vanishes since $H^{\nu 00\mu} = H^{0\mu\nu 0} = H^{\nu 00\mu}$, hence the total angular momentum in a spatial region V is given by yet another flux integral:

$$J^{\mu\nu} = \frac{1}{8\pi G_N} \int_{\partial V} d^2S n_i (x^{[\mu} H^{\nu]\alpha 0i} + H^{[\mu|i 0|\nu]})$$

The integrands in these fluxes are not gauge invariant, since it vanishes in any local free-fall frame. On the contrary, assuming ∂V to be far from any source, the total fluxes are indeed gauge-invariant, and only asymptotic properties are relevant for the computation of total momentum and angular momentum.

§6.2.1.2 Landau–Lifshitz pseudotensor

In the general case, the Einstein field equations can be rewritten in a form similar to Eq. 6.2.1:

$$\partial_\alpha \partial_\beta H^{\mu\alpha\nu\beta} = 16\pi G_N g(T^{\mu\nu} + t^{\mu\nu}) \quad (6.2.4)$$

which requires the Einstein tensor to split like:

$$G^{\mu\nu} = \frac{1}{2g} \partial_\alpha \partial_\beta H^{\mu\alpha\nu\beta} - 8\pi G_N g t^{\mu\nu} \quad (6.2.5)$$

Now, $H^{\mu\alpha\nu\beta}$ is no longer a tensor, but retains the same symmetries Eq. 6.2.3. Then, by total anti-symmetry of the last three indices, $\partial_\nu \partial_\alpha \partial_\beta H^{\mu\alpha\nu\beta} = 0$, meaning that the right-hand side satisfies an ordinary conservation law:

$$\partial_\nu [g(T^{\mu\nu} + t^{\mu\nu})] = 0 \quad (6.2.6)$$

The main problem is that the construction of $t^{\mu\nu}$ by the splitting Eq. 6.2.5 is far from unique. The most widespread choice is the one used by Landau and Lifshitz in [9]:

$$H_{LL}^{\mu\alpha\nu\beta} = g^{\mu\nu} g^{\alpha\beta} - g^{\alpha\nu} g^{\mu\beta} \quad (6.2.7)$$

where the *metric density* $g^{\mu\nu} \equiv \sqrt{g} g^{\mu\nu}$ is not a tensor. The **Landau–Lifshitz pseudo-EMT** is then found to be:

$$t_{LL}^{\alpha\beta} = \frac{1}{16\pi G_N g} \left[\partial_\lambda g^{\alpha\beta} \partial_\mu g^{\lambda\mu} - \partial_\lambda g^{\alpha\lambda} \partial_\mu g^{\beta\mu} + \frac{1}{2} g^{\alpha\beta} g_{\lambda\mu} \partial_\rho g^{\lambda\mu} \partial_\lambda g^{\rho\mu} - 2g^{(\alpha\lambda} g_{\mu\nu} \partial_\rho g^{\beta)\nu} \partial_\lambda g^{\mu\rho} + \right. \\ \left. + g_{\mu\nu} g^{\nu\rho} \partial_\nu g^{\alpha\lambda} \partial_\rho g^{\beta\mu} + \frac{1}{8} (2g^{\alpha\lambda} g^{\beta\mu} - g^{\alpha\beta} g^{\lambda\mu}) (2g_{\nu\rho} g_{\tau\sigma} - g_{\rho\sigma} g_{\nu\tau}) \partial_\lambda g^{\nu\rho} \partial_\mu g^{\sigma\tau} \right] \quad (6.2.8)$$

This expression is symmetric in $\alpha\beta$ and quadratic in ∂g , and indeed the Landau–Lifshitz splitting is the unique choice for which the pseudo-EMT has these properties. Moreover, it has no dependence on second derivatives of the metric, so that the pseudo-EMT vanishes in local inertial frames as expected from the nature of the gravitational field.

By Eq. 6.2.6, it is possible to define an *effective EMT* (which is not a tensor, however) containing information about the energy-momentum distribution of both the gravitational field and the matter fields:

$$T_{\text{eff}}^{\mu\nu} \equiv g(T^{\mu\nu} + t_{LL}^{\mu\nu}) \quad \partial_\mu T_{\text{eff}}^{\mu\nu} = 0 \quad (6.2.9)$$

Then, the effective EMT plays the same role as the standard EMT in flat spacetime, since it is symmetric and $\partial_\mu T_{\text{eff}}^{\mu\nu} = 0$. In flat spacetime, the total 4-momentum is defined as:

$$P^\mu \equiv \int_V d^3x T^{\mu 0} \quad \Rightarrow \quad \frac{dP^\mu}{dt} = \int_V d^3x \partial_0 T^{\mu 0} = - \int_V d^3x \partial_i T^{\mu i} = - \int_{\partial V} d^2S n_i T^{\mu i}$$

which is the usual flux integral. The same equations hold for the effective EMT, hence the total 4-momentum obeys:

$$\frac{dP^\mu}{dt} = - \int_{\partial V} d^2S n_i g(T^{\mu i} + t_{LL}^{\mu i}) \quad (6.2.10)$$

which correctly reduces to the above flat-spacetime expression for $g_{\mu\nu} = \eta_{\mu\nu}$, thus confirming the interpretation as the total 4-momentum of the gravitational fields and the matter fields. Assuming that the spatial region V contains N sources of the gravitational field, each occupying a spatial region $V_i \subseteq V$, then:

$$P^\mu = \int_V d^3x T_{\text{eff}}^{\mu 0} = \sum_{i=1}^N \int_{V_i} d^3x T_{\text{eff}}^{\mu 0} + \int_{\bar{V}} d^3x T_{\text{eff}}^{\mu 0} = \sum_{i=1}^N P_i^\mu + \int_{\bar{V}} d^3x T_{\text{eff}}^{\mu 0}$$

where P_i^μ is the total 4-momentum of the i^{th} source and $\bar{V} \equiv V - \bigcup_{i=1}^N V_i$ is the interbody region between sources. In the majority of cases, the gravitational field is only strong near the sources, hence the contribution of the interbody term is negligible.

Observation 6.2.1 (Energy in asymptotically-flat spacetimes)

In an asymptotically-flat spacetime, taking V to be asymptotically large means that $g_{\mu\nu}|_{\partial V} \simeq \eta_{\mu\nu}$, hence:

$$T_{\text{eff}}^{\mu\nu}|_{\partial V} = g(T^{\mu\nu} + t_{\text{LL}}^{\mu\nu})|_{\partial V} \simeq T^{\mu\nu}|_{\partial V} + t_{\text{LL}}^{\mu\nu}|_{\partial V}$$

Furthermore, assuming the absence of matter at infinity, i.e. $T^{\mu\nu}|_{\partial V} \simeq 0$, reduces the effective EMT to the Landau–Lifshitz pseudo-EMT. Now, note that schematically (working in spherical coordinates):

$$g \sim r^{-n} \quad \Rightarrow \quad \partial g \sim r^{-n-1} \quad \Rightarrow \quad t_{\text{LL}} \sim r^{-2n-2} \quad \Rightarrow \quad \frac{dP}{dt} \sim r^2 t_{\text{LL}} \sim r^{-2n}$$

Therefore, the only non-negligible contribution to the flux integral comes from metric terms with $n \leq 0$: for example, GWs propagating at infinity have $n = 0$, hence their case needs to be studied carefully.

§6.2.2 ADM formalism

The Arnowitt–Deser–Misner (ADM) formalism consists of a Hamiltonian formulation of General Relativity built on a decomposition of spacetime into spacelike hypersurfaces.

Definition 6.2.1 (Foliation)

Let $\mathcal{M}$ be a smooth n -manifold, and let $\mathcal{F}$ be any collection of k -dimensional submanifolds of $\mathcal{M}$. A smooth chart (U, φ) on $\mathcal{M}$ is said to be *flat for $\mathcal{F}$* if $\varphi(U) \subseteq \mathbb{R}^n$ is a cube and each submanifold in $\mathcal{F}$ intersects U in either the empty set or a countable union of k -dimensional slices of the form $x^{k+1} = c^{k+1}, \dots, x^n = c^n$.

Then, $\mathcal{F}$ is a **k -dimensional foliation** of $\mathcal{M}$ if it consists of disjoint connected non-empty immersed k -dimensional submanifolds of $\mathcal{M}$, called the *leaves of the foliation*, whose union is $\mathcal{M}$ and such that in a neighbourhood of each point $p \in \mathcal{M}$ there exists a flat chart for $\mathcal{F}$.

It is convenient to work not with arbitrary foliations of spacetime, but with particular foliations best-suited to express initial-value problems: indeed, the Einstein field equations are hyperbolic, hence fields at any point $p \in \mathcal{M}$ can be influenced only by events in its causal past.

Definition 6.2.2 (Cauchy hypersurface)

Let $(\mathcal{M}, g)$ be a spacetime. A *causal curve* is a piecewise smooth curve whose tangent is every-

where timelike or null and future-directed (or past-directed): such a curve is *inextendible* if it admits no extension to another causal curve^a.

Then, a hypersurface $\Sigma \subseteq \mathcal{M}$ is called a **Cauchy hypersurface** if every inextendible causal curve in $\mathcal{M}$ intersects Σ exactly once.

^aIntuitively, this means that the curve has no past or future endpoints in $\mathcal{M}$.

If Σ is a Cauchy hypersurface of a spacetime $(\mathcal{M}, g)$, the causal past or future of every point $p \in \mathcal{M}$ intersects Σ exactly once, thus initial data for the metric and matter fields on Σ uniquely determine the solution to the Einstein field equations in the entire spacetime. For this reason, a spacetime which admits a Cauchy hypersurface is called **globally hyperbolic**.

Consider a globally hyperbolic spacetime $(\mathcal{M}, g)$ and assume it admits a $(d-1)$ -dimensional foliation into spacelike achronal⁵ hypersurfaces $\{\Sigma_t\}$, where $t \in \mathcal{C}^\infty(\mathcal{M})$ is a smooth time function whose level sets are Cauchy hypersurfaces. On each slice Σ_t introduce spatial coordinates $x^1, \dots, x^{d-1}$ so that $x^\mu = (t, x^1, \dots, x^{d-1})$ defines a coordinate system adapted to the foliation. Let n^μ be the future-directed unit normal to the hypersurface Σ_t , so that $n_\mu n^\mu = -1$ and $n_\mu \propto \nabla_\mu t$, and decompose the time evolution vector field $t^\mu := (\partial_t)^\mu$ as:

$$t^\mu = N n^\mu + N^\mu$$

where N is the *lapse function* and N^μ is the *lapse vector* tangent to Σ_t . Geometrically, N measures the proper time separation between adjacent slices along the normal, while N^i encodes the relative displacement of spatial coordinates from one slice to the next. Then, the metric can be shown to take the form:

$$ds^2 = -N^2 dt^2 + \gamma_{ij} (dx^i + N^i dt)(dx^j + N^j dt) \quad (6.2.11)$$

where γ_{ij} is the induced metric on the spacelike slices. This metric is completely parametrized by N , N^i and γ_{ij} , hence it retains the correct number of degrees of freedom $1 + (d-1) + \frac{1}{2}d(d-1) = \frac{1}{2}d(d+1)$.

This formalism naturally provides definitions for the total energy, momentum and angular momentum of spacetime in terms of boundary intergrals at spatial infinity. In particular, given an asymptotically-flat spacetime with ADM decomposition Eq. 6.2.11 such that $\gamma_{ij} \rightarrow \delta_{ij}$ sufficiently fast at spatial infinity, its **ADM mass** is defined as:

$$M_{\text{ADM}} := \frac{1}{16\pi G_N^{(d)}} \int_{\mathbb{S}_\infty^{d-2}} d^{d-2} S n^i (\partial^j \gamma_{ij} - \partial_i \gamma) \quad (6.2.12)$$

where $\gamma \equiv \gamma^i_i$. In the case of stationary spacetimes, the ADM mass coincides with the Komar mass, i.e. with the Komar charge Eq. 5.1.27 associated to the timelike Killing vector field (which exists by Def. 7.3.1).

§6.2.2.1 Weak gravity

In the case of a small perturbation $h_{\mu\nu}$ propagating on a $d=4$ Minkowski background, the ADM mass Eq. 6.2.12 can be obtained extending the analysis in §6.2.1.1.

Given the perturbed metric $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, at NLO $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + o(h^2)$ and:

$$\sqrt{g} = 1 + \frac{h}{2} + o(h^2) \quad \Rightarrow \quad \mathbf{g}^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + \frac{h}{2} \eta^{\mu\nu} + o(h^2) = \eta^{\mu\nu} - \bar{h}^{\mu\nu} + o(h^2)$$

⁵An hypersurface is *achronal* if no pair of its points can be connected by a timelike curve.

Then, the Landau–Lifshitz split Eq. 6.2.7 becomes:

$$\partial_\alpha H_{\text{LL}}^{\mu\alpha\nu\beta} = -\partial^\beta \bar{h}^{\mu\nu} - \eta^{\mu\nu} \partial_\alpha \bar{h}^{\alpha\beta} + \partial^\nu \bar{h}^{\mu\beta} + \eta^{\mu\beta} \partial_\alpha \bar{h}^{\alpha\nu} + o(h^2)$$

This expression allows to explicitly compute the total energy and momentum of spacetime by direct calculation.

Lemma 6.2.1 (ADM energy and momentum in weak gravity)

$$P^0 = \frac{1}{16\pi G_N} \int_{\mathbb{S}_\infty^2} d^2 S n_i (\partial_j h_{ij} - \partial_i h) \quad (6.2.13)$$

$$P^j = \frac{1}{16\pi G_N} \int_{\mathbb{S}_\infty^2} d^2 S n_i [\delta_i^j (\partial_0 h - \partial_k h_{k0}) + \partial_i h_{j0} - \partial_0 h_{ij}] \quad (6.2.14)$$

§6.3 Propagation of GWs

The perturbative study of GWs is based on the separation of the waves from the background, as in Eq. 6.1.1: the problem is that, in a general setting, there is no unambiguous way to perform this separation, and on top of that it is not gauge invariant either.

A natural splitting between the spacetime background and the gravitational waves arises when there is a clear separation of scales. This is the case, for example, when in some coordinate system the metric can be written as in Eq. 6.1.1, where $\bar{g}$ has a typical scale of spatial variation L_b , while h is characterized by a wavelength⁶ λ such that:

$$\lambda \ll L_b \quad (6.3.1)$$

In this case, h has the physical interpretation of small ripples on a smooth background. Equivalently, in frequency space, if $\bar{g}$ contains frequencies up to f_b , while h is a monochromatic wave peaked around a frequency f , then it must be:

$$f \gg f_b \quad (6.3.2)$$

which is the case of a high-frequency perturbation over a slowly-varying background. Note that the scales L_b and f_b are a priori unrelated, hence the two condition are independent, and it suffices that one of them is satisfied: Eq. 6.3.1 is known as the *short-wave expansion*, but for GWs Eq. 6.3.2 is usually fulfilled instead⁷.

§6.3.1 Back-reaction on the background

To study how GWs modify the background spacetime as they propagate, expand the $\mathcal{R}_{\mu\nu}$ in powers of h :

$$\mathcal{R}_{\mu\nu} = \sum_{n \in \mathbb{N}_0} [\mathcal{R}^{[n]}] = \bar{\mathcal{R}}_{\mu\nu} + [\mathcal{R}^{[1]}]_{\mu\nu} + [\mathcal{R}^{[2]}]_{\mu\nu} + o(h^3)$$

where $\bar{\mathcal{R}}_{\mu\nu}$ is the Ricci tensor of the background, thus containing only low-frequency modes. On the other hand, $[\mathcal{R}^{[1]}]_{\mu\nu}$ is linear in $h_{\mu\nu}$, thus containing only high-frequency modes, while $[\mathcal{R}^{[2]}]_{\mu\nu}$ is quadratic in $h_{\mu\nu}$, thus containing both low- and high-frequency modes: indeed, in a quadratic

⁶For a function $f(x)$ oscillating as e^{ikx} , with $k = 2\pi/\lambda \equiv \lambda^{-1}$, the typical length-scale is λ rather than λ , in the sense that $|f'| = \lambda^{-1} |f|$.

term $\sim h_{\mu\nu}h_{\rho\sigma}$ a mode with high $\mathbf{k}_1$ from $h_{\mu\nu}$ can combine with a mode with high $\mathbf{k}_2 \simeq -\mathbf{k}_1$ from $h_{\rho\sigma}$, yielding a low-frequency mode. This separation into low- and high-frequency modes can be imposed on the Einstein field equations, suitably written as:

$$\mathcal{R}_{\mu\nu} = 8\pi G_N \left(T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T \right)$$

In order to do this, introduce the **Brill–Hartle–Isaacson average** (BHI average) $\langle A \rangle$ which spatially averages the quantity A over a scale of order of a few wavelengths λ , so that:

$$\bar{g}_{\mu\nu} \equiv \langle g_{\mu\nu} \rangle \quad h_{\mu\nu} = g_{\mu\nu} - \langle g_{\mu\nu} \rangle$$

This makes manifest that the BHI average eliminates oscillations on scales $\sim \lambda$, leaving only low-frequency modes: in particular, $\langle h_{\mu\nu} \rangle = 0$, as trivially seen by the above relations. The BHI average can be shown to have the following properties (integration by parts and integration of total derivative):

$$\langle A \bar{\nabla}_\alpha B \rangle = - \langle B \bar{\nabla}_\alpha A \rangle + o\left(\frac{\lambda^2}{L_b^2}\right) \quad \langle \bar{\nabla}_\alpha A \rangle = 0 + o\left(\frac{\lambda}{L_b}\right) \quad (6.3.3)$$

Setting the notation $A^{\text{Low}} \equiv \langle A \rangle$ and $A^{\text{High}} \equiv A - \langle A \rangle$, the above observation on the mode composition of the Ricci tensor at various orders is confirmed by the expansion at NLO:

$$[\mathcal{R}^{[1]}]_{\mu\nu} = \frac{1}{2} (\bar{\nabla}^\alpha \bar{\nabla}_\mu h_{\nu\alpha} + \bar{\nabla}^\alpha \bar{\nabla}_\nu h_{\mu\alpha} - \bar{\square} h_{\mu\nu} - \bar{\nabla}_\nu \bar{\nabla}_\mu h) \quad (6.3.4)$$

which yields $\langle [\mathcal{R}^{[1]}]_{\mu\nu} \rangle = 0$ by $\langle h_{\mu\nu} \rangle = 0$. On the contrary, schematically $[\mathcal{R}^{[2]}]_{\mu\nu} \sim (\nabla h)^2 + h \nabla^2 h$, so $\langle [\mathcal{R}^{[2]}]_{\mu\nu} \rangle \sim \langle (\nabla h)^2 \rangle$ by integration by parts. The Einstein field equations are then split as:

$$\bar{\mathcal{R}}_{\mu\nu} + [\mathcal{R}^{[2]}]_{\mu\nu}^{\text{Low}} = 8\pi G_N \left(T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T \right)^{\text{Low}} \quad (6.3.5)$$

$$[\mathcal{R}^{[1]}]_{\mu\nu} + [\mathcal{R}^{[2]}]_{\mu\nu}^{\text{High}} = 8\pi G_N \left(T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T \right)^{\text{High}} \quad (6.3.6)$$

The low-frequency equation gives the scale of h : in vacuum $\bar{\mathcal{R}}_{\mu\nu} + [\mathcal{R}^{[2]}]_{\mu\nu} = 0$, thus, given $\bar{\mathcal{R}} \sim L_b^{-2}$ and $\mathcal{R}^{[2]} \sim h^2/\lambda^2$, the perturbation scale is $h \sim \lambda/L_b$; in presence of matter, instead, $L_b^{-2} + h^2/\lambda^2$ equals to the matter contribution, which is dominant if $L_b^{-2} \gg h^2/\lambda^2$, i.e. if $h \ll \lambda/L_b$. In general, GWs have typical scale:

$$h \lesssim \frac{\lambda}{L_b} \quad (6.3.7)$$

Notice that $\bar{g}_{\mu\nu} \equiv \eta_{\mu\nu}$ means $L_b^{-1} \rightarrow 0$, hence by the above relation the background spacetime cannot be flat: GWs always back-react on the background. The amplitude of a GW can violate Eq. 6.3.7 only in a non-perturbative description, e.g. in presence of an exact solution.

§6.3.2 Energy carried by a GW

To express the back-reaction of GWs on the background spacetime clearly, define the effective background contribution of matter⁸ as $\langle T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T \rangle \equiv \bar{T}_{\mu\nu} - \frac{1}{2} \bar{g}_{\mu\nu} \bar{T}$, with $\bar{T} \equiv \bar{g}_{\mu\nu} \bar{T}^{\mu\nu}$, so that Eq. 6.3.5 can be rewritten as:

$$\bar{\mathcal{R}}_{\mu\nu} = -[\mathcal{R}^{[2]}]_{\mu\nu}^{\text{Low}} + 8\pi G_N \left(\bar{T}_{\mu\nu} - \frac{1}{2} \bar{g}_{\mu\nu} \bar{T} \right) \quad (6.3.8)$$

This expression directly links the background curvature to the background matter contribution plus an additional gravitational contribution due to the GWs: $\langle \mathcal{R}^{[2]} \rangle$ contains the large-scale back-reaction of small-scale high-frequency GWs.

In order to study the energy carried by a GW, define the pseudo-EMT $t_{\mu\nu}$ as:

$$t_{\mu\nu} := -\frac{1}{8\pi G_N} \left\langle [\mathcal{R}^{[2]}]_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \mathcal{R}^{[2]} \right\rangle \quad (6.3.9)$$

where $\mathcal{R}^{[2]} \equiv g^{\mu\nu} [\mathcal{R}^{[2]}]_{\mu\nu}$. Then, defining the trace $t \equiv g^{\mu\nu} t_{\mu\nu}$, at NNLO:

$$[t^{[2]}]_{\mu\nu} = -\frac{1}{8\pi G_N} \left(\langle [\mathcal{R}^{[2]}]_{\mu\nu} \rangle - \frac{1}{2} \bar{g}_{\mu\nu} \bar{g}^{\alpha\beta} \langle [\mathcal{R}^{[2]}]_{\alpha\beta} \rangle \right) \quad t^{[2]} = \frac{1}{8\pi G_N} \bar{g}^{\alpha\beta} \langle [\mathcal{R}^{[2]}]_{\alpha\beta} \rangle$$

This pseudo-EMT represents the self-consistent contribution of the GW to the background geometry: being defined by through the BHI average, it respects the equivalence principle, since it does not provide a localized notion of gravitational EMT. Inserting the pseudo-EMT into Eq. 6.3.8:

$$\bar{R}_{\mu\nu} = 8\pi G_N \left(t_{\mu\nu} - \frac{1}{2} \bar{g}_{\mu\nu} t + \bar{T}_{\mu\nu} - \frac{1}{2} \bar{g} \bar{T} \right)$$

or, equivalently:

$$\bar{G}_{\mu\nu} = 8\pi G_N (t_{\mu\nu} + \bar{T}_{\mu\nu}) \quad (6.3.10)$$

which makes it clear that this pseudo-EMT encodes the energy-momentum contribution of the GW which back-reacts on the background spacetime. The background Bianchi identities $\bar{\nabla}^\mu \bar{G}_{\mu\nu} = 0$ lead to the covariant conservation $\bar{\nabla}^\mu (t_{\mu\nu} + \bar{T}_{\mu\nu}) = 0$.

Observation 6.3.1 (Quasi-gauge invariance of the pseudo-EMT)

Using the formalism developed in §6.1.1, it is possible to compute $[\mathcal{R}^{[2]}]_{\mu\nu}$, finding:

$$\begin{aligned} [\mathcal{R}^{[2]}]_{\mu\nu} = \frac{1}{2} \bar{g}^{\rho\sigma} \bar{g}^{\alpha\beta} & \left[\frac{1}{2} \bar{\nabla}_\mu h_{\rho\alpha} \bar{\nabla}_\nu h_{\sigma\beta} + 2 \bar{\nabla}_\rho h_{\nu\alpha} \bar{\nabla}_{[\sigma} h_{\mu|\beta]} + \right. \\ & + h_{\rho\alpha} (\bar{\nabla}_\nu \bar{\nabla}_\mu h_{\sigma\beta} + \bar{\nabla}_\beta \bar{\nabla}_\sigma h_{\mu\nu} - \bar{\nabla}_\beta \bar{\nabla}_\nu h_{\mu\sigma} - \bar{\nabla}_\beta \bar{\nabla}_\mu h_{\nu\sigma}) + \\ & \left. + \left(\frac{1}{2} \bar{\nabla}_\alpha h_{\rho\sigma} - \bar{\nabla}_\rho h_{\alpha\sigma} \right) (\bar{\nabla}_\nu h_{\mu\beta} + \bar{\nabla}_\mu h_{\nu\beta} - \bar{\nabla}_\beta h_{\mu\nu}) \right] \end{aligned} \quad (6.3.11)$$

In general, this expression is not gauge invariant, and under a gauge transformation it transforms schematically as:

$$[\mathcal{R}^{[2]}]_{\mu\nu} \mapsto [\mathcal{R}^{[2]}]_{\mu\nu} + \bar{\nabla}_\rho A^\rho_{\mu\nu} \quad (6.3.12)$$

However, recalling Eq. 6.3.3, when computing $t_{\mu\nu}$ the second term only produces a boundary term of order $o(\lambda/L_b)$, which can be neglected. Therefore, the pseudo-EMT is quasi-gauge invariant.

By Obs. 6.3.1, the pseudo-EMT can be computed in the preferred gauge.

⁸By definition, $\bar{T}_{\mu\nu}$ only has low-frequency (or low-momentum) modes, and can be viewed as a smoothed form “macroscopic” form of the fundamental “microscopic” EMT $T_{\mu\nu}$. The EMT generated by a macroscopic matter distribution is already quite smooth, so in this case $\bar{T}_{\mu\nu} \simeq T_{\mu\nu}$.

Lemma 6.3.1 (Pseudo-EMT in Lorentz gauge)

In the Lorentz gauge, the pseudo-EMT becomes:

$$[t^{[2]}]_{\mu\nu} = \frac{1}{32\pi G_N} \langle \bar{\nabla}_\mu \bar{h}_{\mu\beta} \bar{\nabla}_\nu \bar{h}^{\alpha\beta} \rangle \quad (6.3.13)$$

Proof. Recalling the definition of the Lorentz gauge Eq. 6.1.13, the NNLO Ricci tensor Eq. 6.3.11 becomes:

$$[\mathcal{R}^{[2]}]_{\mu\nu}$$

Then, the pseudo-EMT Eq. 6.3.9 □

In a regione where matter is absent, hence far from any source, $\bar{g}_{\mu\nu} \simeq \eta_{\mu\nu}$ and the TT gauge can also be used, yielding a simplified expression:

$$[t^{[2]}]_{\mu\nu} \simeq \frac{1}{32\pi G_N} \langle \partial_\mu h_{ij}^{\text{TT}} \partial_\nu h_{ij}^{\text{TT}} \rangle \quad (6.3.14)$$

As shown in §6.2.1, it is possible to compute the amount of energy extracted from a source producing GWs from the flux of t_{0i} through a spacelike codimension-2 submanifold ∂V far from the source:

$$E_V = \int_V d^3x T_{\text{eff}}^{00} \quad \Rightarrow \quad \frac{dE_V}{dt} = - \int_{\partial V} d^2S n_i T_{\text{eff}}^{0i}$$

In the absence of matter fluxes through ∂V (e.g. photons, neutrinos, ...), the flux is entirely due to the pseudo-EMT:

$$\frac{dE_V}{dt} = - \frac{1}{32\pi G_N} \int_{\partial V} d^2S n^k \langle \partial_0 h_{ij}^{\text{TT}} \partial_k h_{ij}^{\text{TT}} \rangle$$

Far from any source $h_{ij}^{\text{TT}} \sim f_{ij}(t - r)$ (generic wave solution), with $r \equiv |\mathbf{x}|$, so:

$$\partial_k h_{ij}^{\text{TT}} = \partial_0 h_{ij}^{\text{TT}} \partial_k (t - r) = -\dot{h}_{ij}^{\text{TT}} \frac{x_k}{r}$$

Then:

$$\int_{\partial V} d^2S n^k \langle \partial_0 h_{ij}^{\text{TT}} \partial_k h_{ij}^{\text{TT}} \rangle = - \int_{\partial V} d^2S n^k n_k \langle \dot{h}_{ij}^{\text{TT}} \dot{h}_{ij}^{\text{TT}} \rangle = -r^2 \int_{\mathbb{S}_r^2} d\Omega_2 [t^{[2]}]_{00}$$

where in the last step it has been assumed that $\partial V \cong \mathbb{S}_r^2$. The irradiated energy can then be computed as:

$$\frac{dE_V}{dt} = - \frac{1}{32\pi G_N} r^2 \int_{\mathbb{S}_r^2} d\Omega_2 \langle \dot{h}_{ij}^{\text{TT}} \dot{h}_{ij}^{\text{TT}} \rangle \quad (6.3.15)$$

§6.3.3 Propagation in curved spacetime

While the low-frequency equation Eq. 6.3.5 allows to identify the pseudo-EMT of the GW, its propagation through spacetime is determined by the high-frequency equation Eq. 6.3.6.

First of all, consider the limiting case of no external sources, i.e. $T_{\mu\nu} = 0$, so that Eq. 6.3.6 becomes:

$$[\mathcal{R}^{[1]}]_{\mu\nu} = -[\mathcal{R}^{[2]}]_{\mu\nu}^{\text{High}}$$

Performing order-of-magnitude estimates:

$$\mathcal{R}^{[1]} \sim \partial^2 h \sim \frac{h}{\lambda^2} \quad [\mathcal{R}^{[2]}]_{\text{High}} \sim (\partial h)^2 \sim \frac{h^2}{\lambda^2}$$

Recalling Eq. 6.3.7, a small parameter $\varepsilon \in \mathbb{R}^+$ can be defined as $\varepsilon \sim h \sim \lambda/L_b$, so that:

$$\mathcal{R}^{[1]} \sim \frac{1}{\varepsilon L_b} \quad [\mathcal{R}^{[2]}]^{\text{High}} \sim \frac{1}{L_b}$$

Since $\varepsilon > 0$ is small, the second term is subleading when compared to the first, hence at LO:

$$[\mathcal{R}^{[1]}]_{\mu\nu} \simeq 0 \quad (6.3.16)$$

This equation can be written in the Lorentz gauge as:

$$\square \bar{h}_{\mu\nu} + 2\bar{\mathcal{R}}_{\mu\alpha\nu\beta} \bar{h}^{\alpha\beta} - 2\bar{\mathcal{R}}_{\alpha(\nu} \bar{h}_{\mu)}^{\alpha} \simeq 0$$

where $\square \equiv \partial_\mu \partial^\mu$ since $\bar{g}_{\mu\nu} \simeq \eta_{\mu\nu}$ at LO in ε . The two curvature terms are of order $\sim h/L_b^2 \sim \varepsilon^3/\lambda^2$, while the first term is of order $\sim h/\lambda^2 \sim \varepsilon/\lambda^2$, hence the curvature terms are subleading and at LO:

$$\square \bar{h}_{\mu\nu} \simeq 0 \quad (6.3.17)$$

This is just the wave equation derived from the linearized theory, which thus remains valid in the limit of GWs being the only sources of curvature. In the absence of matter (or at least far from sources) the TT gauge can be used, and the wave equation has general solution:

$$h_{ij}^{\text{TT}}(x) = \int_{\mathbb{R}^3} \frac{d^3k}{(2\pi)^3} [\mathcal{A}_{ij}(k) e^{ik \cdot x} + \mathcal{A}_{ij}^*(k) e^{-ik \cdot x}] \quad (6.3.18)$$

with $k \equiv (\omega, \mathbf{k}) : \|\mathbf{k}\| = \omega = 2\pi f$. Typical astrophysical sources are localized in a certain direction $\hat{\mathbf{k}}_0$, so $\mathcal{A}_{ij}(k) = \mathcal{A}_{ij}(f) \delta^{(3)}(\mathbf{k} - \hat{\mathbf{k}}_0)$. Note that for ground detectors the spatial dependence of the wave can often be ignored.

Now, consider the presence of external matter dominating the curvature. In this case, recalling the discussion on Eq. 6.3.7, $h \ll \lambda/L_b \ll 1$, so the expansions in h and in λ/L_b are different: for GWs, terms quadratic in h are extremely negligible, so retain only terms linear in h and expand them in powers of λ/L_b . In this case, $[\mathcal{R}^{[2]}]^{\text{High}} \sim h^2$ is negligible, while the source term is:

$$\left(T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T \right)^{\text{High}} \sim \frac{h}{L_b^2}$$

This is because $T_{\mu\nu}$ is smooth for macroscopic matter sources, hence its high-frequency components depend on the perturbation $h_{\mu\nu}$, thus $T_{\mu\nu} \sim h/L_b^2$, and similarly for $g_{\mu\nu} T = (\bar{g}_{\mu\nu} + h_{\mu\nu}) T$. On the contrary, $\mathcal{R}^{[1]} \sim \partial^2 h \sim h/\lambda^2$, so the source term is smaller by a factor $o(\lambda^2/L_b^2)$, which does not contribute at LO and NLO. The linear equation at NLO in λ/L_b is then:

$$[\mathcal{R}^{[1]}]_{\mu\nu} = 0$$

As discussed above, in the Lorentz gauge this equation leads to:

$$\bar{\square} \bar{h}_{\mu\nu} = 0 \quad (6.3.19)$$

where now $\bar{\square} \equiv \bar{\nabla}_\mu \bar{\nabla}^\mu$. This wave equation therefore determines the propagation of GWs in a curved background, in the short-wave limit $\lambda \ll L_b$ (or equivalently $f \gg f_b$).

In conclusion, the separation of the Einstein field equations into low- and high-frequency components allows to describe both the effects of GWs and external matter on the background spacetime (low-frequency), and the propagation of GWs in the curved background (high-frequency).

§6.4 Generation of GWs

Black Holes

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In this chapter, we develop the concept of black hole (BH). First, black holes need to be rigorously defined (§7.1): in order to do that, it is necessary to study the global structure of spacetime, and in particular its asymptotics, with a technique developed by Penrose [13, 15] known as *conformal completion*. Secondly, we perform a geometric analysis of event horizons (§7.2), in order to study properties of black holes which are independent of the gravitational theory used. Then, we study black holes at equilibrium (sec:bh-stationary), covering the three main stationary black holes solutions: Schwarzschild (§7.4), Reissner–Nordström (§7.5) and Kerr black holes (§7.6). Finally, we study the dynamics of black holes out of equilibrium (§7.7).

§7.1 Definition of black hole

§7.1.1 Naive definition

It is possible to give a naive and intuitive definition of BH: given a spacetime $(\mathcal{M}, g)$, a **black hole** is “a localized region of spacetime from which neither massive nor massless particles can escape”. This definition has two essential features: *localization* and *inescapability*. The latter implies the existence of a boundary which no particle emitted in the black hole can cross: this boundary is known as the **event horizon** (or simply horizon), and it is a *one-way membrane*, in the sense that it can be crossed from the black hole’s “exterior” towards its “interior”, but not in the reverse way. This means that the horizon must be a hypersurface which divides $\mathcal{M}$ into two regions: the interior (i.e. the BH itself) and the exterior of the black hole.

§7.1.1.1 Black hole boundaries

Recall that, on a Lorentzian d -manifold $(\mathcal{M}, g)$, a hypersurface Σ can be locally classified based on its *causal type*, which is determined by the signature of the restricted metric $g|_{\Sigma}$. The hypersurface Σ is said to be:

- **spacelike** if $g|_{\Sigma}$ is positive definite, i.e. if its signature is $(0, d - 1)$, which means that $(\Sigma, g|_{\Sigma})$ is a Riemannian manifold;

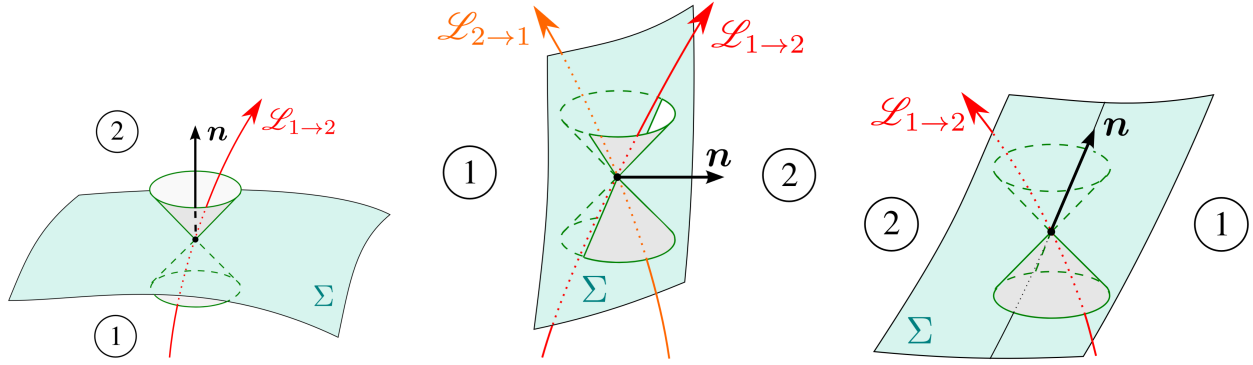


Figure 7.1: The three causal types of hypersurfaces: spacelike (left), timelike (middle) and null (right). Figure from [6].

- **timelike** if $g|_{\Sigma}$ is a Lorentzian metric, i.e. if its signature is $(1, d-2)$, which means that $(\Sigma, g|_{\Sigma})$ is a Lorentzian manifold;
- **null** if $g|_{\Sigma}$ is degenerate, i.e. if its local matrix representation has a vanishing eigenvalue.

All these definitions can be applied locally, i.e. to each point of $p \in \Sigma$, and it is of course possible that Σ has a different causal nature at different points. The causal type of Σ at $p \in \Sigma$ can also be deduced from any normal vector $n_p \in N_p \Sigma$, since the causal type of Σ at p is the opposite of the causal type of n_p , i.e.:

- Σ spacelike at $p \iff n_p$ timelike;
- Σ timelike at $p \iff n_p$ spacelike;
- Σ null at $p \iff n_p$ null;

This can be proved by considering an orthogonal basis adapted to Σ . Note that null hypersurfaces have the distinctive feature that their normals are also tangent to them, as $\langle n, n \rangle_g = 0$ by definition of null vector.

Clearly, a timelike hypersurface is a two-way membrane: it locally divides spacetime in two regions, and a future-directed timelike or null worldline can cross it in both ways, as shown in Fig. 7.1. On the contrary, a spacelike hypersurface is a one-way membrane: a future-directed timelike or null worldline, which is constrained to move inside light cones, can cross it only in the future-directed way. A null hypersurface is also a one-way membrane, since, at most, a null worldline which is not going from the past region to the future region must remain on the hypersurface, as depicted in Fig. 7.1. Therefore, a black hole boundary can in principle be either a null or a spacelike hypersurface: it turns out that, wherever it is smooth, the event horizon is a null hypersurface (Prop. 7.1.7).

§7.1.1.2 Localization

Inescapability is only one of the two naive defining features of black holes, and it alone is not enough to characterize them. Indeed, any region bounded by a null hypersurface cannot be qualified as a “black hole”: for instance, the interior of a future light cone in Minkowski spacetime is such a region. What distinguishes it from a black hole is that any particle moving inside it, although trapped by the light-cone boundary, can travel infinitely far from any “center” in that region.

The necessity of the localization condition is then evident, since it forces the black hole region to have a final spatial extent, so that particles inside it cannot travel to infinitely remote regions. To formalize

this concept in a coordinate-independent way, one introduces a so-called *conformal completion* of the spacetime $(\mathcal{M}, g)$, which brings its infinities to a finite distance in another manifold.

§7.1.2 Conformal completions

The machinery of conformal completions is based on conformal transformations between spacetimes, which turn out to preserve their causal structure, thus allowing to study their “behaviour at infinity” and, eventually, to give a formal definition of black hole.

§7.1.2.1 Conformal transformations

Definition 7.1.1 (Conformal embedding)

Let $(\mathcal{M}, g)$ and $(\tilde{\mathcal{M}}, \tilde{g})$ be two metric manifolds. A **conformal embedding** of $\mathcal{M}$ into $\tilde{\mathcal{M}}$ is a smooth embedding $\Phi : \mathcal{M} \rightarrow \tilde{\mathcal{M}}$ equipped with $\Omega \in \mathcal{C}^\infty(\mathcal{M}, \mathbb{R}^+)$ such that:

$$\tilde{g}|_{\Phi(\mathcal{M})} = \Omega^2 g \quad (7.1.1)$$

Usually, $\Phi(\mathcal{M})$ is simply identified with $\mathcal{M}$. A key property of conformal embeddings is that they preserve orthogonality relations at each point $p \in \mathcal{M}$, since Eq. 7.1.1 clearly implies that $\tilde{g}_p(u, v) = 0 \iff g_p(u, v) = 0$ for any $u, v \in T_p\mathcal{M}$. In particular, for Lorentzian manifolds this means that null vectors for $\tilde{g}$ coincide with null vectors for g , hence the light cones of $(\mathcal{M}, g)$ and $(\mathcal{M}, \tilde{g})$ are identical: this implies that they have the same casual structure.

The same, however, is not true for geodesics, as elucidated with the formalism of Lemma 6.1.1. Denoting as ∇ and $\tilde{\nabla}$ the Levi-Civita connections associated to g and $\tilde{g}$ respectively, they determine a smooth tensor field $C \in \mathfrak{T}_2^1(\mathcal{M})$ which acts on $X, Y \in \mathfrak{X}(\mathcal{M})$ as:

$$C(X, Y) := \tilde{\nabla}_Y X - \nabla_X Y \quad (7.1.2)$$

The components of C in a local frame on $\mathcal{M}$ are given by Eq. 6.1.4. From the conformal relation Eq. 7.1.1 $\nabla_\alpha \tilde{g}_{\mu\nu} = 2\Omega g_{\mu\nu} \nabla_\alpha \log \Omega$, by the metric compatibility of the Levi-Civita connection, thus:

$$C_{\mu\nu}^\alpha = 2\delta_{(\mu}^\alpha \nabla_{\nu)} \log \Omega - g_{\mu\nu} g^{\alpha\beta} \nabla_\beta \log \Omega \quad (7.1.3)$$

Proposition 7.1.1 (Conformal preservation of null geodesics)

Let $(\mathcal{M}, g)$ and $(\mathcal{M}, \tilde{g})$ be two Lorentzian manifolds with conformally-related metrics. Then, a smooth curve γ in $\mathcal{M}$ is a null geodesic for $\tilde{g}$ if and only if it is a null geodesic for g .

Proof. Given a geodesic γ for g , its tangent vector field X satisfied $\nabla_X X = 0$. By Eq. 7.1.2, then:

$$\tilde{\nabla}_X X = \nabla_X X - C(X, X) = C(X, X)$$

which in components becomes:

$$(\nabla_X X)^\alpha = 2X^\alpha \nabla_X \log \Omega - \langle X, X \rangle_g g^{\alpha\beta} \nabla_\beta \log \Omega$$

Therefore, in general geodesics for g are *not* geodesics for $\tilde{g}$, and vice versa. However, if γ is a null geodesic $\langle X, X \rangle_g = 0$, hence the equation above becomes:

$$\nabla_X X = (2\nabla_X \log \Omega) X \quad (7.1.4)$$

which is just the equation for a non-affinely-parametrized geodesic. It can be shown that the affine parameter $\tilde{\lambda}$ of γ for $\tilde{g}$ is related to the affine parameter λ of γ for g by:

$$\frac{d\tilde{\lambda}}{d\lambda} = c\Omega^2$$

for a constant $c \in \mathbb{R}$. □

Having shown that conformal embeddings preserve the causal structure, turn now to the study of curvature.

Lemma 7.1.1 (Conformal transformation of the curvature tensor)

Let $(\mathcal{M}, g)$ and $(\mathcal{M}, \tilde{g})$ be two Lorentzian manifolds with conformally-related metrics. Then, the components $\tilde{\mathcal{R}}_{\mu\nu\alpha}{}^{\beta}$ and $\mathcal{R}_{\mu\nu\alpha}{}^{\beta}$ of the Riemann tensor for $\tilde{g}$ and g , respectively, are related by:

$$\begin{aligned} \tilde{\mathcal{R}}_{\mu\nu\alpha}{}^{\beta} = & \mathcal{R}_{\mu\nu\alpha}{}^{\beta} + 2\delta_{[\mu}^{\beta} \nabla_{\nu]} \nabla_{\alpha} \log \Omega - 2g^{\beta\rho} g_{\alpha[\mu} \nabla_{\nu]} \nabla_{\rho} \log \Omega + 2(\nabla_{[\mu} \log \Omega) \delta_{\nu]}^{\beta} \nabla_{\alpha} \log \Omega + \\ & - 2(\nabla_{[\mu} \log \Omega) g_{\nu]\alpha} g^{\beta\rho} \nabla_{\rho} \log \Omega - 2g_{\alpha[\mu} \delta_{\nu]}^{\beta} g^{\rho\sigma} \nabla_{\rho} \log \Omega \nabla_{\sigma} \log \Omega \end{aligned} \quad (7.1.5)$$

where ∇ is the Levi-Civita connection associated to g .

Contracting 7.1.5, the conformal transformation law of the Ricci scalar curvature is found:

$$\tilde{\mathcal{R}} = \Omega^{-2} [\mathcal{R} - 2(d-1)g^{\mu\nu} \nabla_{\mu} \nabla_{\nu} \log \Omega - (d-2)(d-1)g^{\mu\nu} \nabla_{\mu} \log \Omega \nabla_{\nu} \log \Omega] \quad (7.1.6)$$

where $\tilde{\mathcal{R}} \equiv \tilde{g}^{\mu\nu} \tilde{\mathcal{R}}_{\mu\nu}$ and $\mathcal{R} \equiv g^{\mu\nu} \mathcal{R}_{\mu\nu}$. It is convenient to define the “trace-free part” of the Riemann tensor by subtracting all its possible contractions, yielding the **Weyl tensor** $\text{Weyl} \in \mathfrak{T}_3^1(\mathcal{M})$, which has components:

$$\mathcal{W}_{\mu\nu\alpha\beta} := \mathcal{R}_{\mu\nu\alpha\beta} + \frac{2}{d-2} (g_{\mu[\alpha} \mathcal{R}_{\beta]\nu} - g_{\nu[\alpha} \mathcal{R}_{\beta]\mu}) - \frac{2}{(d-1)(d-2)} g_{\mu[\alpha} g_{\beta]\nu} \mathcal{R} \quad (7.1.7)$$

This expression is only defined for $d \geq 3$, as for $d = 1, 2$ the Riemann tensor is trivial (recall Eq. 3.3.31): in $d = 2$ it has a single independent component, given by the Ricci scalar, while in $d = 1$ it vanishes identically. The Weyl tensor has the same symmetries Eq. 2.5.11 of the Riemann tensor, with the additional property of being trace-free in any pair of its indices. Moreover, a crucial property of Weyl is that it is conformally invariant, as can be shown from Eq. 7.1.5:

$$\tilde{\mathcal{W}}_{\mu\nu\alpha}{}^{\beta} = \mathcal{W}_{\mu\nu\alpha}{}^{\beta}$$

For this reason, Weyl is often called the *conformal (curvature) tensor*. Note that, since indices are naturally raised and lowered with different metrics, the conformal invariance relation depends on the position of the indices: e.g. $\tilde{\mathcal{W}}_{\beta\mu\nu\alpha} \equiv \tilde{g}_{\beta\rho} \tilde{\mathcal{W}}^{\rho}_{\mu\nu\alpha} = \Omega^2 g_{\beta\rho} \mathcal{W}^{\rho}_{\mu\nu\alpha} \equiv \Omega^2 \mathcal{W}_{\beta\mu\nu\alpha}$. The Weyl tensor is then linked to tidal effects, which are indeed conformally invariant, since they remain unchanged when rescaling the volume.

Observation 7.1.1 (Petrov classification)

In the absence of matter, the Einstein field equations nullify the Ricci tensor, hence the only independent components of the Riemann tensor in vacuum are given by the Weyl tensor. For this reason the Weyl tensor is used to classify the vacuum solutions of the Einstein field equations: in particular, in $d = 4$ it is possible to take advantage of the symmetry of Hodge operator $\star$:

$\Omega^2(\mathcal{M}) \rightarrow \Omega^2(\mathcal{M})$, obtaining what is known as the **Petrov classification** of vacuum solutions.

§7.1.2.2 Conformal completion of Minkowski spacetime

Before defining conformal completions of arbitrary spacetimes, it is instructive to first build intuition with the special case of $d = 4$ Minkowski spacetime $(\mathcal{M}, \eta)$.

To study the “far-away” behaviour of Minkowski spacetime, it is convenient to work in spherical coordinates $(t, r, \vartheta, \varphi)$, which are related to the global Cartesian coordinates by:

$$x = r \sin \vartheta \cos \varphi \quad y = r \sin \vartheta \sin \varphi \quad z = r \cos \vartheta$$

These coordinates span $\mathbb{R} \times \mathbb{R}^+ \times (0, \pi) \times (0, 2\pi)$, but they are not a smooth chart on the whole $\mathcal{M}$, but only on $\mathcal{M} - \Pi$, where Π is the closed half-hyperplane defined by $x \geq 0$ and $y = 0$. In these coordinates, the Lorentz–Minkowski metric takes the form:

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega_2^2$$

On the non-angular part $\mathbb{R} \times \mathbb{R}^+$, introduce null coordinates (u, v) , respectively the retarded and advanced time, defined by:

$$u = t - r \quad v = t + r$$

These coordinates span the half-part of $\mathbb{R}^2$ defined by $u < v$, and the metric becomes:

$$ds^2 = -du dv + \frac{1}{4}(v - u)^2 d\Omega_2^2$$

In order to perform the conformal completion, consider the finite-range coordinates defined by¹:

$$U = \arctan u \quad V = \arctan v$$

The arctan function is monotonically increasing, so (U, V) span the half-part of $(-\frac{\pi}{2}, \frac{\pi}{2}) \times (-\frac{\pi}{2}, \frac{\pi}{2})$ defined by $U < V$. Using standard trigonometric relations, the metric takes the form:

$$ds^2 = \frac{1}{4 \cos^2 U \cos^2 V} [-4dU dV + \sin^2(V - U) d\Omega_2^2]$$

Therefore, the Lorentz–Minkowski metric is conformal to a simpler metric:

$$d\bar{s}^2 = -4dU dV + \sin^2(V - U) d\Omega_2^2 \quad (7.1.8)$$

with conformal factor $\Omega : (-\frac{\pi}{2}, \frac{\pi}{2}) \times (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}^+ : \Omega(U, V) = 2 \cos U \cos V$. The coordinates (U, V) are of null type, so consider time-space coordinates (τ, χ) defined by:

$$\tau = V + U \quad \chi = V - U$$

These range of these coordinates is $\chi \in (0, \pi)$ and $\tau \in (\chi - \pi, \pi - \chi)$, hence drawing the Minkowski spacetime in the (τ, χ) results in a half-diamond shape (see Fig. 7.2). Combining all the above coordinate transformations, it is possible to link (t, r) to (τ, χ) directly, obtaining:

$$t = \frac{\sin \tau}{\cos \tau + \cos \chi} \quad r = \frac{\sin \chi}{\cos \tau + \cos \chi} \quad (7.1.9)$$

¹To be precise, note that (u, v) have dimensions of time, i.e. of length, hence finite-range coordinates should be defined as $U = \arctan(u/\ell_0)$ and $V = \arctan(v/\ell_0)$, for some length scale ℓ_0 . However, this length scale is completely arbitrary and is eliminated by the conformal transformation, hence it is implicitly set to $\ell_0 \equiv 1$.

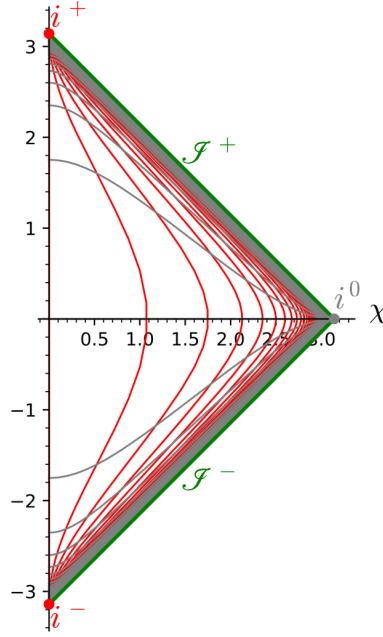


Figure 7.2: Conformal diagram of Minkowski spacetime. Curves $r = \text{const.}$ are drawn in red, while curved $r = \text{const.}$ are drawn in grey. Figure from [6].

Inserting these relations in the Lorentz–Minkowski metric, or directly from Eq. 7.1.8, the conformal metric in the $(\tau, \chi, \vartheta, \varphi)$ coordinates is found to be:

$$d\tilde{s}^2 = -d\tau^2 + d\chi^2 + \sin^2 \chi d\Omega_2^2 \quad (7.1.10)$$

The spatial part of the metric is just the standard metric on $\mathbb{S}^3$ in hyperspherical coordinates $(\chi, \vartheta, \varphi)$. Moreover, the full metric is regular even upon relaxing the condition on τ , letting it span the entire $\mathbb{R}$. Therefore, Eq. 7.1.10 is truly a metric $\tilde{g}$ on $\mathcal{E} := \mathbb{R} \times \mathbb{S}^3$, and $(\mathcal{E}, \tilde{g})$ is a spacetimes known as the **Einstein static universe** (ESU), which is a solution to the Einstein field equations with $\Lambda > 0$ and a cosmological fluid with $w = 0$ and $\rho = \Lambda/(4\pi)$ (see §8.3.1.3 for details on cosmological fluids). Eq. 7.1.9 are thus a conformal embedding of $(\mathfrak{M}, \eta)$ into $(\mathcal{E}, \tilde{g})$.

Since $\mathfrak{M}$ and $\mathcal{E}$ have the same dimension, $\mathfrak{M} \in \mathcal{T}(\mathcal{E})$, and its topological closure in $\mathcal{E}$ is:

$$\overline{\mathfrak{M}} = \mathfrak{M} \sqcup \mathcal{J}^+ \sqcup \mathcal{J}^- \sqcup \{i^0\} \sqcup \{i^+\} \sqcup \{i^-\} \quad (7.1.11)$$

as shown in Fig. 7.2, where the various types of infinity are:

- **future null infinity** $\mathcal{J}^+ \subseteq \mathcal{E}$, which is the hypersurface defined by $\tau = \pi - \chi$ and $\tau \in (0, \pi)$, i.e. $V = \frac{\pi}{2}$ and $U \in (-\frac{\pi}{2}, \frac{\pi}{2})$;
- **past null infinity** $\mathcal{J}^- \subseteq \mathcal{E}$, which is the hypersurface defined by $\tau = \chi - \pi$ and $\tau \in (-\pi, 0)$, i.e. $V = -\frac{\pi}{2}$ and $V \in (-\frac{\pi}{2}, \frac{\pi}{2})$;
- **spacelike infinity** $i^0 \in \mathcal{E}$, which is defined by $(\tau, \chi) = (0, \pi)$, i.e. $(U, V) = (-\frac{\pi}{2}, \frac{\pi}{2})$;
- **future timelike infinity** $i^+ \in \mathcal{E}$, which is defined by $(\tau, \chi) = (\pi, 0)$, i.e. $(U, V) = (\frac{\pi}{2}, \frac{\pi}{2})$;
- **past timelike infinity** $i^- \in \mathcal{E}$, which is defined by $(\tau, \chi) = (-\pi, 0)$, i.e. $(U, V) = (-\frac{\pi}{2}, -\frac{\pi}{2})$.

Note that, since hyperspherical coordinates $(\chi, \vartheta, \varphi)$ on $\mathbb{S}^3$ are singular at $\chi = 0, \pi$, the conditions for i^0 and $i^\pm$ do indeed define unique points of $\mathcal{E}$, whatever the value of $(\tau, \varphi, \vartheta)$, and not hypersurfaces. Moreover, the vertical left boundary of the half-diamond shape in Fig. 7.2, i.e. the segment

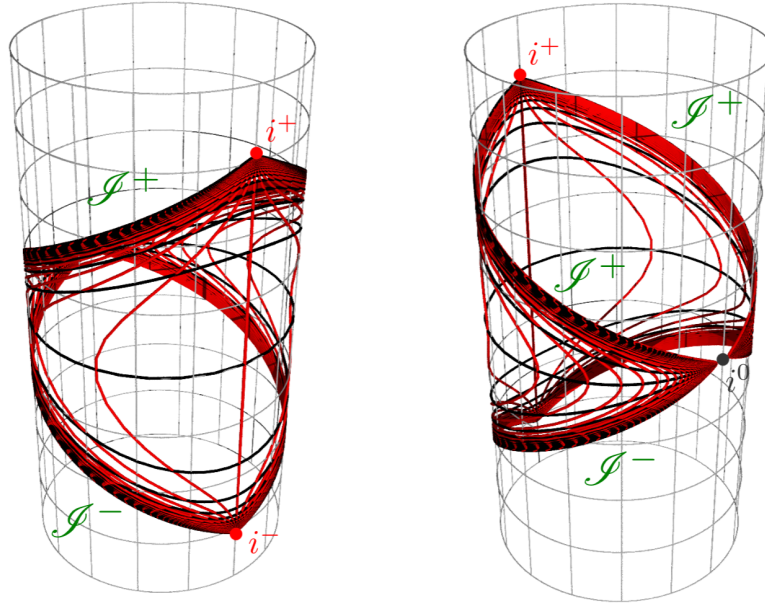


Figure 7.3: Two views of a 2-dimensional section of $\mathcal{E}$ at $\vartheta = \frac{\pi}{2}$ and $\varphi = \{0, \pi\}$ (the “right” and “left” half of each cylinder, respectively). Again, red curves are $r = \text{const.}$ curves and grey curves are $t = \text{const.}$ ones. Figure from [6].

defined by $\chi = 0$ and $\tau \in (-\pi, \pi)$, is not part of the boundary of $\mathfrak{M}$, but merely reflects the coordinate singularity at $\chi = 0$, which corresponds to the usual coordinate singularity at $r = 0$ in ordinary spherical coordinates.

Fig. 7.3 shows the pathological nature of i^0 : indeed, $\overline{\mathfrak{M}}$ is self-intersecting at i^0 , so there is no open neighbourhood of i^0 that is homeomorphic to a neighbourhood in $\mathbb{U}^4$, thus it is not possible to endow it with the structure of manifold with boundary (Def. 1.1.9). On the other hand, $i^\pm$ are points where the boundary is not smooth, hence $\mathfrak{M} - \{i^0\}$ can only be defined as a topological manifold with boundary, not a smooth one. For this reason, let $\mathcal{I} \equiv \mathcal{I}^+ \sqcup \mathcal{I}^-$ and $\tilde{\mathcal{M}} := \mathfrak{M} \sqcup \mathcal{I} \subseteq \mathcal{E}$: then, $\tilde{\mathcal{M}}$ is naturally a smooth manifold with boundary, and its manifold boundary is $\partial\tilde{\mathcal{M}} = \mathcal{I}$.

The nature of the various infinities of $\mathfrak{M}$ is determined by the behaviour of its geodesics: all null geodesics start on $\mathcal{I}^-$ and end on $\mathcal{I}^+$, all timelike geodesics start at i^- and end at i^+ , and all spacelike geodesics start and end at i^0 , as shown in Fig. 7.2.

Proposition 7.1.2 (Null conformal boundary of Minkowski spacetime)

Given the spacetime $(\tilde{\mathcal{M}}, \tilde{g})$ defined above, $\mathcal{I}^+$ and $\mathcal{I}^-$ are null hypersurfaces of $(\tilde{\mathcal{M}}, \tilde{g})$.

Proof. Consider first $\mathcal{I}^+$: it is defined by $V = \frac{\pi}{2}$, so a valid coordinate system on it is (U, ϑ, φ) with $U \in (-\frac{\pi}{2}, \frac{\pi}{2})$. The induced metric $\tilde{g}|_{\mathcal{I}^+}$ is easily obtained setting $V = \frac{\pi}{2}$ in Eq. 7.1.8:

$$d\tilde{s}^2|_{\mathcal{I}^+} = \cos^2 U \, d\Omega_2^2$$

This metric clearly has degenerate signature $(0, +, +)$, hence $\mathcal{I}^+$ is a null hypersurface of $(\tilde{\mathcal{M}}, \tilde{g})$. Similarly, $\mathcal{I}^-$ is defined by $U = -\frac{\pi}{2}$, so a valid coordinate system on it is (V, ϑ, φ) with $V \in (-\frac{\pi}{2}, \frac{\pi}{2})$, and the induced metric $\tilde{g}|_{\mathcal{I}^-}$ is:

$$d\tilde{s}^2|_{\mathcal{I}^-} = \cos^2 V \, d\Omega_2^2$$

This metric has degenerate signature $(0, +, +)$, hence $\mathcal{I}^-$ is a null hypersurface of $(\tilde{\mathcal{M}}, \tilde{g})$. $\square$

The null character of $\mathcal{I}^\pm$ also appears clearly in the conform diagram Fig. 7.2.

§7.1.2.3 Conformal completion of arbitrary spacetimes

Having elucidated the asymptotic structure of Minkowski spacetime, it is now possible to generalize the construction of conformal completions to arbitrary spacetimes.

Definition 7.1.2 (Conformal completion)

A spacetime $(\mathcal{M}, g)$ admits a **conformal completion** if there exist a Lorentzian manifold with boundary $(\tilde{\mathcal{M}}, \tilde{g})$ and a conformal embedding $\Phi : \mathcal{M} \rightarrow \tilde{\mathcal{M}}$ with $\Omega \in \mathcal{C}^\infty(\mathcal{M}, \mathbb{R}^+)$ such that:

- (1) $\tilde{\mathcal{M}} = \Phi(\mathcal{M}) \sqcup \mathcal{I}$, with $\mathcal{I} := \partial\tilde{\mathcal{M}}$;
- (2) $\Omega|_{\mathcal{I}} \equiv 0$ and $d\Omega|_{\mathcal{I}} \neq 0$.

$\mathcal{I}$ is called the **conformal infinity** (or *conformal boundary*) of $(\mathcal{M}, g)$ within $(\tilde{\mathcal{M}}, \tilde{g})$.

Note that $\mathcal{I}$ is the manifold boundary of $\tilde{\mathcal{M}}$, not the topological boundary of $\tilde{\mathcal{M}}$ (which is $\emptyset$, recall Obs. 1.1.1). Moreover, it is not part of the physical spacetime $\mathcal{M}$, since $\mathcal{I} \cap \mathcal{M} = \emptyset$.

Condition (1) expresses that $\mathcal{M}$ is endowed with some boundary by the conformal embedding Φ , which in the following is suppressed setting $\Phi(\mathcal{M}) \equiv \mathcal{M}$. Condition (2) expresses instead that this boundary, which lies “lies at an infinite distance” with respect to g , is brought to a finite distance with respect to $\tilde{g}$ by the conformal transformation, which in terms of length elements reads:

$$ds^2 = \Omega^{-2} d\tilde{s}^2$$

The first condition (2) clarifies the “lies at infinite distance” notion, since $\Omega^{-2} \rightarrow +\infty$ approaching $\mathcal{I}$, while the second condition (2) ensures that $\mathcal{I}$ is a regular hypersurface of $\tilde{\mathcal{M}}$.

In general, unlike the caso of Minkowski spacetime, the conformal boundary $\mathcal{I}$ is not necessarily a null hypersurface. Actually, the causal type of $\mathcal{I}$ is determined by the cosmological constant.

Proposition 7.1.3 (Causal type of conformal boundaries)

Let the spacetime $(\mathcal{M}, g)$ be a solution to the Einstein field equation with a cosmological constant Λ , and let it be conformal to a spacetime $(\tilde{\mathcal{M}}, \tilde{g})$ with conformal factor $\Omega \in \mathcal{C}^\infty(\mathcal{M}, \mathbb{R}^+)$ and conformal boundary $\mathcal{I} := \partial\tilde{\mathcal{M}}$. If the trace of the EMT is $T \rightarrow 0$ as $\Omega \rightarrow 0$, then $\mathcal{I}$ is a:

- (1) null hypersurface of $(\tilde{\mathcal{M}}, \tilde{g})$ if and only if $\Lambda = 0$;
- (2) spacelike hypersurface of $(\tilde{\mathcal{M}}, \tilde{g})$ if and only if $\Lambda > 0$;
- (3) timelike hypersurface of $(\tilde{\mathcal{M}}, \tilde{g})$ if and only if $\Lambda < 0$.

Proof. From Eq. 7.1.6, inverting the roles of $\tilde{g}$ and g using the conformal relation Eq. 7.1.1 and the difference tensor Eq. 7.1.3, it is possible to relate the Ricci scalar curvatures $\tilde{\mathcal{R}}$ and $\mathcal{R}$ of respectively $\tilde{g}$ and g as:

$$\Omega^2 \tilde{\mathcal{R}} = \mathcal{R} - (d-1) \tilde{g}^{\mu\nu} \left[2\Omega \tilde{\nabla}_\mu \tilde{\nabla}_\nu \Omega - 8\partial_\mu \Omega \partial_\nu \Omega \right] \quad (7.1.12)$$

where $\tilde{\nabla}$ is the Levi-Civita connection associated to $\tilde{g}$. Using Eq. 5.2.12 to express $\mathcal{R}$, the relation becomes:

$$\Omega^2 \tilde{\mathcal{R}} = \frac{2}{d-2} \left[d\Lambda - 8\pi G_N^{(d)} T \right] - (d-1) \tilde{g}^{\mu\nu} \left[2\Omega \tilde{\nabla}_\mu \tilde{\nabla}_\nu \Omega - 8\partial_\mu \Omega \partial_\nu \Omega \right]$$

This equation is, in principle, valid only on $\mathcal{M} = \tilde{\mathcal{M}} - \mathcal{I}$. Taking the limit $\Omega \rightarrow 0$, with the assumption that $T \rightarrow 0$ as $\Omega \rightarrow 0$, by continuity it becomes an identity on $\mathcal{I}$:

$$\tilde{g}^{\mu\nu} \partial_\mu \Omega \partial_\nu \Omega|_{\mathcal{I}} = -\frac{d}{4(d-1)(d-2)} \Lambda$$

Now, note that $\mathcal{I}$ corresponds to a constant value of Ω , namely $\Omega = 0$, therefore $\partial_\mu \Omega$ are just the components of the normal vector field $n \in \Gamma(N\mathcal{I})$ to $\mathcal{I}$, which is defined as the metric dual of the 1-form $d\Omega = \partial_\mu \Omega dx^\mu$ (which is non-vanishing by condition (2) of Def. 7.1.2). This means that the left-hand side of the above relation is just $\langle n, n \rangle_g$, and it constraints the sign of $\langle n, n \rangle_g$ to be the opposite of the sign of Λ . Given the link between the causal type of a hypersurface and the causal type of its normal vector field, this completes the proof. $\square$

To generalize the classification of infinities performed in Minkowski spacetime, let $(\mathcal{M}, g)$ be a time-orientable spacetime admitting a conformal completion $(\tilde{\mathcal{M}}, \tilde{g})$ with conformal boundary $\mathcal{I}$. Then, it is possible to distinguish two parts of the conformal boundary: the **future infinity** of $(\mathcal{M}, g)$ is the subset $\mathcal{I}^+ \subseteq \mathcal{I}$ whose points can be reached from any point in $\mathcal{M}$ by a future-directed causal curve in $\tilde{\mathcal{M}}$; while the **past infinity** of $(\mathcal{M}, g)$ is the subset $\mathcal{I}^- \subseteq \mathcal{I}$ whose points can be reached from any point in $\mathcal{M}$ by a past-directed causal curve in $(\tilde{\mathcal{M}})$. If $\mathcal{I}^+$ (or $\mathcal{I}^-$) is a null hypersurface, then it is called the *future* (or *past*) *null infinity*. If $\mathcal{I}^\pm$ are null hypersurfaces and $\mathcal{I} = \mathcal{I}^+ \cup \mathcal{I}^-$, then $(\tilde{\mathcal{M}}, \tilde{g})$ is called a **conformal completion at null infinity** of $(\mathcal{M}, g)$.

Example 7.1.1 (Conformal completion of AdS_4)

The 4-dimensional anti-de Sitter spacetime AdS_4 is $(\mathcal{M}, g)$, with $\mathcal{M} \cong \mathbb{R}^4$. The metric in the so-called *global static coordinates* $(\tau, r, \vartheta, \varphi)$ has components:

$$ds^2 = \ell^2 \left[-(1+r^2)d\tau^2 + \frac{1}{1+r^2}dr^2 + r^2 d\Omega_2^2 \right]$$

where $\ell \in \mathbb{R}^+$ is a constant. Note that $(\tau, r) \in \mathbb{R} \times \mathbb{R}^+$ are dimensionless coordinates, while $(\vartheta, \varphi) \in (0, \pi) \times (0, 2\pi)$ are the standard spherical coordinates on $\mathbb{S}^2$. This metric is a solution to the vacuum Einstein field equations with $\Lambda = -3/\ell^2$, and using the so-called *conformal coordinates* $(\tau, \chi, \vartheta, \varphi)$, with $\chi = \arctan r \in (0, \frac{\pi}{2})$, it becomes:

$$ds^2 = \frac{\ell^2}{\cos^2 \chi} [-d\tau^2 + d\chi^2 + \sin^2 \chi d\Omega_2^2]$$

Note that $\chi \in (0, \frac{\pi}{2})$ only spans one half of the hypersphere $\mathbb{S}^3$, hence $\Omega = \ell^{-1} \cos \chi$ determines a conformal completion of $(\mathcal{M}, g)$ into $(\tilde{\mathcal{M}}, \tilde{g})$, where $\tilde{\mathcal{M}}$ is the part of the ESU $\mathcal{E}$ defined by $\chi \leq \frac{\pi}{2}$, and $\tilde{g}$ is the standard ESU metric Eq. 7.1.10.

The conformal boundary $\mathcal{I} = \partial\tilde{\mathcal{M}}$ is then the hypersurface $\chi = \frac{\pi}{2}$, as shown in Fig. 7.4: its is spanned by the coordinates $(\tau, \vartheta, \varphi)$, hence its topology is $\mathcal{I} \cong \mathbb{R} \times \mathbb{S}^2$, and it is a timelike hypersurface of $(\tilde{\mathcal{M}}, \tilde{g})$, as the induced 3-metric is clearly Lorentzian (confirming Prop. 7.1.3). By the behaviour of radial null geodesics (plotted in Eq. 7.4), it is evident that any point of $\mathcal{I}$ can be connected to $\mathcal{M}$ by a future-directed null geodesics as well as by a past-directed one, therefore $\mathcal{I} = \mathcal{I}^+ = \mathcal{I}^-$, so that $\mathcal{I}^\pm$ are timelike hypersurfaces.

Example 7.1.2 (Conformal completion of dS_4)

The 4-dimensional de Sitter spacetime dS_4 is $(\mathcal{M}, g)$, with $\mathcal{M} \cong \mathbb{R} \times \mathbb{S}^3$. The metric in global

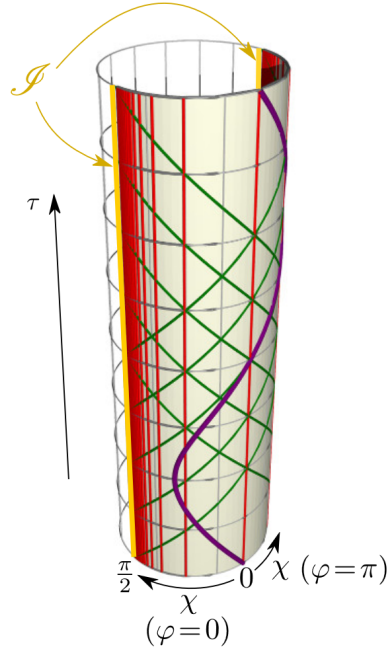


Figure 7.4: Conformal completion of AdS_4 spacetime. The conformal boundary $\mathcal{S}$ is depicted in yellow, red lines are $\chi = \text{const.}$ lines (uniformly sampled in terms of $\tan \chi = \sinh \rho$), green curves are radial null geodesics, and the purple curve is a radial timelike geodesic bouncing around $\chi = 0$. Figure from [6].

coordinates $(t, \chi, \vartheta, \varphi)$ has components:

$$ds^2 = \ell^2 [-dt^2 + \cosh^2 t (d\chi^2 + \sin^2 \chi d\Omega_2^2)]$$

where $\ell \in \mathbb{R}^+$ is a constant. Note that $t \in \mathbb{R}$ is a dimensionless coordinate, while $(\chi, \vartheta, \varphi) \in (0, \pi)^2 \times (0, 2\pi)$ are the standard hyperspherical coordinates on $\mathbb{S}^3$. This metric is a solution to the vacuum Einstein field equations with $\Lambda = 3/\ell^2$, and using coordinates $(\tau, \chi, \vartheta, \varphi)$, with $\tau = 2 \arctan \tanh \frac{t}{2} \in (-\frac{\pi}{2}, \frac{\pi}{2})$, it becomes:

$$ds^2 = \frac{\ell^2}{\cos^2 \tau} [-d\tau^2 + d\chi^2 + \sin^2 \chi d\Omega_2^2]$$

Defining $\Omega = \ell^{-1} \cos \tau = (\ell \cosh t)^{-1}$ determines a conformal completion of $(\mathcal{M}, g)$ into $(\tilde{\mathcal{M}}, \tilde{g})$, where $\tilde{\mathcal{M}}$ is the part of the ESU $\mathcal{E}$ defined by $-\frac{\pi}{2} \leq \tau \leq \frac{\pi}{2}$, and $\tilde{g}$ is Eq. 7.1.10.

The conformal boundary $\mathcal{S} = \partial \tilde{\mathcal{M}}$, found by $\Omega|_{\mathcal{S}} = 0$, has two connected components: $\mathcal{S}^\pm$, which is the hypersurface $\tau = \pm \frac{\pi}{2}$ of $\tilde{\mathcal{M}}$. Both are spanned by the coordinates $(\chi, \vartheta, \varphi)$, hence their topology is $\mathcal{S}^\pm \cong \mathbb{S}^3$, and they are both spacelike hypersurfaces of $(\tilde{\mathcal{M}}, \tilde{g})$, as the induces 3-metric is clearly Riemannian (confirming Prop. 7.1.3). Finally, it is trivial to see that $\mathcal{S}^+$ and $\mathcal{S}^-$ respectively match the definitions of future and past infinity, hence $(\tilde{\mathcal{M}}, \tilde{g})$ is a conformal completion at infinity of dS_4 .

§7.1.2.4 Asymptotic flatness

Following the conventions of Penrose [15], a spacetime $(\mathcal{M}, g)$ is **asymptotically simple** if it admits a conformal completion such that every null geodesic in $\mathcal{M}$ has two endpoints in $\mathcal{S}$.

This condition, although satisfied by Minkowski, de Sitter and anti-de Sitter spacetimes, is rather

restrictive: in particular, it excludes black hole spacetimes, since by definition they contain null geodesics that have no endpoint on $\mathcal{I}^+$. For this reason, define a spacetime $(\mathcal{M}, g)$ to be **weakly asymptotically simple** if there exist $U \in \mathcal{T}(\mathcal{M})$ and an asymptotically simple spacetime $(\mathcal{M}_0, g_0)$ with an open neighbourhood $U_0 \subseteq \tilde{\mathcal{M}}_0$ of $\mathcal{I}_0 = \partial\tilde{\mathcal{M}}_0$ such that $(U_0 \cap \mathcal{M}_0, g_0)$ is isometric to (U, g) (recall Def. 2.1.3). Of course, for a given weakly asymptotically simple spacetime, there may be different non-overlapping regions U satisfying this property (e.g. in the maximally extended Kerr spacetime).

Definition 7.1.3 (Asymptotically flat spacetime)

A spacetime $(\mathcal{M}, g)$ is **asymptotically flat** if it is weakly asymptotically simple and the Ricci tensor of g vanishes in an open neighbourhood of $\mathcal{I}$.

By this definition, Minkowski spacetime is asymptotically simple, while de Sitter and anti-de Sitter spacetimes are not. Penrose [14] has shown that if $(\mathcal{M}, g)$ is asymptotically simple and empty, then the Weyl tensor of g vanishes at $\mathcal{I}$, which implies that $R|_{\mathcal{I}} \equiv 0$: this explains the nomenclature of *asymptotically flat*.

Proposition 7.1.4 (Null boundary of asymptotically flat spacetimes)

The conformal boundary $\mathcal{I}$ of an asymptotically flat spacetime $(\mathcal{M}, g)$ is a null hypersurface of the conformal completion $(\tilde{\mathcal{M}}, \tilde{g})$.

Proof. Recall Eq. 7.1.12. $\mathcal{R} = 0$ in an open neighbourhood of $\mathcal{I}$, by Def. 7.1.3, so the limit $\Omega \rightarrow 0$ results into $\langle n, n \rangle_g = 0$: by the same argument of Prop. 7.1.3, $\mathcal{I}$ is a null hypersurface. $\square$

§7.1.3 Formal definition

§7.1.3.1 Causal structure

Let $(\mathcal{M}, g)$ be a time-orientable spacetime. For any subset $S \subseteq \mathcal{M}$, define:

- the **chronological future** (or **past**) of S as the subset $I^\pm(S) \subseteq \mathcal{M}$ of points that can be reached from a point of S by a future-directed (or past-directed) timelike curve of non-zero extent;
- the **causal future** (or **past**) of S as the subset $J^\pm(S) \subseteq \mathcal{M}$ of points that either are in S or can be reached from a point of S by a future-directed (or past-directed) causal curve (recall Def. 6.2.2).

From these definitions, it is clear that $S \subseteq J^\pm(S)$ and $I^\pm(S) \subseteq J^\pm(S)$. On the contrary, not necessarily $S \subseteq I^\pm(S)$: for instance, if $\mathcal{M}$ does not contain any closed timelike curve, then $S \cap I^\pm(S) = \emptyset$ for $S = \{p\}$, with $p \in \mathcal{M}$.

Lemma 7.1.2 (Topological properties of causal structures)

Let $(\mathcal{M}, g)$ be a time-orientable spacetime, and let $S \subseteq \mathcal{M}$. Then:

- (1) $I^\pm(S) \in \mathcal{T}(\mathcal{M})$, while $J^\pm(S)$ is not necessarily closed;
- (2) $\text{Int } J^\pm(S) = I^\pm(S)$;
- (3) $\overline{J^\pm(S)} = \overline{I^\pm(S)}$;

| *Proof.* See e.g. §6.2 of [8] or Chap. 14 of [12]. □

By (2) and (3) of Lemma 7.1.2, it follows that the topological boundaries² are $\partial J^\pm(S) = \partial I^\pm(S)$. A spacetime $(\mathcal{M}, g)$ is then said to be **causally simple** if $J^\pm(K) \subseteq \mathcal{M}$ are closed for every compact subset $K \subseteq \mathcal{M}$.

If the spacetime $(\mathcal{M}, g)$ admits a conformal completion $(\tilde{\mathcal{M}}, \tilde{g})$ with conformal infinity $\mathcal{I}$, then, by the above definitions, the future and past infinity of $\mathcal{M}$ are expressible as:

$$\mathcal{I}^\pm = \mathcal{I} \cap J^\pm(\mathcal{M}) \quad (7.1.13)$$

Note that $J^\pm(\mathcal{M})$ are the causal future and causal past of $\mathcal{M}$ considered as a subset of $\tilde{\mathcal{M}}$.

§7.1.3.2 Black holes

Consider a spacetime $(\mathcal{M}, g)$ which admits a conformal completion with conformal boundary $\mathcal{I}$: then, the future infinity $\mathcal{I}^+$ is **complete** if it is a null hypersurface such that all its generators are complete null geodesics³. The neighbourhood of $\mathcal{I}^+$ in $\tilde{\mathcal{M}}$ can be considered as the infinitely-far region reached by outgoing null geodesics: hence, if a null geodesic does not reach this region, it can be considered as trapped somewhere else in spacetime, i.e. in a black hole region.

Definition 7.1.4 (Black hole)

Let $(\mathcal{M}, g)$ be a time-oriented spacetime with a conformal completion such that its future infinity $\mathcal{I}^+$ is complete. A **black hole** is the set of points of $\mathcal{M}$ that are not in the causal past of $\mathcal{I}^+$:

$$\mathcal{B} := \mathcal{M} - (J^-(\mathcal{I}^+) \cap \mathcal{M}) \quad (7.1.14)$$

The topological boundary $\mathcal{H}^+ := \partial \mathcal{B}$ is called the **future event horizon** (or simply *event horizon*, if no ambiguity arises).

From Eq. 7.1.14, a black hole is the set of points of $\mathcal{M}$ from which no future-directed causal curve in $\tilde{\mathcal{M}}$ reaches $\mathcal{I}^+$. A useful expression for the future event horizon can be found recalling the topological identity $\partial \mathcal{B} = \overline{\mathcal{B} \cap \mathcal{M}} - \mathcal{B}$:

$$\mathcal{H}^+ = \overline{\mathcal{M} - (J^-(\mathcal{I}^+) \cap \mathcal{M})} \cap \overline{(J^-(\mathcal{I}^+) \cap \mathcal{M})} = \partial(J^-(\mathcal{I}^+) \cap \mathcal{M})$$

Now, the boundary of $J^-(\mathcal{I}^+)$ in $\tilde{\mathcal{M}}$ is $\partial J^-(\mathcal{I}^+) = \partial(J^-(\mathcal{I}^+) \cap \mathcal{M}) \cup \mathcal{I}^+$, so $\partial J^-(\mathcal{I}^+) \cap \mathcal{M} = \partial(J^-(\mathcal{I}^+) \cap \mathcal{M})$ and:

$$\mathcal{H}^+ = \partial J^-(\mathcal{I}^+) \cap \mathcal{M} \quad (7.1.15)$$

This means that the future event horizon is the part of the boundary of the causal past of the future infinity $\mathcal{I}^+$ that lies in $\mathcal{M}$, as illustrated in Fig. 7.5. From Lemma 7.1.2, $\mathcal{H}^+ = \partial I^-(\mathcal{I}^+) \cap \mathcal{M}$.

Example 7.1.3 (No black hole in Minkowski spacetime)

The Minkowski spacetime $(\mathfrak{M}, \eta)$ contains no black hole, since all future-directed null geodesics terminate in $\mathcal{I}^+$: this implies that $\mathcal{M} \subseteq J^-(\mathcal{I}^+)$, so that $\mathcal{B} = \emptyset$. More generally, any asymptotically simple spacetime contains no black hole, as evident from definition (see §7.1.2.4).

If one relaxes the assumption of $\mathcal{I}^+$ being complete in Def. 7.1.4, unphysical or “spurious” black holes emerge. For instance, consider the conformal completion of Minkowski spacetime $(\mathcal{M}, g)$

²Recall that, given a topological space $\mathcal{X}$, the *topological boundary* of $S \subseteq \mathcal{X}$ is $\partial S := \overline{S} - \text{Int } S$.

³A geodesic is *complete* if its affine parameters range on the whole $\mathbb{R}$. In particular, complete geodesics are index-tendible.

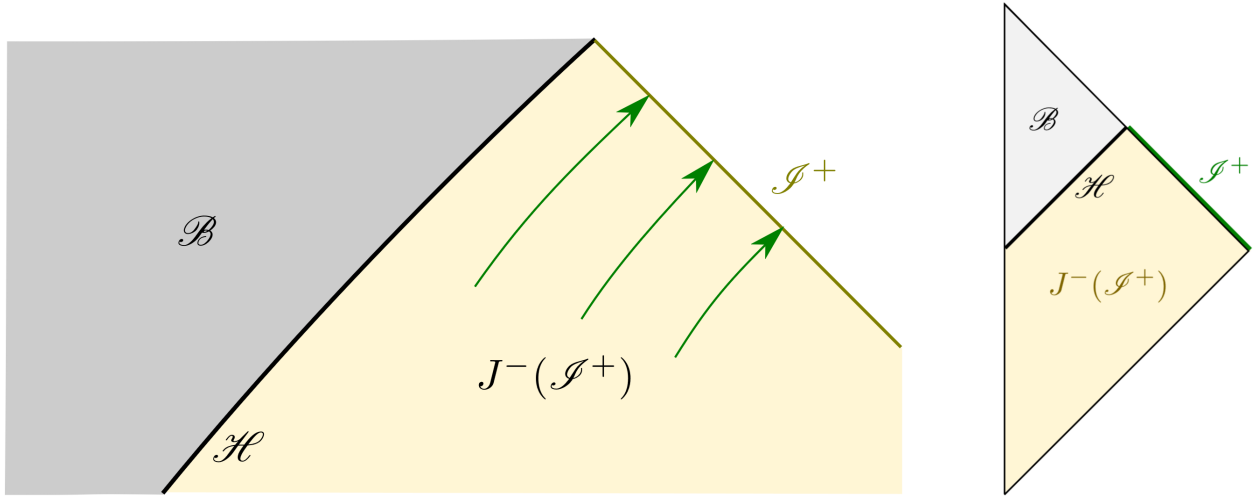


Figure 7.5: On the left, a black hole $\mathcal{B}$ defined as the complement of the causal past of the future infinity, $J^-(\mathcal{I}^+)$. On the right, a spurious black hole $\mathcal{B}$ in Minkowski spacetime resulting from a conformal completion with an incomplete $\mathcal{I}^+$. Figures from [6].

from its embedding $\Phi : \mathcal{M} \rightarrow \mathcal{E}$ into the ESU (Eq. 7.1.9), but define now $\mathcal{I}^+$ as the hypersurface of $\mathcal{E}$ given by $\tau = \pi - \chi$ and $\tau \in (0, \frac{\pi}{2})$, instead of $\tau \in (0, \pi)$, while keeping $\mathcal{I}^-$ unchanged: the manifold with boundary $\tilde{\mathcal{M}} \equiv \mathcal{M} \cup \mathcal{I}^+ \cup \mathcal{I}^-$, equipped with the metric Eq. 7.1.10, is then a conformal completion at null infinity of $(\mathcal{M}, g)$. With such a $\mathcal{I}^+$, the black hole Eq. 7.1.14 is non-empty, as shown in Fig. 7.5.

Example 7.1.4 (Schwarzschild black hole)

The prototype of a black hole is the Schwarzschild black hole, which is studied in §7.4. In particular, the Schwarzschild spacetime contains a region $\mathcal{B}$ satisfying Def. 7.1.4, and it can be studied using a conformal diagram of the maximally-extended Schwarzschild spacetime, also known as a *Carter–Penrose diagram* (see §7.4.3.2), which is pictured in Fig. 7.6. This diagram offers a compactified view of the entire spacetime, with each point being a 2-sphere of metric area $4\pi r^2$.

The maximally-extended Schwarzschild spacetime has two causally-disconnected regions where $2m < r < +\infty$, denoted by $\mathcal{M}_I$ and $\mathcal{M}_{III}$ in Fig. 7.6: both are asymptotically flat, and their large- r parts resemble the conformal diagram for Minkowski spacetime (Eq. 7.2). Their future conformal boundaries $\mathcal{I}_I^+$ and $\mathcal{I}_{III}^+$ are null hypersurfaces of $(\tilde{\mathcal{M}}, \tilde{g})$, confirming Prop. 7.1.3 since the Schwarzschild solution has $\Lambda = 0$: the full future conformal boundary $\mathcal{I}^+ = \mathcal{I}_I^+ \cup \mathcal{I}_{III}^+$ is thus a null hypersurface with two connected components.

From Fig. 7.6, it is clear that $J^-(\mathcal{I}^+) \cap \mathcal{M}$ does not cover the whole $\mathcal{M}$, hence there is a black hole denoted by $\mathcal{B}$. Its event horizon $\mathcal{H}^+$ is a $r = 2m$ hypersurface, where m is the mass parameter of the Schwarzschild solution.

In general, the black hole region in a given spacetime $(\mathcal{M}, g)$ with a disconnected future infinity $\mathcal{I}^+$ may depend on the choice of connected components of $\mathcal{I}^+$, even if each component is assumed complete. For instance, consider the maximally-extended Schwarzschild spacetime: the regions $\mathcal{M}_I$ and $\mathcal{M}_{III}$, in which $r \in (2m, +\infty)$, are both asymptotically flat and causally disconnected, since no causal curve can connect a point in $\mathcal{M}_I$ to one in $\mathcal{M}_{III}$ or vice versa. Then, choosing WLOG an observer in $\mathcal{M}_I$, it is possible to consider a conformal completion $(\tilde{\mathcal{M}}, \tilde{g})$ of $(\mathcal{M}, g)$ such that $\mathcal{I} = \mathcal{I}_I$: the corresponding black hole region $\mathcal{B}$ is then larger from that in Ex. 7.1.4, since it now contains $\mathcal{M}_{III}$ too, as shown in Fig. 7.7. From the point of view of the observer in $\mathcal{M}_I$, these two choices are physically indistinguishable, since no particle can travel from $\mathcal{M}_{III}$ to $\mathcal{M}_I$ anyway.

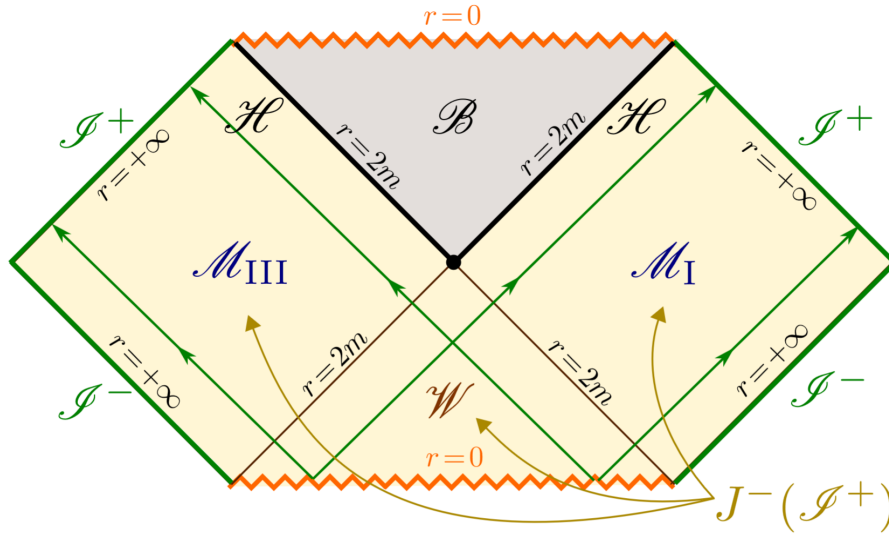


Figure 7.6: Maximally-extended Schwarzschild spacetime, depicted in a Carter–Penrose diagram. The yellow area is the causal past of the future null infinity $J^-(\mathcal{I}^+)$, divided into three parts by $r = 2m$ hypersurfaces, the grey area is the black hole $\mathcal{B}$, green lines are radial null geodesics and orange zigzag lines at $r = 0$ denote curvature singularities. Figure from [6].

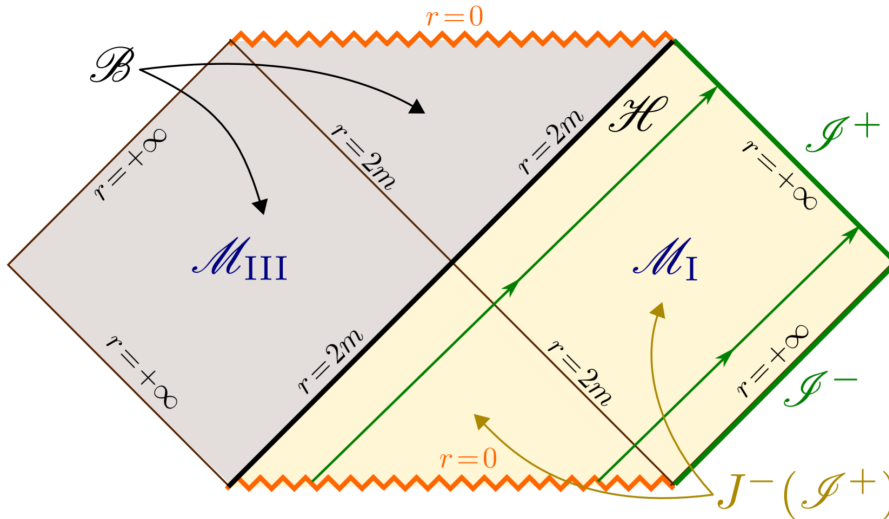


Figure 7.7: Maximally-extended Schwarzschild spacetime, but with a different conformal completion, so that $\mathcal{I}^+$ lies only at the boundary of the asymptotically-flat region $\mathcal{M}_{\text{I}}$. Figure from [6].

§7.1.3.3 White holes

Inverting past and future in the definition of a black hole Eq. 7.1.14, one defines the so-called **white hole** of a spacetime $(\mathcal{M}, g)$ with a conformal completion with complete past infinity $\mathcal{I}^-$:

$$\mathcal{W} := \mathcal{M} - (J^+(\mathcal{I}^-) \cap \mathcal{M}) \quad (7.1.16)$$

The white hole region is thus the set of points of $\mathcal{M}$ that cannot be reached by a future-directed causal curve emanating from $\mathcal{I}^-$. The boundary of this white hole is called the **past event horizon**:

$$\mathcal{H}^- := \partial\mathcal{W} = \partial J^+(\mathcal{I}^-) \cap \mathcal{M} \quad (7.1.17)$$

Carter [3] introduced the concept of **domain of outer communications** as the part of $\mathcal{M}$ that lies neither in a black hole nor in a white hole:

$$\langle\langle\mathcal{M}\rangle\rangle := \mathcal{M} - (\mathcal{B} \cup \mathcal{W}) = (J^-(\mathcal{I}^+) \cap J^+(\mathcal{I}^-)) \cap \mathcal{M} \quad (7.1.18)$$

The last equality shows that the domain of outer communications is the set of points from which it is possible to exchange signals with arbitrarily far regions. Moreover, from the definitions of black and white hole, it follows trivially that:

$$\partial\langle\langle\mathcal{M}\rangle\rangle = \mathcal{H}^+ \cup \mathcal{H}^- \quad (7.1.19)$$

Example 7.1.5 (Schwarzschild white hole)

In the maximally-extended Schwarzschild spacetime depicted in Fig. 7.6, the region $\mathcal{W}$ is a white hole, with the past event horizon $\mathcal{H}^-$ being a null hypersurface $r = 2m$.

§7.1.3.4 Achronal boundaries

From the general definition of black hole Def. 7.1.4, it is possible to derive some geometric properties of its boundary $\mathcal{H}^+$, independently of the gravitational theory being used.

Let $(\mathcal{M}, g)$ be a time-orientable spacetime. Following the conventions of [8], a set of the type $\partial J^\pm(S)$, with $S \subseteq \mathcal{M}$, is called an **achronal boundary** in $\mathcal{M}$. In particular, event horizons are achronal boundaries.

Proposition 7.1.5 (Achronal character of achronal boundaries)

Let $(\mathcal{M}, g)$ be a time-orientable spacetime. Any achronal boundary in $\mathcal{M}$ is an **achronal set**, i.e. no pair of its points can be connected by a timelike curve of $\mathcal{M}$.

This result justifies the used nomenclature. Note that, in this notion of achronal set, it is not required that the timelike curve lies entirely in the set. Accordingly, an analogous statement is: no timelike curve of $\mathcal{M}$ meets an achronal boundary at more than one point.

Since $\mathcal{H}^+ = \partial\mathcal{B}$, where $\dim_{\mathbb{R}} \mathcal{B} = d$, one would expect the future event horizon to be an hypersurface of spacetime. However, $\mathcal{H}^+$ is not necessarily a smooth manifold: there may exist some points where $\mathcal{H}^+$ is not differentiable.

Proposition 7.1.6 (Topological nature of achronal boundaries)

Let $(\mathcal{M}, g)$ be a time-orientable spacetime. Any achronal boundary in $\mathcal{M}$ is a codimension-1 topological submanifold of $\mathcal{M}$.

To be precise, any achronal boundary in $\mathcal{M}$ is a *Lipschitz submanifold* of $\mathcal{M}$, i.e. an intermediate between a topological submanifold, i.e. a submanifold of class $\mathcal{C}^0$, and a differentiable submanifold of class $\mathcal{C}^1$.

Besides being an achronal codimension-1 topological submanifold, the main property of achronal boundaries has been proved by Penrose [16].

Theorem 7.1.1 (Achronal boundaries ruled by null geodesics)

Let $(\mathcal{M}, g)$ be a time-orientable spacetime, and let $\mathcal{H} \subseteq \mathcal{M}$ be a future achronal boundary. Then, $\mathcal{H}$ is ruled^a by a family of null geodesics, called the **generators** of $\mathcal{H}$, that:

- (1) either lie entirely in $\mathcal{H}$ or never leave $\mathcal{H}$ when followed into the future from the point where they enter $\mathcal{H}$;
- (2) have no endpoint in the future.

Moreover, there is exactly one generator through each point of $\mathcal{H}$, except at points where null geodesics enter $\mathcal{H}$, called **crossovers**. A crossover is called *caustic* if it is a point where neighbouring null geodesics focus and converge while arriving in $\mathcal{H}$.

^aAnalogy with a *ruled surface*, i.e. a surface $\mathcal{S}$ in Euclidean space such that there is a straight line lying in $\mathcal{S}$ through each point of $\mathcal{S}$ (e.g. a cylinder or a cone). The analogy holds because straight lines are the geodesics of Euclidean space.

The same result holds exchanging future $\leftrightarrow$ past. As a consequence, once a null geodesic has merged with $\mathcal{H}$ (at a point where it may intersect other null geodesics), it will remain forever on $\mathcal{H}$ and will never intersect any other generator. The set of all crossovers is called the **crease set** of $\mathcal{H}$.

Proposition 7.1.7 (Achronal boundaries as null hypersurfaces)

Let $(\mathcal{M}, g)$ be a time-orientable spacetime, and let $\mathcal{H} \subseteq \mathcal{M}$ be an achronal boundary. Then, wherever it is smooth, $\mathcal{H}$ is a null hypersurface.

Proof. Assume that $\mathcal{H}$ is smooth in some subset $U \in \mathcal{T}(\mathcal{M})$. By Prop. 7.1.6, $\mathcal{H}$ is then a smooth hypersurface in U . According to Th. 7.1.1, there is a null geodesic lying in $\mathcal{H}$ through any point of $\mathcal{H} \cap U$. This implies the existence of null tangent vectors at any point of $\mathcal{H} \cap U$, so that $\mathcal{H}$ must be either a null or a timelike hypersurface in U . But $\mathcal{H}$ is achronal by Prop. 7.1.5, therefore it cannot be timelike. Hence, $\mathcal{H}$ is a null hypersurface in U . $\square$

It can be shown that future event horizons are smooth almost everywhere (e.g. the Schwarzschild black hole in Fig. 7.6): the only subset where they are not differentiable is the crease set, i.e. the set of points where null geodesics cross each other while entering $\mathcal{H}^+$ to become null generators.

§7.2 Event horizons

Having established with Prop. 7.1.7 that the smooth part of the event horizon of a black hole is a null hypersurface, it is convenient to analyze the geometric properties of this type of hypersurfaces, in order to distinguish between true black hole event horizons and other null hypersurfaces, e.g. future light cones.

§7.2.1 Null hypersurfaces

Let $(\mathcal{M}, g)$ be a spacetime. As any hypersurface, a null hypersurface $\mathcal{H}$ can be locally considered as regular a level set (Def. 1.3.5) of a defining map $u \in \mathcal{C}^\infty(U)$, with $U \subseteq \mathcal{M} : U \cap \mathcal{H} \neq \emptyset$. In particular, the smooth function can be chosen so that, for all $p \in U$:

$$p \in \mathcal{H} \iff u(p) = 0 \quad \text{and} \quad du|_{\mathcal{H}} \neq 0$$

The second condition ensures that $\mathcal{H}$ is not self-intersecting.

§7.2.1.1 Null geodesic generators

Let $\ell \in \Gamma(N\mathcal{H})$ be a smooth vector field normal to $\mathcal{H}$. Being $\mathcal{H}$ a null hypersurface, ℓ is null, i.e. $\langle \ell, \ell \rangle_g = 0$; moreover, WLOG choose it future-directed. As $\ell^2 = 0$, there is no natural normalization for null normals, since any rescaling $\ell \mapsto \ell' = \alpha \ell$ with $\alpha \in \mathbb{R}^+$ (to preserve causal orientation) yields an analogous null normal to $\mathcal{H}$.

A priori, the null normal vector field ℓ is defined on $\mathcal{H}$ only. However, it is useful to consider $\ell \in \mathfrak{X}(U)$, with $U \in \mathcal{T}(\mathcal{M}) : U \supseteq \mathcal{H}$, so as to allow for the definition of the covariant derivative $\nabla \ell$. To do so, consider a foliation of $\mathcal{M}$, in general in the vicinity of $\mathcal{H}$, by a family of null hypersurfaces: WLOG this foliation can be seen as a family of level set of the above-defined $u \in \mathcal{C}^\infty(U)$, denoted by $\{\mathcal{H}_c\}_{c \in \mathbb{R}} : \mathcal{H}_c = u^{-1}(c)$, so that $\mathcal{H} \equiv \mathcal{H}_0$. Then, ℓ can be defined on the part of $\mathcal{M}$ foliated by $(\mathcal{H}_c)$ so that $\forall p \in U \exists c \in \mathbb{R} : \ell_p \in N_p \mathcal{H}_c$, i.e. for each point in the region there is a leaf of the foliation to which ℓ is a null normal.

Obviously, the foliation $(\mathcal{H}_c)$ is not unique, but all geometrical quantities defined hereafter do not depend on the chosen foliation once they are evaluated on $\mathcal{H}$. Since $\mathcal{H}$ is a hypersurface on which u is constant, then $v \in T_p \mathcal{M}$ is tangent to $\mathcal{H}$ if and only if $\nabla_v u = 0$, i.e. $\langle \nabla u, v \rangle_g = 0$ since $u \in \mathfrak{T}_0^0(\mathcal{M})$. This means that ∇u is a normal vector field on $\mathcal{H}$, so, by uniqueness of the normal direction to a hypersurface, it must be collinear to ℓ , and it is always possible to choose $\rho \in \mathcal{C}^\infty(U)$ such that:

$$\ell = -e^\rho \nabla u \quad (7.2.1)$$

The strictly-negative sign of the coefficients is always possible to find by a redefinition of u , and it ensures that if u increases towards the future, then ℓ is future-directed.

Lemma 7.2.1 (Frobenius' identity)

$$d\ell^b = d\rho \wedge \ell^b \quad (7.2.2)$$

Proof. Writing the metric dual of Eq. 7.2.1 $\ell^b = -e^\rho du$ in a local coordinate frame yields $\ell_\alpha = -e^\rho \nabla_\alpha u$. Taking the covariant derivative:

$$\nabla_\alpha \ell_\beta = -e^\rho \nabla_\alpha \rho \nabla_\beta u - e^\rho \nabla_\alpha \nabla_\beta u = \nabla_\alpha \rho \ell_\beta - e^\rho \nabla_\alpha \nabla_\beta u$$

Antisymmetrizing and using the torsion-free nature of the Levi-Civita connection:

$$\nabla_{[\alpha} \ell_{\beta]} = \nabla_{[\alpha} \rho \ell_{\beta]} \quad (7.2.3)$$

The left-hand side are the components of $d\ell^b$, again by the torsion-free property, while the right-hand side are the components of $d\rho \wedge \ell^b$. $\square$

This identity is linked to the Frobenius theorem, see e.g. Th. B.3.2 of [20] or Th. C.2 of [17]: a 1-form satisfies Eq. 7.2.2 if and only if it defines hyperplanes that are integrable in some hypersurface ($\mathcal{H}$

in this case). Contracting the component equation Eq. 7.2.3 with ℓ yields:

$$\ell^\mu \nabla_\mu \ell_\alpha - \ell^\mu \nabla_\mu \ell_\alpha = \ell^\mu \nabla_\mu \rho \ell_\alpha - \ell^\mu \ell_\mu \nabla_\alpha \rho = \ell^\mu \nabla_\mu \rho \ell_\alpha$$

Since ℓ is null, $\ell^\mu \nabla_\alpha \ell_\mu = \nabla_\alpha (\ell^\mu \ell_\mu) - \ell_\mu \nabla_\alpha \ell^\mu = -\ell_\mu \nabla_\alpha \ell^\mu$, i.e. $\ell^\mu \nabla_\alpha \ell_\mu = 0$, thus:

$$\nabla_\ell \ell = \kappa \ell \quad \kappa \equiv \nabla_\ell \rho \quad (7.2.4)$$

This is the equation for a non-affinely parametrized geodesic, hence the integral curves of ℓ are null geodesics. κ is often called the *non-affinity coefficient*, and it can be set to $\kappa = 0$ with a suitable⁴ rescaling $\ell \mapsto \ell' = \alpha \ell$. This analysis allows to state a result analogous to Th. 7.1.1.

Proposition 7.2.1 (Geodesic generators of a null hypersurface)

Let $(\mathcal{M}, g)$ be a spacetime. Any null hypersurface $\mathcal{H} \subseteq \mathcal{M}$ is ruled by a family of null geodesics, called **null geodesic generators** of $\mathcal{H}$, and each $\ell \in \Gamma(N\mathcal{H})$ is tangent to these null geodesics. Moreover, the null geodesic generators of $\mathcal{H}$ form a congruence^a in $\mathcal{H}$.

^aLet $\mathcal{M}$ be a smooth manifold, and let $\Sigma \subseteq \mathcal{M}$ be a smooth submanifold. Then, a *congruence* in Σ is a family of curves in Σ such that through each point in Σ there passes one and exactly one such curve.

This result is not trivial: while it is obvious that integral curves of a null normal of $\mathcal{H}$ at null curves tangent to $\mathcal{H}$, keep in mind that not all null curves are null geodesics. For a smooth achronal boundary, the geodesic generators defined in Prop. 7.2.1 are the same as those defined in Th. 7.1.1.

§7.2.1.2 Cross-sections

Definition 7.2.1 (Cross-section of a null hypersurface)

Let $(\mathcal{M}, g)$ be a d -spacetime, with $d \geq 3$, and let $\mathcal{H}$ be a null hypersurface. A **cross-section** of $\mathcal{H}$ is a $(d-2)$ -submanifold $\mathcal{S} \subseteq \mathcal{H}$ such that:

- (1) any $\ell \in \Gamma(N\mathcal{H})$ is nowhere tangent to $\mathcal{S}$;
- (2) each null geodesic generator of $\mathcal{H}$ intersects $\mathcal{S}$ at most once.

If each null geodesic generator of $\mathcal{H}$ intersects $\mathcal{S}$ exactly once, then $\mathcal{S}$ is a **complete cross-section** (see Fig. 7.8). In the following, indices relative to a cross-section range from 2 to $d-1$ and are denoted by Latin letters from the beginning of the alphabet.

No restriction is placed on the topology of cross-section. However, note that if $\mathcal{H}$ is a black hole event horizon, its complete cross-sections are closed manifolds, i.e. compact manifolds without boundary: the simplest example is $\mathcal{S} = \mathbb{S}^{d-2}$, which is the relevant case for $d = 4$ (for $d > 4$ other compact topologies are possible).

⁴Explicitly, if $\ell' = \alpha \ell$:

$$\nabla_{\ell'} \ell' = \alpha (\nabla_\ell \alpha + \kappa \alpha) \ell = 0 \quad \Longleftrightarrow \quad \nabla_\ell \log \alpha = -\kappa \quad \Longleftrightarrow \quad \frac{d}{d\lambda} \log \alpha = -\kappa$$

where λ is the parameter of ℓ . Therefore, setting $\alpha(\lambda) = \exp \left(- \int_0^\lambda d\bar{\lambda} \kappa(\bar{\lambda}) \right)$ along each integral curve ensures that it is an affinely-parametrized geodesic. Moreover, under a rescaling $\ell' = \alpha \ell$, the non-affinity coefficient transforms as:

$$\kappa' = \alpha \kappa + \nabla_\ell \alpha \quad (7.2.5)$$

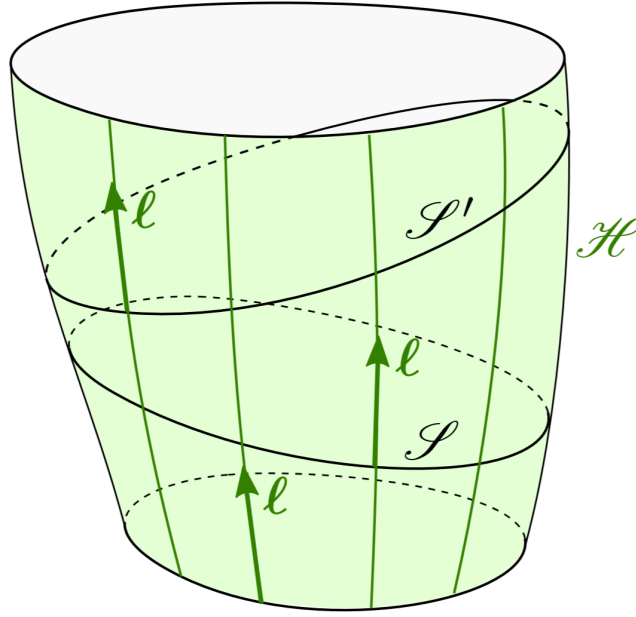


Figure 7.8: Two complete cross-section $\mathcal{S}$ and $\mathcal{S}'$ of a null hypersurface $\mathcal{H}$. Green lines represent null geodesic generators. Figure from [6].

Proposition 7.2.2 (Spacelike character of cross-sections)

Any cross-section of a null hypersurface is spacelike, i.e. all its tangent vectors are spacelike.

Proof. To prove this proposition, first state a useful lemma.

Lemma 7.2.2 (Tangent vectors to a null hypersurface)

Every non-zero tangent vector v to a null hypersurface is either spacelike or null. Moreover, if it is null, v is tangent to a null geodesic generator, i.e. it is normal to the hypersurface.

Let $p \in \mathcal{S}$ and $v \in T_p\mathcal{M}$ be a non-zero tangent vector to $\mathcal{S}$. By Lemma 7.2.2, v is either spacelike or tangent to the null geodesic generator γ going through p , but then γ would be tangent to $\mathcal{S}$, which is not allowed by (1) of Def. 7.2.1. This implies that v must be spacelike, which proves that $\mathcal{S} \subseteq \mathcal{M}$ is a spacelike submanifold. $\square$

Being $\mathcal{S}$ spacelike, the metric induced by g on $\mathcal{S}$ is a Riemannian metric: denote it with $\bar{q} : T\mathcal{S} \times T\mathcal{S} \rightarrow \mathbb{R} : \bar{q} \equiv g|_{\mathcal{S}}$. Then, through the Gram-Schmidt algorithm, it is possible to find at each $p \in \mathcal{S}$ an *orthogonal complement* $T_p^\perp \mathcal{S}$ of $T_p \mathcal{S}$, which is a 2-dimensional vector space such that:

$$T_p \mathcal{M} = T_p \mathcal{S} \oplus T_p^\perp \mathcal{S} \quad (7.2.6)$$

where each vector in $T_p^\perp \mathcal{S}$ is orthogonal to $\mathcal{S}$. Since $\bar{q}$ is Riemannian, the induced metric on $T_p^\perp \mathcal{S}$ must be Lorentzian with signature $(-, +)$, so to ensure that $g_p = g|_{T_p \mathcal{S} \oplus T_p^\perp \mathcal{S}}$ has signature $(-, +, \dots, +)$. Moreover, $\mathcal{S} \subseteq \mathcal{H}$ means that a null normal ℓ of $\mathcal{H}$ is orthogonal to any vector tangent to $\mathcal{S}$, i.e. $\ell_p \in T_p^\perp \mathcal{S} \ \forall p \in \mathcal{S}$: one can then supplement ℓ_p with another null vector to form a null basis of $T_p^\perp \mathcal{S}$. It can be shown that, at each $p \in \mathcal{S}$, there exists a unique such complement $k_p \in T_p \mathcal{M}$ with normalization $\langle k_p, \ell_p \rangle = -1$, so denote with $k \in \mathfrak{X}(\mathcal{M}) : k(p) = k_p$ the associated vector field: note that, while ℓ is independent of the choice of the cross-section $\mathcal{S}$ of $\mathcal{H}$ through

p , $\mathbf{k}$ does depend on $\mathcal{S}$. Then, the induced metric $\bar{q}$ on $\mathcal{S}$ can be extended to a metric on $\mathcal{M}$ as:

$$q := g + \ell^b \otimes \mathbf{k}^b + \mathbf{k}^b \otimes \ell^b \quad (7.2.7)$$

In a local coordinate frame, this metric has components $q_{\alpha\beta} = g_{\alpha\beta} + \ell_\alpha k_\beta + k_\beta \ell_\alpha$. Then, at each $p \in \mathcal{S}$, given $\mathbf{u}, \mathbf{v} \in T_p \mathcal{M}$:

$$q(\mathbf{u}, \mathbf{v}) = q(\mathbf{u}^\parallel, \mathbf{v}^\parallel) \quad (7.2.8)$$

where $\mathbf{u}^\parallel, \mathbf{v}^\parallel \in T_p \mathcal{S}$ are the tangent parts of $\mathbf{u}$ and $\mathbf{v}$ to $\mathcal{S}$, which are uniquely defined by the direct sum Eq. 7.2.6. This means that the metric dual of q is an orthogonal projector onto $\mathcal{S}$:

$$\pi_{\mathcal{S}} := \mathbb{1}_{\mathcal{M}} + \ell \otimes \mathbf{k}^b + \mathbf{k} \otimes \ell^b \quad (7.2.9)$$

In a local coordinate frame, this projector has components $\pi_{\beta}^{\alpha} = \delta_{\beta}^{\alpha} + \ell^{\alpha} k_{\beta} + k^{\alpha} \ell_{\beta}$, and it acts on $T_p \mathcal{M}$ as $\pi_{\mathcal{S}}(\mathbf{v}) = \mathbf{v}^\parallel$. In particular, $\pi_{\mathcal{S}}(\ell) = \pi_{\mathcal{S}}(\mathbf{k}) = \mathbf{0}$.

The null basis $(\ell, \mathbf{k}) \in T_p^\perp \mathcal{S}$ can be orthonormalized as follows:

$$\mathbf{n} = \frac{1}{2}\ell + \mathbf{k} \quad \mathbf{s} = \frac{1}{2}\ell - \mathbf{k} \quad (7.2.10)$$

so that $\langle \mathbf{n}, \mathbf{n} \rangle_g = -1$, $\langle \mathbf{s}, \mathbf{s} \rangle_g = 1$ and $\langle \mathbf{s}, \mathbf{n} \rangle_g = 0$. Substituting into Eq. 7.2.7, the metric becomes:

$$q = g + \mathbf{n}^b \otimes \mathbf{n}^b - \mathbf{s}^b \otimes \mathbf{s}^b \quad (7.2.11)$$

§7.2.1.3 Expansion of cross-sections

Let $(\mathcal{M}, g)$ be a d -spacetime, with $d \geq 3$, let $\mathcal{H} \subseteq \mathcal{M}$ be a null hypersurface with null normal $\ell \in \Gamma(N\mathcal{H})$, and let $\mathcal{S} \subseteq \mathcal{H}$ be a cross-section of $\mathcal{H}$. Consider an infinitesimal $\varepsilon \geq 0$, and consider a pair of points $p, p_\varepsilon \in \mathcal{H} : p \in \mathcal{S}$ separated by the vector $\varepsilon \ell$ (clearly $p_\varepsilon \in \mathcal{H}$ since ℓ is tangent to $\mathcal{H}$ and ε is infinitesimal). Repeating this construction for each point in $\mathcal{S}$, keeping ε fixed, results in a new $(d-2)$ -dimensional surface $\mathcal{S}_\varepsilon$, as in Fig. 7.9: this is called the *Lie dragging* of $\mathcal{S}$ along ℓ by ε . Clearly, $\mathcal{S}_0 = \mathcal{S}$.

Since $p_\varepsilon \in \mathcal{H} \forall p \in \mathcal{S}$ for infinitesimal ε , then $\mathcal{S}_\varepsilon \subseteq \mathcal{H}$. Moreover, being the null direction ℓ transverse to $\mathcal{S}_\varepsilon$ by construction, it follows that $\mathcal{S}_\varepsilon$ is spacelike too (by Lemma 7.2.2). Then, at each point $p \in \mathcal{S}$ it is possible to define the **expansion of $\mathcal{S}$ along ℓ** as the rate of change of the hyper-area δA of a surface element $\delta S \mathcal{S}$ around p (see Fig. 7.9):

$$\theta(\ell) := \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \frac{\delta A_\varepsilon - \delta A}{\delta A} \quad (7.2.12)$$

where δA_ε is the hyper-area of the Lie-dragged surface element $\delta S_\varepsilon \subseteq \mathcal{S}_\varepsilon$. Note that the hyper-area δA is measured with respect to the induced metric q on $\mathcal{S}$, and similarly for δA_ε with the Riemannian metric induced by g on $\mathcal{S}_\varepsilon$. For instance, if $\delta S \subseteq \mathcal{S}$ is defined by some infinitesimal displacements $\delta \mathbf{x}_{(2)}, \dots, \delta \mathbf{x}_{(d-1)}$:

$$\delta A = dV_{\mathcal{S}}(\delta \mathbf{x}_{(2)}, \dots, \delta \mathbf{x}_{(d-1)}) \quad dV_{\mathcal{S}} \equiv \iota_{\mathbf{s}} \iota_{\mathbf{n}} dV_g$$

which means that $\delta A = dV_g(\mathbf{n}, \mathbf{s}, \delta \mathbf{x}_{(2)}, \dots, \delta \mathbf{x}_{(d-1)})$. Choosing adapted coordinates to $\mathcal{S}$ and ℓ , i.e. $(x^\alpha) = (u, v, x^2, \dots, x^{d-1})$ such that $\mathcal{H}$ is locally the set $u = 0$, $\mathcal{S}$ is the set $(u, v) = (0, 0)$ and $\ell = \partial_v$, then $\mathcal{S}_\varepsilon$ is the set $(u, v) = (0, \varepsilon)$. Choosing the infinitesimal displacements defining δS to be along the coordinate lines of this frame, the hyper-area element becomes:

$$\delta A_\varepsilon = \sqrt{q(\varepsilon)} \delta x_{(2)} \dots \delta x_{(d-1)} \quad (7.2.13)$$

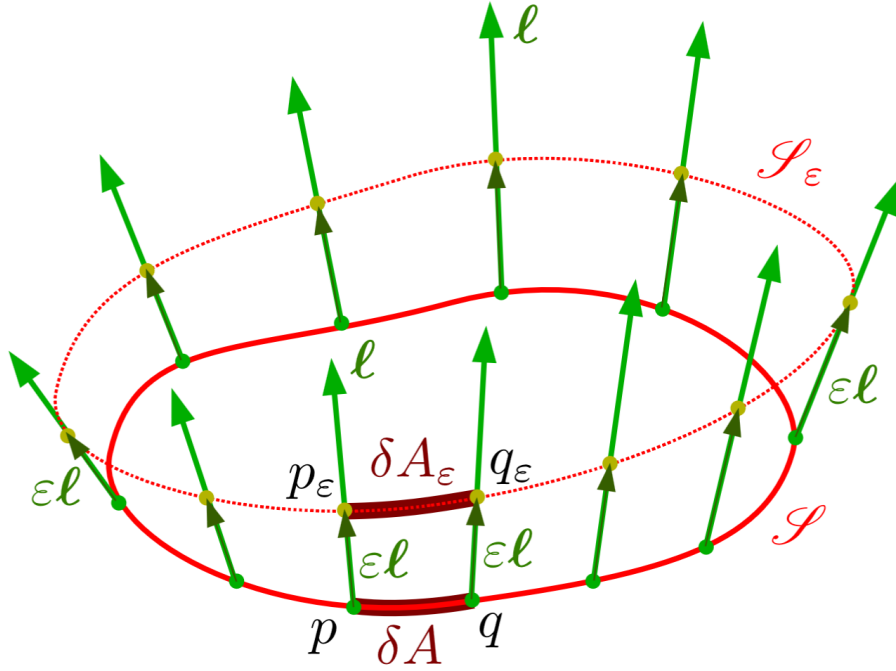


Figure 7.9: Lie dragging of $\mathcal{S}$ along ℓ by ε , here pictured with $d = 3$. Figure from [6].

where $q(\varepsilon)$ is the determinant of the representative matrix of the induced metric q_ε on $\mathcal{S}_\varepsilon$, computed in the chosen coordinate frame. Plugging this into the definition Eq. 7.2.12:

$$\theta_{(\ell)} = \frac{1}{2} \frac{d}{d\varepsilon} \bigg|_{\varepsilon=0} \ln q$$

Since ε is the value of the coordinate v associated to ℓ , i.e. the coordinate along which the Lie dragging happens, this is just the Lie derivative of $\ln q$ along ℓ (recall §1.4.3):

$$\theta_{(\ell)} = \frac{1}{2} \mathcal{L}_\ell \ln q \quad (7.2.14)$$

This is a coordinate-independent definition of the expansion of $\mathcal{S}$ along ℓ , and with further manipulation it can be rewritten in a local coordinate frame on $\mathcal{M}$ as:

$$\theta_{(\ell)} = q^{\mu\nu} \nabla_\mu \ell_\nu \quad (7.2.15)$$

where $q^{\mu\nu}$ are the component of the inverse metric of Eq. 7.2.7. Inserting the expression $q^{\mu\nu} = g^{\mu\nu} = \ell^\mu k_\nu + k^\mu \ell_\nu$ and using the normalizations of ℓ and k (assuming that ℓ remains null on an open neighbourhood of $\mathcal{H}$):

$$\theta_{(\ell)} = \nabla_\mu \ell^\mu - \kappa \quad (7.2.16)$$

This expression is completely independent⁵ of $\mathcal{S}$, thus showing that the expansion at a point $p \in \mathcal{H}$ does not depend on the choice of cross-section through p : this justifies the notation $\theta_{(\ell)}$ in place of $\theta_{(\ell)}(\mathcal{S})$, and from now on $\theta_{(\ell)}$ is called the *expansion of $\mathcal{H}$ along ℓ* . From Eq. 7.2.14, the expansion rescales as $\theta_{(\alpha\ell)} = \alpha\theta_{(\ell)}$.

⁵This can be shown geometrically too. Let $\mathcal{S}$ and $\mathcal{S}'$ be two distinct cross-sections of $\mathcal{H}$ through p , let $q \in \mathcal{S}$ be a point close to p on $\mathcal{S}$, and let $q' \in \mathcal{S}'$ be the point on $\mathcal{S}'$ along the same null geodesic generator as q , so that q and q' are separated by $\varepsilon\ell$. Then, denoting as δx and $\delta x'$ the vectors connecting p to q and q' , respectively, it is clear that $\delta x' = \delta x + \varepsilon\ell$, so that:

$$(\delta x')^2 = (\delta x)^2 + 2\varepsilon \langle \delta x, \ell \rangle_g + \varepsilon^2 \ell^2 = (\delta x)^2$$

since ℓ is null and normal to any tangent vector to $\mathcal{H}$, such as δx . Therefore, the lengths of all segments from p do not depend on the cross-section chosen, as long as their other ends lie on the same null geodesic generator of $\mathcal{H}$.

Definition 7.2.2 (Deformation rate of a cross-section)

Let $\mathcal{S} \subseteq \mathcal{H}$ be a cross-section of a null hypersurface $\mathcal{H}$, and let $\ell \in \Gamma(N\mathcal{H})$ be a null normal to $\mathcal{H}$. The **deformation rate of $\mathcal{S}$ along ℓ** is defined as:

$$\Theta := \frac{1}{2} \pi_{\mathcal{S}} \mathcal{L}_{\ell} q \quad (7.2.17)$$

where the action of the orthogonal projector $\pi_{\mathcal{S}}$ onto $\mathcal{S}$ on the bilinear form $\mathcal{L}_{\ell} q$ is defined by:

$$\pi_{\mathcal{S}} \mathcal{L}_{\ell} q(\mathbf{u}, \mathbf{v}) = \mathcal{L}_{\ell} q(\pi_{\mathcal{S}}(\mathbf{u}), \pi_{\mathcal{S}}(\mathbf{v})) \quad \forall \mathbf{u}, \mathbf{v} \in T_p \mathcal{M} \quad (7.2.18)$$

Since $\pi_{\mathcal{S}}(\ell) = \pi_{\mathcal{S}}(\mathbf{k}) = \mathbf{0}$, in components $\Theta_{\alpha\beta} = \frac{1}{2} \pi_{\alpha}^{\mu} \pi_{\beta}^{\nu} \mathcal{L}_{\ell} g_{\mu\nu}$. Recalling Eq. 5.1.14, then:

$$\Theta_{\alpha\beta} = \pi_{\alpha}^{\mu} \pi_{\beta}^{\nu} \nabla_{\mu} \ell_{\nu} \quad (7.2.19)$$

where Eq. 7.2.1 and $\pi_{\alpha}^{\mu} \ell_{\mu} = 0$ have been used. Comparing this to Eq. 7.2.15 it follows that the trace of Θ is just the expansion $\theta_{(\ell)}$:

$$\theta_{(\ell)} = g^{\mu\nu} \Theta_{\mu\nu} \equiv \Theta^{\mu}_{\mu} \quad (7.2.20)$$

The trace-free part of Θ is then called the **shear tensor of $\mathcal{S}$ along ℓ** :

$$\sigma := \Theta - \frac{1}{d-2} \theta_{(\ell)} q \quad (7.2.21)$$

where the $(d-2)^{-1}$ factor is needed to cancel the trace of the induced metric q on $\mathcal{S}$. Note that, like q , both tensor fields Θ and σ are tangent to $\mathcal{S}$, in the sense that $\Theta(\mathbf{v}, \cdot) = \sigma(\mathbf{v}, \cdot) \equiv 0 \quad \forall \mathbf{v} \in T_p^{\perp} \mathcal{S}$. Moreover, they share the same rescaling property as $\theta_{(\ell)}$ under $\ell \mapsto \ell' = \alpha \ell$: $\Theta' = \alpha \Theta$ and $\sigma' = \alpha \sigma$. These scalings show that both the deformation rate and the shear tensor depend on the choice of the null normal ℓ to $\mathcal{H}$; contrary to the expansion, they also depend on the choice of the cross-section $\mathcal{S} \subseteq \mathcal{H}$.

§7.2.1.4 Evolution of the expansion**Theorem 7.2.1** (Null Raychaudhuri–Landau equation)

Let $(\mathcal{M}, g)$ be a d -spacetime, and let $\mathcal{H} \subseteq \mathcal{M}$ be a null hypersurface. Then, any $\ell \in \Gamma(N\mathcal{H})$ satisfies:

$$\nabla_{\ell} \theta_{(\ell)} = \kappa \theta_{(\ell)} - \frac{1}{d-2} \theta_{(\ell)}^2 - \sigma_{ab} \sigma^{ab} - \text{Ric}(\ell, \ell) \quad (7.2.22)$$

where $\theta_{(\ell)}$ and σ are the expansion and the shear tensor of $\mathcal{H}$ along ℓ , and κ is non-affinity parameter of ℓ .

Note that, although σ depends on the choice of cross-section, the shear square $\sigma_{ab} \sigma^{ab}$ does not, since all other quantities in the null Raychaudhuri–Landau⁶ equation do not depend on it. If ℓ is future-directed, the null Raychaudhuri–Landau equation is an evolution equation for $\theta_{(\ell)}$ along the null generators of $\mathcal{H}$.

⁶Actually, this is a special case of the null Raychaudhuri–Landau equation in which the vorticity of the vector field ℓ vanishes. The general case considers a generic congruence of null geodesics (i.e. a family of null geodesics such that one and exactly one passes through each point of $\mathcal{M}$), and a null vector field tangent to the geodesics of this congruence has a priori a non-vanishing vorticity tensor w , so that the null Raychaudhuri–Landau equation gets a $+w_{ab} w^{ab}$ term on the right-hand side, see e.g. Eq. 4.35 of [8].

Proposition 7.2.3 (Null convergence condition)

Let $(\mathcal{M}, g)$ be a d -spacetime, and let $\mathcal{H} \subseteq \mathcal{M}$ be a null hypersurface. If the **null convergence condition** holds on $\mathcal{H}$, i.e. if $\text{Ric}(v, v) \geq 0$ for any null $v \in T_p \mathcal{M} : p \in \mathcal{H}$, then the expansion $\theta_{(\ell)}$ along any null normal $\ell \in \Gamma(N\mathcal{H})$ associated to an affine parametrization of the null geodesic generators of $\mathcal{H}$ cannot increase along ℓ , i.e.:

$$\nabla_{\ell} \theta_{(\ell)} \leq 0 \quad (7.2.23)$$

Proof. Choosing ℓ associated to an affine parametrization of the null geodesic generators of $\mathcal{H}$ means that $\kappa = 0$, hence the null Raychaudhuri–Landau equation becomes:

$$\nabla_{\ell} \theta_{(\ell)} = -\frac{1}{d-2} \theta_{(\ell)}^2 - \sigma_{ab} \sigma^{ab} - \text{Ric}(\ell, \ell)$$

The first term of the right-hand side is manifestly non-positive. To show that $\sigma_{ab} \sigma^{ab} \geq 0$, consider a q -orthonormal basis of $T_p \mathcal{S}$, so that $q_{ab} = \delta_{ab}$ and $q^{ab} = \delta^{ab}$ in such a basis; then:

$$\sigma_{ab} \sigma^{ab} = \sum_{a,b=2}^{n-1} (\sigma_{ab})^2 \geq 0$$

The third term is non-positive due to the null convergence condition, hence $\nabla_{\ell} \theta_{(\ell)} \leq 0$. $\square$

This property justifies the name *convergence condition*: if $\theta_{(\ell)} < 0$ in some part of $\mathcal{H}$, so that nearby null geodesic generators are converging, then by Eq. 7.2.1.4 they cannot subsequently diverge.

If the spacetime $(\mathcal{M}, g)$ is ruled by General Relativity, from Eq. 5.2.12 $\text{Ric}(\ell, \ell) = 8\pi G_N^{(d)} T(\ell, \ell)$, since $g(\ell, \ell) = 0$ being ℓ null. Therefore, the null convergence condition is equivalent to the null energy condition (see §5.2.3).

§7.2.2 Killing horizons

§7.2.2.1 Non-expanding event horizons

It is now possible to distinguish between event horizons and other null hypersurfaces. Indeed, event horizons are characterized as null hypersurface by the “no-escape” aspect of the naive definition (§7.1.1.1): to enforce the “localized” aspect, instead, it is reasonable to demand that the cross-sections are closed (i.e. compact and without boundary) and with a constant area, i.e. a vanishing expansion, as natural for black holes *at equilibrium*.

Definition 7.2.3 (Non-expanding horizon)

Let $(\mathcal{M}, g)$ be a d -spacetime. A null hypersurface $\mathcal{H} \subseteq \mathcal{M}$ is a **non-expanding horizon** if:

- (1) its topology is $\mathcal{H} \cong \mathbb{R} \times \mathcal{S}$, where $\mathcal{S}$ is a closed $(d-2)$ -submanifold;
- (2) $\theta_{(\ell)} = 0$ along any null normal $\ell \in \Gamma(N\mathcal{H})$.

Note that, given the scaling law $\theta_{(\alpha\ell)} = \alpha\theta_{(\ell)}$, if the expansion vanishes for some null normal, then it vanishes for all other null normals. For black holes out of equilibrium (§7.7), generally $\theta_{(\ell)} > 0$. Given a complete cross-section $\mathcal{S} \subseteq \mathcal{H}$ of a non-expanding horizon $\mathcal{H}$, its area with respect to g

is (recall Eq. 7.2.13):

$$A := \int_{\mathcal{S}} dV_{\mathcal{S}} = \int_{\mathcal{S}} dx^2 \dots dx^{d-1} \sqrt{q} \quad (7.2.24)$$

where $(x^a) = (x^2, \dots, x^{d-1})$ is a coordinate frame on $\mathcal{S}$ and $q \equiv \det q_{ab}$ is the determinant of the induced Riemannian metric on $\mathcal{S}$, computed in this frame. It can be shown that this area is independent of the choice of the complete cross-section $\mathcal{S} \subseteq \mathcal{H}$, hence, $A \equiv A[\mathcal{H}]$ is called the **area** of the non-expanding horizon $\mathcal{H}$ (event though it is a hyper-area, being $\dim_{\mathbb{R}} \mathcal{H} = d-1 \geq 2$).

Proposition 7.2.4 (Invariance of the cross-section metric)

Let $(\mathcal{M}, g)$ be a spacetime, and let $\mathcal{H} \subseteq \mathcal{M}$ be a non-expanding horizon. If the null convergence condition holds on $\mathcal{H}$, the extended induced metric on any cross-section $\mathcal{S}$ is invariant along the geodesic generators of $\mathcal{H}$.

Proof. If $\mathcal{H}$ is a non-expanding horizon, the null Raychaudhuri–Landau equation Eq. 7.2.22 becomes:

$$\sigma_{ab}\sigma^{ab} + \text{Ric}(\ell, \ell) = 0$$

where Ric is the Ricci tensor of the metric g . Assuming the null convergence condition $\text{Ric}(\ell, \ell) \geq 0$, this implies $\sigma_{ab}\sigma^{ab} = 0$, which is possible only if $\sigma \equiv 0$. Since $\theta_{(\ell)} = 0$ too, the deformation rate tensor vanishes identically: $\Theta = \frac{1}{2}\pi_{\mathcal{S}}\mathcal{L}_{\ell}q = 0$, which is the thesis. $\square$

Since $\mathcal{H}$ is a null hypersurface, the induced metric $g|_{\mathcal{H}}$ is degenerate, hence a priori there is no unique Koszul connection on $\mathcal{H}$ associated to $g|_{\mathcal{H}}$. However, if $\mathcal{H}$ is a non-expanding horizon and the null convergence condition holds on $\mathcal{H}$, then it can be shown that the Levi–Civita connection ∇ associated to g induces a unique torsion-free connection $\tilde{\nabla} : \mathfrak{X}(\mathcal{H}) \times \mathfrak{X}(\mathcal{H}) \rightarrow \mathfrak{X}(\mathcal{H})$ on $\mathcal{H}$ defined by $\tilde{\nabla}_u v = \nabla_u v \ \forall u, v \in \mathfrak{X}(\mathcal{H})$. As a consequence, $(\mathcal{H}, g|_{\mathcal{H}})$ is a **totally-geodesic submanifold** of $(\mathcal{M}, g)$, since any geodesic of $(\mathcal{H}, g|_{\mathcal{H}})$ is also a geodesic of $(\mathcal{M}, g)$.

§7.2.2.2 Killing horizons

A special kind of non-expanding horizons, which is of fundamental importance in the theory of stationary black holes, is that of Killing horizons with closed-manifold cross-sections.

Definition 7.2.4

Let $(\mathcal{M}, g)$ be a spacetime, and let $\xi \in \mathfrak{X}(\mathcal{M})$ be a Killing vector field of $\mathcal{M}$. A **Killing horizon** is a connected null hypersurface $\mathcal{H}$ such that, on $\mathcal{H}$, ξ is normal to $\mathcal{H}$.

Since normals to a null hypersurface are tangent to its null geodesic generators, while a Killing vector is tangent to the orbits of the corresponding isometry group⁷, then it is clear that the null geodesic generators of $\mathcal{H}$ are orbits of the isometry group of $(\mathcal{M}, g)$. Moreover, by definition (a normal to a hypersurface cannot vanish), the Killing vector field is non-vanishing and null on the Killing horizon:

$$\xi|_{\mathcal{H}} \neq 0 \quad \xi^2|_{\mathcal{H}} = 0$$

It will be shown (REF-TO-TH) that in a stationary spacetime, under some mild hypotheses, the connected part of a black hole event horizon must be a Killing horizon.

⁷Let G be a Lie group with a group action $\Phi : G \times \mathcal{X} \rightarrow \mathcal{X}$ on some set $\mathcal{X}$: the *orbit* of $p \in \mathcal{X}$ is the set $\{\Phi_g(p) : g \in G\} \subseteq \mathcal{X}$.

Example 7.2.1 (Schwarzschild black hole)

Consider the Schwarzschild spacetime $(\mathcal{M}, g)$, with $\mathcal{M} \cong \mathbb{R}^4$, with the **ingoing Eddington–Finkelstein coordinates** (§7.4.2) $(t, r, \vartheta, \varphi) \in \mathbb{R} \times \mathbb{R}^+ \times (0, \pi) \times (0, 2\pi)$, so that the Schwarzschild metric 5.3.3 takes the form:

$$ds^2 = - \left(1 - \frac{2m}{r}\right) dt^2 + \frac{4m}{r} dt dr + \left(1 + \frac{2m}{r}\right) dt^2 + r^2 d\Omega_2^2$$

Consider $u \in \mathcal{C}^\infty(\mathcal{M})$ defined in this coordinate frame as:

$$u(t, r, \vartheta, \varphi) = \left(1 - \frac{r}{2m}\right) \exp \frac{r-t}{4m} \quad (7.2.25)$$

The hypersurface $u = 0$, which corresponds to the 3-cylinder $r = 2m$ (cylinder as its topology is $\mathbb{R} \times \mathbb{S}^2$), is the *Schwarzschild horizon* $\mathcal{H}$, i.e. the event horizon of the Schwarzschild black hole. Then, the components of $\nabla u = g^{\alpha\mu} \partial_\alpha u \partial_\mu$ are computed to be:

$$\nabla u = \frac{1}{4m} e^{(r-t)/(4m)} \left[- \left(1 + \frac{r}{2m}\right) \frac{\partial}{\partial t} + \left(1 - \frac{r}{2m}\right) \frac{\partial}{\partial r} \right]$$

Recalling Eq. 7.2.1, chooses $\rho \in \mathcal{C}^\infty(\mathcal{M})$ so that $\ell^0 = 1$, i.e.:

$$e^\rho = -\frac{1}{\nabla^0 u} \iff \rho = \frac{t-r}{4m} - \log \left(1 + \frac{r}{2m}\right) + \log(4m)$$

From Eq. 7.2.1, then, the null normal to $\mathcal{H}$ is found to be:

$$\ell = \frac{\partial}{\partial t} + \frac{r-2m}{r+2m} \frac{\partial}{\partial r} \quad (7.2.26)$$

which is easily checked to be null, i.e. $g(\ell, \ell) = 0$. Since $\ell \neq 0$, this proves that all hypersurfaces $\mathcal{H}_c$ determined by u , and in particular $\mathcal{H}_0 \equiv \mathcal{H}$, are null. Moreover, on the horizon $\ell \stackrel{\mathcal{H}}{=} \partial_t$, which clearly is a Killing vector field since no metric component depends on t : therefore, the Schwarzschild horizon is a Killing horizon.

An important feature of Killing horizons is that the deformation rate of any cross-section along any null normal vanishes. Indeed, consider a cross-section $\mathcal{S}$ of a Killing horizon $\mathcal{H}$, and let the null normal coincide with the Killing vector field on the horizon, i.e. $\ell \stackrel{\mathcal{H}}{=} \xi$, so that in a neighbourhood of $\mathcal{H}$ it can be expressed as $\ell = \xi + u\mathcal{C}^\infty(\mathcal{M})$, where u is the defining function of the horizon and $v \in \mathfrak{X}(\mathcal{M})$. Then:

$$\Theta_{\alpha\beta} = \pi_\alpha^\mu \pi_\beta^\nu \nabla_\mu \xi_\nu + u \pi_\alpha^\mu \pi_\beta^\nu \nabla_\mu v_\nu + \pi_\alpha^\mu \pi_\beta^\nu w_\nu \nabla_\mu u \stackrel{\mathcal{H}}{=} \pi_\alpha^\mu \pi_\beta^\nu \nabla_\mu \xi_\nu$$

where $\pi_\alpha^\mu \nabla_\mu u \stackrel{\mathcal{H}}{=} 0$ since u is constant on $\mathcal{H}$, and so on $\mathcal{S} \subseteq \mathcal{H}$. By the Killing equation Eq. 5.1.17 $\pi_\alpha^\mu \pi_\beta^\nu \nabla_\mu \xi_\nu$ is antisymmetric, while $\Theta_{\alpha\beta}$ is symmetric (since $\Theta_{\alpha\beta} = \frac{1}{2} \pi_\alpha^\mu \pi_\beta^\nu \mathcal{L}_\ell g_{\mu\nu}$), hence necessarily $\Theta \equiv 0$. By the scaling law of Θ , it follows that $\Theta \equiv 0$ along any null normal to $\mathcal{H}$.

Proposition 7.2.5 (Killing horizons as non-expanding horizons)

Any Killing horizon with closed-manifold cross-sections is a non-expanding horizon.

Proof. Since the deformation rate of any cross-section along any null normal vanishes, (2) of Def. 7.2.3 is satisfied. (1) is satisfied instead by the requirement of the cross-sections being closed submanifolds. $\square$

Note that Θ vanishes on any Killing horizon, while to get the same result on a generic non-expanding horizon the null convergence condition is needed (Prop. 7.2.4).

§7.2.2.3 Surface gravity

Let κ be the non-affinity coefficient of the null normal $\ell \stackrel{\mathcal{H}}{=} \xi$ to a Killing horizon $\mathcal{H}$:

$$\nabla_{\xi} \xi \stackrel{\mathcal{H}}{=} \kappa \xi$$

Combining the metric dual with the Killing equation Eq. 5.1.17 yields $\xi^{\mu} \nabla_{\alpha} \xi_{\mu} \stackrel{\mathcal{H}}{=} -\kappa \xi_{\alpha}$, but notice that $\xi^{\mu} \nabla_{\alpha} \xi_{\mu} = \frac{1}{2} \nabla_{\alpha} (\xi_{\mu} \xi^{\mu})$, so replacing the covariant derivative with the exterior derivative (since $\xi^{\mu} \xi_{\mu}$ is a scalar field):

$$d \langle \xi, \xi \rangle_g \stackrel{\mathcal{H}}{=} -2\kappa \xi^{\flat} \quad (7.2.27)$$

A relation like this was to be expected: if $u \equiv \langle \xi, \xi \rangle_g$ is regular in an open neighbourhood of $\mathcal{H}$, i.e. if $du \neq 0$, then $\mathcal{H}$ is the hypersurface $u = 0$, since ξ is null on $\mathcal{H}$, so its gradient must be collinear to the normal ξ to $\mathcal{H}$, i.e. $du \stackrel{\mathcal{H}}{=} \alpha \xi^{\flat}$.

Lemma 7.2.3 (Non-affinity coefficient)

$$\kappa^2 \stackrel{\mathcal{H}}{=} -\frac{1}{2} \nabla_{\mu} \xi_{\nu} \nabla^{\mu} \xi^{\nu} \quad (7.2.28)$$

Proof. Since ξ is normal to the hypersurface $\mathcal{H}$, by the Frobenius theorem there exists a 1-form a such that:

$$d\xi^{\flat} \stackrel{\mathcal{H}}{=} a \wedge \xi^{\flat}$$

which in a local coordinate frame becomes:

$$\nabla_{\alpha} \xi_{\beta} - \nabla_{\beta} \xi_{\alpha} \stackrel{\mathcal{H}}{=} a_{\alpha} \xi_{\beta} - a_{\beta} \xi_{\alpha}$$

By the Killing equation Eq. 5.1.17 the left-hand side reduces to $2\nabla_{\alpha} \xi_{\beta}$, so contracting with ξ :

$$2\xi^{\mu} \nabla_{\mu} \xi_{\alpha} \stackrel{\mathcal{H}}{=} a_{\mu} \xi^{\mu} \alpha_{\alpha} - \xi^{\mu} \xi_{\mu} a_{\alpha} = a_{\mu} \xi^{\mu} \xi_{\alpha}$$

By Eq. 7.2.27 the left-hand side is just $2\kappa \xi_{\alpha}$, so $a_{\mu} \xi^{\mu} \stackrel{\mathcal{H}}{=} 2\kappa$. Thus:

$$\begin{aligned} 4\nabla_{\mu} \xi_{\nu} \nabla^{\mu} \xi^{\nu} &\stackrel{\mathcal{H}}{=} (a_{\mu} \xi_{\nu} - a_{\nu} \xi_{\mu})(a^{\mu} \xi^{\nu} - a^{\nu} \xi^{\mu}) \\ &\stackrel{\mathcal{H}}{=} a_{\mu} a^{\mu} \xi_{\nu} \xi^{\nu} - a_{\mu} \xi^{\mu} a_{\nu} \xi^{\nu} - a_{\nu} \xi^{\nu} a_{\mu} \xi^{\mu} + a_{\nu} a^{\nu} \xi_{\mu} \xi^{\mu} = -8\kappa^2 \end{aligned}$$

which completes the proof. $\square$

Although Eq. 7.2.28 is an explicit expression of κ in terms of ξ , in actual calculations Eq. 7.2.27 is usually employed to compute κ , since it does not involve any covariant derivative.

Since $\ell \stackrel{\mathcal{H}}{=} \xi$, the vector field ℓ is a symmetry generator on $\mathcal{H}$, hence one may expect that κ is invariant along ℓ : this turns out to be the case. Indeed, taking the Lie derivative along ξ of Eq. 7.2.27

and using Eq. 2.2.30:

$$\begin{aligned}\mathcal{L}_\xi(d\langle\xi, \xi\rangle_g) &= \iota_\xi d^2\langle\xi, \xi\rangle_g + d\iota_\xi d\langle\xi, \xi\rangle_g = 0 \\ &\stackrel{\mathcal{H}}{=} -2\mathcal{L}_\xi(\kappa\xi^b) = -2(\mathcal{L}_\xi\kappa)\xi^b - 2\kappa\mathcal{L}_\xi\xi^b = -2(\mathcal{L}_\xi\kappa)\xi^b\end{aligned}$$

where $\langle\xi, \xi\rangle_g \stackrel{\mathcal{H}}{=} 0$ being ξ null on $\mathcal{H}$ and $\mathcal{L}_\xi\xi^b = 0 \iff \mathcal{L}_\xi\xi = 0$, as $\mathcal{L}_\xi g = 0$ by definition of Killing vector field. Since $\ell \stackrel{\mathcal{H}}{=} \xi$ and $\xi^b \neq 0$, this shows that the non-affinity coefficient is constant along the null geodesic generators of the Killing horizon:

$$\mathcal{L}_\ell\kappa = 0 \quad (7.2.29)$$

Note that, although κ is constant along any null geodesic generators, it may in principle vary from one generator to the other. In order to study under which conditions κ is constant on the whole horizon, introduce the **null dominance condition**, which states that there exists $f \in C^\infty(\mathcal{M})$ such that $\mathbf{W} := \mathbf{G}(\ell) - f\ell \in \mathfrak{X}(\mathcal{M})$ is either zero or future-directed (null or timelike) for any future-directed null $\ell \in \mathfrak{X}(\mathcal{M})$, where $\mathbf{G} \in \mathfrak{T}_1^1(\mathcal{M})$ is the metric dual of the Einstein tensor, so that in local coordinate frame:

$$W^\alpha := -G^\alpha{}_\mu \ell^\mu - f\ell^\alpha \quad (7.2.30)$$

Note that the null dominance condition implies the null convergence condition, since $\text{Ric}(\ell, \ell) = -\langle \mathbf{W}, \ell \rangle_g \geq 0$ (since $\mathbf{W}$ and ℓ are both future-directed).

Theorem 7.2.2 (Zeroth law of black hole dynamics)

Let $(\mathcal{M}, g)$ be a spacetime, and let $\mathcal{H} \subseteq \mathcal{M}$ be a Killing horizon defined by the Killing vector field $\xi \in \mathfrak{X}(\mathcal{M})$. Then, if the null dominance condition holds on $\mathcal{H}$, the non-affinity coefficient κ associated to the null normal coinciding with ξ on $\mathcal{H}$ is constant over $\mathcal{H}$.

Proof. The relation $\nabla_\xi \xi \stackrel{\mathcal{H}}{=} \kappa\xi$ is only valid at the horizon, so, to study the constancy of κ , it is not possible to apply a generic covariant derivative to it. However, note that the rank-(0, 3) tensor with components $\varepsilon^{\mu\nu\alpha\beta}\xi_\beta$, where $\varepsilon^{\mu\nu\alpha\beta} \equiv \sqrt{g}\epsilon^{\mu\nu\alpha\beta}$ is the Levi-Civita tensor, is indeed tangent to the horizon, since $(\varepsilon^{\mu\nu\alpha\beta}\xi_\mu)\xi_\alpha = 0$, thus $\varepsilon^{\mu\nu\alpha\beta}\xi_\beta\nabla_\alpha$ is a differential operator that can be safely applied to any equation holding on $\mathcal{H}$.

Equivalently, apply $\xi_{[\alpha}\nabla_{\beta]}$ to $\nabla_\xi \xi \stackrel{\mathcal{H}}{=} \kappa\xi$, yielding, in a local coordinate frame:

$$\begin{aligned}\xi_\mu\xi_{[\alpha}\nabla_{\beta]}\kappa + \kappa\xi_{[\alpha}\nabla_{\beta]}\xi_\mu &\stackrel{\mathcal{H}}{=} \xi_{[\alpha}\nabla_{\beta]}(\xi^\nu\nabla_\nu\xi_\mu) \stackrel{\mathcal{H}}{=} (\xi_{[\alpha}\nabla_{\beta]}\xi^\nu)(\nabla_\nu\xi_\mu) + \xi^\nu\xi_{[\alpha}\nabla_{\beta]}\nabla_\nu\xi_\mu \\ &\stackrel{\mathcal{H}}{=} (\xi_{[\alpha}\nabla_{\beta]}\xi^\nu)(\nabla_\nu\xi_\mu) - \xi^\nu\mathcal{R}_{\nu\mu[\beta}{}^\rho\xi_{\alpha]}\xi_\rho\end{aligned}$$

where Eq. 5.1.20 was used in the last equality. Since ξ is hypersurface-orthogonal to the horizon, by Frobenius' theorem $\xi_{[\mu}\nabla_\nu\xi_{\beta]} \stackrel{\mathcal{H}}{=} 0$, so using the Killing equation $\nabla_\nu\xi_\beta = -\nabla_\beta\xi_\nu$:

$$\xi_\beta\nabla_\mu\xi_\nu \stackrel{\mathcal{H}}{=} -2\xi_{[\mu}\nabla_{\nu]}\xi_\beta \quad (7.2.31)$$

This allows to rewrite the first term on the right-hand side of the above equation as:

$$(\xi_{[\alpha}\nabla_{\beta]}\xi^\nu)(\nabla_\nu\xi_\mu) \stackrel{\mathcal{H}}{=} -\frac{1}{2}(\xi^\nu\nabla_\alpha\xi_\beta)\nabla_\nu\xi_\mu \stackrel{\mathcal{H}}{=} -\frac{1}{2}\kappa\xi_\mu\nabla_\alpha\xi_\beta \stackrel{\mathcal{H}}{=} \kappa\xi_{[\alpha}\nabla_{\beta]}\xi_\mu$$

which cancels the second term on the left-hand side, leaving only:

$$\xi_\mu\xi_{[\alpha}\nabla_{\beta]}\kappa \stackrel{\mathcal{H}}{=} \xi^\nu\mathcal{R}_{\mu\nu[\beta}{}^\rho\xi_{\alpha]}\xi_\rho$$

On the other hand, applying $\xi_{[\alpha} \nabla_{\rho]}$ to Eq. 7.2.31:

$$(\xi_{[\alpha} \nabla_{\rho} \xi_{\beta]}) \nabla_{\mu} \xi_{\nu} + \xi_{\beta} \xi_{[\alpha} \nabla_{\rho]} \nabla_{\mu} \xi_{\nu} \stackrel{\mathcal{H}}{=} -2(\xi_{[\alpha} \nabla_{\rho]} \xi_{[\mu} \nabla_{\nu]} \xi_{\beta]} - 2(\xi_{[\alpha} \nabla_{\rho]} \nabla_{[\nu} \xi_{\beta]}) \xi_{\mu]}$$

Using Eq. 7.2.31 repeatedly, the first term on the left-hand side cancels the first term on the right-hand side, thus, using Eq. 5.1.20:

$$-\xi_{\beta} \mathcal{R}_{\mu\nu[\rho}{}^{\sigma} \xi_{\alpha]} \xi_{\sigma} \stackrel{\mathcal{H}}{=} 2\xi_{[\mu} \mathcal{R}_{\nu]\beta[\rho}{}^{\sigma} \xi_{\alpha]} \xi_{\sigma}$$

Contracting β and ρ makes the left-hand side vanish, so:

$$-\xi_{[\mu} \mathcal{R}_{\nu]}{}^{\sigma} \xi_{\sigma} \xi_{\alpha} \stackrel{\mathcal{H}}{=} \xi_{[\mu} \mathcal{R}_{\nu]\beta\alpha}{}^{\sigma} \xi^{\beta} x_{\sigma}$$

The right-hand side of this equation has the same form of the right-hand side of the above equation (since $\mathcal{R}_{\mu\nu\alpha\beta} = \mathcal{R}_{\alpha\nu\mu\beta}$), hence combining them:

$$\xi_{[\alpha} \nabla_{\beta]} \kappa \stackrel{\mathcal{H}}{=} -\xi_{[\alpha} \mathcal{R}_{\beta]\rho}{}^{\sigma} \xi^{\rho} \quad (7.2.32)$$

The right-hand side vanishes by the null dominance condition, so there only remains $\xi_{[\alpha} \nabla_{\beta]} \kappa \stackrel{\mathcal{H}}{=} 0$, which is the thesis. $\square$

In the context of Killing horizon, in light of Th. 7.2.2, the non-affinity coefficient κ is called the horizon's **surface gravity**. Note that, if gravity is described by General Relativity, the null dominance condition with $f = \Lambda$ is equivalent to the *null dominant energy condition*, which states that the vector field with components $-T^{\mu}_{\nu} \ell^{\nu}$ is either zero or future-directed (null or timelike) for any future-directed null $\ell \in \mathfrak{X}(\mathcal{M})$, where $T_{\mu\nu}$ is the energy-momentum tensor of matter fields. The null dominant energy condition is implied by the standard dominant energy condition (§5.2.3). The constancy of κ over $\mathcal{H}$ (assuming the null dominance condition) allows to classify Killing horizons in two categories:

- (1) if $\kappa = 0$, then ξ is a geodesic vector on $\mathcal{H}$, and $\mathcal{H}$ is called a **degenerate Killing horizon**;
- (2) if $\kappa \neq 0$, then $\mathcal{H}$ is called a **non-degenerate Killing horizon**.

Example 7.2.2 (Schwarzschild and Kerr black holes)

For the Schwarzschild black hole, ℓ is given by Eq. 7.2.26 (in ingoing Eddington–Finkelstein coordinates), so its covariant derivative is computed to be:

$$\nabla_{\ell} \ell = \frac{4m}{(r+2m)^2} \frac{\partial}{\partial t} + \frac{4m(r-2m)}{(r+2m)^3} \frac{\partial}{\partial r} = \frac{4m}{(r+2m)^2} \ell \quad (7.2.33)$$

Therefore, for any leave $\mathcal{H}_c$ of the foliation, $\kappa = 4m/(r+2m)^2$. In particular, for the event horizon $\mathcal{H}$:

$$\kappa = \frac{1}{4m} \quad (7.2.34)$$

Note that this value agrees with $\kappa = \nabla_{\ell} \rho$ (see Ex. 7.2.1). Clearly, κ is a constant, a trivial result given that the Schwarzschild black hole is spherically-symmetric. A general result, valid for a Kerr–Newman black hole with mass parameter m , angular momentum parameter a and charge parameter q is (Eq. 12.5.4 of [20]):

$$\kappa = \frac{\sqrt{m^2 - a^2 - q^2}}{2m \left(m + \sqrt{m^2 - a^2 - q^2} \right) - q^2} \quad (7.2.35)$$

which is constant too. Note that, for $a = 0$ and $q = 0$, it reduces to the Schwarzschild case. Moreover, setting $q = 0$, for $a < m$ the Kerr horizon is a non-degenerate Killing horizon, while it becomes degenerate for $a = m$.

To see why κ is interpreted as a surface gravity, consider a non-degenerate Killing horizon $\mathcal{H}$, so that $\kappa \neq 0$, and let $p \in \mathcal{H}$. Given $v \in T_p \mathcal{M}$ transverse to $\mathcal{H}$, i.e. not tangent to $\mathcal{H}$, by Eq. 7.2.27:

$$\nabla_v \langle \xi, \xi \rangle_g = -2\kappa \langle \xi, v \rangle_g$$

The right-hand side does not vanish, as $\kappa \neq 0$ and $\langle \xi, v \rangle_g \neq 0$ (being v not tangent to $\mathcal{H}$), thus $\nabla_v \langle \xi, \xi \rangle_g \neq 0$, the derivative of ξ^2 along any direction transverse to $\mathcal{H}$ does not vanish. Since $\xi^2 \stackrel{\mathcal{H}}{=} 0$, this means that, in the vicinity of $\mathcal{H}$, on one side $\xi^2 < 0$ and on the other side $\xi^2 > 0$. Therefore, in the vicinity of a non-degenerate Killing horizon, the associated Killing vector field is timelike on one side, null on the horizon and spacelike on the other side.

Focus on the side of $\mathcal{H}$ where ξ is timelike, and set the “norm” $V \equiv \sqrt{-\langle \xi, \xi \rangle_g} > 0$. Then, it can be shown that:

$$\kappa^2 = \lim_{\mathcal{H}} \nabla_\mu V \nabla^\mu V \quad (7.2.36)$$

where $\lim_{\mathcal{H}}$ is the limit as one approaches $\mathcal{H}$ from the timelike- ξ side, which implies $V \rightarrow 0$. In this region where ξ is timelike, define:

$$u := \frac{1}{V} \xi \quad (7.2.37)$$

which is a future-directed unit timelike vector field (since by convention ξ is future-directed and null on $\mathcal{H}$, otherwise take $-\xi$). This vector field u can be considered as the d -velocity of an observer whose worldline is a integral curve of ξ , i.e. an orbit of the isometry group generated by ξ . Since spacetime is not changing along this worldline (by the nature of Killing vector fields), this may be called a *stationary observer*. Its d -acceleration then is:

$$a = \nabla_u u = \nabla_{V^{-1}\xi} (V^{-1}\xi) = V^{-1} [-V^{-2}(\nabla_\xi V)\xi + V^{-1}\nabla_\xi \xi]$$

Since ξ is a symmetry generator, $\nabla_\xi = 0$, which can be shown as:

$$\nabla_\xi V = \xi^\mu \nabla_\mu \sqrt{-\xi_\nu \xi^\nu} = -\frac{1}{2\sqrt{-\xi_\nu \xi^\nu}} \xi^\mu \nabla_\mu (\xi_\nu \xi^\nu) = -\frac{1}{V} \xi^\mu \xi^\nu \nabla_\mu \xi_\nu = 0$$

since $\nabla_{(\mu} \xi_{\nu)} = 0$ (Killing's equation). Thus, using again the Killing equation:

$$a_\mu = \frac{1}{V^2} \xi^\alpha \nabla_\alpha \xi_\mu = -\frac{1}{V^2} \xi^\mu \nabla_\alpha \xi_\mu = -\frac{1}{V^2} \nabla_\alpha (\xi_\mu \xi^\mu) = \frac{1}{2V^2} \nabla_\alpha V^2 = \nabla_\alpha \log V$$

Therefore, the d -acceleration can be expressed as a gradient:

$$a = \nabla \log V \quad (7.2.38)$$

Since $\langle a, u \rangle_g = 0$, the d -acceleration must spacelike, i.e. $a_\mu a^\mu > 0$, so it is possible to define its modulus as:

$$a \equiv \sqrt{\langle a, a \rangle_g} = \sqrt{\nabla_\mu \log V \nabla^\mu \log V} = \frac{1}{V} \sqrt{\nabla_\mu V \nabla^\mu V}$$

Recalling Eq. 7.2.36, it is then possible to express κ in terms of a :

$$\kappa = \lim_{\mathcal{H}} V a \quad (7.2.39)$$

where $\lim_{\mathcal{H}}$ means that the worldline of the observer whose acceleration is $\mathbf{a}$ is taken arbitrarily close to the horizon $\mathcal{H}$. Since $V \rightarrow 0$ approaching $\mathcal{H}$, it must be $a \rightarrow +\infty$ in order to have a finite κ : this means that the acceleration felt by an observer (i.e. the “gravity”) diverges as it approaches the horizon. In this sense, the physical surface gravity of $\mathcal{H}$ is infinite. However, the rescaled acceleration Va remains finite approaching $\mathcal{H}$, and it tends to κ : for this reason, κ is regarded as the *surface gravity* of the Killing horizon $\mathcal{H}$.

The distinction between the surface gravity κ and the actual gravity a measured *locally*, i.e. by an observer at rest with respect to $\mathcal{H}$ and infinitely close to it. However, in the special case of a Schwarzschild black hole, κ can be interpreted as a physical force (per unit mass) measured by a *distant* observer, for which ξ is timelike in the entire region outside the Killing horizon⁸: in this case, κ can be identified with the magnitude of the force exerted by an observer “at infinity” to hold in place a particle of unit mass close to $\mathcal{H}$.

§7.3 Stationary black holes

Having defined the general notion of black hole in §7.1, focus now on the equilibrium case, i.e. black holes in stationary spacetimes.

§7.3.1 Stationary spacetimes

Definition 7.3.1 (Stationary spacetime)

A spacetime $(\mathcal{M}, g)$ is called **strictly stationary** if it is invariant under the action of $\mathbb{R}_+$ and the associated Killing vector field $\xi \in \mathfrak{X}(\mathcal{M})$ is everywhere timelike.

If $(\mathcal{M}, g)$ admits a conformal completion and ξ is timelike only in the vicinity of the conformal boundary $\mathcal{I}$, then the spacetime is called **stationary**.

The condition on the Killing vector fields (generator of $\mathbb{R}_+$) is equivalent to requiring that the orbits of the group action are everywhere timelike curves (or only in the vicinity of $\mathcal{I}$). The distinction between strictly stationary and stationary spacetimes is due to the fact that, in presence of a black hole, ξ cannot be timelike everywhere.

A priori, the Killing vector field ξ is not unique: $\mathbb{R}_+$ admits a reparametrization $t \mapsto \alpha t$, with $\alpha \in \mathbb{R} - \{0\}$, which corresponds⁹ to $\xi \mapsto \alpha^{-1}\xi$. When the spacetime admits a conformal boundary $\mathcal{I}$ (required in presence of a black hole), then ξ is uniquely fixed by choosing that near $\mathcal{I}$ it is future-directed and $\langle \xi, \xi \rangle_g \mapsto -1$: indeed, this fixes $\alpha = +1$, in order to preserve the norm and the causal orientation.

If the d -spacetime $(\mathcal{M}, g)$ is stationary, then it is possible to choose adapted coordinates $(x^\alpha) = (t, x^1, \dots, x^{d-1})$ such that $\xi = \partial_t$, which means that t is an ignorable coordinates, i.e. $\partial_t g_{\alpha\beta} = 0$.

⁸This is not true for a rotating Kerr black hole, since ξ becomes null at some “light-cylinder” outside $\mathcal{H}$ and is then spacelike away from it.

⁹Given a connected 1-dimensional Lie group G , its elements can be labelled as $g_t \in G : t \in \mathbb{R}$, with $g_0 = e$. Then, setting the notation for the group action on a smooth manifold $\mathcal{M}$ as $\Phi_t \equiv \Phi_{g_t} : \mathcal{M} \rightarrow \mathcal{M}$, the orbit of $p \in \mathcal{M}$ is a curve $\gamma_p : \mathbb{R} \rightarrow \mathcal{M} : \gamma_p(t) = \Phi_t(p)$ (which may be degenerate if p is a fixed point), and the *generator* $\xi \in \mathfrak{X}(\mathcal{M})$ of G is the velocity vector field $\xi_p := \gamma'_p(0)$. Under a rescaling $t \mapsto \alpha t$, the velocity rescales as $\gamma'_p(0) \mapsto \alpha^{-1}\gamma'_p(0)$, hence the generator rescales as $\xi \mapsto \alpha^{-1}\xi$.

Definition 7.3.2 (Static spacetime)

A spacetime $(\mathcal{M}, g)$ is called (strictly) **static** if it is (strictly) stationary and the Killing vector field $\xi \in \mathfrak{X}(\mathcal{M})$ generating the stationary group action is orthogonal to a family of hypersurfaces.

If the d -spacetime $(\mathcal{M}, g)$ is static, then it is possible to choose adapted coordinates such that $\xi = \partial_t$ and the hypersurfaces $t = \text{const.}$ are orthogonal to ξ , i.e. $\xi^\flat = g(\xi, \xi)dt$. This orthogonality condition means that $g_{0i} = 0 \ \forall i \in \{1, \dots, d-1\}$, hence in these adapted coordinates the metric of the static spacetime reads:

$$ds^2 = g(\xi, \xi)dt^2 + g_{ij}dx^i dx^j \quad (7.3.1)$$

Clearly, this metric is invariant under $t \mapsto -t$, which means that a static spacetime is *time-reflection symmetric*. In loose terms, a spacetime is stationary if “nothing changes with time”, and it is static if in addition “nothing moves”. An example of a stationary spacetime that is not static is a spacetime containing a steadily-rotating body, e.g. a star or a black hole.

§7.3.1.1 Black holes in stationary spacetimes

Consider a spacetime $(\mathcal{M}, g)$ which contains a black hole: in particular, $(\mathcal{M}, g)$ admits a conformal completion $(\tilde{\mathcal{M}}, \tilde{g})$ with a complete future infinity $\mathcal{I}^+$ (Def. 7.1.4). Assuming $(\mathcal{M}, g)$ to be stationary, it is invariant under the isometry group $\mathbb{R}_+$ generated by the Killing vector field $\xi \in \mathfrak{X}(\mathcal{M})$: this can then be extended smoothly to the boundary $\partial\tilde{\mathcal{M}} = \mathcal{I}$, thereby defining a group action $\Phi : \mathbb{R}_+ \times \tilde{\mathcal{M}} \rightarrow \tilde{\mathcal{M}}$ on the whole conformal completion.

Proposition 7.3.1 (Stationary event horizon)

Let $(\mathcal{M}, g)$ be a stationary spacetime containing a black hole. Then, the black hole event horizon $\mathcal{H}$ is globally invariant under the action of the stationary group $\mathbb{R}_+$.

Proof. By the above discussion, the stationary group $\mathbb{R}_+$ acts on the whole conformal completion $\tilde{\mathcal{M}}$, so on the conformal infinity $\mathcal{I}$ too. Let $t \in \mathbb{R}$ and $p \in \mathcal{I}^+$. Since $\mathbb{R}_+$ is an isometry group of g and p is infinitely remote from points of $\mathcal{M}$, then $\Phi_t(p)$ must be infinitely remote from any point of $\mathcal{M}$ too: thus, $\Phi_t(\mathcal{I}^+) \subseteq \mathcal{I}$. Since $\Phi_0(\mathcal{I}^+) = \mathcal{I}^+$, by continuity $\Phi_t(\mathcal{I}^+) \subseteq \mathcal{I}^+$, i.e. $\mathcal{I}^+$ is globally invariant under the stationary group action, which means that its causal past $J^-(\mathcal{I}^+)$ is invariant too. By Eq. 7.1.15, this concludes the proof. $\square$

The fact that $\mathcal{H}$ is *globally* invariant stresses the fact that it is invariant as a whole, not that each point of $\mathcal{H}$ is a fixed point of the group action.

Now, assume that $\mathcal{H}$ is smooth (which can be shown to be the case in a stationary context): by Prop. 7.1.7, it is then a null hypersurface. $\mathcal{H}$ is globally invariant under the isometry group if and only if its Killing generator ξ is tangent to $\mathcal{H}$, but by Lemma 7.2.2 a timelike vector cannot be tangent to a null hypersurface¹⁰, thus ξ is either null or spacelike on $\mathcal{H}$. Given a connected component $\mathcal{H}_0 \subseteq \mathcal{H}$, if ξ is null on all $\mathcal{H}_0$, then $\mathcal{H}_0$ is called **non-rotating**, while if ξ is spacelike on some part of $\mathcal{H}_0$, then $\mathcal{H}_0$ is called **rotating**.

Intuitively, the event horizon of a black hole in a stationary spacetime should have vanishing expansion along its null normals: indeed, if $\theta_{(\ell)} \neq 0$, the area of cross-section would vary when dragged along ℓ , thus defining an “evolution” along $\mathcal{H}$ which would break the invariance of $\mathcal{H}$ under the

¹⁰This justifies the distinction between strictly stationary and stationary spacetimes, since a black hole spacetime cannot be strictly stationary.

action of the stationary group. A rigorous proof is in Prop. 9.3.1 of [8]. Moreover, if the cross-section $\mathcal{S}$ of $\mathcal{H}$ are closed manifolds such that the horizon topology is $\mathcal{H} \cong \mathbb{R} \times \mathcal{S}$, then $\mathcal{H}$ is a non-expanding horizon (Def. 7.2.3).

§7.3.1.2 Topology theorems

In $d = 4$, the topology of the horizon cross-sections can be strongly constrained.

Theorem 7.3.1 (Hawking's topology theorem [7])

Let $(\mathcal{M}, g)$ be a 4-dimensional stationary spacetime containing a black hole with event horizon $\mathcal{H}$. Let $\mathcal{S}$ be a connected component of a complete cross-section of $\mathcal{H}$, and assume that:

- (1) $\mathcal{S}$ is closed (compact withouth boundary) and orientable;
- (2) the null dominance condition (Eq. 7.2.30) is satisfied on $\mathcal{H}$ for some $f \in \mathcal{C}(\mathcal{M}, \mathbb{R}_0^+)$;
- (3) given an ingoing null normal $\mathbf{k}$ to $\mathcal{S}$, when displaced into the black hole exterior along $-\mathbf{k}$, $\mathcal{S}$ becomes a surface with $\theta_{(\ll)} > 0$.

Then, the cross-section has generically the topology $\mathcal{S} \cong \mathbb{S}^2$, or, under specific hypotheses (namely the induced metric $g|_{\mathcal{S}}$ being flat), $\mathcal{S} \cong \mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$.

For General Relativity, (2) with $f = \Lambda$ is equivalent to assuming $\Lambda \geq 0$ and the dominant null energy condition.

This topology theorem is not specific of stationary black hole event horizons, since it relies only on quasi-local properties: in particular, it only requires an hypersurface (not even a null one) sliced by spacelike compact surfaces with $\theta_{(\ell)} = 0$ and $\mathcal{L}_{\mathbf{k}}\theta_{(\ell)} \leq 0$, with a conventional definiton of “interior” region as that pointed towards by $\mathbf{k}$.

Another version of the topology theorem relies on the weaker null convergence condition with global hypotheses, and it excludes the torus topology. In particular, it relies on the notion of global hyperbolicity: given a spacetime $(\mathcal{M}, g)$, a subset $U \subseteq \mathcal{M}$ is **globally hyperbolic** if it admits a *Cauchy hypersurface* (Def. 6.2.2).

Theorem 7.3.2 (Chruściel–Wald topology theorem [5])

Let $(\mathcal{M}, g)$ be a 4-dimensional asymptotically-flat stationary spacetime containing a black hole with event horizon $\mathcal{H}$. Assume that:

- (1) the null convergence condition (Prop. 7.2.3) is satisfied;
- (2) the domain of outer communications $\langle\langle \mathcal{M} \rangle\rangle$ is globally hyperbolic;
- (3) $\langle\langle \mathcal{M} \rangle\rangle$ contains an achronal asymptotically-flat hypersurface Σ such that $\mathcal{H} \cap \Sigma = \mathcal{S}$, where $\mathcal{S}$ is a compact cross-section;
- (4) other technical conditions on the spacetime (see [5]);

Then, $\langle\langle \mathcal{M} \rangle\rangle$ is simply connected and any connected component of $\mathcal{S}$ is homeomorphic to $\mathbb{S}^2$.

Both Th. 7.3.1 and Th. 7.3.2 regard only a connected component of the complete cross-section of the horizon: indeed, there exist stationary (actually static) spacetimes containing a black hole whose event horizon has disconnected complete cross-sections, known as *Majumdar–Papapetrou space-*

times. A generalization of Th. 7.3.1 to spacetimes with $d > 4$ is possible, but the result is much weaker and puts less stringent constraints on the topology of cross-sections: this is due to the fact that, for $d > 4$, $\dim_{\mathbb{R}} \mathcal{S} > 2$, but the proof of Th. 7.3.1 relies on the Gauss–Bonnet theorem¹¹ which is only valid for 2-surfaces.

§7.3.2 Mass and angular momentum

For asymptotically-flat stationary spacetimes, it is possible to make the concept of mass well-defined. With the additional requirement of axisymmetry, the concept of angular momentum can be introduced too.

§7.3.2.1 Weakly-relativistic stationary spacetimes

Definition 7.3.3 (Weakly-relativistic spacetime)

Let $(\mathcal{M}, g)$ be a d -spacetime. Then, it is called **weakly relativistic** if $\mathcal{M}$ is diffeomorphic to $\mathbb{R}^d$ and the metric tensor can be decomposed as:

$$g = f + h \quad (7.3.3)$$

where f is a flat Lorentzian metric on $\mathcal{M}$ and h is small in the following sense: in any Minkowskian coordinates, i.e. (x^α) on $\mathcal{M}$ such that $f_{\alpha\beta} = \eta_{\alpha\beta}$, then $|h_{\alpha\beta}| \ll 1$.

Note that Minkowskian coordinates are not unique, since any Poincaré transformation $(\Lambda, \mathbf{a}) \in \text{ISO}(1, d-1) : x^\alpha \mapsto \Lambda^\alpha_{\mu} x^\mu + a^\alpha$ leads to another Minkowskian coordinate frame. Furthermore, the flat metric itself is not unique. Indeed, consider the change of coordinates $x^\alpha \mapsto y^\alpha = x^\alpha + \zeta^\alpha(x)$, where ζ^α are infinitesimal functions, so that:

$$g'_{\alpha\beta} = \frac{\partial x^\mu}{\partial y^\alpha} \frac{\partial x^\nu}{\partial y^\beta} g_{\mu\nu} = (\delta^\mu_\alpha - \partial_\alpha \zeta^\mu + o(\zeta^2)) (\delta^\nu_\beta - \partial_\beta \zeta^\nu + o(\zeta^2)) (\eta_{\mu\nu} + h_{\mu\nu})$$

At first order in ζ , defining $\zeta_\alpha \equiv \eta_{\alpha\mu} \zeta^\mu$, then:

$$g'_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta} - \partial_\alpha \zeta_\beta - \partial_\beta \zeta_\alpha$$

Recasting this expression as $g' = f' + h'$, where f' is the metric with components $f'_{\alpha\beta} = \eta_{\alpha\beta}$ in the (y^α) coordinate frame, the transformation preserves the flat metric while transforming the “small perturbation” as:

$$h'_{\alpha\beta} = h_{\alpha\beta} - \partial_\alpha \zeta_\beta - \partial_\beta \zeta_\alpha \quad (7.3.4)$$

This can be regarded as a *gauge freedom* of h , which reflects the freedom in the choice of the flat background metric f in the decomposition Eq. 7.3.3.

Lemma 7.3.1 (Lorentz gauge)

Let $(\mathcal{M}, g)$ be a weakly-relativistic spacetime with metric $g = f + h$, where f is a flat Lorentzian

¹¹For a smooth 2-manifold $\mathcal{S}$ with Riemannian metric $q \in \mathfrak{T}_2^0(\mathcal{S})$, the Ricci scalar curvature $\mathcal{R}_{\mathcal{S}}$ associated to q and the Euler characteristic $\chi(\mathcal{S})$ are related by:

$$\int_{\mathcal{S}} dV_q \mathcal{R}_{\mathcal{S}} = 4\pi \chi(\mathcal{S}) \quad (7.3.2)$$

metric. Then, it is always possible to choose Minkowskian coordinates (x^α) such that:

$$\partial_\mu(\eta^{\mu\nu}\bar{h}_{\alpha\nu}) = 0 \quad \bar{h} = h - \frac{1}{2}f \operatorname{tr} h \quad (7.3.5)$$

where $\operatorname{tr} h \equiv \eta^{\mu\nu}h_{\mu\nu}$.

Proof. By Eq. 7.3.4 $\operatorname{tr} h' \equiv \eta^{\mu\nu}h'_{\mu\nu} = \operatorname{tr} h - 2\partial_\mu\zeta^\mu$, so that $\bar{h}'_{\alpha\beta} = \bar{h}_{\alpha\beta} - \partial_\alpha\zeta_\beta - \partial_\beta\zeta_\alpha + \eta_{\alpha\beta}\partial_\mu\zeta^\mu$. Thus:

$$\partial_\mu(\eta^{\mu\nu}\bar{h}'_{\alpha\nu}) = \partial_\mu(\eta^{\mu\nu}\bar{h}_{\alpha\nu}) - \square_f\zeta_\alpha \quad \square_f \equiv \eta^{\mu\nu}\partial_\mu\partial_\nu$$

Therefore, if the Lorentz gauge condition is not fulfilled, it can be enforced with a gauge transformation satisfying $\square_f\zeta_\alpha = \partial_\mu(\eta^{\mu\nu}\bar{h}_{\alpha\nu})$, which always has a solution. $\square$

Recalling Eq. 6.1.12 (in this case with flat background), the Einstein tensor reduces to $G = -\frac{1}{2}\square_f\bar{h}$. Consider now a weakly-relativistic stationary spacetime $(\mathcal{M}, g)$ with stationary Killing vector ξ . The Minkowskian coordinates can be chosen so that x^0 is ignorable, i.e. $\xi = \partial_0$. Let Σ be a $x^0 = \text{const.}$ hypersurface, and let $\gamma = f|_\Sigma$: the coordinates (x^i) form a Cartesian coordinate system on the submanifold (Σ, γ) , since $\gamma = \delta_{ij}dx^i \otimes dx^j$, and the d'Alembertian operator $\square_f$ reduces to the Laplace–Beltrami operator ∇_γ^2 . The Einstein field equations (at linear order in h) then take the form on Σ of $\frac{1}{2}d(d+1)$ independent Poisson equations:

$$\nabla^2\bar{h}_{\alpha\beta} = -16\pi G_N^{(d)}T_{\alpha\beta} \quad (7.3.6)$$

The solutions are obtained using the Green's function of the $(d-1)$ -dimensional Laplace–Beltrami operator:

$$\bar{h}_{\alpha\beta}(x) = \frac{16\pi G_N^{(d)}}{(d-3)\Omega_{d-2}} \int_\Sigma d^{d-1}y T_{\alpha\beta}(y) \|x-y\|^{3-d} \quad (7.3.7)$$

where the norm is the standard Euclidean norm, since γ is a Euclidean metric. Recall that the surface hyper-area Ω_d of $\mathbb{S}^d$ is given by:

$$\Omega_d = \frac{2\pi^{\frac{d+1}{2}}}{\Gamma(\frac{d+1}{2})} \quad \Gamma(z) := \int_0^\infty dt e^{-t} t^{z-1}, \quad z \in \mathbb{C} : \Re z > 0 \quad (7.3.8)$$

Proposition 7.3.2 (Asymptotic metric of weakly-relativistic system)

Let $(\mathcal{M}, g)$ be a weakly-relativistic stationary d -spacetime, with $d \geq 4$, obeying the Einstein field equations Eq. 5.2.10 with $\Lambda = 0$ and an EMT T with compact support (the *source*), where the energy density dominates over spatial stresses (weakly-relativistic matter). In the Lorentz gauge, there exists a coordinate frame (x^α) such that the asymptotic behaviour of the metric tensor for $r^2 \equiv \delta_{ij}x^i x^j \rightarrow \infty$ is:

$$g_{00} = -1 + \alpha_d \frac{M}{r^{d-3}} + o(r^{1-d}) \quad (7.3.9)$$

$$g_{0i} = \frac{d-2}{2} \alpha_d \frac{J_{ij}x^j}{r^{d-1}} + o(r^{1-d}) \quad (7.3.10)$$

$$g_{ij} = \left(1 + \frac{\alpha_d}{d-3} \frac{M}{r^{d-3}}\right) \delta_{ij} + o(r^{2-d}) \quad (7.3.11)$$

where $\alpha_d \equiv 16\pi/((d-2)\Omega_{d-2})$. The **mass** and the **angular momentum** of the source are:

$$M := \int_{\Sigma} d^{d-1}x T_{00}(x) \quad J_{ij} := \int_{\Sigma} d^{d-1}x (x^j T_{0i}(x) - x^i T_{0j}(x)) \quad (7.3.12)$$

where Σ is a $x^0 = \text{const.}$ hypersurface (M and J_{ij} are independent of Σ , i.e. of x^0). Such a coordinate frame is called the **central source rest-frame**, since their origin is in the center of mass of the source, in the sense that:

$$\int_{\Sigma} d^{d-1}x T_{0i}(x) = 0 \quad \int_{\Sigma} d^{d-1}x x^i T_{00}(x) = 0 \quad (7.3.13)$$

The weakly-relativistic nature of the source is expressed in the (x^α) frame as $T_{00} \gg |T_{ij}|$.

§7.3.2.2 Komar mass

To generalize the concept of mass to strongly-relativistic spacetimes in a coordinate-independent way, consider the fact that, in Newtonian gravity, the mass is provided by Gauss's law for the gravitational field $\mathbf{g} = -\nabla\Phi$:

$$M = \frac{1}{4\pi} \int_{\mathcal{S}} d\mathbf{S} \cdot \nabla\Phi$$

where $\mathcal{S}$ is any closed surface surrounding the body. The arbitrariness of $\mathcal{S}$ is possible thanks to the Gauss–Ostrogradsky theorem and to the Laplace equation $\nabla^2\Phi = 0$ (outside the source). Analogously, note that the mass M appears as the coefficient of the dominant r^{3-d} term in the asymptotic expansion of g_{00} in Eq. 7.3.9, so it could be recovered considering the flux of $\partial_r g_{00}$ through a $(d-2)$ -sphere at large r .

In a coordinate-invariant way, it is more appropriate to consider the stationary Killing vector field $\xi \in \mathfrak{X}(\mathcal{M})$, and in particular its metric-dual 1-form $\xi^\flat \in \Omega^1(\mathcal{M})$, since in adapted coordinates $\xi^\alpha = (1, 0, \dots, 0)$ and $\xi_\alpha = g_{0\alpha}$. Then, the concept of flux is naturally generalized by Komar charges (§5.1.1.3), so that, assuming ξ to be normalized as $\langle \xi, \xi \rangle_g \rightarrow -1$ on the asymptotic boundary, the **Komar mass** over a spacelike closed $(d-2)$ -surface $\mathcal{S} \subseteq \mathcal{M}$ is defined as:

$$M_{\mathcal{S}} := \frac{d-2}{d-3} Q_K[\xi; \mathcal{S}] = -\frac{d-2}{d-3} \frac{1}{16\pi G_N^{(d)}} \int_{\mathcal{S}} \star d\xi^\flat \quad (7.3.14)$$

The orientation of $\mathcal{S}$ is set by the $(d-2)$ -form $dV_{\mathcal{S}} \equiv \iota_s \iota_n dV_g$, where $\mathbf{n}$ is any future-directed unit timelike vector field normal to $\mathcal{S}$ and $\mathbf{s}$ is a unit spacelike vector field normal to $\mathbf{n}$, orthogonal to $\mathbf{n}$ and directed towards the “exterior” of $\mathcal{S}$ in the following sense: assuming that $\mathcal{S}$ can be embedded into a spacelike hypersurface $\Sigma \subseteq \mathcal{M}$ that extends towards the asymptotic flat end of $(\mathcal{M}, g)$, then for any point $p \in \mathcal{S}$ there exists a curve entirely lying in Σ , starting from p with tangent vector $\mathbf{s}_p$ and extending to the asymptotic flat end without crossing $\mathcal{S}$ again.

Instead of the integral of a $(d-2)$ -form over a $(d-2)$ -surface, it is possible to express the Komar mass as a flux integral, i.e. the integral of a 2-form contracted with an “area element” normal to the $(d-2)$ -surface. In particular, given a spacelike $(d-2)$ -surface $\mathcal{S} \subseteq \mathcal{M}$, the **area element normal bivector** to $\mathcal{S}$ is the antisymmetric $(2, 0)$ -tensor field defined on $\mathcal{S}$ as:

$$d\mathbf{S} := (\mathbf{s} \wedge \mathbf{n}) \sqrt{q} d^{d-2}x \quad (7.3.15)$$

where $\mathbf{n}$ and $\mathbf{s}$ are the unit timelike and spacelike vector fields normal to $\mathcal{S}$ defined above (in particular, at each $p \in \mathcal{S}$, an orthonormal basis of the timelike plane $T_p^\perp \mathcal{S}$ normal to $\mathcal{S}$ is $(\mathbf{n}_p, \mathbf{s}_p)$), $q \equiv$

$\det q_{ab}$ is the determinant of the induced (Riemannian) metric $q = g|_{\mathcal{S}}$, and $(x^a) = (x^1, \dots, x^{d-2})$ is a coordinate frame on $\mathcal{S}$ such that $(\partial_1, \dots, \partial_{d-2})$ is a right-handed basis for the orientation $dV_{\mathcal{S}}$. The *exterior product* of vectors is analogous to that of forms: $\mathbf{s} \wedge \mathbf{n} := \mathbf{s} \otimes \mathbf{n} - \mathbf{n} \otimes \mathbf{s}$. The bivector $d\mathbf{S}$ can be shown to be independent of the choice of coordinates (x^a) on $\mathcal{S}$ and of orthonormal basis $(\mathbf{n}, \mathbf{s})$ of $T_p^\perp \mathcal{S}$.

Lemma 7.3.2 (Flux integral)

Let $(\mathcal{M}, g)$ be a d -spacetime, let $\omega \in \Omega^2(\mathcal{M})$ and let $\mathcal{S} \subseteq \mathcal{M}$ be a spacelike $(d-2)$ -surface. Then:

$$\int_{\mathcal{S}} \star \omega = \frac{1}{2} \int_{\mathcal{S}} dS^{\mu\nu} \omega_{\mu\nu} \quad (7.3.16)$$

where $d\mathbf{S}$ is the area element normal bivector to $\mathcal{S}$ defined by Eq. 7.3.15.

Using Eq. 7.3.16, since $(d\xi^b)_{\mu\nu} = 2\nabla_\mu \xi_\nu$ (by metric compatibility and Killing's equation), the Komar mass can be rewritten as a flux integral:

$$M_{\mathcal{S}} = -\frac{d-2}{d-3} \frac{1}{16\pi G_N^{(d)}} \int_{\mathcal{S}} dS^{\mu\nu} \nabla_\mu \xi_\nu \quad (7.3.17)$$

Alternatively, without using covariant derivatives:

$$M_{\mathcal{S}} = -\frac{d-2}{d-3} \frac{1}{16\pi G_N^{(d)}} \int_{\mathcal{S}} d^{d-2}x \sqrt{q} \partial_\mu \xi_\nu (s^\mu n^\nu - n^\mu s^\nu) \quad (7.3.18)$$

Example 7.3.1 (Komar mass of Schwarzschild black hole)

Consider the Schwarzschild spacetime $(\mathcal{M}, g)$ in ingoing Eddington–Finkelstein coordinates $(t, r, \vartheta, \varphi)$, so that the metric is Eq. 7.2.33. The stationary Killing vector is $\xi = \partial_t$, and choose $\mathcal{S}$ to be a 2-sphere defined by $(t, r) = \text{const.}$, so that the coordinates spanning $\mathcal{S}$ are $(x^a) = (\vartheta, \varphi)$. For the orthonormal basis of $T_p^\perp \mathcal{S}$, choose:

$$\mathbf{n} = N^{-1} \frac{\partial}{\partial t} - \frac{2mN}{r} \frac{\partial}{\partial r} \quad \mathbf{s} = N \frac{\partial}{\partial r} \quad N \equiv \left(1 + \frac{2m}{r}\right)^{-1/2}$$

which are easily checked to satisfy $\langle \mathbf{n}, \mathbf{n} \rangle_g = -1$, $\langle \mathbf{s}, \mathbf{s} \rangle_g = 1$ and $\langle \mathbf{n}, \mathbf{s} \rangle_g = 0$. Moreover, $\langle \mathbf{n}, \partial_\vartheta \rangle_g = \langle \mathbf{n}, \partial_\varphi \rangle_g = 0$ and $\langle \mathbf{s}, \partial_\vartheta \rangle_g = \langle \mathbf{s}, \partial_\varphi \rangle_g = 0$, which proves that $\mathbf{n}$ and $\mathbf{s}$ are normal to $\mathcal{S}$, and $\mathbf{s}$ points to the exterior of $\mathcal{S}$ (since $N > 0$), so that $(\mathbf{n}, \mathbf{s}, \partial_\vartheta, \partial_\varphi)$ is a right-handed basis. To evaluate the Komar mass, compute:

$$\xi_\alpha = g_{\alpha t} = \left(-1 + \frac{2m}{r}, \frac{2m}{r}, 0, 0\right)$$

which means that the only non-zero derivatives $\partial_\mu \xi_\nu$ are $\partial_r \xi_t = \partial_r \xi_r = -2m/r^2$. The integrand in Eq. 7.3.18 reduces to:

$$\partial_\mu \xi_\nu (s^\mu n^\nu - n^\mu s^\nu) = \partial_r \xi_t (s^r n^t - n^r s^t) = -\frac{2m}{r^2} (N \cdot N^{-1} + 2mN \cdot 0) = -\frac{2m}{r^2}$$

From Eq. 7.2.33, the induced metric on $\mathcal{S}$ is $ds^2 = r^2 d\Omega_2^2$, so that $\sqrt{q} = r^2 \sin \vartheta$ and:

$$M_{\mathcal{S}} = -\frac{1}{8\pi G_N} \int_{\mathcal{S}} d\vartheta d\varphi r^2 \sin \vartheta \left(-\frac{2m}{r^2}\right) = \frac{m}{4\pi G_N} \int_{\mathcal{S}} d\vartheta d\varphi = \frac{m}{G_N}$$

Note that $M_{\mathcal{S}}$ does not depend on r , i.e. it does not depend on the choice of the sphere $\mathcal{S}$.

In general, it can be shown that, for a stationary asymptotically-flat spacetime, the mass in Eq. 7.3.9 is precisely the Komar mass over a sphere at constant (x^0, r) , with $r \rightarrow \infty$: in particular, if the spacetime is weakly relativistic, the Komar mass is given by Eq. 7.3.12 (integral over $\mathcal{S}$). Another useful expression for the Komar mass can be obtained in the form of a volume integral.

Lemma 7.3.3 (Flux and volume integrals)

Let $(\mathcal{M}, g)$ be an asymptotically-flat d -spacetime, and let Σ be a compact spacelike hypersurface bounded by two closed $(d-2)$ -surfaces, an “internal” one $\mathcal{S}_{\text{int}}$ and an “external” one $\mathcal{S}_{\text{ext}}$, where the exterior direction is that of the asymptotic flat end of $(\mathcal{M}, g)$. Let $\mathbf{n}$ be the future-directed unit normal to Σ , and define the **volume element normal vector** to Σ as:

$$d\mathbf{V} := -\mathbf{n}\sqrt{\gamma} d^{d-1}x \equiv -\mathbf{n}dV_{\Sigma} \quad (7.3.19)$$

where $(x^i) = (x^1, \dots, x^{d-1})$ is a coordinate frame on Σ and $\gamma \equiv \det \gamma_{ij}$ is the determinant of the induced metric on Σ . Then, for any $\omega \in \Omega^2(\mathcal{M})$:

$$2 \int_{\Sigma} dV^{\mu} \nabla^{\nu} \omega_{\mu\nu} = \int_{\mathcal{S}_{\text{ext}}} dS^{\mu\nu} \omega_{\mu\nu} - \int_{\mathcal{S}_{\text{int}}} dS^{\mu\nu} \omega_{\mu\nu} \quad (7.3.20)$$

where $dS^{\mu\nu}$ is defined by Eq. 7.3.15, with $\mathbf{s}$ pointing to the asymptotic flat end of $(\mathcal{M}, g)$ for both $\mathcal{S}_{\text{ext}}$ and $\mathcal{S}_{\text{int}}$.

Recalling Eq. 5.1.20, for Killing vector fields $\nabla_{\mu} \nabla^{\alpha} \xi^{\mu} = \mathcal{R}^{\alpha}_{\mu} \xi^{\mu}$, so, considering the 2-form $d\xi^b$, which has components $(d\xi^b)_{\mu\nu} = 2\nabla_{\mu} \xi_{\nu}$, and applying Lemma 7.3.3, for an asymptotically-flat spacetime $(\mathcal{M}, g)$:

$$M_{\mathcal{S}_{\text{ext}}} = M_{\mathcal{S}_{\text{int}}} + \frac{d-2}{d-3} \frac{1}{8\pi G_N} \int_{\Sigma} dV_{\Sigma} \text{Ric}(\xi, \mathbf{n}) \quad (7.3.21)$$

where Σ is a compact spacelike hypersurface like in Lemma 7.3.3. The same holds if Σ is bounded by a single closed $(d-2)$ -surface, provided that the Komar mass of the other (non-existing) surface is set to zero. If $(\mathcal{M}, g)$ is ruled by General Relativity, then in vacuum $\text{Ric} \equiv 0$, which means that the Komar mass $M_{\mathcal{S}}$ is independent of the $(d-2)$ -surface $\mathcal{S}$ chosen. In particular, assuming that the EMT of the matter distribution vanishes for $r \geq r_0$, for some $r_0 \in \mathbb{R}^+$, then the Komar mass of any $(d-2)$ -surface $\mathcal{S}_r$ defined by $(x^0, r) = \text{const.}$ with $r \geq r_0$ is equal to that computed on a $(d-2)$ -sphere at infinity: $M_{\infty} \equiv M_{\mathbb{S}_{\infty}^{d-2}}$.

The Komar mass can be regarded as the *total mass* of a spacetime, with contributions from both matter fields and the gravitational field.

§7.3.2.3 Komar angular momentum

As a stationary Killing vector determines a Komar mass, an axisymmetric Killing vector determines a Komar angular momentum.

Definition 7.3.4 (Axisymmetric spacetime)

A d -spacetime $(\mathcal{M}, g)$ is called **axisymmetric** if it is invariant under the action of $\text{SO}(2) \cong \text{U}(1)$ and the orbits of the group action are everywhere spacelike curves.

Since $\dim_{\mathbb{R}} \text{SO}(2) = 1$, it is generated by a single Killing vector field $\psi \in \mathfrak{X}(\mathcal{M})$, which is unique up to a rescaling constant: since the orbits of $\text{SO}(2)$ are closed curves, the rescaling constant is fixed requiring that the group parameter φ associated with ψ has the standard periodicity 2π . If the set of

points of $\mathcal{M}$ left invariant under $\text{SO}(2)$ is non-empty, it is called the **rotation axis**: equivalently, it is the set of points $p \in \mathcal{M} : \psi_p = \mathbf{0}$. Usually, the rotation axis is a $(d-2)$ -dimensional submanifold of $\mathcal{M}$, and it is timelike in the region where $\text{SO}(2)$ orbits are spacelike.

Given an axisymmetric d -spacetime $(\mathcal{M}, g)$ with $d \geq 3$ and axisymmetric Killing vector field $\psi \in \mathfrak{X}(\mathcal{M})$, the **Komar angular momentum** over a spacelike closed $(d-2)$ -surface $\mathcal{S} \subseteq \mathcal{M}$ is defined as:

$$J_{\mathcal{S}} := -Q_K[\psi, \mathcal{S}] = \frac{1}{16\pi G_N^{(d)}} \int_{\mathcal{S}} \star d\psi^b \quad (7.3.22)$$

Notice the absence of the dimensional factor, when compared to Eq. 7.3.14: this independence of Eq. 7.3.22 from the spacetime dimension is due to the $\text{SO}(2)$ action effective only in a 2-dimensional space, which is the orthogonal complement of the rotation axis. Moreover, $J_{\mathcal{S}}$ is defined for $d \geq 3$, while M_{surf} only for $d \geq 4$.

Since Eq. 7.3.17 and Eq. 7.3.18 only rely on ξ being a Killing vector, they remain valid for ψ (with the appropriate prefactor):

$$J_{\mathcal{S}} = \frac{1}{16\pi G_N^{(d)}} \int_{\mathcal{S}} dS^{\mu\nu} \nabla_{\mu} \psi_{\nu} \quad (7.3.23)$$

$$J_{\mathcal{S}} = \frac{1}{16\pi G_N^{(d)}} \int_{\mathcal{S}} d^{d-2}x \sqrt{q} \partial_{\mu} \psi_{\nu} (s^{\mu} n^{\nu} - n^{\mu} s^{\nu}) \quad (7.3.24)$$

Moreover, it can be shown that, for a stationary axisymmetric asymptotically-flat spacetime, if asymptotically Minkowskian coordinates adapted to ψ are used, so that $\psi = -x^2 \partial_{x^1} + x^1 \partial_{x^2}$ (i.e. the rotation axis is the $(d-2)$ -surface $(x^1, x^2) = (0, 0)$), then the coefficient J_{12} in Eq. 7.3.10 is precisely the Komar angular momentum over a sphere at constant (x^0, r) , with $r \rightarrow \infty$: in particular, if the spacetime is weakly relativistic, the Komar angular momentum is given by Eq. 7.3.12 (J_{12} over $\mathcal{S}$). Eq. 7.3.21 too can be imported to the Komar angular momentum:

$$J_{\mathcal{S}_{\text{ext}}} = J_{\mathcal{S}_{\text{int}}} + \frac{1}{16\pi G_N^{(d)}} \int_{\Sigma} d^{d-1}x \sqrt{\gamma} (-2\mathcal{R}^{\alpha}_{\mu} \psi^{\mu}) n_{\alpha} \quad (7.3.25)$$

where Σ is a compact spacelike hypersurface bounded by two closed $(d-2)$ -surfaces, n is the future directed unit normal to Σ and $\gamma \equiv \det \gamma_{ij}$ is the determinant of the induced metric on Σ . Therefore, in a vacuum spacetime, the Komar angular momentum too is independent of the $(d-2)$ -surface on which it is computed.

The Komar angular momentum can be regarded as the *total angular momentum* of a spacetime, with contributions from both matter fields and the gravitational field.

§7.4 Schwarzschild black holes

The simplest black hole solution is the Schwarzschild metric:

$$ds^2 = - \left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2) \quad (7.4.1)$$

This is a special case of the general metric in §5.1.2.2 with $f(r)^2 = 1 - 2GM/r$: it solves the vacuum Einstein equations $R_{\mu\nu} = 0$.

This solution depends on a single parameter M , which is identified as the mass of the black hole, as highlighted in §5.3. The same can be shown via Komar integrals: note that the Schwarzschild metric admits a timelike Killing vector field $K = \partial_t$, with dual 1-form $K^\flat = g_{00}dt$ and Komar charge:

$$F = dK^\flat = -\frac{2GM}{r^2} dr \wedge dt \quad \Rightarrow \quad M_{\text{Komar}} = -\frac{1}{8\pi G} \int_{\mathbb{S}_R^2} \star F = M$$

for all $R > 2GM$ (radius of the horizon). Although the 2-form F obeys the vacuum Maxwell equations $d \star F = 0$, thus one would expect the associated Komar charge to vanish as there is no current, $M_{\text{Komar}} = M \neq 0$: this is possible because the mass of the black hole is localized at the origin, where F diverges. Moreover, the Schwarzschild solution is mathematically valid for $M \in \mathbb{R}$: the solution $M = 0$ is just Minkowski spacetime, while $M < 0$ will be shown to be unphysical.

§7.4.1 Horizon

The Schwarzschild metric diverges at $r = 0$ and $r = 2GM \equiv R_s$. To distinguish between true singularities and coordinate singularities, the usual way is defining a coordinate-independent scalar quantity and studying it at divergence points.

Due to the vacuum field equations, the simplest scalars R and $R_{\mu\nu}R^{\mu\nu}$ both vanish: the next simplest curvature scalar is the **Krentschmann scalar** $R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}$, which for the Schwarzschild metric is:

$$R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma} = \frac{48G^2M^2}{r^6} \quad (7.4.2)$$

At $r = 2GM$, the Krentschmann scalar is $\sim 1/(GM)^4$, suggesting that this is only a coordinate singularity: it turns out that is the case, and indeed the surface $r = R_s$ is known as the **event horizon** of the black hole. Note that, interestingly, heavier black holes have smaller curvature at the horizon.

On the contrary, $r = 0$ is a true singularity, simply known as the **singularity** of the (classical) black hole.

§7.4.1.1 Near horizon limit

To study the spacetime in the vicinity of the horizon, set $r = 2GM + \eta$, with $|\eta| \ll 2GM$. Moreover, consider $\eta \in \mathbb{R}^+$, thus restricting to the spacetime just outside the horizon. The metric then becomes, at first order in η :

$$ds^2 = -\frac{\eta}{2GM} dt^2 + \frac{2GM}{\eta} d\eta^2 + (2GM)^2 d\Omega_2^2$$

Spacetime is thus decomposed in a direct product of $\mathbb{S}_{2GM}^2$ and a 2-dimensional Lorentzian manifold. Focusing on the latter, set $\rho^2 = 8GM\eta$, so that:

$$ds^2 = - \left(\frac{\rho}{4GM}\right)^2 dt^2 + d\rho^2 \quad (7.4.3)$$

This is called **Rindler metric** and it is, in fact, just Minkowski spacetime:

$$T = \rho \sinh\left(\frac{t}{4GM}\right) \quad X = \rho \cosh\left(\frac{t}{4GM}\right) \quad \Rightarrow \quad ds^2 = -dT^2 + dX^2$$

These are precisely the coordinates Eq. 4.2.5 experienced by an observer undergoing constant acceleration $a = 1/(4GM)$: this makes sense, as an observer sitting at constant ρ , i.e. constant r , must accelerate in order not to fall into the black hole.

The outside of the black hole is $(\rho, t) \in \mathbb{R}^+ \times \mathbb{R}$, corresponding to $X > |T|$. The horizon $\rho = 0$, instead, is mapped to the origin $T = X = 0$ of Minkowski space. However, note that the time coordinate is undefined at the origin since $g_{00} = 0$: scaling $t \rightarrow \infty$ and $\rho \rightarrow 0$ keeping $\rho e^{\pm t/4GM}$ fixed makes it clear that the horizon actually corresponds to the lines:

$$r = 2GM \quad \Rightarrow \quad X = \pm T$$

Therefore, the event horizon is not a timelike surface (like the surface of a star), but a null surface. Although the outside of the event horizon is only described by $X > |T|$, it is nonetheless possible to extend the metric to the whole Minkowski space $X, T \in \mathbb{R}$: this makes it clear that the event horizon is not a real singularity, as nothing particular happens at $X = \pm T$ (see Fig. 7.10). Zooming out, however, the peculiar properties of the horizon start to emerge from a global perspective.

§7.4.2 Eddington–Finkelstein coordinates

Both the Schwarzschild coordinates and the Rindler coordinates do not cover the whole manifold. To find a set of coordinates that does, first introduce a new radial coordinate:

$$dr_*^2 = \left(1 - \frac{2GM}{r}\right)^{-2} dr^2 \quad \Rightarrow \quad r_* = r + 2GM \log\left(\frac{r - 2GM}{2GM}\right) \quad (7.4.4)$$

This is the **Regge–Wheeler radial coordinate** and it maps the region outside the horizon $r \in (2GM, \infty)$ to $r_* \in \mathbb{R}$: as an observer approaches the horizon, r changes increasingly slowly varying r_* , since $\frac{dr}{dr_*} \rightarrow 0$ as $r \rightarrow 2GM$. This coordinate is suited to describe the path of lightrays travelling in the radial direction:

$$ds^2 = 0 \quad \Rightarrow \quad \frac{dr}{dt} = \pm \left(1 - \frac{2GM}{r}\right) \quad \Rightarrow \quad \frac{dr_*}{dt} = \pm 1 \quad \Rightarrow \quad t \pm r_* = \text{const.}$$

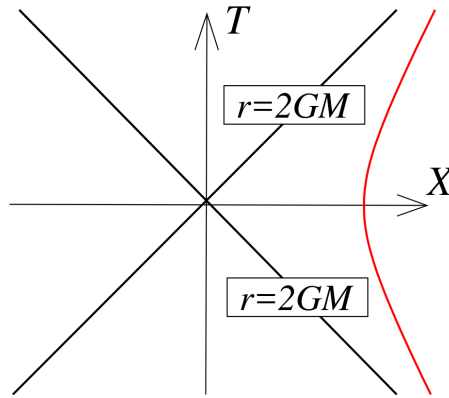


Figure 7.10: Rindler spacetime, with null lines in black and a line at constant $r > 2GM$ in red.

These null radial geodesics correspond to an ingoing lightray for the positive sign and an outgoing lightray for the negative sign (r_* must decrease/increase as t increases). Next, introduce a pair of null coordinates:

$$v = t + r_* \quad u = t - r_* \quad (7.4.5)$$

These allow to extend the Schwarzschild solution beyond the horizon.

§7.4.2.1 Ingoing coordinates

Replace t with $t = v - r_*(r)$:

$$dt = dv - \left(1 - \frac{2GM}{r}\right)^{-1} dr$$

The Schwarzschild metric then becomes, in coordinates (v, r) :

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dv^2 + 2dv dr + r^2 d\Omega_2^2 \quad (7.4.6)$$

This is the Schwarzschild black hole in **ingoing Eddington–Finkelstein coordinates**. Note that there is no more singularity at $r = 2GM$; however, the dv^2 term vanishes at the horizon and becomes negative for $r < 2GM$. Nonetheless, the metric is still non-degenerate thanks to the cross-term:

$$\det g = \det \begin{bmatrix} -(1 - \frac{2GM}{r}) & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \vartheta \end{bmatrix} = -r^4 \sin^2 \vartheta$$

Thus, the metric is non-degenerate and with Lorentzian signature for all values of $r \in \mathbb{R}_+$ (except for the known poles of $\mathbb{S}^2$): this means that the radial coordinate can be extended past the horizon. The globally-timelike Killing vector field $K = \partial_t$ of the Schwarzschild metric is retained in the ingoing Eddington–Finkelstein coordinates as $K = \partial_v$, however now it is no longer globally-timelike: it remains so outside the horizon, where $g_{vv} < 0$, but become spacelike inside it, where $g_{vv} > 0$. This means that the full black hole geometry is not time-independent.

Recall that ingoing lightrays follow geodesics given by $v = \text{const.}$, while outgoing lightrays follow $u = t - r_* = \text{const.}$. As a function of (v, r_*) , the latter reads $v = 2r_* + \text{const.}$, so in (v, r) coordinates:

$$v = 2r + 4GM \log \left(\frac{r - 2GM}{2GM} \right) + \text{const.}$$

This clearly has domain $r \in (2GM, \infty)$, therefore the Regge–Wheeler radial coordinate needs a redefinition:

$$r_* = r + 2GM \log \left| \frac{r - 2GM}{2GM} \right|$$

This means that now r_* is a multi-valued function: $r \in (2GM, \infty)$ is mapped to $r_* \in \mathbb{R}$, while $r \in (0, 2GM)$ is mapped to $r_* \in \mathbb{R}_-$, with the singularity at $r_* = 0$. Outgoing lightrays inside the horizon will then follow geodesics given by:

$$v = 2r + 4GM \log \left(\frac{2GM - r}{2GM} \right) + \text{const.}$$

Outgoing null geodesics at the horizon are easily found: the dv^2 term in the metric Eq. 7.4.6 vanishes and the surface $r = 2GM$ is itself a null geodesic (as event horizons are always null surfaces).

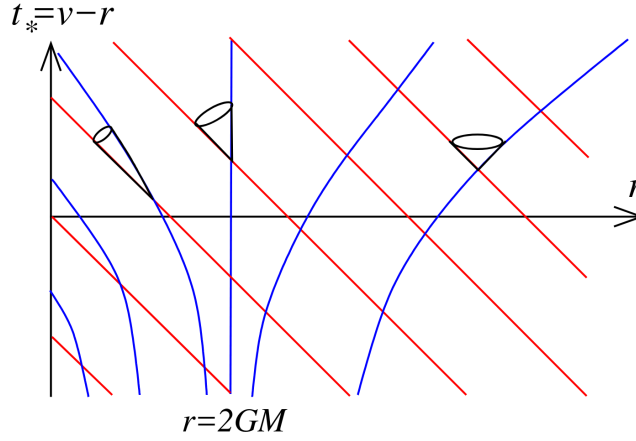


Figure 7.11: Finkelstein diagram in ingoing coordinates: ingoing null geodesics are shown in red, outgoing ones in blue.

All of this information about lightrays can be pictured in a **Finkelstein diagram**, which shows null geodesics in a radial-temporal diagram. Since r_* is multi-valued, r is chosen as radial coordinate: as a consequence, one needs to define a new temporal coordinate $t_* : v = t + r_* = t_* + r$, i.e. $t_* = v - r$. The Finkelstein diagram in ingoing coordinates is shown in Fig. 7.11: ingoing null geodesics are $v = \text{const.}$, i.e. $t_* = \text{const.} - r$, hence travelling at 45° , while outgoing ones are given by the previously found expressions. Note that only outgoing null geodesics outside of the horizon effectively “go out”, since $r \rightarrow \infty$ as $t \rightarrow \infty$ (so $t_* \rightarrow \infty$): those inside the horizon move inexorably towards the curvature singularity at $r = 0$ as time passes, and each one of them reaches it in a finite time t_* . The boundary between these two regions are the null geodesics which run along the horizon at $r = 2GM$.

This analysis can be extended to massive particles: they must move along timelike geodesics inside their future lightcone, which is determined by a pair of ingoing and outgoing future-pointing null geodesics. These future lightcones are also shown in Fig. 7.11: the name “black hole” captures the causal structure of the spacetime, as neither massive particles nor lightrays can escape outside of the event horizon, thus an outside observer can know nothing of the inside of the black hole.

Consider two observers, one at constant $r > 2GM$ and the other moving towards the singularity: as the second approaches the event horizon, light signals emitted by it will take longer and longer to reach the first, and they will appear progressively more redshifted due to the increasingly strong gravitational well. Thus, the outside observer will see the ingoing observer forever, but progressively more slowed down and redshifted, and will never see it passing the event horizon.

§7.4.2.2 Outgoing coordinates

A different extension of the exterior of the Schwarzschild black hole is obtained by replacing t with $u = t - r_*$, i.e. $t = u + r_*(r)$:

$$ds^2 = - \left(1 - \frac{2GM}{r} \right) du^2 - 2du dr + r^2 d\Omega_2^2 \quad (7.4.7)$$

This is the Schwarzschild black hole in **outgoing Eddington–Finkelstein coordinates**. Once again, the metric is smooth and non-degenerate at the horizon, thus it is possible to extend $r \in \mathbb{R}_+$. However, now $r \in (0, 2GM)$ describe a different part of spacetime from the analogous region in ingoing coordinates. To see this, set coordinates $(r, t_*) : t_* = u + r$ and draw the Finkelstein diagram (Fig. 7.12): outgoing null geodesics hence travel at 45° , effectively “going out” regardless of

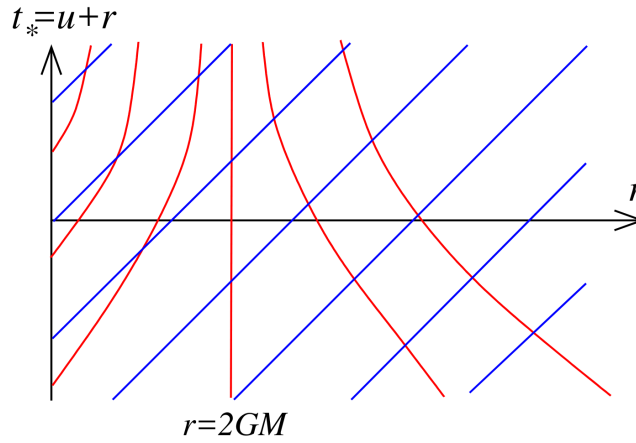


Figure 7.12: Finkelstein diagram in outgoing coordinates: ingoing null geodesics are shown in red, outgoing ones in blue.

their position with respect to the horizon. On the contrary, ingoing null geodesics now do not “go in”: those starting outside of the horizon pile up at the horizon itself, unable to cross it, while those starting inside the horizon move outwards and pile up at it, still unable to cross it.

Studying lightcones, it is easy to see that massive particles inside the horizon are inexorably expelled outside of it in a finite amount of time t_* . This is clearly a very different physics from a black hole: the metric Eq. 7.4.7 is that of a **white hole**, an object which expells any matter inside. A white hole can be viewed as the time reversal of a black hole: this is to be traced to the negative sign in the cross-term and can be seen by flipping Fig. 7.11 upside-down, getting Fig. 7.11.

White holes are perfectly acceptable solutions to the Einstein field equations, and their existence is to be expected from the time-reversal invariance of the field equations. However, they’re not physically relevant, as they cannot be formed from collapsing matter.

§7.4.3 Kruskal spacetime

To better understand how the region parametrized by $r \in (0, 2GM]$ corresponds to two different parts of spacetime, one needs to introduce coordinates which cover the entire spacetime, including both black and white holes.

First, rewrite the Schwarzschild metric in (u, v) coordinates:

$$ds^2 = - \left(1 - \frac{2GM}{r} \right) du dv + r^2 d\Omega_2^2$$

where $r = r(u - v)$. There’s a degeneracy at $r = 2GM$ which can be solved introducing the **Kruskal–Szekeres coordinates**:

$$U = - \exp \left(- \frac{u}{4GM} \right) \quad V = \exp \left(\frac{v}{4GM} \right) \quad (7.4.8)$$

Both U and V are null coordinates, and the exterior of the black hole is parametrized by $U < 0, V > 0$. The metric becomes:

$$ds^2 = - \frac{32(GM)^3}{r} e^{-\frac{r}{2GM}} dU dV + r^2 d\Omega_2^2 \quad (7.4.9)$$

This is known as the metric of *Kruskal spacetime*. The function $r = r(U, V)$ can be found inverting:

$$UV = - \exp \left(\frac{r_*}{2GM} \right) = \frac{2GM - r}{2GM} \exp \left(\frac{r}{2GM} \right) \quad (7.4.10)$$

On the other hand, the function $t = t(U, V)$ can be found inverting:

$$\frac{U}{V} = -\exp\left(-\frac{t}{2GM}\right) \quad (7.4.11)$$

The original Schwarzschild metric, which covers $U < 0, V > 0$, can now be extended to $(U, V) \in \mathbb{R}^2$, without any divergence at $r = 2GM$.

Proposition 7.4.1

Kruskal spacetime is the **maximal extension** of the Schwarzschild metric.

Proof. To check whether a given spacetime can be further extended, one needs to look at all possible geodesics: if they are defined for infinite affine parameter, then they can escape to infinity; however, if they alt at some finite affine parameter, that is either a coordinate singularity, hinting that the spacetime can in fact be extended, or a true singularity. A spacetime is maximally extended if any geodesics that alts at a finite affine parameter does so at a true singularity, and that is the case for Kruskal spacetime. $\square$

The extension process allows to solve the field equations in some particular region (open subset) of spacetime: being the metric components real analytic functions, this is sufficient to determine the metric in all of spacetime.

§7.4.3.1 Kruskal diagram

To see where the event horizon is mapped in Kruskal spacetime:

$$r = 2GM \quad \Rightarrow \quad U = 0 \quad \vee \quad V = 0$$

This means that the event horizon no longer is a null surface, but two null surfaces intersecting at $U = V = 0$, in agreement with the near horizon limit (Rindler space): the null surface $U = 0$ is the horizon of the black hole, known as the *future horizon*; the null surface $V = 0$ is the horizon of the white hole, known as the *past horizon*. On the other hand, the singularity becomes:

$$r = 0 \quad \Rightarrow \quad UV = 1$$

This hyperbola has two disconnected components: one ($U, V > 0$) corresponding to the singularity of the black hole and one ($U, V < 0$) to the singularity of the white hole.

All of this information can be represented in a **Kruskal diagram**, as shown in Fig. 7.13: the U and V axes are at 45° , reflecting that they are null lines, and these are the two horizons. In this diagram, the vertical direction can be viewed as a time coordinate $T = \frac{1}{2}(V + U)$ and the horizontal one as a spatial coordinate $X = \frac{1}{2}(V - U)$. This diagram makes it clear that the black hole and the white hole cohabit the same spacetime. Lines of constant r and t correspond respectively to lines of constant UV and U/V , as from Eq. 7.4.10-7.4.11. The Kruskal diagram makes it clear that, once an observer crosses the event horizon, the singularity is unavoidable: the $r = 0$ are not timelike worldlines, but *the singularity is spacelike*. Therefore, for the in-falling observer, after crossing the horizon the singularity of the black hole does not lie in a spatial direction, but it lies in the future. Similarly, the singularity of the white hole lies in the past. This can be formally seen using the Killing vector field $K = \partial_t$ of the Schwarzschild solution. Outside of the horizon K is timelike, therefore it defines the conserved energy of geodesics; in Kruskal–Szekeres coordinates:

$$K = \partial_t = \frac{\partial V}{\partial t} \partial_V + \frac{\partial U}{\partial t} \partial_U = \frac{1}{4GM} \left(V \frac{\partial}{\partial V} - U \frac{\partial}{\partial U} \right) \quad \Rightarrow \quad g_{\mu\nu} K^\mu K^\nu = - \left(1 - \frac{2GM}{r} \right)$$

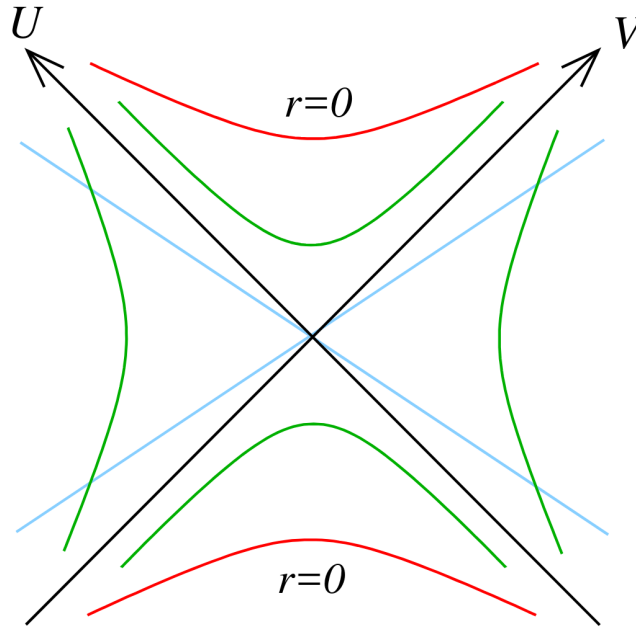


Figure 7.13: Kruskal diagram for the Schwarzschild black hole: green lines are $r = \text{const.}$ and blue lines $t = \text{const.}$, while singularities are in red and horizons in black.

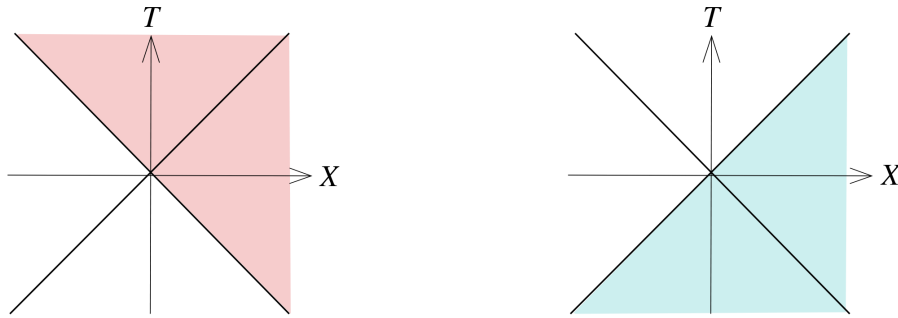


Figure 7.14: Regions of Kruskal spacetime covered by ingoing and outgoing Eddington–Finkelstein coordinates respectively.

Outside of the horizon $K_\mu K^\mu < 0$ as expected. But inside, with $r < 2GM$, the Killing vector field is spacelike: therefore, by Def. 7.3.1 and Prop. 5.3.1, the full Schwarzschild spacetime is not time-independent, but it becomes clear only after crossing the event horizon.

Einstein–Rosen bridge With reference to the Kruskal diagram in Fig. 7.13, the right-hand quadrant is the exterior of the black hole, covered by the Schwarzschild coordinates, while the upper and lower quadrants are respectively the interior of the black hole and of the white hole, covered by ingoing and outgoing Eddington–Finkelstein coordinates (see Fig. 7.14). There remains the left-hand quadrant: this turns out to be just another copy of the exterior of the black hole, now covered by $U > 0, V < 0$.

The full spacetime hence contains two asymptotically-flat regions, joined together by a black hole: note that it is not possible for an observer in one region to send a signal in the other, because the causal structure of the spacetime does not allow this.

To elucidate the spatial geometry connecting these two regions, one needs to study the $t = 0$ hypersurface of Kruskal spacetime: this is a straight horizontal line passing through $U = V = 0$ in

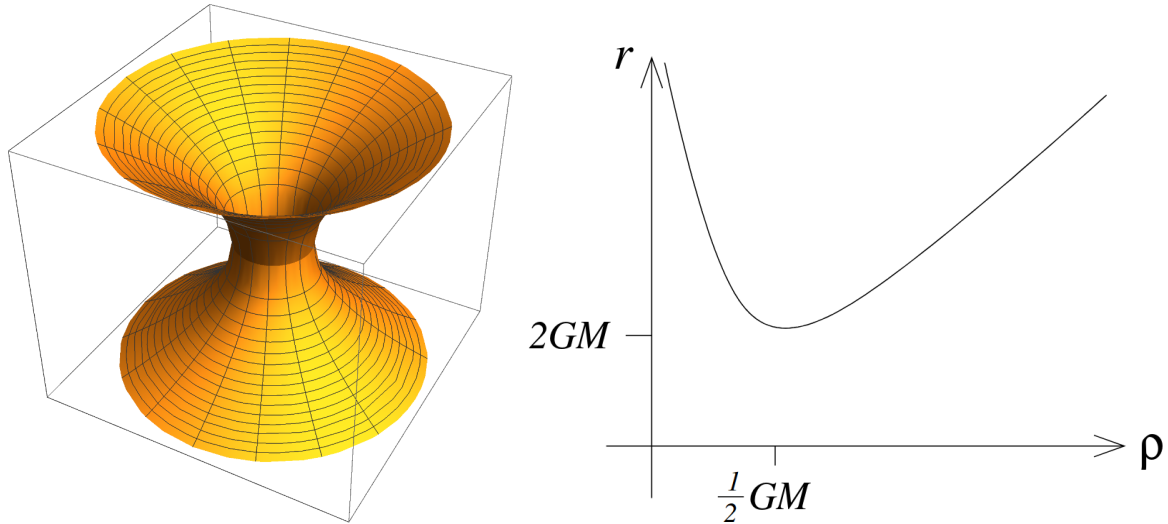


Figure 7.15: The Einstein–Rosen bridge and the ρ coordinate which parametrizes it.

Fig. 7.13. In Schwarzschild coordinates, the geometry of the $t = 0$ hypersurface is given by:

$$ds^2 = \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2) \quad (7.4.12)$$

which is valid for $r > 2GM$. Kruskal spacetime presents two copies of this geometry, one in the right-hand quadrant and one in the left-hand one, which are glued together at $r = 2GM$ to give a wormhole-like geometry as in Fig. 7.15: this is called the **Einstein–Rosen bridge**. Note that it is not possible to travel through it, as the paths are spacelike, not timelike.

It is possible to write a metric which includes both sides. Define the radial coordinate ρ such that:

$$r = \rho \left(1 + \frac{GM}{2\rho}\right)^2 = \rho + GM + \frac{G^2 M^2}{4\rho}$$

Plotting $r = r(\rho)$, as in Fig. 7.15, it is clear that there are two values of ρ for each value of $r > 2GM$, while $r = 2GM$ only corresponds to $\rho = GM/2$: thus, it is possible to parametrize one side of the wormhole with $\rho \in (0, GM/2)$ and the other with $\rho \in (GM/2, \infty)$. The metric in Eq. 7.4.12 becomes:

$$ds^2 = \left(1 + \frac{GM}{2\rho}\right)^4 [d\rho^2 + \rho^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2)] \quad (7.4.13)$$

It is clear that, as $\rho \rightarrow \infty$, the (spatial) metric is that of $\mathbb{R}^3$. But the same is true for $\rho \rightarrow 0$: note that there's a symmetry $r \mapsto r$ for $\rho \mapsto G^2 M^2 / (4\rho)$, which swaps the two asymptotic spacetimes while leaving the midpoint at $\rho = GM/2$ invariant, i.e. it swaps the two sides of the wormhole.

Furthermore, the radius of the spatial $\mathbb{S}^2$ is minimum at $2GM$ in the midpoint $\rho = GM/2$ of the wormhole and increases in either direction: this middle point corresponds to the common point $U = V = 0$ of the two horizons and is called the **bifurcation sphere**.

Although there is no way that an observer in one quadrant can signal an observer in the other quadrant, there is one way for them to communicate: jump into the black hole and meet behind the respective horizons. However, as the white hole is thought to have no physical manifestation, the other universe in the left-hand quadrant of Kruskal space is thought to be a mathematical artifact too.

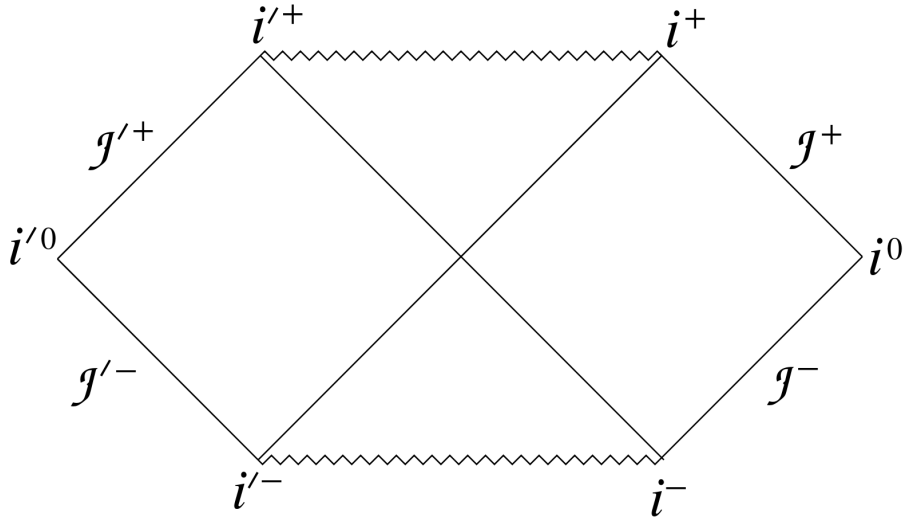


Figure 7.16: The Penrose diagram for Kruskal spacetime.

§7.4.3.2 Penrose diagram

From the Kruskal diagram in Fig. 7.13, it is easy to draw the Penrose diagram for Kruskal spacetime. First, introduce new coordinates which cover the entire spacetime in a finite range:

$$U = \tan \tilde{U} \quad V = \tan \tilde{V}$$

The new coordinates have range $(\tilde{U}, \tilde{V}) \in (-\frac{\pi}{2}, +\frac{\pi}{2})$ and the Kruskal metric becomes:

$$ds^2 = \frac{1}{\cos^2 \tilde{U} \cos^2 \tilde{V}} \left[-\frac{32(GM)^3}{r} e^{-\frac{r}{2GM}} d\tilde{U} d\tilde{V} + r^2 \cos^2 \tilde{U} \cos^2 \tilde{V} d\Omega_2^2 \right]$$

This metric is conformally equivalent to:

$$ds^2 = -\frac{32(GM)^3}{r} e^{-\frac{r}{2GM}} d\tilde{U} d\tilde{V} + r^2 \cos^2 \tilde{U} \cos^2 \tilde{V} d\Omega_2^2$$

However, the singularity at $r = 0$ must be retained: this is mapped to $UV = 1$, thus in the finite-range coordinates:

$$\begin{aligned} \tan \tilde{U} \tan \tilde{V} = 1 &\iff \sin \tilde{U} \sin \tilde{V} - \cos \tilde{U} \cos \tilde{V} = 0 \\ &\iff \cos(\tilde{U} + \tilde{V}) = 0 \iff \tilde{U} + \tilde{V} = \pm \frac{\pi}{2} \end{aligned}$$

These are straight horizontal lines in the Penrose diagram which chop the diamond-shaped diagram: the resulting diagram is shown in Fig. 7.16, with singularities represented by jagged lines.

The Penrose diagram contains the same information as the Kruskal diagram in Fig. 7.13, but it allows to state a more rigorous definition of black hole. Restricting to asymptotically flat spacetimes, the two asymptotic regions contain both null infinities $\mathcal{I}^+$ and $\mathcal{I}^-$: the black hole region is then defined as the set of points which cannot send signals to $\mathcal{I}^+$. The boundary of the black hole region is the *future event horizon* $\mathcal{H}^+$: equivalently, $\mathcal{H}^+$ is the boundary of the causal past of $\mathcal{I}^+$. With reference to Fig. 7.16, the black hole associated to $\mathcal{I}^+$ is the upper quadrant and the left quadrant, while the black hole associated to $\mathcal{I}^-$ is the upper quadrant and the right quadrant.

Importantly, note that to define a black hole one needs to know the whole spacetime, as all lightrays must be run backwards from $\mathcal{I}^+$ to determine $\mathcal{H}^+$ as their boundary: there is no reference to any

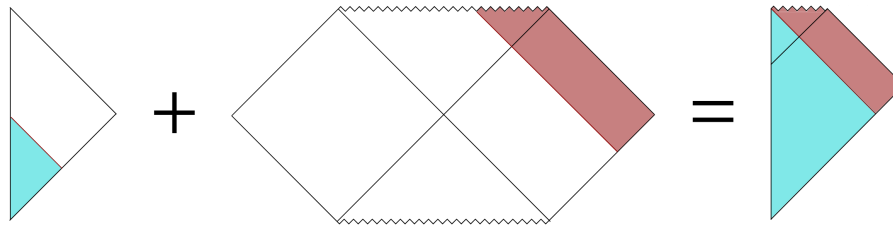


Figure 7.17: Joining two Penrose diagrams.

spacelike hypersurface at constant coordinate time, thus an observer cannot really know if it is inside a black hole unless by knowing the entire future evolution of the spacetime.

Equivalently, it is possible to define the white hole region and the *past event horizon* as the region that cannot receive signals from $\mathcal{I}^-$ and its boundary.

§7.4.4 Weak cosmic censorship

Kruskal spacetime is unphysical in a number of ways. In fact, in reality, black holes do not emerge from white holes, but are formed by collapsing stars: the resulting Penrose diagram is rather different from that in Fig. 7.16. To understand the causal structure of such a black hole, consider the (unrealistic) situation of the spherically-symmetric collapse of a shell of matter: inside the shell, spacetime is flat, meanwhile on the outside it is described by Schwarzschild metric Eq. 7.4.1 (by Birkhoff's theorem, time dependence does not change this fact). Furthermore, assume (unrealistically) that the shell is travelling at the speed of light, so that the Penrose diagrams for Minkowski spacetime and Kruskal spacetime can be glued as in Fig. 7.17: this is the Penrose diagram of a collapsing black hole.

Despite the number of unphysical assumptions, the Penrose diagram thus derived correctly describes the causal structure of a realistic spherically-symmetric collapsing star: as in Eq. 7.4.9, its surface follows a timelike trajectory.

§7.4.4.1 Naked singularities

An important feature of the black hole is that the singularity remains shrouded behind the event horizon, so that an asymptotic observer cannot see it.

On the contrary, Einstein field equations also allow *naked singularities*, i.e. singularities without

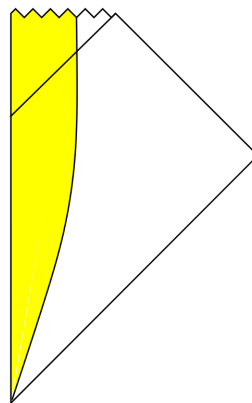


Figure 7.18: Penrose diagram for a collapsing star forming a black hole.



Figure 7.19: Penrose diagram for a negative mass black hole and unphysical collapse of a star.

an horizon. For example, consider Schwarzschild metric with $M < 0$:

$$ds^2 = - \left(1 + \frac{2G|M|}{r} \right) dt^2 + \left(1 + \frac{2G|M|}{r} \right)^{-1} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2)$$

This metric presents no coordinate singularity at $r = 2G|M|$, and so no event horizon. The Penrose diagram for such a spacetime is found analogously to that of Minkowski spacetime, setting null coordinates $u = t - r$ and $v = t + r$. The resulting diagram, shown in Fig. 7.19, is identical to that of Minkowski spacetime (Fig. 7.2), with the difference that now $r = 0$ presents a curvature singularity not shielded by an horizon: it is thus observed from $\mathcal{I}^+$.

An important conjecture in General Relativity, known as the **weak cosmic censorship conjecture**, states the following: given matter which obeys the dominant energy condition Eq. 5.2.26, generic smooth initial conditions on a spatial hypersurface for both the metric and matter fields in an asymptotically-flat spacetime will not evolve to form naked singularities.

If this conjecture is true, then a dynamical evolution such as that in Fig. 7.19 must be ruled out: in this diagram, once the singularity forms, fields can no longer be evolved beyond the shown light-tray; strictly speaking, this means that the dynamical evolution stops at the red line and cannot be extended further, thus abruptly ending $\mathcal{I}^+$.

There is no proof of the weak cosmic censorship conjecture, but only circumstantial evidence. However, there is one naked singularity that seems to be physical: the Big Bang singularity. In fact, since it is in the far past, it does not violate cosmic censorship.

§7.4.5 de Sitter black holes

Schwarzschild metric Eq. 7.4.1 solves the Einstein field equations with $\Lambda = 0$, i.e. in asymptotically Minkowski spacetime. To generalize, consider $\Lambda \neq 0$, so that the field equations become $R_{\mu\nu} = \Lambda g_{\mu\nu}$, and use the following (already seen) ansatz:

$$ds^2 = -f(r)^2 dt^2 + f(r)^{-2} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2)$$

The field equations imply that:

$$f'' + \frac{2f'}{r} + \frac{f'^2}{f} = -\frac{\Lambda}{f} \quad \wedge \quad 1 - 2ff'r - f^2 = \Lambda r^2$$

The most general solution is:

$$f(r)^2 = 1 - \frac{2GM}{r} \mp \frac{r^2}{R^2} \quad R^2 = \frac{3}{|\Lambda|} \quad (7.4.14)$$

where the negative sign is the solution with $\Lambda > 0$ and the positive sign with $\Lambda < 0$, hence describing black holes in de Sitter and anti-de Sitter spacetime respectively: in fact, if $2GM \ll R^2$, then for $r \ll R$ the metric is that of a Schwarzschild black hole in Minkowski spacetime.

§7.5 Reissner–Nordström black holes

§7.6 Kerr black holes

§7.7 Dynamical black holes

Part III

Cosmology

Notation

In the Cosmology part, the metric signature is inverted to a mostly negative $(n - 1, 1)$ one, that is $(+, -, \dots, -)$, therefore a vector $X_p \in T_p \mathcal{M}$ is:

- timelike if $X_p^2 \equiv g_p(X_p, X_p) > 0$;
- null if $X_p^2 \equiv g_p(X_p, X_p) = 0$;
- spacelike if $X_p^2 \equiv g_p(X_p, X_p) < 0$;

Quantities with a subscript '0' are evaluated at the present coordinate time t_0 .

Cosmological Geometrodynamics

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Of the few situations in which Einstein field equations sourced by matter Eq. 5.2.10 need to be solved directly, the one where $T_{\mu\nu}$ plays a crucial role is Cosmology, the study of the universe.

§8.1 Geometry

The key assumption of cosmology is that the universe is *spatially homogeneous and isotropic*, as inferred from empirical observations (at least on large scales): this determines a foliation of spacetime into three-dimensional spatial sections Σ_t at various times $t \in \mathbb{R}$, with each section being homogeneous and isotropic, i.e. maximally-symmetric.

Lemma 8.1.1 (Maximally-symmetric 3-spaces)

The three possible maximally-symmetric 3-spaces are:

- Euclidean space $\mathbb{R}^3$, with vanishing curvature and metric:

$$d\ell^2 = dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2)$$

- the 3-sphere $\mathbb{S}^3$, with uniform positive curvature and metric (implicit unitary radius):

$$d\ell^2 = \frac{1}{1 - r^2} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2)$$

- the 3-hyperboloid $\mathbb{H}^3$, with uniform negative curvature and metric:

$$d\ell^2 = \frac{1}{1 + r^2} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2)$$

These spatial metrics can be unified as:

$$d\ell^2 = \gamma_{ij} dx^i dx^j = \frac{dr^2}{1 - kr^2} + r^2 d\Omega_2^2 \quad (8.1.1)$$

with $k = +1, 0, -1$ on $\mathbb{S}^3, \mathbb{R}^3, \mathbb{H}^3$ respectively.

Proof. First, show that these spaces are homogeneous and isotropic:

- Euclidean space $\mathbb{R}^3$ is clearly invariant under both translations and rotations;
- the 3-sphere $\mathbb{S}^3$ can be embedded in $\mathbb{R}^4$ as:

$$d\ell^2 = d\mathbf{x}^2 + du^2 \quad \mathbf{x}^2 + u^2 = a^2$$

where $a \in \mathbb{R}^+$ is the radius of the 3-sphere. Then, homogeneity and isotropy are inherited from the invariance under 4-rotations of the embedding space;

- the 3-hyperboloid $\mathbb{H}^3$ can be embedded in $\mathbb{R}^{1,3}$ as:

$$d\ell^2 = d\mathbf{x}^2 - du^2 \quad \mathbf{x}^2 - u^2 = -a^2$$

where $a \in \mathbb{R}^+$ is an arbitrary constant and u is a time coordinate. Then, homogeneity and isotropy are inherited from the invariance under pseudo-rotations, i.e. Lorentz transformations, of the embedding space.

In the last two cases, the coordinates can be rescaled as $\mathbf{x} \mapsto a\mathbf{x}$, $u \mapsto au$, so that the spaces can be written as:

$$d\ell^2 = a^2 [d\mathbf{x}^2 + k du^2] \quad \mathbf{x}^2 + k u^2 = k$$

with $k = +1$ for $\mathbb{S}^3$ and $k = -1$ for $\mathbb{H}^3$. The embedding condition gives $\mathbf{x} \cdot d\mathbf{x} = -k u du$, so that:

$$d\ell^2 = a^2 \left[d\mathbf{x}^2 + k \frac{(\mathbf{x} \cdot d\mathbf{x})^2}{1 - k\mathbf{x}^2} \right]$$

which also includes Euclidean space for $k = 0$. Switching to spherical coordinates and using $\mathbf{x} \cdot d\mathbf{x} = r dr$ the thesis is found. $\square$

The existence of these three maximally-symmetric 3-spaces is analogous to the existence of three maximally-symmetric spacetimes as solutions of the vacuum field equations: dS and AdS have constant spacetime curvature, supplied by the cosmological constant, while $\mathbb{S}^3$ and $\mathbb{H}^3$ have constant spatial curvature. Indeed, the metric on $\mathbb{S}^3$ corresponds to the spatial part of the de Sitter metric Eq. 5.1.28, while the metric on $\mathbb{H}^3$ corresponds to the spatial part of the anti-de Sitter metric Eq. 5.1.38.

§8.1.1 FLRW metric

Cosmology studies spacetimes in which space expands as the universe evolves, thus the metric takes the form:

$$ds^2 = dt^2 - a^2(t) \gamma_{ij} dx^i dx^j \quad (8.1.2)$$

This is the **Friedmann–Lemaître–Robertson–Walker metric** and the dimensionless factor $a(t)$ can be viewed as the size of the spatial dimensions, called the *scale factor*.

Example 8.1.1 (de Sitter metric)

de Sitter metric in global coordinates Eq. 5.1.37 is a FLRW metric with $k = +1$.

The FLRW metric has an important scale invariance:

$$a \mapsto \lambda a \quad r \mapsto \lambda^{-1} r \quad k \mapsto \lambda^2 k \quad (8.1.3)$$

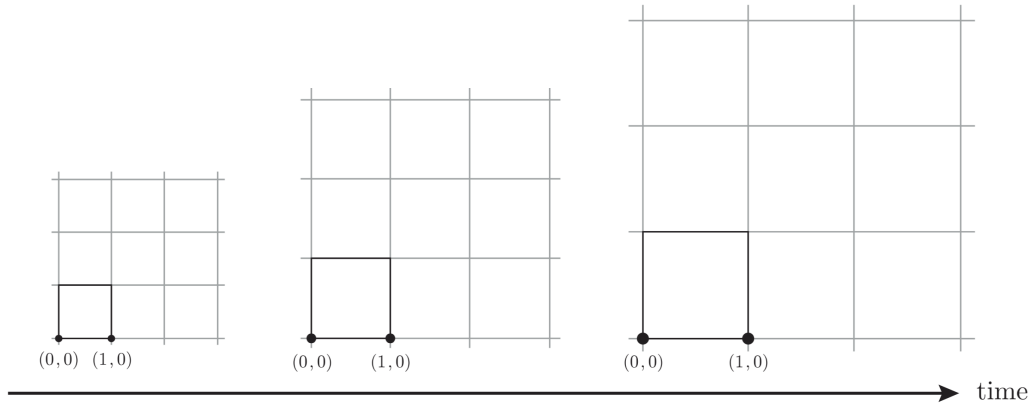


Figure 8.1: Comoving distance and physical distance between two fixed points: as the universe expands, the comoving distance remains constant, while the physical distance varies with $a(t)$.

This symmetry can be used to set $a_0 := a(t_0) \equiv 1$, where t_0 is the present coordinate time. Therefore, the scale factor is dimensionless, while both r and $k^{-1/2}$ have dimensions of a length.

An important clarification must be made on r : this is called a *comoving coordinate*, while the physical results depend only on the *physical coordinate* $r_{\text{phys}} := a(t)r$, as illustrated in Fig. 8.1. Then, the physical velocity of an object clearly is:

$$v_{\text{phys}} = v_{\text{pec}} + Hr_{\text{phys}} \quad (8.1.4)$$

where the peculiar velocity $v_{\text{pec}} = a(t)\dot{r}$ is the velocity measured by a comoving observer, i.e. an observer which follows the Hubble flow, and the **Hubble flow** Hr_{phys} is the contribution due to the expansion of the universe, with the *Hubble parameter* defined as:

$$H := \frac{\dot{a}}{a} \quad (8.1.5)$$

Note that the dot always represents a derivative with respect to coordinate time t .

It is convenient to express the metric in terms of redefined time and radial coordinates, defined as:

$$d\chi \equiv \frac{dr}{\sqrt{1 - kr^2}} \quad d\eta \equiv \frac{dt}{a(t)} \quad (8.1.6)$$

where η is called **conformal time**. The new radial coordinate allows to simplify the expression for g_{rr} , while conformal time determines a factorization of the FLRW metric into a static metric and a time-dependent conformal factor:

$$ds^2 = a^2(\eta) [d\eta^2 - (d\chi^2 + S_k^2(\chi)d\Omega_2^2)] \quad (8.1.7)$$

with:

$$S_k(\chi) = \frac{1}{\sqrt{k}} \begin{cases} \sinh(\sqrt{k}\chi) & k < 0 \\ \sqrt{k}\chi & k = 0 \\ \sin(\sqrt{k}\chi) & k > 0 \end{cases} \quad (8.1.8)$$

§8.2 Kinematics

§8.2.1 Particle motion

In order to study the dynamics of particles in the FLRW metric from the geodesic equation, it is necessary to first compute the Christoffel symbols for this metric. It is straightforward to prove that

the only non-vanishing Christoffel symbols are:

$$\Gamma_{ij}^0 = a\dot{a}\gamma_{ij} \quad \Gamma_{0j}^i = \frac{\dot{a}}{a}\delta_j^i \quad \Gamma_{jk}^i = \frac{1}{2}\gamma^{il}(\partial_j\gamma_{kl} + \partial_k\gamma_{jl} - \partial_l\gamma_{jk}) \quad (8.2.1)$$

Proposition 8.2.1 (Geodesics and 4-momentum)

A particle with 4-momentum p^μ moves along geodesics described by:

$$p^\alpha \partial_\alpha p^\mu = -\Gamma_{\alpha\beta}^\mu p^\alpha p^\beta \quad (8.2.2)$$

Proof. Recall Eq. 2.6.4 and express it in terms of the 4-velocity $u^\mu \equiv \frac{dx^\mu}{d\tau}$:

$$\frac{du^\mu}{d\tau} + \Gamma_{\nu\rho}^\mu u^\nu u^\rho = 0$$

But, using the chain rule:

$$\frac{du^\mu}{d\tau} = \frac{dx^\alpha}{d\tau} \frac{\partial u^\mu}{\partial x^\alpha} = u^\alpha \partial_\alpha u^\mu$$

Substituting $p^\mu = mu^\mu$ yields the thesis (which is valid in the massless case too). $\square$

Note that this is nothing but the condition for p^μ to be parallelly-transported along the geodesic, i.e. $p^\alpha \nabla_\alpha p^\mu = 0$. By the spatial homogeneity of the FLRW metric $\partial_i p^\mu = 0$, hence:

$$p^0 \dot{p}^\mu = -\Gamma_{\alpha\beta}^\mu p^\alpha p^\beta = -(2\Gamma_{0j}^\mu p^0 + \Gamma_{ij}^\mu p^i p^j)$$

First of all, a consequence of this equation is that massive particles which are at rest in the comoving frame, i.e. $p^j = 0$, remain at rest since $\dot{p}^\mu = 0$. Then, consider the $\mu = 0$ component of this equation and set $p^0 \equiv E$:

$$E\dot{E} = -a\dot{a}\gamma_{ij}p^i p^j = -\frac{\dot{a}}{a}\mathbf{p}^2$$

where $\mathbf{p}^2 \equiv -g_{ij}p^i p^j$ so that $p^2 = E^2 - \mathbf{p}^2 = m^2$. Thus, $E dE = \mathbf{p} \cdot d\mathbf{p} \equiv p dp$, with $p \equiv \|\mathbf{p}\|$, and:

$$\frac{\dot{p}}{p} = -\frac{\dot{a}}{a} \implies p \propto \frac{1}{a} \quad (8.2.3)$$

Thus, the physical 3-momentum of any particle decays with the expansion of the universe. In particular, for massless particles $E = p$ decays with the expansion, while for massive particles:

$$p = \frac{mv}{\sqrt{1-\mathbf{v}^2}} \propto \frac{1}{a}$$

where $v^i = \dot{x}^i$ is the comoving peculiar velocity, i.e. the velocity with respect to the comoving frame. Hence, freely-falling particles converge onto the Hubble flow, as per Eq. 8.1.4.

§8.2.2 Redshift

Since the physical properties of the universe are inferred from light signals, it is crucial to quantify how the wavelength of light gets redshifted by the expansions of the universe.

From a quantum point of view, the wavelength of a light signal is linked to the photon's momentum by the de Broglie relation $\lambda = h/p$. From Eq. 8.2.3, then, $\lambda \propto a$, i.e.:

$$\lambda_0 = \frac{a_0}{a(t)} \lambda(t) \quad (8.2.4)$$

where $t < t_0$ is the time of emission of the photon. As $a(t) < a_0$, the light is redshifted. The same result can be derived for classical EM waves. Consider a galaxy at fixed comoving distance d which, at a time η , emits a signal of short conformal duration $\Delta\eta$. The light signal arrives on Earth at $\eta_0 = \eta + d$ with the same conformal duration, but the physical time interval is different:

$$\Delta t = a(\eta)\Delta\eta \quad \Delta t_0 = a_0\Delta\eta$$

Assuming Δt to be the period of the EM wave, then $\lambda(t) = \Delta t$, hence Eq. 8.2.4 is recovered.

Definition 8.2.1 (Redshift parameter)

The **redshift parameter** of a light signal emitted at $t < t_0$ with a wavelength $\lambda(t)$ is

$$z := \frac{\lambda_0 - \lambda(t)}{\lambda(t)} \quad (8.2.5)$$

i.e. as the fractional shift in the photon's wavelength.

By Eq. 8.2.4, setting $a_0 \equiv 1$, the common definition of z is found:

$$1 + z = \frac{1}{a(t)} \quad (8.2.6)$$

For nearby sources, it is possible to expand $a(t)$ as a power series:

$$a(t) = a_0 [1 + H_0(t - t_0) + \dots]$$

where the **Hubble constant** is defined as:

$$H_0 \equiv \frac{\dot{a}(t_0)}{a(t_0)} = 100h \text{ km s}^{-1} \text{ Mpc}^{-1} \quad h \approx 0.67 \pm 0.01 \quad (8.2.7)$$

The dimensionless parameter h is introduced to keep track of how the uncertainties in the measurement of H_0 propagate in other physical quantities, since H_0 normalizes everything. Inserting this expansion in the definition of z yields $z = H_0(t_0 - t) + \dots$, which, for close objects at distance $d = t_0 - t$, becomes:

$$z \simeq H_0 d \quad (8.2.8)$$

Hence, the rate of redshift of nearby objects measures the current expansion rate of the universe.

§8.2.3 Distances

For distant objects, it is important to precisely state which notion of distance is adopted.

Metric distance Rewrite the FLRW metric as:

$$ds^2 = dt^2 - a^2(t) [d\chi^2 + S_k^2(\chi)d\Omega_2^2] \quad S_k(\chi) = \begin{cases} R_0 \sinh(\chi/R_0) & k = -1 \\ \chi & k = 0 \\ R_0 \sin(\chi/R_0) & k = +1 \end{cases}$$

where the length scale R_0 has been introduced thanks to the scale invariance of the metric to compensate for the imposition of $a_0 \equiv 1$. The distance multiplying the solid angle element is the *metric distance*:

$$d_m = S_k(\chi) \quad (8.2.9)$$

In a flat universe, the metric distance is equal to the *comoving distance* χ , which can be computed as a function of redshift:

$$\chi(z) = \int_t^{t_0} \frac{dt'}{a(t')} = \int_0^z \frac{dz'}{H(z')} \quad (8.2.10)$$

which depends on the matter content of the universe through $H(z)$. It must be emphasized that neither the metric distance nor the comoving distance are physically observable.

Luminosity distance Type IA supernovae are called “standard candles”, since they are believed to be objects of known absolute luminosity L , i.e. energy emitted per second: therefore, the observed flux F , i.e. energy per second per receiving area, from a supernova explosion can be used to infer its *luminosity distance*. Considering a source at fixed comoving distance χ , in a static Euclidean space the relation between F and L would be:

$$F = \frac{L}{4\pi\chi^2}$$

However, in a generic FLRW spacetime, this relation needs to be modified for three reasons:

- (1) the proper area of a 3-sphere grows like $4\pi d_m^2$, not like $4\pi\chi^2$;
- (2) the rate of received photons is lower than the rate of emitted photons by a factor of $1/(1+z)$;
- (3) received photons have an energy that is lower than the energy of emitted photons by the same factor of $1/(1+z)$.

Hence, the correct relation is:

$$F = \frac{L}{4\pi d_L^2} \quad d_L = (1+z)d_m \quad (8.2.11)$$

where d_L is the *luminosity distance*.

Angular diameter distance Sometimes, it is possible to make use of “standard rulers”, i.e. objects of known physical size D (e.g. CMB fluctuations). Assuming that a standard ruler is at a fixed comoving distance χ , the *angular diameter distance* can be defined as:

$$d_A = \frac{D}{\delta\theta}$$

where $\delta\theta \ll 1$ is the observed angular size of the object (small for all cosmological objects). In a generic FLRW spacetime, the relation between the physical transverse size of the object and its angular size in the sky instead is:

$$D = a(t)S_k(\chi)\delta\theta = \frac{d_m}{1+z}\delta\theta$$

Hence, the angular diameter distance is:

$$d_A = \frac{d_m}{1+z} \quad (8.2.12)$$

This is the measured distance between Earth and the object when the light was emitted, and in fact it is related to the luminosity distance by:

$$d_A = \frac{d_L}{(1+z)^2} \quad (8.2.13)$$

The redshift dependence of the three distances d_m , d_L and d_A is plotted in Fig. 8.2: note that the presence of a cosmological constant makes distances larger, a fact that has been crucial for the discovery of dark energy.

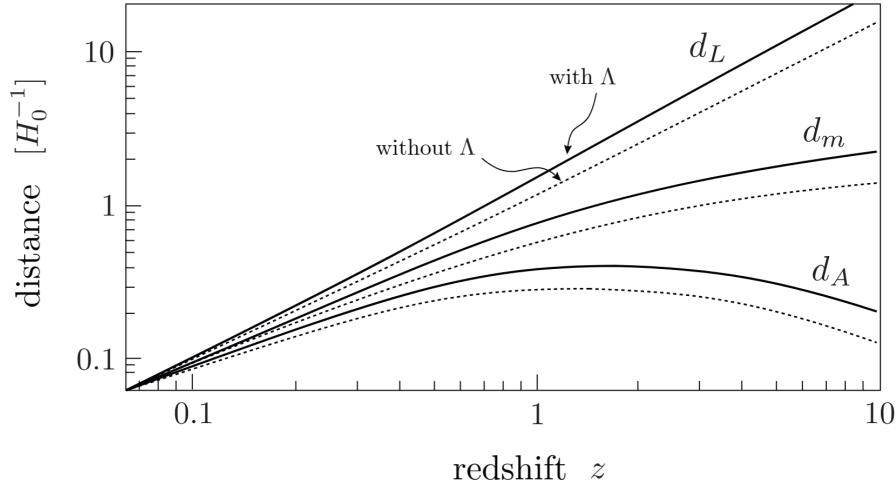


Figure 8.2: Distances measured in a flat universe, with matter only (dotted lines) and with 70% dark energy (solid lines).

§8.3 Dynamics

§8.3.1 Matter sources

The requirements of spatial homogeneity and isotropy impose strict constraints on the energy-momentum tensor, so that it can only assume the form Eq. 5.2.16 for a perfect fluid.

§8.3.1.1 Number density

First, consider the number 4-current N^μ , whose components are the number density N^0 of “particles” (where here particles have a generic nature, e.g. a particle could be a galaxy) and the flux of particles N^i in the direction x^i .

Definition 8.3.1 (Isotropy and homogeneity)

A spacetime is **isotropic** if the mean value of any 3-vector vanishes, while it is **homogeneous** if the mean value of any 3-scalar^a is a function of the time coordinate only.

^aA 3-scalar is a scalar which is invariant under purely spatial coordinate transformations.

The isotropy and homogeneity conditions then impose that a comoving observer measures a number 4-current $N^\mu = (n(t), \mathbf{0})$, where $n(t)$ is the number of particles per proper volume as measured by the comoving observer. A general observer with a relative 4-velocity u^μ with respect to the rest frame of the particle distribution measures instead:

$$N^\mu = n u^\mu \quad (8.3.1)$$

Indeed, for a comoving observer $u^\mu = (1, \mathbf{0})$, while in general $u^\mu = \gamma(1, \mathbf{v})$, which is the correctly boosted result. Since the number of particles has to be conserved, in Minkowski spacetime $\dot{n} = -\partial_i N^i$, i.e. $\partial_\mu N^\mu = 0$, which generalizes to a curved spacetime as:

$$\nabla_\mu N^\mu = 0 \quad (8.3.2)$$

In the comoving frame, then, recalling Eq. 8.2.1:

$$\frac{dn}{dt} + \Gamma_{i0}^i n = 0 \quad \Rightarrow \quad \frac{\dot{n}}{n} = -3 \frac{\dot{a}}{a} \quad \Rightarrow \quad n(t) \propto a^{-3}(t)$$

which means that the number density is proportional to the inverse of the proper volume, as expected.

§8.3.1.2 Energy-momentum tensor

In order to extend this analysis to the energy-momentum tensor $T_{\mu\nu}$, decompose it into a 3-scalar T_{00} , a 3-vector T_{i0} (recall that it is symmetric) and a 3-tensor T_{ij} . By isotropy $\pi_i \equiv T_{i0} = 0$ (energy-momentum flow in the x^i direction), while around a point $\mathbf{x} = \mathbf{0}$ it must be:

$$T_{ij}(\mathbf{x} = \mathbf{0}) \propto \delta_{ij} \propto g_{ij}(\mathbf{x} = \mathbf{0})$$

Since by homogeneity the proportionality constant can only depend on time, the energy-momentum tensor measured by a comoving observer takes the form:

$$T_{00} = \rho(t) \quad \pi_i \equiv T_{i0} = 0 \quad T_{ij} = -P(t)g_{ij}(t, \mathbf{x})$$

or, with mixed indices, $T^\mu_\nu = \text{diag}(\rho, -P, -P, -P)$, which is precisely the energy-momentum tensor for a perfect fluid with energy density $\rho(t)$ and pressure $P(t)$ in its comoving frame. For a general observer with a 4-velocity u^μ with respect to the rest frame of the fluid:

$$T^\mu_\nu = (\rho + P)u^\mu u_\nu - P\delta^\mu_\nu \quad (8.3.3)$$

To study the evolution of $\rho(t)$ and $P(t)$, recall that, in Minkowski spacetime, the fluid is subject to the continuity equation $\dot{\rho} = -\partial_i \pi^i$ and to the Euler equation $\dot{\pi}_i = \partial_i P$, which are combined into $\partial_\mu T^\mu_\nu = 0$. Generalizing to curved spacetime:

$$\nabla_\mu T^\mu_\nu = 0 \quad (8.3.4)$$

i.e. the energy momentum tensor is covariantly-conserved. Recalling Eq. 8.2.1, it is easy to see that the evolution of the energy density is given by this generalized **continuity equation**:

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + P) = 0 \quad (8.3.5)$$

which is equivalent to $dU = -PdV$, with $U \equiv \rho V$ and using $V \propto a^3$.

§8.3.1.3 Matter content

Since the universe is filled with a mixture of different matter components, it is useful to classify the various matter sources by their equation of state $\rho = \rho(P)$, which can be cast in the form:

$$\rho = wP \quad (8.3.6)$$

with $w \in \mathbb{R}$. With this relation, Eq. 8.3.5 implies the scaling relation:

$$\rho \propto a^{-3(1+w)} \quad (8.3.7)$$

Matter Matter is characterized by $w = 0$, i.e. the pressure is negligible when compared to the energy density (to be precise, $P = 0$ for pressureless dust only). Then, the dilution of $\rho \propto a^{-3}$ simply reflects the expansion of the volume $V \propto a^3$. This category has two elements:

- **baryons**, where in Cosmology this term is improperly used to denote both nuclei and electrons¹;
- **dark matter**, which is the dominant form of matter in the universe, although its nature is unknown.

¹Since the mass of the lightest nucleus $^1\text{H} \equiv p^+$ is $m_p \simeq 1836m_e$, this convention is approximately justified, since the mass of baryonic matter is dominated by the mass of the nuclei.

Radiation Radiation is characterized by $w = \frac{1}{3}$, which is the case, for instance, for a gas of relativistic particles, whose energy density is dominated by the kinetic energy (i.e. $p \gg m$). In this case, the dilution $\rho \propto a^{-4}$ includes an additional factor for the redshift of the energy $E \propto a^{-1}$. This category includes:

- **photons**, which are always relativistic (since they are massless particles) and dominated the early universe;
- **neutrinos**, which behaved like radiation for most of the history of the universe, and only recently started behaving like matter due to their small masses;
- **gravitons**, which are believed to have been formed in the early universe as a background of gravitational waves.

Dark energy By experimental observations, the universe today seems to be dominated by a mysterious *negative* pressure $P = -\rho$, commonly referred to as dark energy. This mysterious form of energy has $w = -1$, hence its energy density does not dilute $\rho \propto a^0$, which means that energy has to be created as the universe expands².

A possible candidate as a form of dark energy is the vacuum energy predicted by Quantum Field Theory. Indeed, the ground state of the vacuum has the following energy-momentum tensor:

$$T_{\mu\nu}^{\text{vac}} = \rho_{\text{vac}} g_{\mu\nu}$$

which clearly implies $P_{\text{vac}} = -\rho_{\text{vac}}$. However, the predicted ρ_{vac} is off of a factor of $\sim 10^{120}$ with respect to the observed dark energy density, and this is called the “cosmological constant problem”.

§8.3.2 Curvature tensors

To compute the Einstein tensor for the FLRW metric, recall the Christoffel symbols Eq. 8.2.1.

Proposition 8.3.1 (Ricci tensor)

The Ricci tensor for a FLRW metric has non-vanishing components:

$$\mathcal{R}_{00} = -3\frac{\ddot{a}}{a} \quad \mathcal{R}_{ij} = -\left[\frac{\ddot{a}}{a} + 2\left(\frac{\dot{a}}{a}\right)^2 + 2\frac{k}{a^2}\right]g_{ij} \quad (8.3.8)$$

Proof. First, $\mathcal{R}_{0i} = 0$ due to the isotropy of the FLRW metric. Then, contracting Eq. 2.5.6 and recalling the only non-vanishing Christoffel symbols:

$$\mathcal{R}_{00} = -\partial_0 \Gamma_{i0}^i - \Gamma_{i0}^j \Gamma_{j0}^i = -3\frac{d}{dt}\left(\frac{\dot{a}}{a}\right) - 3\left(\frac{\dot{a}}{a}\right)^2 = -3\frac{\ddot{a}}{a}$$

For the spatial components, consider the spatial metric in Cartesian coordinates (as derived in the proof of Lemma 8.1.1):

$$\gamma_{ij} = \delta_{ij} + \frac{kx_i x_j}{1 - k\mathbf{x} \cdot \mathbf{x}}$$

The Christoffel symbols depend on $\partial\gamma$ and the Ricci tensor on $\partial^2\gamma$, thus to evaluate the latter at

²Recall §5.2.2: since it is not possible to clearly state an energy conservation principle for a gravitational system, the only meaningful equation is Eq. 8.3.5, which is not violated by the creation of additional dark energy as the universe expands.

$\mathbf{x} = 0$ one only needs the metric up to quadratic order:

$$\gamma_{ij} = \delta_{ij} + kx_i x_j + o(x^4) \Rightarrow \gamma^{ij} = \delta^{ij} - kx^i x^j + o(x^4) \Rightarrow \Gamma_{jk}^i = kx^i \delta_{jk} + o(x^3)$$

where i, j indices are raised or lowered by δ^{ij} . The Ricci tensor is then computed as:

$$\begin{aligned} \mathcal{R}_{ij} &= \partial_\rho \Gamma_{ij}^\rho - \partial_j \Gamma_{\rho i}^\rho + \Gamma_{ij}^\lambda \Gamma_{\rho\lambda}^\rho - \Gamma_{\rho i}^\lambda \Gamma_{j\lambda}^\rho \\ &= (\partial_0 \Gamma_{ij}^0 + \partial_k \Gamma_{ij}^k) - \partial_j \Gamma_{ki}^k + (\Gamma_{ij}^0 \Gamma_{k0}^k + \Gamma_{ij}^k \Gamma_{lk}^l) - (\Gamma_{ki}^0 \Gamma_{j0}^k + \Gamma_{0i}^k \Gamma_{jk}^0 + \Gamma_{li}^k \Gamma_{jk}^l) \end{aligned}$$

Evaluating this expression at $\mathbf{x} = \mathbf{0}$ allows to drop the $\Gamma_{ij}^k \Gamma_{lk}^l$ term and to replace any undifferentiated γ_{ij} with δ_{ij} , so that:

$$\begin{aligned} \mathcal{R}_{ij}(\mathbf{x} = \mathbf{0}) &= (\partial_0(a\dot{a}) + 3k - k + 3\dot{a}^2 - \dot{a}^2 - \dot{a}^2) \delta_{ij} + o(x^2) \\ &= (a\ddot{a} + 2\dot{a}^2 + 2k) \delta_{ij} + o(x^2) \end{aligned}$$

Spatial homogeneity and isotropy imply that $\mathcal{R}_{ij} \propto \gamma_{ij}$, thus the general result is:

$$\mathcal{R}_{ij} = (a\ddot{a} + 2\dot{a}^2 + 2k) \gamma_{ij} = -\frac{1}{a^2} (a\ddot{a} + 2\dot{a}^2 + 2k) g_{ij}$$

which is the thesis. $\square$

The Ricci scalar is then easily computed:

$$\mathcal{R} = -6 \left[\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} \right] \quad (8.3.9)$$

Finally, the non-vanishing components of the Einstein tensor are:

$$G_{00} = 3 \left[\left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} \right] \quad G_{ij} = \left[2\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} \right] g_{ij} \quad (8.3.10)$$

§8.3.3 Friedmann equations

With Eq. 8.3.10, the Einstein field equations Eq. 5.2.10 with energy-momentum tensor Eq. 8.3.3 become (respectively the temporal and spatial components):

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2} \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3} (\rho + 3P) \quad (8.3.11)$$

where the cosmological constant has been suppressed since it is accounted for by dark energy as a form of matter content. Note that these equations are not independent, as the latter is the time derivative of the former: nonetheless, these are called the **Friedman equations**. The energy density ρ and the pressure P are to be understood as the sums of the contributions from the various forms of matter: the contribution from radiation $\rho_r = \rho_\gamma + \rho_\nu$ (photons and neutrinos), the one from matter $\rho_m = \rho_b + \rho_c$ (baryons and cold dark matter) and the vacuum energy contribution ρ_Λ . Recalling the definition of the Hubble parameter Eq. 8.1.5, the first Friedmann equation becomes:

$$H^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2} \quad (8.3.12)$$

It is convenient to define dimensionless density parameters. To do so, observe that a flat universe corresponds to the **critical density** today:

$$\rho_{\text{crit},0} = \frac{3H_0^2}{8\pi G} \simeq 1.9 \cdot 10^{-29} h^2 \text{ g cm}^{-3}$$

where $H_0 \equiv 100h \text{ km s}^{-1} \text{ Mpc}^{-1}$. Then, the dimensionless density parameters are defined as:

$$\Omega_a \equiv \frac{\rho_{a,0}}{\rho_{\text{crit},0}} \quad (8.3.13)$$

with $a = \text{r, m, } \Lambda, \dots$ running over all possible matter species. Eq. 8.3.12 can thus be rewritten as:

$$H^2(a) = H_0^2 \left[\Omega_{\text{r}} \left(\frac{a_0}{a} \right)^4 + \Omega_{\text{m}} \left(\frac{a_0}{a} \right)^3 + \Omega_k \left(\frac{a_0}{a} \right)^2 + \Omega_{\Lambda} \right] \quad (8.3.14)$$

where a curvature density parameter was defined as:

$$\Omega_k \equiv -\frac{k}{(a_0 H_0)^2} \quad (8.3.15)$$

Adopting the conventional normalization for the scale factor $a_0 \equiv 1$:

$$\frac{H^2}{H_0^2} = \Omega_{\text{r}} a^{-4} + \Omega_{\text{m}} a^{-3} + \Omega_k a^{-2} + \Omega_{\Lambda} \quad (8.3.16)$$

§8.3.3.1 Λ CDM

Empirical observations show that the universe is filled with radiation, matter and dark energy so that:

$$|\Omega_k| \leq 0.01 \quad \Omega_{\text{r}} \simeq 9.4 \cdot 10^{-5} \quad \Omega_{\text{m}} \simeq 0.32 \quad \Omega_{\Lambda} \simeq 0.68$$

In particular, matter splits into 5% baryonic matter, i.e. with $\Omega_{\text{b}} \simeq 0.05$, and 27% cold dark matter, i.e. with $\Omega_{\text{c}} \simeq 0.27$. Moreover, note that even today curvature only contributes for less than 1% of cosmic energy, and, since $\Omega_k \propto a^{-2}$ while $\Omega_{\text{m}} \propto a^{-3}$ and $\Omega_{\text{r}} \propto a^{-4}$, in the past it was completely negligible: hence, setting $\Omega_k \equiv 0$ is justified, i.e. $k \equiv 0$.

§8.3.3.2 Single-component universe

Given the clearly distinct scalings of $\Omega_{\text{r}} \propto a^{-4}$, $\Omega_{\text{m}} \propto a^{-3}$ and $\Omega_{\Lambda} \propto a^0$, for most of its history the universe was dominated by a single component: first radiation, then matter and finally vacuum energy. This is illustrated in Fig. 8.3.

For a flat single-component universe, the first Friedmann equation becomes:

$$\frac{\dot{a}}{a} = H_0 \sqrt{\Omega_a} a^{-\frac{3}{2}(1+w_a)} \quad (8.3.17)$$

where w_a characterizes the component. Note that $\Omega_a \equiv 1$, since this single-component universe is flat. This equation can be integrated, obtaining the time-dependence of the scale factor:

$$a(t) \propto \begin{cases} t^{\frac{2}{3(1+w_a)}} & w_a \neq -1 \\ e^{Ht} & w_a = -1 \end{cases} \quad (8.3.18)$$

Specializing for a radiation-dominated (RD), matter-dominated (MD) and Λ -dominated (Λ D) universe:

$$a(t) \propto \begin{cases} t^{1/2} & \text{RD} \\ t^{2/3} & \text{MD} \\ e^{Ht} & \Lambda\text{D} \end{cases} \implies a(\eta) \propto \begin{cases} \eta & \text{RD} \\ \eta^2 & \text{MD} \\ -\eta^{-1} & \Lambda\text{D} \end{cases} \quad (8.3.19)$$

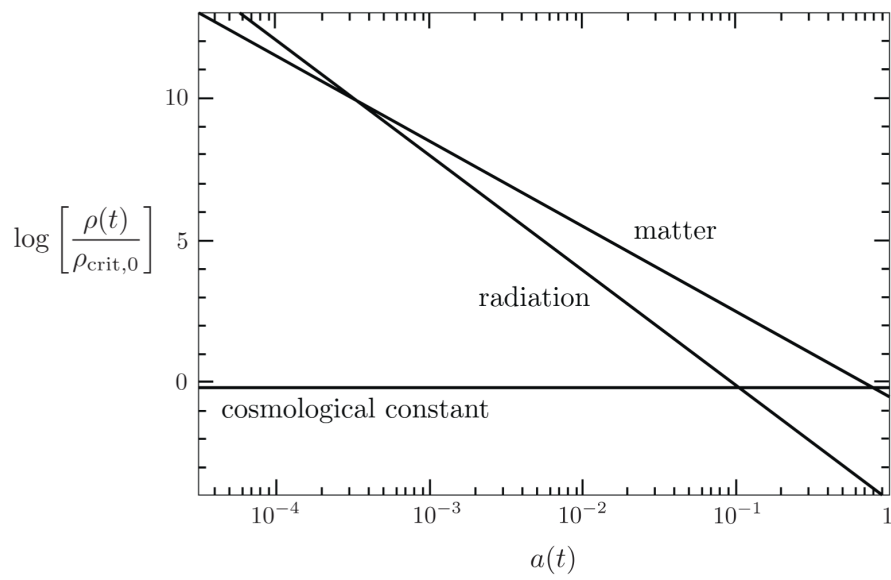


Figure 8.3: Evolution of energy densities in the universe.

Inflation

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§9.1 Horizon problem

§9.1.1 Particle horizon

The propagation of lightrays is best studied using conformal time. Given the spatial isotropy of spacetime, it is always possible to define a coordinate system in which the lightrays move radially, so that their motion is determined by a 2-dimensional line element:

$$ds^2 = a^2(\eta) [d\eta^2 - d\chi^2]$$

where, for a flat spacetime, $\chi \equiv r$ as of Eq. 8.1.6. Since photons are massless, they move along null geodesics with $ds^2 = 0$, hence:

$$\Delta\chi(\eta) = \pm\Delta\eta$$

where the $+$ describes outgoing photons and the $-$ incoming photons. This means that the distance that time travels between η_1 and $\eta_2 > \eta_1$ is simply $\Delta\eta = \eta_2 - \eta_1$. Hence, assuming that the Big Bang started with a singularity¹ at $t_i \equiv 0$, then the greatest distance from which an observer at time t is able to receive signals is:

$$\chi_p(\eta) = \eta - \eta_i = \int_{t_i}^t \frac{dt}{a(t)} \quad (9.1.1)$$

which is called the **comoving particle horizon**. The particle horizon is obtained intersecting the past light-cone of the observer with the spacelike hypersurface $\eta = \eta_i$, as shown in Fig. 9.1: causal influences on the observer can only come from within this region.

The comoving particle horizon χ_p can be expressed in terms of the *comoving Hubble radius* $(aH)^{-1}$ as follows:

$$\chi_p(\eta) = \int_{a_i}^a \frac{da}{a\dot{a}} = \int_{\ln a_i}^{\ln a} \frac{d \ln a}{aH}$$

where a_i corresponds to the Big Bang singularity. For a universe dominated by a single matter component with parameter w , Eq. 8.3.17 determines the evolution of the comoving Hubble radius:

$$(aH)^{-1} = H_0^{-1} a^{\frac{1}{2}(1+3w)} \quad (9.1.2)$$

¹Note that the Big Bang singularity is a moment in time, but *not* a point in space: indeed, in general it is pictured as an extended (possibly infinite) spacelike hypersurface.

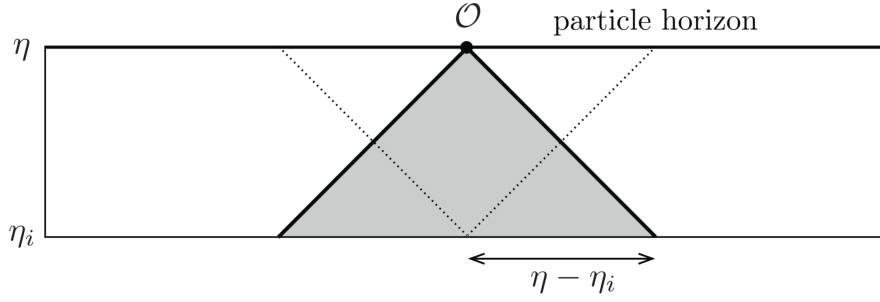


Figure 9.1: Spacetime diagram in the $\chi - \eta$ coordinate system illustrating the particle horizon.

Note that for ordinary matter sources the strong energy condition (SEC) $1 + 3w > 0$ is satisfied, in which case the integral expression of χ_p is dominated by the upper limit, while receiving vanishing contributions from earlier times. Performing the explicit integration:

$$\chi_p(a) = \frac{2H_0^{-1}}{1 + 3w} \left[a^{\frac{1}{2}(1+3w)} - a_i^{\frac{1}{2}(1+3w)} \right] \equiv \eta - \eta_i \quad (9.1.3)$$

Clearly, in presence of the SEC, $\eta_i \rightarrow 0$ as $a_i \rightarrow 0$, hence setting $a_i \equiv 0$ yields a finite comoving particle horizon:

$$\chi_p(t) = \frac{2H_0^{-1}}{1 + 3w} a^{\frac{1}{2}(1+3w)} = \frac{2}{1 + 3w} (aH)^{-1} \quad (9.1.4)$$

In the standard cosmology, then, $\chi_p \sim (aH)^{-1}$, since the Big Bang model postulates $a_i \rightarrow 0$.

§9.1.2 CMB uniformity

About 380'000 years after the Big Bang, the universe had cooled enough to allow the formation of the first hydrogen atoms: in this process, photons decoupled from the primordial plasma, a phenomenon known as Recombination (see §10.3.4) and observed today in the form of the cosmic microwave background (CMB).

The CMB is almost perfectly isotropic, with anisotropies in the order of 10^{-5} , and this is a serious problem for the Big Bang model. Indeed, $a_{\text{CMB}} \simeq 1100^{-1}$ and, since $a(\eta) \sim \eta^2$ for a matter-dominated universe, $\eta_{\text{CMB}} \sim \sqrt{a_{\text{CMB}}}$, which determines a comoving particle horizon of $\chi_{\text{CMB}} = \eta_{\text{CMB}} \sim 1100^{-1/2} \sim 1^\circ$: this means that any two points in the CMB that are separated by more than 1° in the sky are causally-disconnected.

The fact that the $\sim 10^4$ causally-disconnected patches of the CMB are remarkably isotropic is known as the *horizon problem*, since the comoving particle horizon in the Big Bang model is not sufficient to explain the isotropy of the CMB: there is not enough conformal time between the Big Bang and the CMB in the standard cosmology, as pictured in Fig. 9.2.

§9.1.3 SEC violation and Inflation

To explain the uniformity of the CMB, more conformal time between the Big Bang and Recombination is needed. In order to do so, the SEC must be violated: if e.g. $w = -\frac{1}{3}$, then from a certain point in time $(aH)^{-1}$ remains constant going backwards, resulting in an infinite available conformal time for the various patches of the CMB to come into causal contact with each other.

However, the preferred model is one with $w = -1$ (constrained by other observations): in this case, the comoving Hubble radius expands going backwards in the past, as shown in Fig. 9.3. Indeed:

$$\frac{d}{dt}(aH)^{-1} = \frac{1 + 3w}{2} \frac{1}{a} < 0$$

which confirms the shrinking of the comoving Hubble radius, i.e. its expansion going backwards. Moreover:

$$\frac{d}{dt}(aH)^{-1} = \frac{d}{dt}\dot{a}^{-1} = -\frac{\ddot{a}}{\dot{a}^2}$$

which means that a shrinking comoving Hubble radius implies an accelerated expansion, as $\ddot{a} > 0$. For this reason, this is called an *inflationary model*: in this model, the $\eta = 0$ surface is no longer the initial singularity of the Big Bang, but merely the end of the inflationary period, i.e. the transition between inflationary cosmology and standard cosmology, while the Big Bang singularity is pushed to negative conformal time:

$$\eta_i = \frac{2H_0^{-1}}{1+3w} a_i^{\frac{1}{2}(1+3w)} \xrightarrow{a_i \rightarrow 0, w < -\frac{1}{3}} -\infty$$

Moreover, since $w = -1$ in the inflationary period, the energy density of the dominant form of matter ρ_I was (approximately) constant, resulting in a constant Hubble parameter H_I and an exponential evolution of the scale parameter: $a(t) = e^{H_I t}$, from Eq. 8.3.17. The inflationary metric is then (approximately) expressed as:

$$ds^2 = dt^2 - e^{2H_I t} d\mathbf{x}^2$$

which is a de Sitter metric. In fact, Inflation is considered a period of quasi-de Sitter expansion.

§9.1.3.1 Estimates

Measurements on the CMB allow to give some estimates on the inflationary period. In particular, in the power spectrum of the CMB only wavelengths smaller than today's comoving Hubble horizon are observed, meaning that, at the start of inflation, the comoving Hubble radius had to be at least the same as today, if not larger, i.e. $(a_I H_I)^{-1} \gtrsim (a_0 H_0)^{-1}$.

On the other hand, the end of Inflation is conventionally taken to be around the GUT phase transition, at $T_{\text{GUT}} \sim 10^{15}$ GeV. Assuming that, after Inflation, the universe has been dominated by radiation only:

$$\frac{a_0 H_0}{a_e H_e} = \frac{a_0 H_0}{a_e H_0 a_e^{-2}} = a_e = \frac{T_{\text{CMB},0}}{T_e} \sim 10^{-28}$$

where $T_{\text{CMB},0} \simeq 0.2$ meV is the CMB temperature measured today. Then, since during Inflation $(aH)^{-1} \sim a^{-1}$, while after Inflation $(aH)^{-1} \sim a$, logarithmically $(a_I H_I)^{-1} \sim (a_0 H_0)^{-1}$ means that $(\ln a_0 - \ln a_e) \sim 2(\ln a_0 - \ln a_I)$, i.e. $a_I \sim 10^{-56}$.

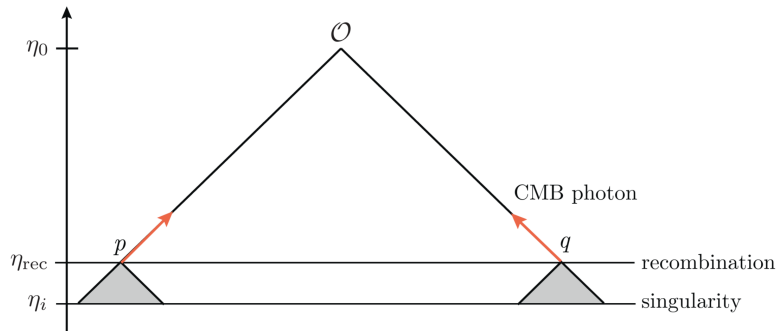


Figure 9.2: The horizon problem in the standard Big Bang model.

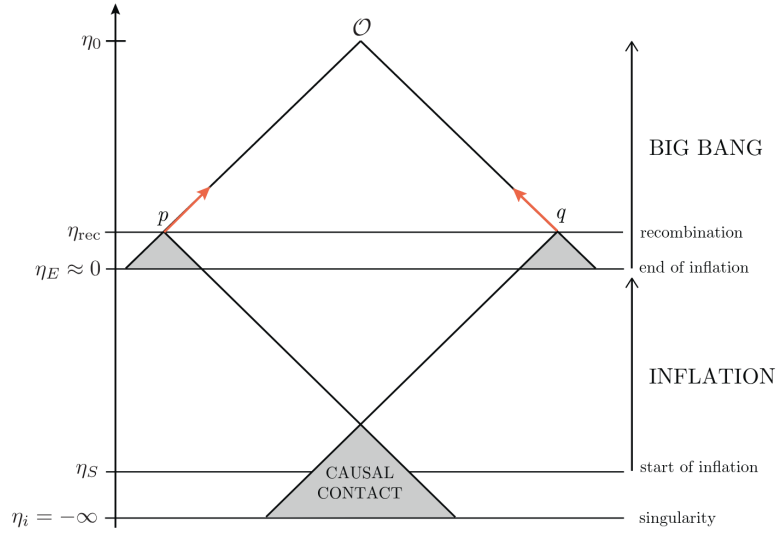


Figure 9.3: Inflationary solution to the horizon problem.

§9.2 Inflationary model

§9.2.1 Scalar field dynamics

As a simple model for Inflation, consider a real scalar field $\phi(t, \mathbf{x})$, the **inflaton**, subject to a potential $V(\phi)$: if the energy-momentum carried by the inflaton field dominates the universe, then it determines the evolution of the FLRW background. From Eq. 5.2.14 (recall the change of signature):

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \left[\frac{1}{2} \partial^\rho \phi \partial_\rho \phi - V(\phi) \right]$$

Consistency with the symmetries of the FLRW spacetime requires that $\phi = \phi(t)$, hence, from $T^0_0 = \rho_\phi$, the energy density of the inflaton field is found to be:

$$\rho_\phi = \frac{1}{2} \dot{\phi}^2 + V(\phi) \quad (9.2.1)$$

which is simply the sum of the kinetic energy density and the potential energy density. Their difference is instead the pressure of the inflaton field, as inferred from $T^i_j = -P_\phi \delta^i_j$:

$$P_\phi = \frac{1}{2} \dot{\phi}^2 - V(\phi) \quad (9.2.2)$$

Since inflation requires $P_\phi < -\frac{1}{3}\rho_\phi$, for the inflaton the potential energy must dominate over the kinetic energy. Inserting ρ_ϕ into Eq. 8.3.12 (rewritten in terms of the reduced Planck mass²):

$$H^2 = \frac{1}{3M_p^2} \left[\frac{1}{2} \dot{\phi}^2 + V(\phi) \right] \quad (9.2.3)$$

Proposition 9.2.1 (Klein–Gordon Equation)

The equation of motion for the inflaton field is:

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0 \quad (9.2.4)$$

Proof. First, from Eq. 8.3.5, 8.3.12:

$$2H\dot{H} = \frac{\dot{\rho}}{3M_{\text{p}}^2} = -\frac{H}{M_{\text{p}}^2}(\rho + P) \quad \Rightarrow \quad \dot{H} = -\frac{\rho + P}{2M_{\text{p}}^2}$$

Since $\rho_\phi + P_\phi = \dot{\phi}$, inserting this equation in the time derivative of Eq. 9.2.3:

$$2H\dot{H} = \frac{1}{3M_{\text{p}}^2} [\dot{\phi}\ddot{\phi} + V_{,\phi}\dot{\phi}] \quad \Rightarrow \quad \ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0$$

which is the thesis. $\square$

Note that the evolution of the inflaton field is sourced by the potential, which acts like a force $V_{,\phi}$, while the expansion of the universe determines a *friction* term $3H\dot{\phi}$.

§9.2.1.1 Slow-roll inflation

Near a minimum ϕ_0 , the potential can be approximated by a harmonic potential:

$$V(\phi) \approx V_0 + \frac{1}{2}m^2(\phi - \phi_0)^2$$

with $V_0 \equiv V(\phi_0)$ and $m^2 \equiv V_{,\phi\phi}(\phi_0)$. Since near the minimum $\dot{\phi} \approx 0$, the energy-momentum tensor can be written as:

$$T^\mu_\nu = \begin{bmatrix} \rho_\phi & 0 & 0 & 0 \\ 0 & -P_\phi & 0 & 0 \\ 0 & 0 & -P_\phi & 0 \\ 0 & 0 & 0 & -P_\phi \end{bmatrix} \approx \begin{bmatrix} V_0 & 0 & 0 & 0 \\ 0 & V_0 & 0 & 0 \\ 0 & 0 & V_0 & 0 \\ 0 & 0 & 0 & V_0 \end{bmatrix}$$

Comparing to Eq. 5.2.11, it is clear that V_0 acts like a cosmological constant Λ . Moreover, near the minimum $w \approx -1$ as sought.

However, with a harmonic potential the inflaton field does not significantly evolve over time, while in the inflationary cosmology Inflation must only occur between a_{I} and a_{e} . To achieve this, a so-called “slow-roll model” is introduced: the potential $V(\phi)$ starts nearly constant, with a slightly negative slope, which imposes the condition $V \gg T$ and results into $w \approx -1$. Then, the potential smoothly changes from linear to parabolic, reaching finally a minimum. An example of potential is drawn in Fig. 9.4: the formulation of a precise model is futile, since it is not possible to empirically probe Inflation.

§9.2.2 Reheating

During Inflation, the energy density of the universe is dominated by the inflaton potential $V(\phi)$. Then, as Inflation ends and the inflaton field gains kinetic energy, the energy of the inflaton sector has to be transferred to the Standard Model sector: this process is called **Reheating**.

²In place of Newton’s constant G , it is useful to use the reduced Planck mass:

$$M_{\text{p}} \equiv \sqrt{\frac{\hbar c}{8\pi G}} \simeq 2.4 \cdot 10^{18} \text{ GeV}$$

so that Eq. 8.3.12 for a flat spacetime can be written as $H^2 = \rho/(3M_{\text{p}}^2)$.

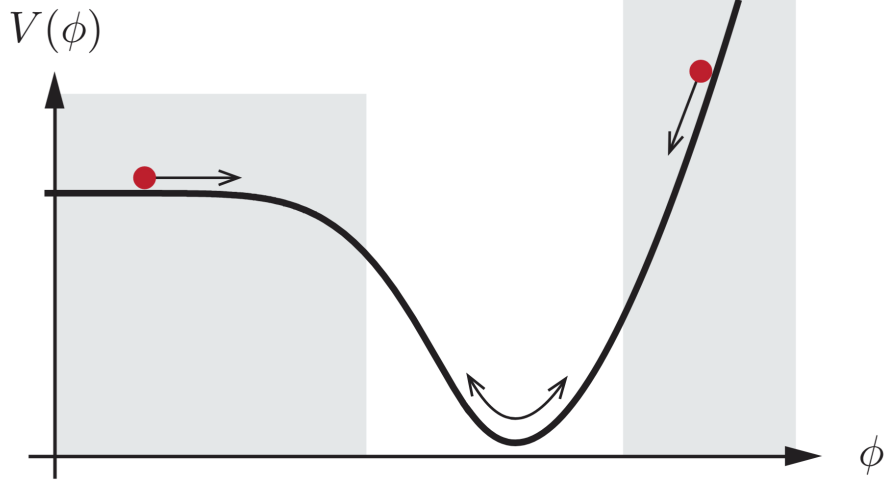


Figure 9.4: Example of slow-roll potential: inflation occurs in the shaded parts of the potential.

Setting WLOG $V_0 \equiv 0$, the homogeneous evolution of the inflaton field $\phi = \phi(t)$ reads:

$$\ddot{\phi} = -m^2\phi - 3H\dot{\phi}$$

The expansion timescale soon becomes much larger than the oscillation period, i.e. $H^{-1} \gg m^{-1}$, hence the friction term can be neglected. Then, the continuity equation can be written as:

$$\dot{\rho}_\phi + 3H\rho_\phi = -3HP_\phi = -\frac{3}{2}H \left[m^2\phi^2 - \dot{\phi}^2 \right]$$

Over a period of oscillation the term in parentheses averages to zero, hence the oscillating inflaton field behaves like pressureless matter with $\rho_\phi \propto a^{-3}$, while the decrease of its energy density is reflected in a decrease of its oscillation amplitude.

After Reheating, the universe is dominated by radiation and its evolution is described by the standard cosmology.

Thermal History

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At early times, the thermodynamical properties of the universe were determined by local equilibrium. However, non-equilibrium dynamics is crucial for massive particles to acquire cosmological abundances and to understand the CMB.

§10.1 Hot Big Bang

To understand the thermal history of the universe, the key is to compare the **rate of interactions** Γ with the **rate of expansion** H (Hubble parameter). In particular, when $\Gamma \gg H$, the timescale of particle interactions is much smaller than the expansion timescale:

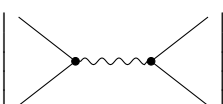
$$t_\Gamma \equiv \frac{1}{\Gamma} \ll \frac{1}{H} \equiv t_H \quad (10.1.1)$$

Therefore, local thermal equilibrium is reached before the effect of the expansion becomes relevant. As the universe cools, however, the rate of interactions typically decreases faster than the expansion rate, and as $t_\Gamma \sim t_H$ the particles decouple from the thermal bath. Note that different species have different interaction rates and decouple at different times.

Consider a generic $1 + 2 \leftrightarrow 3 + 4$ scattering process. In principle, each species has its own rate of interactions, e.g. $\Gamma_1 = n_2 \sigma v_{1,2}$, where n_2 is the number density of the target species 2 and $v_{1,2}$ is the average relative velocity between species 1 and 2, but at high energies it is expected that $n_1 \sim n_2 \equiv n$, hence it is possible to define a single rate of interactions:

$$\Gamma \equiv n \sigma v \quad (10.1.2)$$

where n is the number density of particles, v is their average velocity and σ is the interaction cross-section. For $T \gtrsim 100$ GeV, all known particles are ultra-relativistic, so $v \sim 1$, and their masses can be ignored, leaving the temperature T as the only dimensionfull scale: by dimensional analysis $n \sim T^3$. Moreover, interactions are mediated by gauge bosons, which are massless above the scale of electroweak symmetry breaking, leaving a similar dependence for the cross-section of strong and electroweak processes:

$$\sigma \sim \left| \text{diagram} \right|^2 \sim \frac{\alpha^2}{T^2}$$


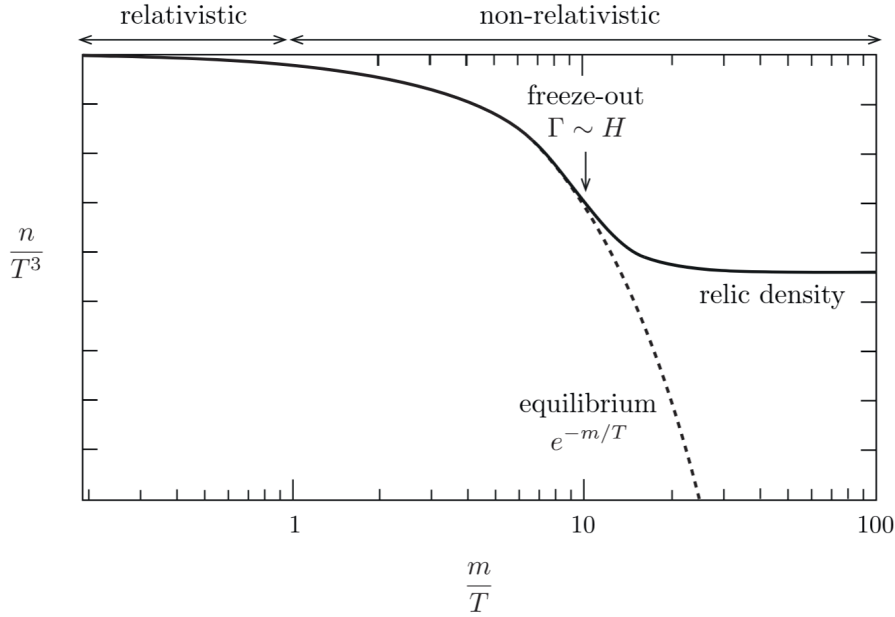


Figure 10.1: Schematic illustration of freeze-out: at high temperature ($T \gg m$) the particle abundance tracks its equilibrium value, while at low temperature ($T \ll m$) particles freeze-out and maintain a density that is much larger than the Boltzmann-suppressed equilibrium abundance.

where $\alpha \equiv g^2/4\pi$ is the generalized structure constant of the considered interaction. Then, dimensional analysis results into:

$$\Gamma = n\sigma v \sim T^3 \cdot \frac{\alpha^2}{T^2} \cdot 1 \sim \alpha^2 T$$

Now, recall Eq. 8.3.12, and in particular express it in terms of the reduced Planck mass:

$$M_p \equiv \sqrt{\frac{\hbar c}{8\pi G}} \simeq 2.4 \cdot 10^{18} \text{ GeV} \quad \Rightarrow \quad H = \frac{\sqrt{\rho}}{3M_p}$$

The same argument as above gives $\rho \sim T^4$, thus $H \sim T^2/M_p$ and:

$$\frac{\Gamma}{H} \sim \frac{\alpha^2 M_p}{T} \sim \frac{10^{16} \text{ GeV}}{T} \quad (10.1.3)$$

where $\alpha \sim 0.1$ has been used as numerical estimate. Therefore, the condition Eq. 10.1.1 is satisfied for $100 \text{ GeV} \lesssim T \lesssim 10^{16} \text{ GeV}$: in this range, all SM particles are in thermal equilibrium. This means that they exchange energy and momentum efficiently, thus reaching a state of maximum entropy: in this state, the number of particles per unit volume in phase space, i.e. the *distribution function*, takes the form of the Bose–Einstein or the Fermi–Dirac distribution, depending on the statistics of the considered species.

On the contrary, when temperature drops below the mass of a particle species, that species becomes sub-relativistic and its distribution function acquires an exponential suppression of the form $\sim e^{-m/T}$: this means that relativistic particles dominate the density and pressure of the primordial plasma. Furthermore, another consequence is that, had equilibrium persisted until today, the universe would be mostly photons, since any massive particle would be exponentially suppressed: this is not the case, due to the **freeze-out** of massive particles caused by deviations from equilibrium (see Fig. 10.1 and Ex. 10.3.1).

Below the scale of electroweak symmetry breaking, i.e. $T \lesssim 100$ GeV, the gauge bosons of weak interactions receive masses through the Higgs mechanism ($M_W \approx 80$ GeV and $M_Z \approx 90$ GeV), and the cross-section of weak processes becomes:

$$\sigma \sim \left| \begin{array}{c} \diagup \quad \diagdown \\ \bullet \\ \diagdown \quad \diagup \end{array} \right|^2 \sim G_F^2 T^2 \quad G_F \sim \frac{\alpha}{M_W^2} \simeq 1.17 \cdot 10^{-5} \text{ GeV}^{-2}$$

The strength of weak interactions now decreases as the temperature of the universe drops, and the ratio Γ/H becomes:

$$\frac{\Gamma}{H} \sim \frac{\alpha^2 M_p T^3}{M_W^4} \sim \left(\frac{T}{1 \text{ MeV}} \right)^3 \quad (10.1.4)$$

The **decoupling** of particles which interact with the primordial plasma only through weak interactions then happens at $T_{\text{dec}} \sim 1$ MeV.

§10.2 Equilibrium

Observational evidence (mainly the perfect blackbody spectrum of the CMB) suggests that the early universe was in *local thermal equilibrium*¹, a fact supported by the above analysis of the Standard Model at $T \gtrsim 100$ GeV.

§10.2.1 Equilibrium thermodynamics

Consider a gas of weakly interacting particles. Quantistically, the momentum eigenstates of a particle in a volume V have a discrete spectrum: in particular, the density of states in momentum space $\{\mathbf{p}\}$ is V/h^3 , and the density of states in phase space $\{\mathbf{x}, \mathbf{p}\}$ is $1/h^3$.

Considering a particle with g internal degrees of freedom (e.g. spin), in natural unit the density of states (which from now on is implicitly assumed to be in phase space) is $g/(2\pi)^3$. In order to obtain the number density, it is necessary to know how the particles are distributed amongst the momentum eigenstates: this information is encoded in the (*phase space*) *distribution function* $f(\mathbf{x}, \mathbf{p}, t)$. Homogeneity restricts f to be independent of $\mathbf{x}$, while isotropy implies that the momentum-dependence is only through the magnitude $p \equiv \|\mathbf{p}\|$. The time dependence is typically left implicit, as it manifests itself in terms of the temperature dependence of the distribution function. Putting everything together, the number density of particles in real space is found to be:

$$n(T) = \frac{g}{(2\pi)^3} \int d^3p f(p, T) \quad (10.2.1)$$

The energy density is instead found by weighting the integration over momentum eigenstates with their energy. In particular, the assumption of the particles being weakly-interacting allows to approximate the energy as $E(p) = \sqrt{m^2 + p^2}$, so that:

$$\rho(T) = \frac{g}{(2\pi)^3} \int d^3p f(p, T) E(p) \quad (10.2.2)$$

Pressure, on the other hand, requires more care.

¹Note that the universe can never be in perfect equilibrium, strictly speaking, since the FLRW metric does not have a time-like Killing vector. However, if the expansion is slow enough, particles have enough time to settle close to local equilibrium, and, since the universe is homogeneous, local values of thermodynamic quantities are also global values.

Proposition 10.2.1 (Pressure of a weakly-interacting gas)

$$P = \frac{g}{(2\pi)^3} \int d^3p f(p, T) \frac{p^2}{3E} \quad (10.2.3)$$

Proof. Consider a small area element dA with normal versor $\hat{\mathbf{n}}$. All particles with velocity $v \equiv \|\mathbf{v}\|$, striking this area element between t and $t + dt$, were located at $t = 0$ on a spherical shell of radius $R = vt$ and width $v dt$, hence a solid angle $d\Omega$ of this shell defines a volume $dV = R^2 v dt d\Omega$. Given Eq. 10.2.1, the number of particles in this volume is:

$$dN = \frac{g}{(2\pi)^3} f(E, T) R^2 v dt d\Omega$$

Not all particles in dV reach the target, only those with velocities directed to the area element do. Taking into account the isotropy of the velocity distribution, the total number of these striking particles is:

$$dN_A = \frac{|\hat{\mathbf{v}} \cdot \hat{\mathbf{n}}| dA}{4\pi R^2} dN = \frac{g}{(2\pi)^3} f(E, T) v \frac{|\hat{\mathbf{v}} \cdot \hat{\mathbf{n}}|}{4\pi} dA dt d\Omega$$

where $\hat{\mathbf{v}} \cdot \hat{\mathbf{n}} < 0$ since the striking particles are directed towards the area element. Assuming elastic reflections, each particles transfers a momentum $2|\mathbf{p} \cdot \hat{\mathbf{n}}|$ to the target, hence the contribution of particles with velocity v to the pressure is:

$$P(v) = \int dN_A \frac{2|\mathbf{p} \cdot \hat{\mathbf{n}}|}{dA dt} = \frac{g}{(2\pi)^3} f(E, T) \frac{p^2}{2\pi E} \int_0^{2\pi} d\varphi \int_{-1}^0 d\cos\theta \cos^2\theta = \frac{g}{(2\pi)^3} f(E, T) \frac{p^2}{3E}$$

where $v = p/E$ was used, and θ was defined as $\hat{\mathbf{v}} \cdot \hat{\mathbf{n}} = -\cos\theta < 0$. Integrating over energy or momentum yields the thesis. $\square$

The form of the distribution function depends on the thermodynamical state of the system. Assuming that the gas is in *kinetic equilibrium*, i.e. if the particles can exchange energy and momentum efficiently, then the system is in a state of maximal entropy and the distribution function is either the Fermi–Dirac distribution (fermions) or the Bose–Einstein distribution (bosons):

$$f(p, T) = \left[e^{\frac{E(p) - \mu(T)}{T}} \pm 1 \right]^{-1} \quad (10.2.4)$$

with the $+$ sign for fermions and the $-$ sign for bosons. At low temperatures, i.e. $T < E - \mu$, both distributions reduce to the Maxwell–Boltzmann distribution:

$$f(p, T) \approx e^{-\frac{E(p) - \mu(T)}{T}} \quad (10.2.5)$$

The chemical potential $\mu(T)$ characterizes the response of a system to a change in particle number: the second law of thermodynamics implies that, given a reaction, particles flow towards the side of the reaction with the lower total chemical potential, until *chemical equilibrium* is achieved:

$$\sum_{\text{reactants}} \mu_i = \sum_{\text{products}} \mu_j$$

Note that, since the number of photons is not conserved (e.g. double Compton scattering $e^- \gamma \leftrightarrow e^- \gamma \gamma$), then $\mu_\gamma = 0$ and $\mu_{\bar{X}} = -\mu_X$, due to particle-antiparticle annihilation $X \bar{X} \leftrightarrow \gamma \gamma$.

A system is in **thermal equilibrium** if its species are both in kinetic and chemical equilibrium: these species then share a common temperature $T_i = T$, which is often identified with the photon temperature T_γ (the “temperature of the universe”).

§10.2.2 Density and pressure

At early times, the chemical potentials of all particles are so small that they can be neglected, thus Eq. 10.2.1-10.2.2 can be written as:

$$n(T) = \frac{g}{2\pi^2} T^3 I_{\pm}(x) \quad \rho(T) = \frac{g}{2\pi^2} T^4 J_{\pm}(x) \quad (10.2.6)$$

with $x \equiv m/T$, $\xi \equiv p/T$ and:

$$I_{\pm}(x) \equiv \int_0^{\infty} d\xi \frac{\xi^2}{\exp \sqrt{\xi^2 + x^2} \pm 1} \quad J_{\pm}(x) \equiv \int_0^{\infty} d\xi \frac{\xi^2 \sqrt{\xi^2 + x^2}}{\exp \sqrt{\xi^2 + x^2} \pm 1} \quad (10.2.7)$$

These integrals need, in general, to be evaluated numerically, but analytic expressions are possible in the ultra-relativistic and sub-relativistic limits.

Lemma 10.2.1 (Ultra-relativistic limit)

In the limit of $T \gg m$, i.e. $x \rightarrow 0$:

$$n(T) = \frac{\zeta(3)}{\pi^2} g T^3 \begin{cases} 1 & \text{bosons} \\ \frac{3}{4} & \text{fermions} \end{cases} \quad \rho(T) = \frac{\pi^2}{30} g T^4 \begin{cases} 1 & \text{bosons} \\ \frac{7}{8} & \text{fermions} \end{cases} \quad (10.2.8)$$

Proof. For $x \rightarrow 0$, the integrals in Eq. 10.2.7 reduce to:

$$I_{\pm}(0) = \int_0^{\infty} d\xi \frac{\xi^2}{e^{\xi} \pm 1} \quad J_{\pm}(0) = \int_0^{\infty} d\xi \frac{\xi^3}{e^{\xi} \pm 1}$$

These integrals can be solved using the standard integral:

$$\int_0^{\infty} d\xi \frac{\xi^n}{e^{\xi} - 1} = \zeta(n+1) \Gamma(n+1) \quad (10.2.9)$$

In the bosonic case they are straightforward: $I_-(0) = 2\zeta(3)$ and $J_-(0) = 6\zeta(4) = \frac{\pi^4}{15}$. In the fermionic case, use instead:

$$\frac{1}{e^{\xi} + 1} = \frac{1}{e^{\xi} - 1} - \frac{2}{e^{2\xi} - 1}$$

so that:

$$I_+(0) = I_-(0) - 2 \left(\frac{1}{2} \right)^3 I_-(0) = \frac{3}{2} I_-(0)$$

$$J_+(0) = J_-(0) - 2 \left(\frac{1}{2} \right)^4 J_-(0) = \frac{7}{8} J_-(0)$$

which concludes the proof. $\square$

From Eq. 10.2.3 with $E \simeq p$ in the relativistic limit, the equation of state for a relativistic gas $P = \frac{1}{3}\rho$ is recovered. Moreover, recalling Eq. 8.3.13, using $T_0 = 2.73$ K it is possible to compute the density parameter of radiation:

$$\rho_{\gamma,0} = \frac{\pi^2}{15} T_0^4 \simeq 4.6 \cdot 10^{-34} \text{ g cm}^{-3} \quad \Rightarrow \quad \Omega_{\gamma} h^2 \simeq 2.5 \cdot 10^{-5}$$

Lemma 10.2.2 (Sub-relativistic limit)

In the limit $T \ll m$, i.e. $x \gg 1$:

$$n(T) = g \left(\frac{mT}{2\pi} \right)^{3/2} e^{-m/T} \quad \rho(T) = mn(T) + \frac{3}{2}n(T)T \quad (10.2.10)$$

Proof. For $x \gg 1$:

$$I_{\pm}(x) \approx \int_0^{\infty} d\xi \xi^2 e^{-\sqrt{\xi^2+x^2}}$$

Most of the contribution comes from the $\xi \ll x$ regime, hence, at lowest order in ξ :

$$I_{\pm}(x) \approx \int_0^{\infty} d\xi \xi^2 \exp\left(-x - \frac{\xi^2}{2x}\right) = (2x)^{3/2} e^{-x} \int_0^{\infty} d\xi \xi^2 e^{-\xi^2}$$

Recalling the standard integral:

$$\int_0^{\infty} d\xi \xi^n e^{-\xi^2} = \frac{1}{2} \Gamma\left(\frac{n+1}{2}\right) \quad (10.2.11)$$

with $\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$:

$$I_{\pm}(x) = \sqrt{\frac{\pi}{2}} x^{3/2} e^{-x} \quad \Rightarrow \quad n(T) = g \left(\frac{mT}{2\pi} \right)^{3/2} e^{-m/T}$$

For the energy density, use the sub-relativistic expansion at lowest order in ξ in Eq. 10.2.7:

$$\begin{aligned} J_{\pm}(x) &\approx \int_0^{\infty} d\xi \xi^2 \left(x + \frac{\xi^2}{2x} \right) \exp\left(-x - \frac{\xi^2}{2x}\right) = (2x)^{3/2} e^{-x} \int_0^{\infty} d\xi \xi^2 (x + \xi^2) e^{-\xi^2} \\ &= (2x)^{3/2} e^{-x} \frac{1}{2} \left[x \Gamma\left(\frac{3}{2}\right) + \Gamma\left(\frac{5}{2}\right) \right] = \sqrt{\frac{\pi}{2}} x^{3/2} e^{-x} \left(x + \frac{3}{2} \right) \end{aligned}$$

Inserting $x = m/T$ yields the thesis. □

In a similar way, from Eq. 10.2.3 it can be shown that $P = nT$, which means that a non-relativistic gas behaves like pressureless dust since $P = nT \ll nm = \rho$.

As expected, in the sub-relativistic limit $T \ll m$ the number density, the energy density and the pressure of massive particle species fall exponentially (Boltzmann suppression). This is interpreted as the annihilation of particles with antiparticles: at high temperature (i.e. energy), matter-antimatter annihilation is balanced by pair production, but as T decreases below the mass of the particle the thermal energy is no longer sufficient for pair production and the number density decreases exponentially.

§10.2.3 Effective number of relativistic species

From now on, the temperature of the universe is identified with the temperature of the photon gas: $T \equiv T_{\gamma}$.

Definition 10.2.1 (Effective number of relativistic degrees of freedom)

The total radiation density in the universe is defined as:

$$\rho_r := \sum_i \rho_i \equiv \frac{\pi^2}{30} g_\star(T) T^4 \quad (10.2.12)$$

where the sum runs over all relativistic particle species. The **effective number of relativistic degrees of freedom** is denoted by $g_\star(T)$.

Note that $g_\star(T)$ receives different contributions from relativistic species in thermal equilibrium with the photons, i.e. with $T = T_i \gg m_i$, and from decoupled relativistic species, i.e. with $T \neq T_i \gg m_i$. In particular, from Eq. 10.2.8:

$$g_\star^{\text{eq}}(T) = \sum_{\text{bosons}} g_i + \frac{7}{8} \sum_{\text{fermions}} g_j \quad g_\star^{\text{dec}}(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T}\right)^4 + \frac{7}{8} \sum_{\text{fermions}} g_j \left(\frac{T_j}{T}\right)^4 \quad (10.2.13)$$

When the temperature drops below the mass m_i of a particle species, it becomes non-relativistic and no longer contributes to $g_\star(T)$.

Example 10.2.1 ($g_\star(T)$ above 100 GeV)

For $T \gtrsim 100$ GeV, all SM particles are relativistic (as noted in §10.1), hence all species contribute to $g_\star^{\text{eq}}(T)$. The bosons have:

$$g_b = g_\gamma + g_g + g_{W,Z} + g_H = 2_{\text{spin}} + 8_{\text{colour}} \cdot 2_{\text{spin}} + 3_{\text{SU}(2)} \cdot 3_{\text{spin}} + 1_{\text{spin}} = 28$$

since massless vector bosons only have transverse polarization. The fermions, on the other hand, have:

$$\begin{aligned} g_f &= g_q + g_{e,\mu,\tau} + g_\nu \\ &= 6_{\text{species}} \cdot 2_{\text{spin}} \cdot 2_{\text{anti}} \cdot 3_{\text{colour}} + 3_{\text{species}} \cdot 2_{\text{spin}} \cdot 2_{\text{anti}} + 3_{\text{species}} \cdot 1_{\text{spin}} \cdot 2_{\text{anti}} = 90 \end{aligned}$$

since neutrinos only have one helicity state (left-handed neutrinos and right-handed antineutrinos). The effective number of relativistic degrees of freedom then is:

$$g_\star = g_b + \frac{7}{8} g_f = 106.75 \quad (10.2.14)$$

As the temperature drops, various particle species become non-relativistic and annihilate: the evolution of $g_\star(T)$ is plotted in Fig. 10.2. At $T \sim 150$ MeV, before the strange quarks have time to annihilate, the QCD phase transition takes place: quarks combine into baryons and mesons, which are all non-relativistic under 150 MeV, except for the pions ($\pi^\pm, \pi^0$). The only relativistic species left are then pions, electrons, muons, neutrinos and photons: since the pions are spin-0 particles, they contribute to g_b with $g_\pi = 3_{\text{species}} \cdot 1_{\text{spin}}$, resulting in $g_\star = 2 + 3 + \frac{7}{8} \cdot (4 + 4 + 6) = 17.25$.

Moreover, it is important to note that the transition from relativistic to non-relativistic behaviour is not instantaneous: about 80% of the particle-antiparticle annihilations takes place in the interval $T = m \rightarrow T = \frac{1}{6}m$.

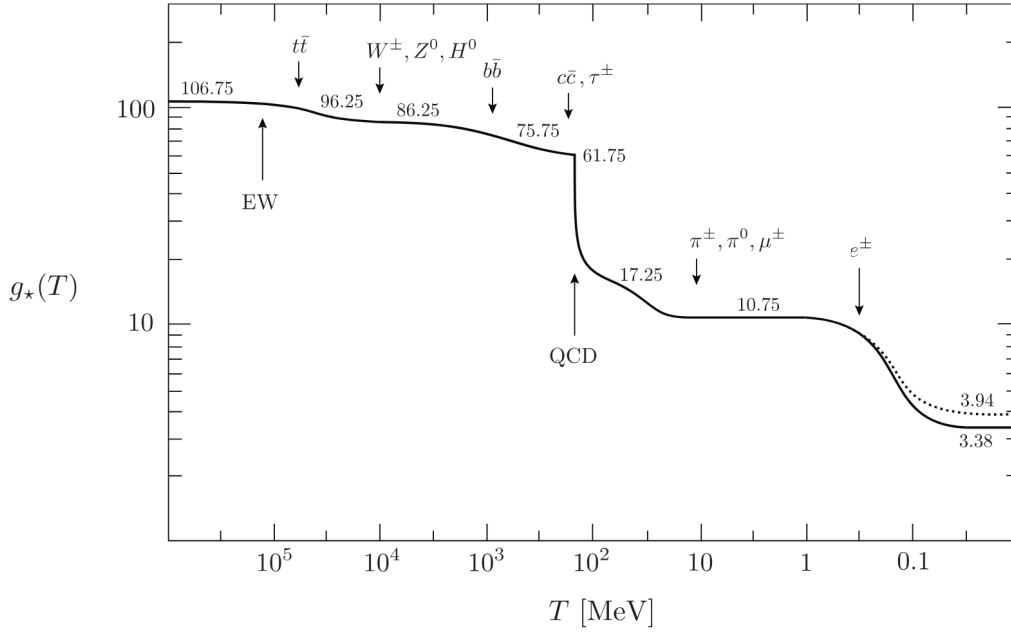


Figure 10.2: Evolution of $g_*(T)$ with temperature, assuming Standard Model particles. The dotted line is the effective number of degrees of freedom in entropy $g_{*S}(T)$.

§10.2.3.1 Entropy

According to the second law of thermodynamics, the total entropy of the universe can only increase or stay constant. Since the fraction of baryons to photons is measured to be:

$$\eta_b \equiv \frac{n_b}{n_\gamma} \simeq 5.5 \cdot 10^{-10} \frac{\Omega_b h^2}{0.020} \sim 10^{-9} \quad (10.2.15)$$

the entropy of the universe is dominated by the entropy of the photon bath (at least as long as the universe is sufficiently uniform): any entropy production from non-equilibrium processes is then insignificant relative to the total entropy, and therefore the expansion of the universe can be treated as *adiabatic*, so that the total entropy stays constant, even beyond equilibrium.

Lemma 10.2.3 (Entropy at equilibrium)

The entropy of a system at equilibrium stays constant.

Proof. Consider the second law of thermodynamics: $TdS = dU + PdV$. Using $U = \rho V$:

$$dS = \frac{d[(\rho + P)V]}{T} - \frac{VdP}{T} = \frac{V}{T} \frac{\partial \rho}{\partial T} dT + \frac{\rho + P}{T} dV$$

Since the entropy function must exist, this differential form must be exact. Since $d^2 = 0$, an exact form is also closed, hence:

$$\frac{\partial}{\partial V} \left[\frac{V}{T} \frac{\partial \rho}{\partial T} \right] = \frac{\partial}{\partial T} \left[\frac{\rho + P}{T} \right] \quad \Rightarrow \quad \frac{\partial P}{\partial T} = \frac{\rho + P}{T} \quad (10.2.16)$$

Rearranging the second law of thermodynamics, then:

$$dS = \frac{d[(\rho + P)V]}{T} - \frac{(\rho + P)V}{T^2} dT = d \left[\frac{\rho + P}{T} V \right]$$

To show that entropy is conserved at equilibrium, consider:

$$\frac{dS}{dt} = \frac{d}{dt} \left[\frac{\rho + P}{T} V \right] = \frac{V}{T} \left[\dot{\rho} + \frac{\dot{V}}{V} (\rho + P) \right] + \frac{V}{T} \left[\dot{P} - \frac{\rho + P}{T} \dot{T} \right]$$

Since $V \propto a^3$, then $\dot{V}/V = 3H$ and the first term vanishes by Eq. 8.3.5, while the second vanishes by Eq. 10.2.16. Therefore, $\dot{S} = 0$ at equilibrium. $\square$

It is convenient to define the **entropy density**, which ignoring an integration constant is (by the above proof):

$$s = \frac{\rho + P}{T} \quad (10.2.17)$$

which, by adiabaticity (i.e. $dQ = 0 \implies dS = 0$), scales like $s \propto a^{-3}$. Since for a relativistic gas $P = \frac{1}{3}\rho$, recalling Eq. 10.2.8 it is possible to express the entropy density in the ultra-relativistic limit as:

$$s = \frac{2\pi^2}{45} g(T) T^3 \quad (10.2.18)$$

Definition 10.2.2 (Effective number of relativistic degrees of freedom in entropy)

The total entropy density in the universe is defined as:

$$s := \sum_i s_i \equiv \frac{2\pi^2}{45} g_{*S}(T) T^3 \quad (10.2.19)$$

where the sum runs over all relativistic particle species. The **effective number of degrees of freedom in entropy** is denoted by $g_{*S}(T)$.

As for $g_*(T)$, there are two different contributions to $g_{*S}(T)$ too: one from species in thermal equilibrium with the photon bath and one from decoupled species. Note that $g_{*S}^{\text{eq}}(T) = g_*^{\text{eq}}(T)$, while, since $s_i \propto T_i^3$, for decoupled species:

$$g_{*S}^{\text{dec}}(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T} \right)^3 + \frac{7}{8} \sum_{\text{fermions}} g_j \left(\frac{T_j}{T} \right)^3 \quad (10.2.20)$$

which is $g_{*S}^{\text{dec}}(T) \neq g_*^{\text{dec}}(T)$. It follows that $g_*(T) = g_{*S}(T)$ only when *all* relativistic species are in thermal equilibrium with the photon bath: in the real universe, this is the case until $t \approx 1$ sec (see Fig. 10.2). The conservation of entropy has two important consequences:

- since $s \propto a^{-3}$, if the number of particles in a comoving volume N_i is conserved (i.e. $n_i \propto a^{-3}$), then $n_i \propto s$, with $N_i = n_i/s$ since it is a scale-invariant quantity. This is the case, for example, for the net baryon number after baryogenesis, which is conserved, hence $N_{\text{net}} \equiv (n_B - n_{\bar{B}})/s$ is constant;
- from the definition of s , it follows that $T \propto g_{*S}^{-1/3} a^{-1}$. Away from mass thresholds, g_{*S} is approximately constant and $T \propto a^{-1}$ as expected, but when a particle species becomes non-relativistic the factor g_{*S} accounts for the transfer of its entropy to other relativistic species, making T change in the scaling as a consequence.

§10.2.3.2 Neutrino decoupling

Neutrinos are coupled to the thermal bath only through weak interaction processes (e.g. $\nu_e \bar{\nu}_e \leftrightarrow e^+ e^-$), whose cross-section is estimated as $\sigma \sim G_F^2 T^2$. By Eq. 10.1.2, then, $\Gamma \sim G_F^2 T^5$, which allows to estimate the temperature for neutrino decoupling (with $H \sim T^2/M_p$):

$$\frac{\Gamma}{H} \sim \left(\frac{T}{1 \text{ MeV}} \right)^3 \quad \Rightarrow \quad T_{\text{dec}} \sim 1 \text{ MeV}$$

A more accurate computation gives $T_{\text{dec}} \sim 0.8 \text{ MeV}$. After decoupling, neutrinos move freely along geodesics. Since $p \propto a^{-1}$ (by Eq. 8.2.3), it is convenient to define the time-independent combination $q \equiv ap$, so that the neutrino number density after decoupling can be written as:

$$n_\nu \propto a^{-3} \int d^3 q \left[\exp \frac{q}{aT_\nu} + 1 \right]^{-1}$$

But, after decoupling, particle number conservation requires $n_\nu \propto a^{-3}$, hence the neutrino temperature must scale as $T_\nu \propto a^{-1}$. The neutrinos then follow a relativistic Fermi–Dirac distribution with decreasing temperature, even after they become non-relativistic at later times (when $T_\nu \ll m_\mu$), due to the Liouville theorem².

§10.2.3.3 Electron-positron annihilation

Shortly after neutrinos decouple, the temperature drops below the electron mass ($m_e \simeq 0.511 \text{ MeV}$), so that electron-positron annihilation is no longer balanced by the $\gamma\gamma \rightarrow e^- e^+$ pair production. The energy density and entropy of electrons and positrons are then transferred to the photons, but not to the decoupled neutrinos: the photons are thus heated relative to the neutrinos (see Fig. 10.3). To quantify this effect, consider the change in the effective number of degrees of freedom in entropy. Neglecting neutrinos and other decoupled species³:

$$g_{\star S}^{\text{eq}} = \begin{cases} 2_\gamma + \frac{7}{8} \cdot 4_e = \frac{11}{2} & T \gtrsim m_e \\ 2_\gamma = 2 & T < m_e \end{cases}$$

Since, in equilibrium, $g_{\star S}^{\text{eq}} T^3 a^3$ remains constant, the photon temperature $aT_\gamma \equiv aT$ must increase by a factor of $(11/4)^{1/3}$ after electron-positron annihilation, while aT_ν remains the same. This means that, for $T < m_e$:

$$T_\nu = \sqrt[3]{\frac{4}{11}} T_\gamma \quad (10.2.21)$$

For $T \ll m_e$, the only remaining relativistic species are photons and neutrinos, and the above equation implies that $g_\star \neq g_{\star S}$:

$$g_\star = 2_\gamma + \frac{7}{8} \cdot 2_{\text{anti}} \cdot N_{\text{eff}} \left(\frac{4}{11} \right)^{4/3} = 3.36 \quad g_{\star S} = 2_\gamma + \frac{7}{8} \cdot 2_{\text{anti}} \cdot N_{\text{eff}} \left(\frac{4}{11} \right) = 3.94$$

where N_{eff} is the **effective number of neutrino species**. If neutrino decoupling was instantaneous, then $N_{\text{eff}} = 3$; however, neutrino decoupling was not complete when $e^+ e^-$ annihilation began, so

²The Liouville theorem states that, in a Hamiltonian system, the phase-space probability density (i.e. distribution function) is constant along the trajectories defined by the Hamilton equations of motion.

³Entropy is conserved separately for the thermal bath and the decoupled species, since they are non-interacting systems.

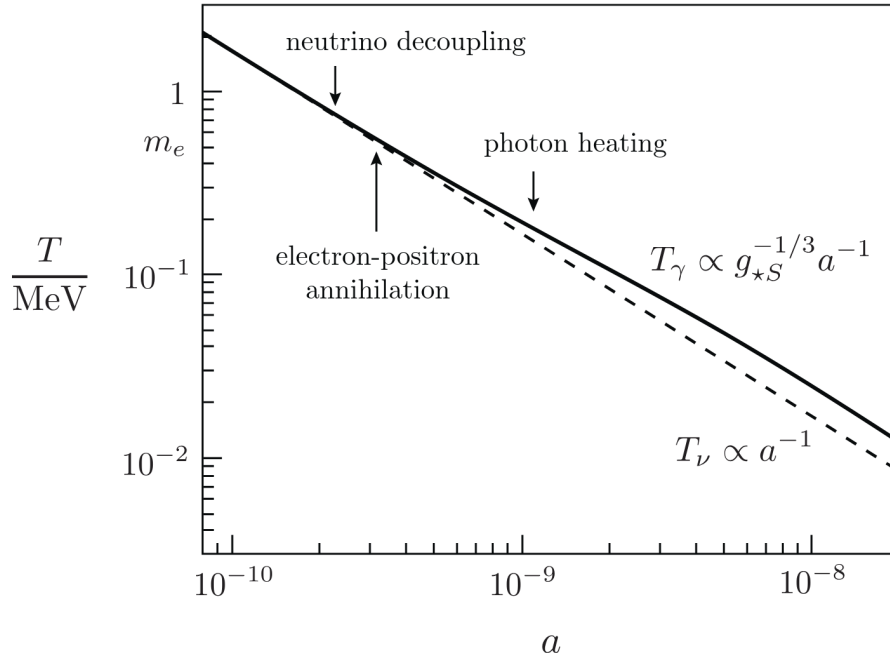


Figure 10.3: Difference in redshift between neutrino temperature $T_\nu \propto a^{-1}$ and photon temperature $T_\gamma \propto g_{\star S}^{-1/3} a^{-1}$.

a fraction of the released energy and entropy did leak to the neutrinos, resulting into $N_{\text{eff}} = 3.046$ (the current bound from the Planck satellite is $N_{\text{eff}} = 3.04 \pm 0.18$).

Note that Eq. 10.2.21 holds until the present, thus allowing to estimate the temperature of the **cosmic neutrino background** (CνB): given the present CMB temperature $T_{\text{CMB}} = 2.73 \text{ K} = 0.24 \text{ meV}$, it is $T_{\text{CνB}} = 1.95 \text{ K} = 0.17 \text{ meV}$. From Eq. 10.2.8, the number density of neutrinos is:

$$n_\nu = \frac{3}{4} N_{\text{eff}} \cdot \frac{4}{11} n_\gamma \quad (10.2.22)$$

which today corresponds to $n_\nu/N_{\text{eff}} \simeq 112 \text{ cm}^{-3}$ per flavour, as $n_\gamma(2.73 \text{ K}) \simeq 5 \text{ K}^3 \simeq 415 \text{ cm}^{-3}$. On the other hand, the present energy density of neutrinos depends on whether they are relativistic or non-relativistic today:

- massless neutrinos: massless neutrinos are always relativistic, hence, by Eq. 10.2.8:

$$\rho_\nu = \frac{7}{8} N_{\text{eff}} \left(\frac{4}{11} \right)^{4/3} \rho_\gamma \quad \Rightarrow \quad \Omega_\nu h^2 \simeq 1.7 \cdot 10^{-5}$$

were $\Omega_\gamma h^2 \simeq 2.5 \cdot 10^{-5}$ was used;

- massive neutrinos: experiments on neutrino oscillations impose constraints on the sum of neutrino masses: $60 \text{ meV} < \sum_i m_{\nu_i} < 1 \text{ eV}$. This means that neutrinos behave as radiation-like particles in the early universe ($T \gg m_\nu$) and as matter-like particles in the late universe ($T \ll m_\nu$), which means that $\rho_{\nu,0} \approx \sum_i m_{\nu_i} n_{\nu_i,0}$ (by Eq. 10.2.10). Using the estimate $n_{\nu_i} \simeq 112 \text{ cm}^{-3}$:

$$\Omega_\nu h^2 \approx \frac{\sum_i m_{\nu_i}}{95 \text{ eV}} \quad (10.2.23)$$

Hence neutrinos are a subdominant component, with $0.001 \lesssim \Omega_\nu \lesssim 0.02$ (using $h = 0.7$).

§10.2.4 Thermal history

In radiation era $a \propto \sqrt{t}$, hence $H = \frac{1}{2t}$ and, using Eq. 10.2.12:

$$\frac{1}{2t} = H \simeq \sqrt{\frac{\rho_r}{3M_p^2}} \simeq \frac{\pi}{3} \sqrt{\frac{g_*}{10}} \frac{T^2}{M_p} \quad \Rightarrow \quad \frac{T}{1 \text{ MeV}} \simeq 1.5 g_*^{-1/4} \sqrt{\frac{1 \text{ sec}}{t}} \quad (10.2.24)$$

Therefore, 1 second after the Big Bang the universe was about 1 MeV. It is also possible to state an approximated thermal history of the universe (recalling Eq. 8.2.6):

Event	time t	redshift z	temperature T
Singularity	0	∞	∞
Inflation	$\sim 10^{-35} \text{ s}$	$\sim 10^{28}$	$\sim 10^{15} \text{ GeV}$
Baryogenesis	$\lesssim 20 \text{ ps}$	$> 10^{15}$	$> 100 \text{ GeV}$
EW phase transition	20 ps	10^{15}	100 GeV
QCD phase transition	$20 \mu\text{s}$	10^{12}	150 MeV
Dark matter freeze-out	?	?	?
Neutrino decoupling	1 s	$6 \cdot 10^9$	0.8 MeV
e^-e^+ annihilation	6 s	$2 \cdot 10^9$	500 keV
Big Bang nucleosynthesis	3 min	$4 \cdot 10^8$	100 keV
Matter-radiation equality	60 kyr	3400	0.75 eV
Recombination	$260 - 380 \text{ kyr}$	$1100 - 1400$	$0.26 - 0.33 \text{ eV}$
Photon decoupling (CMB)	380 kyr	1100	0.26 eV
Reionization	$100 - 400 \text{ Myr}$	$10 - 30$	$2.6 - 7.0 \text{ meV}$
Λ -matter equality	9 Gyr	0.4	0.33 meV
Today	13.8 Gyr	0	0.24 meV

Note that the expansion of the universe during Inflation had a very different timescale with respect to today, although in both cases it is exponential ($a \propto \exp Ht$):

$$H_I \sim \frac{T_I^2}{M_p} \sim 10^{11} \text{ GeV} \quad \Rightarrow \quad t_{H_I} \equiv \frac{1}{H_I} \sim 10^{-35} \text{ sec}$$

$$H_0 \approx \frac{70 \text{ km s}^{-1}}{1 \text{ Mpc}} \quad \Rightarrow \quad t_{H_0} \equiv \frac{1}{H_0} \sim 4.5 \cdot 10^{17} \text{ sec}$$

The expansion during Inflation was $\sim 10^{52}$ times faster than it is today.

§10.3 Beyond equilibrium

§10.3.1 Boltzmann equation

In the absence of interactions, the number of particles in a fixed physical volume $V \propto a^3$ is conserved, thus $n_i \propto a^{-3}$ and:

$$\frac{\dot{n}_i}{n_i} = -3 \frac{\dot{a}}{a} \quad \Longleftrightarrow \quad \frac{1}{a^3} \frac{d(n_i a^3)}{dt} = 0$$

To include the effect of interactions, a collision term $C_i(\{n_j\})$ is added, which depends on the number density of all particle species:

$$\frac{1}{a^3} \frac{d(n_i a^3)}{dt} = C_i(\{n_j\}) \quad (10.3.1)$$

This is the **Boltzmann equation**, which describes the behaviour of the system beyond equilibrium. The form of the collision term depends on the specific interactions under consideration: interactions between three or more particles are very unlikely, so it is reasonable to assume only single-particle decays and two-particle scatterings/annihilations.

Example 10.3.1 ($2 \leftrightarrow 2$ scattering)

Consider a generic process $1 + 2 \leftrightarrow 3 + 4$, and suppose to track the number density n_1 of species 1. The rate of change in the abundance of species 1 is given by the difference between the rates of annihilation (direct process) and production (inverse process) of the species, hence:

$$\frac{1}{a^3} \frac{d(n_1 a^3)}{dt} = -\alpha n_1 n_2 + \beta n_3 n_4$$

The first term describes the direct process, and the parameter $\alpha = \langle \sigma v \rangle$ is the *thermally-averaged cross-section*^a. To determine the parameter β , note that the collision term must vanish at (chemical) equilibrium, so that:

$$\beta = \left(\frac{n_1 n_2}{n_3 n_4} \right)_{\text{eq}} \alpha$$

where the equilibrium number densities are given by Eq. 10.2.1. The Boltzmann equation then becomes:

$$\frac{1}{a^3} \frac{d(n_1 a^3)}{dt} = -\langle \sigma v \rangle \left[n_1 n_2 - \left(\frac{n_1 n_2}{n_3 n_4} \right)_{\text{eq}} n_3 n_4 \right] \quad (10.3.2)$$

Rewriting this equation in terms of the number of particles in a comoving volume $N_i \equiv n_i/s$:

$$\frac{d \ln N_1}{d \ln a} = -\frac{\Gamma_1}{H} \left[1 - \left(\frac{N_1 N_2}{N_3 N_4} \right)_{\text{eq}} \frac{N_3 N_4}{N_1 N_2} \right] \quad (10.3.3)$$

with the interaction rate $\Gamma_1 \equiv n_2 \langle \sigma v \rangle$. The r.h.s. contains two different factors: the Γ_1/H factor, describing the **interaction efficiency**, and the $[1 - \dots]$ factor, describing the **deviation from equilibrium**. There are two main regimes for the reaction:

- $\Gamma_1 \gg H$: the natural state of the system is chemical equilibrium since, given the large interaction rate, if $N_1 \gg N_1^{\text{eq}}$ the r.h.s. is negative and N_1 reduces towards N_1^{eq} , while if $N_1 \ll N_1^{\text{eq}}$ the r.h.s. is positive and N_1 increases towards N_1^{eq} . At equilibrium, then, the r.h.s. vanishes and particles assume their equilibrium abundances;
- $\Gamma_1 < H$: when the reaction rate drops below the Hubble scale, the r.h.s. gets suppressed and the comoving density of particles approaches a constant *relic density*, a phenomenon known as **freeze-out**.

^aThe angle brackets denote an average over the relative velocity v between particles 1 and 2.

§10.3.2 Dark matter relics

In this section, the hypothesis that dark matter is composed of weakly interacting massive particles (WIMPs) is assumed. WIMPs were in close contact with the rest of the cosmic plasma at high temperatures, but then experiences freeze-out at a critical temperature T_f , which can be estimated using the Boltzmann equation.

§10.3.2.1 Dark matter freeze-out

Some assumptions on the WIMP interactions have to be made. In particular, assume that a heavy dark matter particle X and its antiparticle $\bar{X}$ can annihilate to produce two light (essentially massless) particle:

$$X + \bar{X} \leftrightarrow \ell + \bar{\ell}$$

Moreover, assume the light particles to be tightly coupled to the cosmic plasma (e.g. if they are charged), so that they maintain their equilibrium densities $n_\ell = n_\ell^{\text{eq}}$, and also that there are no initial asymmetries between X and $\bar{X}$, i.e. $n_X = n_{\bar{X}}$. Eq. 10.3.2 for the number of WIMPs in a comoving volume $N_X \equiv n_X/s$ then is:

$$\frac{dN_X}{dt} = -s \langle \sigma v \rangle [N_X^2 - (N_X^{\text{eq}})^2]$$

Since the interesting dynamics takes place at $T \sim M_X$, define $x \equiv M_X/T$, so that:

$$\frac{dx}{dt} = \frac{d}{dt} \frac{M_X}{T} = -\frac{1}{T} \frac{dT}{dt} x \simeq Hx$$

where $T \propto a^{-1}$ was used (assuming $g_{*S} \simeq \text{const.} \equiv g_{*S}(M_X)$) for the times relevant to freeze-out. Assuming radiation domination⁴, i.e. $H = H(M_X)/x^2$, the Boltzmann equation reduces to the so-called **Riccati equation**:

$$\frac{dN_X}{dx} = -\frac{\lambda}{x^2} [N_X^2 - (N_X^{\text{eq}})^2] \quad (10.3.4)$$

with:

$$\lambda \equiv \frac{2\pi^2}{45} g_{*S} \frac{M_X^3 \langle \sigma v \rangle}{H(M_X)} \quad (10.3.5)$$

This parameter can be treated as a constant in most WIMP theories, but, despite this, the Riccati equation has no analytic solutions. Numerical solutions for two different values of λ are pictured in Fig. 10.4: at high temperatures ($x < 1$) $N_X \approx N_X^{\text{eq}} \simeq 1$, while at low temperatures ($x \gg 1$) the equilibrium abundance is exponentially suppressed as $N_X^{\text{eq}} \sim e^{-x}$. Ultimately, WIMPs will become so rare that they will not be able to find each other fast enough to maintain the equilibrium abundance: numerically, this freeze-out happens at $x_f \sim 10$, where the solution N_X to the Riccati equation starts to deviate significantly from the Boltzmann-suppressed equilibrium abundance N_X^{eq} . The final relic abundance $N_X^\infty \equiv N_X(x = \infty)$ determines the freeze-out density of dark matter, and it can be estimated as a function of λ . For $x \gg x_f$ (after freeze-out), N_X will be much larger than N_X^{eq} , hence the Riccati equation reduces to:

$$\frac{dN_X}{dx} \simeq -\frac{\lambda}{x^2} N_X^2$$

⁴From Eq. 10.2.24 (with $g_* \simeq \text{const.}$) $H/T^2 = \text{const.}$, hence $H/T^2 = H(M_X)/M_X^2$, i.e. $H = H(M_X)/x^2$.

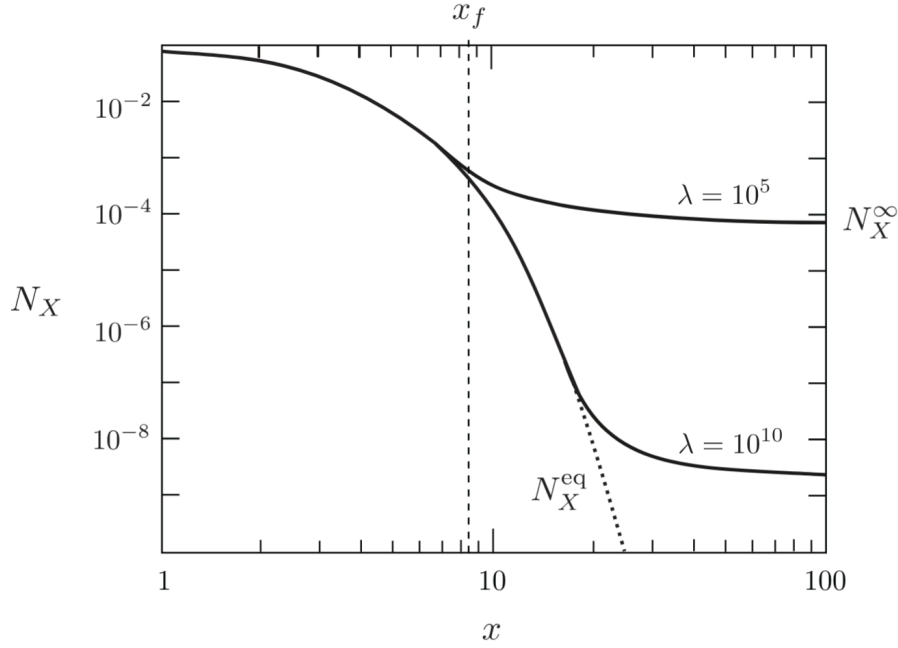


Figure 10.4: Abundance of WIMPs as the temperature drops below their mass.

Integrating over the domain (x_f, ∞) , the following solution is found:

$$\frac{1}{N_X^\infty} - \frac{1}{N_X^f} = \frac{\lambda}{x_f}$$

with $N_X^f \equiv N_X(x_f)$. Typically $N_X^f \gg N_X^\infty$ (see Fig. 10.4), hence an estimate for the relic abundance is:

$$N_X^\infty \simeq \frac{x_f}{\lambda} \quad (10.3.6)$$

This relation predicts that the relic abundance decreases as the interaction rate λ increases: this makes sense, since larger interaction rates maintain equilibrium longer, i.e. deeper in the Boltzmann-suppressed regime.

Note that the value of $x_f \sim 10$ is not terribly sensitive to the precise value of λ : indeed, $x_f \propto |\log \lambda|$.

§10.3.2.2 WIMP miracle

The freeze-out abundance of dark matter relics can be related to the dark matter density today:

$$\Omega_X \equiv \frac{\rho_{X,0}}{\rho_{\text{crit},0}} = \frac{M_X n_{X,0}}{3M_p^2 H_0^2} = \frac{M_X N_{X,0} s_0}{3M_p^2 H_0^2} = M_X N_X^\infty \frac{s_0}{3M_p^2 H_0^2}$$

as the number of WIMPs is conserved after freeze-out: $N_{X,0} = N_X^\infty$. Inserting Eq. 10.2.19 and Eq. 10.3.6:

$$\Omega_X \simeq \frac{M_X x_f}{\lambda} \frac{2\pi^2}{45} \frac{g_{*S}(T_0) T_0^3}{3M_p^2 H_0^2} = \frac{H(M_X)}{M_X^2} \frac{x_f}{\langle \sigma v \rangle} \frac{g_{*S}(T_0)}{g_{*S}(M_X)} \frac{T_0^3}{3M_p^2 H_0^2}$$

Using Eq. 10.2.24 for $H(M_X)$ gives:

$$\Omega_X \simeq \frac{\pi}{9} \frac{x_f}{\langle \sigma v \rangle} \sqrt{\frac{g_*(M_X)}{10}} \frac{g_{*S}(T_0)}{g_{*S}(M_X)} \frac{T_0^3}{M_p^3 H_0^2} \quad (10.3.7)$$

Substituting the measured value $T_0 = 2.73 \text{ K}$ and using $g_{\star S}(T_0) = 3.91$ and $g_{\star S}(M_X) \simeq g_{\star}(M_X)$:

$$\Omega_X h^2 \simeq \frac{x_f}{100} \sqrt{\frac{10}{g_{\star}(M_X)}} \frac{10^{-8} \text{ GeV}^{-2}}{\langle \sigma v \rangle}$$

This reproduces the observed dark matter density if $\sqrt{\langle \sigma v \rangle} \sim 10^{-4} \text{ GeV}^{-1} \sim 0.1 \sqrt{G_F}$: the fact that a thermal relic with a cross-section typical of the weak interaction gives the right dark matter abundance is known as the *WIMP miracle*.

§10.3.3 Big Bang nucleosynthesis

At $T \sim 1 \text{ MeV}$, photons, electrons and positrons are in equilibrium, neutrinos are about to decouple and baryons are non-relativistic, hence much fewer in number than relativistic particles. However, due to baryon number conservation, the total number of nucleons remains constant: weak nuclear reactions may convert neutrons and protons into each other, while strong nuclear reactions may build nuclei from them. This is known as **Big Bang nucleosynthesis** (BBN).

In principle, BBN is a very complicated process involving many coupled Boltzmann equations. In practice, it is convenient to make two simplifications:

- (1) only consider hydrogen and helium: no heavier element is produced at appreciable levels in BBN, hence one only needs to track $^1\text{H} \equiv \text{H}$, $^2\text{H} \equiv \text{D}$, $^3\text{H} \equiv \text{T}$, ^3He and ^4He ;
- (2) only consider protons and neutrons for $T \gtrsim 0.1 \text{ MeV}$: at these temperatures, only free protons and neutrons exist, while other light nuclei have yet to form.

§10.3.3.1 Equilibrium abundances

To show the validity of these assumptions, recall that at $T \gtrsim 0.8 \text{ MeV}$ protons and neutrons are coupled via weak interactions (β -decay and inverse β -decay):

$$n + \nu_e \leftrightarrow p^+ + e^- \quad n + e^+ \leftrightarrow p^+ + \bar{\nu}_e$$

Assume $\mu_\nu, \mu_e \simeq 0$, so that $\mu_p = \mu_n$. Then, generalizing Eq. 10.2.10 to include the chemical potential:

$$n_i^{\text{eq}} = g_i \left(\frac{m_i T}{2\pi} \right) e^{\frac{\mu_i - m_i}{T}} \implies \left(\frac{n_n}{n_p} \right)_{\text{eq}} = \left(\frac{m_n}{m_p} \right)^{3/2} e^{-(m_n - m_p)/T} \simeq e^{-Q/T} \quad (10.3.8)$$

where $Q \equiv m_n - m_p \simeq 1.30 \text{ MeV} \ll m_n, m_p$. This shows that for $T \gg 1 \text{ MeV}$ there are as many neutrons as protons, while for $T < 1 \text{ MeV}$ the neutron fraction gets smaller⁵.

Next, consider deuterium, which is produced by the following relation:

$$n + p^+ \leftrightarrow D + \gamma$$

Since $\mu_\gamma = 0$, then $\mu_D = \mu_n + \mu_p$, recalling that $g_D = 4$ and $g_n = g_p = 2$:

$$\left(\frac{n_D}{n_n n_p} \right)_{\text{eq}} = \left(\frac{m_D}{m_n m_p} \frac{2\pi}{T} \right)^{3/2} e^{-(m_D - m_n - m_p)/T} \quad (10.3.9)$$

⁵Were the up quark heavier than the down quark, then $m_n < m_p$ and, with the decrease of temperature, protons would weakly decay into neutrons, producing a much different universe.

In the prefactor $m_D \approx 2m_n \approx 2m_p \simeq 1.9 \text{ GeV}$, while the exponent can be expressed in terms of the binding energy of deuterium $B_D \equiv m_n + m_p - m_D \simeq 2.22 \text{ MeV}$. Hence, as long as chemical equilibrium holds:

$$\left(\frac{n_D}{n_p}\right)_{\text{eq}} = n_n^{\text{eq}} \left(\frac{4\pi}{m_p T}\right)^{3/2} e^{B_D/T}$$

For an order-of-magnitude estimate, approximate the neutron density with the baryon density, so that:

$$n_n \sim n_b = \eta_b n_\gamma = \eta_b \frac{2\zeta(3)}{\pi^2} T^3 \quad \Rightarrow \quad \left(\frac{n_D}{n_p}\right)_{\text{eq}} \sim \eta_b \left(\frac{T}{m_p}\right)^{3/2} e^{B_D/T}$$

The smallness of $\eta_b \sim 10^{-9}$ inhibits deuterium production until the temperature drops well below the binding energy B_D , and the same applies to all other nuclei: at $T \gtrsim 0.1 \text{ MeV}$, then, virtually all baryons are in the form of neutrons and protons. At this time, deuterium and helium are produced, but the reaction rates are too slow to produce any heavier nucleus.

§10.3.3.2 Neutron freeze-out and decay

The primordial ratio of neutrons is particularly important for BBN, since essentially all neutrons became incorporated into ^4He . Defining the neutron fraction as $X_n \equiv n_n/(n_n + n_p)$, with Eq. 10.3.8:

$$X_n^{\text{eq}}(T) = \frac{e^{-Q/T}}{1 + e^{-Q/T}} \quad (10.3.10)$$

Neutrons follow this equilibrium abundance until neutrinos decouple⁶ at $T_{\text{dec}} \simeq 0.8 \text{ MeV}$, when weak processes like the β -decay effectively shut off. At this moment $X_n^{\text{eq}}(T_{\text{dec}}) \simeq 0.17 \simeq \frac{1}{6}$, which can be taken as a rough estimate of the final freeze-out abundance, so that $X_n^\infty \sim \frac{1}{6}$ (a more precise estimate with QFT gives $X_n^\infty = 0.15$).

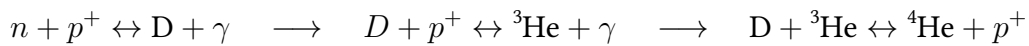
At $T \lesssim 0.2 \text{ MeV}$, i.e. $t \gtrsim 100 \text{ sec}$, the finite lifetime of the neutron becomes important, and indeed neutron decay must be accounted for:

$$X_n(t) = X_n^\infty e^{-t/\tau_n} \sim \frac{1}{6} e^{-t/\tau_n} \quad (10.3.11)$$

where $\tau_n = 886.7 \pm 0.8 \text{ sec}$ is the neutron's lifetime.

§10.3.3.3 Helium fusion

At this point, the universe is mostly made of protons and neutrons. Helium cannot form directly, since the density is too low and the time available is too short for reactions involving three or more nuclei to occur (at least at an appreciable rate). Heavier nuclei are instead formed sequentially, according to the following chain of reactions:



Since deuterium is formed directly from neutrons and protons, it can follow its equilibrium abundance as long as there are enough free neutrons available. However, since B_D is rather small, the deuterium abundance becomes large rather late (at $T < 100 \text{ keV}$), so, although heavier nuclei have larger binding energy and hence would have larger equilibrium abundances, they cannot be formed

⁶Note that $Q \sim T_{\text{dec}}$ is just a coincidence, since the former is determined by the strong and electromagnetic interactions, while the latter by the weak interaction.

until sufficient deuterium has become available, a phenomenon known as **deuterium bottleneck**. Only when there is enough deuterium can helium be produced.

To get a rough estimate for the time of nucleosynthesis, first estimate the temperature T_{nuc} when the deuterium fraction in equilibrium would be of order 1, i.e. $(n_{\text{D}}/n_{\text{p}})_{\text{eq}} \sim 1$:

$$\eta_{\text{b}} \left(\frac{T_{\text{nuc}}}{m_{\text{p}}} \right)^{3/2} e^{B_{\text{D}}/T} \sim 1 \quad \Rightarrow \quad T_{\text{nuc}} \sim 0.06 \text{ MeV}$$

From Eq. 10.2.24 with $g_{\star} = 3.38$, the time of nucleosynthesis is found:

$$t_{\text{nuc}} \simeq 120 \text{ sec} \left(\frac{0.1 \text{ MeV}}{T_{\text{nuc}}} \right)^2 \sim 330 \text{ sec}$$

A better estimate would be imposing $(n_{\text{D}}/n_{\text{p}})_{\text{eq}} \sim 10^{-3}$, as suggested by Fig. 10.6, resulting in $t_{\text{nuc}} \sim 250 \text{ sec}$: in both cases $t_{\text{nuc}} \ll \tau_{\text{n}}$, so Eq. 10.3.11 is not very sensitive to a precise estimate of t_{nuc} . In both cases:

$$X_{\text{n}}(t_{\text{nuc}}) \sim \frac{1}{8} \quad (10.3.12)$$

Since $B_{\text{He}} > B_{\text{D}}$, the Boltzmann factor $e^{B/T}$ favours helium over deuterium, so helium is produced almost immediately after deuterium: virtually all remaining neutrons at $t \sim t_{\text{nuc}}$ are then processed into ${}^4\text{He}$. Since two neutrons go into one nucleus of ${}^4\text{He}$, its final abundance is equal to half the neutron abundance at t_{nuc} , hence:

$$\frac{n_{\text{He}}}{n_{\text{H}}} = \frac{n_{\text{He}}}{n_{\text{p}}} \simeq \frac{\frac{1}{2}X_{\text{n}}(t_{\text{nuc}})}{1 - X_{\text{n}}(t_{\text{nuc}})} \sim \frac{1}{16} \quad (10.3.13)$$

This prediction is consistent with the observed helium in the universe, as shown in Fig. 10.5. The abundances of other light elements are determined numerically solving the system of coupled Boltzmann equation, and the result is shown in Fig. 10.6 and Fig. 10.7: all these theoretical abundances have quantitative agreement with observational data, with the exception of lithium (the so-called *lithium problem*).

§10.3.3.4 Constraints on BSM physics

The result in Eq. 10.3.13 depends on several input parameters:

- $g_{\star}$: during radiation era $H \propto g_{\star}^{1/2} T^2$, hence variations in $g_{\star}$ affect the neutron freeze-out temperature $T_{\text{f}} \sim T_{\text{dec}}$, since the neutrino decoupling temperature is determined by $G_{\text{F}} T_{\text{dec}}^5 \sim \sqrt{G_{\text{N}} g_{\star}} T_{\text{dec}}^2$, i.e. $T_{\text{f}} \sim T_{\text{dec}} \propto g_{\star}^{1/6}$. An increase of $g_{\star}$ causes an increase of T_{f} , which in turn increases the $n_{\text{n}}/n_{\text{p}}$ ratio at freeze-out, hence increasing the final helium abundance;
- τ_{n} : a larger neutron lifetime would reduce the amount of neutron decay after freeze-out, thus increasing the final helium abundance;
- Q : a larger mass difference between neutrons and protons would decrease the $n_{\text{n}}/n_{\text{p}}$ ratio at freeze-out, so decreasing the final helium abundance;
- η_{b} : the final helium abundance increases with an increase of η_{b} , as nucleosynthesis starts earlier for larger baryon density (i.e. larger deuterium density);
- G_{N} : increasing the strength of gravity would increase the neutron freeze-out temperature $T_{\text{f}} \sim T_{\text{dec}} \propto G_{\text{N}}^{1/6}$, therefore increasing the final helium abundance;

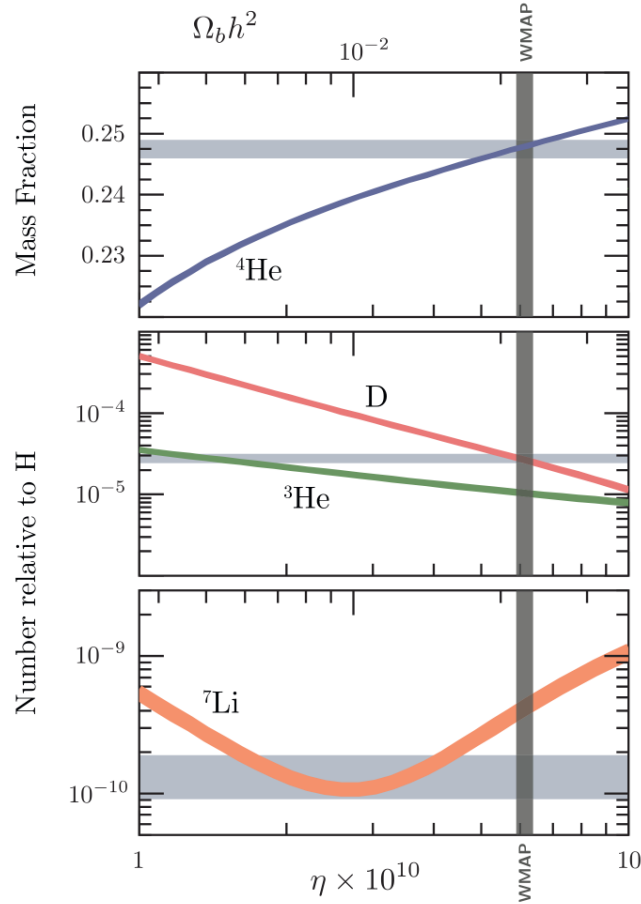


Figure 10.5: Theoretical predictions (coloured) and observational constraints (grey) on today abundances of light nuclei.

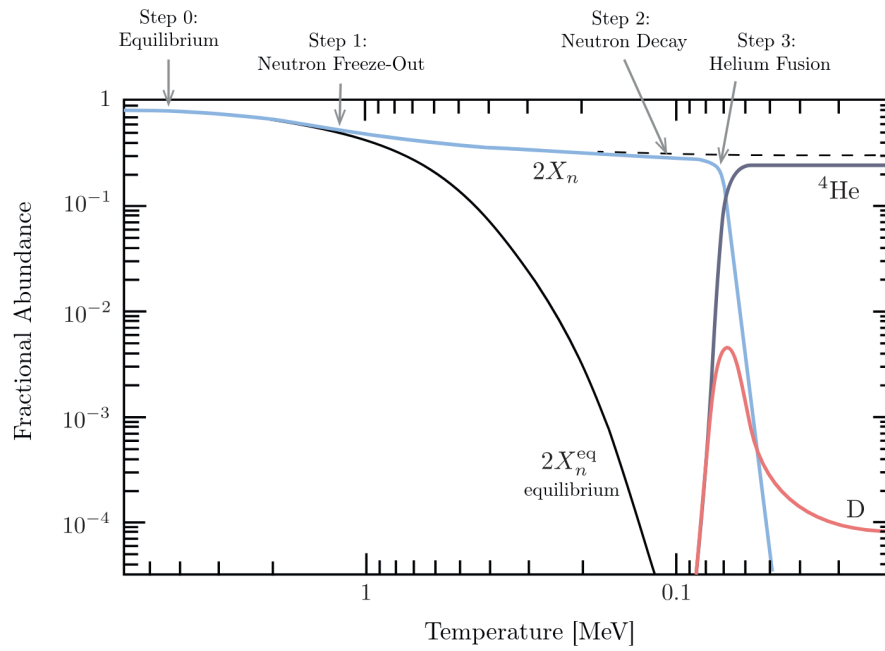


Figure 10.6: Numerical results for helium production in the early universe.

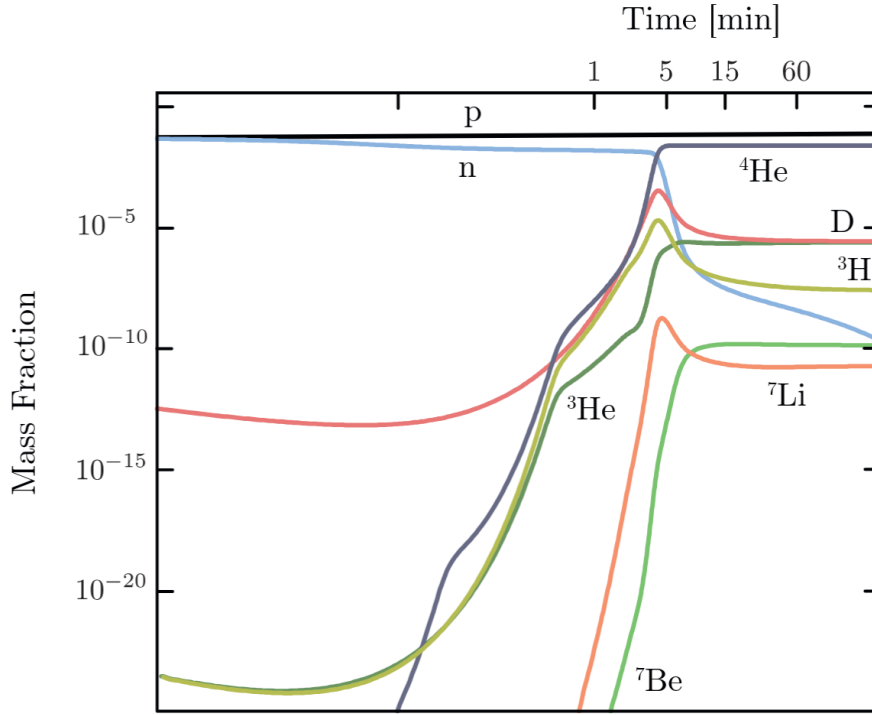


Figure 10.7: Numerical results for the evolution of light element abundances.

- G_F : increasing the strength of the weak nuclear force would decrease the neutron freeze-out temperature $T_f \sim T_{\text{dec}} \propto G_F^{-2/3}$, thus decreasing the final helium abundance.

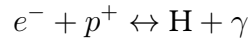
The dependence of BBN on SM parameters makes it a probe (and a constraint) of BSM physics.

§10.3.4 Recombination

At $T \gtrsim 1$ eV, the universe still consisted of a plasma of free electrons and nuclei: photons were tightly coupled to electrons through Compton scattering, while electrons were strongly coupled to protons via Coulomb scattering. When temperature became low enough, electrons and nuclei combined to form neutral atoms: this phenomenon is known as **Recombination**. As a consequence, the electron density fell sharply, thus causing a growth of the photon mean free path, which became larger than the horizon distance: photons decoupled from matter and the universe became transparent. Today, these photons are the cosmic microwave background.

§10.3.4.1 Saha equilibrium

At $T > 1$ eV, baryons and photons are still in equilibrium through electromagnetic reactions, e.g.:



Since $T < m_e < m_p < m_H$, from Eq. 10.2.10 the equilibrium abundances of the various species are:

$$n_i^{\text{eq}} = g_i \left(\frac{m_i T}{2\pi} \right)^{3/2} \exp \frac{\mu_i - m_i}{T} \quad (10.3.14)$$

with $\mu_p + \mu_e = \mu_H$ (since $\mu_\gamma = 0$). Then:

$$\left(\frac{n_H}{n_e n_p} \right)_{\text{eq}} = \frac{g_H}{g_e g_p} \left(\frac{m_H}{m_e m_p} \frac{2\pi}{T} \right)^{3/2} \exp \frac{m_p + m_e - m_H}{T}$$

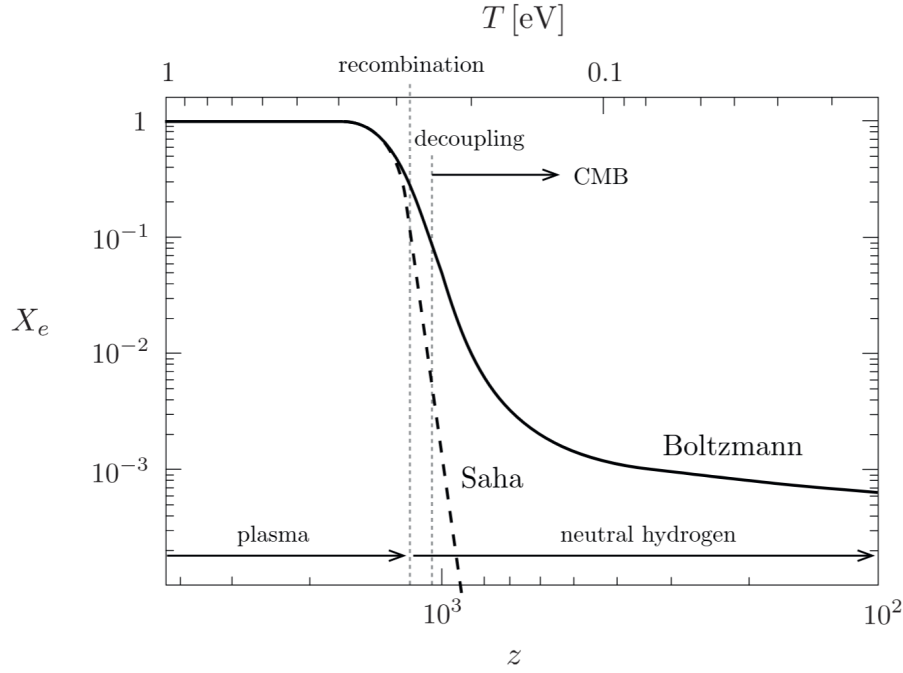


Figure 10.8: Free electron-to-baryon ratio as a function of redshift.

Introducing the binding energy of hydrogen $B_H \equiv m_p + m_e - m_H \simeq 13.6 \text{ eV}$ and the spin factors $g_H = 4$, $g_p = g_e = 2$, and noting that $n_e = n_p$ since the universe is not electrically charged (at least according to current observations), this equation becomes:

$$\left(\frac{n_H}{n_e^2} \right)_{\text{eq}} = \left(\frac{2\pi}{m_e T} \right)^{3/2} e^{B_H/T}$$

Now, define the *free electron-to-baryon fraction* as $X_e \equiv n_e/n_b$, where the baryon number can be written as $n_b = \eta_b (2\zeta(3)/\pi^2) T^3$, and note that protons are over 90% (in number) of all nuclei, so that $n_b \approx n_p + n_H = n_e + n_H$, so that:

$$\frac{1 - X_e}{X_e^2} = \frac{n_H}{n_e^2} n_b \quad (10.3.15)$$

Putting everything together, the **Saha equation** is found:

$$\left(\frac{1 - X_e}{X_e^2} \right)_{\text{eq}} = \frac{2\zeta(3)}{\pi^2} \eta_b \left(\frac{2\pi T}{m_e} \right)^{3/2} e^{B_H/T} \quad (10.3.16)$$

The solution of this equation as a function of redshift is plotted in Fig. 10.8: although the Saha approximation correctly identifies the onset of Recombination, it is clearly insufficient to determine the relic densities of electrons after freeze-out.

§10.3.4.2 Hydrogen recombination

To study Recombination, define the recombination temperature T_{rec} as the temperature at which 90% of electrons have combined with protons to form hydrogen, i.e.:

$$X_e(T_{\text{rec}}) = 0.1 \quad \Rightarrow \quad T_{\text{rec}} \approx 0.3 \text{ eV} \simeq 3600 \text{ K} \quad (10.3.17)$$

Note that $T_{\text{rec}} \ll B_{\text{H}}$: the reason is that there are many photons for each hydrogen atom ($\eta_{\text{b}} \sim 10^{-9}$), hence, even when $T < B_{\text{H}}$, the high-energy tail of the photon distribution contains a sufficient number of photons with energy $E > B_{\text{H}}$ which can ionize the hydrogen atoms.

This estimate puts $z_{\text{rec}} \simeq 1320$, since $T_{\text{rec}} = T_0(1 + z_{\text{rec}})$, which puts Recombination in the matter-dominated era which started at $z_{\text{eq}} \simeq 3500$. Then, using $a(t) = (t/t_0)^{2/3}$, the time of Recombination is found:

$$t_{\text{rec}} = \frac{t_0}{(1 + z_{\text{rec}})^{3/2}} \simeq 290'000 \text{ yrs} \quad (10.3.18)$$

§10.3.4.3 Photon decoupling

At $T = T_{\text{rec}}$ there are still free electrons which are coupled to the photon bath. Indeed, photons are coupled to matter mostly through their interactions with electrons:

$$e^- + \gamma \leftrightarrow e^- + \gamma$$

with an interaction rate $\Gamma_{\gamma} \approx n_e \sigma_{\text{T}}$ (as $v \approx c = 1$), where $\sigma_{\text{T}} \simeq 2 \cdot 10^{-3} \text{ MeV}^{-2}$ is the *Thomson cross-section*. The interaction rate therefore decreases as the density of free electrons drops, and the decoupling of photons happens roughly when it becomes smaller than the expansion rate, i.e.:

$$\Gamma_{\gamma}(T_{\text{CMB}}) \sim H(T_{\text{CMB}}) \quad (10.3.19)$$

Writing $n_e = n_{\text{b}} X_e$, the interaction rate becomes:

$$\Gamma_{\gamma}(T) = n_{\text{b}} X_e(T) \sigma_{\text{T}} = \frac{2\zeta(3)}{\pi^2} \eta_{\text{b}} \sigma_{\text{T}} X_e(T) T^3$$

On the other hand, in matter era the Ω_{m} term in Eq. 8.3.16 dominates, so that:

$$H(T) \simeq H_0 \sqrt{\Omega_{\text{m}}} \left(\frac{T}{T_0} \right)^{3/2}$$

Combining this equation, T_{CMB} is found numerically as:

$$X_e(T_{\text{CMB}}) T_{\text{CMB}}^{3/2} \sim \frac{\pi^2}{2\zeta(3)} \frac{H_0 \sqrt{\Omega_{\text{m}}}}{\eta_{\text{b}} \sigma_{\text{T}} T_0^{3/2}} \implies T_{\text{CMB}} \sim 0.27 \text{ eV} \quad (10.3.20)$$

which corresponds to $z_{\text{CMB}} \sim 1100$ and $t_{\text{CMB}} \sim 380 \text{ kyrs}$. Note that, although T_{CMB} is not far from T_{rec} , the ionization fraction decreases significantly between recombination and photon decoupling: $X_e(T_{\text{rec}}) \simeq 0.1 \rightarrow X_e(T_{\text{CMB}}) \simeq 0.01$. This shows that a large degree of neutrality is necessary for the universe to become transparent to photon propagation: after decoupling, photons stream freely through the universe, and are observable today as the CMB.

Structure Formation

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Observations of the CMB reveal that, at least until z_{CMB} , the universe was extremely homogeneous, with temperature variations of the order $\delta T/T \sim 10^{-5}$. However, to understand the formation and evolution of large-scale structures, it is necessary to introduce inhomogeneities: today these perturbations of the homogeneous background are huge (e.g. galaxies and voids), but up to z_{CMB} they can be treated with perturbation theory.

Fig. 11.1 shows the evolution of the comoving Hubble radius with the key events of the history of the universe highlighted. Cosmic perturbations (assuming a wake-like ansatz) are pictured: these perturbations evolve in a linear regime up to a_{CMB} , which can be treated analytically, but then enter a non-linear regime which lasts to these days and has to be treated computationally.

§11.1 Newtonian perturbation theory

Since the perturbations are assumed small, the metric perturbations with respect to the background metric can be treated in the Newtonian approximation.

§11.1.1 Flat background

First, consider the evolution of cosmological perturbations in a flat universe. In the Newtonian approximation, the gravitational potential is just:

$$\Phi(t, \mathbf{x}) = \delta\Phi(t, \mathbf{x}) \quad (11.1.1)$$

since a flat background has $\bar{\Phi}(t) \equiv 0$. The matter sources too must be perturbed: assuming the background to be at rest, i.e. $\bar{\mathbf{u}}(t) \equiv 0$:

$$\rho(t, \mathbf{x}) = \bar{\rho}(t) + \delta\rho(t, \mathbf{x}) \quad P(t, \mathbf{x}) = \bar{P}(t) + \delta P(t, \mathbf{x}) \quad \mathbf{u}(t, \mathbf{x}) = \delta\mathbf{u}(t, \mathbf{x}) \quad (11.1.2)$$

The equations that dictate the physical evolution of these perturbations are the continuity equation, the Euler equation and the Poisson equation:

$$\dot{\rho} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (11.1.3)$$

$$(\partial_t + \mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla P - \nabla \Phi \quad (11.1.4)$$

$$\nabla^2 \Phi = 4\pi G \rho \quad (11.1.5)$$

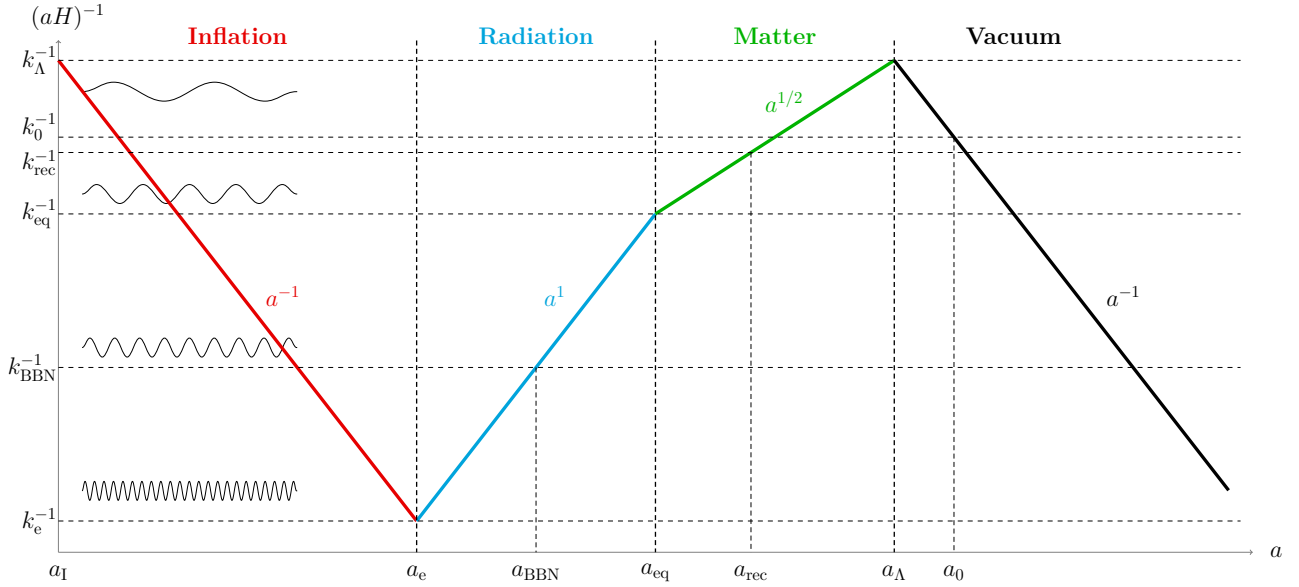


Figure 11.1: Evolution of the comoving Hubble radius. Four examples of cosmological perturbations with different wavelengths are drawn.

§11.1.1.1 Matter perturbations

To study how matter perturbations evolve, consider the complete absence of gravity, i.e. $\Phi(t, \mathbf{x}) \equiv 0$, and a static background $\bar{\rho}$, $\bar{P} = \text{const.}$, so that the (linear) continuity and Euler equations read:

$$\delta\dot{\rho} + \bar{\rho} \nabla \cdot \delta\mathbf{u} = 0 \quad \bar{\rho} \delta\dot{\mathbf{u}} = -\nabla \delta P$$

Introducing the density contrast $\delta \equiv \delta\rho/\bar{\rho}$, these equations are rewritten as:

$$\dot{\delta} = -\nabla \cdot \delta\mathbf{u} \quad \delta\dot{\mathbf{u}} = -c_s^2 \nabla \delta$$

where the **sound velocity** is defined as:

$$c_s^2 \equiv \frac{\delta p}{\delta \rho} \quad (11.1.6)$$

Combining the above equation, a wave equation for the density contrast is found:

$$\ddot{\delta} - c_s^2 \nabla^2 \delta = 0 \quad (11.1.7)$$

Therefore, perturbations evolve as sound waves with sound velocity c_s . Since for cosmological matter sources $P = w\rho$, the sound velocity is $c_s^2 = w$: in matter ($w = 0$) no sound waves propagate since there is no pressure. The resulting dispersion relation is:

$$\omega^2(k) = c_s^2 k^2 \quad (11.1.8)$$

§11.1.1.2 Gravitational perturbations

Introducing gravitational perturbations in the universe, i.e. $\delta\Phi(t, \mathbf{x}) \neq 0$ (which evolves as $\nabla^2 \delta\Phi = 4\pi G \delta\rho$), the wave equation for the density contrast gains a source term:

$$\ddot{\delta} - c_s^2 \nabla^2 \delta = 4\pi G \bar{\rho} \delta \quad (11.1.9)$$

The dispersion relation is then modified as:

$$\omega^2(k) = c_s^2 k^2 - 4\pi G \bar{\rho} \quad (11.1.10)$$

The source term then acts as a *tachyonic mass term* (i.e. with $m^2 < 0$), which introduces a characteristic scale of perturbations, the Jeans scale:

$$k_J^2 \equiv \frac{4\pi G \bar{\rho}}{c_s^2} \quad (11.1.11)$$

For $k > k_J$, i.e. for small-scale perturbations, $\omega \in \mathbb{R}$ and the behavior of the perturbations is oscillating (sound waves), while for $k < k_J$, i.e. large-scale perturbations, $\omega \in \mathbb{C}$ and the perturbations experience an exponential evolution, which is known as ***Jeans' instability***: this is the expected collapse of large structures.

After the decoupling of photons (i.e. the emission of the CMB), matter starts behaving with $w = 0$, i.e. $c_s = 0$: since there are no waves, $\omega \in \mathbb{C} \forall k \in \mathbb{R}_0^+$ (from Eq. 11.1.10), i.e. there is a collapse and consequent formation of structures at all scales (stars, galaxies, galaxy clusters, ...).

§11.1.2 FLRW background

Now, consider the evolution of cosmological perturbations in a FLRW background, which introduces the distinction between comoving coordinates $\mathbf{x}$ and physical coordinates $\mathbf{r} := a(t)\mathbf{x}$. As a consequence, the physical velocity is $\mathbf{u} \equiv \dot{\mathbf{r}} = H\mathbf{r} + \mathbf{v}$, where $\mathbf{v} \equiv a(t)\dot{\mathbf{x}}$ is the physical peculiar velocity.

Lemma 11.1.1 (Derivatives in a FLRW background)

$$\nabla_r = a^{-1}(t) \nabla_x \quad \partial_t|_r = \partial_t|_x - H\mathbf{x} \cdot \nabla_x \quad (11.1.12)$$

Proof. Consider a function $f(t, \mathbf{r}) = f(t, a(t)\mathbf{x})$. Then, $\nabla_r = a^{-1}(t) \nabla_x$ by definition, and:

$$\left(\frac{\partial f}{\partial t} \right)_x = \frac{\partial}{\partial t} \bigg|_x f(t, a(t)\mathbf{x}) = \frac{\partial}{\partial t} \bigg|_r f(t, \mathbf{r}) + \frac{\partial(a(t)\mathbf{x})}{\partial t} \cdot \nabla_r f(t, \mathbf{r}) = \left[\frac{\partial}{\partial t} \bigg|_r + H\mathbf{x} \cdot \nabla_x \right] f$$

which is the thesis. $\square$

This shows that, while in a static spacetime ∂_t and ∇ are independent, in an expanding spacetime this is no longer true. Then, assuming that the quantities in Eq. 11.1.3-11.1.5 are physical quantities, they are modified when expressed in terms of comoving coordinates.

Theorem 11.1.1 (Newtonian approximation in FLRW background)

Cosmological perturbations in a FLRW background are governed by:

$$\dot{\rho} + 3H\rho + \frac{1}{a} \nabla \cdot (\rho \mathbf{v}) = 0 \quad (11.1.13)$$

$$\left(\frac{\partial}{\partial t} + \frac{\mathbf{v}}{a} \cdot \nabla \right) \mathbf{u} = -\frac{1}{a\rho} \nabla P - \frac{1}{a} \nabla \Phi \quad (11.1.14)$$

$$\nabla^2 \Phi = 4\pi G a^2 \rho \quad (11.1.15)$$

where $\nabla \equiv \nabla_x$ and $\partial_t \equiv \partial_t|_x$.

Proof. First, consider the continuity equation Eq. 11.1.3:

$$\begin{aligned} 0 &= \left(\frac{\partial \rho}{\partial t} \right)_r + \nabla_r \cdot (\rho \mathbf{u}) = \left(\frac{\partial}{\partial t} \right)_x - H \mathbf{x} \cdot \nabla_x \rho + \frac{1}{a} \nabla_x \cdot (\rho \dot{a} \mathbf{x} + \rho \mathbf{v}) \\ &\equiv \dot{\rho} - H \mathbf{x} \cdot \nabla \rho + H \mathbf{x} \cdot \nabla \rho + H \rho \nabla \cdot \mathbf{x} + \frac{1}{a} \nabla \cdot (\rho \mathbf{v}) = \dot{\rho} + 3H\rho + \frac{1}{a} \nabla \cdot (\rho \mathbf{v}) \end{aligned}$$

Then, consider the Euler equation Eq. 11.1.4:

$$\begin{aligned} \left(\frac{\partial}{\partial t} \right)_r + \mathbf{u} \cdot \nabla_r \mathbf{u} &= -\frac{1}{\rho} \nabla_r P - \nabla_r \Phi \\ \left(\frac{\partial}{\partial t} - H \mathbf{x} \cdot \nabla + H \mathbf{x} \cdot \nabla + \frac{\mathbf{v}}{a} \cdot \nabla \right) \mathbf{u} &= -\frac{1}{a\rho} \nabla P - \frac{1}{a} \nabla \Phi \end{aligned}$$

which is the thesis. Finally, the Poisson equation is trivially modified by $\nabla_r^2 = a^{-2}(t)\nabla^2$. $\square$

Introducing the density contrast $\delta \ll 1$, these equations can be linearized¹. The continuity equation at zeroth order and at first order reads:

$$\dot{\bar{\rho}} + 3H\bar{\rho} = 0 \quad \frac{\partial(\bar{\rho}\delta)}{\partial t} + 3H\bar{\rho}\delta + \frac{1}{a} \nabla \cdot (\bar{\rho} \mathbf{v}) = 0$$

Using the zeroth order equation for the evolution of the background density, the evolution of the density contrast is found to be:

$$\dot{\delta} = -\frac{1}{a} \nabla \cdot \mathbf{v} \quad (11.1.16)$$

At linear order, the growth of density perturbations is sourced by the divergence of the fluid velocity. Then, the linearized Euler equation reads:

$$\frac{\partial(H\mathbf{r})}{\partial t} = -\frac{1}{a\bar{\rho}} \nabla \bar{P} - \frac{1}{a} \nabla \bar{\Phi} \quad \frac{\partial(H\mathbf{r} + \mathbf{v})}{\partial t} + \frac{\mathbf{v}}{a} H \cdot \nabla \mathbf{r} = -\frac{1}{a\bar{\rho}} \nabla(\bar{P} + \delta P) - \frac{1}{a} \nabla(\bar{\Phi} + \delta \Phi)$$

which, using $\mathbf{v} \cdot \nabla \mathbf{x} = \mathbf{v}$, reduces to:

$$\dot{\mathbf{v}} + H\mathbf{v} = -\frac{1}{a\bar{\rho}} \nabla \delta P - \frac{1}{a} \nabla \delta \Phi \quad (11.1.17)$$

In the absence of perturbations, this relation gives the scaling $\mathbf{v} \propto a^{-1}(t)$. Now, combine Eq. 11.1.16 with Eq. 11.1.17 and substitute the Poisson equation $\nabla^2 \delta \Phi = 4\pi G a^2 \bar{\rho} \delta$ to obtain:

$$\ddot{\delta} + 2H\dot{\delta} - \frac{c_s^2}{a^2} \nabla^2 \delta = 4\pi G \bar{\rho} \delta \quad (11.1.18)$$

This equation is equivalent to Eq. 11.1.9, but with two important differences caused by the Hubble flow: a redshift of the sound velocity $a^{-1}(t)c_s$ and the presence of friction term, which makes this a damped wave equation.

Note that the Jeans scale is still present, although now it is affected by the redshift due to the Hubble flow: if $k > ak_j$, the perturbations behave like damped waves, while for $k < ak_j$ the system presents Jeans' instability, which is no longer exponential due to the friction term (now it is a power-law instability).

¹Recall that, for the velocity field, the perturbation is the peculiar velocity $\mathbf{v}$.

For baryons, the sound velocity, and so the Jeans scale, depends on time. Before recombination, baryons are strongly coupled to photons, forming a relativistic photon-baryon fluid with $c_s \approx 1/\sqrt{3}$: the Jeans scale is then of the order of the comoving Hubble radius, suppressing the growth of baryonic fluctuations at subhorizon scales (oscillatory regime). After recombination, the baryonic fluid starts behaving like pressureless dust and baryonic fluctuations at all scales start to grow. On the other hand, for cold dark matter (CDM) the sound speed is negligible at all times, so that subhorizon fluctuations can start to grow earlier.

§11.1.2.1 Matter era

In matter era $a \propto t^{2/3}$ and $H(t) = \frac{2}{3t}$. The density contrast is just $\delta \simeq \delta_m \equiv \delta\rho_m/\bar{\rho}_m$, since:

$$\delta = \frac{\delta\rho}{\bar{\rho}} \simeq \frac{\delta\rho_r + \delta\rho_m}{\bar{\rho}_m} \equiv \delta_m + \delta_r \frac{\bar{\rho}_r}{\bar{\rho}_m} \simeq \delta_m$$

By Eq. 8.3.12, then, $4\pi G\bar{\rho}_m = \frac{3}{2}H^2 = \frac{2}{3t^2}$, and Eq. 11.1.18 for subhorizon modes (i.e. $\lambda < (aH)^{-1}$) becomes:

$$\ddot{\delta}_m + \frac{4}{3t}\dot{\delta}_m - \frac{2}{3t^2}\delta_m = 0 \quad (11.1.19)$$

since $c_s^2 \approx w = 0$ in matter era. With a power-law ansatz $\delta_m \propto t^\alpha$, two solutions are found:

$$\delta_m \propto \begin{cases} t^{-1} \propto a^{-2/3} \\ t^{2/3} \propto a \end{cases} \quad (11.1.20)$$

The first solution is a *decaying solution*, while the second one is a *growing solution*. The growing solution is $\delta \propto a$, which during matter era varies from $a_{\text{eq}} \simeq 1/3401$ to $a_\Lambda \simeq 1/1.4$, i.e. by $\sim 10^3$ times: assuming initial perturbations of order $\delta_m \sim 10^{-5}$ (from CMB), at the end of matter era these perturbations have grown to $\delta_m \sim 10^{-2}$, still not enough to explain the cosmic structures observed today. This is due to the fact that, after the CMB emission, cosmological perturbations enter the non-linear regime, and non-linear collapse is faster than the linear one.

Moreover, note that a smaller matter era determines a smaller growth of fluctuations: this means that the existence of dark matter (i.e. an additional dominant contribution to Ω_m other than baryonic matter) is necessary to explain the current cosmic structures.

§11.1.2.2 Radiation era

In radiation era $a \propto t^{1/2}$ and $H(t) = \frac{1}{2t}$. Now, the density contrast is $\delta \simeq \delta_r + \delta_m \frac{\bar{\rho}_m}{\bar{\rho}_r}$: since the radiation fluid has a large sound velocity $c_s \approx 1/\sqrt{3}$, it is expected to oscillate over scales smaller than the comoving Hubble horizon, so that, averaging over many oscillation periods, $\langle \delta_r \rangle = 0$ and the time-average density contrast reduces to the matter contribution. Then, Eq. 11.1.18 becomes:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G\bar{\rho}_m\delta_m = 0$$

Since $\dot{\delta}_m \sim H\delta_m$, the first two terms are $\sim H^2 \sim \bar{\rho}_r \gg \bar{\rho}_m$, hence the last term is negligible and the evolution of δ_m is given by:

$$\ddot{\delta}_m + \frac{1}{t}\dot{\delta}_m = 0 \quad (11.1.21)$$

With a power-law ansatz, two solution are found:

$$\delta_m \propto \begin{cases} \text{const.} \\ \ln t \propto \ln a \end{cases} \quad (11.1.22)$$

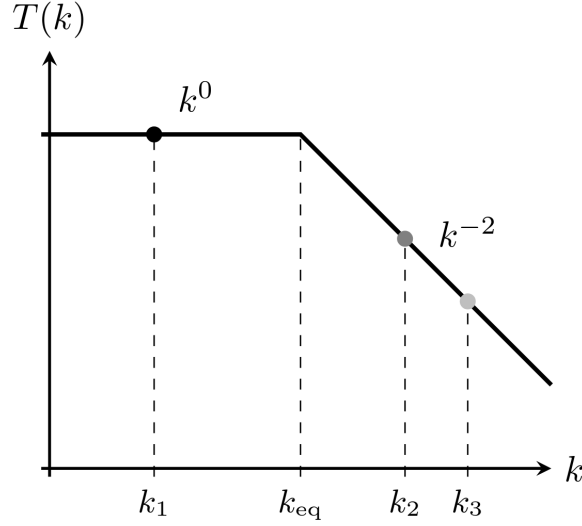


Figure 11.2: Transfer function for matter perturbation: the shape of $\mathcal{T}(k)$ is a consequence of the suppressed growth of subhorizon modes during the radiation era.

Therefore, matter perturbations grow slower in radiation era, compared to matter era (*Mészáros effect*). On the other hand, superhorizon modes (i.e. $\lambda > (aH)^{-1}$) cannot causally interact, and their evolution can be studied with a relativistic analysis.

To show how matter perturbations at different scales evolve, it is necessary to study their power spectrum $\delta_m(k)$. In particular, Inflation predicts that, when perturbations enter the comoving Hubble horizon, they have the primordial power spectrum $\delta_{m,\text{initial}}^2(k) \propto k^n$ with spectral index $n \simeq 1$ (*Harrison–Zel’dovich spectrum*), and then they evolve according to a transfer function of the form (see Fig. 11.2):

$$\mathcal{T}(k) \simeq \begin{cases} 1 & k < k_{\text{eq}} \\ \left(\frac{k_{\text{eq}}}{k}\right)^2 \ln \frac{k}{k_{\text{eq}}} & k > k_{\text{eq}} \end{cases} \quad (11.1.23)$$

where $k_{\text{eq}} \sim 60$ is the wavenumber of perturbations entering the comoving Hubble horizon at radiation-matter equality ($\lambda_{\text{eq}} \sim 100$ Mpc, approximately the size of a galaxy cluster). This means that large-scale perturbations grow less than small-scale perturbations, since they enter the Hubble radius later (see Fig. 11.1).

Combining these, it is possible to compute the predicted (linear) matter power spectrum for matter fluctuations observed today:

$$\delta_m^2 = (\mathcal{T}(k)\delta_{m,\text{initial}})^2 \propto \begin{cases} k & k < k_{\text{eq}} \\ k^{-3} \ln^2 \frac{k}{k_{\text{eq}}} & k > k_{\text{eq}} \end{cases} \quad (11.1.24)$$

which has excellent agreement with experimental observations (Fig. 11.3), except for the high- k (i.e. small- λ) regime, a discrepancy which is due to the limited applicability of the linear approximation to small-scale fluctuations. Indeed, the linear analysis tends to underestimate the formation of small-scale structure fuelled by the faster non-linear collapse.

§11.1.2.3 Vacuum era

Finally, in vacuum era $a \propto e^{Ht}$, with $H \approx \text{const.}$ during the dark-matter-dominated era. Again $4\pi G\bar{\rho}_m \ll H^2 \propto \rho_\Lambda$ (the density of dark energy does not fluctuate), hence Eq. 11.1.18 reduces to:

$$\ddot{\delta}_m + 2H\dot{\delta}_m = 0 \quad (11.1.25)$$

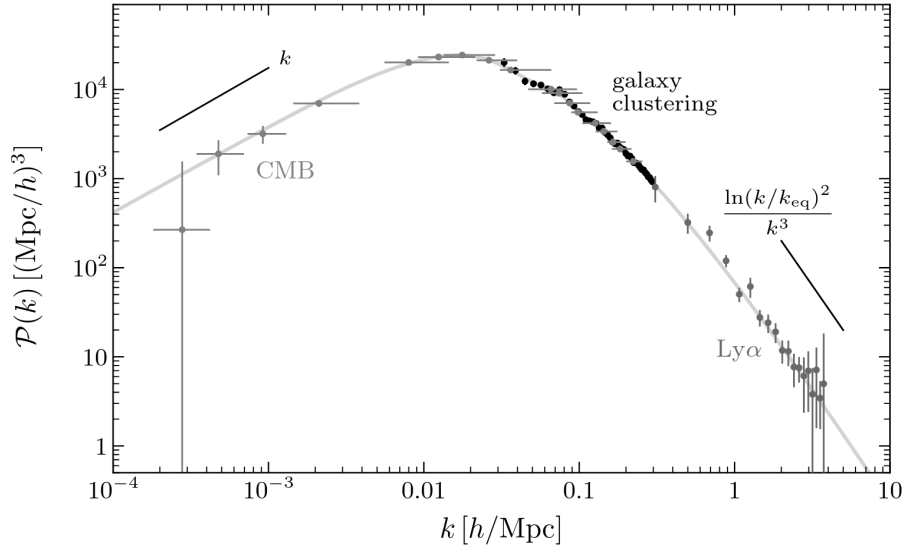


Figure 11.3: Measurements of the linear matter power spectrum.

which, with a power-law ansatz, has two solutions:

$$\delta_m \propto \begin{cases} e^{-2Ht} \propto a^{-2} \\ \text{const.} \end{cases} \quad (11.1.26)$$

which shows that, during the vacuum era, the growth of cosmological perturbations (i.e. of structures) halts.

§11.2 Relativistic perturbation theory

The Newtonian analysis is inadequate on scales larger than the Hubble radius and for relativistic fluids (e.g. photons and neutrinos). Before photon decoupling, photons, electrons and protons are strongly coupled by electromagnetic interactions, forming a single (relativistic) photon-baryon fluid: the matter perturbations then oscillate, since gravity makes perturbations collapse, causing an increase in radiation density that makes them re-expand. At photon decoupling, modes with different wavelengths are captured at different phases in their evolution, giving rise to a characteristic pattern in the CMB. After photon decoupling, the baryons lose the pressure support of the photons and fall into the gravitational potential wells created by dark matter, eventually forming stars and galaxies. These dark matter clusters only start to form after radiation-matter equality, since before that they are suppressed by the large radiation pressure of photons.

§11.2.1 Perturbations

The Einstein field equations Eq. 5.2.10 couple perturbations in the stress-energy tensor to those in the metric, which then need to be studied simultaneously. In particular, working in conformal time, the perturbations with respect to the homogeneous background can be written as:

$$g_{\mu\nu}(\eta, \mathbf{x}) = \bar{g}_{\mu\nu}(\eta) + \delta g_{\mu\nu}(\eta, \mathbf{x}) \quad (11.2.1)$$

$$T_{\mu\nu}(\eta, \mathbf{x}) = \bar{T}_{\mu\nu}(\eta) + \delta T_{\mu\nu}(\eta, \mathbf{x}) \quad (11.2.2)$$

To avoid clutter, from now on the dependence of perturbations from $(\eta, \mathbf{x})$ is suppressed. Note that the unperturbed metric is assumed to be the flat FLRW metric, so that:

$$d\bar{s}^2 \equiv \bar{g}_{\mu\nu} dx^\mu dx^\nu = a^2(\eta) [d\eta^2 - d\mathbf{x}^2]$$

§11.2.1.1 Metric perturbations

Since the metric tensor of a 4D manifold has 10 independent components, $\delta g_{\mu\nu}$ can be written as:

$$ds^2 \equiv g_{\mu\nu} dx^\mu dx^\nu = a^2(\eta) [(1 + 2A) d\eta^2 - 2B_i dx^i d\eta - (\delta_{ij} + h_{ij}) dx^i dx^j] \quad (11.2.3)$$

where A is a scalar, B_i is a spatial 3-vector and h_{ij} is a symmetric spatial 3-tensor. Then, by convention $B^i = \delta^{ij} B_j$ and $h^i_j = \delta^{ik} h_{kj}$.

Lemma 11.2.1 (SVT decomposition of vectors)

Given a 3-vector B_i , it can be decomposed as:

$$B_i = \partial_i B + \hat{B}_i \quad (11.2.4)$$

where B is a scalar and $\hat{B}_i : \partial^i \hat{B}_i = 0$ is a divergenceless 3-vector.

Lemma 11.2.2 (SVT decomposition of symmetric tensors)

Given a symmetric 3-tensor h_{ij} , it can be decomposed as:

$$h_{ij} = 2C\delta_{ij} + 2\partial_{\langle i}\partial_{j\rangle} E + 2\partial_{\langle i}\hat{E}_{j\rangle} + 2\hat{E}_{ij} \quad (11.2.5)$$

where C, E are scalars, $\hat{E}_i : \partial^i \hat{E}_i = 0$ is a divergenceless 3-vector, $\hat{E}_{ij} : \partial^i \hat{E}_{ij} = 0 \wedge \hat{E}^i_i = 0$ is a divergenceless traceless symmetric 3-tensor and:

$$\partial_{\langle i}\partial_{j\rangle} := \partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2 \quad (11.2.6)$$

With the SVT decomposition, the 3 degrees of freedom of B_i are decomposed into 1 + 2 and the 6 of h_{ij} into 1 + 1 + 2 + 2, resulting in a complete decomposition of the 10 degrees of freedom of $\delta g_{\mu\nu}$ into 4 + 4 + 2: scalars A, B, C, E , vectors $\hat{B}_i, \hat{E}_i$ and tensor $\hat{E}_{ij}$.

The SVT decomposition is convenient since, at linear order, the Einstein equations for scalars, vectors and tensors do not mix and can therefore be treated separately. In particular, in the following vector perturbations are ignored, since they are not produced by Inflation, and even if they were, they would decay quickly with the expansion of the universe. On the other hand, $\hat{E}_{ij}$ is a transverse traceless tensor, hence tensor perturbations are truly gravitational waves (Eq. 6.1.19) and will be analyzed later.

Gauge fixing The remaining scalar perturbations can be further simplified with gauge invariance. Indeed, the perturbations in Eq. 11.2.3 are not uniquely defined, but depend on the choice of coordinates, i.e. on the choice of gauge: the chosen form of the perturbed metric has been determined by the implicit introduction of a specific time slicing of the spacetime and definition of specific coordinates on these time slices. In general, choosing different coordinates introduces “fictitious perturbations”, known as *gauge modes*.

In order to avoid these gauge problems, there are two solutions: either define special combinations of the metric perturbations that do not transform under a change of coordinates, called *Bardeen*

variables, or fix a particular gauge. In the following, the **Newtonian gauge** is used, where two of the four scalar perturbations are set to zero:

$$B = E \equiv 0 \quad (11.2.7)$$

The perturbed metric then becomes:

$$ds^2 = a^2(\eta) [(1 + 2\Psi) d\eta^2 - (1 - 2\Phi) \delta_{ij} dx^i dx^j] \quad (11.2.8)$$

where the Bardeen potentials $\Psi \equiv A$ and $\Phi \equiv -C$ have been introduced. For perturbations that decay at spatial infinity, the Newtonian gauge is unique (i.e. the gauge is fixed completely). Note that, in this gauge, time slices are orthogonal to the worldlines of observers at rest in these coordinates (since $B = 0$), and the induced geometry on time slices is isotropic (since $E = 0$), thus making the physics transparent.

The metric in the Newtonian gauge bears similarity to the weak-field limit in Minkowski space: this suggests that Ψ plays the role of a gravitational potential.

§11.2.1.2 Matter perturbations

The perturbations of the stress-energy tensor can be written in term of the perturbed energy density $\rho(\eta, \mathbf{x}) = \bar{\rho}(\eta) + \delta\rho(\eta, \mathbf{x})$ and pressure $P(\eta, \mathbf{x}) = \bar{P}(\eta) + \delta P(\eta, \mathbf{x})$:

$$\delta T^0_0 = \delta\rho \quad (11.2.9)$$

$$\delta T^i_0 = (\bar{\rho} + \bar{P}) v^i \quad (11.2.10)$$

$$\delta T^i_j = -\delta P \delta^i_j - \Pi^i_j \quad (11.2.11)$$

where v^i is a 3-vector, known as *bulk velocity*, and Π^i_j is a symmetric traceless 3-tensor, known as *anisotropic stress*. The 10 degrees of freedom of δT^μ_ν are then correctly decomposed as $1 + 1 + 3 + 5$. In the following analysis, the anisotropic stress is ignored: $\Pi^i_j \equiv 0$.

§11.2.2 Evolution

Lemma 11.2.3 (Christoffel symbols in perturbed FLRW)

The Christoffel symbols associated to the metric Eq. 11.2.8 are:

$$\Gamma^0_{00} = \mathcal{H} + \Psi' \quad (11.2.12)$$

$$\Gamma^0_{i0} = \partial_i \Psi \quad (11.2.13)$$

$$\Gamma^i_{00} = \delta^{ij} \partial_j \Psi \quad (11.2.14)$$

$$\Gamma^0_{ij} = \mathcal{H} \delta_{ij} - [\Phi' + 2\mathcal{H}(\Phi + \Psi)] \delta_{ij} \quad (11.2.15)$$

$$\Gamma^i_{j0} = [\mathcal{H} - \Phi'] \delta^i_j \quad (11.2.16)$$

$$\Gamma^i_{jk} = -2\delta^i_{(j} \partial_{k)} \Phi + \delta_{jk} \delta^{i\ell} \partial_\ell \Phi \quad (11.2.17)$$

where $\mathcal{H} \equiv \frac{a'}{a}$ and $a' := \frac{da}{d\eta}$.

The equations of motion of matter perturbations follow from the covariant conservation of the stress-energy tensor: $\nabla_\mu T^\mu_\nu = 0$. The $\nu = 0$ component is the continuity equation, which at zeroth and linear order in perturbations is:

$$\bar{\rho}' + 3\mathcal{H}(\bar{\rho} + \bar{P}) = 0 \quad (11.2.18)$$

$$\delta\rho' + 3\mathcal{H}(\delta\rho + \delta P) - 3\Phi'(\bar{\rho} + \bar{P}) + \nabla \cdot [(\bar{\rho} + \bar{P})\mathbf{v}] = 0 \quad (11.2.19)$$

The first equation is just the continuity equation Eq. 8.3.5 for the background, while the second equation describes the evolution of the density perturbations: in particular, the first term is the universal dilution due to the expansion, the second term is a local dilution due to the “local scale factor” $(1 - \Phi)a$ in the spatial part of the metric, and the third term is the local divergence of the fluid flow. It is also possible to rewrite the linearized continuity equation in terms of the density contrast $\delta \equiv \delta\rho/\bar{\rho}$.

Proposition 11.2.1 (Continuity equation for the density contrast)

$$\delta' = (1 + c_s^2)(\Phi' - \nabla \cdot \mathbf{v}) \quad (11.2.20)$$

Proof. Since the background is homogeneous, i.e. $\bar{\rho} = \bar{\rho}(\eta)$ and $\bar{P} = \bar{P}(\eta)$, Eq. 11.2.19 becomes:

$$\delta\rho' + 3(1 + c_s^2)(\mathcal{H}\delta\rho - \Phi'\bar{\rho}) + (1 + c_s^2)\bar{\rho}\nabla \cdot \mathbf{v} = 0$$

where the sound velocity $c_s^2 \equiv \bar{P}/\bar{\rho} = w$ has been introduced. Then, recall Eq. 11.2.18, so that:

$$\delta' = \frac{\delta\rho'}{\bar{\rho}} - \delta\frac{\bar{\rho}'}{\bar{\rho}} = \frac{\delta\rho'}{\bar{\rho}} + 3(1 + c_s^2)\mathcal{H}\delta$$

Inserting the above expression for $\delta\rho'$ yields the thesis. $\square$

The $\nu = i$ component is the Euler equation, which reads (using Eq. 11.2.18):

$$\mathbf{v}' + (1 - 3c_s^2)\mathcal{H}\mathbf{v} = -\frac{1}{\bar{\rho} + \bar{P}}\nabla\delta P - \nabla\Psi \quad (11.2.21)$$

where $c_s^2 \equiv \bar{p}'/\bar{\rho}' = w$. Note the equivalence with Eq. 11.1.4 and the fact that, in absence of energy density and pressure, this equation is equivalent to $v \propto a^{-1}$ as expected.

The equations of motion of metric perturbations follow instead from the Einstein field equations: $G^\mu{}_\nu = 8\pi GT^\mu{}_\nu$. The (00)-component at zeroth and linear order is:

$$\mathcal{H}^2 = \frac{8\pi G}{3}a^2\bar{\rho} \quad (11.2.22)$$

$$\nabla^2\Psi - 3\mathcal{H}(\Phi' + \mathcal{H}\Phi) = 4\pi Ga^2\delta\rho \quad (11.2.23)$$

The first equation is just the Friedmann equation Eq. 8.3.12, while the second is the relativistic generalization of a sourced Poisson equation for Ψ , coupled to Φ : for subhorizon modes, i.e. with $k \gg aH = \mathcal{H}$ (since $\partial_\eta = a\partial_t$), it reduces to the Poisson equation in the Newtonian limit $\nabla^2\Psi = 4\pi Ga^2\delta\rho$. Then, the (0i)-component reads:

$$\partial_i(\Phi' + \mathcal{H}\Phi) = -4\pi Ga^2(\bar{\rho} + \bar{P})v_i$$

Applying a SVT decomposition to $v^i = \partial^i v + \hat{v}^i$ and ignoring vector perturbations (as already stated), this equation reduces to a scalar equation:

$$\Phi' + \mathcal{H}\Phi = -4\pi Ga^2(\bar{\rho} + \bar{P})v \quad (11.2.24)$$

which is again a relativistic sourced Poisson equation for Φ . There only remain the spatial components, which are best studied in terms of their trace and traceless parts. The traceless component $(ij - \frac{1}{3}ii)$ simply is:

$$\Phi = \Psi \quad (11.2.25)$$

while the trace component (ii) is:

$$\Phi'' + 3\mathcal{H}\Phi' + (2\mathcal{H}' + \mathcal{H}^2)\Phi = 4\pi G a^2 \delta P \quad (11.2.26)$$

Note that these equations are only valid with $\Pi_j^i \equiv 0$: if the anisotropic stress cannot be ignored, then $\Phi \neq \Psi$ and the last equation has an additional source term.

§11.2.2.1 Metric evolution

The evolution of the potential Φ differs between radiation era and matter era.

Radiation era In radiation era $\mathcal{H} = \eta^{-1}$ (Eq. 8.3.19), so that $2\mathcal{H}' + \mathcal{H}^2 = -\mathcal{H}^2$. Moreover, since $\delta P = \frac{1}{3}\delta\rho$, combining Eq. 11.2.23 and Eq. 11.2.26 results into:

$$\Phi'' + 4\mathcal{H}\Phi' - \frac{1}{3}\nabla^2\Phi = 0 \quad (11.2.27)$$

which is a damped wave equation with $c_s = 1/\sqrt{3}$ (as expected from $c_s^2 = w$). Now, insert the Fourier expansion of the metric perturbation:

$$\Phi(\eta, \mathbf{x}) = \int_{\mathbb{R}^3} \frac{d^3k}{(2\pi)^{3/2}} \Phi_{\mathbf{k}}(\eta) e^{i\mathbf{k}\cdot\mathbf{x}} \quad (11.2.28)$$

so that the wave equation becomes an ODE:

$$\Phi_{\mathbf{k}}'' + \frac{4}{\eta}\Phi_{\mathbf{k}}' + \frac{k^2}{3}\Phi_{\mathbf{k}} = 0$$

The solutions to this equation have a distinct behavior in the superhorizon and subhorizon regimes:

$$\Phi_{\mathbf{k}}(\eta) \approx \begin{cases} \text{const.} & \eta \ll \frac{1}{k} \\ -G\Phi_{\mathbf{k}}(0) \frac{1}{(k\eta)^2} \cos \frac{k\eta}{\sqrt{3}} & \eta \gg \frac{1}{k} \end{cases} \quad (11.2.29)$$

This shows that outside the comoving Hubble horizon no linear evolution happens, and the metric perturbations only evolve with small non-linear terms. On the other hand, inside the comoving Hubble horizon, in radiation era the metric perturbations oscillate with a damped amplitude that decreases as a^{-2} (since $a \propto \eta$).

Matter era In matter era $\mathcal{H} = 2\eta^{-1}$, so that $2\mathcal{H}' + \mathcal{H}^2 = 0$. Moreover, since $w = 0$, Eq. 11.2.26 reduces to:

$$\Phi'' + 3\mathcal{H}\Phi' = 0 \quad (11.2.30)$$

which has two solutions:

$$\Phi(\eta) \propto \begin{cases} \text{const.} \\ \eta^{-5} \propto a^{-5/2} \end{cases} \quad (11.2.31)$$

In matter era, then, the growing solution is constant independently on the wavelength of the fluctuation.

Putting everything together, it is possible to outline the linear evolution of curvature perturbations. Modes with $k > k_{\text{eq}}$, which enter the Hubble radius in radiation era, are affected by the damped oscillations described by Eq. 11.2.29 before freezing in matter era. On the contrary, modes with $k < k_{\text{eq}}$ enter the Hubble radius in matter era, thus freezing immediately (see Fig. 11.4).

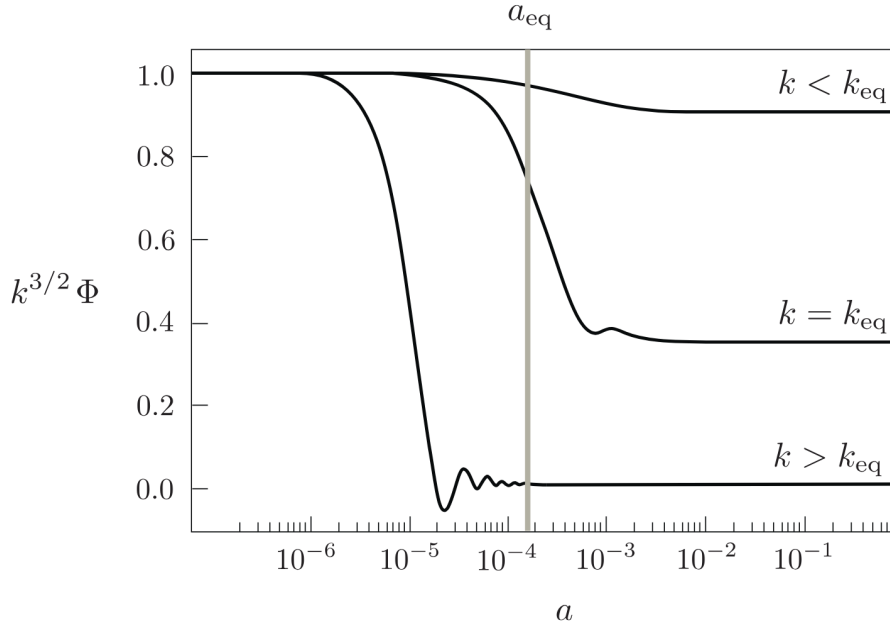


Figure 11.4: Linear evolution of metric perturbations. The small suppression of modes $k > k_{\text{eq}}$ is a minor effect due to the radiation-to-matter transition.

§11.2.2.2 Radiation evolution

The evolution of perturbations in the radiation energy density, encapsulated in the radiation density contrast $\delta_r \equiv \delta\rho_r/\bar{\rho}_r$, is of particular interest, since it predicts the shape of the CMB power spectrum (which is still described by the linear approximation): indeed, since $\rho_r \propto T^4$, then $\delta\rho \propto 4T^3\delta T$ and:

$$\delta_r = 4\delta_T \quad (11.2.32)$$

where $\delta_T \equiv \delta T/\bar{T}$ is the temperature contrast, which is measured in CMB anisotropies. To study the evolution of δ_r , combine Eq. 11.2.22-11.2.23:

$$\delta = \frac{2}{3} \frac{\nabla^2 \Phi}{\mathcal{H}^2} - \frac{2\Phi'}{\mathcal{H}} - 2\Phi \quad (11.2.33)$$

After introducing the Fourier expansion Eq. 11.2.28, this equation can be studied both in radiation era and in matter era.

Radiation era In radiation era δ_r is dominant, i.e. $\delta \approx \delta_r$, and $\mathcal{H} = \eta^{-1}$, so that:

$$\delta_r = -\frac{2}{3}(k\eta)^2\Phi_{\mathbf{k}} - 2\eta\Phi'_{\mathbf{k}} - 2\Phi_{\mathbf{k}}$$

Recalling the expression Eq. 11.2.29 for the metric perturbation, the solution for δ_r is found:

$$\delta_r(\eta) \approx \begin{cases} \text{const.} & \eta \ll \frac{1}{k} \\ 4\Phi_{\mathbf{k}}(0) \cos \frac{k\eta}{\sqrt{3}} & \eta \gg \frac{1}{k} \end{cases} \quad (11.2.34)$$

Note that superhorizon modes are still frozen in the linear regime, while subhorizon radiation fluctuations oscillate without damping (contrary to metric perturbations, which are damped).

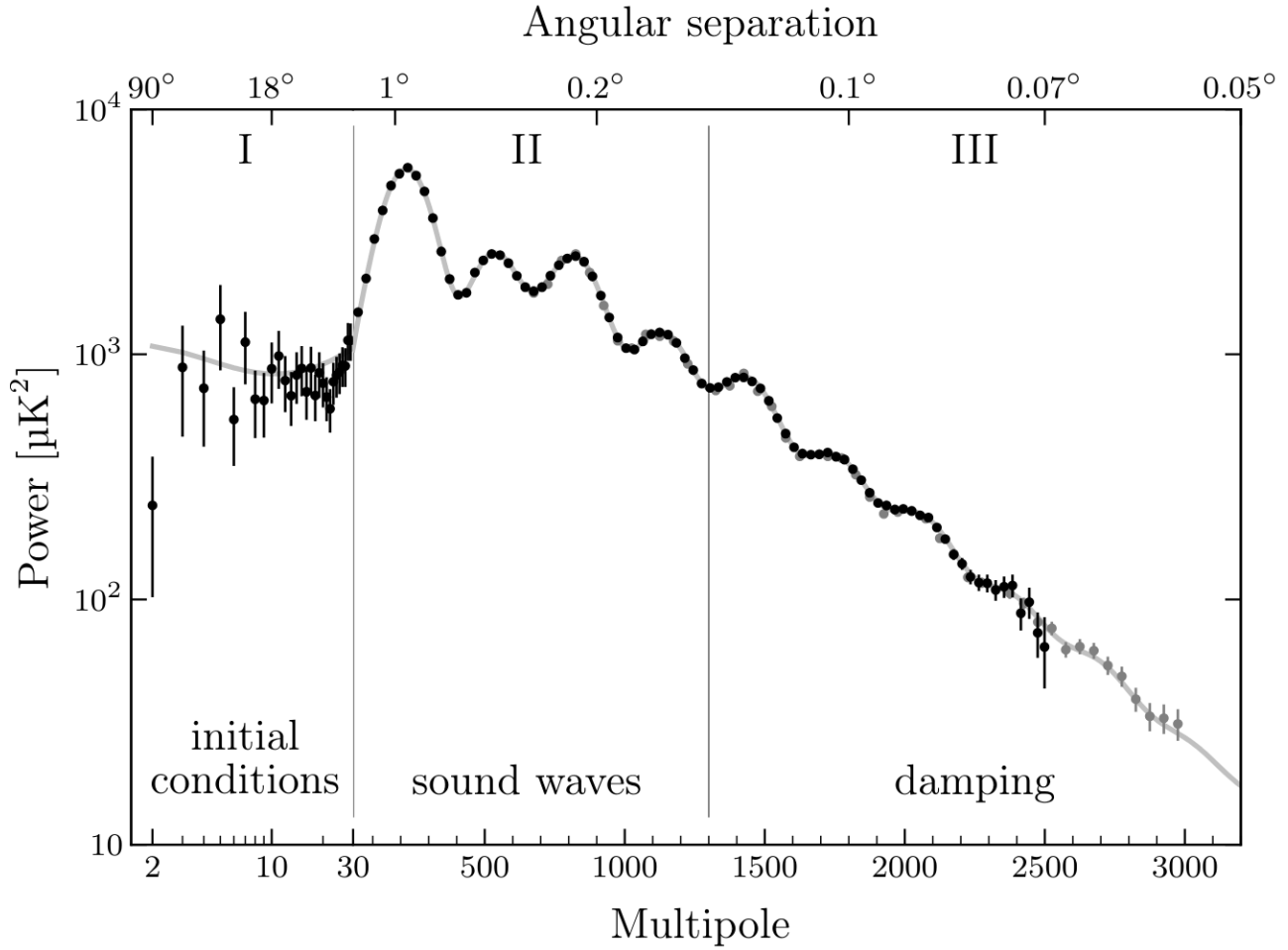


Figure 11.5: Power spectrum of the CMB anisotropies.

Matter era In matter era δ_r is subdominant, and its evolution has to be inferred from the continuity and the Euler equations: subhorizon modes still oscillate with the same frequency as in radiation era, however now with a decreased amplitude and a shift due to the gravitational potential of the dominant matter:

$$\delta_r(\eta) \approx \begin{cases} \text{const.} & \eta \ll \frac{1}{k} \\ -4\Phi_m + A \cos \frac{k\eta}{\sqrt{3}} & \eta \gg \frac{1}{k} \end{cases} \quad (11.2.35)$$

As seen in Fig. 11.1, radiation modes with $k > k_{\text{eq}}$, i.e. those which enter the comoving Hubble horizon in matter era, do not evolve significantly before photon decoupling, while radiation modes with $k < k_{\text{eq}}$ are affected by oscillations, thus arriving at the last-scattering surface with different phases. The power spectrum of the CMB is then expected to have a plateau at low $k < k_{\text{eq}}$ (i.e. at high wavelengths): this is experimentally observed and is known as the **Sachs–Wolfe plateau** (see Fig. 11.6).

§11.2.2.3 CMB power spectrum

The radiation density contrast is not the only contribution to the temperature anisotropies of the CMB. Indeed, the complete expression for δ_T as a function of the direction of observation is:

$$\delta_T(\hat{\mathbf{n}}) = \left(\frac{1}{4}\delta_r + \Psi \right)_* - (\hat{\mathbf{n}} \cdot \mathbf{v}_b)_* + \int_{\eta_*}^{\eta_0} d\eta (\Phi' + \Psi') \quad (11.2.36)$$

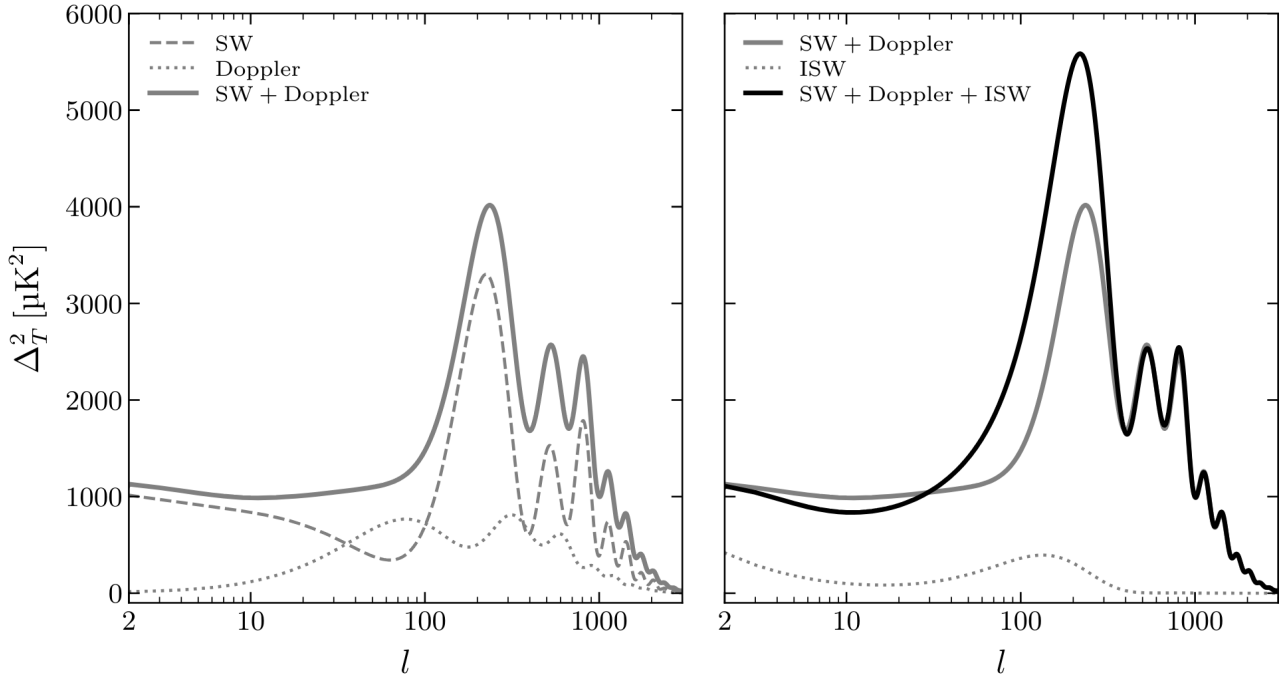


Figure 11.6: Illustration of the different contributions to the CMB power spectrum.

where $\mathbf{v}_b$ is the bulk velocity of baryons relative to the observer RF, and the subscript $*$ denotes the evaluation at η_* , i.e. at last scattering. These three terms are separate contributions:

- the first term is the *Sachs–Wolfe term*, which combines the intrinsic temperature fluctuations associated to the radiation density contrast with the induced temperature perturbations arising from the gravitational redshift of the photons. With the sign convention adopted in Eq. 11.2.8, $\Psi < 0$ corresponds to an overdensity (since, in the Newtonian limit, $\Psi \propto -M$), hence $\Psi_* < 0$ leads as expected to a temperature decrement, since a photon loses energy when climbing out of a potential well;
- the second term is a *Doppler term*, due to the relative motion of the last-scattering surface with respect to the observer (see Fig. 11.7). This contribution is subdominant on superhorizon scales, while its oscillations are out of phase with the oscillations in the Sachs–Wolfe term, thus reducing the contrast between the peaks in the spectrum;
- the last term describes the additional gravitational redshift due to the evolution of the metric potentials along the line-of-sight, determining what is known as the *integrated Sachs–Wolfe effect* (ISW). For most of the universe’s history (namely, when it was matter dominated) these potential were constants, thus yielding no ISW effect. On the contrary, at early times the residual amount of radiation gives time-dependent gravitational potentials, resulting in an *early ISW effect*: this contribution raises the height of the first peak in the power spectrum (see Fig. 11.6).

The CMB power spectrum Fig. 11.5 imposes some constraints on the amount of matter, and in particular of dark matter (since baryonic matter is observed), in the universe: indeed, the extension of the Sachs–Wolfe plateau is determined by k_{eq} , and changing the amount of matter changes the time at which matter–radiation equality happens. Moreover, the second and third peaks of the power spectrum are sensible to the gravitational potential of matter: varying the amount of matter causes a variation in the height difference between the peaks.



Figure 11.7: Doppler effect on the last-scattering surface.

Appendices

Complements of Topology

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§A.1 Homotopy

Definition A.1.1 (Homotopy)

Let X, Y be topological spaces, and let $f_0, f_1 : X \rightarrow Y$ be continuous maps. A **homotopy** from f_0 to f_1 is a continuous map $H : X \times [0, 1] \rightarrow Y$ such that, for all $x \in X$:

$$H(x, 0) = f_0(x) \quad H(x, 1) = f_1(x)$$

If there exists a homotopy from f_0 to f_1 , these maps are said to be *homotopic*, which is seldom written as $f_0 \simeq f_1$. If $H(x, t) = f_0(x) = f_1(x) \forall t \in [0, 1]$ at all points $x \in A \subseteq X$, then f_0 and f_1 are said to be *homotopic relative to A* . Both of these relations are equivalence relations on the set of all continuous maps from X to Y .

Definition A.1.2 (Path homotopy)

Let X be a topological space. Two paths $f_0, f_1 : [0, 1] \rightarrow X$ are **path-homotopic** if they are homotopic relative to $\{0, 1\}$.

Explicitly, two paths $f_0, f_1 : [0, 1] \rightarrow X$ are path-homotopic if there exists a continuous map $H : [0, 1]^2 \rightarrow X$ such that, for all $s, t \in [0, 1]$:

$$\begin{aligned} H(s, 0) &= f_0(s) & H(0, t) &= f_0(0) = f_1(0) \\ H(s, 1) &= f_1(s) & H(1, t) &= f_0(1) = f_1(1) \end{aligned}$$

For any given $p, q \in X$, path-homotopy is an equivalence relation on the set of all paths from p to q , denoted by $f_0 \sim f_1$: the equivalence class of a path f is called its *path class* $[f]$.

Given two paths $f, g : [0, 1] \rightarrow X$ such that $f(1) = g(0)$, their product is the path $f \cdot g : [0, 1] \rightarrow X$ defined as:

$$f \cdot g(s) = \begin{cases} f(2s) & 0 \leq s \leq \frac{1}{2} \\ g(2s - 1) & \frac{1}{2} \leq s \leq 1 \end{cases} \quad (\text{A.1.1})$$

If $f \sim f'$ and $g \sim g'$, then $f \cdot g \sim f' \cdot g'$, hence it makes sense to define the product of path classes: $[f] \cdot [g] = [f \cdot g]$. Note that this product is not associative, so the convention of evaluation from left to right $f \cdot g \cdot h = (f \cdot g) \cdot h$ is adopted; however, it is associative up to path-homotopy, hence $([f] \cdot [g]) \cdot [h] = [f] \cdot ([g] \cdot [h])$.

Given $q \in X$, a *loop* in X at q is a path in X from q to q , i.e. a continuous map $f : [0, 1] \rightarrow X$ such that $f(0) = f(1) = q$. The set of path classes of loops at q is denoted by $\pi_1(X, q)$, and it has a group structure with the product Eq. A.1.1, hence it is called the **fundamental group** of X at q : the identity element is the path class of the *constant paths* $c_q(s) \equiv q$, while the inverse of $[f]$ is the path class of the *reverse path* $\bar{f}(s) = f(1 - s)$. It can be shown that, for a path-connected topological space, the fundamental groups at different points are isomorphic.

Definition A.1.3 (Simply connected space)

If X is a path-connected topological space and, for some $q \in X$, the fundamental group $\pi_1(X, q)$ is the trivial group consisting of $[c_q]$ alone, then X is said to be **simply connected**.

This means that in a simply connected space every loop is path-homotopic to a constant path (i.e. to a point).

Proposition A.1.1

Let X be a path-connected topological space. Then, X is simply connected if and only if every pair of paths in X with the same starting and ending points are path-homotopic.

A key feature of homotopy is that it is preserved by composition.

Proposition A.1.2

Let X, Y, Z be topological spaces, and let $f_0, f_1 : X \rightarrow Y$ and $g_0, g_1 : Y \rightarrow Z$ be continuous maps such that $f_0 \simeq f_1$ and $g_0 \simeq g_1$. Then, $g_0 \circ f_0 \simeq g_1 \circ f_1$.

Similarly, given $f_0 \sim f_1$ paths in X and a continuous map $F : X \rightarrow Y$, then $F \circ f_0 \sim F \circ f_1$. Thus, for each $q \in X$ there is a well defined map:

$$F_* : \pi_1(X, q) \rightarrow \pi_1(Y, F(q)) : [f] \mapsto [F \circ f] \quad (\text{A.1.2})$$

which is a group homomorphism, known as the *homomorphism induced by F* .

Lemma A.1.1 (Properties of the induced homomorphism)

- (1) Let $F : X \rightarrow Y$ and $G : Y \rightarrow Z$ be continuous maps. Then, $(G \circ F)_* = G_* \circ F_* : \pi_1(X, q) \rightarrow \pi_1(Z, G(F(q)))$ for each $q \in X$.
- (2) For each topological space X and each $q \in X$, $(\mathbb{1}_X)_* = \mathbb{1}_{\pi_1(X, q)}$.
- (3) If $F : X \rightarrow Y$ is a homeomorphism, then $F_* : \pi_1(X, q) \rightarrow \pi_1(Y, F(q))$ is an isomorphism.

By (3) of Lemma A.1.1, homeomorphic topological spaces have isomorphic fundamental groups.

Example A.1.1 (Fundamental groups of spheres)

$\pi_1(\mathbb{S}^1, (1, 0))$ is the infinite cyclic group generated by the path class of the loop $\omega : [0, 1] \rightarrow \mathbb{S}^1$ defined as $\omega(s) = (\cos 2\pi s, \sin 2\pi s)$. On the contrary, $\mathbb{S}^n$ is simply connected for all $n > 1$.

Proposition A.1.3 (Fundamental group of product spaces)

Let $X_1, \dots, X_k$ be topological spaces, and let $p_i : X_1 \times \dots \times X_k \rightarrow X_i$ be the i^{th} projection map.

For any points $q_i \in X_i$, with $i = 1, \dots, k$, the following map is an isomorphism:

$$\begin{aligned} P : \pi_1(X_1 \times \cdots \times X_k, (q_1, \dots, q_k)) &\rightarrow \pi_1(X_1, q_1) \times \cdots \times \pi_1(X_k, q_k) \\ [f] &\mapsto (p_{1*}[f], \dots, p_{k*}[f]) \end{aligned}$$

A continuous map $F : X \rightarrow Y$ between topological spaces is said to be a **homotopy equivalence** if there is a continuous map $G : Y \rightarrow X$ such that $F \circ G$ is homotopic to $\mathbb{1}_Y$ and $G \circ F$ is homotopic to $\mathbb{1}_X$: such a map G is called the *homotopy inverse* of F . If there exists a homotopy equivalence between two topological spaces, they are said to be *homotopy equivalent*.

Example A.1.2 (Homotopy equivalence of spheres)

The inclusion map $\iota : \mathbb{S}^{n-1} \hookrightarrow \mathbb{R}^n - \{0\}$ is a homotopy equivalence with homotopy inverse $r(x) = x/\|x\|$, since $r \circ \iota = \mathbb{1}_{\mathbb{S}^{n-1}}$ and $\iota \circ r$ is homotopic to $\mathbb{1}_{\mathbb{R}^n - \{0\}}$ via a straight-line homotopy:

$$H(x, t) = tx + (1 - t) \frac{x}{\|x\|}$$

Theorem A.1.1 (Homotopy invariance)

If $F : X \rightarrow Y$ is a homotopy equivalence, then $F_* : \pi_1(X, p) \rightarrow \pi_1(Y, F(p))$ is an isomorphism for each $p \in X$.

Covariant Maxwell Theory

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§B.1 Maxwell action

Consider spacetime as a manifold $\mathcal{M}$. The electromagnetic field can be described by a form on this manifold: indeed, the electromagnetic gauge field $A_\mu = (\phi, \mathbf{A})$ is to be thought as the components of a 1-form $A = A_\mu(x)dx^\mu$. The exterior derivative of this form is a 2-form $F = dA$:

$$F = \frac{1}{2}F_{\mu\nu}dx^\mu \wedge dx^\nu = \frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu)dx^\mu \wedge dx^\nu$$

The components $F_{\mu\nu}$ are in reality the components of a tensor, the Faraday tensor. By construction, a useful identity holds, sometimes called the **Bianchi identity**:

$$dF = 0 \tag{B.1.1}$$

From this identity derive two Maxwell equations: $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$. Moreover, note that the gauge field is not unique: the gauge transformation $A \mapsto A + d\alpha$, which equals $A_\mu \mapsto A_\mu + \partial_\mu \alpha$, leaves F unchanged.

To study the dynamics of these fields, an action is needed: Differential Geometry allows very few actions to be written down.

For example, suppose that on the considered manifold no metric is defined. To integrate over $\mathcal{M}$ a 4-form is needed, but F is a 2-form, thus the only possible action is:

$$\mathcal{S}_{\text{top}} = -\frac{1}{2} \int F \wedge F \tag{B.1.2}$$

The integrand become $dx^0 dx^1 dx^2 dx^3 \mathbf{E} \cdot \mathbf{B}$. Actions of this kind, independent of the metric, are called **topological actions** and are of no interest in classical physics: in fact, $F \wedge F = d(A \wedge F)$, so the action is a total derivative and does not affect the equations of motion.

To construct an action of classical interest, a metric is needed. This allows to introduce a second 2-form, $\star F$, so to construct the Maxwell action:

$$\mathcal{S}_M = -\frac{1}{2} \int F \wedge \star F \tag{B.1.3}$$

The integrand can then be expanded as:

$$\mathcal{S}_M = -\frac{1}{4} \int d^4x \sqrt{g} g^{\mu\nu} g^{\rho\sigma} F_{\mu\rho} F_{\nu\sigma} = -\frac{1}{4} \int d^4x \sqrt{g} F^{\mu\nu} F_{\mu\nu}$$

In flat spacetime $F^{\mu\nu}F_{\mu\nu} = 2(\mathbf{B}^2 - \mathbf{E}^2)$. In a general curved spacetime, the equation of motion resulting from the variation of the Maxwell action is $d \star F = 0$.

To complete the theory, consider a gauge field coupled to a current, described by a 1-form J . The Maxwell action then becomes:

$$\mathcal{S}_M = \int -\frac{1}{2} F \wedge \star F + A \wedge \star J \quad (\text{B.1.4})$$

This action must retain its gauge invariance, but under $A \mapsto A + d\alpha$ it transforms as $\mathcal{S}_M \mapsto \mathcal{S}_M + \int d\alpha \wedge \star J$, therefore, after integrating by parts, the condition of gauge invariance translates to:

$$d \star J = 0 \quad (\text{B.1.5})$$

This is current conservation in the language of forms. Varying the action in Eq. B.1.3 now leads to the Maxwell equations with source terms:

$$d \star F = \star J \quad (\text{B.1.6})$$

To define electric and magnetic charges, integrate over submanifolds. Consider a 3-dimensional spatial submanifold Σ : the electric charge in Σ is defined as:

$$Q_e(\Sigma) := \int_{\Sigma} \star J \quad (\text{B.1.7})$$

This agrees with the usual definition in flat spacetime $Q_e = \int_{\Sigma} d^3x J^0$. Using the equations of motion and Stokes' theorem, a general form of Gauss' law is obtained:

$$Q_e(\Sigma) = \int_{\partial\Sigma} \star F \quad (\text{B.1.8})$$

Similarly, the magnetic charge in Σ is defined as:

$$Q_m(\Sigma) := \int_{\partial\Sigma} F \quad (\text{B.1.9})$$

The non-existence of magnetic charges, following from Bianchi identity, can be evaded in topologically interesting manifolds.

From charge conservation in Eq. B.1.5, it follows that the electric charge in a region cannot change, unless current flows in or out of that region. Consider a cylindrical region of spacetime V , ending in two spacelike hypersurfaces Σ_1 and Σ_2 : its boundary is $\partial V = \Sigma_1 \cup \Sigma_2 \cup B$, where B is a cylindrical timelike hypersurface. The statement that no current flows in or out of V means that $J|_B = 0$. Then:

$$Q_e(\Sigma_1) - Q_e(\Sigma_2) = \int_{\Sigma_1} \star J - \int_{\Sigma_2} \star J = \int_{\partial V} \star J - \int_B \star J = \int_{\partial V} \star J = \int_V d \star J = 0$$

Thus, electric charge in remains constant in time.

§B.1.1 Maxwell equations from connections

First note that, given the gauge field $A \in \Omega^1(\mathcal{M})$, the field strength can be expressed via covariant derivatives:

$$F_{\mu\nu} = \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu} = \nabla_{\mu}A_{\nu} - \nabla_{\nu}A_{\mu}$$

The Christoffel symbols cancel out due to anti-symmetry: this is what allows to define the exterior derivative without introducing connections first.

Proposition B.1.1 (Current conservation)

Current conservation can be written as: $d \star J = 0 \iff \nabla_\mu J^\mu = 0$.

Proof. Recalling Lemma 2.5.1:

$$\nabla_\mu J^\mu = \partial_\mu J^\mu + \Gamma_{\mu\rho}^\mu J^\rho = \partial_\mu J^\mu + \partial_\rho (\log \sqrt{g}) J^\rho = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} J^\mu) \propto d \star J$$

Hence, $d \star J = 0 \iff \nabla_\mu J^\mu = 0$. □

As an aside, in general the divergence in different coordinate systems can be computed using the formula $\nabla_\mu J^\mu = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} J^\mu)$.

Proposition B.1.2

$$d \star F = \star J \iff \nabla_\mu F^{\mu\nu} = J^\nu \quad (\text{B.1.10})$$

Proof. Recalling Lemma 2.5.1:

$$\nabla_\mu F^{\mu\nu} = \partial_\mu F^{\mu\nu} + \Gamma_{\mu\rho}^\mu F^{\rho\nu} + \Gamma_{\mu\rho}^\nu F^{\mu\rho} = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} F^{\mu\nu})$$

where $\Gamma_{\mu\rho}^\nu F^{\mu\rho} = 0$ because $\Gamma_{\mu\rho}^\nu$ is symmetric in μ and ρ , while $F^{\mu\rho}$ is anti-symmetric. The proof follows recalling the definition of the Hodge dual in Def. 3.2.1. □

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